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Alatancang Chen
Deyu Wu
Junjie Huang
Guolin Hou

Spectral Analysis of Infinite-Dimensional Hamiltonian Operators

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Guolin Hou

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Alatancang Chen
School of Mathematical Sciences
Inner Mongolia Normal University
Hohhot, Nei Mongol, China

Deyu Wu
School of Mathematical Sciences
Inner Mongolia University
Hohhot, Nei Mongol, China

Junjie Huang
School of Mathematical Sciences
Inner Mongolia University
Hohhot, Nei Mongol, China

Guolin Hou
School of Mathematical Sciences
Inner Mongolia University
Hohhot, Nei Mongol, China

Translated by Xiaoli Wang and Fan Gao

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Preface

Hamiltonian principle has become the cornerstone of modern physics.

—Erwin Schrödinger

Linear operator theory in Hilbert space, in particular, the spectral theory of self-adjoint operators is one of the most important achievements in the field of mathematical sciences in the twentieth century. The founder of the spectral theory of bounded self-adjoint operators is D. Hilbert. He expounded the idea of the spectral theory of bounded self-adjoint operators in six papers on integral equations from 1904 to 1910. Then, from 1927 to 1929, J. von Neumann extended the spectral theory of bounded self-adjoint operators to the unbounded case, in order to study quantum mechanics problems. In the 1930s, F. Riesz and M. S. Stone further improved the spectral theory framework of unbounded self-adjoint operators. Currently, the spectral theory of bounded and unbounded self-adjoint operators has formed a relatively complete framework system. Representative results include the Hilbert-Schmidt theorem, Sturm-Liouville problem, and so on. The well-known separation of variables method solving partial differential equations, also known as Fourier series method, is based on the spectral theory of self-adjoint operators.

It is worth mentioning that when the corresponding state operator of the system is a self-adjoint operator, the energy of the system is conserved. However, there are many closed systems with energy conservation in practical problems whose state operators are non-self-adjoint. Cases in point are that the continuous spectral perturbation problem in the fields of scattering theory and quantum field theory and the eigenvalue perturbation problem in quantum mechanics all belong to the spectral theory of non-self-adjoint operators. It therefore becomes very necessary to study the spectral theory of non-self-adjoint operators and extend the application of Fourier series method. From 1908 to 1913, G. D. Birkhoff discussed the eigenfunction expansion of ordinary differential operators with non-self-adjoint boundary conditions, and initiated the research field of the spectral theory of non-self-adjoint operators. At that time, G. D. Birkhoff used the resolvent operator method. Until the 1950s, M. V. Keldyš

established the completeness theory of eigenvectors and related vectors of non-self-adjoint operators, which obtained not only the spectral decomposition theory of non-self-adjoint operators but also some important results on the completeness of the root vector system of non-self-adjoint operators. Later, T. Kato, F. Wolf et al. founded the perturbation theory of closed linear operators, and solved the problems such as the stability of the spectrum of non-self-adjoint operators, which provides a theoretical guarantee for the study of partial differential operators with non-self-adjoint boundary conditions and lays a foundation for the study of spectral theory of general non-self-adjoint operators. At present, the research methods of spectral theory of non-self-adjoint operators mainly include resolvent operator method, block operator matrix method, and perturbation theory. Although some important breakthroughs have been made, a perfect framework system has not yet been formed. The spectral theory of non-self-adjoint operators has become one of the important theoretical problems to be solved in the field of contemporary mathematics and mechanics.

As mentioned above, the theoretical research concerning spectral theory of general non-self-adjoint operators is much more complicated than that of self-adjoint operators. In view of this, we can study the operators whose research difficulty is between self-adjoint operators and general non-self-adjoint operators. The infinite-dimensional Hamiltonian operator belongs to this class of operators, which is a block operator matrix with a profound mechanical background derived from a linear infinite-dimensional Hamiltonian canonical system. As we know, the problem of coefficient calculation in the eigenvector expansion can be solved by using the orthogonality of the eigenvector system of self-adjoint operators. The eigenvector system of general non-self-adjoint operators is not orthogonal, so the calculation of coefficients in the eigenvector expansion is still a difficult problem. However, the eigenvector system of the infinite-dimensional Hamiltonian operator has symplectic orthogonality, and the symplectic orthogonality can be used to solve the calculation of coefficients in the eigenvector expansion, which is the theoretical basis of a new systematic methodology of elasticity founded by academician Wanxie Zhong.

The classical Hamiltonian system has to be mentioned when we discuss the creation and development of the infinite-dimensional Hamiltonian canonical system. In the 1830s, W. R. Hamilton creatively studied the classical mechanics according to the profound relationship between geometrical optics and mechanics, and obtained the classical Hamiltonian system (see [1])

$$\begin{cases} \dot{\mathrm{q}} = \frac{\partial \mathscr{H}}{\partial \mathrm{p}}, \\ \dot{\mathrm{p}} = -\frac{\partial \mathscr{H}}{\partial \mathrm{q}}, \end{cases}$$

where $\dot{\mathrm{q}} = \frac{\mathrm{dq}}{\mathrm{d}t}$. Let $\mathscr{H}(Z) = \frac{1}{2} Z^T S(t) Z$ with $S(t) = S(t)^T$, then we obtain the linear Hamiltonian system (see [2])

$$\dot{Z} = H(t) Z, \tag{1}$$

where $Z \in \mathbb{R}^{2n}, t \in \mathbb{R}$, the coefficient matrix $H(t)$ is a bounded measurable real matrix valued function and satisfies

$$(J_{2n}H)(t)^T = J_{2n}H(t)$$

with $J_{2n} = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$. The matrix $H(t)$ is called a Hamiltonian matrix, and has block form of

$$H(t) = \begin{bmatrix} A(t) & B(t) \\ C(t) & -A(t)^T \end{bmatrix},$$

where $A(t), B(t), C(t) \in \mathbb{R}^{n\times n}$, $B(t)^T = B(t), C(t)^T = C(t)$, and A^T denote the transpose of A.

Until the 1960s, in order to study the partial differential equations derived from continuum mechanical problems such as the KdV equation, Schrödinger equation, and Maxwell's equations, scholars such as F. Magri and V. I. Arnold introduced the infinite-dimensional Hamiltonian canonical system

$$\dot{u} = J\frac{\delta \mathscr{H}}{\delta u}, \tag{2}$$

where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$, $\mathscr{H}$ is a Hamiltonian functional, and $\frac{\delta \mathscr{H}}{\delta u}$ is its variational derivative. The introduction of infinite-dimensional Hamiltonian canonical system has far-reaching significance for it opens up a new way of solving applied partial differential equations. If extending the finite-dimensional linear Hamiltonian canonical system to the infinite-dimensional case, we have the linear infinite-dimensional Hamiltonian canonical system (see [3])

$$\dot{U} = HU, \tag{3}$$

where H is called an infinite-dimensional Hamiltonian operator which is a block operator matrix of the form

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}, \tag{4}$$

A is a densely defined closed linear operator in an infinite-dimensional Hilbert space X, A^* is the adjoint of A, and B, C are self-adjoint operators in X. The following will list some important application fields of Hamiltonian matrices and infinite-dimensional Hamiltonian operators.

Hamiltonian Matrices in Control Theory. Linear quadratic regulator problem is an important topic in control theory, which can be solved by Hamiltonian matrices.

The so-called linear quadratic regulator problem is to find a control

$$\mathrm{u}(t) : [0, \tau] \to \mathbb{R}^m$$

such that

$$L_{\mathrm{x}_0}(\mathrm{x}(t), \mathrm{u}(t)) = (\mathrm{x}(t), S\mathrm{x}(t)) + \int_0^\tau [(\mathrm{x}(t), G(t)\mathrm{x}(t)) + (\mathrm{u}(t), R\mathrm{u}(t))]\mathrm{d}t \quad (5)$$

attains the minimum value, where $\tau \in (0, \infty]$, S is a symmetric positive semi-definite matrix, $G(t)$, $R(t)$ are bounded continuous matrix-valued functions with $G(t)^T = G(t) \geq 0, R(t)^T = R(t) > 0$ for any $t \in \mathbb{R}$, and $\mathrm{x}(t)$ satisfies the linear control problem

$$\begin{cases} \dot{\mathrm{x}}(t) = A(t)\mathrm{x}(t) + B(t)\mathrm{u}_0, \mathrm{x}(t) \in \mathbb{R}^n, \mathrm{u}_0 \in \mathbb{R}^m, \\ \mathrm{x}(0) = \mathrm{x}_0, \end{cases} \quad (6)$$

with $A(t) \in \mathbb{R}^{n\times n}, B(t) \in \mathbb{R}^{n\times m}$ given bounded continuous matrix-valued functions. By Pontryagin's maximum principle (see [2]), we know that $\mathrm{u}(t)$, which minimizes the expression (5), is bound to maximize the Hamiltonian functional

$$\mathscr{H}(t, \mathrm{x}, \mathrm{y}, \mathrm{u}) = (\mathrm{y}, A(t)\mathrm{x}(t) + B(t)\mathrm{u}(t)) - \frac{1}{2}[(\mathrm{x}(t), G(t)\mathrm{x}(t)) + (\mathrm{u}(t), R(t)\mathrm{u}(t))] \quad (7)$$

and hence

$$\mathrm{u}(t) = R^{-1}(t)B(t)^T\mathrm{y}.$$

Then considering the Hamiltonian canonical equation

$$\begin{cases} \dot{\mathrm{x}} = \frac{\partial \mathscr{H}}{\partial \mathrm{y}}, \\ \dot{\mathrm{y}} = -\frac{\partial \mathscr{H}}{\partial \mathrm{x}} \end{cases}$$

gives the linear Hamiltonian system

$$\dot{\mathrm{z}} = \begin{bmatrix} A(t) & B(t)R^{-1}(t)B(t)^T \\ G(t) & -A(t)^T \end{bmatrix} \mathrm{z}, \quad (8)$$

where $\mathrm{z} = [\mathrm{x}\,\mathrm{y}]^T$. When the system (6) satisfies the standard controllability and observability conditions, there exists a negative definite symmetric matrix $M(t) \in \mathbb{R}^n$ such that

$$\mathrm{z} = \begin{bmatrix} \mathrm{x} \\ M(t)\mathrm{x} \end{bmatrix},$$

which means

$$\mathrm{u}(t) = R^{-1}(t)B(t)^T M(t)\mathrm{x}$$

is exactly the unique solution of the quadratic regulator problem (5).

Hamiltonian Matrix and Schrödinger Equation. Consider the linear Hamiltonian system

$$\dot{Z} = H(t)Z, \tag{9}$$

where $Z = [\mathrm{x}\ \mathrm{y}]^T$, $\mathrm{x}, \mathrm{y} \in \mathbb{R}^n$ and

$$H(t) = \begin{bmatrix} 0 & I_n \\ G(t) & 0 \end{bmatrix}$$

with matrix-valued function $G(t) \in \mathbb{R}^{n\times n}$ satisfies $G(t)^T = G(t)$, so the system (9) is equivalent to the second-order differential equation

$$\frac{\mathrm{d}^2\mathrm{x}}{\mathrm{d}t^2} = G(t)\mathrm{x}. \tag{10}$$

The differential equation (10) often appears in mechanical systems close to equilibrium. Further, let $n = 1$ and

$$H(t) = \begin{bmatrix} 0 & \frac{1}{p(t)} \\ g(t) - \lambda q(t) & 0 \end{bmatrix},$$

where $p(t)$, $g(t)$, $q(t)$ are real-valued functions satisfying certain conditions, then the system (9) is equivalent to the Sturm-Liouville problem

$$-\frac{\mathrm{d}}{\mathrm{d}t}\left(p(t)\frac{\mathrm{dx}}{\mathrm{d}t}\right) + g(t)\mathrm{x} = \lambda q(t)\mathrm{x}. \tag{11}$$

Let $p(t) = q(t) = 1$, then Eq. (11) is reduced to the Schrödinger equation

$$-\frac{\mathrm{d}^2\mathrm{x}}{\mathrm{d}t^2} + g(t)\mathrm{x} = \lambda\mathrm{x}. \tag{12}$$

Equation (12) has important applications in one-dimensional quantum mechanics.

Infinite-dimensional Hamiltonian Operators and Riccati Algebraic Equations. Algebraic Riccati equation

$$-\Pi B\Pi + \Pi A + A^*\Pi + C = 0 \tag{13}$$

has important applications in optimal filter design and control theory (see [4]), and its solution problem has attracted much attention, where Π is the operator to be determined, $B^* = B$, $C^* = C$ and A are given operators. However, the solving of equation (13) is closely related to the infinite-dimensional Hamiltonian operator. In fact, if there exists operators $P,\ Q,\ R$ such that

$$\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\begin{bmatrix} P \\ Q \end{bmatrix} = \begin{bmatrix} P \\ Q \end{bmatrix}R \tag{14}$$

and P is invertible, then $\Pi = QP^{-1}$ is a solution of equation (13).

Infinite-dimensional Hamiltonian Operators and New Systematic Methodology for Theory of Elasticity. Academician Wanxie Zhong transformed the equations in elasticity into Hamiltonian systems and created a new systematic methodology for theory of elasticity by using the minimum potential energy variational principle in elasticity to choose the state and its dual variables as well as simulating time independent variable, based on the simulation theory of structural mechanics and optimal control found by himself. The new solving system extends the Strum-Liouville problem and its expansion method in terms of the eigenfunction system. It not only transforms the equation in elasticity whose variables cannot be separated into the Hamiltonian canonical system with variable separation form but also, by using the symplectic orthogonality of the eigenfunction system of the infinite-dimensional Hamiltonian operator, solves the calculation of coefficients of the eigenfunction expansion of non-self-adjoint operators. It is therefore praised by Qian Lingxi as a new beginning of elasticity.

Take the thin plate bending equation (see [5])

$$\frac{\partial^2 M_x}{\partial x^2} - 2\frac{\partial^2 M_{xy}}{\partial x \partial y} + \frac{\partial^2 M_y}{\partial y^2} = 0 \tag{15}$$

as an example. The corresponding infinite-dimensional Hamiltonian operator is

$$H = \begin{bmatrix} 0 & v\frac{\mathrm{d}}{\mathrm{d}y} & D(1-v^2) & 0 \\ -\frac{\mathrm{d}}{\mathrm{d}y} & 0 & 0 & 2D(1-v) \\ 0 & 0 & 0 & -\frac{\mathrm{d}}{\mathrm{d}y} \\ 0 & -\frac{\mathrm{d}^2}{D\mathrm{d}y^2} & v\frac{\mathrm{d}}{\mathrm{d}y} & 0 \end{bmatrix},$$

where $D > 0$ is a constant and

$$\mathscr{D}(H) = \left\{ \begin{bmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \\ \psi_4(x) \end{bmatrix} \right.$$

$$\in X : \left. \begin{array}{l} \psi_1(0) = \psi_1(b) = 0, \ \frac{\mathrm{d}\psi_2(0)}{D\mathrm{d}x} - v\psi_3(0) = 0, \\ \frac{\mathrm{d}\psi_2(b)}{D\mathrm{d}x} - v\psi_3(b) = 0, \\ \psi_i(x) \quad \text{is absolutely continuous, } \psi_i'(x) \in L^2(0, b), i = 1, 3, 4, \\ \psi_2(x), \psi_2'(x) \quad \text{are absolutely continuous, } \psi_2''(x) \in L^2(0, b) \end{array} \right\}.$$

Therefore, through the spectral properties of the infinite-dimensional Hamiltonian operator H, such as the symplectic orthogonality and completeness of the eigenfunction system, the problem of solving the above thin plate bending equation can be solved.

Infinite-dimensional Hamiltonian Operators and Complete Indefinite Inner Product Spaces. The operator theory on indefinite inner product spaces is not a simple logical extension of that on Hilbert spaces, but has a profound theoretical basis. Its application involves physics, mathematics and mechanics. The space-time space in Einstein's theory of relativity is just an indefinite inner product space. If we study its spectral theory on the indefinite inner product space, we will achieve half the work with double results due to the special structure of the infinite-dimensional Hamiltonian operator.

For example, consider the complete indefinite inner product space $\{X \times X, [\cdot, \cdot]_{\mathfrak{J}_1}\}$, where

$$\mathfrak{J}_1 = \begin{bmatrix} 0 & \mathrm{i}I \\ -\mathrm{i}I & 0 \end{bmatrix}$$

and define the indefinite inner product

$$[\cdot, \cdot]_{\mathfrak{J}_1} = (\mathfrak{J}_1 \cdot, \cdot),$$

then the infinite-dimensional Hamiltonian operator is an anti-symmetric operator in the indefinite inner product space $\{X \times X, [\cdot, \cdot]_{\mathfrak{J}_1}\}$. Moreover, let

$$\mathfrak{J}_2 = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$$

and define

$$[\cdot, \cdot]_{\mathfrak{J}_2} = (\mathfrak{J}_2 \cdot, \cdot),$$

then the non-negative Hamiltonian operator in the indefinite inner product space $\{X \times X, [\cdot, \cdot]_{\mathfrak{J}_2}\}$ satisfies

$$Re[Hu, u]_{\mathfrak{J}_2} \geq 0, \quad u \in \mathscr{D}(H),$$

that is, H is an accretive operator. Therefore, by using the anti-symmetry and accretive properties of infinite-dimensional Hamiltonian operators in a class of complete indefinite inner product spaces, it is expected to solve the spectral theory of non-self-adjoint operators that need to be solved urgently in the field of mathematics and mechanics.

At present, great progress has been made in the study of the spectral theory of infinite-dimensional Hamiltonian operators. Scholars abroad, such as T. Y. Azizov and G. A. Kruina, and the research group of infinite-dimensional Hamiltonian operators led by Professor Alatancang in China, have made some important achievements in the study of the completeness of the eigenvector system and root subspace, the symplectic self-adjointness, and the quadratic numerical range and the classical numerical range of infinite-dimensional Hamiltonian operators, as well as the invertibility of non-negative Hamiltonian operators. The purpose of this book is to systematically introduce the theory and method of infinite-dimensional Hamiltonian operators to the reader. Taking the spectral analysis of infinite-dimensional Hamiltonian operators as the main line, some of the most basic structures and properties of infinite-dimensional Hamiltonian operators are formulated.

The content of this book is arranged as follows.

Chapter 1 briefly introduces the intrinsic relationship between the finite-dimensional and infinite-dimensional Hamiltonian systems, the Hamiltonian canonical system and the Newtonian two-body equations of motion, and also introduces the inverse problem of the infinite-dimensional Hamiltonian canonical system, which is closely related to Wu's method in the field of mathematics mechanization proposed by academician Wenjun Wu. The program package of pseudo-division algorithm of multivariate polynomial matrices is also given.

Chapter 2 introduces the spectral theory of infinite-dimensional Hamiltonian operators, including the point spectrum, residual spectrum, continuous spectrum and essential spectrum of diagonal, skew diagonal and upper triangular Hamiltonian operators, as well as non-negative Hamiltonian operators.

Chapter 3 describes the operator theory in symplectic spaces and the completeness of the eigenfunction system of infinite-dimensional Hamiltonian operators, and gives a series of conditions for the completeness of the eigenfunction system of infinite-dimensional Hamiltonian operators in the sense of Cauchy principal value.

Chapter 4 is devoted to the symplectic self-adjointness of infinite-dimensional Hamiltonian operators. By using perturbation theory and product operator theory, the adjoint of infinite-dimensional Hamiltonian operators is characterized, which lays a foundation for solving the spectral symmetry problem of infinite-dimensional Hamiltonian operators.

Chapter 5 is concerned with the numerical range of infinite-dimensional Hamiltonian operators, including the classical numerical range, quadratic numerical range, essential numerical range, polynomial numerical range, and numerical radius inequality, which lays a foundation for the establishment of stability analysis framework based on the numerical radius inequality and polynomial numerical range.

In Chap. 6, we introduce the spectral theory of infinite-dimensional Hamiltonian operators and the J–numerical range theory in complete indefinite inner product spaces, and give the sufficient conditions for the existence of maximal definite invariant subspaces for infinite-dimensional Hamiltonian operators. We also study the dissipativity and boundedness of infinite-dimensional Hamiltonian operators in Krein spaces, which lays a foundation for the establishment of the spectral theory framework of linear operators in Krein spaces.

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Any advice and criticism is warmly and gratefully welcomed about this book from experts and readers.

Hohhot, China
June 2019

Alatancang Chen
Deyu Wu
Junjie Huang
Guolin Hou

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Chapter 1
Hamiltonian Systems



The theoretical model of Hamiltonian systems was established by W. R. Hamilton from geometric optics, and then he creatively studied classical mechanics according to the profound relationship between optics and mechanics, and obtained a third formation of mechanics—Hamiltonian principle, which is equivalent to the variational principles of Newtonian system and Lagrangian system in classical mechanical description. The Hamiltonian principle is used in almost every aspect of daily life, including structural biology, pharmacology, semiconductors, plasma, celestial mechanics, and so on. One of the founders of quantum mechanics Schrödinger once said that the Hamiltonian principle has become the cornerstone of modern physics. If you want to use modern theory to solve any physical problem, you must first express it as a Hamiltonian form [6]. The Hamiltonian system is a mathematical representation of the Hamiltonian principle. So far, Hamiltonian systems can be divided into the classical Hamiltonian system, the generalized Hamiltonian system, and the infinite-dimensional Hamiltonian system. As the beginning of the book, this chapter briefly reviews the classical Hamiltonian system, and the generation, research, and development of infinite-dimensional Hamiltonian systems are introduced. Secondly, the application of the Hamiltonian matrices corresponding to linear classical Hamiltonian systems in the optimal control problem and the application of the infinite-dimensional Hamiltonian operators corresponding to linear infinite-dimensional Hamiltonian systems in the new systematic methodology of elasticity are briefly introduced [5]. Finally, the inverse problem of the infinite-dimensional Hamiltonian canonical systems and the pseudo-division algorithm of multivariate polynomial matrices are introduced, and the program package of the pseudo-division algorithm of multivariate polynomial matrices is given.

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1.1 Finite-Dimensional Hamiltonian Systems and Hamiltonian Matrices

1.1.1 Classical Hamiltonian Systems

When it comes to the variational principle in mechanics, it should be traced back to the Hamiltonian principle of least action (i.e., the Hamiltonian principle). Define

$$\mathfrak{q} = \left[q_1(t)\ q_2(t) \cdots q_n(t) \right]^T,$$

$$\dot{\mathfrak{q}} = \left[\tfrac{\mathrm{d}q_1(t)}{\mathrm{d}t}\ \tfrac{\mathrm{d}q_2(t)}{\mathrm{d}t} \cdots \tfrac{\mathrm{d}q_n(t)}{\mathrm{d}t} \right]^T,$$

where $\{q_i(t)\}_{i=1}^n$ denotes the generalized displacement and $\{\frac{\mathrm{d}q_i(t)}{\mathrm{d}t}\}_{i=1}^n$ denotes the generalized velocity, then the Lagrangian function of the dynamical system is

$$\mathfrak{L} = \mathfrak{L}(\mathbf{q}, \dot{\mathbf{q}}). \tag{1.1}$$

The Hamiltonian principle can be expressed as: a conservative system moves from the initial point $(\mathbf{q}_0, t_0)$ to the terminal point $(\mathbf{q}_e, t_e)$, and its real trajectory can be expressed as a functional

$$\Phi(\mathbf{q}) = \int_{t_0}^{t_e} \mathfrak{L}(\mathbf{q}, \dot{\mathbf{q}}, t)\mathrm{d}t \tag{1.2}$$

and hence

$$\delta\Phi = 0. \tag{1.3}$$

Then, computing the variation of equation (1.2) and using integration by parts, we obtain

$$\begin{aligned}
\delta\Phi &= \int_{t_0}^{t_e} \left(\left(\frac{\partial \mathfrak{L}}{\partial \mathbf{q}} \right)^T \cdot \delta\mathbf{q} + \left(\frac{\partial \mathfrak{L}}{\partial \dot{\mathbf{q}}} \right)^T \cdot \delta\dot{\mathbf{q}} \right) \mathrm{d}t \\
&= \int_{t_0}^{t_e} \left(\frac{\partial \mathfrak{L}}{\partial \mathbf{q}} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial \mathfrak{L}}{\partial \dot{\mathbf{q}}} \right) \right)^T \cdot \delta\mathbf{q}\mathrm{d}t \\
&= 0,
\end{aligned}$$

in which the fact that $\delta\mathbf{q} = 0$ at time t_0 and t_e is used. Since $\delta\mathbf{q}$ is an arbitrary variational for $t_0 < t < t_e$, the Lagrangian equation

$$\frac{\partial \mathfrak{L}}{\partial \mathbf{q}} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial \mathfrak{L}}{\partial \dot{\mathbf{q}}} \right) = 0 \tag{1.4}$$

can be derived.

Introduce the generalized momentum $\mathbf{p} = \frac{\partial \mathfrak{L}}{\partial \dot{\mathbf{q}}}$ and Hamiltonian energy function (kinetic energy + potential energy).

$$\mathscr{H} = \mathbf{p}^T \cdot \dot{\mathbf{q}} - \mathfrak{L}(\mathbf{q}, \dot{\mathbf{q}}), \tag{1.5}$$

where $\dot{\mathbf{q}}$ should be regarded as a function of $\mathbf{p}, \mathbf{q}$ in (1.5). Computing the partial derivatives with respect to $\mathbf{p}, \mathbf{q}$ on both sides of the equation gives

$$\frac{\partial \mathscr{H}}{\partial \mathbf{q}} = \mathbf{p}^T \frac{\partial \dot{\mathbf{q}}}{\partial \mathbf{q}} - \frac{\partial \mathfrak{L}}{\partial \mathbf{q}} - \left(\frac{\partial \mathfrak{L}}{\partial \dot{\mathbf{q}}}\right)^T \frac{\partial \dot{\mathbf{q}}}{\partial \mathbf{q}} = -\frac{\partial \mathfrak{L}}{\partial \mathbf{q}} = -\dot{\mathbf{p}}, \tag{1.6}$$

and

$$\frac{\partial \mathscr{H}}{\partial \mathbf{p}} = \dot{\mathbf{q}} + \mathbf{p}^T \frac{\partial \dot{\mathbf{q}}}{\partial \mathbf{q}} - \left(\frac{\partial \mathfrak{L}}{\partial \dot{\mathbf{q}}}\right)^T \frac{\partial \dot{\mathbf{q}}}{\partial \mathbf{q}} = \dot{\mathbf{q}}, \tag{1.7}$$

respectively. Combining (1.4), (1.6) and (1.7), we can obtain

$$\begin{cases} \dot{\mathbf{q}} = \frac{\partial \mathscr{H}}{\partial \mathbf{p}}, \\ \dot{\mathbf{p}} = -\frac{\partial \mathscr{H}}{\partial \mathbf{q}}, \end{cases} \tag{1.8}$$

which is the classical Hamiltonian canonical equation.

For the classical Hamiltonian system, the following reverse thinking can also be carried out. What condition does the energy function $\mathscr{H} = \mathscr{H}(\mathbf{q}, \mathbf{p})$ satisfy, it will not change with time? That is, when can an equation

$$\frac{\mathrm{d}\mathscr{H}}{\mathrm{d}t} = \frac{\partial \mathscr{H}}{\partial \mathbf{q}} \dot{\mathbf{q}} + \frac{\partial \mathscr{H}}{\partial \mathbf{p}} \dot{\mathbf{p}} = 0$$

be established? Obviously, if you select

$$\dot{\mathbf{q}} = \frac{\partial \mathscr{H}}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial \mathscr{H}}{\partial \mathbf{q}},$$

then the equation holds. This may be the original idea when Hamilton introduced the classical Hamiltonian system.

Remark 1.1.1 The partial derivative of function $\mathscr{H}$ with respect to $\mathbf{p}, \mathbf{q}$ is also called the Gateax directional derivative, which is a generalization of the classical directional derivative. Let $f = f(\mathrm{p}, \mathrm{q})$, then the Gateaux directional derivative of f with respect to p and q is defined as

$$f'_{\mathrm{p}}(\mathrm{p}, \mathrm{q}; d) = \lim_{t \to 0} \frac{f(\mathrm{p} + td, \mathrm{q}) - f(\mathrm{p}, \mathrm{q})}{t}, \tag{1.9}$$

$$f'_{\mathrm{q}}(\mathrm{p}, \mathrm{q}; d) = \lim_{t \to 0} \frac{f(\mathrm{p}, \mathrm{q} + td) - f(\mathrm{p}, \mathrm{q})}{t}, \tag{1.10}$$

where $d \in \mathbb{R}^n$ is an arbitrary vector. For example,

$$f(\mathtt{p}, \dot{\mathbf{q}}) = \mathtt{p}^T \cdot \dot{\mathbf{q}} = (\mathtt{p}, \dot{\mathbf{q}}),$$

where $(\cdot, \cdot)$ is a real inner product, then the Gateaux directional derivative of f with respect to $\mathtt{p}$ is

$$f'_{\mathtt{p}}(\mathtt{p}, \dot{\mathbf{q}}; d) = \lim_{t \to 0} \frac{f(\mathtt{p}+t\mathtt{d}, \dot{\mathbf{q}}) - f(\mathtt{p}, \dot{\mathbf{q}})}{t} = (d, \dot{\mathbf{q}}) = (\dot{\mathbf{q}}, d).$$

Hence the partial derivative of f with respect to $\mathtt{p}$ is $\frac{\partial f}{\partial \mathtt{p}} = \dot{\mathbf{q}}$. Other partial derivatives are completely similar.

Introduce the vector $Z = \left[\mathbf{q}\ \mathbf{p} \right]$, where

$$\mathbf{q} = \left[q_1(t) \cdots q_n(t) \right], \ \mathbf{p} = \left[p_1(t) \cdots p_n(t) \right],$$

then the classical Hamiltonian canonical equation can also be written as

$$\frac{dZ}{dt} = J_{2n} \cdot \nabla \mathscr{H}, \tag{1.11}$$

where $J_{2n} = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$ is called a symplectic metric matrix that satisfies

$$J_{2n} = -J_{2n}^{-1} = -J_{2n}^T,$$

and $\nabla \mathscr{H}$ represents the gradient vector of $\mathscr{H}$. Equation (1.11) is also called the symplectic form of the Hamiltonian system. The system (1.8) has the following two important properties.

Proposition 1.1.2 (Energy Conservation Theorem) *The Hamiltonian systems are energy conserved, that is,* $\frac{\mathrm{d}\mathscr{H}}{\mathrm{d}t} = 0$.

Proof According to the derivation chain rule, we can have

$$\begin{aligned} \frac{\mathrm{d}\mathscr{H}}{\mathrm{d}t} &= \frac{\partial \mathscr{H}}{\partial \mathbf{q}} \dot{\mathbf{q}} + \frac{\partial \mathscr{H}}{\partial \mathbf{p}} \dot{\mathbf{p}} \\ &= -\dot{\mathbf{p}} \dot{\mathbf{q}} + \dot{\mathbf{q}} \dot{\mathbf{p}} \\ &= 0. \end{aligned}$$

Proposition 1.1.3 (Liouville Theorem) *Let the Hamiltonian function* $\mathscr{H} = \mathscr{H}(\mathbf{q}, \mathbf{p})$ *be smooth and*

$$\mathbf{F} = \left[\dot{\mathbf{q}}\ \dot{\mathbf{p}} \right] = \begin{bmatrix} \frac{\partial \mathscr{H}}{\partial \mathbf{p}} & -\frac{\partial \mathscr{H}}{\partial \mathbf{q}} \end{bmatrix}$$

be the Hamiltonian vector field. Then the volume of the phase space remains unchanged under the Hamiltonian phase flow, that is, $div\mathbf{F} = 0$.

Proof According to the definition of divergence and the commutativity of mixed partial derivative for smooth function,

$$\begin{aligned} div\mathbf{F} &= \frac{\partial \dot{\mathbf{q}}}{\partial \mathbf{q}} + \frac{\partial \dot{\mathbf{p}}}{\partial \mathbf{p}} \\ &= \frac{\partial}{\partial \mathbf{q}}\left(\frac{\partial \mathscr{H}}{\partial \mathbf{p}}\right) - \frac{\partial}{\partial \mathbf{p}}\left(\frac{\partial \mathscr{H}}{\partial \mathbf{q}}\right) \\ &= 0. \end{aligned}$$

1.1.2 Hamiltonian Systems and Newtonian Two-Body Equations of Motion

After the idea and method of Hamiltonian principle were applied to optics and mechanics, Hamilton conjectured that the Hamiltonian principle should also have other applications. However, the close connection between the Newtonian two-body equations of motion and Hamiltonian canonical systems opened up an important application of Hamiltonian canonical systems in the field of celestial mechanics. The correctness of Kepler's three laws, known as the "space constitution," can be verified by Newtonian two-body equations of motion in polar coordinates

$$\begin{cases} \ddot{\mathbf{r}} - \mathbf{r}\dot{\theta}^2 = -\frac{k^2}{\mathbf{r}^2}, \\ 2\dot{\mathbf{r}}\dot{\theta} + \mathbf{r}\ddot{\theta} = 0, \end{cases} \tag{1.12}$$

where $\dot{\mathbf{r}}$, $\ddot{\mathbf{r}}$ denote the first and the second derivatives of $\mathbf{r} = \mathbf{r}(t)$ respectively, that is, Kepler's three laws are contained in the Newtonian two-body equations of motion. On the other hand, taking the dual variables of $\mathbf{r}$, θ

$$R = \dot{\mathbf{r}}, \quad \Theta = \mathbf{r}^2\dot{\theta}$$

respectively, we then have the Hamiltonian energy function

$$\mathscr{H} = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{\mathbf{r}^2}\right) - \frac{k}{\mathbf{r}}$$

of Newtonian two-body equations of motion in polar coordinates. At this point, the Newtonian two-body equations of motion can be transformed into the following Hamiltonian canonical system

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} \mathbf{r} \\ \theta \\ R \\ \Theta \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{\partial \mathscr{H}}{\partial \mathbf{r}} \\ \frac{\partial \mathscr{H}}{\partial \theta} \\ \frac{\partial \mathscr{H}}{\partial R} \\ \frac{\partial \mathscr{H}}{\partial \Theta} \end{bmatrix}. \tag{1.13}$$

Considering the third equation of the system (1.13), we obtain

$$\begin{aligned}\ddot{\mathbf{r}} = \dot{R} &= -\frac{\partial \mathscr{H}}{\partial \mathbf{r}} \\ &= -\frac{k^2}{\mathbf{r}^2} + \frac{\Theta^2}{\mathbf{r}^3} \\ &= -\frac{k^2}{\mathbf{r}^2} + \mathbf{r}\dot{\theta}^2,\end{aligned}$$

that is,

$$\ddot{\mathbf{r}} - \mathbf{r}\dot{\theta}^2 = -\frac{k^2}{\mathbf{r}^2}.$$

This shows that the first equation of Newtonian two-body equations of motion can be derived from Hamiltonian canonical system (1.13). Similarly, considering the fourth equation of the system (1.13), we obtain

$$\dot{\Theta} = -\frac{\partial \mathscr{H}}{\partial \theta} = 0,$$

and hence

$$\begin{aligned}\dot{\Theta} &= \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{r}^2\dot{\theta}) \\ &= 2\dot{\mathbf{r}}\dot{\theta} + \mathbf{r}\ddot{\theta} = 0,\end{aligned}$$

that is, the second equation of Newtonian two-body equations of motion is also contained in the Hamiltonian canonical system (1.13). Therefore, the Newtonian two-body equations of motion are contained in the Hamiltonian canonical system (1.13).

If the energy function $\mathscr{H}$ in the system (1.8) is chosen as

$$\mathscr{H} = \frac{1}{2} Z^T S Z,$$

where $Z = Z(t) \in \mathbb{R}^{2n}$, $t \in \mathbb{R}$, and $S(t) = \begin{bmatrix} C(t) & -A(t)^T \\ -A(t) & -B(t) \end{bmatrix}$ is a self-adjoint matrix function, that is, $A(t), B(t), C(t) \in \mathbb{R}^{n\times n}$ with $B(t)^T = B(t)$, $C(t)^T = C(t)$, then computing Gateaux directional derivative with respect to $\mathbf{p}$, $\mathbf{q}$ yields

$$\frac{\partial \mathscr{H}}{\partial \mathbf{p}} = C(t)\mathbf{q} - A(t)^T\mathbf{p},$$

$$\frac{\partial \mathscr{H}}{\partial \mathbf{q}} = -A(t)\mathbf{q} - B(t)\mathbf{p},$$

and thus the following linear Hamiltonian system (see [2]) can be obtained:

$$\dot{Z} = \begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{bmatrix} = \begin{bmatrix} A(t) & B(t) \\ C(t) & -A(t)^T \end{bmatrix} Z. \tag{1.14}$$

Here the coefficient matrix $H(t)$ is a bounded measurable real matrix-valued function and satisfies

$$(J_{2n}H)(t)^T = J_{2n}H(t)$$

for any $t \in \mathbb{R}$, where $J_{2n} = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$. Matrix $H(t) = \begin{bmatrix} A(t) & B(t) \\ C(t) & -A(t)^T \end{bmatrix}$ is called a Hamiltonian matrix. Hamiltonian matrices have important applications in the field such as control theory and differential equations.

Example 1.1.4 The so-called linear quadratic regulator problem is to find a control $\mathbf{u}(t) : [0, \tau] \to \mathbb{R}^m$ such that

$$L_{\mathbf{x}_0}(\mathbf{x}(t), \mathbf{u}(t)) = (\mathbf{x}(t), R\mathbf{x}(t)) + \int_0^\tau [(\mathbf{x}(t), G(t)\mathbf{x}(t)) + (\mathbf{u}(t), S\mathbf{u}(t))]\mathrm{d}t \tag{1.15}$$

attains the minimum value, where $\tau \in (0, \infty]$, S is a symmetric positive semi-definite matrix, $G(t)$, $R(t)$ are bounded continuous matrix-valued functions with $G^T(t) = G(t) \geq 0$, $R(t)^T = R(t) > 0$ for any $t \in \mathbb{R}$, and $\mathbf{x}(t)$ satisfies the linear control problem

$$\begin{cases} \dot{\mathbf{x}}(t) = A(t)\mathbf{x}(t) + B(t)\mathbf{u}_0, \quad \mathbf{x}(t) \in \mathbb{R}^n, \mathbf{u}_0 \in \mathbb{R}^m, \\ \mathbf{x}(0) = \mathbf{x}_0. \end{cases} \tag{1.16}$$

Here $A(t) \in \mathbb{R}^{n\times n}$, $B(t) \in \mathbb{R}^{n\times m}$ are given bounded continuous matrix-valued functions. By Pontryagin's maximum principle (see [2]), we know that if (1.15) attains minimum value, $\mathbf{u}(t)$ is bound to maximize Hamiltonian functional

$$\mathscr{H}(t, \mathbf{x}, \mathbf{y}, \mathbf{u}) = (\mathbf{y}, A(t)\mathbf{x}(t) + B(t)\mathbf{u}(t)) - \frac{1}{2}[(\mathbf{x}(t), G(t)\mathbf{x}(t)) + (\mathbf{u}(t), R(t)\mathbf{u}(t))] \tag{1.17}$$

so

$$\mathbf{u}(t) = R^{-1}(t)B^T(t)\mathbf{y}.$$

Then from the Hamiltonian system

$$\begin{cases} \dot{\mathbf{x}} = \frac{\partial \mathscr{H}}{\partial \mathbf{y}}, \\ \dot{\mathbf{y}} = -\frac{\partial \mathscr{H}}{\partial \mathbf{x}}, \end{cases}$$

it follows that the linear Hamiltonian system

$$\dot{\mathbf{z}} = \begin{bmatrix} A(t) & B(t)R^{-1}(t)B(t)^T \\ G(t) & -A(t)^T \end{bmatrix} \mathbf{z}, \tag{1.18}$$

where $\mathbf{z} = \begin{bmatrix} \mathbf{x} & \mathbf{y} \end{bmatrix}^T$. If the control problem (1.16) satisfies the standard controllability and observability conditions, there exists a negative-definite symmetric matrix $M(t) \in \mathbb{R}^n$ such that

$$\mathbf{z} = \begin{bmatrix} \mathbf{x} \\ M(t)\mathbf{x} \end{bmatrix},$$

that is,

$$\mathbf{u}(t) = R^{-1}(t)B(t)^T M(t)\mathbf{x}$$

is the unique solution of the quadratic regulator problem (1.15).

1.2 Linear Infinite-Dimensional Hamiltonian Canonical Systems and Infinite-Dimensional Hamiltonian Operators

The research object of infinite-dimensional Hamiltonian systems is the continuous medium problem, which has a wide range of applications in mathematical physics, soliton theory, mechanics, and engineering technology. The general concept of infinite-dimensional Hamiltonian systems appeared for the first time in the work of Magri [11, 12], Vinogradov [13], Kupershmidt [14], and Manin [15] between 1978 and 1980. On this basis, Gel'fand and Dorfman [16], Olver [17–21], Kosmann-Schwarzbach [22] among others made further research. For the detailed discussion of infinite-dimensional Hamiltonian systems, please refer to [3, 20].

By extending the system of equations (1.11) to the general infinite-dimensional Hilbert spaces $X \times X$, the following infinite-dimensional Hamiltonian canonical system can be obtained.

Definition 1.2.1 The following evolution equation is called an infinite-dimensional Hamiltonian canonical system

$$\dot{U} = J^{-1}\frac{\delta \mathscr{H}}{\delta U},$$

where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$, $\frac{\delta \mathscr{H}}{\delta U}$ is the variational derivative, and $\mathscr{H}$ is the Hamiltonian functional.

According to the Vainberg theorem [23] and Definition 1.2.1, the linear infinite-dimensional Hamiltonian system can be simplified as the following form of separation of variables.

Definition 1.2.2 Let X be a Hilbert space and $H : \mathscr{D}(H) \subseteq X \times X \to X \times X$ be a linear operator. If

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$$

with $B = B^*$ and $C = C^*$, then the evolution equation

$$\dot{U} = HU$$

is a linear Hamiltonian canonical system, and H is called an infinite-dimensional Hamiltonian operator.

Remark 1.2.3 A simple explanation of promoting the finite-dimensional Hamiltonian canonical systems to the infinite-dimensional Hamiltonian canonical systems: If the finite-dimensional space $\mathbb{R}^n$ is replaced by an infinite-dimensional Hilbert space X, the transpose of a matrix is replaced by the adjoint operation of a linear operator, and the symmetry of a matrix is replaced by the self-adjointness of a linear operator, the infinite-dimensional Hamiltonian operator can be obtained.

Infinite-dimensional Hamiltonian operators are also widely used in optimal control theory. In the optimal control theory of infinite-dimensional systems, many problems can be solved by the stationary solutions of the following algebraic Riccati equation [24]

$$(Az_1, \Pi z_2) + (\Pi z_1, Az_2) + (z_1, Wz_2) - (z_1, Sz_2) = 0,\ z_1,\ z_2 \in \mathscr{D}(A), \quad (1.19)$$

where A is a densely defined closed linear operator in a Hilbert space X, $W \in \mathscr{B}(U, Y)$, S, W are self-adjoint operators, and U, Y are Hilbert spaces.

In [24], Curtain and Zwart pointed out that a necessary and sufficient condition for a self-adjoint operator $\Pi \in \mathscr{B}(X)$ to be a solution of the Riccati equation (1.19) is $\Pi\mathscr{D}(A) \subset \mathscr{D}(A^*)$ and

$$H\left[\begin{pmatrix} X \\ \Pi X \end{pmatrix} \bigcap \mathscr{D}(H)\right] \subset \begin{bmatrix} X \\ \Pi X \end{bmatrix},$$

that is, $\mathscr{R}\begin{bmatrix} I \\ \Pi \end{bmatrix}$ is an invariant subspace of H where

$$H = \begin{bmatrix} A & -S \\ -W & -A^* \end{bmatrix}$$

is an infinite-dimensional Hamiltonian operator. Therefore, there is a deep connection between the solution of the algebraic Riccati equation and the invariant subspace of the corresponding infinite-dimensional Hamiltonian operator. H. Langer et.al discussed the solution of the following algebraic Riccati equation in [25]:

$$A^*\Pi + \Pi A + Q - \Pi D\Pi = 0,\ Q \geq 0, D \geq 0. \quad (1.20)$$

By giving the angular operator representation of the invariant subspace of the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & -D \\ -Q & -A^* \end{bmatrix}$$

they proved that the angular operator is a bounded self-adjoint operator under some conditions, which satisfies the Riccati equation (1.20).

Remark 1.2.4 The properties of solutions and well-posedness of infinite-dimensional Hamiltonian systems can be characterized by the properties of infinite-dimensional Hamiltonian operators. It is known from the definition that infinite-dimensional Hamiltonian operators are generally unbounded non-self-adjoint operator matrices, which have the following structure characteristics: (i) The diagonal elements are the densely defined closed linear operator A and its adjoint A^*, and the difference is a minus sign; (ii) Skew diagonal elements are all self-adjoint operators. For the study of some problems related to infinite-dimensional Hamiltonian operators, such as spectral distribution and completeness problems, if these characteristics can be fully utilized, some featured conclusions of infinite-dimensional Hamiltonian operators will be obtained. However, these properties also bring difficulties to the study of other problems related to infinite-dimensional Hamiltonian operators, such as spectral perturbation. Therefore, how to solve those related problems becomes quite significant.

1.3 Hamiltonian Canonical Systems and the Pseudo-Division Algorithm of Multivariate Polynomial Matrices

Pseudo-division algorithm of multivariate polynomial matrices [31] plays a very important role in Wu's method. One of the mathematical foundations is the pseudo-division algorithm of multivariate polynomials. In [33], authors also studied the pseudo-division algorithm of multivariate polynomials from different viewpoints. If extending Wu's method to multivariate polynomial matrices, then, first of all, one should have the algorithm and its implementation of the pseudo-division algorithm of multivariate polynomial matrices. The pseudo-division of multivariate polynomials is extended to the matrix case in [36], in which the authors proposed the pseudo-division algorithm of multivariate polynomial matrices. These methods are used to solve the problem of transforming partial differential equations with constant coefficients into infinite-dimensional Hamiltonian systems. However, they are all manual calculations and are not implemented by programs. In [34, 35], the pseudo-division algorithm of multivariate polynomial matrices is applied in constructing the general solution of a class of partial differential equations. In view of this, in what follows, we introduce the pseudo-division algorithm of multivariate polynomial matrices and its applications.

Based on the constructive proof of Theorem 1 in [36], we give an algorithm for the program realization of the pseudo-division algorithm of multivariate polynomial matrices in Sect. 1.3.1. The algorithm is implemented on the computer algebra system

Mathematica [37, 38], and the software package of the pseudo-division algorithm of multivariate polynomial matrices is also provided. On this basis, further implementation of its application to the inverse problem of infinite-dimensional Hamiltonian systems is realized. To a certain extent, the realization of this provides a good tool for the researchers engaged in the study of Hamiltonian mechanics, so that they can save a lot of time by getting rid of tedious calculations. As a further application of the software package, in Sects. 1.4 and 1.5, the matrix solution of the monic factorization of bivariate polynomial matrices and the characteristic sequence of multivariate polynomials in the algebraic cases are given, respectively.

1.3.1 Pseudo-Division Algorithm of Multivariate Polynomial Matrices

Multivariate polynomial matrices are also known as matrix multivariate polynomials, whose exact definition is given below.

Definition 1.3.1 Let $p = (a_{ij})_{l\times m}$ be an $l \times m$ matrix, where $a_{ij} \in p\,[x_1, x_2, \ldots, x_n]$ ($p\,[x_1, x_2, \ldots, x_n]$ is an n-variable polynomial ring). Then p is called an n-variable polynomial matrix.

Definition 1.3.2 In $p = (a_{ij})_{l\times m}$, the variable with the largest subscript actually appearing in a_{ij} is called the principal variable of an n-variable polynomial matrix, denoted as x_k, and the highest power of the principal variable is called the degree of the n-variable polynomial matrix with respect to the principal variable, denoted as $d_{x_k} p$.

Definition 1.3.3 The n-variable polynomial matrix p can always be written in the following form:

$$p = (a_{ij})_{l\times m} = \sum_{s=0}^{d_{x_k}} p_s x_k^s,$$

where x_k is the principal variable of p, $p_s = (a_{ij}^s), a_{ij}^s \in p\,[x_1, x_2, \ldots, x_{k-1}, x_{k+1}, \ldots, x_n]$, and we call $p_{d_{x_k} p}$ the initial formula of p, denoted as I_p.

Theorem 1.3.4 *For the multivariate polynomial matrices A, B, there exist unique smallest non-negative integers α and n ($n \le \alpha$) and multivariate polynomial matrices $Q_i (i = 1, 2, \ldots, n)$ and R such that*

$$I_B^\alpha A = \sum_{i=0}^{n} I_B^i B Q_i + R, \tag{1.21}$$

where $d_{x_k} R < d_{x_k} B$ or $R = 0$.

Proof Let $B = \sum_{l=0}^{s} B_l x_k^l$, $B_l = (b_{ij}^l)$, $b_{ij}^l \in p[x_1, x_2, \ldots, x_{k-1}]$ and $A = \sum_{l=0}^{m} A_l x_k^l$, $A_l = (a_{ij}^l)$, $a_{ij}^l \in p[x_1, x_2, \ldots, x_{k-1}, x_{k+1}, \ldots, x_n]$. We may as well let $m \geq s$, then

$$I_B^{\alpha_1} A = B\widetilde{Q}_1 + R_1, \tag{1.22}$$

where $m_1 = d_{x_k} R_1 < m$ and $\alpha_1 \in \{0, 1\}$ are the smallest non-negative integers such that (1.22) holds.

In fact, $\widetilde{Q}_1$ and α_1 can be found mechanically by the following process.

Step 1: Determine whether there is Q_1 such that $A_m = I_B Q_1$. If there is, then take $\widetilde{Q}_1 = Q_1 x_k^{m-s}$, $\alpha_1 = 0$. If it does not exist, turn to the second step.

Step 2: Determine whether there are Q_1 and Q_2, $Q_2 \in \mathscr{X} \triangleq \{Q | I_B Q = 0\}$, such that $A_m = I_B Q_1 + B_{s-1} Q_2$. If they exist, then take $\widetilde{Q}_1 = I_B Q_1 x_k^{m-s} + B_{s-1} Q_2 x_k^{m-s+1}$, $\alpha_1 = 0$. And if they do not exist, turn to the third step.

Step 3: Check whether there are Q_1, Q_2 and Q_3, $Q_3 \in \mathscr{X}$, $Q_2 \in \mathscr{Y} \triangleq \{Q | I_B Q + Q_3 B_{s-1} = 0, Q_3 \in \mathscr{X}\}$, such that $A_m = I_B Q_1 + B_{s-1} Q_2 + B_{s-2} Q_3$. If they exist, take $\widetilde{Q}_1 = I_B Q_1 x_k^{m-s} + B_{s-1} Q_2 x_k^{m-s+1} + Q_3 x_k^{m-s+2}$, $\alpha_1 = 0$, if not, go to the next step.

Follow the above method until $Q_1, Q_2, \ldots, Q_{s+1}$, if none exists, then take $\widetilde{Q}_1 = A_m x_k^{m-s}$, $\alpha_1 = 1$.

If $m_1 < s$ or $R_1 = 0$, then the lemma holds. If $m_1 \geq s$, then replacing A by R_1 and repeating the steps above we get

$$I_B^{\alpha_2} R_1 = B\widetilde{Q}_2 + R_2, \tag{1.23}$$

where $m_2 = d_{x_k} R_2 < m_1$ and $\alpha_2 \in \{0, 1\}$ are the smallest non-negative integers satisfying (1.23), and $\widetilde{Q}_2$ and α_2 is uniquely determined. By (1.22) and (1.23) we can get:

$$I_B^{\alpha_1+\alpha_2} A = I_B^{\alpha_2} B\widetilde{Q}_1 + B\widetilde{Q}_2 + R_2.$$

When $m_2 < s$ or $R_2 = 0$, the lemma holds. If $m_2 \geq s$, then replacing A by R_2 and repeating the steps above, we can get in turn the multivariate polynomial matrices

$$R_1, R_2, \ldots, R_i,$$

since the degree reduces at least once in every step, we have

$$m > m_1 > \cdots > m_i,$$

so there must be some R_i such that $d_{x_k} R_i < s$ or $R_i = 0$.

Corollary 1.3.5 *If I_B is a full rank matrix, (1.21) in Theorem 1.3.4 can be changed into*

$$A = BQ + R. \tag{1.24}$$

Remark 1.3.6 If we denote $B_1 = \sum_{i=0}^{n} I_B^i$, $Q_1 = \sum_{i=0}^{n} Q_i$, $R_1 = R$, then (1.21) in Theorem 1.3.4 can be written as

$$I_B^{\alpha} A = B_1 B Q_1 + R_1, \tag{1.25}$$

where $d_{x_k} R_1 < d_{x_k} B$ or $R_1 = 0$.

Algorithm Description

Here denote $s = d_{x_k} B, \alpha = \{\}, QS = \{\}$, and the symbols ":=" and "=" represent the meaning of assignment and equals sign, respectively.

Algorithm 1 DWR[A, B, x_k]**:** Given any polynomial matrices $A_{p\times q}$ and $B_{p\times p}$, the algorithm outputs a multivariate polynomial matrix A with x_k as the principal variable divided by B.

(D1) $m := d_{x_k} A$; if $m < s$, then $R := A$ execute (D4); otherwise execute (D2);

(D2) Call algorithm 2 (the corresponding program segment is denoted as MQuet[ ]);

k :=MQuet[R, B, x_k];

If $k =$ False, then $R := I_B A - B A_m x_k^m$; adds 1 in an orderly manner after the elements in the table α; otherwise $R := A - Bk$; add 0, k to the table α, QS after the elements in an ordered manner;

(D3) $A := R$; turn(D1);

(D4) Output the pseudo-division algorithm of multivariate polynomial matrices.

According to the proof of Lemma 1.3.4, the algorithm above can be terminated in finite steps.

Algorithm 2 MQuet[R, B, x_k**]:** Calculate the quotient of each step of the division of polynomial matrices R and B, if the remainder is 0, return its quotient, otherwise return the logical value False.

(M1) If there exists Q such that $A_m = I_B Q$, then return Q.

Otherwise, if $Q_2 \in \mathscr{X} \triangleq \{Q | I_B Q = 0\}$ doesn't exist, execute (M2); if Q_2 exists, check whether there is Q_1 such that $A_m = I_B Q_1 + B_{s-1} Q_2$, if Q_1 exists, then $Q := Q_1 x_k^{m-s} + Q_2 x_k^{m-s+1}$, return Q; Otherwise check whether there is $Q_2 \in \{Q | I_B Q + B_{s-1} Q_3 = 0, Q_3 \in \mathscr{X}\}$, if not, turn (M2); otherwise, check whether there is Q_1' such that $A_m = I_B Q_1' + B_{s-1} Q_2 + B_{s-2} Q_3$, if there exists Q_1' (for simplicity still denoted as Q_1), then $Q := Q_1 x_k^{m-s} + Q_2 x_k^{m-s+1} + Q_3 x_k^{m-s+2}$, return Q; this goes on until $Q_1, Q_2, \ldots, Q_{s+1}$, if none exists, then turn (M2); otherwise return Q.

(M2) Return False.

The function of the second algorithm is to find $\widetilde{Q_1}$ in (1.22) or $\widetilde{Q_2}$ in (1.23), and the proof of Lemma 1.3.4 shows that the algorithm can be terminated in finite steps.

1.3.2 Software Package of Pseudo-Division Algorithm of Multivariate Polynomial Matrices

From the description of the algorithm, it is known that the algorithm can be realized in the computer algebra system. As the coefficients of the multivariate polynomial matrices are generally symbol polynomial matrices (multivariate polynomial matrix), the program implementation of the algorithm is therefore more complicated. After

repeated debugging, the algorithm is finally implemented in the Mathematica system, which mainly solves the following problems:

1. The storage of multivariate polynomial matrices in a computer.
Because the coefficient of the multivariate polynomials is also a polynomial, which is of the same kind, and the coefficient in front of the multivariate polynomial matrix is a matrix, which is not of the same kind, so the multivariate polynomial matrices in the computer cannot do polynomial operations the way multivariate polynomials work after being input. Here we take the following storage method to achieve a similar function, which is convenient and highly efficient. It is described as follows:

[Input] Multivariate polynomial matrix $A = (a_{ij})_{l\times m}$ and principal variable y.

[Output] The coefficient table C of the multivariate polynomial matrix.

[Step 1] For each row of A, the element a_{ij} is taken out in the order of columns, and the coefficient table t_{ij} of the polynomial a_{ij} with y as the principal variable is separated. In addition, each coefficient table t_{ij} is extended to a table with the same length as the longest coefficient table. In the expansion, the higher power part corresponding to the short coefficient table is supplemented with 0. Then, a new matrix T is made with t_{ij} as the row vector.

[Step 2] Takes each column t_k of the matrix T in turn, and divides t_k into l segments with m elements as a segment, which is used as the row vector of the matrix C_k in turn, so as to obtain the coefficient C_k of y^k, and repeat the process $d_y A$ times. Then C_k is stored in a table C in turn and returns C.

2. The problem of solving the matrix equation $A_{m\times m}v = b_{m\times 1}$ (where A, b are multivariate polynomial matrices).
In Sect. 2.1.1, the core of algorithm 2 is to solve the sign matrix equation, so the effectiveness of this part is very important. In the Mathematica system, the equation $Av = b$ is sometimes solvable, but in general, it is not a multivariate polynomial matrix solution. Let $v = (v_1, v_2, \ldots, v_m)^T$. Because $v_i (i = 1, 2, \ldots, m)$ does not appear in A and b, $Av = b$ can be regarded as a system of linear equations in variable v. In order to obtain the general solution of the multivariate polynomial matrix, we adopt the following approach:

[Step 1] $B := (A|\, b)$, B are transformed into upper triangular matrix B_0 by pseudo-division algorithm combined with Gauss elimination method;

[Step 2] Obtain the general solution v of multivariate polynomial matrices from B_0;

[Step 3] Verify whether v satisfies $Av = b$.

It has been proven that this implementation is more effective, and the speed of program execution is faster.

The following is the program description of the software package: The main program of the Mathematica software package is DWR $[A, B, x]$, and its function is to output the form of pseudo-division algorithm of multivariate polynomial matrix $A_{p\times q}$ divided by $B_{p\times p}$ in principal variable x. The main program has the function of checking the input data. When it is correct, it calls other subprograms for calculation.

1.3.3 Application of Software Package to the Inverse Problem of Infinite-Dimensional Hamiltonian Systems

The software package of the pseudo-division algorithm of multivariate polynomial matrices is a general program, which can be called directly and will play an important role in many fields. This section only takes its application in the inverse problem of infinite-dimensional Hamiltonian systems as a demonstration. Many problems found in physics [39, 40] can be transformed into infinite-dimensional Hamiltonian systems, but the calculation is often more complicated. In particular, it is difficult to realize the complicated manual calculation. In [36], the pseudo-division algorithm of multivariate polynomial matrices is applied to infinite-dimensional Hamiltonian systems, but they are all manual calculations. Here, the method of program implementation is given.

Replace $\frac{\partial}{\partial x}$ by x, similarly, replace $\frac{\partial}{\partial x}$ by y, replace $\frac{\partial^n}{\partial x^n} with \quad y^n$, so that a class of differential polynomials can be transformed into polynomials, and matrix differential operators can be transformed into multivariate polynomial matrices. It can be known by Definition 1.2.2, the multivariate polynomial matrices corresponding to infinite-dimensional Hamiltonian canonical systems can be set as

$$x \times I_{2n} - H,$$

where I_{2n} denotes the unit matrix of order $2n$,

$$H = \begin{bmatrix} a_{11} \cdots a_{1n} & b_{11} \cdots b_{1n} \\ \vdots \quad \cdots \quad \vdots & \vdots \quad \cdots \quad \vdots \\ a_{n1} \cdots a_{nn} & b_{n1} \cdots b_{nn} \\ c_{11} \cdots c_{1n} & -d_{11} \cdots -d_{1n} \\ \vdots \quad \cdots \quad \vdots & \vdots \quad \cdots \quad \vdots \\ c_{n1} \cdots c_{nn} & -d_{n1} \cdots -d_{nn} \end{bmatrix}$$

with $a_{ij}, b_{ij}, c_{ij}, d_{ij}$ polynomials to be determined satisfying $b_{ij} = b_{ji}^*, c_{ij} = c_{ji}^*, d_{ij} = a_{ji}^*$.

After transforming the differential polynomial matrices (differential polynomials) into polynomial matrices, if the power of the principal variable x is $2m$, then the divisor is set to a $2m$-order square matrix; if the degree of the principal variable x is $2m + 1$, then the divisor is set to a square matrix of order $2(m + 1)$. It is assumed that the degree of the transformed polynomial matrices with respect to the principal variable x is $2m$, so the multivariate polynomial matrices $x \times I - H$ (as shown above) corresponding to the infinite-dimensional Hamiltonian canonical systems to be determined should be a $2m$-order square matrix, abbreviated as B. Because the initial of B is a unit matrix, we have the equality in the form of $A = BQ + R$ by calling the software package. The inverse problem of infinite-dimensional Hamiltonian systems refers to finding the conditions for the parameters of B to

satisfy when $R = 0$, and meanwhile, the differential equations are transformed into infinite-dimensional Hamiltonian systems.

The entire process is as follows:

[Step 1] Call the package to get $A = BQ + R$.

[Step 2] By solving the equation $R = 0$, the matrix polynomial solution S is obtained.

[Step 3] Replace the parameters in B with the corresponding solutions in S (denote by $\widetilde{B}$ the new B after replacement), so $A = \widetilde{B}\widetilde{Q}$.

Since the equation $R = 0$ is generally nonlinear, it can be solved by Wu's method in theory, but for the specific R, it is often solved by the combination of a computer algebra system and manual intervention.

The following three examples are given to illustrate the application of the software package. Some examples in [36] will be calculated in Sect. 1.4.

Example 1.3.7 Consider the Schrödinger equation [41] satisfied by the plane wave of a free particle in quantum mechanics

$$\mathtt{i}\hbar\frac{\partial\varphi(r,t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\varphi(r,t),$$

where $\hbar$ is the Planck constant and $\varphi(r,t)$ is the plane wave function of free particles.

Denote $e = -\frac{\hbar}{2m}\mathtt{i}$, then the equation above can be written as

$$\left(\frac{\partial}{\partial t} + e\frac{\partial^2}{\partial x^2} + e\frac{\partial^2}{\partial y^2} + e\frac{\partial^2}{\partial z^2}\right)\varphi(r,t) = 0 \tag{1.26}$$

Let $A = \begin{bmatrix} 0 \\ t + ex^2 + ey^2 + ez^2 \end{bmatrix}$ be the multivariate polynomial matrix corresponding to (1.26), $B = \begin{bmatrix} x - a11 & -b11 \\ -c11 & x + d11 \end{bmatrix}$ be the polynomial matrix corresponding to the undetermined Hamiltonian systems, and call the software package DWR$[A, B, x]$. Then we get

$$\begin{bmatrix} 0 \\ t + ex^2 + ey^2 + ez^2 \end{bmatrix} = \begin{bmatrix} x - a11 & -b11 \\ -c11 & x + d11 \end{bmatrix}\begin{bmatrix} e\,b11 \\ -e\,d11 + ex \end{bmatrix} + \begin{bmatrix} e\,b11(a11 - d11) \\ e\,b11c11 + e\,d11^2 + t + ey^2 + ez^2 \end{bmatrix}.$$

Let the remainder be equal to zero, that is:

$$\begin{cases} b11(a11 - d11) = 0, \\ e\,b11c11 + d11^2 + t + ey^2 + ez^2 = 0, \end{cases}$$

which has infinitely many solutions. In particular, let $a11 = d11 = d11^* = 0$, and then get $-c11b11 = \frac{1}{e}t + (y^2 + z^2)$. Since we also require $c11 = c11^*$, $b11 =$

$b11^*$, at least one of $c11$ and $b11$ is a constant. Without loss of generality, let $c11 = -1$. Then $b11 = \frac{1}{e}t + (y^2 + z^2)$, and hence Eq. (1.26) has the following Hamiltonian canonical form:

$$\frac{\partial}{\partial x}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{e}\frac{\partial}{\partial t} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \\ -1 & 0 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix},$$

where $u = \frac{\partial \varphi}{\partial t} + e\frac{\partial^2 \varphi}{\partial y^2} + e\frac{\partial^2 \varphi}{\partial z^2}$, $v = e\frac{\partial \varphi}{\partial x}$; the factorization of the following differential operators can also be obtained

$$\begin{bmatrix} 0 \\ \frac{\partial}{\partial t} + e\nabla^2 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} & -\frac{1}{e}\frac{\partial}{\partial t} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2} \\ 1 & \frac{\partial}{\partial x} \end{bmatrix}\begin{bmatrix} \frac{\partial}{\partial t} + e(\frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}) \\ e\frac{\partial}{\partial x} \end{bmatrix}.$$

Example 1.3.8 The Maxwell equations [42] in the vacuum are

$$\begin{cases} \varepsilon_0\mu_0\frac{\partial E}{\partial t} - rotB = -\mu_0 j, \\ \frac{\partial B}{\partial t} + rotE = 0, \end{cases}$$

where $B = (B_x, B_y, B_z)$ is the electromagnetic field strength, $E = (E_x, E_y, E_z)$ is the electric field strength, $j = (j_x, j_y, j_z)$ is the current density, ε_0, μ_0 are non-zero constants, and $rotF$ represents the curl of F.

Denote $u = \varepsilon_0\mu_0$, and let $U = (E_x, E_y, E_z, B_x, B_y, B_z)^T$. Then the Maxwell equations can be written as follows:

$$MU = N,$$

where

$$M = \begin{bmatrix} u\frac{\partial}{\partial t} & 0 & 0 & 0 & \frac{\partial}{\partial z} & -\frac{\partial}{\partial y} \\ 0 & u\frac{\partial}{\partial t} & 0 & -\frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \\ 0 & 0 & u\frac{\partial}{\partial t} & \frac{\partial}{\partial y} & -\frac{\partial}{\partial x} & 0 \\ 0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y} & \frac{\partial}{\partial t} & 0 & 0 \\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x} & 0 & \frac{\partial}{\partial t} & 0 \\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 & 0 & 0 & \frac{\partial}{\partial t} \end{bmatrix}, \quad N = -\mu_0(j_x, j_y, j_z, 0, 0, 0)^T.$$

Let

$$A = \begin{bmatrix} ut & 0 & 0 & 0 & z & -y \\ 0 & ut & 0 & -z & 0 & x \\ 0 & 0 & ut & y & -x & 0 \\ 0 & -z & y & t & 0 & 0 \\ z & 0 & -x & 0 & t & 0 \\ -y & x & 0 & 0 & 0 & t \end{bmatrix}$$

be the multivariate polynomial matrix corresponding to M and the polynomial matrix corresponding to the Hamiltonian system be

$$B = \begin{bmatrix} -a11+t & -a12 & -a13 & -b11 & -b12 & -b13 \\ -a21 & -a22+t & -a23 & -b21 & -b22 & -b23 \\ -a31 & -a32 & -a33+t & -b31 & -b32 & -b33 \\ -c11 & -c12 & -c13 & a11+t & a21 & a31 \\ -c21 & -c22 & -c23 & a12 & a22+t & a32 \\ -c31 & -c32 & -c33 & a13 & a23 & a33+t \end{bmatrix}.$$

Selecting t as the principal variable and calling the software package DWR$[A, B, t]$, we get

$$\begin{bmatrix} ut & 0 & 0 & 0 & z & -y \\ 0 & ut & 0 & -z & 0 & x \\ 0 & 0 & ut & y & -x & 0 \\ 0 & -z & y & t & 0 & 0 \\ z & 0 & -x & 0 & t & 0 \\ -y & x & 0 & 0 & 0 & t \end{bmatrix} =$$

$$\begin{bmatrix} -a11+t & -a12 & -a13 & -b11 & -b12 & -b13 \\ -a21 & -a22+t & -a23 & -b21 & -b22 & -b23 \\ -a31 & -a32 & -a33+t & -b31 & -b32 & -b33 \\ -c11 & -c12 & -c13 & a11+t & a21 & a31 \\ -c21 & -c22 & -c23 & a12 & a22+t & a32 \\ -c31 & -c32 & -c33 & a13 & a23 & a33+t \end{bmatrix} \begin{bmatrix} u & 0 & 0 & 0 & 0 & 0 \\ 0 & u & 0 & 0 & 0 & 0 \\ 0 & 0 & u & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$+ \begin{bmatrix} u\,a11 & u\,a12 & u\,a13 & b11 & b12+z & b13-y \\ u\,a21 & u\,a22 & u\,a23 & b21-z & b22 & b23+x \\ u\,a31 & u\,a32 & u\,a33 & b31+y & b32-x & b33 \\ u\,c11 & u\,c12-z & u\,c13+y & -a11 & -a21 & -a31 \\ u\,c21+z & u\,c22 & u\,c23-x & -a12 & -a22 & -a32 \\ u\,c31-y & u\,c32+x & u\,c33 & -a13 & -a23 & -a33 \end{bmatrix}.$$

Obviously, the remainder is equal to zero, if and only if $a11 = 0, a12 = 0, a13 = 0, a21 = 0, a22 = 0, a23 = 0, a31 = 0, a32 = 0, a33 = 0, b11 = 0, b12 = -z, b13 = y, b21 = z, b22 = 0, b23 = -x, b31 = -y, b32 = x, b33 = 0, c11 = 0, c12 = z/u, c13 = -y/u, c21 = -z/u, c22 = 0, c23 = x/u, c31 = y/u, c32 = -x/u, c33 = 0$. Therefore, the Maxwell equation has the following Hamiltonian canonical representation

$$\frac{\partial V}{\partial t} = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ 0 & 0 & 0 & \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x} \\ 0 & 0 & 0 & -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial z} & -\frac{\partial}{\partial y} & 0 & 0 & 0 \\ -\frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} & 0 & 0 & 0 \\ \frac{\partial}{\partial y} & -\frac{\partial}{\partial x} & 0 & 0 & 0 & 0 \end{bmatrix} V + N,$$

where $V = (u\,E_x, u\,E_y, u\,E_z, B_x, B_y, B_z)^T$, $N = -\mu_0(j_x, j_y, j_z, 0, 0, 0)^T$.

Example 1.3.9 Consider the following wave equation:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}. \tag{1.27}$$

Let $A = \begin{bmatrix} t^2 - x^2 \\ 0 \end{bmatrix}$ be the multivariate polynomial matrix corresponding to (1.27), and $B = \begin{bmatrix} t - a11 & -b11 \\ -c11 & t + d11 \end{bmatrix}$ be the polynomial matrix corresponding to the undetermined Hamiltonian system. By calling the software package DWR $[A, B, t]$, we get

$$\begin{bmatrix} t^2 - x^2 \\ 0 \end{bmatrix} = \begin{bmatrix} t - a11 & -b11 \\ -c11 & t + d11 \end{bmatrix} \begin{bmatrix} a11 + t \\ c11 \end{bmatrix} + \begin{bmatrix} a11^2 + b11c11 - x^2 \\ c11(a11 - d11) \end{bmatrix}.$$

Let the remainder be equal to zero. Then there are two possible cases to consider:

(1) $c11 = 0$ and $a11 = \pm x$. In particular, let $b11 = \sum_{i=0}^{k} b_i x^{2i}$ be an even degree polynomial with respect to x. Since $d11 = a11^*$ implies $d11 = \mp x$, Eq. (1.27) has the following Hamiltonian canonical representation

$$\frac{\partial}{\partial t} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \pm\frac{\partial}{\partial x} & \sum_{i=0}^{k} b_i \frac{\partial^{2i}}{\partial x^{2i}} \\ 0 & \pm\frac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

where $v_1 = \mp\frac{\partial u}{\partial x} + \frac{\partial u}{\partial t}$, $v_2 = 0$.

(2) $a11 = d11 = a11^*$ and $a11^2 + b11c11 = x^2$. Assume that $b11 \neq 0$, $c11 \neq 0$ (otherwise it is a special case of case (1)), then $b11 = \frac{1}{c11}(x^2 - a11^2)$, and hence equation (1.27) has infinite Hamiltonian canonical representations. In particular, if $c11 = 1$, $a11 = \sum_{i=0}^{m} a_i x^{2i}$, we have

$$\frac{\partial}{\partial t} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \sum_{i=0}^{m} a_i \frac{\partial^{2i}}{\partial x^{2i}} & -\hat{a}_0 + (1 - \hat{a}_1)\frac{\partial^2}{\partial x^2} - \sum\limits_{j=2}^{2m} \hat{a}_j \frac{\partial^{2j}}{\partial x^{2j}} \\ 1 & -\sum_{i=0}^{m} a_i \frac{\partial^{2i}}{\partial x^{2i}} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

where $v_1 = \frac{\partial u}{\partial t} + \sum_{i=0}^{m} a_i \frac{\partial^{2i}}{\partial x^{2i}}$, $v_2 = u$; $\hat{a}_j (0 \le j \le m)$ can be represented by the coefficients of $\hat{a}_0 = a_0^2$, $\hat{a}_1 = 2a_0a_1$, $\hat{a}_2 = a_1^2 + 2a_0a_2$, $\hat{a}_3 = 2a_0a_3 + 2a_1a_2$, $\hat{a}_4 = a_2^2 + 2a_0a_4 + 2a_1a_3$, ….

If $c11 = x^2$, $a11 = \sum_{i=1}^{m} a_i x^{2i}$, we have

$$\frac{\partial}{\partial t} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{m} a_i \frac{\partial^{2i}}{\partial x^{2i}} & (1 - \hat{a}_1) - \sum\limits_{j=1}^{2(m-1)} \tilde{a}_j \frac{\partial^{2j}}{\partial x^{2j}} \\ \frac{\partial^2}{\partial x^2} & -\sum_{i=1}^{m} a_i \frac{\partial^{2i}}{\partial x^{2i}} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

where $v_1 = \frac{\partial u}{\partial t} + \sum_{i=1}^{m} a_i \frac{\partial^{2i}}{\partial x^{2i}}$, $v_2 = \frac{\partial^2 u}{\partial x^2}$; $\tilde{a}_j (1 \le j \le m-1)$ can be expressed by the coefficient of $a11$, $\tilde{a}_1 = a_1^2$, $\tilde{a}_2 = 2a_1a_2$, $\tilde{a}_3 = a_2^2 + 2a_1a_3$, $\tilde{a}_4 = 2a_1a_4 + 2a_2a_3$, $\tilde{a}_5 = a_3^2 + 2a_2a_4$.

1.4 Monic Factorization of Bivariate Polynomial Matrices

In Sect. 1.3.3, by transforming the partial differential equations with constant coefficients into multivariate polynomial matrices, and by using the monic multivariate polynomial matrices as the divisor, the partial differential equations are transformed into infinite-dimensional Hamiltonian systems, and a class of monic factorization of polynomial matrices is also obtained. Here the method can be extended and the concrete method of the monic factorization of bivariate polynomial matrices (in theory, this method can be extended to the monic factorization of n-variable polynomial matrices), which is also the specific application of the software package of the pseudo-division algorithm of multivariate polynomial matrices. For linear monic factorization, this method can be applied to the inverse problem of infinite-dimensional Hamiltonian systems, and examples are given to verify the effectiveness of the method.

Definition 1.4.1 Let A be an $l \times m$ matrix n-variable polynomial, and B be an $l \times m$ matrix n-variable polynomial with the same principal variable as A and $I_B = I_l$. If there is a polynomial matrix Q such that $A = BQ$, then A is said to be factorized monically, and B is said to be a (left) monic factor of A.

1.4.1 Specific Method of the Monic Factorization of Bivariate Polynomial Matrices

Let $A = A_n x^n + \cdots + A_1 x + A_0$ be an $l \times m$ bivariate polynomial matrix and x be the principal variable, where $A_k = (a_{ij}^k)_{l\times m}$, $a_{ij}^k \in K[y]$ ($K[y]$ is the polynomial ring of variable y in the complex field), $k = 1, \ldots, n$, $A_n \neq 0$. Then the specific method of the monic factorization of A is as follows.

According to Definition 1.4.1, the monic factor of A may be 1, 2, ..., n order, so it must be searched from 1 to n order. The concrete steps are as follows:

[Step 1] The design of divisor B.

Let $B = I_l x^s + B_{s-1} x^{s-1} + \cdots + B_1 x + B_0$, where $B_k = (b_{ij}^k)$ is an $l \times l$ polynomial matrix, and $b_{ij}^k \in K[y]$, $k = 0, 1, \ldots, s$. Let s change from 1 to n in turn and perform Step 2 and Step 3 for each specific s.

[Step 2] Divide A by B, with x as the principal variable. According to 1.3.5, we have

$$A = BQ + R,$$

turn to Step 3.

[Step 3] Solve the equation $R = 0$. If there is no solution, turn to Step 1; otherwise, B and Q can be further obtained, and hence give a monic factorization of A, then turn to Step 1.

Since A is given, the degree of A with respect to the principal variable x is finite, so the process will terminate after finite steps. Sometimes, for a specific R, the Step 3 can call Step 2 several times, and some conditions of $R = 0$ can also be obtained, see Example 1.4.2.

Example 1.4.2 $A = \begin{bmatrix} 0 \\ 2x^2 + xy + 2y^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} x^2 + \begin{bmatrix} 0 \\ y \end{bmatrix} x + \begin{bmatrix} 0 \\ 2y^2 \end{bmatrix}.$

Firstly, we consider the linear monic factorization of A, so let

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} x + \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}.$$

From the software package of pseudo-division algorithm of multivariate polynomial matrices, it follows that

$$\begin{bmatrix} 0 \\ 2x^2 + xy + 2y^2 \end{bmatrix} = \begin{bmatrix} c_{11} + x & c_{12} \\ c_{21} & c_{22} + x \end{bmatrix} \begin{bmatrix} -2c_{12} \\ -2c_{22} + 2x + y \end{bmatrix} + \begin{bmatrix} c_{12}(2c_{11} + 2c_{22} - y) \\ 2c_{12}c_{21} + 2c_{22}^2 - c_{22}y + 2y^2 \end{bmatrix}.$$

If $R = 0$ is required, the first case is: $c_{12} = 0$, $2c_{22}^2 - c_{22}y + 2y^2 = 0$, that is, $c_{22} = \frac{1}{4}(y \pm \sqrt{15}y\mathtt{i})$, substituting it into the formula above we get

$$\begin{bmatrix} 0 \\ 2x^2 + xy + 2y^2 \end{bmatrix} = \begin{bmatrix} c_{11} + x & 0 \\ c_{21} & x + \frac{1}{4}(y \pm \sqrt{15}y\mathtt{i}) \end{bmatrix} \begin{bmatrix} 0 \\ 2x + \frac{1}{2}(y \mp \sqrt{15}y\mathtt{i}) \end{bmatrix}.$$

By analyzing the equation above, it can be obtained that if $c_{11} = \frac{1}{4}(y \mp \sqrt{15}y\mathtt{i})$, we have infinitely many Hamiltonian systems of the equation

$$2\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} + 2\frac{\partial^2 u}{\partial y^2} = 0.$$

The second case is: $2c_{11} + 2c_{22} - y = 0$, that is, $c_{22} = \frac{y}{2} - c_{11}$, and $c_{11} = \frac{1}{4}y \pm \triangle$, where $\triangle = \frac{1}{4}\sqrt{-16c_{12}c_{21} - 15y^2}$, so there is the monic factorization of the form:

$$\begin{bmatrix} 0 \\ 2x^2+xy+2y^2 \end{bmatrix} = \begin{bmatrix} x+\frac{1}{4}y \pm \triangle & c_{12} \\ c_{21} & x+\frac{1}{4}y \mp \triangle \end{bmatrix} \begin{bmatrix} -2c_{12} \\ 2x+\frac{1}{2}y \pm 2\triangle \end{bmatrix}.$$

It can be obtained by analyzing the result above when $\triangle$ (the variable y in $\triangle$ is regarded as $\frac{\partial}{\partial y}$) is a self-adjoint operator. The equation above corresponds to many infinite-dimensional Hamiltonian systems of the equation

$$2\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} + 2\frac{\partial^2 u}{\partial y^2} = 0.$$

In particular, let $\triangle = 0$, we can get $c_{12}c_{21} = -\frac{15}{16}y^2$, if additionally let $c_{12} = \frac{1}{2}$, $c_{21} = -\frac{15}{8}y^2$, we have

$$\begin{bmatrix} 0 \\ 2x^2+xy+2y^2 \end{bmatrix} = \begin{bmatrix} x+\frac{1}{4}y & \frac{1}{2} \\ -\frac{15}{8}y^2 & x+\frac{1}{4}y \end{bmatrix} \begin{bmatrix} -1 \\ 2x+\frac{1}{2}y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which is consistent with the results in [36]. However, the results above are more abundant than those in [36], and we find that the equation $2\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} + 2\frac{\partial^2 u}{\partial y^2} = 0$ can only have two Hamiltonian forms as shown.

In addition, consider the quadratic monic factorization of A, and let B be the following form:

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} x^2 + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} x + \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}.$$

By calling the software package of the pseudo-division algorithm of multivariate polynomial matrices, we get

$$\begin{aligned} \begin{bmatrix} 0 \\ 2x^2+xy+2y^2 \end{bmatrix} &= \begin{bmatrix} c_{11}+b_{11}x+x^2 & c_{12}+b_{12}x \\ c_{21}+b_{21}x & c_{22}+b_{22}x+x^2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} \\ &\quad + \begin{bmatrix} -2(c_{12}+b_{12}x) \\ -2c_{22}-2b_{22}x+y(x+2y) \end{bmatrix}, \end{aligned}$$

and to get $R = 0$ if and only if

$$\begin{cases} c_{12} = -b_{12}x, \\ c_{22} = -b_{22}x + y^2 + \frac{1}{2}xy. \end{cases}$$

Due to $c_{12} \in K[y]$, it follows from $c_{12} = -b_{12}x$ $c_{12} = b_{12} = 0$, and letting $c_{22} := -b_{22}x + y^2 + \frac{1}{2}xy$, we obtain

$$\begin{bmatrix} 0 \\ 2x^2+xy+2y^2 \end{bmatrix} = \begin{bmatrix} c_{11}+b_{11}x+x^2 & 0 \\ c_{21}+b_{21}x & x^2+\frac{1}{2}xy+y^2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix},$$

which gives the general form of the quadratic monic factorization of A by the pseudo-division algorithm of multivariate polynomial matrices.

In Example 1.4.2, b_{ij}, c_{ij} belong to $K[y]$. It can be seen from the discussions of Example 1.4.2 that it is impossible for A to have the linear and quadratic monic factorizations, that is, the multivariate polynomial matrix A cannot be factorized in this sense. It's interesting that, if we add a zero row to A, it can be factorized, see the following example.

Example 1.4.3 Let

$$A = \begin{bmatrix} x & y & 0 \\ 0 & x & y \\ x^2+y^2 & 0 & x^2+y^2 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0&0&0 \\ 0&0&0 \\ 1&0&1 \\ 0&0&0 \end{bmatrix} x^2 + \begin{bmatrix} 1&0&0 \\ 0&1&0 \\ 0&0&0 \\ 0&0&0 \end{bmatrix} x + \begin{bmatrix} 0&y&0 \\ 0&0&y \\ y^2&0&y^2 \\ 0&0&0 \end{bmatrix}.$$

Similar to Example 1.4.2 in discussing the linear monic factorization of A, there is a step like this:

$$\begin{bmatrix} x & y & 0 \\ 0 & x & y \\ x^2+y^2 & 0 & x^2+y^2 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} x & y & c_{13} & c_{14} \\ -y & x & c_{23} & c_{24} \\ 0 & 0 & c_{33}+x & c_{34} \\ 0 & 0 & c_{43} & -c_{33}+x \end{bmatrix} \begin{bmatrix} 1-c_{33} & 0 & -c_{13} \\ -c_{23} & 1 & -c_{23} \\ -c_{33}+x & 0 & -c_{33}+x \\ -c_{43} & 0 & -c_{43} \end{bmatrix}$$
$$+ \begin{bmatrix} c_{13}c_{33}+c_{14}c_{43}+c_{23}y & 0 & c_{13}c_{33}+c_{14}c_{43}+c_{23}y \\ c_{23}c_{33}+c_{24}c_{43}+y-c_{13}y & 0 & c_{23}c_{33}+c_{24}c_{43}+y-c_{13}y \\ {c_{33}}^2+c_{34}c_{43}+y^2 & 0 & {c_{33}}^2+c_{34}c_{43}+y^2 \\ 0 & 0 & 0 \end{bmatrix},$$

then $R = 0$ if and only if

$$\begin{cases} c_{13}c_{33}+c_{14}c_{43}+c_{23}y = 0, \\ c_{23}c_{33}+c_{24}c_{43}+y-c_{13}y = 0, \\ c_{33}^2+c_{34}c_{43}+y^2 = 0, \end{cases}$$

and hence $c_{14} = \frac{c_{34}(c_{23}y+c_{33}c_{13})}{c_{33}^2+y^2}$, $c_{13} = \frac{c_{23}c_{33}+c_{24}c_{43}+y}{y}$. However, since c_{14}, c_{13} should be polynomials with respect to the variable y, and as can be seen from the formula above, they have infinitely many forms. In order to make the monic factorization of A become a Hamiltonian system, it must be required that $c_{33} = 0, c_{43} = y, c_{34} = -y$, and then it can be obtained

$$\begin{bmatrix} x & y & 0 \\ 0 & x & y \\ x^2+y^2 & 0 & x^2+y^2 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} x & y & c_{13} & c_{14} \\ -y & x & c_{23} & c_{24} \\ 0 & 0 & x & -y \\ 0 & 0 & y & x \end{bmatrix} \begin{bmatrix} 1-c_{33} & 0 & -c_{13} \\ -c_{23} & 1 & -c_{23} \\ x & 0 & x \\ -y & 0 & -y \end{bmatrix}$$
$$+ \begin{bmatrix} (c_{14}+c_{23})y & 0 & (c_{14}+c_{23}y) \\ (1-c_{13}+c_{24})y & 0 & (1-c_{13}+c_{24})y \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

by the pseudo-division algorithm of the multivariate polynomial matrices. Since the monic factorization of A is a Hamiltonian system, we have $c_{14} = c_{23}$, and hence $c_{14} = c_{23} = 0$, $c_{24} = -1 + c_{13}$. Therefore,

$$\begin{bmatrix} x & y & 0 \\ 0 & x & y \\ x^2+y^2 & 0 & x^2+y^2 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} x & y & c_{13} & 0 \\ -y & x & 0 & -1+c_{13} \\ 0 & 0 & x & -y \\ 0 & 0 & y & x \end{bmatrix} \begin{bmatrix} 1-c_{33} & 0 & -c_{13} \\ 0 & 1 & 0 \\ x & 0 & x \\ -y & 0 & -y \end{bmatrix},$$

which is consistent with the results in [36]. It is further known that the formula above gives all forms of infinite-dimensional Hamiltonian systems corresponding to the two-dimensional equations in elasticity.

Now let $B = I_4x^2 + B_1x + B_0$, where $B_1 = (b_{ij})_{4\times4}$ and $B_0 = (c_{ij})_{4\times4}$. For the quadratic monic factorization of A, it can be known from the division of multivariate polynomial matrices that

$$\begin{bmatrix} x & y & 0 \\ 0 & x & y \\ x^2+y^2 & 0 & x^2+y^2 \\ 0 & 0 & 0 \end{bmatrix} =$$

$$\begin{bmatrix} c_{11}+b_{11}x+x^2 & c_{12}+b_{12}x & c_{13}+b_{13}x & c_{14}+b_{14}x \\ c_{21}+b_{21}x & c_{22}+b_{22}x+x^2 & c_{23}+b_{23}x & c_{24}+b_{24}x \\ c_{31}+b_{31}x & c_{32}+b_{32}x & c_{33}+b_{33}x+x^2 & c_{34}+b_{34}x \\ c_{41}+b_{41}x & c_{42}+b_{42}x & c_{43}+b_{43}x & c_{44}+b_{44}x+x^2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} +$$

$$\begin{bmatrix} -c_{13}+x-b_{13}x & y & -c_{13}-b_{13}x \\ -c_{23}-b_{23}x & x & -c_{23}-b_{23}x+y \\ -c_{33}-b_{33}x+y^2 & 0 & -c_{33}-b_{33}x+y^2 \\ -c_{43}-b_{43}x & 0 & -c_{43}-b_{43}x \end{bmatrix}.$$

It can be seen that A cannot have a quadratic monic factorization due to the expression of the remainder R.

1.5 Matrix Method of Solving Characteristic Sequences of the System of Polynomials

The cause of mathematical mechanization advocated by Mr. Wenjun Wu has made remarkable achievements, and the importance of Wu's method has been well known in the world. The primary purpose of Wu's method is to find the characteristic sequences of the system of polynomials, that is, Wu's ordering principle, which has many specific methods. The package charset ( ) is also provided in the automatic reasoning platform of mathematical mechanization in the Mathematical Mechanization Research Center of the Institute of System Science and Mathematics, Chinese Academy of Sciences. In this section, a new method for finding the characteristic sequences of polynomial systems in the algebraic case is given by using the software package of the pseudo-division algorithm of multivariate polynomial matrices, and the effectiveness of this method is verified by examples. From the examples, we can

see that this method can obtain different characteristic sequences from the package charset ( ), and sometimes it is simpler than the calculation results of charset ( ).

Our idea is to consider the system of polynomials as a polynomial matrix, that is, to consider it as a whole.

1.5.1 Expression of Remainder Based on Pseudo-Division Algorithm of Multivariate Polynomial Matrices

Let $A = \{a_1, \ldots, a_r\}$ be an ascending sequence, where $a_i = a(u, x_1, x_2, \ldots, x_i), i = 1, 2, \ldots, r$. Denote $B_i = \text{diag}(a_i, a_i, \ldots, a_i)_{n\times n}$, and the remainder expression of the system of polynomials $P = \{p_1, p_2, \ldots, p_n\}$ with respect to A is given below. For this reason, P is written in matrix form $\text{diag}(p_1, p_2, \ldots, p_n)$, which is still written as P.

Firstly, the system of P is divided by B_r, and according to (1.25), we get

$$I_r^{\alpha_r} P = B_r Q_1 + R_1,$$

taking the remainder of R_1 with respect to B_{r-1}, we get

$$I_{r-1}^{\alpha_{r-1}} R_1 = B_{r-1} Q_2 + R_2.$$

In general, letting R_k be divided by B_{r-k} yields

$$I_{r-k}^{\alpha_{r-k}} R_k = B_{r-k} Q_{k+1} + R_{k+1}, \tag{1.28}$$

where I_{r-k} represents the initial of the polynomial a_{r-k} with respect to the principal variable x_{r-k}, $k = 0, 1, \ldots, r-1$ and $R_0 = P$. From (1.28), it can be obtained that

$$I_1^{\alpha_1} \ldots I_{r-1}^{\alpha_{r-1}} I_r^{\alpha_r} P = I_1^{\alpha_1} \ldots I_{r-1}^{\alpha_{r-1}} B_r Q_1 + I_1^{\alpha_1} \ldots I_{r-2}^{\alpha_{r-2}} B_{r-1} Q_2 + \cdots + B_1 Q_r + R_r. \tag{1.29}$$

Expression (1.29) is called the remainder formula of the polynomial matrix case of the system of polynomials $P = \{p_1, p_2, \ldots, p_n\}$ being divided by the ascending sequence A, simplified as the remainder formula.

Remark 1.5.1 (i) The formula (1.29) achieves the effect of "$1 + 0 = 1$" which considers P as a whole [31].

(ii) Recalling the practice of [31], the polynomial systems $P = \{p_1, p_2, \ldots, p_n\}$ are divided by pairs of ascending sequences $A = \{a_1, \ldots, a_r\}$, which should be carried out in this way: Each polynomial in P needs to be divided by A, and for $p_i (i = 1, 2, \ldots, n)$, according to (2.2.4) in [32, p32]

$$I_1^{s_{i1}} \dots I_{r-1}^{s_{i,r-1}} I_r^{s_{ir}} p_i = I_1^{s_{i1}} \dots I_{r-1}^{s_{i,r-1}} a_r \hat{Q}_1 + I_1^{s_{i1}} \dots I_{r-2}^{s_{i,r-2}} a_{r-1} \hat{Q}_2 + \dots + a_1 \hat{Q}_r + \hat{R}_r, \quad i = 1, 2, \cdots, n.$$

In total, we need to do the pseudo-division algorithm of the multivariate polynomial matrices for $n \times r$ times. The formula (1.29) only needs to do the pseudo-division algorithm of multivariate polynomial matrices for r times. It is known that in the ordering process of the system of polynomials P, the ascending sequence A is selected from P. In general, r is less than n (unless P is its own characteristic sequence). In other words, using (1.29) to calculate the characteristic sequence involves less calculation, but it increases the demand for storage space. In addition, according to the minimum of $s_{ij} (j = 1, 2, \dots, r;\ i = 1, 2, \dots, n)$ and $\alpha_j (j = 1, 2, \dots, r)$, it is easy to prove that α_j is the least common multiple of $s_{1j}, s_{2j}, \dots, s_{nj}, (j = 1, 2, \dots, r)$.

(iii) The degenerate case of the remainder formula (1.29), that is, $n \times n$ polynomial matrix P and $B_i (i = 1, 2, \dots, r)$ are both reduced to multivariate polynomials, and the formula (1.29) is the same as the remainder formula in [31]. To a certain extent, the remainder formula in [31] is extended to the matrix case.

1.5.2 *Matrix Method of Solving Characteristic Sequences of the System of Polynomials*

Here the method of calculating characteristic sequences of the system of polynomial $PS = \{p_1, p_2, \dots, p_n\}$ by using the remainder formula above is given (only one method is given here). The specific steps are as follows:

[Step 1] Select a base sequence BS in PS, write PS as a diagonal matrix P, and make a diagonal matrix with each specific element in BS as a diagonal element. In this way, we can get the set B of base sequences consisting of polynomial matrices, and remainder R is obtained by taking the remainder of P with respect to B for remainder based on (1.29).

[Step 2] The non-zero elements in R are added to PS to obtain a new system of polynomials PS_1, and rewrite PS_1 as a diagonal matrix P_1. Because BS is taken out from PS,

$$Zero(PS_1) = Zero(PS, BS) = Zero(PS),$$
$$Zero(P_1) = Zero(P, B) = Zero(P).$$

Then we select a base sequence BS_1 from PS_1, and construct a diagonal matrix by using each element of BS_1 as the diagonal elements. In this way, the base sequence set B_1 of polynomial matrices is obtained. Let P_1 be divided by B_1 by using (1.29), and then we obtain the remainder R_1.

[Step 3] The steps above are repeated inductively. For $i > 1$, the base sequence BS_{i-1} is selected in the system of polynomials PS_{i-1}. PS_{i-1} is written as a diagonal

matrix P_{i-1}, and each specific element in BS_{i-1} is used as a diagonal element to make a diagonal matrix. Thus we get the base sequence B_{i-1} of polynomial matrices. Taking the remainder of P_{i-1} with respect to B_{i-1} by (1.29), we obtain the remainder R_{i-1}. By adding the non-zero polynomials in R_{i-1} to PS_{i-1}, a new system of polynomials PS_i is obtained.

Since PS is given, PS is a multivariate polynomial matrix of finite degree. Therefore, after finite steps of repeated elimination, we can obtain the system of polynomials PS_k and the base sequence BS_k selected by it, so let PS_k be divided by BS_k on the whole, and the remainder then is zero. If BS_k is a non-contradictory ascending sequence, then BS_k is the characteristic sequence of PS.

1.5.3 Software Package Description

By calling the software package of the pseudo-division algorithm of multivariate polynomial matrices, the method of finding the characteristic sequence in Sect. 1.5.2 has been compiled into a program package in the symbolic software system, and the package is SEARCHPS[ps, t], where ps is a given system of polynomial, t is the variable order table, and the variable order after t is high. The function of the system of polynomials package is to calculate ps the characteristic sequence with respect to variable order table t, and to output the result. "Charset is:" is the prompt information before the output of the result.

Example 1.5.2 p1= $x^2 - 1$; p2 = $y^3 * x + 3$; p3 = $z^2 - y * x$; p4 = $x - 1$;
ps = {p1, p2, p3, p4}; t = {x, y, z}; SEARCHPS[ps, t]
Charset is:
$\{-1 + x, 3 + y^2, -y + z^2\}$.

Example 1.5.3 p1 = $(x^3 - 1) * (y^2 - 1)$; p2 = $(x^4 * z^2 - z^3) * (3 * x - 3)$;p3 = $y^2 - x^2$; ps = {p1, p2, p3}; t = {x, y, z}; SEARCHPS[ps, t]
Charset is:
$\{1 - x^2 - x^3 + x^5, -x^2 + y^2, z^2 - x^2z^2 - x^3z^2 + x^4z^2 - z^3 + xz^3\}$.

Example 1.5.4 p1 = $y^2 - 1$; p2 = $x^4 * z^2 - z^3$; p3 = $y^2 - x^2$;
ps = {p1, p2, p3}; t = {x, y, z};
SEARCHPS[ps, t]
Charset is:
$\{-1 + x^2, -1 + y^2, z^2 - z^3\}$.

Example 1.5.5 p1 = $x3 * (x5^2 - (x4 - x1)^2) - 2 * x1 * x5 * (x4 - x1)$; p2 = $x3 * (x5^2 - (x4 - x2)^2) - 2 * x2 * x5 * (x4 - x2)$; p3 = $x4 - x1 - x2$;
ps ={p1, p2, p3}; t = {$x3$, $x4$, $x5$};
SEARCHPS[ps, t]
Charset is:
$\{x1^2x3 - x2^2x3, -x1 - x2 + x4, x1x2x5\}$.

Example 1.5.6 p1 = $(u1-x1)^2-x1^2$; p2 = $(u2-x1)^2+(u3-x2)^2-x1^2-x2^2$); p3 = $(u4-x1)^2+(x3-u2)^2-x1^2-x2^2$; p4 = $(x3-x5)*u3+(u4-x4)*(u2-u1)$; p5 = $x5*(u2-u1)-u3*(x4-x1)$; p6 = $(x3-x7)*u3+(u4-x6)*u2$; p7 = $x7*u2-x6*u3$;

ps ={p1, p2, p3, p4, p5, p6, p7}; t = {$x1, x2, x3, x4, x5, x6, x7$};

SEARCHPS[ps, t]

Charset is:

$\{u1^2-2u1x1, -u1u2+u2^2+u3^2-2u3x2, -u1^4u2^2+4u1^3u2^3-6u1^2u2^4$
$+4u1u2^5-u2^6+2u1^3u2u3^2-3u1^2u2^2u3^2+u2^4u3^2-u1^2u3^4+4u1u2u3^4$
$+u2^2u3^4-u3^6-4u1^3u3^2u4+8u1^2u2u3^2u4-4u1u2^2u3^2u4-4u1u3^4u4$
$+4u1^2u3^2u4^2-8u1u2u3^2u4^2+4u2^2u3^2u4^2+4u3^4u4^2-8u1^2u2u3^2x3$
$+16u1u2^2u3^2x3-8u2^3u3^2x3-8u2u3^4x3+4u1^2u3^2x3^2-8u1u2u3^2x3^2$
$+4u2^2u3^2x3^2+4u3^4x3^2, -u1u3^2/2-u1^2u4+2u1u2u4-u2^2u4+u1u3x3$
$-u2u3x3+u1^2x4-2u1u2x4+u2^2x4+u3^2x4, -u1^2u3^3+u1u2u3^3$
$+2u1u3^3u4-2u2u3^3u4-2u3^4x3+2u1^2u3^2x5-4u1u2u3^2x5+$
$2u2^2u3^2x5+2u3^4x5, u2^2u4+u2u3x3-u2^2x6-u3^2x6, -u1^2u2^2u3^2u4+$
$2u1u2^3u3^2u4$
$-u2^4u3^2u4-u2^2u3^4u4-u1^2u2u3^3x3+2u1u2^2u3^3x3-u2^3u3^3x3$
$-u2u3^5x3+u1^2u2^3u3x7-2u1u2^4u3x7+u2^5u3x7+u1^2u2u3^3x7$
$-2u1u2^2u3^3x7+2u2^3u3^3x7+u2u3^5x7\}$.

Chapter 2
Spectra of Infinite-Dimensional Hamiltonian Operators



The study of solving partial differential equations with the purpose of applications or the background of problems in physics, mechanics, and other areas is not only the most important content of traditional applied mathematics and mechanics but also an important part of contemporary mathematics and mechanics. It is an important bridge between theory and practical applications. However, it is often seen that although the partial differential equations with the background of practical problems are established, it is very difficult to solve them. It is worth noting that for the linear operators corresponding to linear systems, the spectral theory of linear operators can describe the structure of linear operators themselves, and characterize the construction of the solutions of the corresponding systems. For example, through the eigenvalue problem of matrices, the invariant subspace of a matrix can be characterized, its standard form can be written, and thus the construction of the solutions of the corresponding homogeneous or equations can be thoroughly clarified. The general form of the linear infinite-dimensional Hamiltonian canonical system is

$$\dot{U} = HU.$$

The linear operator H corresponding to the system is infinite-dimensional Hamiltonian operator, which is the block operator matrix in the form of

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subseteq X \times X \to X \times X,$$

where X is an infinite-dimensional Hilbert space, A is a densely defined closed linear operator in X, A^* is the adjoint of A, and B, C are self-adjoint operators in X. Therefore, by studying the spectral theory of infinite-dimensional Hamiltonian operators, we can characterize the construction of solutions of linear Hamiltonian canonical systems. From now on, we will discuss the spectral theory of infinite-dimensional Hamiltonian operators.

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2.1 Spectra of Linear Operators

2.1.1 Closedness and Closability of Unbounded Linear Operators

The graph of linear operators plays an important role in studying the closedness and closability of unbounded linear operators. This concept was first introduced by J. von Neumann.

Definition 2.1.1 The graph $G(T)$ of linear operator T in a Hilbert space X can be defined as

$$G(T) = \{(x, Tx) : x \in \mathscr{D}(T)\}.$$

$G(T)$ is a subset of the product space $X \times X$, where $X \times X$ is also a Hilbert space, and its inner product is defined as

$$((\varphi_1, \psi_1), (\varphi_2, \psi_2)) = (\varphi_1, \varphi_2) + (\psi_1, \psi_2).$$

The operator T is said to be closed , if its graph $G(T)$ is the closed subset in $X \times X$.

Then, what kind of set is a graph of an operator in space $X \times X$? We have the following conclusion, which can be seen in [8].

Lemma 2.1.2 *A subset G in the product space $X \times X$ is a graph of a linear operator T, if and only if G is a linear subspace such that $(0, y) \in G$ implies $y = 0$.*

Theorem 2.1.3 *A Linear operator T in a Hilbert space X is closed, if and only if for any $\{x_n\} \subset \mathscr{D}(T)$ with $x_n \to x$ and $Tx_n \to y (n \to \infty)$, it can be concluded that $x \in \mathscr{D}(T)$ and $y = Tx$.*

Proof Let $(x, y) \in \overline{G(T)}$. There exists $\{x_n\} \subset \mathscr{D}(T)$ such that

$$(x_n, Tx_n) \to (x, y)$$

as $n \to \infty$, so

$$\|(x_n - x, Tx_n - y)\|^2 = \|x_n - x\|^2 + \|Tx_n - y\|^2 \to 0.$$

Therefore, $x_n \to x$, $Tx_n \to y$, $x \in \mathscr{D}(T)$, and $y = Tx$, and hence $(x, y) \in G(T)$, that is, T is a closed operator.

Conversely, let $\{x_n\} \subset \mathscr{D}(T)$, $x_n \to x$, and $Tx_n \to y$ $(n \to \infty)$, then

$$(x_n, Tx_n) \to (x, y).$$

Because $G(T)$ is a closed, $(x, y) \in G(T)$, that is, $x \in \mathscr{D}(T)$ and $y = Tx$. □

Remark 2.1.4 It is easy to prove that the null space of a closed operator is closed, which is similar to the properties of the null space of a bounded linear operator.

Theorem 2.1.5 (Banach's closed graph theorem) *Let X be a Hilbert space. If $T : \mathscr{D}(T) \subset X \to X$ is a bounded linear operator. Then T is a closed operator if and only if $\mathscr{D}(T)$ is closed.*

Proof Sufficiency. Let $\{x_n\} \subset \mathscr{D}(T)$, $x_n \to x$ and $Tx_n \to y$. Since $\mathscr{D}(T)$ is closed, $x \in \mathscr{D}(T)$. Because T is continuous, $Tx_n \to Tx$, and therefore $y = Tx$, that is, T is a closed operator.

Necessity. For any $x \in \overline{\mathscr{D}(T)}$, there exists a point sequence $\{x_n\} \subset \mathscr{D}(T)$, such that $x_n \to x$ as $n \to \infty$. Because T is continuous, $\{Tx_n\}$ is a Cauchy sequence, so there exists $y \in X$ such that $Tx_n \to y$. Since T is a closed operator, we obtain $x \in \mathscr{D}(T)$ and $Tx = y$, that is, $\mathscr{D}(T)$ is closed. □

Definition 2.1.6 Let T_1, T be linear operators in a Hilbert space X. If $G(T_1) \supset G(T)$, then T_1 is called an extension of T, denoted as $T_1 \supset T$. Equivalently, $T_1 \supset T$ if and only if $\mathscr{D}(T_1) \supset \mathscr{D}(T)$ and $T_1 x = Tx$ for all $x \in \mathscr{D}(T)$.

Definition 2.1.7 A linear operator T is closable, if it has a closed extension. Every closable operator has the smallest closed extension, which is called its closure, denoted as $\overline{T}$.

Theorem 2.1.8 *A linear operator T in a Hilbert space X is closable, if and only if for any $\{x_n\} \subset \mathscr{D}(T)$, $x_n \to 0$ and $Tx_n \to y$ $(n \to \infty)$, we have $y = 0$.*

Proof Let $(0, y) \in \overline{G(T)}$. Then there exists $\{x_n\} \subset \mathscr{D}(T)$ such that

$$(x_n, Tx_n) \to (0, y),$$

so

$$\|(x_n, Tx_n - y)\|^2 = \|x_n\|^2 + \|Tx_n - y\|^2 \to 0,$$

therefore $x_n \to 0$, $Tx_n \to y$ as $n \to \infty$. According to the hypothesis, we have $y = 0$, that is, $\overline{G(T)}$ is a graph of an operator. From Lemma 2.1.2, we know that T is closable.

Conversely, let $\{x_n\} \subset \mathscr{D}(T)$, $x_n \to 0$ and $Tx_n \to y (n \to \infty)$, then

$$(x_n, Tx_n) \to (0, y),$$

and therefore $(0, y) \in \overline{G(T)}$. As T is closable, $\overline{G(T)}$ is a graph of an operator, so $y = 0$. □

2.1.2 Definition and Classification of Spectra of Linear Operators

Quantum mechanics is one of the principal application fields of spectral analysis of linear operators. In particular, the spectral theory of unbounded self-adjoint operators can be used to accurately explain some physical phenomena in quantum mechanics that cannot be explained by classical mechanics. For example, from the perspective of quantum mechanics, the eigenvalues of the state operator describe the energy level of the quantum system. In other words, the state of the entire energy level of the quantum system can be characterized by the point spectrum of the state operator. Therefore, we will discuss the basic properties of certain spectra of the diagonal, skew diagonal and upper triangular infinite-dimensional Hamiltonian operator, the non-negative Hamiltonian operator and the general infinite-dimensional Hamiltonian operator.

In order to study the spectral theory of linear operators, the concept of invertibility of linear operators is first mentioned. If a linear operator $T : \mathscr{D}(T) \subset X \to X$ is injective, its inverse operator T^{-1} can be defined. It is well known that the existence and continuity of T^{-1} are quite important. T is said to be invertible if its inverse is bounded and the domain is the whole space (namely, $\mathscr{R}(T) = X$). There are only two possibilities for the operator $T - \lambda I$ on a finite-dimensional space: not injective or invertible. If $T - \lambda I$ is not injective, the complex number λ is the eigenvalue of T. But in infinite-dimensional spaces, the situation is very complicated. The operator $T : \mathscr{D}(T) \subset X \to X$ in an infinite-dimensional space X is not necessarily only the above two possibilities, there may be other cases.

Example 2.1.9 Let $X = \ell^2[1, \infty)$, and define the right shift operator $S_r : \ell^2 \to \ell^2$, that is,

$$y = S_r x,$$

for any $x = (\xi_1, \xi_2, \xi_3, \ldots) \in X$ where $y = (0, \xi_1, \xi_2, \xi_3, \ldots)$. Then it is easy to prove that S_r is a bounded linear operator such that $\|S_r x\|^2 = \|x\|^2$, that is, $\|S_r\| = 1$. But, S_r is injective and not invertible. In fact, for any $y = (\eta_1, \eta_2, \eta_3, \ldots)$, $\eta_1 \neq 0$ in X, it is easy to prove that y is not contained in $\mathscr{R}(S_r)$, that is, S_r is not surjection. Therefore, S_r is injective but is not a invertible operator.

In view of the complexity of the infinite-dimensional case, according to the classification for the complex number λ, the following definition of regular points can be given. Hereafter, we always assume that X is an infinite-dimensional complex Hilbert space and I is a unit (identity) operator in X.

Definition 2.1.10 Let $\lambda \in \mathbb{C}$, and T be a densely defined linear operator in X. If the operator $T - \lambda I$ is injective, $(T - \lambda I)^{-1}$ is bounded and $\mathscr{R}(T - \lambda I) = X$, then we say $\lambda \in \mathbb{C}$ is a regular point of T, and the set of all regular points is the resolvent set, denoted as $\rho(T)$. Sometimes, $R(T, \lambda) = (T - \lambda I)^{-1}$ is regarded as the resolvent of T.

Remark 2.1.11 For the general linear operator T, there are different definitions for the invertibility of T in different references. For example, [8, 43], and we adopt the definition in [43]. However, for densely defined closed linear operators, the above two definitions are unified into T being a bijection.

The complement of the resolvent set $\rho(T)$ is called the spectrum of T, denoted as $\sigma(T)$; $\sup\{|\lambda| : \lambda \in \sigma(T)\}$ is called the spectral radius of T, denoted as $r(T)$. When $\lambda \in \sigma(T)$, there are three possible cases:

(i) $T - \lambda I$ is not injective. In this case λ is called the point spectrum of T, and the set of the point spectrum is denoted as $\sigma_p(T)$;
(ii) $T - \lambda I$ is injective, but $\overline{\mathscr{R}(T - \lambda I)} \neq X$. In this case λ is a residual spectrum of T, and the set of the residual spectrum is denoted as $\sigma_r(T)$;
(iii) $T - \lambda I$ is injective, $\overline{\mathscr{R}(T - \lambda I)} = X$ but $\mathscr{R}(T - \lambda I) \neq X$, in this case λ is the continuous spectrum of T, and the set of the continuous spectrum is denoted as $\sigma_c(T)$.

Obviously, $\sigma_p(T)$, $\sigma_r(T)$ and $\sigma_c(T)$ are disjoint sets, and

$$\sigma(T) = \sigma_p(T) \cup \sigma_r(T) \cup \sigma_c(T).$$

Besides, if $\mathscr{R}(T - \lambda I) \neq X$, λ is called a defect spectrum of T, and the defect spectrum is denoted as $\sigma_\delta(T)$. It is obvious that

$$\sigma_r(T) \cup \sigma_c(T) \subset \sigma_\delta(T), \quad \sigma(T) = \sigma_p(T) \cup \sigma_\delta(T).$$

In addition, according to the density and closedness of the range, the point spectrum and residual spectrum of a linear operator T in X can be further subdivided:

$$\begin{aligned}
\sigma_{p,1}(T) &= \{\lambda \in \sigma_p(T) : \mathscr{R}(T - \lambda) = X\};\\
\sigma_{p,2}(T) &= \{\lambda \in \sigma_p(T) : \overline{\mathscr{R}(T - \lambda)} = X, \mathscr{R}(T - \lambda) \text{ is not closed}\};\\
\sigma_{p,3}(T) &= \{\lambda \in \sigma_p(T) : \overline{\mathscr{R}(T - \lambda)} \neq X, \mathscr{R}(T - \lambda) \text{ is closed}\};\\
\sigma_{p,4}(T) &= \{\lambda \in \sigma_p(T) : \overline{\mathscr{R}(T - \lambda)} \neq X, \mathscr{R}(T - \lambda) \text{ is not closed}\};\\
\sigma_{r,1}(T) &= \{\lambda \in \sigma_r(T) : \mathscr{R}(T - \lambda) \text{ is closed}\};\\
\sigma_{r,2}(T) &= \{\lambda \in \sigma_r(T) : \mathscr{R}(T - \lambda) \text{ is not closed}\}.
\end{aligned}$$

By further subdividing the point spectrum and the residual spectrum, some necessary and sufficient conditions between the spectrum and the residual spectrum can be obtained.

Lemma 2.1.12 *Let T be a densely defined closed linear operator in a Hilbert space X. Then*

(i) $\lambda \in \sigma_{p,1}(T)$ *if and only if* $\overline{\lambda} \in \sigma_{r,1}(T^*)$;
(ii) $\lambda \in \sigma_{p,2}(T)$ *if and only if* $\overline{\lambda} \in \sigma_{r,2}(T^*)$;

(iii) $\lambda \in \sigma_{p,3}(T)$ *if and only if* $\overline{\lambda} \in \sigma_{p,3}(T^*)$*;*
(iv) $\lambda \in \sigma_{p,4}(T)$ *if and only if* $\overline{\lambda} \in \sigma_{p,4}(T^*)$.

Proof (i) If $\lambda \in \sigma_{p,1}(T)$, $T - \lambda I$ is not injective, so there exists $x \neq 0$ such that

$$((T - \lambda I)x, x^*) = (x, (T^* - \overline{\lambda} I)x^*) = 0$$

for any $x^* \in \mathscr{D}(T^*)$, so $\mathscr{R}(T^* - \overline{\lambda} I)$ is not dense. Since $T - \lambda I$ is surjective, $\mathscr{R}(T - \lambda I)$ is a closed set, and hence $\mathscr{R}(T^* - \overline{\lambda} I)$ is also a closed set. Now we prove that $T^* - \overline{\lambda} I$ is injective. Let

$$(T^* - \overline{\lambda} I)x^* = 0,$$

then

$$(x, (T^* - \overline{\lambda} I)x^*) = ((T - \lambda I)x, x^*) = 0$$

for any $x \in \mathscr{D}(T)$. Because $\mathscr{R}(T - \lambda I) = X$, $x^* = 0$. In summary, $\overline{\lambda} \in \sigma_{r,1}(T^*)$.

Conversely, let $\overline{\lambda} \in \sigma_{r,1}(T^*)$, then from the fact that $\mathscr{R}(T^* - \overline{\lambda} I)$ is a closed set we know that so is $\mathscr{R}(T - \lambda I)$. As $\mathscr{R}(T^* - \overline{\lambda} I)$ is not dense, there exists $x \neq 0$ such that

$$(x, (T^* - \overline{\lambda} I)y) = 0,$$

for any $y \in \mathscr{D}(T^*)$, so $x \in \mathscr{D}(T)$ and $((T - \lambda I)x, y) = 0$. Because of the density of $\mathscr{D}(T^*)$, we get $\lambda \in \sigma_{p,1}(T) \cup \sigma_{p,3}(T)$. Therefore, we can assert that $\mathscr{R}(T - \lambda I)$ is dense. If not, assume there exists $0 \neq x^* \in X$ such that

$$((T - \lambda I)x, x^*) = 0,$$

then $x^* \in \mathscr{D}(T^*)$ and $(x, (T^* - \overline{\lambda} I)x^*) = 0$, so $(T^* - \overline{\lambda} I)x^* = 0$, which contradicts the fact $T^* - \overline{\lambda} I$ being injective. So, $\lambda \in \sigma_{p,1}(T)$.

The proof of (ii) is completely similar to (i). (iii) If $\lambda \in \sigma_{p,3}(T)$, from the fact that $\mathscr{R}(T - \lambda I)$ is a closed set we know that so is $\mathscr{R}(T^* - \overline{\lambda} I)$. Let x be the eigenvector corresponding to λ, then

$$((T - \lambda I)x, y) = (x, (T^* - \overline{\lambda} I)y) = 0$$

for any $y \in \mathscr{D}(T^*)$. Therefore, $\mathscr{R}(T^* - \overline{\lambda} I)$ is not dense. Since $\mathscr{R}(T - \lambda I)$ is not dense, there exists $\widetilde{x} \neq 0$ such that

$$((T - \lambda I)x, \widetilde{x}) = 0$$

for any $x \in \mathscr{D}(T)$. Therefore, $\widetilde{x} \in \mathscr{D}(T^*)$ and $(T^* - \overline{\lambda} I)\widetilde{x} = 0$, that is, $\overline{\lambda} \in \sigma_{p,3}(T^*)$. Conversely, the proof is similar.

The proof of (iv) is completely similar to (iii). □

For a densely defined closed linear operator T in X, the fact that T^{-1} is continuous is not equivalent to that of T is invertible. In fact, if T^{-1} is continuous but we do not

necessarily have $\mathscr{R}(T) = X$, that is, T^{-1} is not necessarily a bounded linear operator defined on the whole space. According to Banach's closed range theorem, it is easy to prove that T^{-1} is bounded if and only if there exists $M > 0$ such that

$$\|Tx\| \geq M\|x\|, \quad x \in \mathscr{D}(T). \tag{2.1}$$

If T satisfies (2.1), then it is said to be bounded below, and the set of all complex numbers λ such that $T - \lambda I$ is bounded below is therefore called the regular region field of T, denoted as

$$\Gamma(T) = \{\lambda \in \mathbb{C} : \mathscr{R}(T^* - \overline{\lambda} I) = X\}.$$

According to the definition of 1-class residual spectrum, it is not difficult to know

$$\Gamma(T) = \sigma_{r,1}(T) \cup \rho(T).$$

For normal operators, the boundedness below and invertibility are equivalent.

The following is a necessary and sufficient condition for a densely defined closed linear operator T in X to be bounded below.

Lemma 2.1.13 *Let T be a densely defined closed linear operator in a Hilbert space X. Then the following assertions are equivalent:*

(i) *T is bounded below;*
(ii) *T is injective and $\mathscr{R}(T)$ is a closed set;*
(iii) *$\mathscr{R}(T^*) = X$;*
(iv) *There exists $R \in \mathscr{B}(X)$ such that $T^*R = I$;*
(v) *There exists $L \in \mathscr{B}(X)$ such that $LT = I_{\mathscr{D}(T)}$.*

Proof (i)⇒(ii): If T is bounded below, $0 \in \sigma_{r,1}(T) \cup \rho(T)$, which implies that T is injective and $\mathscr{R}(T)$ is a closed set.

(ii)⇒(iii): If T is injective and $\mathscr{R}(T)$ is closed, $0 \in \sigma_{r,1}(T) \cup \rho(T)$, which implies that $0 \in \sigma_{p,1}(T^*) \cup \rho(T^*)$, so $\mathscr{R}(T^*) = X$.

(iii)⇒(iv): Let $X = \mathscr{N}(T^*) \oplus \mathscr{N}(T^*)^{\perp}$. Then T^* has block form

$$T^* = \begin{bmatrix} 0 & T_1^* \end{bmatrix} : \mathscr{N}(T^*) \oplus (\mathscr{N}(T^*)^{\perp} \cap \mathscr{D}(T^*) \to X.$$

It is easy to prove that $T_1^* : \mathscr{N}(T^*)^{\perp} \cap \mathscr{D}(T^*) \to X$ is closed, injective, and $\mathscr{R}(T_1^*) = \mathscr{R}(T^*) = X$, that is, T_1^* is invertible. Let

$$R = \begin{bmatrix} 0 \\ (T_1^*)^{-1} \end{bmatrix},$$

then

$$T^*R = \begin{bmatrix} 0 & T_1^* \end{bmatrix} \begin{bmatrix} 0 \\ (T_1^*)^{-1} \end{bmatrix} = I.$$

(iv)⇒(v): Because $(R^*T)^* = T^*R = I$, we have $\overline{R^*T} = I$. Therefore, there exists $L \in \mathscr{B}(X)$ such that $LT = I_{\mathscr{D}(T)}$.

(v)⇒(i): For any $x \in \mathscr{D}(T)$,

$$\|x\| = \|LTx\| \leq \|L\|\|Tx\|.$$

Since $\|L\| \neq 0$, the conclusion is established. □

It is obvious that the densely defined closed linear operator T in X is invertible if and only if T is bounded below and $\overline{\mathscr{R}(T - \lambda I)} = X$. In this way, another method of spectral point classification has been obtained: for a densely defined closed linear operator T in X, if $T - \lambda I$ is not bounded below, that is, there exists a vector sequence $\{x_n\}_{n=1}^{+\infty}$, $\|x_n\| = 1, n = 1, 2, \ldots$ such that

$$\|(T - \lambda I)x_n\| \to 0, \tag{2.2}$$

as $n \to \infty$, then we say λ belongs to the approximate point spectrum of T, and the set of all such λ are denoted as $\sigma_{ap}(T)$. It is easy to know

$$\sigma_{ap}(T) = \sigma(T) \backslash \sigma_{r,1}(T).$$

If $\overline{\mathscr{R}(T - \lambda I)} \neq X$, we say λ belongs to the compression spectrum of T, and the set of al such λ is denoted as $\sigma_{com}(T)$. Obviously,

$$\sigma(T) = \sigma_{ap}(T) \cup \sigma_{com}(T).$$

According to the definition, it is not difficult to find that the set of eigenvalues in the finite-dimensional space corresponds to the set of point spectra in the infinite-dimensional space. In a finite-dimensional space, if $T - \lambda I$ is injective, there will be no $\overline{\mathscr{R}(T - \lambda I)} \neq X$, $\overline{\mathscr{R}(T - \lambda I)} = X$, $\mathscr{R}(T - \lambda I) \neq X$, and hence $\sigma_r(T) = \emptyset$ and $\sigma_c(T) = \emptyset$. Some specific examples will be given to illustrate the spectral point classification of linear operators.

Example 2.1.14 Let $X = \ell^2[1, \infty)$. Define the operator $A : \ell^2 \to \ell^2$ by

$$y = Ax,$$

where for any $x = (\xi_1, \xi_2, \xi_3, \ldots) \in X$, let $y = (\frac{\xi_1}{1}, \frac{\xi_2}{2}, \frac{\xi_3}{3}, \ldots)$. The corresponding sets $\sigma_p(A)$, $\sigma_r(A)$, $\sigma_c(A)$ and $\rho(A)$ are as follows.

For $\lambda \neq \frac{1}{n} (n = 1, 2, \ldots)$, it is easy to get $(A - \lambda I)x = 0$ if and only if $x = 0$, and $A - \lambda I$ is surjective, and hence $\rho(A) = \{\lambda : \lambda \neq \frac{1}{n}, n = 1, 2, \ldots\}$.

Obviously, A is injective, $A - \lambda$ is not injective for $\lambda = \frac{1}{n} (n = 1, 2, \ldots)$, and $\mathscr{R}(A - \lambda I)$ is closed but not dense, so

$$\sigma_p(A) = \sigma_{p,3}(A) = \{\lambda : \lambda = \frac{1}{n}, n = 1, 2, \ldots\}.$$

It can also be proven that $\mathscr{R}(A)$ is dense but not surjective, and hence $\sigma_c(A) = \{0\}$, $\sigma_r(A) = \emptyset$ and

$$\sigma_{ap}(A) = \{\lambda : \lambda = \frac{1}{n}, n = 1, 2, \ldots\} \cup \{0\},$$

$$\sigma_{\delta}(A) = \{\lambda : \lambda = \frac{1}{n}, n = 1, 2, \ldots\} \cup \{0\},$$

$$\sigma_{com}(A) = \{\lambda : \lambda = \frac{1}{n}, n = 1, 2, \ldots\}.$$

Example 2.1.15 Let $X = \ell^2[1, \infty)$. Define the right shift operator $S_r : \ell^2 \to \ell^2$, that is, for any $x = (\xi_1, \xi_2, \xi_3, \ldots)$,

$$y = S_r x,$$

where $y = (0, \xi_1, \xi_2, \xi_3, \ldots) \in X$.

In fact, for any $\lambda \neq 0$, let $(S_r - \lambda I)x = 0$, and it is easy to see that $x = 0$, that is $\lambda \notin \sigma_p(S_r)$. Let $S_r x = 0$, then we have $x = 0$. Therefore, $\sigma_p(S_r) = \emptyset$.

If $|\lambda| > 1$, for any $y = (\eta_1, \eta_2, \eta_3, \ldots) \in X$, taking

$$x = (-\frac{\eta_1}{\lambda}, -(\frac{\eta_1}{\lambda^2} + \frac{\eta_2}{\lambda}), -\frac{\eta_1}{\lambda^3} - \frac{\eta_2}{\lambda^2} - \frac{\eta_3}{\lambda}, \ldots)$$

we can know $x \in X$ and

$$(S_r - \lambda I)x = y.$$

Therefore, $\{\lambda : |\lambda| > 1\} \subset \rho(S_r)$. For the converse, when $\lambda \in \rho(S_r)$, it is easy to prove $|\lambda| > 1$. So, $\rho(S_r) = \{\lambda : |\lambda| > 1\}$, and $\sigma(S_r) = \sigma_r(S_r) \cup \sigma_c(S_r) = \{\lambda : |\lambda| \leq 1\}$.

Remark 2.1.16 For operator S_r, we also have

$$\sigma_r(S_r) = \sigma_{com}(S_r) = \{\lambda : |\lambda| < 1\},$$

$$\sigma_c(S_r) = \sigma_{ap}(S_r) = \{\lambda : |\lambda| = 1\},$$

$$\sigma_{\delta}(S_r) = \{\lambda : |\lambda| \leq 1\}.$$

For further details, see [8–10].

The following are some basic properties of the spectral sets of densely defined closed linear operators.

Lemma 2.1.17 *Let T be a densely defined closed linear operator in a Hilbert space X. Then*

(i) *If* $\lambda \in \sigma_p(T)$, *then* $\overline{\lambda} \in \sigma_p(T^*) \cup \sigma_r(T^*)$;
(ii) *If* $\lambda \in \sigma_r(T)$, *then* $\overline{\lambda} \in \sigma_p(T^*)$;
(iii) $\lambda \in \sigma(T)$ *if and only if* $\overline{\lambda} \in \sigma(T^*)$;
(iv) $\lambda \in \sigma_c(T)$ *if and only if* $\overline{\lambda} \in \sigma_c(T^*)$;
(v) $\lambda \in \sigma_{ap}(T)$ *if and only if* $\overline{\lambda} \in \sigma_\delta(T^*)$;
(vi) $\lambda \in \sigma_{com}(T)$ *if and only if* $\overline{\lambda} \in \sigma_p(T^*)$.

Proof (i) Let $\lambda \in \sigma_p(T)$ and x_0 is a corresponding eigenvector. If $T^* - \overline{\lambda}$ is not injective, then the conclusion is obviously true.

If $T^* - \overline{\lambda}$ is injective, then

$$((T - \lambda)x_0, y) = (x_0, (T^* - \overline{\lambda})y) = 0$$

for any $y \in X$ which implies $\overline{R(T^* - \overline{\lambda})} \neq X$, so $\overline{\lambda} \in \sigma_r(T^*)$.

(ii) Let $\lambda \in \sigma_r(T)$. Since

$$\overline{R(T - \lambda)}^{\perp} = N(T^* - \overline{\lambda}),$$

we have $\mathscr{N}(T^* - \overline{\lambda}) \neq \{0\}$, which implies $\overline{\lambda} \in \sigma_p(T^*)$ by the definition of the residual spectrum.

(iii) We only need to prove $\lambda \in \rho(T)$ if and only if $\overline{\lambda} \in \rho(T^*)$. If $\lambda \in \rho(T)$, from

$$\overline{R(T - \lambda)}^{\perp} = N(T^* - \overline{\lambda}),$$

we have that $(T^* - \overline{\lambda}I)^{-1}$ exists and is bounded. Noting that

$$(T^* - \overline{\lambda}I)^{-1} = ((T - \lambda I)^{-1})^*,$$

so $\overline{\lambda} \in \rho(T^*)$.

For $\overline{\lambda} \in \rho(T^*)$, the proof of $\lambda \in \rho(T)$ is similar.

(iv) Using the fact that $\mathscr{R}(T - \lambda I)$ is closed if and only if $\mathscr{R}(T^* - \overline{\lambda}I)$ is closed, the conclusion is obvious.

(v) When $\lambda \in \sigma_{ap}(T)$, $\lambda \in \sigma_p(T) \cup \sigma_{r,2}(T) \cup \sigma_c(T)$, and from the proof of (i)-(v) we know that $\mathscr{R}(T^* - \overline{\lambda}I)$ is not surjective, which implies $\overline{\lambda} \in \sigma_\delta(T^*)$.

Conversely, when $\overline{\lambda} \in \sigma_\delta(T^*)$, $\lambda \in (\sigma_p(T) \backslash \sigma_{p,1}(T)) \cup \sigma_r(T) \cup \sigma_c(T)$, so from the proof of (i)–(v) we know $\lambda \in \sigma_p(T) \cup \sigma_{r,2}(T) \cup \sigma_c(T)$, which implies $\lambda \in \sigma_{ap}(T)$.

(vi) If $\lambda \in \sigma_{com}(T)$, then $\mathscr{R}(T - \lambda I)$ is not dense, $\lambda \in \sigma_{p,3}(T) \cup \sigma_{p,4}(T) \cup \sigma_r(T)$, from the proof of (i)–(iv), we have $\overline{\lambda} \in \sigma_p(T^*)$.

Conversely, if $\overline{\lambda} \in \sigma_p(T^*)$, then $\lambda \in \sigma_{p,3}(T) \cup \sigma_{p,4}(T) \cup \sigma_r(T)$ and $\mathscr{R}(T - \lambda I)$ is not dense, so $\lambda \in \sigma_{com}(T)$. The proof is completed. □

Remark 2.1.18 If T is a self-adjoint operator, then

$$\sigma_{ap}(T) = \sigma_\delta(T), \quad \sigma_{com}(T) = \sigma_p(T).$$

In addition to the spectra mentioned above, a very important type of spectra that needs to be mentioned is the essential spectrum. The essential spectrum theory of linear operators is an important part of the spectral analysis of linear operators. It has a wide range of applications in many fields of mathematics and physics, including matrix theory, function theory, complex analysis, differential and integral equations, and quantum physics. The original definition of essential spectrum dates back to the work of Weyl [45]. Let T be a self-adjoint operator in a Hilbert space X, then all the accumulation points and infinite-dimensional isolated eigenvalues in $\sigma(T)$ are called the essential spectrum of T, denoted by $\sigma_{ess}(T)$, that is,

$$\sigma_{ess}(T) = \sigma(T)\backslash\sigma_{dis}(T),$$

where $\sigma_{dis}(T)$ represents the set of isolated eigenvalues with finite multiplicity (see Lemma B.1 in [50] for more details). Weyl not only proved $\lambda \notin \sigma_{ess}(T)$ if and only if $\mathscr{R}(T-\lambda I)$ is closed and $\mathtt{dim}\mathscr{N}(T-\lambda I) = \mathtt{dim}\mathscr{R}(T-\lambda I)^{\perp} < \infty$ but also proved that the essential spectrum remains unchanged under compact perturbation, that is,

$$\sigma_{ess}(T+K) = \sigma_{ess}(T)$$

for any compact operator K. Later, based on the closedness of $\mathscr{R}(T-\lambda I)$ and whether $\mathtt{dim}\mathscr{N}(T-\lambda I)$, $\mathtt{dim}\mathscr{R}(T-\lambda I)^{\perp}$ are finite or not, the definitions of Fredholm operators and semi-Fredholm operators are introduced, and the definitions of essential spectra of non self-adjoint operators are given. It is worth noting that the definitions of essential spectra is different in many literatures. In order to give the definitions of different essential spectra, the definitions of Fredholm operators and semi-Fredholm operators are introduced.

Definition 2.1.19 Let T be a densely defined closed linear operator. If $\mathscr{R}(T)$ is closed and $\mathtt{dim}\mathscr{N}(T) < \infty$, then we say T is a left semi-Fredholm operator; if $\mathscr{R}(T)$ is closed and $\mathtt{dim}\mathscr{R}(T)^{\perp} < \infty$, then T is a right semi-Fredholm operator. If T is both left semi-Fredholm and right semi-Fredholm, then T is a Fredholm operator, anf if T is left semi-Fredholm or right semi-Fredholm operator, then T is a semi-Fredholm operator.

Theorem 2.1.20 *Let T be a densely defined closed linear operator. If $\mathscr{R}(T)$ is closed set and $\mathtt{dim}\mathscr{N}(T) = 0$, then T is a left invertible operator; If $\mathscr{R}(T) = X$, then T is a right invertible operator.*

Remark 2.1.21 A left invertible operator is left semi-Fredholm, and a right invertible operator is right semi-Fredholm. T is a left invertible operator if and only if T^* is a right invertible operator, and T is a left semi-Fredholm operator if and only if T^* is a right semi-Fredholm operator.

Definition 2.1.22 If T is a semi-Fredholm operator, we say $\mathtt{ind}(T) = nul(T) - def(T)$ is the index of the operator T, where $nul(T) =: \mathtt{dim}\mathscr{N}(T)$, $def(T) =: \mathtt{dim}\mathscr{R}(T)^{\perp}$. Fredholm operator with index 0 is called a Weyl operator.

The essential spectra defined in the following eight forms are discussed below:

$$\sigma_{e,i}(T) = \mathbb{C}\backslash\triangle_i(T), \quad i = 1, 2, \ldots, 6,$$
$$\sigma_{e,7}(T) = \bigcap_{K\in\mathscr{K}(X)} \sigma_{ap}(T+K),$$
$$\sigma_{e,8}(T) = \bigcap_{K\in\mathscr{K}(X)} \sigma_{\delta}(T+K),$$

where $\triangle_i(T), i = 1, 2, \ldots, 6$, are defined as

$$\triangle_1(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is a left semi-Fredholm operator}\},$$
$$\triangle_2(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is a right semi-Fredholm operator}\},$$
$$\triangle_3(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is a semi-Fredholm operator}\},$$
$$\triangle_4(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is a Fredholm operator}\},$$
$$\triangle_5(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is a Weyl operator}\},$$
$$\triangle_6(T) = \{\lambda \in \triangle_5(T) : \lambda \text{ some decentric areacontained in}\rho(T)\}.$$

Remark 2.1.23 The sets $\sigma_{e,1}(T), \sigma_{e,2}(T), \sigma_{e,3}(T), \sigma_{e,4}(T)$ are called the Gutasfson essential spectrum, Weidmann essential spectrum, Kato essential spectrum, Wolf essential spectrum, respectively; $\sigma_{e,5}(T)$ is the Schechter essential spectrum or Weyl spectrum, and $\sigma_{e,6}(T)$ is called the Browder essential spectrum; $\sigma_{e,7}(T), \sigma_{e,8}(T)$ are called essential approximate point spectra and essential deficiency spectrum, respectively (for more details see [46–49]).

It is not hard to find

$$\sigma_{e,3}(T) = \sigma_{e,1}(T) \cap \sigma_{e,2}(T) \subset \sigma_{e,4}(T) \subset \sigma_{e,5}(T) \subset \sigma_{e,6}(T).$$

Besides,

$$\triangle_6(T) = \rho(T) \cup \{\lambda \in \mathbb{C} : \lambda \text{ is an isolated spectral point such that } T - \lambda I \text{ is Weyl}\},$$

and hence

$$\sigma_{e,6}(T) = \sigma(T)\backslash\sigma_{dis}(T).$$

In addition, there are also the following relationships for the various essential spectra.

Lemma 2.1.24 *Let T be a densely defined closed linear operator in a Hilbert space X, then*

(i) $(\sigma_c(T) \cup \sigma_{p,2}(T) \cup \sigma_{p,4}(T) \cup \sigma_{r,2}(T)) \subset \bigcap_{i=1}^{8} \sigma_{ess,i}(T)$;
(ii) $\lambda \in \sigma_{e,1}(T)$ *if and only if* $\overline{\lambda} \in \sigma_{e,2}(T^*)$;
(iii) $\lambda \in \sigma_{e,i}(T)$ *if and only if* $\overline{\lambda} \in \sigma_{e,i}(T^*)$, i=3, 4, 5, 6;

(iv) $\lambda \in \sigma_{e,7}(T)$ *if and only if* $\overline{\lambda} \in \sigma_{e,8}(T^*)$.

Proof Taking into account the definitions of various essential spectra, the proof of this Lemma is obvious. □

The following are about the properties of $\sigma_{e,7}(T)$ and $\sigma_{e,8}(T)$.

Lemma 2.1.25 *Let T be a densely defined closed linear operator in a Hilbert space X. Then*

(i) $\lambda \notin \sigma_{e,7}(T)$ *if and only if* $\lambda \in \triangle_1(T)$ *and* $\mathtt{ind}(T - \lambda I) \leq 0$;
(ii) $\lambda \notin \sigma_{e,8}(T)$ *if and only if* $\lambda \in \triangle_2(T)$ *and* $\mathtt{ind}(T - \lambda I) \geq 0$;
(iii) $\sigma_{e,5}(T) = \sigma_{e,7}(T) \cup \sigma_{e,8}(T)$;
(iv) $\sigma_{e,1}(T) \subset \sigma_{e,7}(T)$ *and* $\sigma_{e,2}(T) \subset \sigma_{e,8}(T)$.

Proof (i) If $\lambda \in \triangle_1(T)$ and $\mathtt{ind}(T - \lambda I) \leq 0$, we have for any $K \in \mathscr{K}(X)$ that $\lambda \in \triangle_1(T + K)$ and $\mathtt{ind}(T + K - \lambda I) \leq 0$. So, $\lambda \in \sigma_{p,3}(T + K) \cup \sigma_{r,1}(T + K) \cup \rho(T + K)$. Assume $\lambda \in \sigma_{p,3}(T + K)$, then $T + K - \lambda I$ has the following block forms from $X = \mathscr{N}(T + K - \lambda I) \oplus \mathscr{N}(T + K - \lambda I)^{\perp}$ into $X = \mathscr{R}(T + K - \lambda I)^{\perp} \oplus \mathscr{R}(T + K - \lambda I)$:

$$T + K - \lambda I = \begin{bmatrix} 0 & 0 \\ 0 & (T + K)_{\lambda} \end{bmatrix},$$

which means

$$T - \lambda I = \begin{bmatrix} I_{\mathscr{N}(T+K-\lambda I)} & 0 \\ 0 & (T + K)_{\lambda} \end{bmatrix} - \left(\begin{bmatrix} I_{\mathscr{N}(T+K-\lambda I)} & 0 \\ 0 & 0 \end{bmatrix} + K \right),$$

where $I_{\mathscr{N}(T+K-\lambda I)}$ is the identity operator on $\mathscr{N}(T + K - \lambda I)$, $I_{\mathscr{N}(T+K-\lambda I)}$ is a compact by $\mathtt{dim}\mathscr{N}(T + K - \lambda I) < \infty$, and therefore $\widetilde{K} = \begin{bmatrix} I_{\mathscr{N}(T+K-\lambda I)} & 0 \\ 0 & 0 \end{bmatrix} + K$ is a compact. Noting that $(T + K)_{\lambda}$ is a Fredholm operator, we get $\lambda \notin \sigma_{ap}(T + \widetilde{K})$, which implies $\lambda \notin \sigma_{e,7}(T)$.

Conversely, if $\lambda \notin \sigma_{e,7}(T)$, there exists $K \in \mathscr{K}(X)$ such that $\lambda \in \sigma_{r,1}(T + K) \cup \rho(T + K)$, and it is obvious that $\lambda \in \triangle_1(T)$ and $\mathtt{ind}(T - \lambda I) \leq 0$.

(ii) According to conclusion (i) and by Lemma 2.1.24, the conclusion follows easily.

In view of (i) and (ii), the proofs of conclusions (iii) and (iv) are trivial. □

Remark 2.1.26 By Lemma 2.1.25, we have

$$\sigma_{e,7}(T) = \sigma_{e,1}(T) \cup \{\lambda \in \mathbb{C} : \mathtt{ind}(T - \lambda I) > 0\},$$
$$\sigma_{e,8}(T) = \sigma_{e,2}(T) \cup \{\lambda \in \mathbb{C} : \mathtt{ind}(T - \lambda I) < 0\}.$$

If T is a self-adjoint operator, the above essential spectra are the same.

Corollary 2.1.27 *Let T be a self-adjoint operator in a Hilbert space X. Then*

$$\sigma_{e,i}(T) = \sigma_{ess}(T), \quad i = 1, 2, \ldots, 8.$$

Proof If T is a self-adjoint operator X, the proofs of

$$\sigma_{ess,i}(T) = \sigma_{ess}(T), \quad i = 1, 2, 3, 4, 5$$

and

$$\sigma_{ess,7}(T) = \sigma_{ess,8}(T)$$

are obvious. Since $\sigma_{e,5}(T) = \sigma_{e,7}(T) \cup \sigma_{e,8}(T)$, we have $\sigma_{e,5}(T) = \sigma_{e,7}(T) = \sigma_{e,8}(T)$ which ends the proof. □

Theorem 2.1.28 *Let T be a densely defined closed linear operator in a Hilbert space X.*

(i) If the complement of $\sigma_{e,3}(T)$ is connected and $\rho(T) \neq \emptyset$, then $\sigma_{e,3}(T) = \sigma_{e,4}(T)$;
(ii) If the complement of $\sigma_{e,1}(T)$ is connected and $\rho(T) \neq \emptyset$, then $\sigma_{e,1}(T) = \sigma_{e,7}(T)$;
(iii) If the complement of $\sigma_{e,2}(T)$ is connected and $\rho(T) \neq \emptyset$, then $\sigma_{e,2}(T) = \sigma_{e,8}(T)$;
(iv) If the complement of $\sigma_{e,4}(T)$ is connected and $\rho(T) \neq \emptyset$, then $\sigma_{e,4}(T) = \sigma_{e,5}(T)$;
(iii) If the complement of $\sigma_{e,5}(T)$ is connected and $\rho(T) \neq \emptyset$, then $\sigma_{e,5}(T) = \sigma_{e,6}(T)$.

Proof It is only necessary to prove (i), other conclusions are proved to be exactly the same. Since $\lambda_0 \in \triangle_3(T) \cap \sigma_{e,4}(T)$, $T - \lambda_0 I$ is a semi-Fredholm operator but not Fredholm operator. Since $\rho(T) \neq \emptyset$, there exists $\lambda_1 \in \mathbb{C}$ such that $\lambda_1 \in \rho(T)$, that is, $\lambda_1 \in \triangle_3(T)$ and

$$\mathtt{dim}\mathscr{N}(T - \lambda_1 I) = \mathtt{dim}\mathscr{R}(T - \lambda_1 I)^{\perp} = 0.$$

By Theorem 5.31 of [54], there exists $\varepsilon > 0$ such that $|\mu| \leq \varepsilon$, $T - \lambda_1 I - \mu I$ is Fredholm operator and

$$\mathtt{dim}\mathscr{N}(T - \lambda_1 I) = \mathtt{dim}\mathscr{N}(T - \lambda_1 I - \mu I) = 0,$$

$$\mathtt{dim}\mathscr{R}(T - \lambda_1 I)^{\perp} = \mathtt{dim}\mathscr{R}(T - \lambda_1 I - \mu I)^{\perp} = 0.$$

Considering the connectivity of $\triangle_3(T)$, we get

$$\mathtt{dim}\mathscr{N}(T - \lambda_0 I) = \mathtt{dim}\mathscr{N}(T - \lambda_1 I) = 0,$$

$$\mathtt{dim}\mathscr{R}(T - \lambda_0 I)^{\perp} = \mathtt{dim}\mathscr{R}(T - \lambda_1 I)^{\perp} = 0.$$

This is contradictory to $\lambda_0 \in \sigma_{e,4}(T)$, so $\triangle_3(T) \cap \sigma_{e,4}(T) = \emptyset$, that is $\sigma_{e,3}(T) \supset \sigma_{e,4}(T)$. □

2.2 Spectra of Diagonal Infinite-Dimensional Hamiltonian Operators

For the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subseteq X \times X \to X \times X,$$

if $B = C = 0$, then H is the diagonal infinite-dimensional Hamiltonian operator, which is a relatively simple and common infinite-dimensional Hamiltonian operator in practical problems. In this section, we will discuss the basic properties of the point spectrum, residual spectrum, continuous spectrum and essential spectrum of the diagonal infinite-dimensional Hamiltonian operator. The following is a simple application of diagonal infinite-dimensional Hamiltonian operators.

Example 2.2.1 For operator equation

$$\Pi A + A^* \Pi = 0, \tag{2.3}$$

the problem of solving Eq. (2.3) is closely related to the diagonal infinite-dimensional Hamiltonian operator. In fact, if there exist operators P, Q, R such that

$$\begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} \begin{bmatrix} P \\ Q \end{bmatrix} = \begin{bmatrix} P \\ Q \end{bmatrix} R \tag{2.4}$$

and P is invertible, then $\Pi = QP^{-1}$ is the solution of Eq. (2.3).

2.2.1 Point Spectra of Diagonal Infinite-Dimensional Hamiltonian Operators

Using the definition of point spectrum, it is easy to prove that the point spectra of diagonal infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

satisfies

$$\sigma_p(H) = \sigma_p(A) \cup \sigma_p(-A^*).$$

For the finite-dimensional case, the set $\sigma_p(A)$ and the set $\sigma_p(-A^*)$ are symmetric with respect to the imaginary axis, thus the point spectrum of the diagonal Hamiltonian matrix is symmetric with respect to the imaginary axis. Therefore, a natural question arises: is the point spectrum of the diagonal infinite-dimensional Hamiltonian operator symmetric with respect to the imaginary axis? The answer is negative.

Example 2.2.2 Let $X = L^2[0, \infty)$, $A = \frac{\mathrm{d}}{\mathrm{d}t}$, $Ax = \frac{\mathrm{d}x}{\mathrm{d}t}$,

$$\mathscr{D}(A) = \{x \in X : x \text{ is locally absolutely continuous}, x' \in X, x(0) = 0\},$$

then $A^* = -\frac{\mathrm{d}}{\mathrm{d}t}$,

$$\mathscr{D}(A^*) = \{x \in X : x \text{ is locally absolutely continuous}, x' \in X\}.$$

Consider the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} \frac{\mathrm{d}}{\mathrm{d}t} & 0 \\ 0 & \frac{\mathrm{d}}{\mathrm{d}t} \end{bmatrix},$$

then it is easy to get $\sigma_p(A) = \emptyset$. In fact, let $\lambda \in \sigma_p(A)$, then solving the differential equation

$$x'(t) = \lambda x(t),$$

we get $x(t) = Ce^{\lambda t}$, which together with $x(0) = 0$ implies $C = 0$. Therefore, $\sigma_p(A) = \emptyset$, and thus $\sigma_r(A^*) = \emptyset$, which implies

$$\sigma_p(H) = \sigma_p(-A^*) = \{\lambda \in \mathbb{C} : Re(\lambda) < 0\}.$$

Obviously, the point spectrum of the diagonal infinite-dimensional Hamiltonian operator is not necessarily symmetric with respect to the imaginary axis.

Theorem 2.2.3 *The point spectrum of the diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

is symmetric with respect to the imaginary axis if and only if $(\sigma_r(A) \setminus \sigma_p(-A^*)) \cup (\sigma_r(-A^*) \setminus \sigma_p(A)) = \emptyset$.

Proof Without loss of generality, let $\sigma_p(H) \neq \emptyset$ be symmetric with respect to the imaginary axis. Assume that $\sigma_r(A) \setminus \sigma_p(-A^*) \neq \emptyset$. Then there exists $\lambda \in \sigma_r(A)$ such that $\lambda \notin \sigma_p(-A^*)$. By $\lambda \in \sigma_r(A)$, we know $-\overline{\lambda} \in \sigma_p(-A^*)$, that is, $-\overline{\lambda} \in \sigma_p(H)$. Since $\sigma_p(H)$ is symmetric with respect to the imaginary axis and $\lambda \notin \sigma_p(-A^*)$, we get $\lambda \in \sigma_p(A)$. This is contradictory to $\lambda \in \sigma_r(A)$. Similarly, $\sigma_r(-A^*) \setminus \sigma_p(A) = \emptyset$.

Conversely, if $(\sigma_r(A) \setminus \sigma_p(-A^*)) \cup (\sigma_r(-A^*) \setminus \sigma_p(A)) = \emptyset$, it is easy to get $\sigma_r(A) \subset \sigma_p(-A^*)$ and $\sigma_r(-A^*) \subset \sigma_p(A)$. We might as well let $\sigma_p(H) \neq \emptyset$, and let $\lambda \in \sigma_p(H)$, then $\lambda \in \sigma_p(A) \cup \sigma_p(-A^*)$. Id $\lambda \in \sigma_p(A)$, $-\overline{\lambda} \in \sigma_p(-A^*) \cup \sigma_r(-A^*)$. From

$$(\sigma_p(-A^*) \cup \sigma_r(-A^*)) \subset (\sigma_p(-A^*) \cup \sigma_p(A)) = \sigma_p(H),$$

we get $-\overline{\lambda} \in \sigma_p(H)$. If $\lambda \in \sigma_p(-A^*)$, similarly it can be proved that $-\overline{\lambda} \in \sigma_p(H)$. So, $\sigma_p(H)$ is symmetric with respect to the imaginary axis. □

Remark 2.2.4 Another equivalent description of Theorem 2.2.3 is: $\sigma_p(H)$ is symmetric about imaginary axis if and only if $\sigma_r(A) \subset \sigma_p(-A^*)$ and $\sigma_r(-A^*) \subset \sigma_p(A)$.

Corollary 2.2.5 *Let $\sigma_r(A) = \emptyset$ and $\sigma_r(A^*) = \emptyset$. Then the point spectrum of the diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

is empty set or symmetric with respect to the imaginary axis.

Proof If $\sigma_r(A) = \emptyset$ and $\sigma_r(A^*) = \emptyset$, then $(\sigma_r(A) \setminus \sigma_p(-A^*)) \cup (\sigma_r(-A^*) \setminus \sigma_p(A)) = \emptyset$. It can be seen from Theorem 2.2.3 that $\sigma_p(H)$ is symmetric with respect to the imaginary axis. □

Remark 2.2.6 For the finite-dimensional case, it is obvious that $\sigma_r(A) = \emptyset$ and $\sigma_r(-A^*) = \emptyset$. Then the eigenvalue set $\sigma_p(H)$ of the Hamiltonian matrix is symmetric with respect to the imaginary axis.

Remark 2.2.7 If the condition in the above conclusion is changed to $\sigma_r(A) = \emptyset$ or $\sigma_r(A^*) = \emptyset$, it is not enough to ensure that the point spectrum of the diagonal infinite-dimensional Hamiltonian operator is symmetric with respect to the imaginary axis.

Example 2.2.8 Let $X = L^2[0, \infty)$, $A = \frac{\mathrm{d}}{\mathrm{d}t}$, $Ax = \frac{\mathrm{d}x}{\mathrm{d}t}$,

$$\mathscr{D}(A) = \{x \in X : x \text{ is locally absolutely continuous}, x' \in X, x(0) = 0\},$$

then $A^* = -\frac{\mathrm{d}}{\mathrm{d}t}$,

$$\mathscr{D}(A^*) = \{x \in X : x \text{ is locally absolutely continuous}, x' \in X\}.$$

By Example 2.2.2 we know that the point spectrum of the diagonal infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} \frac{\mathrm{d}}{\mathrm{d}t} & 0 \\ 0 & \frac{\mathrm{d}}{\mathrm{d}t} \end{bmatrix},$$

is

$$\sigma_p(H) = \{\lambda \in \mathbb{C} : Re(\lambda) < 0\},$$

that is, the point spectrum is not symmetric with respect to the imaginary axis.

Next, we discuss the problem of when the eigenvalues of the diagonal infinite-dimensional Hamiltonian operator will only have pure imaginary point spectrum.

Theorem 2.2.9 *If $\sigma_p(A) \subset \mathrm{i}\mathbb{R}$, then the diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

has only the pure imaginary point spectrum if and only if $\sigma_r(A) \subset \mathrm{i}\mathbb{R}$.

Proof When $\sigma_r(A) \subset \mathrm{i}\mathbb{R}$, let $\lambda \in \sigma_p(H)$, then $\lambda \in \sigma_p(A) \cup \sigma_p(-A^*)$. Let $\lambda \in \sigma_p(-A^*)$. Since $\sigma_r(A) \subset \mathrm{i}\mathbb{R}$, we get $-\overline{\lambda} \in \sigma_p(A)$, so $Re(\lambda) = 0$.

Conversely, if $\lambda \in \sigma_r(A)$, then $-\overline{\lambda} \in \sigma_p(-A^*)$. From $\sigma_p(H) \subset \mathrm{i}\mathbb{R}$ we get $-\overline{\lambda} \in \mathrm{i}\mathbb{R}$, that is, $\lambda \in \mathrm{i}\mathbb{R}$. The proof is completed. □

Similarly, the following conclusions can be proved.

Theorem 2.2.10 *If $\sigma_p(A^*) \subset \mathrm{i}\mathbb{R}$, the diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

has the pure imaginary point spectrum if and only if $\sigma_r(A^*) \subset \mathrm{i}\mathbb{R}$.

Corollary 2.2.11 *If $\mathscr{D}(A^*) \subset \mathscr{D}(A)$ and $(Ax, x) \subset \mathrm{i}\mathbb{R}$ for any $x \in \mathscr{D}(A)$, then the point spectrum of the diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

is an empty set or is located on the imaginary axis.

Proof We may let the point spectrum of the diagonal infinite-dimensional Hamiltonian operator be non-empty. Let $\lambda \in \sigma_p(H)$, then there exists $\begin{bmatrix} f & g \end{bmatrix}^T \neq 0$ such that

$$Af = \lambda f, \quad A^* g = -\lambda g.$$

If $f \neq 0$, $(Af, f) = \lambda(f, f) \in \mathrm{i}\mathbb{R}$, so $\lambda \in \mathrm{i}\mathbb{R}$. If $g \neq 0$, $(A^* g, g) = (g, Ag) = -\lambda(g, g) \in \mathrm{i}\mathbb{R}$, so $\lambda \in \mathrm{i}\mathbb{R}$. The proof is completed. □

Similarly, the following conclusions can be proved.

Corollary 2.2.12 *If $\mathscr{D}(A) \subset \mathscr{D}(A^*)$ and $(A^* y, y) \subset \mathrm{i}\mathbb{R}$ for any $y \in \mathscr{D}(A^*)$, then the point spectrum of the diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

is an empty set or is located on the imaginary axis.

With the above conclusion, a natural question is when the point spectrum of the diagonal infinite-dimensional Hamiltonian operator is distributed on the real axis? It is easy to prove the following conclusion.

Corollary 2.2.13 *If $\sigma_p(A) \subset \mathbb{R}$, for the diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X,$$

we have $\sigma_p(H) \subset \mathbb{R}$ if and only if $\sigma_r(A) \subset \mathbb{R}$.

Proof If $\lambda \in \sigma_p(H)$, then $\lambda \in \sigma_p(A) \cup \sigma_p(-A^*)$. If $\lambda \in \sigma_p(A)$, from $\sigma_p(A) \subset \mathbb{R}$ we get $\lambda \in \mathbb{R}$. If $\lambda \in \sigma_p(-A^*)$, then $-\overline{\lambda} \in \sigma_p(A) \cup \sigma_r(A)$. Hence, $\lambda \in \mathbb{R}$. Conversely, the proof is similar. □

Corollary 2.2.14 *If $\sigma_p(A) \subset \mathbb{R}$ and $\sigma_r(A) = \emptyset$, then the point spectrum of the diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

is distributed on the real axis.

More generally, we can find the following conclusion.

Theorem 2.2.15 *Let $\sigma_r(A) = \emptyset$ and $\sigma_p(A)$ be symmetric with respect to the imaginary axis. Then the point spectrum of diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

satisfies $\sigma_p(H) = \sigma_p(A)$.

Proof It is only necessary to prove $\sigma_p(H) \subset \sigma_p(A)$. Let $\lambda \in \sigma_p(H)$, then $\lambda \in \sigma_p(A) \cup \sigma_p(-A^*)$. If $\lambda \in \sigma_p(A)$, the conclusion is clearly established. If $\lambda \in \sigma_p(-A^*)$, $-\overline{\lambda} \in \sigma_p(A) \cup \sigma_r(A)$. Taking into account that $\sigma_r(A) = \emptyset$ and $\sigma_p(A)$ with respect to the imaginary axis, we can get $\lambda \in \sigma_p(A)$. The proof is completed.

□

In the following, we will characterize four classes of point spectra of diagonal infinite-dimensional Hamiltonian operators.

Theorem 2.2.16 *Let $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ be a diagonal infinite-dimensional Hamiltonian operator. Then*

(i) $\sigma_{p,1}(H) = (\sigma_{p,1}(A) \cap \sigma_{p,1}(-A^*)) \cup (\sigma_{p,1}(A) \cap \rho(-A^*)) \cup (\rho(A) \cap \sigma_{p,1}(-A^*))$;

(ii) $\sigma_{p,2}(H) = \Delta_{11} \cup \Delta_{12} \cup \Delta_{13} \cup \Delta_{14}$;

(iii) $\sigma_{p,3}(H) = \Delta_{21} \cup \Delta_{22} \cup \Delta_{23} \cup \Delta_{24}$;
(iv) $\sigma_{p,4}(H) = \Delta_{31} \cup \Delta_{32} \cup \Delta_{33} \cup \Delta_{34}$.

Here

$$\begin{aligned}
\Delta_{11} &= \sigma_{p,1}(A) \cap \sigma_c(-A^*),\\
\Delta_{12} &= \sigma_{p,2}(A) \setminus \sigma_{com}(-A^*),\\
\Delta_{13} &= \sigma_{p,1}(-A^*) \cap \sigma_c(A),\\
\Delta_{14} &= \sigma_{p,2}(-A^*) \setminus \sigma_{com}(A),\\
\Delta_{21} &= \sigma_{p,1}(A) \cap \sigma_{r,1}(-A^*),\\
\Delta_{22} &= \sigma_{p,3}(A) \setminus \sigma_{cr}(-A^*),\\
\Delta_{23} &= \sigma_{p,1}(-A^*) \cap \sigma_{r,1}(A),\\
\Delta_{24} &= \sigma_{p,3}(-A^*) \setminus \sigma_{cr}(A),\\
\Delta_{31} &= \sigma_{p,4}(A) \cup \sigma_{p,4}(-A^*),\\
\Delta_{32} &= (\sigma_{p,1}(A) \cap \sigma_{r,2}(-A^*)) \cup (\sigma_{p,1}(-A^*) \cap \sigma_{r,2}(A)),\\
\Delta_{33} &= (\sigma_{p,2}(A) \cap \sigma_{com}(-A^*)) \cup (\sigma_{p,2}(-A^*) \cap \sigma_{com}(A)),\\
\Delta_{34} &= (\sigma_{p,3}(A) \cap \sigma_{cr}(-A^*)) \cup (\sigma_{p,3}(-A^*) \cap \sigma_{cr}(A)),
\end{aligned}$$

and $\sigma_{cr}(T) = \{\lambda \in \mathbb{C} : \mathscr{R}(T - \lambda I)$*is not closed*$\}$.

Proof (i) The proof of the expression

$$\sigma_{p,1}(H) \supset (\sigma_{p,1}(A) \cap \sigma_{p,1}(-A^*)) \cup (\sigma_{p,1}(A) \cap \rho(-A^*)) \cup (\rho(A) \cap \sigma_{p,1}(-A^*))$$

is obvious. The following proves the reverse inclusion. Let $\lambda \in \sigma_{p,1}(H)$, then $\lambda \in \sigma_{p,1}(A) \cup \rho(A)$. If $\lambda \in \sigma_{p,1}(A)$, it follows from $\mathscr{R}(H - \lambda I) = X \times X$ that $\mathscr{R}(-A^* - \lambda I) = X$, so $\lambda \in \sigma_{p,1}(-A^*) \cup \rho(-A^*)$.

If $\lambda \notin \sigma_{p,1}(A)$, taking into account $\mathscr{R}(A - \lambda I) = X$ we get $\lambda \in \rho(A)$, and from $\lambda \in \sigma_{p,1}(H)$ we further get $\lambda \in \sigma_{p,1}(-A^*)$. Thus the reverse-inclusion relationship is also established.

(ii) The proof of the expression

$$\sigma_{p,2}(H) \supset \Delta_{11} \cup \Delta_{12} \cup \Delta_{13} \cup \Delta_{14}$$

is obvious. In order to complete the proof, we only need to prove $\lambda \in \Delta_{14}$ when $\lambda \in \sigma_{p,2}(H)$ and $\lambda \notin \Delta_{11} \cup \Delta_{12} \cup \Delta_{13}$. By $\overline{\mathscr{R}(H - \lambda I)} = X \times X$ we know

$$\overline{\mathscr{R}(A - \lambda I)} = X, \quad \overline{\mathscr{R}(-A^* - \lambda I)} = X,$$

that is, $\lambda \notin \sigma_{com}(A)$ and $\lambda \notin \sigma_{com}(-A^*)$. Furthermore, since $\lambda \notin \Delta_{12}$ we get $\lambda \notin \sigma_{p,2}(A)$. On the other hand, by $\lambda \notin \Delta_{11}$, we know

$$\lambda \in \sigma_c(A) \cup \rho(A) \quad \text{or} \quad \lambda \in \sigma_{p,1}(-A^*) \cup \sigma_{p,2}(-A^*) \cup \rho(-A^*). \tag{2.5}$$

It can be asserted that $\lambda \in \sigma_{p,2}(-A^*)$. In fact, assume $\lambda \notin \sigma_{p,2}(-A^*)$. Then

$$\lambda \in \sigma_{p,1}(-A^*) \cup \rho(-A^*). \tag{2.6}$$

So, taking $\mathscr{R}(H - \lambda I) \neq X \times X$ into account, we get $\lambda \in \sigma_c(A)$. This is contradictory to $\lambda \notin \Delta_{13}$. Thus $\lambda \in \Delta_{14}$.

The proofs of (iii) and (iv) are completely similar to (ii). □

According to the above conclusions, the following inferences are obvious.

Corollary 2.2.17 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. Then*

(i) $\sigma_{p,1}(H) \cap i\mathbb{R} = \emptyset$;
(ii) $\sigma_{p,2}(H) \cap i\mathbb{R} = \emptyset$.

Corollary 2.2.18 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. If* $\sigma(A) \cap \sigma(-A^*) = \emptyset$, *then* $\sigma_{p,i}(H) = \sigma_{p,i}(A) \cup \sigma_{p,i}(-A^*)$, $i = 1, 2, 3, 4$.

The following example is given to illustrate the effectiveness of the above theorem.

Example 2.2.19 Let S_r, S_l be linear operators in the Hilbert space $X = \ell^2[1, \infty)$ and

$$\begin{aligned} S_r x &= (0, x_1, x_2, x_3, \ldots), \\ S_l x &= (x_2, x_3, x_4, \ldots), \end{aligned}$$

where $x = (x_1, x_2, x_3, \ldots) \in X$. It is easy to get

$$S_l^* = S_r$$

and

$$\begin{aligned} \sigma_p(S_r) &= \emptyset, \\ \sigma_p(S_l) &= \sigma_{p,1}(S_l) = \{\lambda \in \mathbb{C} : |\lambda| < 1\}, \\ \sigma_c(S_l) &= \sigma_c(S_r) = \{\lambda \in \mathbb{C} : |\lambda| = 1\}. \end{aligned}$$

Define the diagonal infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} S_l & 0 \\ 0 & -S_r \end{bmatrix},$$

and it is easy to have that

$$\sigma_p(H) = \sigma_{p,3}(H) = \{\lambda \in \mathbb{C} : |\lambda| < 1\}.$$

On the other hand, since $\sigma_p(-A^*) = \emptyset$ and $\sigma_{p,1}(A) \cap \sigma_c(-A^*) = \emptyset$, we know

$$\sigma_{p,i}(H) = \emptyset, \quad i = 1, 2, 4$$

by Theorem 2.2.16. Similarly, $\Delta_{22} = \Delta_{23} = \Delta_{24} = \emptyset$ and $\Delta_{21} = \{\lambda \in \mathbb{C} : |\lambda| < 1\}$. Therefore

$$\sigma_p(H) = \sigma_{p,3}(H) = \{\lambda \in \mathbb{C} : |\lambda| < 1\},$$

which is coincide with the above calculation.

2.2.2 Residual Spectra of Diagonal Infinite-Dimensional Hamiltonian Operators

In this section, we will discuss some basic properties of the residual spectra of diagonal infinite-dimensional Hamiltonian operators.

Theorem 2.2.20 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. Then*

$$\sigma_r(H) = (\sigma_r(A) \setminus \sigma_p(-A^*)) \cup (\sigma_r(-A^*) \setminus \sigma_p(A)).$$

Proof The proof of

$$\sigma_r(H) \supset (\sigma_r(A) \setminus \sigma_p(-A^*)) \cup (\sigma_r(-A^*) \setminus \sigma_p(A))$$

is obvious.

The following proves the reverse inclusion. Let $\lambda \in \sigma_r(H)$. If $\lambda \in \sigma_r(A)$, then it is easy to get $\lambda \notin \sigma_p(-A^*)$, that is, $\lambda \in \sigma_r(A) \setminus \sigma_p(-A^*)$. If $\lambda \notin \sigma_r(A)$, it follows from $\lambda \in \sigma_r(H)$ that $\lambda \notin \sigma_p(A)$, and thus $\lambda \in \sigma_c(A) \cup \rho(A)$. Since $\mathscr{R}(H - \lambda I) \neq X \times X$, we get $\lambda \in \sigma_r(-A^*) \setminus \sigma_p(A)$. Therefore,

$$\sigma_r(H) = (\sigma_r(A) \setminus \sigma_p(-A^*)) \cup (\sigma_r(-A^*) \setminus \sigma_p(A)).$$

The proof is completed. □

The following corollaries answer the question of when the residual spectrum of diagonal infinite-dimensional Hamiltonian operators is an empty set.

Corollary 2.2.21 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. Then* $\sigma_r(H) = \emptyset$ *if and only if* $\sigma_r(A) \subset \sigma_p(-A^*)$ *and* $\sigma_r(-A^*) \subset \sigma_p(A)$.

Remark 2.2.22 From the above corollary, we have $\sigma_r(H) = \emptyset$ if $\sigma_r(A) = \emptyset$ and $\sigma_r(A^*) = \emptyset$. In general, $\sigma_r(A) = \emptyset$ doesn't necessarily imply $\sigma_r(A^*) = \emptyset$. For example, let $X = l^2[1, \infty)$ and define

$$Ax = (x_2, x_3, \ldots)$$

for any $x = (x_1, x_2, \ldots) \in X$, then we get by calculation that $\sigma_r(A) = \emptyset$ and $\sigma_r(A^*) = \{\lambda \in \mathbb{C} : |\lambda| < 1\}$.

Corollary 2.2.23 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. If* $\sigma(A) \cap \sigma(-A^*) = \emptyset$, *then* $\sigma_r(H) = \sigma_r(A) \cup \sigma_r(-A^*)$.

In general, the residual spectrum of the diagonal infinite-dimensional Hamiltonian operator is not necessarily empty set, but it has no pure imaginary residual spectrum.

Theorem 2.2.24 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. Then* $\sigma_r(H) \cap \mathtt{i}\mathbb{R} = \emptyset$.

Proof Let $\lambda \in \sigma_r(H) \cap \mathtt{i}\mathbb{R}$. Then $\lambda = -\overline{\lambda}$ and $\overline{\lambda} \in \sigma_p(H^*)$. Based on the similarity of $-H$ and H^* we get $-\overline{\lambda} \in \sigma_p(H)$, that is $\lambda \in \sigma_p(H)$, this is contradictory to $\lambda \in \sigma_r(H)$. Hence, $\sigma_r(H) \cap \mathtt{i}\mathscr{R} = \emptyset$. The proof is completed. □

The distribution of the residual spectrum of the diagonal infinite-dimensional Hamiltonian operator will be discussed as follows.

Theorem 2.2.25 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. Then* $\sigma_r(H) \subset \mathbb{C}_+$ *if and only if*

$$(\sigma_{p,12}(A) \setminus \sigma_{com}(-A^*)) \cup (\sigma_{p,12}(-A^*) \setminus \sigma_{com}(A)) \subset \mathbb{C}_-,$$

where $\mathbb{C}_-, \mathbb{C}_+$ *represent the left and right half-open planes, respectively, and* $\sigma_{p,12}(A) = \sigma_{p,1}(A) \cup \sigma_{p,2}(A)$.

Proof By Lemma 2.1.12 we know $\lambda \in \sigma_r(H)$ if and only if $-\overline{\lambda} \in \sigma_{p,12}(H)$. On the other hand, it is easy to prove

$$\sigma_{p,12}(H) = (\sigma_{p,12}(A) \setminus \sigma_{com}(-A^*)) \cup (\sigma_{p,12}(-A^*) \setminus \sigma_{com}(A)).$$

In fact, $\sigma_{p,12}(H) \supset (\sigma_{p,12}(A) \setminus \sigma_{com}(-A^*)) \cup (\sigma_{p,12}(-A^*) \setminus \sigma_{com}(A))$ is obvious. In order to prove the reverse inclusion, let $\lambda \in \sigma_{p,12}(H)$, then

$$\sigma_p(H) = \sigma_{p,12}(A) \cup \sigma_{p,34}(A) \cup \sigma_{p,12}(-A^*) \cup \sigma_{p,34}(-A^*),$$

where $\sigma_{p,34}(A) = \sigma_{p,3}(A) \cup \sigma_{p,4}(A)$. If $\lambda \in \sigma_{p,12}(A)$, then $\mathscr{R}(-A^* - \lambda I)$ is dense by $\lambda \in \sigma_{p,12}(H)$, and hence $\lambda \notin \sigma_{com}(-A^*)$.

If $\lambda \notin \sigma_{p,12}(A)$, taking $\lambda \in \sigma_{p,12}(H)$ into account we have that $\lambda \in \sigma_{p,12}(-A^*)$ and $\mathscr{R}(A - \lambda I)$ is dense, that is, $\lambda \in \sigma_{p,12}(-A^*) \setminus \sigma_{com}(A)$. The proof is completed. □

In the following, we will give the characterization of two classes of residual spectra of diagonal infinite-dimensional Hamiltonian operators.

Theorem 2.2.26 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. Then*

(i) $\sigma_{r,1}(H) = (\sigma_{r,1}(A) \setminus \sigma_{ap}(-A^*)) \cup (\sigma_{r,1}(-A^*) \setminus \sigma_{ap}(A))$;
(ii) $\sigma_{r,2}(H) = \Delta_1 \cup \Delta_2$,

where

$$\Delta_1 = (\sigma_{r,1}(A) \cap \sigma_c(-A^*)) \cup (\sigma_{r,2}(A) \setminus \sigma_p(-A^*)),$$

$$\Delta_2 = (\sigma_{r,1}(-A^*) \cap \sigma_c(A)) \cup (\sigma_{r,2}(-A^*) \setminus \sigma_p(A)).$$

Proof (i) Since $\lambda \notin \sigma_{ap}(A)$, $\lambda \in \rho(A) \cup \sigma_{r,1}(A)$, and hence the proof of the expression

$$\sigma_{r,1}(H) \supset (\sigma_{r,1}(A) \setminus \sigma_{ap}(-A^*)) \cup (\sigma_{r,1}(-A^*) \setminus \sigma_{ap}(A))$$

is obvious. We now prove the reverse inclusion. Let $\lambda \in \sigma_{r,1}(H)$ and $\lambda \notin (\sigma_{r,1}(A) \setminus \sigma_{ap}(-A^*))$, then we can assert $\lambda \notin \sigma_{r,1}(A)$. Because $A - \lambda I$, $-A^* - \lambda I$ are injective and $\mathscr{R}(A - \lambda I)$, $\mathscr{R}(-A^* - \lambda I)$ are closed, we get $\lambda \in \rho(A)$, that is, $\lambda \notin \sigma_{ap}(A)$. Since $\mathscr{R}(H - \lambda I)$ is not dense, considering $\lambda \in \rho(A)$ we can get $\lambda \in \sigma_{r,1}(-A^*)$, that is, $\lambda \in (\sigma_{r,1}(-A^*) \setminus \sigma_{ap}(A))$.

(ii) The proof of

$$\sigma_{r,2}(H) \supset \Delta_1 \cup \Delta_2$$

is obvious. The following proves the reverse inclusion. Let $\lambda \in \sigma_{r,2}(H)$ and $\lambda \notin \Delta_1$, then from $\lambda \notin (\sigma_{r,2}(A) \setminus \sigma_p(-A^*))$ we get $\lambda \notin \sigma_{r,2}(A)$. Since $\lambda \notin (\sigma_{r,1}(A) \cap \sigma_c(-A^*))$, $\lambda \in (\rho(A) \cup \sigma_c(A)) \cup (\sigma_r(-A^*) \cup \rho(-A^*))$. Note that $\mathscr{R}(H - \lambda I)$ is not dense, so $\lambda \in \sigma_r(-A^*)$. If $\lambda \in \sigma_{r,1}(-A^*)$, since $\mathscr{R}(H - \lambda I)$ is not closed, we get $\lambda \in \sigma_c(A)$. If $\lambda \in \sigma_{r,2}(-A^*)$, since $H - \lambda I$ is injective, we get $\lambda \notin \sigma_p(A)$, that is, $\lambda \in \Delta_2$. The proof is completed. □

The following inference is obvious according to the conclusion above.

Corollary 2.2.27 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. If* $\sigma(A) \cap \sigma(-A^*) = \emptyset$, *then*

(i) $\sigma_{r,1}(H) = \sigma_{r,1}(A) \cup \sigma_{r,1}(-A^*)$;
(ii) $\sigma_{r,2}(H) = \sigma_{r,2}(A) \cup \sigma_{r,2}(-A^*)$.

Here is an example to illustrate the effectiveness of the above result.

Example 2.2.28 Let S_r, S_l be linear operators in the Hilbert space $X = \ell^2[1, \infty)$ and

$$\begin{aligned} S_r x &= (0, x_1, x_2, x_3, \ldots), \\ S_l x &= (x_2, x_3, x_4, \ldots), \end{aligned}$$

where $x = (x_1, x_2, x_3, \ldots) \in X$. It is easy to get $S_l^* = S_r$ as before and

$$\begin{aligned} &\sigma_r(S_l) = \emptyset, \\ &\sigma_p(S_r) = \emptyset, \\ &\sigma_r(S_r) = \sigma_{r,1}(S_r) = \{\lambda \in \mathbb{C} : |\lambda| < 1\}. \end{aligned}$$

Define the diagonal infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} S_l & 0 \\ 0 & -S_r \end{bmatrix},$$

and clearly $\sigma_r(H) = \emptyset$. On the other hand, since

$$\sigma_r(A) = \sigma_p(-A^*) = \emptyset$$

and

$$\sigma_r(-A^*) = \sigma_p(A) = \{\lambda \in \mathbb{C} : |\lambda| < 1\}$$

we have $\sigma_r(A) \setminus \sigma_p(-A^*) = \emptyset$ and $\sigma_r(-A^*) \setminus \sigma_p(A) = \emptyset$. By Theorem 2.2.20 we then get $\sigma_r(H) = \emptyset$. As is shown, the above calculations are the same.

2.2.3 *Continuous Spectra of Diagonal Infinite-Dimensional Hamiltonian Operators*

In this section, we will discuss the basic properties of the continuous spectrum of the diagonal infinite-dimensional Hamiltonian operator. For diagonal infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X,$$

taking into account

$$(JH)^* = JH$$

with $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$, we can easily have $\lambda \in \sigma_c(H)$ if and only if $-\overline{\lambda} \in \sigma_c(H)$, that is, the continuous spectrum is symmetric with respect to the imaginary axis. The following will give the specific distribution of the continuous spectrum of the diagonal infinite-dimensional Hamiltonian operator.

Theorem 2.2.29 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. Then*

$$\sigma_c(H) = (\sigma_c(A) \cap \rho(-A^*)) \cup (\sigma_c(-A^*) \cap \rho(A)) \cup (\sigma_c(A) \cap \sigma_c(-A^*)).$$

Proof The proof of

$$\sigma_c(H) \supset (\sigma_c(A) \cap \rho(-A^*)) \cup (\sigma_c(-A^*) \cap \rho(A)) \cup (\sigma_c(A) \cap \sigma_c(-A^*))$$

is obvious. We now prove the reverse inclusion. Let $\lambda \in \sigma_c(H)$ and $\lambda \notin (\sigma_c(A) \cap \rho(-A^*)) \cup (\sigma_c(-A^*) \cap \rho(A))$, then from $\lambda \in \sigma_c(H)$ we know

$$\lambda \in (\sigma(A) \cap \sigma(-A^*)).$$

Because $\overline{\mathscr{R}(A - \lambda I)} = \overline{\mathscr{R}(-A^* - \lambda I)} = X$, $\lambda \in (\sigma_c(A) \cap \sigma_c(-A^*))$. The proof is completed. □

If $\sigma(A) \cap \sigma(-A^*) = \emptyset$, it is easy to prove

$$\sigma_c(A) \cap \rho(-A^*) = \sigma_c(A) \backslash \sigma(-A^*) = \sigma_c(A),$$

$$\sigma_c(-A^*) \cap \rho(A) = \sigma_c(-A^*) \backslash \sigma(A) = \sigma_c(-A^*).$$

Thus the following inferences are obtained.

Corollary 2.2.30 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. If* $\sigma(A) \cap \sigma(-A^*) = \emptyset$, *then* $\sigma_c(H) = \sigma_c(A) \cup \sigma_c(-A^*)$.

The following is another characterization of the continuous spectrum of the diagonal infinite-dimensional Hamiltonian operator.

Theorem 2.2.31 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. If* $\sigma_c(A) \subset \sigma(-A^*)$ *or* $\sigma_c(-A^*) \subset \sigma(A)$, *then* $\sigma_c(H) = \sigma_c(A) \cap \sigma_c(-A^*)$.

Proof Since $\sigma_c(A) \subset \sigma(-A^*)$, it is easy to prove $\sigma_c(-A^*) \subset \sigma(A)$. In fact, let $\lambda \in \sigma_c(-A^*)$. Then $-\overline{\lambda} \in \sigma_c(A)$. Considering $\sigma_c(A) \subset \sigma(-A^*)$ we get $-\overline{\lambda} \in \sigma(-A^*)$, that is, $\lambda \in \sigma(A)$. Thus, $\sigma_c(-A^*) \subset \sigma(A)$. Furthermore,

$$\sigma_c(A) \cap \rho(-A^*) = \sigma_c(A) \backslash \sigma(-A^*) = \emptyset,$$

$$\sigma_c(-A^*) \cap \rho(A) = \sigma_c(-A^*) \backslash \sigma(A) = \emptyset.$$

By Theorem 2.2.29, we then get $\sigma_c(H) = \sigma_c(A) \cap \sigma_c(-A^*)$. Similarly, if $\sigma_c(-A^*) \subset \sigma(A)$, it is easy to prove $\sigma_c(A) \subset \sigma(-A^*)$. The proof is completed. □

The continuous spectrum of a linear operator is very unstable, even under degenerate perturbations, and this property is inconvenient for theoretical research and solving practical problems. Therefore, the problem of whether the continuous spectrum of the linear operator corresponding to the system is an empty set is particularly important. Next, we will answer the question of when the continuous spectrum of the diagonal infinite-dimensional Hamiltonian operator is an empty set.

Theorem 2.2.32 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. Then* $\sigma_c(H) = \emptyset$ *if and only if* $\sigma_c(A) \subset \sigma(-A^*)$ *and* $\sigma_c(A) \cap \sigma_c(-A^*) = \emptyset$. *Equivalently, we also have that* $\sigma_c(H) = \emptyset$ *if and only if* $\sigma_c(-A^*) \subset \sigma(A)$ *and* $\sigma_c(A) \cap \sigma_c(-A^*) = \emptyset$. *Particularly, if* $\sigma_c(A) = \emptyset$, *then* $\sigma_c(H) = \emptyset$.

Proof Note that $\sigma_c(A) \subset \sigma(-A^*)$ if and only if $\sigma_c(-A^*) \subset \sigma(A)$ and $\sigma_c(A) = \emptyset$, and if and only if $\sigma_c(-A^*) = \emptyset$. By Theorem 2.2.29, the conclusion can be proved. □

By Corollary 2.2.30, it is easy to get the following conclusion.

Corollary 2.2.33 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. If* $\sigma(A) \cap \sigma(-A^*) = \emptyset$, *then* $\sigma_c(H) = \emptyset$ *if and only if* $\sigma_c(A) = \emptyset$.

Remark 2.2.34 Combining the definition of regular points and approximate point spectrum, it is easy to prove for the diagonal infinite-dimensional Hamiltonian operator H that

$$\sigma_{ap}(H) = \sigma_{ap}(A) \cup \sigma_{ap}(-A^*),$$
$$\sigma(H) = \sigma(A) \cup \sigma(-A^*).$$

Therefore, the spectral set of the diagonal infinite-dimensional Hamiltonian operator is symmetric with respect to the imaginary axis.

The following example is given to illustrate the effectiveness of the criteria.

Example 2.2.35 Let S_r, S_l be linear operators in the Hilbert space $X = \ell^2[1, \infty)$ and

$$\begin{aligned} S_r x &= (0, x_1, x_2, x_3, \ldots), \\ S_l x &= (x_2, x_3, x_4, \ldots), \end{aligned}$$

where $x = (x_1, x_2, x_3, \ldots) \in X$. It is easy to get $S_l^* = S_r$ as before and

$$\begin{aligned} \sigma_c(S_r) = \sigma_c(S_r) &= \{\lambda \in \mathbb{C} : |\lambda| = 1\}, \\ \rho(S_r) = \rho(S_r) &= \{\lambda \in \mathbb{C} : |\lambda| > 1\}. \end{aligned}$$

Define the diagonal infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} S_l & 0 \\ 0 & -S_r \end{bmatrix},$$

and it is clear that

$$\sigma_c(H) = \{\lambda \in \mathbb{C} : |\lambda| = 1\}.$$

On the other hand, since $\sigma_c(A) \cap \rho(-A^*) = \emptyset$ and $\sigma_c(-A^*) \cap \rho(A) = \emptyset$, we have

$$\sigma_c(H) = \sigma_c(A) \cap \sigma_c(-A^*) = \{\lambda \in \mathbb{C} : |\lambda| = 1\}$$

by Theorem 2.2.29. As is shown, the above calculations are the same.

2.2.4 *Essential Spectra of Diagonal Infinite-Dimensional Hamiltonian Operators*

The following part begins to discuss the essential spectrum of the diagonal infinite-dimensional Hamiltonian operator.

Theorem 2.2.36 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. If there exists* $\lambda \in i\mathbb{R}$ *such that* $A - \lambda I$ *is a Fredholm operator, then* $H - \lambda I$ *is a Weyl operator.*

Proof For any $\lambda \in i\mathbb{R}$,

$$H - \lambda I = \begin{bmatrix} A - \lambda I & 0 \\ 0 & -A^* - \lambda I \end{bmatrix} = \begin{bmatrix} A - \lambda I & 0 \\ 0 & -(A - \lambda I)^* \end{bmatrix}.$$

Let $A - \lambda I$ be a Fredholm operator, and it is easy to prove $H - \lambda I$ is a Fredholm operator and

$$\begin{aligned}\dim\mathscr{N}(H-\lambda I) &= \dim\mathscr{N}(A-\lambda I)+\dim\mathscr{N}(A-\lambda I)^*\\ &= \dim\mathscr{N}(A-\lambda I)+\dim\mathscr{R}(A-\lambda I)^\perp<\infty,\end{aligned}$$

$$\begin{aligned}\dim\mathscr{R}(H-\lambda I)^\perp &= \dim\mathscr{R}(A-\lambda I)^\perp+\dim\mathscr{R}(A^*-\overline{\lambda}I)^\perp\\ &= \dim\mathscr{R}(A-\lambda I)^\perp+\dim\mathscr{N}(A-\lambda I)<\infty,\end{aligned}$$

that is, $\mathrm{ind}(H-\lambda I)=0$. The proof is completed. □

Considering the symplectic self-adjointness of the diagonal infinite-dimensional Hamiltonian operator, the following conclusions are obvious.

Theorem 2.2.37 *Let* $H=\begin{bmatrix}A & 0\\ 0 & -A^*\end{bmatrix}:\mathscr{D}(A)\times\mathscr{D}(A^*)\to X\times X$ *be a diagonal infinite-dimensional Hamiltonian operator, then*

(i) $\lambda\in\sigma_{e,1}(H)$ *if and only if* $-\overline{\lambda}\in\sigma_{e,2}(H)$*;*
(ii) $\lambda\in\sigma_{e,i}(H)$ *if and only if* $-\overline{\lambda}\in\sigma_{e,i}(H)$, $i=3,4,5,6$*;*
(iii) $\lambda\in\sigma_{e,7}(H)$ *if and only if* $-\overline{\lambda}\in\sigma_{e,8}(H)$.

Theorem 2.2.38 *Let* $H=\begin{bmatrix}A & 0\\ 0 & -A^*\end{bmatrix}:\mathscr{D}(A)\times\mathscr{D}(A^*)\to X\times X$ *be a diagonal infinite-dimensional Hamiltonian operator. Then*

(i) $\sigma_{e,i}(H)=\sigma_{e,i}(A)\cup\sigma_{e,i}(-A^*)$, $i=1,2,4$*;*
(ii) $\sigma_{e,3}(H)\supset\sigma_{e,3}(A)\cup\sigma_{e,3}(-A^*)$*;*
(iii) $\sigma_{e,i}(H)\subset\sigma_{e,i}(A)\cup\sigma_{e,i}(-A^*)$, $i=5,6,7,8$.

Proof (i) We only prove $\sigma_{e,1}(H)=\sigma_{e,1}(A)\cup\sigma_{e,1}(-A^*)$, the other cases can be similarly obtained. Let $\lambda\in\triangle_1(A)\cap\triangle_1(-A^*)$, it is easy to know that $\mathscr{R}(H-\lambda I)$ is a closed set and

$$\dim\mathscr{N}(H-\lambda I)=\dim\mathscr{N}(A-\lambda I)+\dim\mathscr{N}(-A^*-\lambda I)<\infty,$$

so $\lambda\in\triangle_1(H-\lambda I)$, that is, $\sigma_{e,1}(H)\subset\sigma_{e,1}(A)\cup\sigma_{e,1}(-A^*)$.

Conversely, let $\lambda\in\triangle_1(H)$. Then we have $\mathscr{R}(A-\lambda I)$, $\mathscr{R}(-A^*-\lambda I)$ is closed, which together with

$$\dim\mathscr{N}(H-\lambda I)=\dim\mathscr{N}(A-\lambda I)+\dim\mathscr{N}(-A^*-\lambda I)$$

implies $\dim\mathscr{N}(A-\lambda I)<\infty$ and $\dim\mathscr{N}(-A^*-\lambda I)<\infty$, so $\sigma_{e,1}(H)\supset\sigma_{e,1}(A)\cup\sigma_{e,1}(-A^*)$.

(ii) Let $\lambda\in\triangle_3(H)$. Then from (i) we know that $\lambda\in\triangle_3(A)\cap\triangle_3(-A^*)$, so the conclusion is established.

(iii) We only prove $\sigma_{e,5}(H)\subset\sigma_{e,5}(A)\cup\sigma_{e,5}(-A^*)$, and the other cases can be similarly obtained. Let $\lambda\in\triangle_5(A)\cap\triangle_5(-A^*)$. Then it is easy to see that $\mathscr{R}(H-\lambda I)$ is closed and

$$\begin{aligned}\dim\mathscr{N}(H-\lambda I) &= \dim\mathscr{N}(A-\lambda I)+\dim\mathscr{N}(-A^*-\lambda I)<\infty,\\ \dim\mathscr{R}(H-\lambda I)^{\perp} &= \dim\mathscr{R}(A-\lambda I)^{\perp}+\dim\mathscr{R}(-A^*-\lambda I)^{\perp}<\infty,\\ \operatorname{ind}(H-\lambda I) &= \operatorname{ind}(A-\lambda I)+\operatorname{ind}(-A^*-\lambda I)=0.\end{aligned}$$

Thus $\sigma_{e,5}(H)\subset\sigma_{e,5}(A)\cup\sigma_{e,5}(-A^*)$. The proof is completed. □

Remark 2.2.39 For the diagonal infinite-dimensional Hamiltonian operator, $\operatorname{ind}(H)=0$ doesn't necessarily mean $\operatorname{ind}(A)=0$. So, the equals sign of conclusions (ii) and (iii) of the above theorem do not necessarily hold. For example, let

$$H=\begin{bmatrix}A & 0\\ 0 & -A^*\end{bmatrix},$$

where the linear operator $A:\ell^2\to\ell^2$ is defined as

$$Ax=(x_1,x_3,x_5,\ldots),$$

for any $x=(x_1,x_2,x_3,\ldots,)\in\ell^2$. Then it is easy to know

$$\dim\mathscr{N}(A)=+\infty,\ \dim\mathscr{R}(A)^{\perp}=0,$$

$$\dim\mathscr{N}(H)=\dim\mathscr{R}(H)^{\perp}=+\infty,$$

that is, $0\in\triangle_3(A)\cap\triangle_3(-A^*)$. But, $0\notin\triangle_3(H)$.

Again, let

$$H=\begin{bmatrix}A & 0\\ 0 & -A^*\end{bmatrix}=\begin{bmatrix}S_l & 0\\ 0 & -S_r\end{bmatrix},$$

where S_l, S_r are the left shift and right shift operators, respectively. It is easy to have $\operatorname{ind}(H)=0$ contradicting $\operatorname{ind}(A)\neq 0$.

2.3 Spectra of Skew-Diagonal Infinite-Dimensional Hamiltonian Operators

For the infinite-dimensional Hamiltonian operator

$$H=\begin{bmatrix}A & B\\ C & -A^*\end{bmatrix}:\mathscr{D}(H)\subseteq X\times X\to X\times X$$

if $A=0$, then H is called a skew-diagonal infinite-dimensional Hamiltonian operator. In this section, we will discuss basic properties of the point spectrum, residual spectrum, continuous spectrum and essential spectrum of skew-diagonal infinite-

dimensional Hamiltonian operators. The following are the specific examples of skew-diagonal infinite-dimensional Hamiltonian operators.

Example 2.3.1 The corresponding infinite-dimensional Hamiltonian system of wave equation

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} = \dfrac{\partial^2 u}{\partial x^2}, \\ u(0,t) = u(1,t) = 0, \\ u(x,0) = \varphi(x), u_t(x,0) = \phi(x) \end{cases}$$

can be given by

$$\frac{\partial}{\partial t}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & I \\ \dfrac{\partial^2}{\partial x^2} & 0 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix},$$

where $\frac{\partial v}{\partial t} = \frac{\partial^2 u}{\partial x^2}$. Let $X = L^2[0,1]$, then we obtain the following skew-diagonal infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} = \begin{bmatrix} 0 & I \\ \dfrac{\mathrm{d}^2}{\mathrm{d}x^2} & 0 \end{bmatrix},$$

where $\mathscr{D}(C) = \{x \in X : x, x' \text{ are absolutely continuous }, x'' \in X, x(0)=x(1) = 0\}$.

Theorem 2.3.2 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then the operators* H *and* $-H$, H^*, $-H^*$ *are similar.*

Proof It is easy to prove

$$H = J_1(-H)J_1, \quad H = J_2(H^*)J_2, \quad H = J_3(-H^*)J_3,$$

where

$$J_1 = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}, \quad J_2 = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}, \quad J_3 = \begin{bmatrix} 0 & iI \\ -iI & 0 \end{bmatrix}$$

satisfying $J_1 = J_1^{-1}$, $J_2 = J_2^{-1}$,$J_3 = J_3^{-1}$, and thus the conclusion is proved. □

2.3.1 Point Spectra of Skew-Diagonal Infinite-Dimensional Hamiltonian Operators

Definition 2.3.3 The symmetric operator T is called a non-negative operator, if $(Tx, x) \geq 0$ for all $x \in \mathscr{D}(T)$; if T or $-T$ is a non-negative operator, then T is called a semi-definite operator.

In order to discuss the point spectrum of skew-diagonal infinite-dimensional Hamiltonian operators, we firstly give the following Lemma.

Lemma 2.3.4 *Let $T : \mathscr{D}(T) \subset X \to X$ be a non-negative operator in a Hilbert space X, and there exists $x_0 \in \mathscr{D}(T)$ such that $(Tx_0, x_0) = 0$, then $Tx_0 = 0$.*

Proof Define $[x, y]_T = (Tx, y)$, and it is easy to prove that $[x, y]_T$ is pre-inner product. Denote $Y = \mathscr{D}(T)$, then

$$[x, x]_T \geq 0$$

for any $x \in Y$. Since the Schwarz inequality holds in the pre-inner product space $\{Y, [\cdot, \cdot]_T\}$,

$$|[x, y]_T|^2 \leq [x, x]_T [y, y]_T$$

for any $x, y \in Y$. Since $[x_0, x_0]_T = 0$,

$$|[x_0, y]_T|^2 \leq [x_0, x_0]_T [y, y]_T = 0$$

for any $y \in Y$, that is,

$$(Tx_0, y) = 0.$$

Note that $\overline{\mathscr{D}(T)} = X$, so $Tx_0 = 0$. The proof is completed. □

Lemma 2.3.5 *Let B, C be linear operators (not necessarily bounded) in a Hilbert space X. Then $\sigma_p(BC)\backslash\{0\} = \sigma_p(CB)\backslash\{0\}$.*

Proof Let $\lambda \in \sigma_p(BC)\backslash\{0\}$. Then there exists $x_0 \neq 0$ such that

$$BCx_0 = \lambda x_0,$$

which means that $BCx_0 \in \mathscr{D}(C)$ and

$$CB(Cx_0) = C(BCx_0) = \lambda Cx_0.$$

Since $\lambda \neq 0$ and $Cx_0 \neq 0$, $\lambda \in \sigma_p(CB)\backslash\{0\}$ and the corresponding eigenvector is Cx_0. If $\lambda \in \sigma_p(CB)\backslash\{0\}$, the proof of $\lambda \in \sigma_p(BC)\backslash\{0\}$ is completely similar. □

In general, $0 \in \sigma_p(CB)$ doesn't necessarily mean $0 \in \sigma_p(BC)$.

Example 2.3.6 Let $X = l^2$ and define

$$\begin{aligned} Bx &= (x_2, x_3, \ldots), \\ Cx &= (0, x_1, x_2, x_3, \ldots) \end{aligned}$$

for any $x = (x_1, x_2, x_3, \ldots) \in l^2$, then

$$CBx = (0, x_2, x_3, \ldots),$$
$$BCx = (x_1, x_2, x_3, \ldots).$$

Obviously, $0 \in \sigma_p(CB)$ contradicting $0 \notin \sigma_p(BC)$.

Remark 2.3.7 If B, C are bounded linear operators defined on the whole space, then

$$\begin{bmatrix} \lambda I - CB & 0 \\ 0 & I \end{bmatrix} = F(\lambda) \begin{bmatrix} \lambda I - BC & 0 \\ 0 & I \end{bmatrix} E(\lambda)$$

for any $\lambda \neq 0$, where

$$E(\lambda) = \begin{bmatrix} -\lambda^{-1}B & \lambda^{-1}I \\ \lambda I - CB & C \end{bmatrix}, \quad F(\lambda) = \begin{bmatrix} -C & I - \lambda^{-1}CB \\ I & \lambda^{-1}B \end{bmatrix}$$

and

$$E(\lambda)^{-1} = \begin{bmatrix} -C & \lambda^{-1}I \\ \lambda I - BC & \lambda^{-1}B \end{bmatrix}, \quad F(\lambda)^{-1} = \begin{bmatrix} \lambda^{-1}B & \lambda^{-1}BC + I \\ I & C \end{bmatrix}.$$

So the non-zero spectrum of CB and BC are the same. However, for unbounded operators, the above conclusion does not necessarily hold. For example, let

$$B = \begin{bmatrix} I & S \\ 0 & S \end{bmatrix}, \quad C = \begin{bmatrix} I & 0 \\ 0 & S^{-1} \end{bmatrix},$$

where S is an unbounded densely defined closed operator and $0 \in \rho(S)$, then

$$BC = \begin{bmatrix} I & I \\ 0 & I \end{bmatrix}, \quad CB = \begin{bmatrix} I & S \\ 0 & I \end{bmatrix}.$$

By calculation, $\sigma(BC)=\{1\}$, $\sigma(CB) = \mathbb{C}$, and obviously $\sigma(BC)\backslash\{0\} \neq \sigma(CB)\backslash\{0\}$.

In order to discuss when the relation $\sigma(BC)\backslash\{0\} = \sigma(CB)\backslash\{0\}$ is established, several lemmas will be given below.

Lemma 2.3.8 *Let* $T = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be closed,* BC, CB *are dense and* $\lambda \in \rho(T)$. *Then*

$$\lambda^2 \in \rho(BC) \cap \rho(CB).$$

Also,

$$(CB - \lambda^2)^{-1}C, \quad B(CB - \lambda^2)^{-1}C$$

are bounded on their domains $\mathscr{D}(C)$,

$$(BC - \lambda^2)^{-1}B, \quad C(BC - \lambda^2)^{-1}B$$

are bounded on their domains $\mathscr{D}(B)$*, and if* $\lambda \neq 0$*,*

$$(T-\lambda I)^{-1} = \begin{bmatrix} -\frac{1}{\lambda} + \frac{1}{\lambda}\overline{B(CB-\lambda^2 I)^{-1}C} & B(CB-\lambda^2 I)^{-1} \\ \overline{(CB-\lambda^2 I)^{-1}C} & \lambda(CB-\lambda^2 I)^{-1} \end{bmatrix}$$
$$= \begin{bmatrix} \lambda(BC-\lambda^2 I)^{-1} & \overline{(BC-\lambda^2 I)^{-1}B} \\ C(BC-\lambda^2 I)^{-1} & -\frac{1}{\lambda} + \frac{1}{\lambda}\overline{C(BC-\lambda^2 I)^{-1}B} \end{bmatrix}.$$

Proof We may let $\lambda \in \rho(T)\backslash\{0\}$. Since

$$T = J(-T)J,$$

where $J = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}$ and $J^{-1} = J^* = J$, we obtain $-\lambda \in \rho(T)$. Note that

$$(T-\lambda I)(T+\lambda I) = (T+\lambda I)(T-\lambda I) = \begin{bmatrix} BC-\lambda^2 I & 0 \\ 0 & CB-\lambda^2 I \end{bmatrix},$$

so $\lambda^2 \in \rho(BC) \cap \rho(CB)$.

Because

$$C(BC-\lambda^2 I) = (CB-\lambda^2 I)C,$$

multiplying $(BC-\lambda^2 I)^{-1}$ on the right and multiplying $(CB-\lambda^2 I)^{-1}$ on the left, we obtain

$$(CB-\lambda^2 I)^{-1}C = C(BC-\lambda^2 I)^{-1}|_{\mathscr{D}(C)}. \tag{2.7}$$

Since $C(BC-\lambda^2 I)^{-1}$ is bounded, $(CB-\lambda^2 I)^{-1}C$ is bounded on the domain $\mathscr{D}(C)$. Multiplying B on the left in (2.7) yields

$$B(CB-\lambda^2 I)^{-1}C = BC(BC-\lambda^2 I)^{-1} = I|_{\mathscr{D}(C)} + \lambda^2(BC-\lambda^2 I)^{-1}|_{\mathscr{D}(C)},$$

that is, $B(CB-\lambda^2 I)^{-1}C$ is bounded on $\mathscr{D}(C)$. The proof of operators $(BC-\lambda^2 I)^{-1}B$, $C(BC-\lambda^2 I)^{-1}B$ on their domain $\mathscr{D}(B)$ is similar.

Note that

$$(T-\lambda)\begin{bmatrix} -\frac{1}{\lambda} + \frac{1}{\lambda}\overline{B(CB-\lambda^2 I)^{-1}C} & B(CB-\lambda^2 I)^{-1} \\ \overline{(CB-\lambda^2 I)^{-1}C} & \lambda(CB-\lambda^2 I)^{-1} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

and

$$\begin{bmatrix} -\frac{1}{\lambda} + \frac{1}{\lambda}\overline{B(CB-\lambda^2 I)^{-1}C} & B(CB-\lambda^2 I)^{-1} \\ \overline{(CB-\lambda^2 I)^{-1}C} & \lambda(CB-\lambda^2 I)^{-1} \end{bmatrix}(T-\lambda I) = \begin{bmatrix} I|_{\mathscr{D}(C)} & 0 \\ 0 & I|_{\mathscr{D}(B)} \end{bmatrix},$$

so considering the closedness of $T-\lambda I$ implies

$$(T-\lambda I)^{-1}=\begin{bmatrix} -\frac{1}{\lambda}+\frac{1}{\lambda}\overline{B(CB-\lambda^2 I)^{-1}C} & B(CB-\lambda^2 I)^{-1} \\ \overline{(CB-\lambda^2 I)^{-1}C} & \lambda(CB-\lambda^2 I)^{-1} \end{bmatrix}.$$

The proof is completed. □

Corollary 2.3.9 *Let* $T=\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *with* $\rho(T)\neq\emptyset$. *If* BC *is densely defined. Then* $(BC)^*=C^*B^*$, *and if* CB *is densely defined, then* $(CB)^*=B^*C^*$.

Proof Since $\rho(T)\neq\emptyset$, let $\lambda\in\rho(T)$, and then $\lambda^2\in\rho(BC)\cap\rho(CB)$. Note that $\lambda\in\rho(T)$ if and only if $\overline{\lambda}\in\rho(T^*)$ and $T^*=\begin{bmatrix} 0 & C^* \\ B^* & 0 \end{bmatrix}$, similar to proof of Lemma 2.3.8, we can prove

$$\overline{\lambda^2}\in\rho(B^*C^*)\cap\rho(C^*B^*).$$

We may let BC be densely defined. Then to prove $(BC)^*=C^*B^*$, it suffices to prove $\mathscr{D}((BC)^*)\subset\mathscr{D}(C^*B^*)$ by $(BC)^*\supset C^*B^*$. Let $x^*\in\mathscr{D}((BC)^*)$. Because $\overline{\lambda^2}\in\rho(C^*B^*)$, there exists $\widetilde{x}\in\mathscr{D}(C^*B^*)$ such that

$$((BC)^*-\overline{\lambda^2}I)x^*=(C^*B^*-\overline{\lambda^2}I)\widetilde{x},$$

and as $(BC)^*|_{\mathscr{D}((C^*B^*)}=C^*B^*$, $(C^*B^*-\overline{\lambda^2}I)\widetilde{x}=((BC)^*-\overline{\lambda^2}I)\widetilde{x}$ we have

$$((BC)^*-\overline{\lambda^2}I)(x^*-\widetilde{x})=0.$$

From $\lambda^2\in\rho(BC)$, it follows that $\overline{\lambda^2}\in\rho(BC)^*)$, and therefore $x^*=\widetilde{x}$, that is, $\mathscr{D}((BC)^*)\subset\mathscr{D}(C^*B^*)$. □

Lemma 2.3.10 *Let* $T=\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a densely defined and closed operator, and* BC, CB *are dense. Then* $\rho(T)=\{\lambda\in\mathbb{C}:\lambda^2\in\rho(BC)\cap\rho(CB)\}$.

Proof From the proof of Lemma 2.3.8, we know

$$\rho(T)\subset\{\lambda\in\mathbb{C}:\lambda^2\in\rho(BC)\cap\rho(CB)\}.$$

Conversely, let $\lambda\in\{\lambda\in\mathbb{C}:\lambda^2\in\rho(BC)\cap\rho(CB)\}$. Since

$$(T-\lambda)(T+\lambda)=\begin{bmatrix} BC-\lambda^2 I & 0 \\ 0 & CB-\lambda^2 I \end{bmatrix},$$

we obtain that $T-\lambda$ is surjective. Again

$$(T+\lambda I)(T-\lambda I)=\begin{bmatrix} BC-\lambda^2 I & 0 \\ 0 & CB-\lambda^2 I \end{bmatrix},$$

which implies that $T-\lambda$ is injective. As T is closed, $\lambda\in\rho(T)$. □

Theorem 2.3.11 *If $\rho(BC) \cap \rho(CB) \neq \emptyset$, then $\sigma(BC)\backslash\{0\} = \sigma(CB)\backslash\{0\}$.*

Proof If $\rho(BC) \cap \rho(CB) \neq \emptyset$, then we know $\rho(T) \neq \emptyset$ by Lemma 2.3.10. Since $\rho(T)$ is an open set, suppose $\rho(T)\backslash\{0\} \neq \emptyset$, and we may let $\lambda \in \rho(T)\backslash\{0\}$, then $\lambda^2 \in \rho(CB)\backslash\{0\}$, that is,

$$\rho(T)\backslash\{0\} \subset \{\lambda \in \mathbb{C} : \lambda^2 \in \rho(CB)\}\backslash\{0\}.$$

Conversely, let $\lambda \in \{\lambda \in \mathbb{C} : \lambda^2 \in \rho(CB)\}\backslash\{0\}$, then

$$(T-\lambda)^{-1} = \begin{bmatrix} -\frac{1}{\lambda} + \frac{1}{\lambda}\overline{B(CB-\lambda^2 I)^{-1}C} & B(CB-\lambda^2 I)^{-1} \\ \overline{(CB-\lambda^2 I)^{-1}C} & \lambda(CB-\lambda^2 I)^{-1} \end{bmatrix},$$

namely, $\lambda \in \rho(T)$, so

$$\rho(T)\backslash\{0\} = \{\lambda \in \mathbb{C} : \lambda^2 \in \rho(CB)\}\backslash\{0\}.$$

Similarly, we can obtain

$$\rho(T)\backslash\{0\} = \{\lambda \in \mathbb{C} : \lambda^2 \in \rho(BC)\}\backslash\{0\}.$$

To sum up, $\rho(BC)\backslash\{0\} = \rho(CB)\backslash\{0\}$. The proof is completed. □

Theorem 2.3.12 *If $\rho(BC) \neq \emptyset$ and $\rho(CB) \neq \emptyset$, then $\sigma(BC)\backslash\{0\} = \sigma(CB)\backslash\{0\}$.*

Proof If $\rho(BC) \neq \emptyset$ and $\rho(CB) \neq \emptyset$, then $\rho(BC)\backslash\{0\} \neq \emptyset$ and $\rho(CB)\backslash\{0\} \neq \emptyset$. Since $\rho(BC)\backslash\{0\} = \rho(CB)\backslash\{0\}$, we know

$$\rho(CB) \cap \rho(BC) \neq \emptyset.$$

By Theorem 2.3.11, we know

$$\sigma(BC)\backslash\{0\} = \sigma(CB)\backslash\{0\}.$$

The proof is completed. □

Theorem 2.3.13 *Let $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ be a skew-diagonal infinite-dimensional Hamiltonian operator. Then*

(i) $\sigma_p(H) = \sigma_p(-H) = \sigma_p(H^*) = \sigma_p(-H^*)$;
(ii) $\sigma_{p,i}(H) = \sigma_{p,i}(-H) = \sigma_{p,i}(H^*) = \sigma_{p,i}(-H^*)$, $i = 1, 2, 3, 4$;
(iii) $\lambda \in \sigma_p(H)$ *if and only if* $\lambda^2 \in \sigma_p(H^2)$;
(iv) $\lambda \in \sigma_{p,i}(H)$ *if and only if* $\lambda^2 \in \sigma_{p,i}(H^2)$, $i = 1, 2$;
(v) $\lambda \in (\sigma_{p,3}(H) \cup \sigma_{p,4}(H))$ *if and only if* $\lambda^2 \in (\sigma_{p,3}(H^2) \cup \sigma_{p,4}(H^2))$.

Proof By Theorem 2.3.2, we know that H and $-H$, H^* and $-H^*$ are similar operators, therefore the conclusions (i) and (ii) are obvious.

(iii) Let $\lambda \in \sigma_p(H)$. Then $-\lambda \in \sigma_p(H)$ by (i), which together with

$$H^2 - \lambda^2 I = (H + \lambda I)(H - \lambda I)$$

means $\lambda^2 \in \sigma_p(H^2)$. Conversely, let $\lambda^2 \in \sigma_p(H^2)$, and note that

$$H^2 - \lambda^2 I = (H + \lambda I)(H - \lambda I) = (H - \lambda I)(H + \lambda I),$$

so we can obtain $\lambda \in \sigma_p(H)$ by the similarity of H and $-H$.

(iv) The proof of $\lambda \in \sigma_{p,1}(H)$ if and only if $\lambda^2 \in \sigma_{p,1}(H^2)$ is obvious. Let $\lambda \in \sigma_{p,2}(H)$. Then $\overline{\lambda} \in \sigma_{r,2}(H^*)$ and $\mathscr{R}(H - \lambda I)$ is dense but not surjective. From conclusion (i), we know that $\mathscr{R}(H + \lambda I)$ is dense but not surjective, and hence we can assert that $\mathscr{R}(H^2 - \lambda^2 I)$ is dense but not surjective. In fact, it is obvious that $\mathscr{R}(H^2 - \lambda^2 I)$ is not surjective, so assume $\mathscr{R}(H^2 - \lambda^2 I)$ is not dense, then $\overline{\lambda^2} \in \sigma_p((H^*)^2)$, that is, $\overline{\lambda} \in \sigma_p(H^*)$, this is contradictory to $\overline{\lambda} \in \sigma_{r,2}(H^*)$. Thus $\lambda \in \sigma_{p,2}(H)$ if and only if $\lambda^2 \in \sigma_{p,2}(H^2)$.

The proof of (v) and (iv) are completely similar. □

Theorem 2.3.14 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then*

(i) $0 \in \sigma_p(H)$ *if and only if* $0 \in \sigma_p(B) \cup \sigma_p(C)$;
(ii) $\sigma_p(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in \sigma_p(CB) \cup \sigma_p(BC)\}$;
(iii) If B *or* C *is a uniformly positive definite operator, then* $\sigma_p(H) \subset \mathbb{R} \cup i\mathbb{R}$.

Proof The proof of (i) is trivial. Next prove (ii). From $\mathscr{D}(H^2) = \mathscr{D}(BC) \times \mathscr{D}(CB)$,

$$H^2 = \begin{bmatrix} BC & 0 \\ 0 & CB \end{bmatrix},$$

and

$$H^2 - \lambda^2 I = (H + \lambda I)(H - \lambda I)$$

it follows that

$$\sigma_p(H) \subset \{\lambda \in \mathbb{C} : \lambda^2 \in \sigma_p(CB) \cup \sigma_p(BC)\}$$

can be proven directly.

Let $\lambda^2 \in \sigma_p(CB) \cup \sigma_p(BC)$. Then $\lambda^2 \in \sigma_p(H^2)$ and $\lambda \in \sigma_p(H) \cup \sigma_p(-H)$. By Theorem 2.3.2, we have $\sigma_p(H) = \sigma_p(-H)$, and therefore $\lambda \in \sigma_p(H)$, that is, $\sigma_p(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in \sigma_p(CB) \cup \sigma_p(BC)\}$.

(iii) We may let B be uniformly positive definite. Let $\lambda \in \sigma_p(H)$ and the corresponding eigenvector be $\begin{bmatrix} f & g \end{bmatrix}^T$. Then it is easy to obtain $f \neq 0$ and

$$\lambda^2 = \frac{(Cf, f)}{B^{-1}f, f} \in \mathbb{R},$$

so the conclusion holds. □

The following will give characterizations of 1-class and 2-class point spectra.

Theorem 2.3.15 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be the skew-diagonal infinite-dimensional Hamiltonian operator. Then*

(i) $(\sigma_{p,1}(H) \cup \sigma_{p,2}(H)) \cap \mathbb{R} = \emptyset$ *and* $(\sigma_{p,1}(H) \cup \sigma_{p,2}(H)) \cap i\mathbb{R} = \emptyset$;
(ii) $\sigma_{p,1}(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in \sigma_{p,1}(CB) \cap \sigma_{p,1}(BC)\}$;
(iii) $\sigma_{p,2}(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{p,1}(BC) \cap \sigma_{p,2}(CB)) \cup (\sigma_{p,2}(BC) \cap \sigma_{p,1}(CB)) \cup (\sigma_{p,2}(BC) \cap \sigma_{p,2}(CB))\}$.

Proof (i) We only prove $\sigma_{p,1}(H) \cap \mathbb{R} = \emptyset$, the proof of the other are completely similar. Let $\lambda \in \sigma_{p,1}(H) \cap \mathbb{R}$, then $\lambda = \bar{\lambda}$ and $\lambda \in \sigma_{r,1}(H^*)$, that is, $\lambda \in \sigma_{r,1}(H^*)$. On the other hand, due to the similarity of H and H^*, $\lambda \in \sigma_{p,1}(H^*)$, contradiction is inferred.

(ii) Let $\lambda \in \sigma_{p,1}(H)$. Since

$$(H - \lambda I)(H + \lambda I) = (H + \lambda)(H - \lambda) = \begin{bmatrix} BC - \lambda^2 I & 0 \\ 0 & CB - \lambda^2 I \end{bmatrix}, \tag{2.8}$$

$\lambda^2 \in \sigma_p(BC) \cup \sigma_p(CB)$, $\mathscr{R}(BC - \lambda^2 I) = X$ and $\mathscr{R}(CB - \lambda^2 I) = X$. Suppose $\lambda^2 \in \sigma_p(BC)$, we know $\lambda^2 \in \sigma_{p,1}(BC)$ by $\mathscr{R}(BC - \lambda^2 I) = X$. Since $\lambda \neq 0$, we know $\lambda^2 \in \sigma_p(CB)$ by Lemma 2.3.5, which together with $\mathscr{R}(CB - \lambda^2 I) = X$ implies $\lambda^2 \in \sigma_{p,1}(CB)$, so

$$\sigma_{p,1}(H) \subset \{\lambda \in \mathbb{C} : \lambda^2 \in \sigma_{p,1}(CB) \cap \sigma_{p,1}(BC)\}.$$

Conversely, let $\lambda^2 \in \sigma_{p,1}(CB) \cap \sigma_{p,1}(BC)$, then obviously $\lambda \in \sigma_p(H)$. Since $\mathscr{R}(BC - \lambda^2 I) = X$ and $\mathscr{R}(CB - \lambda^2 I) = X$, we know $H - \lambda I$ is surjective by the equality (2.8), that is, $\lambda \in \sigma_{p,1}(H)$.

(iii) Let $\lambda \in \sigma_{p,2}(H)$. Then $-\lambda \in \sigma_{p,2}(H)$, and it is easy to see $\mathscr{R}(H^2 - \lambda^2 I) \neq X \times X$. Next we prove $\mathscr{R}(H^2 - \lambda^2 I)$ is dense. If not, assume that $\mathscr{R}(H^2 - \lambda^2 I)$ is not dense, and then $\overline{\lambda^2} \in \sigma_p((H^*)^2)$, that is, $\bar{\lambda} \in \sigma_p(H^*)$. On the other hand, we know $\bar{\lambda} \in \sigma_{r,2}(H^*)$ by $\lambda \in \sigma_{p,2}(H)$, contradiction is inferred. So $\mathscr{R}(H^2 - \lambda^2 I)$ is dense.

If $\lambda^2 \in \sigma_{p,1}(BC)$, since $\lambda \neq 0$, we know $\lambda^2 \in \sigma_{p,2}(CB)$ by Lemma 2.3.5. If $\lambda^2 \in \sigma_{p,2}(BC)$, it follow from $\lambda \neq 0$ that $\lambda^2 \in \sigma_{p,1}(CB) \cup \sigma_{p,2}(CB)$ by Lemma 2.3.5, and hence

$$\sigma_{p,2}(H) \subset \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{p,1}(BC) \cap \sigma_{p,2}(CB)) \cup (\sigma_{p,2}(BC) \cap \sigma_{p,1}(CB)) \cup (\sigma_{p,2}(BC) \cap \sigma_{p,2}(CB))\}.$$

The proof of reverse inclusion is obvious. □

Remark 2.3.16 It is not hard to find that, when the point spectrum of skew-diagonal infinite-dimensional Hamiltonian operator H is not empty, if $B \geq 0, C \geq 0$ or $-B \geq 0, -C \geq 0$, then $\sigma_p(H) \subset \mathscr{R}$ and $\lambda \in \sigma_p(H)$ if and only if $-\lambda \in \sigma_p(H)$; if $B \geq 0, -C \geq 0$ or $-B \geq 0, C \geq 0$, then $\sigma_p(H) \subset \mathtt{i}\mathscr{R}$ and $\lambda \in \sigma_p(H)$ if and only if $-\lambda \in \sigma_p(H)$, that is, when the signs of B, C are identical, we have only real point spectra, when the symbols of B, C are different, we have only pure imaginary point spectra, and both appear in positive and negative pairs.

2.3.2 *Residual Spectra of Skew-Diagonal Infinite-Dimensional Hamiltonian Operators*

Theorem 2.3.17 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then*

(i) $\sigma_r(H) = \sigma_r(-H) = \sigma_r(H^*) = \sigma_r(-H^*)$;
(ii) $\sigma_{r,i}(H) = \sigma_{r,i}(-H) = \sigma_{r,i}(H^*) = \sigma_{r,i}(-H^*)$, *i=1, 2;*
(iii) $\lambda \in \sigma_r(H)$ *if and only if* $\lambda^2 \in \sigma_r(H^2)$.

Proof By Theorem 2.3.2, H and $-H$, H^* and $-H^*$ are similar, therefore the conclusions (i) and (ii) are obvious.

(iii) Let $\lambda \in \sigma_r(H)$, $-\lambda \in \sigma_r(H)$. Then $H^2 - \lambda^2 I$ is injective. Since $\mathscr{R}(H - \lambda I)$ is not dense, $\mathscr{R}(H^2 - \lambda^2 I)$ is not dense, that is, $\lambda^2 \in \sigma_r(H^2)$. Conversely, let $\lambda^2 \in \sigma_r(H^2)$. It is easy to know that $H - \lambda I$ and $H + \lambda I$ are injective. Assume $\overline{\mathscr{R}(H - \lambda I)} = X \times X$, then $\overline{\mathscr{R}(H + \lambda I)} = X \times X$, so $\lambda \in \sigma_c(H)$. Therefore, $\overline{\lambda} \in \sigma_c(H^*)$ and $-\overline{\lambda} \in \sigma_c(H^*)$, that is, $(H^*)^2 - \overline{\lambda}^2$ is injective. However, form $\lambda^2 \in \sigma_r(H^2)$, we know that $(H^*)^2 - \overline{\lambda}^2$ is not injective, which is a contradiction. Hence, $\overline{\mathscr{R}(H - \lambda I)} \neq X \times X$, that is, $\lambda \in \sigma_r(H)$. The proof is completed. □

Theorem 2.3.18 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then*

(i) The residual spectrum $\sigma_r(H)$ *has no intersection with the real axis and the imaginary axis, that is,* $\sigma_r(H) \cap \mathbb{R} = \emptyset$ *and* $\sigma_r(H) \cap \mathtt{i}\mathbb{R} = \emptyset$;
(ii) $\sigma_r(H) = \{\lambda \in \mathbb{C} : \overline{\lambda^2} \in \sigma_p(BC), \lambda^2 \notin \sigma_p(BC)\}$;
(iii) $\sigma_r(H) = \{\lambda \in \mathbb{C} : \overline{\lambda^2} \in \sigma_p(CB), \lambda^2 \notin \sigma_p(CB)\}$.

Proof (i) Suppose there exists $\lambda \in \mathbb{R}$ such that $\lambda \in \sigma_r(H)$, then $\lambda \in \sigma_p(H^*)$, By Theorem 2.3.2, we obtain

$$\lambda \in \sigma_p(H),$$

which is contradictory to $\lambda \in \sigma_r(H)$.

Similarly, let $\lambda \in \mathrm{i}\mathbb{R}$ and $\lambda \in \sigma_r(H)$. Then $-\lambda \in \sigma_p(H^*)$, that is, $\lambda \in \sigma_p(-H^*)$. By Theorem 2.3.2, we know $\lambda \in \sigma_p(H)$, which is a contradiction.

(ii) Let $\lambda \in \sigma_r(H)$. Then $\lambda \notin \mathbb{R}$ and $\overline{\lambda} \in \sigma_p(H^*)$, so $\overline{\lambda} \in \sigma_p(H)$. By Theorem 2.3.14, we know $\overline{\lambda^2} \in \sigma_p(BC) \cup \sigma_p(CB)$. Noting that $\lambda \neq 0$ and

$$\sigma_p(BC) \setminus \{0\} = \sigma_p(CB) \setminus \{0\},$$

we obtain $\overline{\lambda^2} \in \sigma_p(BC)$. On the other hand, $\lambda \in \sigma_r(H)$ implies $\lambda \notin \sigma_p(H)$, therefore $\lambda^2 \notin \sigma_p(BC)$. To sum up, $\sigma_r(H) \subset \{\lambda \in \mathbb{C} : \overline{\lambda^2} \in \sigma_p(BC), \lambda^2 \notin \sigma_p(BC)\}$.

Let $\overline{\lambda^2} \in \sigma_p(BC)$, $\lambda^2 \notin \sigma_p(BC)$. Then $\lambda \notin \mathscr{R}$, and we know $\lambda^2 \notin \sigma_p(CB)$ by $\sigma_p(BC) \setminus \{0\} = \sigma_p(CB) \setminus \{0\}$, that is,

$$\lambda \notin \sigma_p(H).$$

Note that

$$\begin{bmatrix} CB - \overline{\lambda^2} I & 0 \\ 0 & BC - \overline{\lambda^2} I \end{bmatrix} = (H^* - \overline{\lambda} I)(H^* + \overline{\lambda} I) = (H^* + \overline{\lambda} I)(H^* - \overline{\lambda} I),$$

so $\overline{\lambda} \in \sigma_p(H^*) \cup \sigma_p(-H^*)$. From $\sigma_p(H^*) = \sigma_p(-H^*)$, it follows that $\overline{\lambda} \in \sigma_p(H^*)$ and $\lambda \in \sigma_p(H) \cup \sigma_r(H)$, so $\lambda \in \sigma_r(H)$.

The proof of (iii) and (ii) are completely similar. □

Theorem 2.3.19 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator with* B *or* C *semi-definite. Then* $\sigma_r(H) = \emptyset$.

Proof Let $\lambda \in \sigma_r(H)$. Then $Re(\lambda) \neq 0$, $Im(\lambda) \neq 0$ and $\overline{\lambda} \in \sigma_p(H)$, so there exist $f \in \mathscr{D}(C)$, $g \in \mathscr{D}(B)$ (not both zero vectors) such that

$$Bg = \overline{\lambda} f, \quad Cf = \overline{\lambda} g.$$

Taking the inner product of the first equality by g on the right and of the second equality by f on the left, we have

$$(f, Cf) + (Bg, g) = 2Re(\lambda)(f, g), \tag{2.9}$$

$$(Bg, g) - (f, Cf) = -2\mathrm{i}Im(\lambda)(f, g). \tag{2.10}$$

Because B, C are self-adjoint operators, we know $(f, g) \in \mathbb{R}$ by (2.9) and $(f, g) \in \mathrm{i}\mathbb{R}$ by (2.10), that is, $(f, g) = 0$, so $(Bg, g) = 0$ and $(f, Cf) = 0$. Since B or C is semi-definite, we may suppose B is non-negative, so $Bg = 0$. This means that $f = 0$ and $g = 0$, which is a contradiction. Therefore, $\sigma_r(H) = \emptyset$. The proof is completed. □

The following will give characterization of other residual spectra.

Theorem 2.3.20 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then*

(i) $\sigma_{r,1}(H) = \{\lambda \in \mathbb{C} : \overline{\lambda^2} \in \sigma_{p,1}(CB) \cap \sigma_{p,1}(BC)\}$;

(ii) $\sigma_{r,2}(H) = \{\lambda \in \mathbb{C} : \overline{\lambda^2} \in (\sigma_{p,1}(BC) \cap \sigma_{p,2}(CB)) \cup (\sigma_{p,2}(BC) \cap \sigma_{p,1}(CB)) \cup (\sigma_{p,2}(BC) \cap \sigma_{p,2}(CB))\}$.

Proof Note that $\lambda \in \sigma_{r,i}(H)$ if and only if $\overline{\lambda} \in \sigma_{p,i}(H^*)$, and if and only if $\overline{\lambda} \in \sigma_{p,i}(H)$, $i = 1, 2$, and the conclusion is thus established by Theorem 2.3.15. □

Lemma 2.3.21 *Let* B, C *be densely defined closed operators such that* CB, BC *are densely defined and* $\rho(BC) \cap \rho(CB) \neq \emptyset$. *Then* $\sigma_r(BC)\backslash\{0\} = \sigma_r(CB)\backslash\{0\}$.

Proof If $\rho(BC) \cap \rho(CB) \neq \emptyset$ and CB, BC are densely defined, we can assert

$$(BC)^* = C^*B^*, \quad (CB)^* = B^*C^*.$$

In fact, when $\rho(BC) \cap \rho(CB) \neq \emptyset$, let $\lambda^2 \in \rho(BC) \cap \rho(CB)$, then considering

$$(T - \lambda I)(T + \lambda I) = (T + \lambda I)(T - \lambda I) = \begin{bmatrix} BC - \lambda^2 I & 0 \\ 0 & CB - \lambda^2 I \end{bmatrix},$$

we obtain $\lambda \in \rho(T) \cap \rho(-T)$, where $T = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$, so $\overline{\lambda} \in \rho(T^*) \cap \rho(-T^*)$, that is, $\overline{\lambda^2} \in \rho(C^*B^*) \cap \rho(B^*C^*)$. Let BC be densely defined, then to prove $(BC)^* = C^*B^*$, we only need to show $\mathscr{D}((BC)^*) \subset \mathscr{D}(C^*B^*)$ by $(BC)^* \supset C^*B^*$. Let $x^* \in \mathscr{D}((BC)^*)$, and as $\overline{\lambda} \in \rho(C^*B^*)$, there exists $\widetilde{x} \in \mathscr{D}(C^*B^*)$ such that

$$((BC)^* - \overline{\lambda^2} I)x^* = (C^*B^* - \overline{\lambda^2} I)\widetilde{x}.$$

Since $(BC)^*|_{\mathscr{D}(C^*B^*)} = C^*B^*$,

$$(C^*B^* - \overline{\lambda^2} I)\widetilde{x} = ((BC)^* - \overline{\lambda^2} I)\widetilde{x},$$

that is,

$$((BC)^* - \overline{\lambda^2} I)(x^* - \widetilde{x}) = 0.$$

Since $\lambda^2 \in \rho(BC)$, $\overline{\lambda^2} \in \rho((BC)^*)$, and therefore $x^* = \widetilde{x}$, that is, $\mathscr{D}((BC)^*) \subset \mathscr{D}(C^*B^*)$. Similarly $(CB)^* = B^*C^*$.

Next we prove $\sigma_r(BC)\backslash\{0\} = \sigma_r(CB)\backslash\{0\}$. Let $\lambda \in \sigma_r(BC)\backslash\{0\}$. Then $\overline{\lambda} \in \sigma_p(C^*B^*)\backslash\{0\}$, that is, $\overline{\lambda} \in \sigma_p(B^*C^*)\backslash\{0\}$, so $\lambda \in \sigma_r(CB)\backslash\{0\} \cup \sigma_p(CB)\backslash\{0\}$, and noting $\lambda \notin \sigma_p(CB)\backslash\{0\}$, we have $\lambda \in \sigma_r(CB)\backslash\{0\}$. The converse is similar. □

Theorem 2.3.22 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. If* $\rho(H) \neq \emptyset$ *and* CB, BC *are densely defined, then*

$$\sigma_r(H) = \{\lambda \in \mathbb{C}\backslash\{0\} : \lambda^2 \in \sigma_r(BC)\} = \{\lambda \in \mathbb{C}\backslash\{0\} : \lambda^2 \in \sigma_r(CB)\}.$$

Proof Let $\lambda \in \sigma_r(H)$. Then $\lambda \neq 0$ and $\lambda^2 \in \sigma_r(H^2)$, and the conclusion is proven by Lemma 2.3.21. □

2.3.3 Continuous Spectra of Skew-Diagonal Infinite-Dimensional Hamiltonian Operators

Theorem 2.3.23 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then* $\sigma_c(H) = \sigma_c(-H) = \sigma_c(H^*) = \sigma_c(-H^*)$.

Proof By Theorem 2.3.2 we know that H and $-H$, H^* and $-H^*$ are similar, and therefore the conclusion is obvious. □

Lemma 2.3.24 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. If* BC, CB *are densely defined closed operators, then* $\lambda \in \sigma_c(H)$ *if and only if* $\lambda^2 \in \sigma_c(H^2)$.

Proof If $\lambda \in \sigma_c(H)$, we have $-\lambda \in \sigma_c(H)$ and $\overline{\lambda} \in \sigma_c(H)$, and hence $H^2 - \lambda^2 I$ is injective. Since $\mathscr{R}(H - \lambda I)$ is dense and not surjective, it is easy to know that $\mathscr{R}(H^2 - \lambda^2 I)$ is dense and not surjective. In fact, it is obvious that $\mathscr{R}(H^2 - \lambda^2 I)$ is not surjective. Assuming that $\mathscr{R}(H^2 - \lambda^2 I)$ is not dense yields $\overline{\lambda^2} \in \sigma_p((H^*)^2)$, then $\overline{\lambda} \in \sigma_p(H)$ by Theorem 2.3.14, which is contradictory to $\overline{\lambda} \in \sigma_c(H)$. Hence, $\lambda^2 \in \sigma_c(H^2)$. Conversely, the proof is similar. □

Theorem 2.3.25 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. If* BC, CB *are densely defined closed operators, then*

$$\begin{aligned}\sigma_c(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_c(BC) \cap \sigma_c(CB)) \cup (\rho(BC) \\ \cap \sigma_c(CB)) \cup (\sigma_c(BC) \cap \rho(CB))\}.\end{aligned}$$

Proof The proof of relation

$$\begin{aligned}\sigma_c(H) \supset \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_c(BC) \cap \sigma_c(CB)) \cup (\rho(BC) \\ \cap \sigma_c(CB)) \cup (\sigma_c(BC) \cap \rho(CB))\}\end{aligned}$$

is obvious. Conversely, if $\lambda \in \sigma_c(H)$, we have $\lambda^2 \in \sigma_c(H^2)$ by Lemma 2.3.24, that is, $H^2 - \lambda^2 I$ is densely defined, the range is dense but not surjective. If $\lambda^2 \in \sigma_c(BC)$, it follows from $\overline{\mathscr{R}(H^2 - \lambda^2 I)} = X \times X$ that $\lambda^2 \in (\sigma_c(CB) \cup \rho(CB)$. If $\lambda^2 \notin \sigma_c(BC)$, it follows from $\overline{\mathscr{R}(BC - \lambda^2 I)} = X$ that $\lambda^2 \in \rho(BC)$. So, from the fact that $\mathscr{R}(H^2 - \lambda^2 I)$ is dense but not surjective, we get $\lambda^2 \in \sigma_c(CB)$. The proof is completed. □

Theorem 2.3.26 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator with* $0 \in \rho(B) \cap \rho(C)$. *If* BC, CB *are densely defined closed operators, then* $\sigma_c(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_c(BC) \cap \sigma_c(CB))\}$.

Proof If $0 \in \rho(B) \cap \rho(C)$, then for any $\lambda \in \sigma_c(H)$ we have $\lambda \neq 0$ and

$$(\rho(BC) \cap \sigma_c(CB)) \cup (\sigma_c(BC) \cap \rho(CB)) = \emptyset.$$

So, conclusion is established. □

2.3.4 Essential Spectra of Skew-Diagonal Infinite-Dimensional Hamiltonian Operators

In order to discuss the essential spectrum of skew-diagonal infinite-dimensional Hamiltonian operators, the following Lemma is firstly given, for specific proof see Theorem XVII1.3. of [55].

Lemma 2.3.27 *Let* T, S *be dense closed Fredholm operators in a Hilbert space* X. *Then* TS *is Fredholm operator and* $\mathtt{ind}(TS) = \mathtt{ind}(T) + \mathtt{ind}(S)$.

Now it is natural to ask whether the product of two densely defined closed semi-Fredholm operators is also a semi-Fredholm operator? In order to answer this question, we firstly give the following results.

Theorem 2.3.28 *Let* T *be a densely defined closed operator in a Hilbert space* X. *Then*

(i) T *is a left semi-Fredholm operator if and only if there exist* $T_1 \in \mathscr{B}(X)$ *and finite rank operator* K_1 *such that* $T_1 Tx = (I + K_1)x$ *for any* $x \in \mathscr{D}(T)$;
(ii) T *is a right semi-Fredholm operator if and only if there exist* $T_2 \in \mathscr{B}(X)$ *and finite rank operator* K_2 *such that* $TT_2 = I + K_2$.

Proof (i) If there exists $T_1 \in \mathscr{B}(X)$ and finite rank operator K_1 such that $T_1 Tx = (I + K_1)x$ for any $x \in \mathscr{D}(T)$, from $\mathscr{N}(T) \subset \mathscr{N}(T_1 T)$ we obtain

$$\mathtt{dim}\mathscr{N}(T) \leq \mathtt{dim}\mathscr{N}(I + K_1).$$

As K_1 is a compact operator, $\mathtt{dim}\mathscr{N}(T) < \infty$. Next we need to prove that $\mathscr{R}(T)$ is a closed set. Assume that $\mathscr{R}(T)$ is not closed, then there exists sequence $\{x_n\}_{n=1}^{+\infty} \subset \mathscr{D}(T) \cap \mathscr{N}(T)^{\perp}$, $\|x_n\| = 1 (n = 1, 2, 3, \ldots)$, such that

$$Tx_n \to 0$$

as $n \to \infty$, that is,

$$x_n + K_1 x_n \to 0.$$

Because $\{K_1 x_n\}$ has a convergent subsequence, $\{x_n\}$ has convergent subsequence. Let $x_n \to x_0$. Since T is a closed operator, we get $x_0 \in \mathscr{D}(T)$ and $Tx_0 = 0$. Since $\mathscr{N}(T)^\perp$ is closed, $x_0 \in \mathscr{N}(T)^\perp$ and $x_0 \in \mathscr{N}(T)$, and therefore $x_0 = 0$, which is contradictory to $\|x_n\| = 1 (n = 1, 2, 3, \ldots)$. Hence, T is a left semi-Fredholm operator.

Conversely, if T is a left semi-Fredholm operator, we have the following block forms:

$$T = \begin{bmatrix} 0 & 0 \\ 0 & T_0 \end{bmatrix} : \mathscr{N}(T) \oplus (\mathscr{N}(T)^\perp \cap \mathscr{D}(T)) \to \mathscr{R}(T)^\perp \oplus \mathscr{R}(T),$$

under space decomposition $X = \mathscr{N}(T) \oplus \mathscr{N}(T)^\perp$ and $X = \mathscr{R}(T)^\perp \oplus \mathscr{R}(T)$, that is,

$$T = \begin{bmatrix} -I_{\mathscr{N}(T)} & 0 \\ 0 & T_0 \end{bmatrix} + \begin{bmatrix} I_{\mathscr{N}(T)} & 0 \\ 0 & 0 \end{bmatrix},$$

where $I_{\mathscr{N}(T)}$ represents the identity operator on $\mathscr{N}(T)$. From $\dim \mathscr{N}(T) < \infty$, it follows that $I_{\mathscr{N}(T)}$ is of finite rank. Let

$$T_1 = \begin{bmatrix} -I_{\mathscr{N}(T)} & 0 \\ 0 & T_0 \end{bmatrix}^{-1}, \quad K_1 = \begin{bmatrix} -I_{\mathscr{N}(T)} & 0 \\ 0 & T_0 \end{bmatrix}^{-1} \begin{bmatrix} I_{\mathscr{N}(T)} & 0 \\ 0 & 0 \end{bmatrix}.$$

Then T_1 is bounded on the whole space, K_1 is of finite rank and $T_1 T x = (I + K_1)x$ for any $x \in \mathscr{D}(T)$.

(ii) If $TT_2 = I + K_2$, we have

$$T_2^* T^* x = (I + K_2^*)x$$

for any $x \in \mathscr{D}(T^*)$, that is, T^* is a left semi-Fredholm operator, and therefore T is a right semi-Fredholm operator. Conversely, proof is completely similar. □

Corollary 2.3.29 *Let T, S, TS be the densely defined closed operators in a Hilbert space X. Then*

(i) If T, S are left semi-Fredholm operators, then TS is a left semi-Fredholm operator;
(ii) T, S are right semi-Fredholm operators, then TS is a right semi-Fredholm operator;
(iii) T, S are Fredholm operators, then TS is a Fredholm operator;
(iv) T, S are Weyl operators, then TS is a Weyl operator.

Proof (i) Since T, S are left semi-Fredholm operators, we know by Lemma 2.3.28 that there exists bounded operators T_1, S_1 on the whole space and finite rank operators K_1, K_2 such that

$$T_1 T x = (I + K_1)x,$$

$$S_1 S y = (I + K_2) y$$

for any $x \in \mathscr{D}(T)$ and $y \in \mathscr{D}(S)$. So

$$\begin{aligned} S_1 T_1 T S z &= S_1 (I + K_1) S z \\ &= S_1 S z + S_1 K_1 S z \\ &= [I + (K_2 + S_1 K_1 S)] z \\ &= [I + K_2 + \overline{S_1 K_1 S}] z \end{aligned}$$

for any $z \in \mathscr{D}(TS)$. Since $\overline{K_2 + S_1 K_1 S} = K_2 + S_1 \overline{K_1 S}$ is the compact operator on the whole space, TS is a left semi-Fredholm operator.

The proof of (ii) to that of (i) is completely similar. By Lemma 2.3.27, conclusions (iii) and (iv) are obvious. □

It's noteworthy that TS is not necessarily a semi-Fredholm operator when T is a left semi-Fredholm operator, and S is a right semi-Fredholm operator.

Example 2.3.30 Let $X = l^2$ and define

$$\begin{aligned} Tx &= (x_1, 0, x_2, 0, x_3, \ldots), \\ Sx &= (x_2, x_4, x_6, \ldots) \end{aligned}$$

for any $x = (x_1, x_2, x_3, \ldots) \in X$. Then

$$TSx = (x_2, 0, x_4, 0, x_6, \ldots).$$

It is easy to know that T is a left semi-Fredholm operator, S is a right semi-Fredholm operator, but

$$\dim \mathscr{N}(TS) = \infty, \quad \dim \mathscr{R}(TS)^{\perp} = \infty,$$

that is, TS doesn't have semi-Fredholm property.

Remark 2.3.31 For Corollary 2.3.29, Theorem 7.32 in [56] also gives a different proof.

Theorem 2.3.32 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then* $\sigma_{e,i}(H) = \sigma_{e,i}(-H) = \sigma_{e,i}(H^*) = \sigma_{e,i}(-H^*)$, $i = 1, 2, \ldots, 8$.

Proof By Theorem 2.3.2 we know that H and $-H$, H^* and $-H^*$ are similar, so the conclusion is obvious. □

Theorem 2.3.33 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator, and* BC, CB *be densely closed operators. Then*

(i) $\sigma_{e,i}(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{e,i}(BC) \cup \sigma_{e,i}(CB))\}, i = 2, 4$;

(ii) $\sigma_{e,1}(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{e,1}((BC)^*) \cup \sigma_{e,1}((CB)^*))\} = \{\lambda \in \mathbb{C} : \overline{\lambda}^2 \in (\sigma_{e,2}(BC) \cup \sigma_{e,2}(CB))\}$;

(iii) $\sigma_{e,5}(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{e,4}(BC) \cup \sigma_{e,4}(CB))\} \cup \{\lambda \in \mathbb{C} : ind(BC - \lambda^2) + ind(CB - \lambda^2) \neq 0\}$;

(iv) $\sigma_{e,7}(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{e,1}(BC) \cup \sigma_{e,1}(CB))\} \cup \{\lambda \in \mathbb{C} : ind(BC - \lambda^2) + ind(CB - \lambda^2) > 0\}$;

(v) $\sigma_{e,8}(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{e,2}(BC) \cup \sigma_{e,2}(CB))\} \cup \{\lambda \in \mathbb{C} : ind(BC - \lambda^2) + ind(CB - \lambda^2) < 0\}$.

Proof (i) We only prove $\sigma_{e,2}(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{e,2}(BC) \cup \sigma_{e,2}(CB))\}$, the proof of $\sigma_{e,4}(H) = \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{e,4}(BC) \cup \sigma_{e,4}(CB))\}$ is completely similar. If $\lambda \in \triangle_2(H)$, $H - \lambda I$ is a right semi-Fredholm operator, we know that $H + \lambda I$ is also a right semi-Fredholm operator by Theorem 2.3.2, so

$$(H - \lambda I)(H + \lambda I) = \begin{bmatrix} BC - \lambda^2 I & 0 \\ 0 & CB - \lambda^2 I \end{bmatrix}$$

is also right semi-Fredholm operator, that is, $BC - \lambda^2 I$ and $CB - \lambda^2 I$ are right semi-Fredholm operators.

Conversely, if $BC - \lambda^2 I$ and $CB - \lambda^2 I$ are right-semi-Fredholm operators, then

$$\mathscr{R}(H - \lambda I) \supset \mathscr{R}(BC - \lambda^2 I) \oplus \mathscr{R}(CB - \lambda^2 I)$$

by

$$(H - \lambda I)(H + \lambda I) = (H + \lambda I)(H - \lambda I) = \begin{bmatrix} BC - \lambda^2 I & 0 \\ 0 & CB - \lambda^2 I \end{bmatrix}.$$

So $\dim X/\mathscr{R}(H - \lambda I) < \infty$, and hence $\mathscr{R}(H - \lambda I)$ is closed and $\dim \mathscr{R}(H - \lambda I)^{\perp} < \infty$, that is, $H - \lambda I$ is a right-semi-Fredholm operator.

(ii) Let $\lambda \in \sigma_{e,1}(H)$. Then the similarity of H and H^* implies $\lambda \in \sigma_{e,1}(H^*)$, that is, $\overline{\lambda} \in \sigma_{e,2}(H)$. From conclusion (i), we have $\overline{\lambda}^2 \in \sigma_{e,2}(BC) \cup \sigma_{e,2}(CB)$, that is, $\lambda^2 \in \sigma_{e,1}((BC)^*) \cup \sigma_{e,1}((CB)^*)$. Conversely, the proof is completely similar.

(iii) Let $\lambda \notin \sigma_{e,5}(H)$, which together the similarity of H and $-H$ implies $H^2 - \lambda^2 I$ is a Weyl operator, that is, $H^2 - \lambda^2 I$ is a Fredholm operator and $\mathrm{ind}(H^2 - \lambda^2 I) = 0$. Then $\lambda \notin \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{e,4}(BC) \cup \sigma_{e,4}(CB))\} \cup \{\lambda \in \mathbb{C} : \mathrm{ind}(BC - \lambda^2) + \mathrm{ind}(CB - \lambda^2) \neq 0\}$.

Conversely, if $\lambda \notin \{\lambda \in \mathbb{C} : \lambda^2 \in (\sigma_{e,4}(BC) \cup \sigma_{e,4}(CB))\} \cup \{\lambda \in \mathbb{C} : \mathrm{ind}(BC - \lambda^2 I) + \mathrm{ind}(CB - \lambda^2 I) \neq 0\}$, then $H - \lambda I$ and $H + \lambda I$ are Fredholm operators by (i), which combining the similarity of H and $-H$ proves

$$\mathrm{ind}(H - \lambda I) = \mathrm{ind}(H + \lambda I).$$

So, from

$$\mathtt{ind}(H^2 - \lambda^2 I) = \mathtt{ind}(H - \lambda I) + \mathtt{ind}(H + \lambda I),$$

it follows that $\mathtt{ind}(H - \lambda I) = \mathtt{ind}(H + \lambda I) = 0$, that is, $\lambda \notin \sigma_{e,5}(H)$.

By Lemma 2.1.25, the proof of (iv) and (v) is obvious. □

2.4 Spectra of Upper Triangular Infinite-Dimensional Hamiltonian Operators

For the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subseteq X \times X \to X \times X,$$

if $C = 0$, then H is called an upper triangular infinite-dimensional Hamiltonian operator. The natural domain of the upper triangular infinite-dimensional Hamiltonian operator is

$$\mathscr{D}(H) = \mathscr{D}(A) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*)).$$

Starting from this section on, we will explore the basic properties of the point spectrum, residual spectrum, continuous spectrum and approximate point spectrum of upper triangular infinite-dimensional Hamiltonian operators whose domains are

$$\mathscr{D}(H) = \mathscr{D}(A) \times \mathscr{D}(A^*) \text{ (diagonally dominant)}$$

and

$$\mathscr{D}(H) = \mathscr{D}(A) \times \mathscr{D}(B) \text{ (upper dominant)}.$$

The following is a concrete example of the upper triangular infinite-dimensional Hamiltonian operators appearing in the problems from elasticity (see [44]).

Example 2.4.1 The problem of simply supported rectangular plates

$$D\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)^2 w = 0$$

with two opposite sides is considered, where the boundary conditions are

$$w(x, 0) = w(x, 1) = 0, \quad \frac{\partial^2 w}{\partial y^2}(x, 0) = \frac{\partial^2 w}{\partial y^2}(x, 1) = 0.$$

Let $\theta = \frac{\partial w}{\partial x}$, $q = D(\frac{\partial^3 w}{\partial x^3} + \frac{\partial^3 w}{\partial x \partial y^2})$, $m = -D(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2})$, and then the corresponding infinite-dimensional Hamiltonian system can be written as

$$\frac{\partial}{\partial x}\begin{bmatrix} w \\ \theta \\ q \\ m \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{\partial^2}{\partial y^2} & 0 & 0 & -\frac{1}{D} \\ 0 & 0 & 0 & \frac{\partial^2}{\partial y^2} \\ 0 & 0 & -1 & 0 \end{bmatrix}\begin{bmatrix} w \\ \theta \\ q \\ m \end{bmatrix}. \tag{2.11}$$

Let $X = L^2[0, 1]$. Then the infinite-dimensional Hamiltonian operator corresponding to the system is

$$H = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{\mathrm{d}^2}{\mathrm{d}y^2} & 0 & 0 & -\frac{1}{D} \\ 0 & 0 & 0 & \frac{\mathrm{d}^2}{\mathrm{d}y^2} \\ 0 & 0 & -1 & 0 \end{bmatrix},$$

where

$$\mathscr{D}(H) = \left\{ \begin{bmatrix} w \\ \theta \\ q \\ m \end{bmatrix} \in X^4 : \begin{array}{c} w(0) = w(1) = q(0) = q(1) = 0, w, w', m, m' \\ \text{absolutely continuous and } w'', m'' \in X \end{array} \right\}.$$

The diagonally dominant upper triangular infinite-dimensional Hamiltonian operator is symplectic self-adjoint, so it is a closed operator.

Lemma 2.4.2 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant upper triangular infinite-dimensionalHamiltonian operator. Then*

$$H^* = \begin{bmatrix} A^* & 0 \\ B & -A \end{bmatrix},$$

that is, $(JH)^* = JH$, *where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$.

Proof We only need to prove $\mathscr{D}(H^*) \subset \mathscr{D}(A^*) \times \mathscr{D}(A)$. Let $\begin{bmatrix} u & v \end{bmatrix}^T \in \mathscr{D}(H^*)$. Then there exists $\begin{bmatrix} f & g \end{bmatrix}^T \in X \times X$ such that

$$(Ax + By, u) - (A^*y, v) = (x, f) + (y, g)$$

for any $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$. Taking $y = 0$, then

$$(Ax, u) = (x, f)$$

for any $x \in \mathscr{D}(A)$, that is, $u \in \mathscr{D}(A^*)$. Because $\mathscr{D}(A^*) \subset \mathscr{D}(B)$, $u \in \mathscr{D}(B)$. By taking $x = 0$, it follows that

$$(By, u) - (A^*y, v) = (y, g)$$

for any $y \in \mathscr{D}(A^*)$, that is

$$(A^*y, v) = (y, Bu - g),$$

so $v \in \mathscr{D}(A)$. To sum up, $\mathscr{D}(H^*) \subset \mathscr{D}(A^*) \times \mathscr{D}(A)$. □

The upper dominant upper triangular infinite-dimensional Hamiltonian operator is symplectic symmetric, so it is closable, but not necessarily closed.

Example 2.4.3 Consider the upper dominant upper triangular infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} A & AA^* \\ 0 & -A^* \end{bmatrix},$$

where A is a densely defined closed operator with $\mathscr{D}(A) \neq X$, $0 \in \rho(A)$, then $0 \in \rho(AA^*)$ and

$$H = \begin{bmatrix} I & 0 \\ -A^{-1} & I \end{bmatrix} \begin{bmatrix} 0 & AA^* \\ I_{\mathscr{D}(A)} & 0 \end{bmatrix} \begin{bmatrix} I & 0 \\ (A^*)^{-1} & I \end{bmatrix}.$$

Since $\begin{bmatrix} I & 0 \\ -A^{-1} & I \end{bmatrix}$ and $\begin{bmatrix} I & 0 \\ (A^*)^{-1} & I \end{bmatrix}$ are invertible bounded linear operators, and $I_{\mathscr{D}(A)}$ is closable but not closed, H is not closed.

Lemma 2.4.4 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(B) \to X \times X$ *be an upper dominant upper triangular infinite-dimensional Hamiltonian operator. Then* H *is a closed operator if and only if* $A^*(B - \mathrm{i}I)^{-1}A$ *is closed.*

Proof Since $\mathscr{D}(B) \subset \mathscr{D}(A^*)$ and B is a self-adjoint operator, $\mathrm{i} \in \rho(B)$ and $A^*(B - \mathrm{i}I)^{-1}$ is bounded on the whole space,

$$\begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} I & 0 \\ A^*(B - \mathrm{i}I)^{-1} & I \end{bmatrix} \begin{bmatrix} 0 & B - \mathrm{i}I \\ S_1 & 0 \end{bmatrix} \begin{bmatrix} I & 0 \\ (A^*(B - \mathrm{i}I)^{-1})^* & I \end{bmatrix} + \begin{bmatrix} 0 & \mathrm{i}I \\ 0 & 0 \end{bmatrix},$$

where $S_1 = A^*(B - \mathrm{i}I)^{-1}A$. Since $\begin{bmatrix} I & 0 \\ A^*(B - \mathrm{i}I)^{-1} & I \end{bmatrix}$, $\begin{bmatrix} I & 0 \\ (A^*(B - \mathrm{i}I)^{-1})^* & I \end{bmatrix}$ are bounded on the whole space and invertible, $\begin{bmatrix} 0 & \mathrm{i}I \\ 0 & 0 \end{bmatrix}$ is bounded on the whole space, and $B - \mathrm{i}I$ is a closed operator, H is closed if and only if $A^*(B - \mathrm{i}I)^{-1}A$ is a closed operator. □

According to Lemma 2.4.2, the proof of the following Corollary is obvious.

Corollary 2.4.5 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(B) \to X \times X$ *be an upper dominant upper triangular infinite-dimensional Hamiltonian operator. Then* $H^* = \begin{bmatrix} A^* & 0 \\ B & -A \end{bmatrix}$ *if and only if* $(A^*(B - \mathrm{i}I)^{-1}A)^* = A^*(B + \mathrm{i}I)^{-1}A$.

2.4.1 Point Spectra of Upper Triangular Infinite-Dimensional Hamiltonian Operators

In the case of finite-dimension, the eigenvalue set $\sigma_p(H_{2n})$ of the upper triangle Hamiltonian matrix $H_{2n} = \begin{bmatrix} A_{n\times n} & B_{n\times n} \\ 0 & -A_{n\times n}^T \end{bmatrix}$ satisfies

$$\sigma_p(H_{2n}) = \sigma_p(A_{n\times n}) \cup \sigma_p(-A_{n\times n}^T).$$

Hence, the point spectrum of the Hamiltonian matrix is symmetric with respect to the imaginary axis. However, the above conclusion may not hold for the infinite-dimensional case.

Example 2.4.6 Let $X = l^2[1, \infty)$ and define

$$S_l x = (x_2, x_3, \ldots),$$

$$S_r x = (0, x_1, x_2, x_3, \ldots)$$

for any $x = (x_1, x_2, x_3, \ldots) \in l^2$, then it is easy to get by calculation that $\sigma_p(S_l) = \{\lambda : |\lambda| \leq 1\}$ and $\sigma_p(S_r) = \emptyset$, that is, $0 \in \sigma_p(S_l) \cup \sigma_p(-S_r)$. On the other hand, define the upper triangular infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} S_r & I - S_r S_l \\ 0 & -S_l \end{bmatrix},$$

then it is easy to prove $0 \in \rho(H)$. Hence, $\sigma_p(H) \neq \sigma_p(A) \cup \sigma_p(-A^*)$.

Theorem 2.4.7 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator. Then*

$$(\sigma_p(A) \cup \sigma_p(-A^*))\backslash(\sigma_\delta(A) \cap \sigma_p(-A^*)) \subset \sigma_p(H) \subset (\sigma_p(A) \cup \sigma_p(-A^*)),$$

$$\sigma_p(H) = (\sigma_p(A) \cup \sigma_p(-A^*))\backslash W,$$

where W *is the subset of* $\sigma_\delta(A) \cap \sigma_p(-A^*)$.

Proof The proof of $\sigma_p(H) \subset (\sigma_p(A) \cup \sigma_p(-A^*))$ is obvious. Next we prove $(\sigma_p(A) \cup \sigma_p(-A^*))\backslash(\sigma_\delta(A) \cap \sigma_p(-A^*)) \subset \sigma_p(H)$. Let $\lambda \in (\sigma_p(A) \cup \sigma_p(-A^*))\backslash(\sigma_\delta(A) \cap \sigma_p(-A^*))$. Then $\lambda \in (\sigma_p(A) \cup \sigma_p(-A^*))$ and $\lambda \notin (\sigma_\delta(A) \cap \sigma_p(-A^*))$. If $\lambda \in \sigma_p(A)$ and $\lambda \notin (\sigma_\delta(A) \cap \sigma_p(-A^*))$, $\lambda \in \sigma_p(H)$ is obvious. If $\lambda \in \sigma_p(-A^*)$ and $\lambda \notin (\sigma_\delta(A) \cap \sigma_p(-A^*))$, then it is easy to get $\lambda \in (\sigma_p(-A^*) \cap (\rho(A) \cup \sigma_{p,1}(A)))$. If $\lambda \in (\sigma_p(-A^*) \cap \sigma_{p,1}(A))$, then $\lambda \in \sigma_p(H)$ is obvious. If $\lambda \in (\sigma_p(-A^*) \cap \rho(A))$, noting that

$$\begin{bmatrix} A-\lambda I & B \\ 0 & -A^*-\lambda I \end{bmatrix} = \begin{bmatrix} A-\lambda I & 0 \\ 0 & -A^*-\lambda I \end{bmatrix} \begin{bmatrix} I & \overline{(A-\lambda I)^{-1}B} \\ 0 & I \end{bmatrix},$$

we have $\lambda \in \sigma_p(H)$. Because

$$\sigma_p(H) = (\sigma_p(A) \cup \sigma_p(-A^*)) \backslash ((\sigma_p(A) \cup \sigma_p(-A^*)) \backslash \sigma_p(H))$$

and $\sigma_p(A) \cup \sigma_p(-A^*)) \backslash \sigma_p(H) \subset (\sigma_\delta(A) \cap \sigma_p(-A^*))$, there exists $W \subset (\sigma_\delta(A) \cap \sigma_p(-A^*))$ such that

$$\sigma_p(H) = (\sigma_p(A) \cup \sigma_p(-A^*)) \backslash W.$$

The proof is completed. □

Obviously, if $\sigma(A) \cap \sigma(-A^*) = \emptyset$, then

$$\sigma_p(H) = \sigma_p(A) \cup \sigma_p(-A^*). \tag{2.12}$$

Hence, the question of when expression (2.12) holds is particularly important.

Theorem 2.4.8 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant upper triangular infinite-dimensional Hamiltonian operator. Then*

$$\sigma_p(H) = \sigma_p(A) \cup \sigma_p(-A^*)$$

holds if and only if one of the following propositions holds:

(i) $\sigma_p(-A^*) \subset \sigma_p(A)$;
(ii) $\mathscr{N}(B) \cap \mathscr{N}(-A^* - \lambda I) \neq \{0\}$ *for any* $\lambda \in \sigma_p(-A^*) \setminus \sigma_p(A)$;
(iii) $B\mathscr{N}(-A^* - \lambda I) \cap \mathscr{R}(A - \lambda I) \neq \{0\}$ *for any* $\lambda \in \sigma_p(-A^*) \setminus \sigma_p(A)$.

Proof If $\sigma_p(-A^*) \subset \sigma_p(A)$, then considering $\sigma_p(A) \subset \sigma_p(H)$ we have

$$\sigma_p(H) = \sigma_p(A) \cup \sigma_p(-A^*)$$

holds. If $\mathscr{N}(B) \cap \mathscr{N}(-A^* - \lambda I) \neq \{0\}$ or $B\mathscr{N}(-A^* - \lambda I) \cap \mathscr{R}(A - \lambda I) \neq \{0\}$ for any $\lambda \in \sigma_p(-A^*) \setminus \sigma_p(A)$, then from

$$\sigma_p(-A^*) = (\sigma_p(-A^*) \setminus \sigma_p(A)) \cup (\sigma_p(-A^*) \cap \sigma_p(A))$$

we obtain $\sigma_p(-A^*) \subset \sigma_p(H)$. Hence

$$\sigma_p(H) = \sigma_p(A) \cup \sigma_p(-A^*).$$

Conversely, if

$$\sigma_p(H) = \sigma_p(A) \cup \sigma_p(-A^*)$$

and let $\sigma_p(-A^*) \subset \sigma_p(A)$ not hold, then $\sigma_p(-A^*) \setminus \sigma_p(A) \neq \emptyset$ and $\lambda \in \sigma_p(H)$ for any $\lambda \in \sigma_p(-A^*) \setminus \sigma_p(A)$, that is, there exists $0 \neq \begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$ such that

$$\begin{cases} Ax + By = \lambda x, \\ -A^* y = \lambda y. \end{cases} \tag{2.13}$$

Because $\lambda \notin \sigma_p(A)$, $y \neq 0$. If $x = 0$, then $By = 0$, so $\mathscr{N}(B) \cap \mathscr{N}(-A^* - \lambda I) \neq \{0\}$. If $x \neq 0$, then $By = -(A - \lambda I)x$, that is, $B\mathscr{N}(-A^* - \lambda I) \cap \mathscr{R}(A - \lambda I) \neq \{0\}$. The proof is completed. □

The necessary and sufficient condition will be given for a diagonally dominant upper triangular infinite-dimensional Hamiltonian operator to be injective.

Theorem 2.4.9 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant upper triangular infinite-dimensional Hamiltonian operator, then* H *is injective if and only if the following conditions are satisfied:*

(i) A *is injective;*
(ii) $B|_{\mathscr{N}(A^*)}$ *is injective;*
(iii) $B\mathscr{N}(A^*) \cap \mathscr{R}(A) = \{0\}$.

Proof If the upper triangular infinite-dimensional Hamiltonian operator H is injective, A being injective is obvious. Assume $B|_{\mathscr{N}(A^*)}$ is not injective, then there exists $0 \neq y \in \mathscr{N}(A^*)$ such that

$$By = 0.$$

At this moment, $0 \neq \begin{bmatrix} 0 & y \end{bmatrix}^T \in \mathscr{D}(H)$ and

$$\begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} \begin{bmatrix} 0 \\ y \end{bmatrix} = 0,$$

which is contradictory to H being injective. In the same way, assume $B\mathscr{N}(A^*) \cap \mathscr{R}(A) \neq \{0\}$, then there exists $0 \neq y \in \mathscr{N}(A^*)$ and $x \in \mathscr{D}(A)$ such that

$$By = Ax.$$

Now, $0 \neq \begin{bmatrix} -x & y \end{bmatrix}^T \in \mathscr{D}(H)$ and

$$\begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} \begin{bmatrix} -x \\ y \end{bmatrix} = \begin{bmatrix} -Ax + By \\ -A^* y \end{bmatrix} = 0,$$

which is contradictory to H being injective.

Conversely, let $\begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$, then it is easy to prove that $x = 0$ and $y = 0$. In fact, assume $x \neq 0$, then by A being injective we know $Ax \neq 0$ and by

$$Ax = -By \tag{2.14}$$

we know $y \neq 0$, that is, $y \in \mathscr{N}(A^*)\backslash\{0\}$. Hence, $B(\mathscr{N}(A^*)) \cap \mathscr{R}(A) \neq \{0\}$ by (2.14), contradiction is derived. Similarly we can prove $y = 0$. □

According to the conclusions above, the following proposition is obvious.

Corollary 2.4.10 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant upper triangular infinite-dimensional Hamiltonian operator, then* $0 \in \sigma_p(H)$ *if and only if one of the following conditions is satisfied:*

(i) A *is not injective;*
(ii) $B|_{\mathscr{N}(A^*)}$ *is not injective;*
(iii) $B\mathscr{N}(A^*) \cap \mathscr{R}(A) \neq \{0\}$.

According to the Schur factorization

$$\begin{bmatrix} A - \lambda I & B \\ 0 & -A^* - \lambda I \end{bmatrix} = \begin{bmatrix} I & -B(A^* - \lambda I)^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} A - \lambda I & 0 \\ 0 & -A^* - \lambda I \end{bmatrix}$$

when $\lambda \in \rho(-A^*)$ or

$$\begin{bmatrix} A - \lambda I & B \\ 0 & -A^* - \lambda I \end{bmatrix} = \begin{bmatrix} A - \lambda I & 0 \\ 0 & -A^* - \lambda I \end{bmatrix} \begin{bmatrix} I & \overline{(A - \lambda I)^{-1} B} \\ 0 & I \end{bmatrix}$$

when $\lambda \in \rho(A)$, the following conclusion is obvious.

Corollary 2.4.11 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant upper triangular infinite-dimensional Hamiltonian operator. Then*

(i) $\sigma_p(H) \cap \rho(-A^*) = \sigma_p(A) \cap \rho(-A^*)$;
(ii) $\sigma_p(H) \cap \rho(A) = \sigma_p(-A^*) \cap \rho(A)$;
(iii) $\sigma_{p,i}(H) \cap \rho(A) = \sigma_{p,i}(-A^*) \cap \rho(A), i = 1, 2, 3, 4$;
(iv) $\sigma_{p,i}(H) \cap \rho(-A^*) = \sigma_{p,i}(A) \cap \rho(-A^*), i = 1, 2, 3, 4$.

For the case of upper dominant, the equation

$$\sigma_p(H) = \sigma_p(A) \cup \sigma_p(-A^*)$$

is still valid under some conditions, the proof is completely similar to diagonal dominant case.

Theorem 2.4.12 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(B) \to X \times X$ *be the an upper dominant upper triangular infinite-dimensional Hamiltonian operator. Then*

$$\sigma_p(H) = \sigma_p(A) \cup \sigma_p(-A^*)$$

holds if and only one of the following propositions is satisfied:

(i) $\sigma_p(-A^*|_{\mathscr{D}(B)}) \subset \sigma_p(A)$;
(ii) $\mathscr{N}(B) \cap \mathscr{N}(-A^*|_{\mathscr{D}(B)} - \lambda I) \neq \{0\}$ *for any* $\lambda \in \sigma_p(-A^*|_{\mathscr{D}(B)}) \setminus \sigma_p(A)$;
(iii) $B\mathscr{N}(-A^*|_{\mathscr{D}(B)} - \lambda I) \cap \mathscr{R}(A - \lambda I) \neq \{0\}$ *for any* $\lambda \in \sigma_p(-A^*|_{\mathscr{D}(B)}) \setminus \sigma_p(A)$.

For the upper dominant upper triangular infinite-dimensional Hamiltonian operator, it can be transformed into the case of skew-diagonal operator under some conditions by using the particularity of its structure.

Theorem 2.4.13 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(B) \to X \times X$ *be an upper dominant upper triangular infinite-dimensional Hamiltonian operator with* B *invertible, then* $\lambda \in \sigma_p(H)$ *if and only if* $0 \in \sigma_p((A^* + \lambda I)B^{-1}(A - \lambda I))$.

Proof If B is invertible,

$$\begin{bmatrix} A - \lambda I & B \\ 0 & -A^* - \lambda I \end{bmatrix} = \begin{bmatrix} I & 0 \\ -(A^* + \lambda I)B^{-1} & I \end{bmatrix} \begin{bmatrix} 0 & B \\ S(\lambda) & 0 \end{bmatrix} \begin{bmatrix} I & 0 \\ B^{-1}(A - \lambda I) & I \end{bmatrix}, \tag{2.15}$$

where $S(\lambda) = (A^* + \lambda I)B^{-1}(A - \lambda I)$. So, $\lambda \in \sigma_p(H)$ if and only if $0 \in \sigma_p((A^* + \lambda I)B^{-1}(A - \lambda I))$. □

2.4.2 Residual Spectra of Upper Triangular Infinite-Dimensional Hamiltonian Operators

The following conclusion shows that the residual spectrum of the diagonally dominant upper triangular infinite-dimensional Hamiltonian operator has no intersection with the imaginary axis.

Theorem 2.4.14 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant upper triangular infinite dimension Hamiltonian operator. Then* $\sigma_r(H) \cap i\mathbb{R} = \emptyset$.

Proof Assume $\lambda \in \sigma_r(H) \cap \mathrm{i}\mathscr{R}$, then $\overline{\lambda} = -\lambda$and $\overline{\lambda} \in \sigma_p(H^*)$. By Lemma 2.4.2 we know

$$H^* = JHJ,$$

where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. Since $J^2 = -I$ we can obtain $\lambda \in \sigma_p(H)$, which is a contradiction. Hence, $\sigma_r(H) \cap \mathrm{i}\mathscr{R} = \emptyset$. □

The necessary and sufficient condition for non-density of the range $\mathscr{R}(H - \lambda I)$ of the diagonally dominant upper triangular operator matrix will be given as follows.

Theorem 2.4.15 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant upper triangular infinite-dimensional Hamiltonian operator, then* $\overline{\mathscr{R}(H - \lambda I)} \neq X \times X$ *if and only if one of the following conditions its satisfied:*

(i) $A + \overline{\lambda} I$ *is not injective;*
(ii) $B|_{\mathscr{N}(A^* - \overline{\lambda} I)}$ *is not injective;*
(iii) $B\mathscr{N}(A^* - \overline{\lambda} I) \cap \mathscr{R}(A + \overline{\lambda} I) \neq \{0\}$.

Proof Because $\overline{\mathscr{R}(H - \lambda I)} \neq X \times X$ if and only if $0 \in \sigma_p(H^* - \overline{\lambda} I)$. The equation

$$H^* - \overline{\lambda} I = \begin{bmatrix} A^* - \overline{\lambda} I & 0 \\ B & -A - \overline{\lambda} I \end{bmatrix} = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} -A - \overline{\lambda} I & B \\ 0 & A^* - \overline{\lambda} I \end{bmatrix} \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$$

can be obtained by Lemma 2.4.2. Then the conclusion can be proved from Corollary 2.4.10. □

A necessary and sufficient condition for the diagonally dominant upper triangular operator matrix $H - \lambda I$ to be injective can be given as follows.

Theorem 2.4.16 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant upper triangular infinite-dimensional Hamiltonian operator. Then* $H - \lambda I$ *is injective if and only if the following conditions are satisfied:*

(i) $A - \lambda I$ *is injective;*
(ii) $B|_{\mathscr{N}(A^* + \lambda I)}$ *is injective;*
(iii) $B\mathscr{N}(A^* + \lambda I) \cap \mathscr{R}(A - \lambda I) = \{0\}$.

Proof The conclusion is obvious according to Proposition 2.4.10. □

According to Theorems 2.4.15 and 2.4.16, it is easy to get the following conclusion.

Theorem 2.4.17 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant upper triangular infinite-dimensional Hamiltonian operator, then* $\lambda \in \sigma_r(H)$ *if and only if one of the following conditions is satisfied:*

(i) $A + \overline{\lambda} I$ *is not injective,* $A - \lambda I$ *is injective,* $B|_{\mathscr{N}(A^* + \lambda I)}$ *is injective, and* $B\mathscr{N}(A^* + \lambda I) \cap \mathscr{R}(A - \lambda I) = \{0\}$;
(ii) $B|_{\mathscr{N}(A^* - \overline{\lambda} I)}$ *is injective,* $A - \lambda I$ *is injective,* $B|_{\mathscr{N}(A^* + \lambda I)}$ *is injective, and* $B\mathscr{N}(A^* + \lambda I) \cap \mathscr{R}(A - \lambda I) = \{0\}$;
(iii) $B\mathscr{N}(A^* - \overline{\lambda} I) \cap \mathscr{R}(A + \overline{\lambda} I) \neq \{0\}$, $A - \lambda I$ *is injective,* $B|_{\mathscr{N}(A^* + \lambda I)}$ *is injective, and* $B\mathscr{N}(A^* + \lambda I) \cap \mathscr{R}(A - \lambda I) = \{0\}$.

The following conclusion is obvious.

Corollary 2.4.18 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant upper triangular infinite-dimensional Hamiltonian operator. Then*

(i) $\sigma_r(H) \cap \rho(-A^*) = \sigma_r(A) \cap \rho(-A^*)$;
(ii) $\sigma_r(H) \cap \rho(A) = \sigma_r(-A^*) \cap \rho(A)$;
(iii) $\sigma_{r,i}(H) \cap \rho(A) = \sigma_{r,i}(-A^*) \cap \rho(A), i = 1, 2$;
(iv) $\sigma_{r,i}(H) \cap \rho(-A^*) = \sigma_r(A, i) \cap \rho(-A^*), i = 1, 2$.

The residual spectrum of the upper dominant upper triangular infinite-dimensional Hamiltonian operator will be described as follows.

Theorem 2.4.19 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(B) \to X \times X$ *be an upper dominant upper triangular infinite-dimensional Hamiltonian operator with B invertible. Then* $\lambda \in \sigma_r(H)$ *if and only if* $0 \in \sigma_r((A^* + \lambda I)B^{-1}(A - \lambda I))$.

Proof Since

$$\begin{bmatrix} A - \lambda I & B \\ 0 & -A^* - \lambda I \end{bmatrix} = \begin{bmatrix} I & 0 \\ -(A^* + \lambda I)B^{-1} & I \end{bmatrix} \begin{bmatrix} 0 & B \\ S(\lambda) & 0 \end{bmatrix} \begin{bmatrix} I & 0 \\ B^{-1}(A - \lambda I) & I \end{bmatrix}$$

where $S(\lambda) = (A^* + \lambda I)B^{-1}(A - \lambda I)$, the conclusion is obvious. □

2.4.3 Continuous Spectra of Upper Triangular Infinite-Dimensional Hamiltonian Operators

In this section, we will discuss some properties of the continuous spectrum of the upper triangular infinite-dimensional Hamiltonian operator. Let

$$H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

be an upper triangular infinite-dimensional Hamiltonian operator. Then it is easy to prove

$$(\sigma_c(A) \cap \sigma_c(-A^*)) \cup (\rho(A) \cap \sigma_c(-A^*)) \cup (\sigma_c(A) \cap \rho(-A^*)) \subset \sigma_c(H).$$

Therefore, one question that naturally arises is when equals sign in the above-mentioned expression holds?

Theorem 2.4.20 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator. Then the expression*

$$\sigma_c(H) = (\sigma_c(A) \cap \sigma_c(-A^*)) \cup (\rho(A) \cap \sigma_c(-A^*)) \cup (\sigma_c(A) \cap \rho(-A^*))$$

holds if one of the following conditions is satisfied:

(i) There exists a bounded linear operator Z defined on the whole space such that $B = AZ + ZA^*$;
(ii) $\sigma(A) \cap \sigma(-A^*) = \emptyset$;
(iii) $(\sigma(A)\backslash\sigma_p(A)) \cap (\sigma(-A^*)\backslash\sigma_{com}(-A^*)) = \emptyset$.

Proof (i) If there exists a bounded linear operator Z defined on the whole space such that $B = AZ + ZA^*$, from

$$\mathscr{D}(B) = \{y : Zy \in \mathscr{D}(A), y \in \mathscr{D}(A^*)\},$$

it is easy to prove that

$$\begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} I & -Z \\ 0 & I \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} \begin{bmatrix} I & Z \\ 0 & I \end{bmatrix}.$$

Therefore, according to the properties of the similarity operator, the conclusion is immediately proved.

(ii) For

$$(\sigma_c(A) \cap \sigma_c(-A^*)) \cup (\rho(A) \cap \sigma_c(-A^*)) \cup (\sigma_c(A) \cap \rho(-A^*)) \subset \sigma_c(H),$$

let $\lambda \in \sigma_c(H)$ and $\lambda \notin (\sigma_c(A) \cap \sigma_c(-A^*)) \cup (\rho(A) \cap \sigma_c(-A^*)) \cup (\sigma_c(A) \cap \rho(-A^*))$. Then $\lambda \in (\sigma(A) \cap \sigma(-A^*))$. In fact, assume $\lambda \notin (\sigma(A) \cap \sigma(-A^*))$, then $\lambda \in (\rho(A) \cup \rho(-A^*))$. If $\lambda \in \rho(A)$,

$$H - \lambda I = \begin{bmatrix} I & B(A - \lambda I)^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} A - \lambda I & 0 \\ 0 & -A^* - \lambda I \end{bmatrix},$$

therefore $\lambda \in \sigma_c(-A^*)$, which is a contradiction. Similarly, a contradiction can be obtained if $\lambda \in \rho(-A^*)$. Thus, $\lambda \in (\sigma(A) \cap \sigma(-A^*))$. Therefore, if $\sigma(A) \cap \sigma(-A^*) = \emptyset$, the equation

$$\sigma_c(H) = (\sigma_c(A) \cap \sigma_c(-A^*)) \cup (\rho(A) \cap \sigma_c(-A^*)) \cup (\sigma_c(A) \cap \rho(-A^*))$$

holds.

(iii) Let $\lambda \in \sigma_c(H)$ and $\lambda \notin (\sigma_c(A) \cap \sigma_c(-A^*)) \cup (\rho(A) \cap \sigma_c(-A^*)) \cup (\sigma_c(A) \cap \rho(-A^*))$. Then $\lambda \in (\sigma(A)\backslash\sigma_p(A)) \cap (\sigma(-A^*)\backslash\sigma_{com}(-A^*))$, so if $(\sigma(A)\backslash\sigma_p(A)) \cap (\sigma(-A^*)\backslash\sigma_{com}(-A^*)) = \emptyset$, the equation

$$\sigma_c(H) = (\sigma_c(A) \cap \sigma_c(-A^*)) \cup (\rho(A) \cap \sigma_c(-A^*)) \cup (\sigma_c(A) \cap \rho(-A^*))$$

holds. The proof is completed. □

In the following, we will discuss when the continuous spectrum of the general upper triangular infinite-dimensional Hamiltonian operator is empty.

Lemma 2.4.21 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times (\mathscr{D}(A^*) \cap \mathscr{D}(B)) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator, and* $\sigma_p(H)^c \subset \rho(A) \cap \rho(A_1)$. *Then* $1 \in \rho(K_\lambda)$ *for any* $\lambda \in \sigma_p(H)^c$, *where* $K_\lambda = \widehat{B} A_\lambda^{-1}$, $A_\lambda = \begin{bmatrix} \lambda I - A & 0 \\ 0 & \lambda I - A_1 \end{bmatrix}$, $\widehat{B} = \begin{bmatrix} 0 & B \\ 0 & 0 \end{bmatrix}$, A_1 *is the restriction of* $-A^*$ *on* $\mathscr{D}(B) \cap \mathscr{D}(A^*)$, *and* $\sigma_p(H)^c$ *represents the complement of* $\sigma_p(H)$.

Proof For any $\lambda \in \sigma_p(H)^c$, we can know that $\lambda \in \rho(A) \cap \rho(A_1)$, therefore $\lambda I - A$, $\lambda I - A_1$ are surjective. Because A_1 is the restriction of $-A^*$ on $\mathscr{D}(B) \cap \mathscr{D}(A^*)$, $\mathscr{D}(A_1) \subset \mathscr{D}(B)$, and then $\mathscr{D}(I - K_\lambda) = X \times X$. If $(I - K_\lambda)u = 0$ for $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in X \times X$, that is,

$$\begin{bmatrix} I & -B(\lambda I - A_1)^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tag{2.16}$$

which means that $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, that is, $u = 0$, so $I - K_\lambda$ is injective. In the following, let us consider whether $I - K_\lambda$ is surjective or not, that is, for any $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in X \times X$, whether exists $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in X \times X$ such that $(I - K_\lambda)u = v$, that is,

$$\begin{bmatrix} I & -B(\lambda I - A_1)^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}. \tag{2.17}$$

The equality (2.17) is equivalent to $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} v_1 + B(\lambda I - A_1)^{-1} v_2 \\ v_2 \end{bmatrix}$. Since we has proved $D(I - K_\lambda) = X \times X, u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} v_1 + B(\lambda I - A_1)^{-1} v_2 \\ v_2 \end{bmatrix} \in D(I - K_\lambda)$, and thus $I - K_\lambda$ is surjective. Therefore, we proves $1 \in \rho(K_\lambda)$. □

Lemma 2.4.22 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times (\mathscr{D}(A^*) \cap \mathscr{D}(B)) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator, and* $\sigma_p(H)^c \subset \rho(A) \cap \rho(A_1)$. *Then* $\sigma_p(H)^c \subset \rho(H)$, *where* A_1 *is the restriction of* $-A^*$ *on* $\mathscr{D}(B) \cap \mathscr{D}(A^*)$.

Proof For any $\lambda \in \sigma_p(H)^c$, $\lambda I - H$ is injective by the definition of the point spectrum. Now let us prove that $\lambda I - H$ is surjective. For any $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in X \times X$, we need to find $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in \mathscr{D}(H)$ such that $(\lambda I - H)u = v$, that is,

$$\begin{bmatrix} \lambda I - A & -B \\ 0 & \lambda I + A^* \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}. \tag{2.18}$$

Because $\mathscr{D}(H) = \mathscr{D}(A) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*))$, Eq. (2.18) is equivalent to

$$\begin{bmatrix} \lambda I - A & -B \\ 0 & \lambda I - A_1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}. \tag{2.19}$$

By writing (2.19) as

$$\begin{bmatrix} \lambda I - A & 0 \\ 0 & \lambda I - A_1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} - \begin{bmatrix} 0 & B \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \tag{2.20}$$

(2.20) can be changed into

$$A_\lambda u - \overline{B} u = v \tag{2.21}$$

by Lemma 2.4.21. From the assumption, we know that $\lambda \in \rho(A) \cap \rho(A_1)$. It means $0 \in \rho(A_\lambda)$, that is, A_λ has bounded inverse on $X \times X$, and hence letting A_λ^{-1} act on both sides of (2.21), we can get

$$u - A_\lambda^{-1} \overline{B} u = A_\lambda^{-1} v. \tag{2.22}$$

Let $w = \overline{B} u$, $K_\lambda = \overline{B} A_\lambda^{-1}$, and $\overline{B}$ be used to act on both ends of (2.22), then

$$w - K_\lambda w = K_\lambda v. \tag{2.23}$$

From $\lambda \in \sigma_p(H)^c$ and Lemma 2.4.21, it follows that $1 \in \rho(K_\lambda)$, and thus from (2.23) we can get $w = (I - K_\lambda)^{-1} K_\lambda v$, which together with (2.22) implies that

$$u = A_\lambda^{-1} w + A_\lambda^{-1} v = A_\lambda^{-1} [(I - K_\lambda)^{-1} K_\lambda + I] v.$$

Therefore, for any $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in X \times X$, there exists $u = A_\lambda^{-1} [(I - K_\lambda)^{-1} K_\lambda + I] v$ such that $u \in \mathscr{D}(H)$ and satisfies (2.18), that is, $\lambda I - H$ is surjective. The above proved that if $\lambda \in \sigma_p(H)^c$, $\lambda I - H$ is injective and surjective, that is, $\lambda \in \rho(H)$. Therefore, $\sigma_p(H)^c \subset \rho(H)$. □

Theorem 2.4.23 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times (\mathscr{D}(A^*) \cap \mathscr{D}(B)) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator with* $\sigma_p(H)^c \subset \rho(A) \cap \rho(A_1)$. *Then* $\sigma_c(H) = \emptyset$, *where* A_1 *is the restriction of* $-A^*$ *on* $\mathscr{D}(B) \cap \mathscr{D}(A^*)$.

Proof For any $\lambda \in \mathbb{C}$, let us discuss two cases as follows: if $\lambda \in \sigma_p(H)$, $\lambda \notin \sigma_c(H)$ can be obtained by the definition of spectrum; if $\lambda \in \sigma_p(H)^c$, we know $\sigma_p(H)^c \subset \rho(H)$ by Lemma 2.4.22, so $\lambda \notin \sigma_c(H)$. Thus $\lambda \notin \sigma_c(H)$ for any $\lambda \in \mathbb{C}$, that is $\sigma_c(H) = \emptyset$. □

Theorem 2.4.24 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times (\mathscr{D}(A^*) \cap \mathscr{D}(B)) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator with* $\sigma_r(H) = \sigma_c(H) = \emptyset$ *and* $A_1 = A$. *Then* $\sigma_p(H)^c \subset \rho(A)$, *where* A_1 *is the restriction of* $-A^*$ *on* $\mathscr{D}(B) \cap \mathscr{D}(A^*)$.

Proof Since $\sigma_r(H) = \sigma_c(H) = \emptyset$, $\lambda \in \sigma_p(H)^c$ implies $\lambda \in \rho(H)$, so $\lambda I - H$ is both injective and surjective. By $\lambda I - H$ being injective, we know that

$$\begin{bmatrix} \lambda I - A & -B \\ 0 & \lambda I + A^* \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.24}$$

has only zero solution for any $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in \mathscr{D}(H)$. Since $\mathscr{D}(H) = \mathscr{D}(A) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*))$, (2.24) is equivalent to

$$\begin{bmatrix} \lambda I - A & -B \\ 0 & \lambda I - A_1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

that is,

$$\begin{cases} (\lambda I - A)u_1 - Bu_2 = 0, \\ (\lambda I - A_1)u_2 = 0. \end{cases}$$

Hence, for $u_1 \in \mathscr{D}(A)$, $u_2 \in \mathscr{D}(A^*) \cap \mathscr{D}(B)$, the equations have only zero solution. Let $u_2 = 0$ for any $u_1 \in \mathscr{D}(A)$, substitute them into equations and get

$$\begin{cases} (\lambda I - A)u_1 = 0, \\ 0 = 0. \end{cases}$$

It is obvious that $(\lambda I - A)u_1 = 0$ have only zero solution for any $u_1 \in \mathscr{D}(A)$, that is, $\lambda I - A$ is injective. Since $\lambda I - H$ is surjective , there exists $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in \mathscr{D}(H)$ such that

$$\begin{cases} (\lambda I - A)u_1 - Bu_2 = v_1, \\ (\lambda I - A_1)u_2 = v_2, \end{cases}$$

for any $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in X \times X$. Thus $\lambda I - A_1$ is surjective. Taking note of the known condition $A_1 = A$, we can get $\lambda I - A$ is surjective. So $\lambda I - A$ is both injective and surjective, and therefore $\lambda \in \rho(A)$. □

The following will construct a specific example to illustrate the effectiveness of the results.

Example 2.4.25 Suppose $X = L^2(0, 1)$, and A, B are both second-order ordinary differential operator $A = B = \frac{\mathrm{d}^2}{\mathrm{d}x^2}$, whose domains are

$$\mathscr{D}(A) = \mathscr{D}(B) = \left\{u_1 \in X : u_1, u_1' \text{ are absolute continuous}, u_1'' \in X, u_1(0) = u_1(1) = 0\right\},$$

then

$$H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix}$$

is an infinite-dimensional Hamiltonian operator and its continuous spectrum $\sigma_c(H) = \emptyset$. In fact, under the given conditions, A, B are both Sturm-Liouville operators and are self-adjoint. Thus

$$H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} \frac{\mathrm{d}^2}{\mathrm{d}x^2} & \frac{\mathrm{d}^2}{\mathrm{d}x^2} \\ 0 & -\frac{\mathrm{d}^2}{\mathrm{d}x^2} \end{bmatrix}$$

is an infinite-dimensional Hamiltonian operator. By solving the following equations

$$\begin{cases} \lambda u_1 - \frac{\mathrm{d}^2 u_1}{\mathrm{d}x^2} - \frac{\mathrm{d}^2 u_2}{\mathrm{d}x^2} = 0, & u_1(0) = u_1(1) = 0, \\ \lambda u_2 + \frac{\mathrm{d}^2 u_2}{\mathrm{d}x^2} = 0, & u_2(0) = u_2(1) = 0, \end{cases}$$

the point spectrum $\sigma_p(H) = \left\{ \lambda \in \mathbb{C} : \lambda = \pm k^2\pi^2, k = 1, 2, \dots \right\}$ of H and $\sigma_c(H) = \emptyset$ are obtained.

Next we try to explain $\sigma_c(H) = \emptyset$ by the applying Theorem 2.4.23. Through calculation we can get

$$\sigma(A) = \sigma_p(A) = \left\{ \lambda \in \mathbb{C} : \lambda = -k^2\pi^2, k = 1, 2, \dots \right\},$$

that is, the spectrum of A contains the point spectrum only. Note that in this example $\mathscr{D}(B) = \mathscr{D}(A^*) = \mathscr{D}(A)$, so $A_1 = -A^* = -A$, and thus similarly we know the spectrum of A_1 contains only the point spectrum and

$$\sigma(A_1) = \sigma_p(A_1) = \left\{ \lambda \in \mathbb{C} : \lambda = k^2\pi^2, k = 1, 2, \dots \right\}.$$

Therefore, it follows from $\lambda \in \sigma_p(H)^c$ that $\lambda \notin \sigma_p(A)$ and $\lambda \notin \sigma_p(A_1)$, so $\lambda \in \rho(A) \cap \rho(A_1)$. In summary the conditions of Theorem 2.4.23 are satisfied, so $\sigma_c(H) = \emptyset$. In this way, the results obtained by a direct calculation are consistent with the results derived by applying Theorem 2.4.23.

2.4.4 *Spectral Equalities of Upper Triangular Infinite-Dimensional Hamiltonian Operators*

In the previous sections, we discuss the point spectrum, residual spectrum and continuous spectrum of the upper triangular infinite-dimensional Hamiltonian operator. In this section, we will discuss when the spectrum of the upper triangular infinite-dimensional Hamiltonian operator satisfies the spectral equality.

$$\sigma(H) = \sigma(A) \cup \sigma(-A^*) \tag{2.25}$$

Lemma 2.4.26 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator. Then* $0 \in \rho(H)$ *if and only if the following conditions are satisfied:*

(i) A *is bounded below;*
(ii) $\mathscr{N}(B) \cap \mathscr{N}(-A^*) = \{0\}$;
(iii) $\mathscr{R}(A) \oplus B\mathscr{N}(-A^*) = X$.

Proof Since H is a closed operator, if $0 \in \rho(H)$, H is bounded below, that is, there exists $M > 0$ such that

$$\|Hu\| \geq M\|u\|$$

for any $u = \begin{bmatrix} f & g \end{bmatrix}^T \in \mathscr{D}(H)$. Obviously, A is bounded below. Because H is injective, the proof of $\mathscr{N}(B) \cap \mathscr{N}(-A^*) = \{0\}$ is trivial. In order to prove $B\mathscr{N}(-A^*) \oplus \mathscr{R}(A) = X$, let $y_0 \in B\mathscr{N}(-A^*) \cap \mathscr{R}(A)$, then there exist $x_1 \in D(A)$, $x_2 \in N(A^*)$ such that

$$Ax_1 = Bx_2 = y_0,$$

that is,

$$\begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} \begin{bmatrix} -x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

so $x_1 = x_2 = 0$. On the other hand, since H is injective, we can obtain $B\mathscr{N}(-A^*) + \mathscr{R}(A) = X$, therefore $B\mathscr{N}(-A^*) \oplus \mathscr{R}(A) = X$.

Conversely, to prove $0 \in \rho(H)$, we only need to show that H is a bijection. From (i) and (ii), it is obvious that H is injective. For any $\begin{bmatrix} f & g \end{bmatrix}^T \in X \times X$, from A being bounded below we can get that A^* is surjective, that is, there exists $x_3 \in \mathscr{D}(A^*)$ such that

$$-A^*x_3 = g.$$

Because $B\mathscr{N}(-A^*) \oplus \mathscr{R}(A) = X$, there exists $x_1 \in \mathscr{D}(A)$, $x_2 \in N(-A^*)$ such that

$$Ax_1 + Bx_2 = f - Bx_3.$$

Therefore,

$$\begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 + x_3 \end{bmatrix} = \begin{bmatrix} f \\ g \end{bmatrix},$$

that is, H is surjective. The proof is complected. □

Theorem 2.4.27 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator. Then* $\sigma(H) = \sigma(A) \cup \sigma(-A^*)$ *holds if and only if*

$$(\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*)) = \emptyset$$

or one of the following conditions is satisfied for any $\lambda \in (\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*))$*:*

(i) $\mathscr{N}(B) \cap \mathscr{N}(-A^* - \lambda I) \neq \{0\}$*;*
(ii) $B\mathscr{N}(-A^* - \lambda I) \cap \mathscr{R}(A - \lambda I) \neq \{0\}$*;*
(iii) $B\mathscr{N}(-A^* - \lambda I) + \mathscr{R}(A - \lambda I) \neq X$.

Proof If $\sigma(H) = \sigma(A) \cup \sigma(-A^*)$, suppose $(\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*)) \neq \emptyset$. For any $\lambda \in (\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*))$, assume it doesn't satisfy any condition of (i), (ii) and (iii), then $\lambda \in \rho(H)$ by Lemma 2.4.26, and then considering

$$\rho(H) = \rho(A) \cap \rho(-A^*)$$

we have $\lambda \in \rho(A) \cap \rho(-A^*)$, which contradicts $\lambda \in (\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*))$. So, for any $\lambda \in (\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*))$, one of conditions (i), (ii)and (iii) must be satisfied.

Conversely, when $(\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*)) = \emptyset$, the proof of $\sigma(H) = \sigma(A) \cup \sigma(-A^*)$ is obvious. If any $\lambda \in (\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*))$ satisfies one of (i), (ii) and (iii), by assuming $\sigma(H) \neq \sigma(A) \cup \sigma(-A^*)$, there exists $\mu \in (\sigma(A) \cup \sigma(-A^*))$ such that $\mu \notin \sigma(H)$, that is, $\mu \in \rho(H)$. By Lemma 2.4.26, we can know that $\mu \in (\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*))$, which means that $\mu \notin \rho(H)$, a contradiction. Therefore, $\sigma(H) = \sigma(A) \cup \sigma(-A^*)$. □

Corollary 2.4.28 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix}$ *be a bounded upper triangular infinite-dimensional Hamiltonian operator on the whole space. Then* $\sigma(H) = \sigma(A) \cup \sigma(-A^*)$ *if one of the following conditions is satisfied:*

(i) $AB = -BA^*$,
(ii) There exists $\mu \in \mathbb{C}$ *such that* $(A - \mu I)B = 0$ *or* $B(A^* + \mu I) = 0$,
(iii) $int(\sigma(A) \cup \sigma(-A^*)) = \emptyset$,
(iv) $AA^* \geq A^*A$ *or* $AA^* \leq A^*A$,

where $int(G)$ *represents the Interior of the set* G.

Proof (i) Suppose $(\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*)) \neq \emptyset$, then for any $\lambda \in (\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*))$ we can get

$$(A - \lambda I)B = -B(A^* + \lambda I).$$

So $(A - \lambda I)By = 0$ for any $y \in \mathscr{N}(-A^* - \lambda I)$. By $\lambda \in \sigma_{r,1}(A)$ we know $By = 0$, so

$$B\mathscr{N}(-A^* - \lambda I) + \mathscr{R}(A - \lambda I) = \mathscr{R}(A - \lambda I) \neq X.$$

Therefore, $\sigma(H) = \sigma(A) \cup \sigma(-A^*)$.

(ii) Suppose $(\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*)) \neq \emptyset$, then

$$(A - \lambda I)B + (\lambda - \mu)B = 0$$

for any $\lambda \in (\sigma_{r,1}(A) \cap \sigma_{p,1}(-A^*))$. So, $(A-\lambda I)B\mathscr{N}(-A^*-\lambda I)+(\lambda-\mu)B\mathscr{N}(-A^*-\lambda I)=\{0\}$, that is, $B\mathscr{N}(-A^*-\lambda I)\subset\mathscr{R}(A-\lambda I)$. Therefore

$$B\mathscr{N}(-A^*-\lambda I)+\mathscr{R}(A-\lambda I)=\mathscr{R}(A-\lambda I)\neq X.$$

By Theorem 2.4.27, we can get that $\sigma(H)=\sigma(A)\cup\sigma(-A^*)$. The proof for $B(A^*+\mu I)=0$ is exactly the same as above.

(iii) From $\partial\sigma(A)\subset\sigma_{ap}(A)$, we obtain $\sigma_{r,1}(A)\subset int(\sigma(A))$. So, if $int(\sigma(A)\cup\sigma(-A^*))=\emptyset$, $\sigma_{r,1}(A)=\emptyset$. By Theorem 2.4.27, we have $\sigma(H)=\sigma(A)\cup\sigma(-A^*)$.

(iv) If $AA^*\geq A^*A$, it is easy to prove that $\sigma_{p,1}(-A^*)=\emptyset$. Then $\sigma_{r,1}(A)=\emptyset$ when $AA^*\leq A^*A$. So, we can get $\sigma(H)=\sigma(A)\cup\sigma(-A^*)$ by Theorem 2.4.27. □

Theorem 2.4.29 *Let* $H=\begin{bmatrix}A & B\\ 0 & -A^*\end{bmatrix}:\mathscr{D}(A)\times\mathscr{D}(A^*)\to X\times X$ *be an upper triangular infinite-dimensional Hamiltonian operator. Then* $\sigma_\delta(H)=\sigma_\delta(A)\cup\sigma_\delta(-A^*)$ *holds if and only if*

$$(\sigma_\delta(A)\cap\sigma_{p,1}(-A^*))=\emptyset$$

or $B\mathscr{N}(-A^*-\lambda I)+\mathscr{R}(A-\lambda I)\neq X$ *for any* $\lambda\in(\sigma_\delta(A)\cap\sigma_{p,1}(-A^*))$.

Proof If $\sigma_\delta(H)=\sigma_\delta(A)\cup\sigma_\delta(-A^*)$, we can suppose $(\sigma_\delta(A)\cap\sigma_{p,1}(-A^*))\neq\emptyset$. If there exists $\lambda\in(\sigma_\delta(A)\cap\sigma_{p,1}(-A^*))$ such that

$$B\mathscr{N}(-A^*-\lambda I)+\mathscr{R}(A-\lambda I)=X,$$

then it can be asserted that $\mathscr{R}(H-\lambda I)=X\oplus X$. In fact, for any $\begin{bmatrix}f & g\end{bmatrix}^T\in X\oplus X$ there exists $y\in\mathscr{D}(A^*)$ such that

$$-(A^*+\lambda I)y=g \tag{2.26}$$

since $\lambda\in\sigma_{p,1}(-A^*)$. From $B\mathscr{N}(-A^*-\lambda I)+\mathscr{R}(A-\lambda I)=X$, there exists $x\in\mathscr{D}(A)$, $y_1\in\mathscr{N}(-A^*-\lambda I)$ such that

$$(A-\lambda I)x+By_1=f-By. \tag{2.27}$$

Combining (2.26) and (2.27), we can get

$$\begin{bmatrix}A-\lambda I & B\\ 0 & -A^*-\lambda I\end{bmatrix}\begin{bmatrix}x\\ y+y_1\end{bmatrix}=\begin{bmatrix}f\\ g\end{bmatrix},$$

that is, $\mathscr{R}(H-\lambda I)=X\oplus X$. On the other hand, from $\sigma_\delta(H)=\sigma_\delta(A)\cup\sigma_\delta(-A^*)$ we know $\lambda\in\sigma_\delta(H)$, contradiction is derived. So, $B\mathscr{N}(-A^*-\lambda I)+\mathscr{R}(A-\lambda I)\neq X$.

Conversely, let $\rho_\delta(T)=\{\lambda\in\mathbb{C}:\mathscr{R}(T-\lambda I)=X\}$, then let us consider

$$\rho_\delta(A)\cup\rho_\delta(-A^*)\subset\rho_\delta(H),$$

and it is only necessary to prove the reverse inclusion. Let $\lambda \in \rho_\delta(H)$, then $\lambda \in \rho_\delta(-A^*) = \rho(-A^*) \cup \sigma_{p,1}(-A^*)$. If $\lambda \in \rho(-A^*)$, considering

$$H - \lambda I = \begin{bmatrix} I & -B(A^* - \lambda I)^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} A - \lambda I & 0 \\ 0 & -A^* - \lambda I \end{bmatrix},$$

we have $\lambda \in \rho_\delta(A)$. When $\lambda \in \sigma_{p,1}(-A^*)$, if $(\sigma_\delta(A) \cap \sigma_{p,1}(-A^*)) \neq \emptyset$, then by given conditions we know $\lambda \in \sigma_\delta(H)$, which is contradictory to $\lambda \in \rho_\delta(H)$, so $(\sigma_\delta(A) \cap \sigma_{p,1}(-A^*)) = \emptyset$, and hence $\lambda \in \rho_\delta(A)$. □

Since $\lambda \in \sigma_{ap}(H)$ if and only if $\overline{\lambda} \in \sigma_\delta(H^*)$, and

$$\begin{bmatrix} -A & B \\ 0 & A^* \end{bmatrix} = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} A^* & 0 \\ B & -A \end{bmatrix} \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix},$$

the proof of the following conclusion is obvious.

Theorem 2.4.30 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator. Then* $\sigma_{ap}(H) = \sigma_{ap}(A) \cup \sigma_{ap}(-A^*)$ *holds if and only if*

$$(\sigma_{r,1}(A) \cap \sigma_{ap}(-A^*)) = \emptyset$$

or $B\mathscr{N}(A^* - \overline{\lambda} I) + \mathscr{R}(A + \overline{\lambda} I) \neq X$ *for any* $\lambda \in (\sigma_{r,1}(A) \cap \sigma_{ap}(-A^*))$.

Corollary 2.4.31 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator with* $\sigma(A) \cap \sigma(-A^*) = \emptyset$. *Then*

(i) $\sigma(H) = \sigma(A) \cup \sigma(-A^*)$;
(ii) $\sigma_\delta(H) = \sigma_\delta(A) \cup \sigma_\delta(-A^*)$;
(iii) $\sigma_{ap}(H) = \sigma_{ap}(A) \cup \sigma_{ap}(-A^*)$.

It is worth mentioning that the invertible complement problem of upper triangular operator matrices has important applications in related disciplines such as in interpolation theory, control theory, system theory (see [57]). The so-called upper triangular operator matrix invertible complement problem refers to the consideration of partial operator matrices

$$M_x = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix},$$

complementing the unknown operator B such that M_B has invertibility. The following will give the invertible complement of the upper triangular infinite-dimensional Hamiltonian operator.

Theorem 2.4.32 *The upper triangular infinite-dimensional Hamiltonian operator* $H_B = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix}$ *has a bounded self-adjoint operator* B *such that* H_B *is invertible if and only if* A *is left invertible.*

Proof If there exists a bounded self-adjoint operator B such that H_B is invertible, then A is left invertible by Lemma 2.25. On the contrary, if A is left invertible, $\mathscr{R}(A) = \mathscr{N}(A^*)^\perp$ is closed and $\mathscr{R}(A^*) = \mathscr{N}(A)^\perp = X$. Selecting a bounded self-adjoint operator B so that PBP is an invertible operator, where $P : X \to \mathscr{N}(A^*)$ is orthogonal projection, then under space decompositions $X \oplus X = X \oplus \mathscr{R}(A) \oplus \mathscr{R}(A)^\perp = \mathscr{R}(A) \oplus \mathscr{R}(A)^\perp \oplus X$, H_B has the following block form:

$$H_B = \begin{bmatrix} A_1 & B_1 & B_2 \\ 0 & B_3 & B_4 \\ 0 & \widetilde{A^*} & 0 \end{bmatrix} : \mathscr{D}(A) \oplus (\mathscr{R}(A) \cap \mathscr{D}(A^*) \oplus \mathscr{R}(A)^\perp \to \mathscr{R}(A) \oplus \mathscr{R}(A)^\perp \oplus X.$$

Since $B_4 = PBP$ is invertible, $A_1 : \mathscr{D}(A) \to \mathscr{R}(A)$ and $\widetilde{A^*} : (\mathscr{R}(A) \cap \mathscr{D}(A^*) \to X$ are invertible, H_B is invertible. The proof is completed. □

2.5 Spectra of Non-Negative Hamiltonian Operators

For the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subseteq X \times X \to X \times X,$$

if $C \geq 0$, $B \geq 0$, then H is called a non-negative Hamiltonian operator. The non-negative Hamiltonian operator is the generalization of non-negative Hamiltonian matrix from finite-dimension to infinite-dimension, which has a wide range of applications in control theory and elasticity.

Example 2.5.1 Consider the differential equations corresponding to the problem of simply supported rectangular plates on opposite sides

$$D\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)^2 w = 0,$$

where the boundary conditions are

$$w(x, 0) = w(x, 1) = 0, \quad \frac{\partial^2 w}{\partial y^2}(x, 0) = \frac{\partial^2 w}{\partial y^2}(x, 1) = 0.$$

Let $\theta = \frac{\partial w}{\partial x}$, $q = D(\frac{\partial^3 w}{\partial x^3} + \frac{\partial^3 3}{\partial x \partial y^2})$, $m = -D(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2})$, then the corresponding infinite-dimensional Hamiltonian system is obtained as

$$\frac{\partial}{\partial x}\begin{bmatrix} w \\ \theta \\ q \\ m \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{\partial^2}{\partial y^2} & 0 & 0 & -\frac{1}{D} \\ 0 & 0 & 0 & \frac{\partial^2}{\partial y^2} \\ 0 & 0 & -1 & 0 \end{bmatrix}\begin{bmatrix} w \\ \theta \\ q \\ m \end{bmatrix}. \tag{2.28}$$

Let $X = L^2[0, 1]$, then the infinite-dimensional Hamiltonian operator corresponding to the system is

$$H = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{\mathrm{d}^2}{\mathrm{d}y^2} & 0 & 0 & -\frac{1}{D} \\ 0 & 0 & 0 & \frac{\mathrm{d}^2}{\mathrm{d}y^2} \\ 0 & 0 & -1 & 0 \end{bmatrix},$$

where

$$\mathscr{D}(H) = \left\{ \begin{bmatrix} w \\ \theta \\ q \\ m \end{bmatrix} \in X^4 : \begin{array}{c} w(0) = w(1) = m(0) = m(1) = 0, w, w', m, m' \\ \text{are absolutely continuous and } w'', m'' \in X \end{array} \right\},$$

and $-H$ is a non-negative Hamiltonian operator.

In order to study the properties of non-negative Hamiltonian operators, we firstly introduce the definition of accretive operators.

Definition 2.5.2 A linear operator T in a Hilbert space X is an accretive operator, if the numerical range $W(T)$ is contained in the right half-closed plane, that is,

$$Re(Tu, u) \geq 0$$

for any $u \in \mathscr{D}(T)$. If T is an accretive operator and any accretive extension of T coincides with T, then T is called a maximal accretive operator.

Remark 2.5.3 According to Zorn's Lemma, there must be a maximal accretive extension of the accretive operator.

Lemma 2.5.4 *Let T be densely closed operator in a Hilbert space X, and T be a maximal accretive operator. Then $\mathscr{R}(T + \lambda I) = X$ for any $Re(\lambda) > 0$.*

Proof If T is a maximal accretive operator, for any $Re(\lambda) > 0$ and any $u \in \mathscr{D}(T)$, $\|u\| = 1$

$$\|(T + \lambda I)u\| \geq |((T + \lambda I)u, u)| = |(Tu, u) - (-\lambda)| \geq Re(\lambda).$$

Then $(T + \lambda I)^{-1}$ exists. The following only discusses the case of $\lambda = 1$, and the other cases are completely similar. Let

$$S = (T - I)(T + I)^{-1},$$

then $\mathscr{D}(S) = \mathscr{R}(T + I)$ and S is a contractive operator. In other words, the accretivity of T corresponds to the contractility of S. In fact, for any $y_0 \in \mathscr{D}(T)$, let $u_0 = Ty_0 + y_0$, then $Su_0 = Ty_0 - y_0$ and

$$\begin{aligned}\|Su_0\|^2 =& \|Ty_0\|^2 + \|y_0\|^2 - 2Re(Ty_0, y_0)\\ \leq& \|Ty_0\|^2 + \|y_0\|^2 + 2Re(Ty_0, y_0)\\ =& \|u_0\|^2.\end{aligned}$$

According to the Banach-Steinhaus Theorem, a contractive operator has a contractive extension defined on the whole space. Therefore, S is the maximal contractive if T is maximal accretive, and from $\mathscr{D}(S) = \mathscr{R}(T + I)$, we can obtain $\mathscr{R}(T + I) = X$. The proof is completed. □

Lemma 2.5.5 *Let T, S be densely defined linear operators in a Hilbert space X. If $\rho(T) \neq \emptyset$, $(T - \lambda)^{-1}S$ is bounded if and only if $\mathscr{D}(T^*) \subset \mathscr{D}(S^*)$, where $\lambda \in \rho(T)$.*

Proof If $\mathscr{D}(T^*) \subset \mathscr{D}(S^*)$, then $S^*(T^* - \overline{\lambda})^{-1}$ is bounded and

$$(S^*(T^* - \overline{\lambda})^{-1})^* \supset (T - \lambda)^{-1}S,$$

so $(T - \lambda)^{-1}S$ is bounded.

Conversely, if $(T - \lambda)^{-1}S$ is bounded in $\mathscr{D}(S)$, then $\overline{(T - \lambda)^{-1}S}$ is a bounded operator defined on the whole space and

$$(S^*(T^* - \overline{\lambda})^{-1})^* \supset \overline{(T - \lambda)^{-1}S},$$

so $\mathscr{D}(T^*) \subset \mathscr{D}(S^*)$. The proof is completed. □

Theorem 2.5.6 *Let T be a densely defined closed operator in a Hilbert space X. Then T is maximal accretive if and only if $(T + \lambda I)^{-1} \in \mathscr{B}(X)$ and $\|(T + \lambda)^{-1}\| \leq \frac{1}{Re(\lambda)}$ for any $Re(\lambda) > 0$.*

Proof If T is a maximal accretive operator, then $(T + \lambda I)^{-1}$ exists and $\mathscr{R}(T + \lambda I) = X$ for any $Re(\lambda) > 0$. Then $(T + \lambda I)^{-1} \in \mathscr{B}(X)$. Furthermore, for any $Re(\lambda) > 0$ and $u \in \mathscr{D}(T)$, $\|u\| = 1$,

$$\begin{aligned}\|(T + \lambda I)u\| \geq& |((T + \lambda I)u, u)|\\ =& |(Tu, u) - (-\lambda)|\\ \geq& Re(\lambda),\end{aligned}$$

which implies $\|(T + \lambda I)^{-1}\| \leq \frac{1}{Re(\lambda)}$. Having noticed that $T + \lambda I$ is accretive if and only if $(T + \lambda I)^{-1}$ is accretive operator, the converse proof is obvious. □

By Theorem 2.5.6, the proof of the following inference is obvious.

Corollary 2.5.7 *Let T be a densely defined closed operator in a Hilbert space X, then T is maximal accretive if and only if T^* is maximal accretive operator.*

Theorem 2.5.8 *Let $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ be an infinite-dimensional Hamiltonian operator. Then H is a non-negative Hamiltonian operator if and only if $\mathfrak{J}H$ is an accretive operator, where $\mathfrak{J} = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$.*

Proof Since $\mathscr{D}(H) = \mathscr{D}(C) \times \mathscr{D}(B)$, it is easy to prove $B \geq 0, C \geq 0$ if and only if $Re(\mathfrak{J}H) \geq 0$, so the conclusion is obvious. □

Theorem 2.5.9 *Let $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ be a non-negative Hamiltonian operator. Then $\mathfrak{J}H$ is a maximal accretive operator if and only if $(JH)^* = JH$, that is, H is symplectic self-adjoint, where $\mathfrak{J} = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$, $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$.*

Proof If $\mathfrak{J}H$ is a maximal accretive operator, it follow from $H \subset JH^*J$ that

$$\mathfrak{J}H \subset \mathfrak{J}JH^*J.$$

And for any $u \in \mathscr{D}(H)$ we have

$$Re(\mathfrak{J}JH^*Ju, u) = Re(\mathfrak{J}Hu, u) \geq 0,$$

which implies that $\mathfrak{J}JH^*J$ is the accretive extension of $\mathfrak{J}H$. Note that $\mathfrak{J}H$ is maximal accretive we have

$$\mathfrak{J}H = \mathfrak{J}JH^*J,$$

that is, $(JH)^* = JH$.

Conversely, if $(JH)^* = JH$, it is obvious that $B \geq 0, C \geq 0$, and $\mathfrak{J}H$ are accretive operators. For any $Re(\lambda) > 0$, it is easy to prove $-\lambda \in \Gamma(\mathfrak{J}H)$. Assume $-\lambda \in \sigma_{r,1}(\mathfrak{J}H)$, then $-\overline{\lambda} \in \sigma_{p,1}(H^*\mathfrak{J})$, that is, there exists $u = \begin{bmatrix} f & g \end{bmatrix}^T \in \mathscr{D}(H^*\mathfrak{J})$, $\|u\|^2 = 1$ such that

$$(H^*\mathfrak{J} + \overline{\lambda}I)u = \begin{bmatrix} C + \overline{\lambda} & A^* \\ -A & B + \overline{\lambda} \end{bmatrix} \begin{bmatrix} f \\ g \end{bmatrix} = 0,$$

that is,

$$Cf + \overline{\lambda}f + A^*g = 0$$

and

$$-Af + Bg + \overline{\lambda}g = 0.$$

After taking the inner product of the first equation by f on the right and of the second equation by g on the left and adding the resulting equations, we have

$$(Cf, f) + (g, Bg) + \overline{\lambda}(f, f) + \lambda(g, g) = 0.$$

By $\|f\|^2 + \|g\|^2 = 1$, we obtain $Im(\lambda) = 0$ and

$$(Cf, f) + (g, Bg) + \lambda = 0.$$

This is contradictory to $Re(\lambda) > 0$. Thus $-\lambda \in \rho(\mathfrak{J}H)$, that is, $(\mathfrak{J}H + \lambda I)^{-1} \in \mathscr{B}(X \times X)$. Because for any $Re(\lambda) > 0$ and $u \in \mathscr{D}(T)$, $\|u\| = 1$ we can have

$$\begin{aligned}\|(\mathfrak{J}H + \lambda I)u\| \geq & |((\mathfrak{J}H + \lambda I)u, u)| \\ = & |(\mathfrak{J}Hu, u) - (-\lambda)| \\ \geq & Re(\lambda),\end{aligned}$$

$\|(\mathfrak{J}H + \lambda)^{-1}\| \leq \frac{1}{Re(\lambda)}$. By Theorem 2.5.6, $\mathfrak{J}H$ is a maximal accretive operator. □

2.5.1 *Point Spectra of Non-negative Hamiltonian Operators*

In the finite-dimensional case, for a non-negative Hamiltonian matrices $H = \begin{bmatrix} A_{n\times n} & B_{n\times n} \\ C_{n\times n} & -A^T_{n\times n} \end{bmatrix}$, if there exists an eigenvalue with non-zero real part, then the non-negative Hamiltonian matrix is Schur reducible, that is, there exists a unitary matrix

$$Q = \begin{bmatrix} Q_{11} & Q_{12} \\ -Q_{12} & Q_{11} \end{bmatrix}, \quad Q_{11}, Q_{12} \in \mathbb{C}^{n\times n},$$

such that

$$Q^T H Q = \begin{bmatrix} L_{n\times n} & R_{n\times n} \\ 0 & -L^T_{n\times n} \end{bmatrix}, \tag{2.29}$$

where $L_{n\times n}$ is an upper triangular matrix, $R^T_{n\times n} = R_{n\times n}$, and Eq. (2.29) is also called Schur-Hamilton factorization (see [63]). Then, it is natural to ask whether the above conclusions can be extended to the case of non-negative Hamiltonian operators. In order to answer this question, we will in the first place discuss the properties of the point spectrum of non-negative Hamiltonian operators.

In order to study the spectral properties of non-negative Hamiltonian operator, we firstly give the following Lemma.

Theorem 2.5.10 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian operator. Then* H *is injective if and only if* $\mathscr{N}(A) \cap \mathscr{N}(C) = \{0\}$ *and* $\mathscr{N}(B) \cap \mathscr{N}(-A^*) = \{0\}$.

Proof If $\mathscr{N}(A) \cap \mathscr{N}(C) = \{0\}$ and $\mathscr{N}(B) \cap \mathscr{N}(-A^*) = \{0\}$, assume that there exists $0 \neq u = \begin{bmatrix} f & g \end{bmatrix}^T \in \mathscr{D}(H)$ such that $Hu = 0$, that is,

$$Af + Bg = 0, \quad Cf - A^*g = 0.$$

After taking the inner product of the first equation by g and of the second equation by f and adding the resulting equations, we have

$$(Bg, g) + (Cf, f) = 0.$$

Because B,C are non-negative operators,

$$(Bg, g) = 0, \quad (Cf, f) = 0.$$

Furthermore, we obtain

$$Bg = 0, \quad Cf = 0$$

by Lemma 2.3.4. After substituting, we obtain

$$Af = 0, \quad A^*g = 0.$$

This is contradictory to $\mathscr{N}(A) \cap \mathscr{N}(C) = \{0\}$ and $\mathscr{N}(B) \cap \mathscr{N}(-A^*) = \{0\}$. Thus H is injective.

Conversely, if H is injective, we may suppose $\mathscr{N}(A) \cap \mathscr{N}(C) \neq \{0\}$, and there exists $f \neq 0$ such that $f \in \mathscr{N}(A) \cap \mathscr{N}(C)$, and hence

$$\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} \begin{bmatrix} f \\ 0 \end{bmatrix} = \begin{bmatrix} Af \\ Cf \end{bmatrix} = 0.$$

This is contradictory to H being injective. Similarly we can prove $\mathscr{N}(B) \cap \mathscr{N}(-A^*) = \{0\}$. The proof is completed. □

According to Theorem 2.5.10, the proof of the following conclusion is obvious.

Corollary 2.5.11 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian operator. If* B, C *are injective, then* $H - \lambda I$ *is also injective for arbitrary* $\lambda \in \mathrm{i}\mathbb{R}$ *, that is,* $\sigma_p(H) \cap \mathrm{i}\mathbb{R} = \emptyset$.

Proof For any $\lambda \in \mathrm{i}\mathbb{R}$, $H - \lambda I$ is a non-negative Hamiltonian operator, and therefore $H - \lambda I$ is also injective if B, C are injective. □

Corollary 2.5.12 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian operator. Then the following propositions hold*

(i) If A, A^* *are injective, then* H *is injective;*
(ii) If B, C *are injective, then* H *is injective;*
(iii) If A, B *are injective, then* H *is injective;*
(iv) If C, A^* *are injective, then* H *is injective.*

Theorem 2.5.13 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian operator. Then*

$$\sigma_{Imp}(H) \subset \sigma_p(A) \cup \sigma_p(-A^*),$$

where $\sigma_{Imp}(H) = \{\lambda \in \sigma_p(H) : Re(\lambda) = 0\}$.

Proof Let $\lambda \in \sigma_{Imp}(H)$, then there exists $u = \begin{bmatrix} f & g \end{bmatrix}^T \neq 0$ such that

$$Af + Bg = \lambda f, \quad Cf - A^*g = \lambda g \tag{2.30}$$

After taking inner product of the first equation by g and of the second equation by f adding the resulting equations, we have

$$(f, Cf) + (Bg, g) = 0.$$

Because B, C are non-negative operators,

$$(f, Cf) = 0 \text{ and } (Bg, g) = 0.$$

Again by Lemma 2.3.4, we know

$$Cf = 0 \text{ and } Bg = 0. \tag{2.31}$$

Furthermore, substituting Eq. (2.30), we have

$$Af = \lambda f, \quad -A^*g = \lambda g.$$

As at least one of f and g is a non-zero vector,

$$\lambda \in \sigma_p(A) \cup \sigma_p(-A^*),$$

and thus $\sigma_{Imp}(H) \subset \sigma_p(A) \cup \sigma_p(-A^*)$. □

Theorem 2.5.14 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian operator. If* $0 \in \rho(B)$, $\mathscr{D}(B) \subset \mathscr{D}(A^*)$ *and* $(B^{-1}Ax, x) \in \mathbb{R}$

for any $x \in \mathscr{D}(A)$, then H has only real eigenvalues (if not empty), appearing in positive and negative pairs.

Proof Suppose $\lambda \in \sigma_p(H)$, and let $u = \begin{bmatrix} f & g \end{bmatrix}^T$ be the corresponding eigenvector, then

$$Af + Bg = \lambda f,$$

$$Cf - A^*g = \lambda g.$$

Because $0 \in \rho(B)$, B^{-1} is a positive definite operator, so from the first equation we obtain

$$g = \lambda B^{-1} f - B^{-1} Af,$$

substituting into the second equation we have

$$\lambda^2 B^{-1} f + \lambda A^* B^{-1} f - \lambda B^{-1} Af - Cf - A^* B^{-1} Af = 0.$$

It is easy to know $f \neq 0$, otherwise, it can be inferred that $g = 0$ which is contradictory to the concept of eigenvector. Thus taking inner product on both sides of the equation, we obtain

$$\lambda^2 (B^{-1} f, f) + \lambda (A^* B^{-1} f, f) - \lambda (B^{-1} Af, f) - (Cf, f) - (A^* B^{-1} Af, f) = 0.$$

By $(B^{-1} Af, f) \in \mathscr{R}$, we know

$$\lambda (A^* B^{-1} f, f) - \lambda (B^{-1} Af, f) = 0.$$

Therefore

$$\lambda = \pm \sqrt{\frac{(Cf, f) + (B^{-1} Af, Af)}{(B^{-1} f, f)}},$$

that is, H has real eigenvalues only.

Next we prove $-\lambda \in \sigma_p(H)$. In fact, let $\widetilde{u} = \begin{bmatrix} -f \\ \lambda B^{-1} f + B^{-1} Af \end{bmatrix}$, then

$$H\widetilde{u} + \lambda \widetilde{u} = 2\lambda \begin{bmatrix} 0 \\ B^{-1} Af - A^* B^{-1} f \end{bmatrix}.$$

It can be known that $B^{-1} A$ is a symmetric operator and $B^{-1} A \subset A^* B^{-1}$ for any $x \in \mathscr{D}(A)$ with $(B^{-1} Ax, x) \in \mathbb{R}$, and therefore $B^{-1} Af - A^* B^{-1} f = 0$, that is, $-\lambda \in \sigma_p(H)$ and the corresponding eigenvector is

$$\widetilde{u} = \begin{bmatrix} -f \\ \lambda B^{-1} f + B^{-1} Af \end{bmatrix}.$$

The proof is completed. □

Remark 2.5.15 The conditions of the above theorem are changed to $0 \in \rho(C)$, $\mathscr{D}(C) \subset \mathscr{D}(A)$ and $(C^{-1}A^*x, x) \in \mathbb{R}$ for any $x \in \mathscr{D}(A^*)$, then in the same way, it can be proved that H has only real eigenvalues (if not empty), appearing in positive and negative pairs.

The above results show that when operators B and C are of the same sign (i.e., B, C are both non-negative or $-B, -C$ are both non-negative), eigenvalues of the infinite-dimensional Hamiltonian operator H under certain conditions are only real numbers and appear in positive and negative pairs. Then, if B and C are heterogeneous (i.e., one of B and C is positive definite and the other is negative definite), what will happen? The following conclusion will answer this question.

Theorem 2.5.16 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If* $B, -C$ *are non-negative operators,* $0 \in \rho(B)$, $\mathscr{D}(B) \subset \mathscr{D}(A^*)$, A^*B^{-1} *are symmetric operators and* $C + A^*B^{-1}A \leq 0$, *then*

(i) The infinite-dimensional Hamiltonian operator H *has only pure imaginary eigenvalues and appears in positive and negative pairs;*

(ii) Suppose $\lambda \in \sigma_p(H)$, *and let* $u = \begin{bmatrix} f & g \end{bmatrix}^T$ *be the corresponding eigenvector, then* $(Bg, g) = -(Cf, f)$.

Proof (i) Suppose $\lambda \in \sigma_p(H)$, and let $u = \begin{bmatrix} f & g \end{bmatrix}^T$ be the corresponding egienvector, then

$$\lambda = \pm\sqrt{\frac{(Cf + A^*B^{-1}Af, f)}{(B^{-1}f, f)}}$$

according to the assumptions, so $\sigma_p(H) \subset \mathrm{i}\mathbb{R}$. Next prove $-\lambda \in \sigma_p(H)$. In fact, for $\widetilde{u} = \begin{bmatrix} -f \\ \lambda B^{-1}f + B^{-1}Af \end{bmatrix}$, it follows from $(B^{-1}Ax, x) \in \mathbb{R}$ that

$$(H + \lambda)\widetilde{u} = 0,$$

that is, $-\lambda \in \sigma_p(H)$, and the corresponding eigenvector is $\widetilde{u} = \begin{bmatrix} -f \\ \lambda B^{-1}f + B^{-1}Af \end{bmatrix}$.

(ii) Suppose $\lambda \in \sigma_p(H)$, and let $u = \begin{bmatrix} f & g \end{bmatrix}^T$ be the corresponding eigenvector, then

$$Af + Bg = \lambda f, \tag{2.32}$$

$$Cf - A^*g = \lambda g. \tag{2.33}$$

After taking the inner product of Eq. (2.32) by g and of Eq. (2.33) by f and adding the resulting equations, we have

$$(Bg, g) + (Cf, f) = 2Re(\lambda)(f, g).$$

From (i) we know $Re(\lambda) = 0$, and therefore

$$(Bg, g) = -(Cf, f).$$

The proof is completed. □

Remark 2.5.17 The conditions in the above conclusion can be changed to that $-B, C$ are non-negative operators, $0 \in \rho(C)$ and $\mathscr{D}(C) \subset \mathscr{D}(A)$, AC^{-1} is a symmetric operator and $B + AC^{-1}A^* \leq 0$, then we can also prove that H has only pure imaginary eigenvalues.

2.5.2 *Invertibility of Non-negative Hamiltonian Operators*

If the operator derived by the system has a bounded inverse time (i.e., $T^{-1} : \mathscr{R}(T) \to X$ is bounded), which provides a strong guarantee for the semi-analytical method. In this case, the problem of solving the system will become relatively simple. Thus, in order to solve the problem of solving the infinite-dimensional Hamiltonian system. The question of when the infinite-dimensional Hamiltonian operator has a bounded inverse is particularly important.

Theorem 2.5.18 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian operator. Then*

(i) If $0 \in \rho(B) \cap \rho(C)$*, then* H *has a bounded inverse;*
(ii) If $0 \in \rho(A)$ *and* $A^{-1}B^{\frac{1}{2}}$ *and* $(A^*)^{-1}C^{\frac{1}{2}}$ *are bounded linear operators, then* H *has a bounded inverse.*

Proof (i) Assume that H doesn't have bounded inverse, then there exists orthogonalizing sequence $\{x_n = \begin{bmatrix} x_n^{(1)} \\ x_n^{(2)} \end{bmatrix}\}_{n=1}^{\infty}$, $(\|x_n\| = 1, n = 1, 2, \ldots)$ such that

$$Hx_n \to 0$$

as $n \to \infty$, and further we obtain

$$Ax_n^{(1)} + Bx_n^{(2)} \to 0, \quad Cx_n^{(1)} - A^*x_n^{(2)} \to 0.$$

Taking the inner product of the first equation by $x_n^{(2)}$ and of the second equation by $x_n^{(1)}$ and adding the resulting equations, we have

$$(Bx_n^{(2)}, x_n^{(2)}) + (Cx_n^{(1)}, x_n^{(1)}) \to 0.$$

Because B,C are non-negative operators,

$$(Bx_n^{(2)}, x_n^{(2)}) \to 0, \quad (Cx_n^{(1)}, x_n^{(1)}) \to 0.$$

As B,C are self-adjoint and non-negative operators, there exist unique square root operators $B^{\frac{1}{2}}$ and $C^{\frac{1}{2}}$, and hence

$$B^{\frac{1}{2}}x_n^{(2)} \to 0, \quad C^{\frac{1}{2}}x_n^{(1)} \to 0.$$

If $0 \in \rho(B) \cap \rho(C)$, denoting $by B^{-\frac{1}{2}}$ and $C^{-\frac{1}{2}}$ the square root operators of B^{-1} and C^{-1}, respectively, we have

$$B^{-\frac{1}{2}}(B^{\frac{1}{2}}x_n^{(2)}) = x_n^{(2)} \to 0, \quad C^{-\frac{1}{2}}(C^{\frac{1}{2}}x_n^{(1)}) = x_n^{(1)} \to 0.$$

This is contradictory to $\|x_n\| = 1, (n = 1, 2, \ldots)$. Thus, $0 \in \rho(B) \cap \rho(C)$, and hence H has a bounded inverse.

(ii) If $0 \in \rho(A)$, $A^{-1}B^{\frac{1}{2}}$ and $(A^*)^{-1}C^{\frac{1}{2}}$ are bounded linear operators. By the equation above, we obtain

$$x_n^{(1)} + A^{-1}B^{\frac{1}{2}}(B^{\frac{1}{2}}x_n^{(2)}) \to 0, x_n^{(2)} - (A^*)^{-1}C^{\frac{1}{2}}(C^{\frac{1}{2}}x_n^{(1)}) \to 0,$$

Because $B^{\frac{1}{2}}x_n^{(2)} \to 0, C^{\frac{1}{2}}x_n^{(1)} \to 0,$

$$A^{-1}Bx_n^{(2)} \to 0, (A^*)^{-1}Cx_n^{(1)} \to 0,$$

and furthermore we have

$$x_n^{(1)} \to 0, x_n^{(2)} \to 0.$$

This is contradictory to $\|x_n\| = 1, (n = 1, 2, \ldots)$. Thus, $0 \in \rho(A)$, $A^{-1}B^{\frac{1}{2}}$ and $(A^*)^{-1}C^{\frac{1}{2}}$ are bounded linear operators, then H has a bounded inverse. □

Corollary 2.5.19 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a non-negative Hamiltonian operator with* $0 \in \rho(A)$. *Then* H *has a bounded inverse.*

Proof If B, C are non-negative, $\mathscr{D}(B) \subset \mathscr{D}(B^{\frac{1}{2}})$ and $\mathscr{D}(C) \subset \mathscr{D}(C^{\frac{1}{2}})$ are trivial. So, if $\mathscr{D}(A) \subset \mathscr{D}(C)$, $\mathscr{D}(A^*) \subset \mathscr{D}(B)$ and $0 \in \rho(A)$, we know $A^{-1}B^{\frac{1}{2}}$ and $(A^*)^{-1}C^{\frac{1}{2}}$ are bounded linear operators by Lemma 2.5.5. By Theorem 2.5.18, it follows that H has a bounded inverse. □

Corollary 2.5.20 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian operator with* $0 \in \rho(B) \cap \rho(C)$. *Then* $H - \lambda I$ *has bounded inverse for any* $\lambda \in i\mathbb{R}$.

Proof According to the definition of non-negative Hamiltonian operator, $H - \lambda I$ is still a non-negative Hamiltonian operator for any $\lambda \in i\mathbb{R}$. If $0 \in \rho(B) \cap \rho(C)$, it follows form the theorem above that $H - \lambda I$ has a bounded inverse. □

Corollary 2.5.21 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian with* $A \in \mathscr{B}(X)$ *or* $B, C \in \mathscr{B}(X)$.

(i) If A is invertible, then H is invertible;
(ii) If B, C are invertible, then H is invertible;
(iii) If A, A are surjective, then H is surjective;*
(iv) If B, C are surjective, then H is surjective.

Proof If $A \in \mathscr{B}(X)$ or $B, C \in \mathscr{B}(X)$,

$$H^* = \begin{bmatrix} A^* & C \\ B & -A \end{bmatrix},$$

that is, H^* also is a non-negative Hamiltonian operator and satisfies $(JH)^* = JH$, so the residual spectrum of H has no intersection with the imaginary axis. Therefore, if A is invertible or B, C is invertible, then H is invertible.

If A, A^* are surjective or B, C are surjective, $0 \in \rho(A)$ or $0 \in \rho(B) \cap \rho(C)$. Then H^* is bounded below, that is, H is surjective. The proof is completed. □

Corollary 2.5.22 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian operator with* $A \in \mathscr{B}(X)$ *or* $B, C \in \mathscr{B}(X)$. *If B, C are surjective, then* $H - \lambda I$ *is surjective for any* $\lambda \in i\mathbb{R}$.

The existence of bounded inverses of non-negative Hamiltonian operators is discussed above. In fact, the invertibility of the infinite-dimensional Hamiltonian operator can also be characterized by the invertibility of the internal elements. The following conclusion is obtained by Denisov (see [58]).

Theorem 2.5.23 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian operator. If* $\mathfrak{I}H$ *is a maximal accretive operator, then* $0 \in \rho(B) \cap \rho(C)$ *implies that* $0 \in \rho(H)$.

Proof It follows from Theorem 2.5.9 that $(JH)^* = JH$, so $0 \notin \sigma_r(H)$. Assume that $0 \notin \rho(H)$, then $0 \in \sigma_{ap}(H)$, that is, there exists $\{x_n = \begin{bmatrix} x_n^{(1)} \\ x_n^{(2)} \end{bmatrix}\}_{n=1}^{\infty}$, $(\|x_n\| = 1, n = 1, 2, \ldots)$ such that

$$Hx_n \to 0$$

as $n \to \infty$. Then we obtain

$$Ax_n^{(1)} + Bx_n^{(2)} \to 0, \quad Cx_n^{(1)} - A^*x_n^{(2)} \to 0.$$

Taking the inner product of the first equation by $x_n^{(2)}$ and of the second equation by $x_n^{(1)}$ and adding the resulting equations, we have

$$(Bx_n^{(2)}, x_n^{(2)}) + (Cx_n^{(1)}, x_n^{(1)}) \to 0.$$

Because B,C are non-negative operators,

$$(Bx_n^{(2)}, x_n^{(2)}) \to 0, \quad (Cx_n^{(1)}, x_n^{(1)}) \to 0.$$

As B,C are self-adjoint and non-negative operators, there are unique square root operators $B^{\frac{1}{2}}$ and $C^{\frac{1}{2}}$ such that

$$B^{\frac{1}{2}} x_n^{(2)} \to 0, \quad C^{\frac{1}{2}} x_n^{(1)} \to 0,$$

which together with $0 \in \rho(B) \cap \rho(C)$ implies that

$$B^{-\frac{1}{2}}(B^{\frac{1}{2}} x_n^{(2)}) = x_n^{(2)} \to 0, \quad C^{-\frac{1}{2}}(C^{\frac{1}{2}} x_n^{(1)}) = x_n^{(1)} \to 0.$$

This is contradictory to $\|x_n\| = 1$, $(n = 1, 2, \ldots)$. So, $0 \in \rho(B) \cap \rho(C)$, and then H is invertible. The proof is completed. □

A natural question is whether the invertibility of non-negative Hamiltonian operators can be characterized by the invertibility of diagonal elements.

Theorem 2.5.24 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a non-negative Hamiltonian operator. If* $\mathfrak{J}H$ *is a maximal accretive operator, then* $0 \in \rho(A)$ *implies that* $0 \in \rho(H)$.

Proof It follows from Theorem 2.5.9 that $(JH)^* = JH$, so $0 \notin \sigma_r(H)$. Assume that $0 \notin \rho(H)$, then $0 \in \sigma_{ap}(H)$, that is, there exists $\{x_n = \begin{bmatrix} x_n^{(1)} \\ x_n^{(2)} \end{bmatrix}\}_{n=1}^{\infty}$, $(\|x_n\| = 1, n = 1, 2, \ldots)$ such that

$$Hx_n \to 0.$$

Then we obtain

$$Ax_n^{(1)} + Bx_n^{(2)} \to 0, \tag{2.34}$$

$$Cx_n^{(1)} - A^* x_n^{(2)} \to 0. \tag{2.35}$$

Taking the inner product of the first equation by $x_n^{(2)}$ and of the second equation by $x_n^{(1)}$ and adding the resulting equations, we obtain

$$(Bx_n^{(2)}, x_n^{(2)}) + (Cx_n^{(1)}, x_n^{(1)}) \to 0.$$

Because B,C are non-negative operators,

$$(Bx_n^{(2)}, x_n^{(2)}) \to 0, \quad (Cx_n^{(1)}, x_n^{(1)}) \to 0.$$

Because B,C are self-adjoint and non-negative operators, there exist unique square root operators $B^{\frac{1}{2}}$ and $C^{\frac{1}{2}}$ such that

$$B^{\frac{1}{2}}x_n^{(2)} \to 0, C^{\frac{1}{2}}x_n^{(1)} \to 0.$$

If $0 \in \rho(A)$, since $\mathscr{D}(A^*) \subset \mathscr{D}(B) \subset \mathscr{D}(B^{\frac{1}{2}})$ and $\mathscr{D}(A) \subset \mathscr{D}(C) \subset \mathscr{D}(C^{\frac{1}{2}})$, we know $A^{-1}B^{\frac{1}{2}}$ and $(A^*)^{-1}C^{\frac{1}{2}}$ are bounded by Lemma 2.5.5, so

$$A^{-1}Bx_n^{(2)} \to 0, \quad (A^*)^{-1}Cx_n^{(1)} \to 0.$$

On the other hand, from (2.34) and (2.35), we know

$$x_n^{(1)} + A^{-1}Bx_n^{(2)} \to 0, \tag{2.36}$$

$$(A^*)^{-1}Cx_n^{(1)} - x_n^{(2)} \to 0. \tag{2.37}$$

Furthermore, we obtain

$$x_n^{(2)} \to 0, \quad x_n^{(1)} \to 0,$$

which is contradictory to $\|x_n\| = 1$. Therefore, H is invertible. The proof is completed □

According to the definition of structural stable matrices in [59], we define structurally stable operators.

Definition 2.5.25 An operator without spectral points on the imaginary axis is a structural stable operator.

By Theorems 2.5.23 and 2.5.24, it is easy to draw the following conclusions.

Corollary 2.5.26 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a non-negative Hamiltonian operator, and* $\mathfrak{J}H$ *is a maximal accretive operator.*

(i) If $0 \in \rho(B) \cap \rho(C)$, *then* H *is structural stable;*
(ii) If $\mathscr{D}(A) \subset \mathscr{D}(C)$, $\mathscr{D}(A^*) \subset \mathscr{D}(B)$ *and* A *is structural stable, then* H *is structural stable.*

Obviously, it is not easy to determine the maximal accretive of $\mathfrak{J}H$ (or symplectic self-adjointness of H). Therefore, the following will give some criteria for the invertibility of the non-negative Hamiltonian operator.

Lemma 2.5.27 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If there exists* $M > 0$ *such that*

$$|(By, y) + (Cx, x)| \geq M \left\| \begin{bmatrix} x \\ y \end{bmatrix} \right\|^2$$

for arbitrary $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$, *then* H *has a bounded inverse.*

Proof If there exists $M > 0$ such that

$$|(By, y) + (Cx, x)| \geq M \| \begin{bmatrix} x \\ y \end{bmatrix} \|^2$$

for arbitrary $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$, then

$$|(\mathfrak{J}Hu, u)| = |(By, y) + (Cx, x) + 2\mathrm{i}Im(Ax, y))| \geq |(By, y) + (Cx, x)| \geq M \| \begin{bmatrix} x \\ y \end{bmatrix} \|^2,$$

and thus $\mathfrak{J}H$ has a bounded inverse, that is, H has a bounded inverse, where $\mathfrak{J} = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$, $u = \begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$. □

Theorem 2.5.28 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If there exists* $M > 0$ *such that*

$$|(By, y) + (Cx, x)| \geq M \| \begin{bmatrix} x \\ y \end{bmatrix} \|^2$$

for arbitrary $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$ *and* $0 \in \rho(A)$, *and* $I + (CA^{-1}B(A^*)^{-1})^*$ *is injective, then* $0 \in \rho(H)$.

Proof Since $\mathscr{D}(A) \subset \mathscr{D}(C)$, $\mathscr{D}(A^*) \subset \mathscr{D}(B)$ and $0 \in \rho(A)$, it is easy to get

$$H = \begin{bmatrix} I & B(A^*)^{-1} \\ CA^{-1} & -I \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & A^* \end{bmatrix}.$$

Because $T = \begin{bmatrix} I & B(A^*)^{-1} \\ CA^{-1} & -I \end{bmatrix}$ is bounded such that

$$T^* = \begin{bmatrix} I & \overline{(A^*)^{-1}C} \\ \overline{A^{-1}B} & -I \end{bmatrix},$$

we have

$$H^* = \begin{bmatrix} A^* & 0 \\ 0 & A \end{bmatrix} \begin{bmatrix} I & \overline{(A^*)^{-1}C} \\ \overline{A^{-1}B} & -I \end{bmatrix}.$$

By Lemma 2.5.27, H has a bounded inverse, that is, $0 \in \rho(H) \cup \sigma_{r,1}(H)$, and hence

$$0 \in \rho(H^*) \cup \sigma_{p,1}(H^*).$$

Because $0 \in \rho(H)$ if and only if $0 \in \rho(H^*)$, we only need to show $0 \notin \sigma_p(H^*)$ in order to prove $0 \in \rho(H)$. Assume $0 \in \sigma_p(H^*)$, then there exists vectors $x_0, y_0 \in X$ which are not all zero such that

$$\begin{bmatrix} A^* & 0 \\ 0 & A \end{bmatrix} \begin{bmatrix} I & \overline{(A^*)^{-1}C} \\ \overline{A^{-1}B} & -I \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = 0,$$

and hence

$$(I + (CA^{-1}BA(A^*)^{-1})^*)x_0 = 0,$$

which is a contradiction to the fact that $I + (CA^{-1}B(A^*)^{-1})^*$, so $0 \notin \sigma_p(H^*)$ is injective, that is, $0 \in \rho(H)$. □

Theorem 2.5.29 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If there exists* $M > 0$ *such that*

$$|(By, y) + (Cx, x)| \geq M \| \begin{bmatrix} x \\ y \end{bmatrix} \|^2$$

for arbitrary $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$ *and* $I + (AC^{-1}A^*B^{-1})^*$ *is injective, then* $0 \in \rho(H)$.

Proof It is easy to know

$$H = \begin{bmatrix} I & AC^{-1} \\ -A^*B^{-1} & I \end{bmatrix} \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix},$$

and

$$H^* = \begin{bmatrix} 0 & C \\ B & 0 \end{bmatrix} \begin{bmatrix} I & \overline{-B^{-1}A} \\ \overline{C^{-1}A^*} & I \end{bmatrix}.$$

By Lemma 2.5.27, we know that H has a bounded inverse, i.e., $0 \in \rho(H) \cup \sigma_{r,1}(H)$, so

$$0 \in \rho(H^*) \cup \sigma_{p,1}(H^*).$$

Because $0 \in \rho(H)$ if and only if $0 \in \rho(H^*)$, we only need to prove $0 \notin \sigma_p(H^*)$ in order to prove $0 \in \rho(H)$. Assume $0 \in \sigma_p(H^*)$, then there exist $x_0, y_0 \in X$ which are not all zero such that

$$\begin{bmatrix} 0 & C \\ B & 0 \end{bmatrix} \begin{bmatrix} I & \overline{-B^{-1}A} \\ \overline{C^{-1}A^*} & I \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = 0.$$

Considering $0 \in \rho(B) \cap \rho(C)$, $x_0 \neq 0$ and

$$(I + (AC^{-1}A^*B^{-1})^*)x_0 = 0,$$

which contradicts the injectiveness of $I + (AC^{-1}A^*B^{-1})^*$, so $0 \notin \sigma_p(H^*)$, that is, $0 \in \rho(H)$. □

Applying the above conclusions, we can solve the inverse compactness problem of the infinite-dimensional Hamiltonian operator (that is, the inverse operator is a compact operator).

Corollary 2.5.30 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be an infinite-dimensional Hamiltonian operator, there exists* $M > 0$ *such that*

$$|(By, y) + (Cx, x)| \geq M \| \begin{bmatrix} x \\ y \end{bmatrix} \|^2,$$

for arbitrary $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$ *and* $I + (AC^{-1}A^*B^{-1})^*$ *is injective, and* B^{-1}, C^{-1} *are compact operators, then* H^{-1} *is also a compact operator.*

Proof Because

$$H = \begin{bmatrix} I & AC^{-1} \\ -A^*B^{-1} & I \end{bmatrix} \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix},$$

$0 \in \rho(H)$ and $0 \in \rho(B) \cap \rho(C)$, $\begin{bmatrix} I & AC^{-1} \\ -A^*B^{-1} & I \end{bmatrix}$ is invertible and

$$H^{-1} = \begin{bmatrix} 0 & C^{-1} \\ B^{-1} & 0 \end{bmatrix} \begin{bmatrix} I & AC^{-1} \\ -A^*B^{-1} & I \end{bmatrix}^{-1},$$

and H^{-1} is clearly a compact operator. □

Corollary 2.5.31 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If there exists* $M > 0$ *such that*

$$|(By, y) + (Cx, x)| \geq M \| \begin{bmatrix} x \\ y \end{bmatrix} \|^2$$

for any $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$*, operator* $I + (CA^{-1}B(A^*)^{-1})^*$ *is injective, and* A^{-1} *is a compact operator, then* H^{-1} *is a compact operator.*

Proof According to the given condition, we know $0 \in \rho(H)$ and

$$H^{-1} = \begin{bmatrix} A^{-1} & 0 \\ 0 & (A^*)^{-1} \end{bmatrix} \begin{bmatrix} I & B(A^*)^{-1} \\ CA^{-1} & -I \end{bmatrix}^{-1}$$

by Theorem 2.5.28. It is obvious that H^{-1} is a compact operator. □

2.6 Spectra of the General Infinite-Dimensional Hamiltonian Operators

For the general infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix},$$

its natural domain is

$$\mathscr{D}(H) = (\mathscr{D}(A) \cap \mathscr{D}(C)) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*)).$$

The operation of the general infinite-dimensional Hamiltonian operator needs not only to satisfy the operation rules of block operator matrices but also to satisfy those of general unbounded linear operators. The operation of general infinite-dimensional Hamiltonian operators is more complex than that of unbounded operators. The primary purpose of studying the general infinite-dimensional Hamiltonian operators is to use the information of the entries A, B, C to characterize the relevant information of the block operator matrix H. However, this is not easy, and the main difficulties are reflected in the following aspects:

(a) General linear operators are required to be closed or closable when they are studied, but, for general infinite-dimensional Hamiltonian operators, the closedness of the block operator matrix H cannot be guaranteed, even if each matrix entry is a closed linear operator. Besides, the domain of general infinite-dimensional Hamiltonian operators can also be not dense.

(b) The closure and conjugate operators of general densely defined closable linear operators can be characterized in general. However, the forms of the closure and conjugate operators of general infinite-dimensional Hamiltonian operators are very complicated and even hard to have block forms. For example, let A be an unbounded self-adjoint operator, then there is no bock forms for the conjugate operator of the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} = \begin{bmatrix} A & -A \\ A & -A \end{bmatrix}.$$

(c) For the bounded infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix},$$

under commutative conditions, the spectral properties (e.g., invertibility) of H can be characterized by the operator $AA^* + BC$. For the unbounded case, the domain of the operator $AA^* + BC$ may not be dense even.

(d) The numerical range of the general linear operator T is defined as $W(T) = \{(Tx, x) : \|x\| = 1, x \in \mathscr{D}(T)\}$ which is an important tool to characterize the spectral distribution of linear operators. The numerical range of general infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$$

is given by

$$W(H) = \{\lambda \in \mathbb{C} : \lambda = (Ax, x) + (By, x) + (Cx, y) - (A^*y, y), \\ \|x\|^2 + \|y\|^2 = 1, x \in \mathscr{D}(A) \cap \mathscr{D}(C), y \in \mathscr{D}(B) \cap \mathscr{D}(A^*)\},$$

and it becomes very complicated. In view of this, in this section, we will give the characterization of closure for infinite-dimensional Hamiltonian operators, and then use the quadratic complement of block operator matrices to study the spectral properties of infinite-dimensional Hamiltonian operators.

2.6.1 Closure of Infinite-Dimensional Hamiltonian Operators

In quantum mechanics, the operator corresponding to the system represents a measure of the total energy (if measurable) of the system. The energy change of the system can be characterized by the spectra. For example, if the operator corresponding to the system is closed, then the energy of the system is conserved. For the densely defined infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : (\mathscr{D}(A) \cap \mathscr{D}(C)) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*)) \to X \times X,$$

it is easy to prove

$$H \subset JH^*J,$$

where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. From the invertibility of J and the closedness of H^*, we obtain that JH^*J is a closed extension of H, that is, H is a closable operator and

$$\overline{H} = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}^{**} \subset \begin{bmatrix} A^* & C \\ B & -A \end{bmatrix}^*.$$

However, H is not necessarily closed.

Example 2.6.1 Let the infinite-dimensional Hamiltonian operator be

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} = \begin{bmatrix} A & -A \\ A & -A \end{bmatrix},$$

where A is an unbounded self-adjoint operator, then it can be asserted that H is not a closed operator. In fact, taking $x_0 \notin \mathscr{D}(A)$, then there exists $\{x_n\} \subset \mathscr{D}(A)$ such that $x_n \to x_0$ as $n \to \infty$, so $\{\begin{bmatrix} x_n & x_n \end{bmatrix}^T\} \subset \mathscr{D}(H)$, $\begin{bmatrix} x_n & x_n \end{bmatrix}^T \to \begin{bmatrix} x_0 & x_0 \end{bmatrix}^T$ and

$$\begin{bmatrix} A & -A \\ A & -A \end{bmatrix} \begin{bmatrix} x_n \\ x_n \end{bmatrix} = 0.$$

Assume H is a closed operator, then we know $\begin{bmatrix} x_0 & x_0 \end{bmatrix}^T \in \mathscr{D}(H)$ by Theorem 2.1.3, which is contradictory to $x_0 \notin \mathscr{D}(A)$.

Lemma 2.6.2 *Let R, T be bounded operators on the whole space and $0 \in \rho(R) \cap \rho(T)$.*

(i) RST is closed if and only if S is closed;
(ii) RST is closable if and only if S is closable and $\overline{RST} = R\overline{S}T$.

Proof (i) For any $\{x_n\} \subset \mathscr{D}(RST) = \{x : Tx \in \mathscr{D}(S)\}$ such that $x_n \to x_0$ and $RSTx_n \to y_0$, we have

$$Tx_n \to Tx_0, \quad STx_n \to R^{-1}y_0,$$

since S is closed and R, T are bounded on the whole space and invertible. Note that S is a closed operator, $Tx_0 \in \mathscr{D}(S)$ and $STx_0 = R^{-1}y_0$, that is, $x_0 \in \mathscr{D}(RST)$ and $RSTx_0 = y_0$. By Theorem 2.1.3, we obtain that RST is closed.

Conversely, if RST is closed, then let $\mathscr{A} = RST$, and hence $S = R^{-1}\mathscr{A}T^{-1}$. Considering that R^{-1} and T^{-1} are bounded on the whole space and invertible, completely similar to the above proof, we can prove that S is closed.

(ii) Since S is closable, for any $\{x_n\} \subset \mathscr{D}(RST) = \{x : Tx \in \mathscr{D}(S)\}$ such that $x_n \to 0$, $RSTx_n \to y_0$, and by considering R, T are bounded and invertible, it can be proved that

$$Tx_n \to 0, \quad STx_n \to R^{-1}y_0.$$

Because S is closable, $R^{-1}y_0 = 0$, that is, $y_0 = 0$. By Theorem 2.1.8, we obtain that RST is closable.

Conversely, when RST is closable, let $\mathscr{A} = RST$ and then $S = R^{-1}\mathscr{A}T^{-1}$. From the fact R^{-1}, T^{-1} are bounded and invertible, completely similar to the above proof, it can be proved that S is closed. Because $RST \subset R\overline{S}T$ and $\overline{RST}$ is the minimal closed extension of RST,

$$\overline{RST} \subset R\overline{S}T.$$

For any $x \in \mathscr{D}(R\overline{S}T)$, there exists $\{u_n\} \subset \mathscr{D}(S)$ such that

$$u_n \to Tx, \quad Su_n \to \overline{S}Tx.$$

By $Tx \in \mathscr{D}(\overline{S})$, let $u_n = Tx_n$, then

$$x_n \to x, \quad STx_n \to \overline{S}Tx,$$

that is, $x_n \to x$, $RSTx_n \to R\overline{S}Tx$. Therefore, $x \in \mathscr{D}(\overline{RST})$, that is, $\mathscr{D}(R\overline{S}T) \subset \mathscr{D}(\overline{RST})$. Thus $\overline{RST} = R\overline{S}T$. □

Theorem 2.6.3 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be an infinite-dimensional Hamiltonian operator.*

(i) If $\mathscr{D}(B) \subset \mathscr{D}(A^*)$, *then* $S_1(\lambda) = C + A^*(B - \lambda I)^{-1}A$ *is a closable operator and*

$$\overline{H} = \begin{bmatrix} I & 0 \\ -A^*(B-\lambda I)^{-1} & I \end{bmatrix} \begin{bmatrix} 0 & B-\lambda I \\ \overline{S_1(\lambda)} & 0 \end{bmatrix} \begin{bmatrix} I & 0 \\ \overline{(B-\lambda I)^{-1}A} & I \end{bmatrix} + \begin{bmatrix} 0 & \lambda I \\ 0 & 0 \end{bmatrix}$$

for any $\lambda \in \mathbb{C}\backslash\mathbb{R}$, *where* $\overline{S_1(\lambda)}$ *represents the closure of operator* $S_1(\lambda)$. *If it satisfies* $\mathscr{D}(\overline{S_1(\lambda)}) \subset \mathscr{D}((B-\lambda I)\overline{(B-\lambda I)^{-1}A})$, *then*

$$\overline{H} = \begin{bmatrix} (B-\lambda I)\overline{(B-\lambda I)^{-1}A} & B \\ -A^*\overline{(B-\lambda I)^{-1}A} + \overline{S_1(\lambda)} & -A^* \end{bmatrix}.$$

(ii) If $\mathscr{D}(C) \subset \mathscr{D}(A)$, *then* $S_2(\lambda) = B + A(C - \lambda I)^{-1}A^*$ *is a closable operator and*

$$\overline{H} = \begin{bmatrix} I & A(C-\lambda I)^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} 0 & \overline{S_2(\lambda)} \\ C-\lambda I & 0 \end{bmatrix} \begin{bmatrix} I & -\overline{(C-\lambda I)^{-1}A^*} \\ 0 & I \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \lambda I & 0 \end{bmatrix}$$

for any $\lambda \in \mathbb{C}\backslash\mathbb{R}$, *where* $\overline{S_2(\lambda)}$ *represents the closure of operator* $S_2(\lambda)$. *If* $\mathscr{D}(\overline{S_2(\lambda)}) \subset \mathscr{D}((C-\lambda I)\overline{(C-\lambda I)^{-1}A^*})$, *then*

$$\overline{H} = \begin{bmatrix} A & \overline{S_2(\lambda)} - A\overline{(C-\lambda I)^{-1}A^*} \\ C & -(C-\lambda I)\overline{(C-\lambda I)^{-1}A^*} \end{bmatrix}.$$

Proof The proofs of conclusions (i) and (ii) are completely similar. Because B is a self-adjoint operator, there is $\lambda \in \rho(B)$ and

$$H = \begin{bmatrix} I & 0 \\ -A^*(B-\lambda I)^{-1} & I \end{bmatrix} \begin{bmatrix} 0 & B-\lambda I \\ S_1(\lambda) & 0 \end{bmatrix} \begin{bmatrix} I & 0 \\ \overline{(B-\lambda I)^{-1}A} & I \end{bmatrix} + \begin{bmatrix} 0 & \lambda I \\ 0 & 0 \end{bmatrix}$$

for any $\lambda \in \mathbb{C}\backslash\mathbb{R}$, where $S_1(\lambda) = C + A^*(B - \lambda I)^{-1}A$. By Lemma 2.6.2, we obtain

$$\overline{H} = \begin{bmatrix} I & 0 \\ -A^*(B-\lambda I)^{-1} & I \end{bmatrix} \begin{bmatrix} 0 & B-\lambda I \\ \overline{S_1(\lambda)} & 0 \end{bmatrix} \begin{bmatrix} I & 0 \\ \overline{(B-\lambda I)^{-1}A} & I \end{bmatrix} + \begin{bmatrix} 0 & \lambda I \\ 0 & 0 \end{bmatrix}.$$

If $\mathscr{D}(\overline{S_1(\lambda)}) \subset \mathscr{D}((B-\lambda I)\overline{(B-\lambda I)^{-1}A})$, then

$$\begin{bmatrix} 0 & B-\lambda I \\ \overline{S_1(\lambda)} & 0 \end{bmatrix}\begin{bmatrix} I & 0 \\ \overline{(B-\lambda I)^{-1}A} & I \end{bmatrix} = \begin{bmatrix} (B-\lambda I)\overline{(B-\lambda I)^{-1}A} & B-\lambda I \\ \overline{S_1(\lambda)} & 0 \end{bmatrix},$$

so

$$\overline{H} = \begin{bmatrix} (B-\lambda I)\overline{(B-\lambda I)^{-1}A} & B \\ -A^*\overline{(B-\lambda I)^{-1}A} + \overline{S_1(\lambda)} & -A^* \end{bmatrix}.$$

The proof is completed. □

Remark 2.6.4 In the above factorizations

$$S_1(\lambda) = C + A^*(B-\lambda I)^{-1}A, \lambda \in \mathbb{C}\backslash\mathbb{R},$$

$$S_2(\lambda) = B + A(C-\lambda I)^{-1}A^*, \lambda \in \mathbb{C}\backslash\mathbb{R},$$

are called the quadratic complements of infinite-dimensional Hamiltonian operators.

Using the above conclusions, it is easy to get the following inferences.

Corollary 2.6.5 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator.*

(i) If $\mathscr{D}(B) \subset \mathscr{D}(A^*)$*, then* H *is a closed operator if and only if there exists* $\lambda \in \mathbb{C}\backslash\mathbb{R}$ *such that* $S_1(\lambda) = C + A^*(B-\lambda I)^{-1}A$ *is a closed operator;*

(ii) If $\mathscr{D}(C) \subset \mathscr{D}(A)$*, then* H *is a closed operator if and only if there exists* $\lambda \in \mathbb{C}\backslash\mathbb{R}$ *such that* $S_2(\lambda) = B + A(C-\lambda I)^{-1}A^*$ *is a closed operator.*

Theorem 2.6.6 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be an infinite-dimensional Hamiltonian operator.*

(i) If $\mathscr{D}(B) \subset \mathscr{D}(A^*)$ *and* B *is invertible, then* $\sigma_*(H-\lambda I) = \sigma_*(S_3(\lambda))$*;*

(ii) If $\mathscr{D}(C) \subset \mathscr{D}(A)$ *and* C *is invertible, then* $\sigma_*(H-\lambda I) = \sigma_*(S_4(\lambda))$*, where* $\sigma_* \in \{\sigma_p, \sigma_r, \sigma_c, \sigma_{ap}, \sigma_{e,i}, i = 1, 2, \ldots, 8\}$ *and* $S_3(\lambda) = C + (A^* + \lambda I)B^{-1}(A - \lambda I)$*,* $S_4(\lambda) = B + (A - \lambda I)C^{-1}(A^* + \lambda I)$*.*

Proof Only prove (i), and the proof of conclusion (ii) is completely similar to that of (i). If $\mathscr{D}(B) \subset \mathscr{D}(A^*)$ and B is invertible, then

$$H - \lambda I = \begin{bmatrix} I & 0 \\ -(A^* + \lambda I)B^{-1} & I \end{bmatrix}\begin{bmatrix} 0 & B \\ S_3(\lambda) & 0 \end{bmatrix}\begin{bmatrix} I & 0 \\ B^{-1}(A-\lambda I) & I \end{bmatrix},$$

so the conclusion holds. □

From the above theorem, it is easy to obtain that the quadratic complements $S_3(\lambda)$ and $S_4(\lambda)$ are symmetric operators when $\lambda \in \mathtt{i}\mathbb{R}$, which is convenient for characterizing the spectral properties of $S_3(\lambda)$ and $S_4(\lambda)$. In view of this, the problem when there is only pure imaginary spectra for infinite-dimensional Hamiltonian operators will be discussed below.

Theorem 2.6.7 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be an infinite-dimensional Hamiltonian operator with* $(JH)^* = JH \leq 0$*, then* $(\mathbb{C}_+ \cup \mathbb{C}_-) \subset \rho(H)$*, that is,* H *only have pure imaginary spectrum, where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$*, and* $\mathbb{C}_-, \mathbb{C}_+$ *represent the left and right half-open planes, respectively.*

Proof It will take 4 steps to complete the proof. First of all, we need to prove $\sigma_p(H) \cap (\mathbb{C}_+ \cup \mathbb{C}_-) = \emptyset$. Assume $\lambda \in \sigma_p(H) \cap (\mathbb{C}_+ \cup \mathbb{C}_-)$ and x_0 is the corresponding eigenvector, then

$$(JHx_0, x_0) = \lambda(Jx_0, x_0) \leq 0.$$

On the other hand, $(Jx_0, x_0) \in \mathtt{i}\mathbb{R}$ and $Re(\lambda) \neq 0$, then we can obtain

$$(Jx_0, x_0) = 0,$$

an thus $(JHx_0, x_0) = 0$. From Lemma 2.3.4, it follows that $JHx_0 = 0$, that is, $Hx_0 = \lambda x_0 = 0$, which is contradictory to $x_0 \neq 0$. This proves $\sigma_p(H) \cap (\mathbb{C}_+ \cup \mathbb{C}_-) = \emptyset$.

Secondly we prove $\sigma_r(H) = \emptyset$. If there exists $\lambda = -\overline{\lambda}$, such that $\lambda \in \sigma_r(H)$, then $\overline{\lambda} \in \sigma_p(H^*)$. Taking into account $H^* = JHJ$ and $J^2 = -I$, we get $\lambda \in \sigma_p(H)$, which is contradictory to $\lambda \in \sigma_r(H)$. If there exists $\lambda \neq -\overline{\lambda}$ such that $\lambda \in \sigma_r(H)$, then similarly we obtain $-\overline{\lambda} \in \sigma_p(H)$. This is contradictory to $\sigma_p(H) \cap (\mathbb{C}_+ \cup \mathbb{C}_-) = \emptyset$, so $\sigma_r(H) = \emptyset$.

Then we prove if H is a continuous operator, it can be derived that $(\mathbb{C}_+ \cup \mathbb{C}_-) \subset \rho(H)$. Assume there exists $-\overline{\lambda}_0 \neq \lambda_0 \in \sigma(H)$, then we know $\lambda_0 \in \sigma_c(H)$ by the proof of last two parts. Thus there exists an orthogonalizing sequence $\{x_n\}(\|x_n\| = 1, n = 1, 2, \ldots)$, such that $(H - \lambda_0 I)x_n \to 0$ as $n \to \infty$, that is,

$$(J(H - \lambda_0 I)x_n, x_n) = (J(Hx_n - Im(\lambda_0)I)x_n, x_n) - \mathtt{i}Re(\lambda_0)(Jx_n, x_n) \to 0.$$

It follows from $Re(\lambda_0) \neq 0$ that $(Jx_n, x_n) \to 0$, so

$$(JHx_n, x_n) \to 0.$$

Because of

$$(-JHx_n, x_n) = ((-JH)^{\frac{1}{2}}x_n, (-JH)^{\frac{1}{2}}x_n),$$

$(-JH)^{\frac{1}{2}}x_n \to 0$. Due to

$$Hx_n = J(-JH)^{\frac{1}{2}}(-JH)^{\frac{1}{2}}x_n,$$

we have $Hx_n \to 0$, that is, $\lambda x_n \to 0$, which is contradictory to $\|x_n\| = 1$. This proves that if H is a continuous operator, $(\mathbb{C}_+ \cup \mathbb{C}_-) \subset \rho(H)$.

Finally we prove for the general H that $(\mathbb{C}_+ \cup \mathbb{C}_-) \subset \rho(H)$. Let $\mu \in \rho(H)$, then $-\overline{\mu} \in \rho(H)$. Define

$$B = (H + \overline{\mu}I)^{-1}H(H - \mu I)^{-1},$$

then it is easy to prove $(JB)^* = JB \geq 0$, we know

$$\sigma(B) \cap (\mathbb{C}_+ \cup \mathbb{C}_-) = \emptyset$$

by the proof before. Because

$$\sigma(B) = \left\{ \frac{\lambda}{(\lambda + \overline{\mu})(\lambda - \mu)} : \lambda \in \sigma(H) \right\},$$

if $|\lambda_0| \neq |\mu|$, $\dfrac{\lambda_0}{(\lambda_0 + \overline{\mu})(\lambda_0 - \mu)} \notin \mathtt{i}\mathbb{R}$, then $\lambda_0 \in \rho(B)$ for any $Re(\lambda_0) \neq 0$, that is, $\lambda_0 \in \rho(H)$. Therefore

$$(\mathbb{C}_+ \cup \mathbb{C}_-)\backslash\{\lambda : |\lambda| = |\mu|\} \subset \rho(H). \tag{2.38}$$

Let $\mu_1 \in \rho(H)$, $|\mu_1| \neq |\mu|$, then we can similarly prove

$$(\mathbb{C}_+ \cup \mathbb{C}_-)\backslash\{\lambda : |\lambda| = |\mu_1|\} \subset \rho(H). \tag{2.39}$$

From (2.38) and (2.39), it follows that $(\mathbb{C}_+ \cup \mathbb{C}_-) \subset \rho(H)$. □

Similarly the following conclusion can be proved.

Theorem 2.6.8 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be an infinite-dimensional Hamiltonian operator with* $(JH)^* = JH \geq 0$, *then* $(\mathbb{C}_+ \cup \mathbb{C}_-) \subset \rho(H)$, *that is, H has pure imaginary spectrum only, where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$, $\mathbb{C}_-, \mathbb{C}_+$ *represent the left and right half-open planes, respectively.*

2.6.2 Dichotomy of the Spectra of Infinite-Dimensional Hamiltonian Operators

Given a densely defined closed linear operator T, if $\sigma(T)$ has no intersection with the imaginary axis, then $\sigma(T)$ is said to have dichotomy . The dichotomy of $\sigma(T)$ is closely related to the solution of differential equations. Therefore, the dichotomy of

$\sigma(T)$ has received extensive attention. In this section, we will discuss the dichotomy of the spectrum $\sigma(H)$ for infinite-dimensional Hamiltonian operators.

Theorem 2.6.9 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator, and* B, C *be non-negative self-adjoint operators defined in the whole space. If there exist* $c, w_0 > 0$ *such that*

$$\sup_{|Re(z)|<w_0} \|(A - z)^{-1}\| \le c < \infty,$$

then there exist $0 < w < w_0$ *and* $\widehat{c}$ *such that*

$$\{z : |Re(z)| < w\} \subset \rho(H), \quad \sup_{|Re(z)|<w} \|(H - z)^{-1}\| \le \widehat{c}.$$

Proof Take $w > 0$ such that

$$w + \max\{\|B\|^{\frac{1}{2}}, \|C\|^{\frac{1}{2}}\}\sqrt{w} < \frac{1}{\sqrt{2}c}. \tag{2.40}$$

First of all, we prove there exists $d > 0$ such that

$$\|(H - z)^{-1}u\| \ge d$$

for any $z \in \{z \in \mathbb{C} : |Re(z)| < w\}$ and all $u \in \mathscr{D}(H)$, $\|u\| = 1$. Otherwise, $z \in \sigma_{ap}(H)$, and there exists $\begin{bmatrix} x_n & y_n \end{bmatrix}^T \in \mathscr{D}(H)$ and $z_n = s_n + \mathtt{i}t_n$, $|s_n| < w, t_n \in \mathscr{R}$, $n = 1, 2, \ldots$, such that $\|x_n\|^2 + \|y_n\|^2 = 1$ and

$$(A - s_n - \mathtt{i}t_n)x_n + By_n \to 0, \quad Cx_n - (A^* + s_n + \mathtt{i}t_n)x_n \to 0$$

as $n \to \infty$. Taking the inner product of the first equation by y_n and of the second equation by x_n and adding the resulting equations, we have

$$(By_n, y_n) + (Cx_n, x_n) - 2s_n(x_n, y_n) \to 0.$$

Then

$$\begin{aligned} 0 \le & \|B^{\frac{1}{2}}y_n\|^2 + \|C^{\frac{1}{2}}x_n\|^2 \\ \le & 2|s_n||(x_n, y_n)| + |\|B^{\frac{1}{2}}y_n\|^2 + \|C^{\frac{1}{2}}x_n\|^2 - 2s_n(x_n, y_n)| \\ \le & |s_n| + |\|B^{\frac{1}{2}}y_n\|^2 + \|C^{\frac{1}{2}}x_n\|^2 - 2s_n(x_n, y_n)|, \end{aligned}$$

so

$$\|B^{\frac{1}{2}}y_n\| \le \sqrt{|s_n| + |\|B^{\frac{1}{2}}y_n\|^2 + \|C^{\frac{1}{2}}x_n\|^2 - 2s_n(x_n, y_n)|}$$

and

$$\|C^{\frac{1}{2}}x_n\| \le \sqrt{|s_n| + |\|B^{\frac{1}{2}}y_n\|^2 + \|C^{\frac{1}{2}}x_n\|^2 - 2s_n(x_n, y_n)|}.$$

On the other hand, considering $\|x_n\|^2 + \|y_n\|^2 = 1$, suppose

$$\liminf_{n\to\infty} \|x_n\|^2 \ge \frac{1}{2},$$

then

$$\begin{aligned}
\frac{1}{\sqrt{2}c} &\le \liminf_{n\to\infty} \frac{1}{c}\|x_n\| \\
&\le \liminf_{n\to\infty} \|(A - \mathrm{i}t_n)x_n\| \\
&\le \liminf_{n\to\infty} (\|s_n x_n\| + \|By_n\|) \\
&\le \liminf_{n\to\infty} (|s_n|\|x_n\| + \|B^{\frac{1}{2}}\|\|B^{\frac{1}{2}}y_n\|) \\
&\le \liminf_{n\to\infty} (|s_n| + \|B^{\frac{1}{2}}\|\sqrt{|s_n|}) \le w + \|B^{\frac{1}{2}}\|\sqrt{w}.
\end{aligned}$$

This is contradictory to (2.40), so $z \notin \sigma_{ap}(H)$.

Similarly, considering

$$H^* = \begin{bmatrix} A^* & C \\ B & -A \end{bmatrix},$$

we can find $d > 0$ such that

$$\|(H^* - zI)^{-1}u\| \ge d,$$

for any $z \in \{z \in \mathbb{C} : |Re(z)| < w\}$ and $u \in \mathscr{D}(H^*)$, $\|u\| = 1$. This proves $z \notin \sigma_{ap}(H^*)$. Therefore,

$$\{z : |Re(z)| < w\} \subset \rho(H), \qquad \sup_{|Re(z)|<w} \|(H - zI)^{-1}\| \le \widehat{c}.$$

The proof is completed. □

For an infinite-dimensional Hamiltonian operator with unbounded skew diagonal entries, the following conclusions can be obtained about the dichotomy of its spectrum.

Theorem 2.6.10 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If* $\sigma_r(H) \cap \mathrm{i}\mathbb{R} = \emptyset$, $B \ge 0, C \ge 0$, *and there exists* $w > 0$ *such that*

$$\Delta_w = \{z : |Re(z)| < w\} \subset \rho(A)$$

and

$$\sup_{\lambda\in\Delta_w}\|C^{\frac{1}{2}}(A-\lambda I)^{-1}\|<\infty,\ \sup_{\lambda\in\Delta_w}\|B^{\frac{1}{2}}(A^*+\lambda I)^{-1}\|<\infty,$$

then there exists $0<w_0\le w$ *such that* $\Delta_{w_0}\subset\rho(H)$.

Proof Since A is invertible and $\sigma_r(H)\cap \mathtt{i}\mathbb{R}=\emptyset$,

$$H^*=\begin{bmatrix}A^* & C\\ B & -A\end{bmatrix}. \tag{2.41}$$

Let

$$M=\max\{\sup_{\lambda\in\Delta_w}\|C^{\frac{1}{2}}(A-\lambda I)^{-1}\|,\ \sup_{\lambda\in\Delta_w}\|B^{\frac{1}{2}}(A^*+\lambda I)^{-1}\|\},$$

then suppose $M>0$. If $M=0$, then $B=C=0$, the conclusion is clearly established. Take $w_0=\frac{1}{2M^2}$, then it can be asserted that

$$\Delta_{w_0}=\{\lambda:|Re(\lambda)|<w_0\}\subset\rho(H).$$

In fact, let there exist $\lambda\in\Delta_{w_0}$ such that $\lambda\in\sigma_{ap}(H)$, then there exists sequence $\|x_n\|^2+\|y_n\|^2=1, n=1,2,\ldots,$ such that

$$(A-\lambda I)x_n+By_n\to 0 \tag{2.42}$$

and

$$Cx_n-(A^*+\lambda I)y_n\to 0 \tag{2.43}$$

as $n\to\infty$, so

$$(By_n,y_n)+(Cx_n,x_n)-2Re(\lambda)(x_n,y_n)\to 0,$$

and thus

$$\liminf_{n\to\infty}(\|B^{\frac{1}{2}}y_n\|^2+\|C^{\frac{1}{2}}x_n\|^2)\le|Re(\lambda)|. \tag{2.44}$$

Without loss of generality, let $\liminf_{n\to\infty}\|x_n\|^2\ge\frac{1}{2}$, then it follows from (2.42) and (2.44) that

$$\begin{aligned}\frac{1}{\sqrt{2}}&\le\liminf_{n\to\infty}\|x_n\|\\&\le\liminf_{n\to\infty}\|(A-\lambda I)^{-1}B^{\frac{1}{2}}\|\|B^{\frac{1}{2}}y_n\|\\&\le M\sqrt{|Re(\lambda)|}<\frac{1}{\sqrt{2}}.\end{aligned}$$

The contradiction is derived, so

$$\Delta_{w_0} \subset \rho(H) \cup \sigma_{r,1}(H). \tag{2.45}$$

On the other hand, since the relation (2.41), we have

$$\Delta_{w_0} \subset \rho(H^*) \cup \sigma_{r,1}(H^*). \tag{2.46}$$

Combining (2.45) and (2.46), we obtain $\Delta_{w_0} \subset \rho(H)$. The proof is completed. □

Chapter 3
Completeness of Eigenvector System of Infinite-Dimensional Hamiltonian Operators



3.1 Infinite-Dimensional Symplectic Spaces

All the conserved physical processes can be expressed as proper Hamiltonian systems, and their common mathematical basis is the symplectic space. Unlike the Euclidean space, which investigates metric properties such as length, the symplectic space studies the area or the work instead [28]. The eigenfunction system of an infinite-dimensional Hamiltonian operator is symplectic orthogonal. Therefore, this section takes the symplectic structure as the main line, and briefly introduces the infinite-dimensional complex symplectic space and some other basic properties.

3.1.1 Infinite-Dimensional Complex Symplectic Spaces and Examples

Definition 3.1.1 Let X be an infinite-dimensional linear space over $\mathbb{C}$, and define a bivariate function (or functional) $Q(x, y) =< x, y >$ for any $x, y \in X$. Then $< \cdot, \cdot >$ is called a symplectic structure (or symplectic inner product), if the following four conditions are satisfied:
(i) (non-degeneracy) If $Q(x, y) = 0$ for all $y \in X$, then $x = 0$,
(ii) (symplectic conjugate symmetry) $Q(x, y) = -\overline{Q(y, x)}$,
(iii) (homogeneity) $Q(\alpha x, y) = \alpha Q(x, y)$,
(iv) (additivity) $Q(x_1 + x_2, y) = Q(x_1, y) + Q(x_2, y)$,
where $\alpha \in \mathbb{C}$, $x_1, x_2, x, y \in X$. After endowing with a symplectic structure, $\{X, < \cdot, \cdot >\}$ is said to be an infinite-dimensional complex symplectic space.

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Remark 3.1.2 Let $\{X \times X, (\cdot, \cdot)\}$ be an infinite-dimensional complex inner product space over $\mathbb{C}$, and take $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$ where I is the identical operator on X. Define

$$< x, y >= (Jx, y),$$

then $\{X \times X, < \cdot, \cdot >\}$ is an infinite-dimensional complex symplectic space. According to the definition of inner product, it is easy to prove that $< \cdot, \cdot >$ satisfies the properties of (i), (iii), and (iv). Since $J^* = -J$, we get

$$(Jx, y) = -\overline{(Jy, x)},$$

which means $< \cdot, \cdot >$ satisfies the property (ii). Therefore, $\{X \times X, < \cdot, \cdot >\}$ is a symplectic space. Similarly, we can define a bivariate function

$$< x, y >= (\mathfrak{J}x, y)$$

with $\mathfrak{J} = \begin{bmatrix} \mathtt{i}I & 0 \\ 0 & -\mathtt{i}I \end{bmatrix}$, then $\{X \times X, < \cdot, \cdot >\}$ is also an infinite-dimensional complex symplectic space.

Some examples of symplectic spaces will be given next.

Example 3.1.3 Let $X = \ell^2$, the set of all complex-valued sequences $x = (x_1, x_2, \ldots)$ satisfying $\sum_{i=1}^{\infty} |x_i|^2 < \infty$, with the inner product given by

$$(x, y) = \sum_{i=1}^{\infty} x_i \overline{y_i}, x = (x_1, x_2, \ldots), \quad y = (y_1, y_2, \ldots) \in X.$$

Define a bivariate function

$$< x, y >= (S_\infty x, y),$$

where the matrix S_∞ is

$$S_\infty = \begin{bmatrix} J_2 & 0 & 0 \\ 0 & J_2 & 0 \\ 0 & 0 & \ddots \end{bmatrix}$$

and $J_2 = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. Then $\{X, < \cdot, \cdot >\}$ is an infinite-dimensional symplectic space.

Firstly, to prove that $< \cdot, \cdot >$ is well-defined, we only need to show $S_\infty x \in X$ for any $x \in X$. In fact,

$$\|S_\infty x\|^2 = (S_\infty x, S_\infty x) = \|x\|^2,$$

and then $< \cdot, \cdot >$ is well defined. Secondly, the proof that $< \cdot, \cdot >$ satisfies the property (i)-(iv) is obvious. Hence $\{X, < \cdot, \cdot >\}$ is an infinite-dimensional symplectic space.

Definition 3.1.4 Let $\{X, < \cdot, \cdot >\}$ be a symplectic space. If $< x, y >= 0$ for $x, y \in X$, then we say x and y are symplectic orthogonal. If the vector system $\{\alpha_i\}_{i=1}^{\infty}$ satisfies

$$< \alpha_i, \alpha_j >= 0, \quad i \neq j,$$

then $\{\alpha_i\}_{i=1}^{\infty}$ is said to be a symplectic orthogonal vector system. If the vector systems $\{\alpha_i\}_{i=1}^{\infty}$, $\{\beta_i\}_{i=1}^{\infty}$ satisfy
(i) $< \alpha_i, \alpha_j >=< \beta_i, \beta_j >= 0, i, j = 1, 2, \ldots,$
(ii) $< \alpha_i, \beta_j >= 0, i \neq j,$
(iii) $< \alpha_i, \beta_j >\neq 0, i = j,$
then $\{\alpha_1, \alpha_2, \ldots\}$ and $\{\beta_1, \beta_2, \ldots\}$ are called conjugate symplectic orthogonal vector systems, and if we further have

$$\overline{Span\{\alpha_1, \alpha_2, \ldots, \beta_1, \beta_2, \ldots\}} = X,$$

then $\{\alpha_1, \alpha_2, \ldots\}$ and $\{\beta_1, \beta_2, \ldots\}$ are called complete conjugate symplectic orthogonal vector systems.

Remark 3.1.5 In an inner product space $\{X, (\cdot, \cdot)\}$, if $\{\alpha_i\}_{i=1}^{\infty}$ and $\{\beta_i\}_{i=1}^{\infty}$ satisfy
(i) $(\alpha_i, \alpha_j) = (\beta_i, \beta_j) = 0, i \neq j, i, j = 1, 2, \ldots,$
(ii) $(\alpha_i, \beta_j) = 0, i \neq j,$
(iii) $(\alpha_i, \beta_j) \neq 0, i = j,$
then $\{\alpha_1, \alpha_2, \ldots\}$ and $\{\beta_1, \beta_2, \ldots\}$ are called conjugate orthogonal vector systems.

Example 3.1.6 Let $X = L^2[0, 1]$, and the inner product on X and symplectic inner product on $X \times X$ are respectively denoted as $\int_0^1 f(x)\overline{g(x)}dx$ and

$$< \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}, \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} >= (f_2, g_1) - (f_1, g_2).$$

It is easy to see that

$$\{\begin{bmatrix} \sin \pi x \\ \pi \sin \pi x \end{bmatrix}, \begin{bmatrix} \sin 2\pi x \\ 2\pi \sin 2\pi x \end{bmatrix}, \ldots\}$$

and

$$\{\begin{bmatrix} -\sin \pi x \\ \pi \sin \pi x \end{bmatrix}, \begin{bmatrix} -\sin 2\pi x \\ 2\pi \sin 2\pi x \end{bmatrix}, \ldots\}$$

are conjugate symplectic orthogonal vector systems.

From the definition, it is not difficult to find that $< u, u >= 0$ is not always true for $u \in X$ in the complex symplectic space $\{X, < \cdot, \cdot >\}$. This is the essential difference between the real symplectic space and the complex symplectic space. In addition, the inner product can be induced by the symplectic structure under certain conditions. Let $\{X, < \cdot, \cdot >\}$ be an infinite-dimensional complex symplectic space, and there are submanifolds

$$X_+ = \{u \in X : 0 < Im < u, u >\} \cup \{0\},$$

$$X_- = \{u \in X : 0 > Im < u, u >\} \cup \{0\}$$

of X, such that

$$X = X_+ \oplus X_-, \tag{3.1}$$

that is, $< X_+, X_- >= 0$ and there exists the only $u_\pm \in X_\pm$ such that

$$u = u_+ + u_-$$

for any $u \in X$. Now define

$$\begin{aligned}(u_-, u_-)_- &= \mathrm{i} < u_-, u_- >,\\ (u_+, u_+)_+ &= -\mathrm{i} < u_+, u_+ >,\\ (u, u) &= (u_+, u_+)_+ + (u_-, u_-)_-,\end{aligned}$$

then it is easy to prove that $\{X, (\cdot, \cdot)\}$ is an infinite-dimensional complex inner product space, and thus there is a topology on X which is independent of the decomposition (3.1) of X by the norm equivalence theorem. The concepts of "boundedness," "continuity," and "closedness" can be introduced into $\{X, < \cdot, \cdot >\}$. Furthermore, if $\{X_\pm, (\cdot, \cdot)_\pm\}$ is a Hilbert space, then $\{X, < \cdot, \cdot >\}$ is said to be a complete symplectic space.

Definition 3.1.7 Let $L \subset X$ be a submanifold of a complex symplectic space $\{X, < \cdot, \cdot >\}$. If L satisfies

$$< L, L >= 0,$$

that is, $< u, v >= 0$ for any $u, v \in L$, then L is said to be a Lagrangian submanifold, and if $u \in X$ and $< u, L >= 0$ imply that $u \in L$, then we say the Lagrange submanifold $L \subset X$ is complete.

Obviously, a complete Lagrangian submanifold is a maximal Lagrangian submanifold. If X has the decomposition (3.1), then the closure $\overline{L}$ of a Lagrangian submanifold L is still a Lagrangian submanifold. A complete Lagrangian submanifold is closed.

3.1.2 Linear Operators in Symplectic Spaces

Let $\{X, <\cdot,\cdot>\}$ be a complete symplectic space, and

$$X = X_+ \oplus X_-$$

is the corresponding decomposition. If we use $P_\pm$ to represent the projection from X onto $X_\pm$ respectively, then

$$(x, y) =< Jx, y >= - < x, Jy >$$

with $J = \mathrm{i}P_- - \mathrm{i}P_+$. The operator J has the following property:

$$J^* = J^{-1} = -J. \tag{3.2}$$

Now we can introduce conjugate operators in the sense of symplectic structure: Let $T : \mathscr{D}(T) \subset X \to X$ be a densely defined linear operator, then the connection between the conjugate operators T^s of T in the symplectic space and T^* of T in the induced Hilbert space is as follows:

$$T^s = -J^{-1}T^*J = JT^*J. \tag{3.3}$$

If $T \subset T^s$, then T is called a symplectic symmetry operator, and if $T = T^s$, then T is called a symplectic self-adjoint operator.

The following are the spectral properties of densely defined closed linear operators in complete symplectic spaces.

Lemma 3.1.8 *Let T be a densely defined closed linear operator in a complete symplectic space $\{X, <\cdot,\cdot>\}$, then*
(i) $\lambda \in \rho(T)$ if and only if $-\overline{\lambda} \in \rho(T^s)$;
(ii) $\lambda \in \sigma_{r,1}(T)$ if and only if $-\overline{\lambda} \in \sigma_{p,1}(T^s)$;
(iii) $\lambda \in \sigma_{r,2}(T)$ if and only if $-\overline{\lambda} \in \sigma_{p,2}(T^s)$;
(iv) $\lambda \in \sigma_c(T)$ if and only if $-\overline{\lambda} \in \sigma_c(T^s)$;
(v) $\lambda \in \sigma_{p,3}(T)$ if and only if $-\overline{\lambda} \in \sigma_{p,3}(T^s)$;
(vi) $\lambda \in \sigma_{p,4}(T)$ if and only if $-\overline{\lambda} \in \sigma_{p,4}(T^s)$;
(vii) $\lambda \in \sigma_{ap}(T)$ if and only if $-\overline{\lambda} \in \sigma_\delta(T^s)$;
(viii) $\lambda \in \sigma_{com}(T)$ if and only if $-\overline{\lambda} \in \sigma_p(T^s)$.

Proof Only prove (i), and the other proofs are similar. Let $\lambda \in \rho(T)$, then $\overline{\lambda} \in \rho(T^*)$, which together with

$$T^s = JT^*J$$

and $J^2 = -I$ gives $-\overline{\lambda} \in \rho(T^s)$. Conversely, the proof is similar. □

Lemma 3.1.9 *Let T be a symplectic self-adjoint operator in a complete symplectic space $\{X, <\cdot,\cdot>\}$, then $\sigma_r(T) \cap \mathrm{i}\mathbb{R} = \emptyset$.*

Proof Suppose there exists $\lambda \in \mathrm{i}\mathbb{R}$ such that $\lambda \in \sigma_r(T)$. Then we get $\lambda \in \sigma_p(T^s) = \sigma_p(T)$ by Lemma 3.1.8, a contradiction is derived. □

Remark 3.1.10 Introduce a complete symplectic space $\{X \times X, < \cdot, \cdot >\}$, and the symplectic structure on $X \times X$ is defined as

$$< x, y >= (Jx, y)$$

where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. Then the infinite-dimensional Hamiltonian operator $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ is a symplectic symmetric operator.

3.2 Symplectic Orthogonality of Eigenvector System of Infinite-Dimensional Hamiltonian Operators

In equations of mathematical physics, the traditional separation of variables method is very effective in solving partial differential equations, reducing to Sturm-Liouville problems after separating variables, such as the bounded string vibration equation and the harmonic equation. But for non-Strum-Liouville problems, the orthogonality and completeness of the eigenfunction system cannot be guaranteed, which makes the traditional separation of variables method may fail to work. In order to overcome this difficulty, Academician Zhong Wanxie combined the infinite-dimensional Hamiltonian operator with the separation of variables method and introduced the symplectic orthogonal system simultaneously, and then established a new systematic methodology for elasticity. The new methodology extends the Strum-Liouville problem and its eigenfunction expansion, and it is praised by Qian Lingxi as a new beginning of elasticity. In the new methodology of elasticity, the symplectic orthogonality of the eigenfunction system of infinite-dimensional Hamiltonian operators replaces the orthogonality of the eigenfunction system of Strum-Liouville operators, which solves the problem that the expansion coefficient of the eigenfunction system of non-self-adjoint operators cannot be easily calculated.

Definition 3.2.1 Given a linear operator T, the smallest non-negative integer k which makes

$$\mathscr{N}(T - \lambda I)^k = \mathscr{N}(T - \lambda I)^{k+1}$$

hold is called the zero chain length (or algebraic index) of the operator T at λ, denoted by $P_\lambda(T)$. The vector $v \neq 0$ is said to be the first-order generalized eigenfunction of T, if $(T - \lambda)v = u$, where $u \in \mathscr{N}(T - \lambda)$. Similarly, the k-order generalized eigenfunction can be defined. The eigenfunction and the generalized eigenfunction are collectively referred to as the root vector system. When $P_\lambda(T) = 1$ for all $\lambda \in \sigma_p(T)$, the eigenfunction system and the generalized eigenfunction system of T are the same.

Example 3.2.2 Consider 6×6 matrix $K = \begin{bmatrix} 1&0&0&0&0&0 \\ 1&1&0&0&0&0 \\ 0&0&2&0&0&0 \\ 0&0&0&3&0&0 \\ 0&0&0&1&3&0 \\ 0&0&0&0&1&3 \end{bmatrix}$, then there are eigenvalues $\lambda_1 = 1$, $\lambda_2 = 2$ and $\lambda_3 = 3$. Now, the length of Jordan chain of $\lambda_1 = 1$ is 2, and

$$U = \begin{bmatrix} 0\ 1\ 0\ 0\ 0\ 0 \end{bmatrix}$$

$$V_1 = \begin{bmatrix} 1\ 0\ 0\ 0\ 0\ 0 \end{bmatrix}$$

are the corresponding eigenfunction and first-order generalized eigenfunction. The length of Jordan chain of $\lambda_3 = 3$ is 3, and

$$U = \begin{bmatrix} 0\ 0\ 0\ 0\ 0\ 1 \end{bmatrix},$$

$$V_1 = \begin{bmatrix} 0\ 0\ 0\ 0\ 1\ 0 \end{bmatrix},$$

$$V_2 = \begin{bmatrix} 0\ 0\ 0\ 1\ 0\ 0 \end{bmatrix}.$$

are the corresponding eigenfunction, first-order generalized eigenfunction, and second-order generalized eigenfunction.

For the infinite-dimensional Hamiltonian operator, its eigenfunction and generalized eigenfunction both have symplectic orthogonality.

Lemma 3.2.3 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator, and let* $\lambda_1, \lambda_2 \in \sigma_p(H)$, u_1, u_2 *and* v_1, v_2 *are the corresponding eigenfunction and generalized eigenfunction. If* $\lambda_1 + \overline{\lambda}_2 \neq 0$, *then*

$$(u_1, Ju_2) = 0, \quad (u_1, Jv_2) = 0, \quad (v_1, Ju_2) = 0, \quad (v_1, Jv_2) = 0,$$

where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. *Since the operator can induce a symplectic structure, the above weighted orthogonality is also said to be symplectic orthogonality.*

Proof Firstly, prove the symplectic orthogonality of the eigenfunction. From the definition of eigenfunction and eigenvalue, it can be seen that

$$Hu_1 = \lambda_1 u_1, \tag{3.4}$$

$$Hu_2 = \lambda_2 u_2. \tag{3.5}$$

Taking inner product of Eq. (3.4) by Ju_2 on the right and (3.5) by Ju_1 on the left and subtracting the resulting equations, we obtain

$$(\lambda_1 + \overline{\lambda_2})(u_1, Ju_2) = 0.$$

From $\lambda_1 + \overline{\lambda_2} \neq 0$, it follows that $(u_1, Ju_2) = 0$.

Secondly, prove the symplectic orthogonality of the eigenfunction u_1 and the generalized eigenfunction v_2. It can be seen from the definition of the eigenfunction and generalized eigenfunction that

$$(H - \lambda_2)^{k+1} v_2 = 0, \quad (H - \lambda_2)^k v_2 \neq 0. \tag{3.6}$$

Let $(H - \lambda_2)^k v_2 = v_2^{(1)}$, then $v_2^{(1)} \in \mathcal{N}(H - \lambda_2)$ and

$$(v_2^{(1)}, Ju_1) = 0.$$

Now let $(H - \lambda_2)^{k-1} v_2 = v_2^{(2)}$, then

$$(H - \lambda_2) v_2^{(2)} = v_2^{(1)}. \tag{3.7}$$

After taking inner product of the left equation in (3.6) by $Jv_2^{(2)}$ and Eq. (3.7) by Ju_1 and subtracting the resulting equations, it follows from $(u_1, Jv_2^{(1)}) = 0$ that

$$(\lambda_1 + \overline{\lambda_2})(u_1, Jv_2^{(2)}) = 0.$$

Since $\lambda_1 + \overline{\lambda}_2 \neq 0$, we get $(u_1, Jv_2^{(2)}) = 0$. In the same way, let $(H - \lambda_2)^{k-2} v_2 = v_2^{(3)}$, then $(u_1, Jv_2^{(3)}) = 0$. Repeating the above process, we get that $(u_1, Jv_2) = 0$. The proof of $(v_1, Ju_2) = 0$ is similar.

Finally, the symplectic orthogonality of the generalized eigenfunction will be proven. Without loss of generality, only the symplectic orthogonality of the first-order generalized eigenfunction is proved, and other proofs are similar. By definition, we know that

$$Hv_1 = \lambda_1 v_1 + u_1, \tag{3.8}$$

$$Hv_2 = \lambda_2 v_2 + u_2, \tag{3.9}$$

After taking inner product of Eq. (3.8) by Jv_2 and (3.9) by Jv_1 and subtracting the resulting equations, since $(u_1, Jv_2) = 0$ and $(u_2, Jv_1) = 0$ we get

$$(\lambda_1 + \overline{\lambda_2})(v_1, Jv_2) = 0,$$

which together with $\lambda_1 + \overline{\lambda}_2 \neq 0$ implies that $(v_1, Jv_2) = 0$. □

On the other hand, the infinite-dimensional Hamiltonian operator is a kind of non-self-adjoint operator. Therefore, the completeness of the eigenfunction system is another problem to be solved, after the problem of coefficient calculation is solved by using symplectic orthogonality, in order to use the separation of variables method

under the infinite-dimensional Hamiltonian system. However, it is not easy to achieve the completeness of the eigenfunction system of the infinite-dimensional Hamiltonian operator.

Example 3.2.4 Consider the boundary value problem of harmonic equation

$$\begin{cases} \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0, \\ u|_{x=0} = 0, u|_{x=1} = 0, \\ u|_{y=0} = \varphi(x), u|_{y=1} = \phi(x) \end{cases}$$

and introduce the Hamiltonian system. Then we can get the Hamiltonian canonical equation

$$\frac{\partial}{\partial y}\begin{bmatrix} u \\ \sigma \end{bmatrix} = \begin{bmatrix} 0 & I \\ -\dfrac{\partial^2}{\partial x^2} & 0 \end{bmatrix}\begin{bmatrix} u \\ \sigma \end{bmatrix},$$

where $\sigma = \frac{\partial u}{\partial y}$. Let $X = L^2[0, 1]$, then the Hamiltonian operator derived in $X \times X$ is

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} = \begin{bmatrix} 0 & I \\ -\dfrac{\mathrm{d}^2}{\mathrm{d}x^2} & 0 \end{bmatrix},$$

where $\mathscr{D}(C) = \{x \in X : x, x' \text{are absolutely continuous}, x, x'' \in X, x(0) = x(1) = 0\}$. Through calculation, the eigenvalue of H is $\lambda_k = k\pi$, and $u_k = \begin{bmatrix} \sin k\pi x \\ k\pi \sin k\pi x \end{bmatrix}, k = \pm 1, \pm 2, \ldots$ is the corresponding eigenfunction. It is therefore easy to prove that $\{u_k\}_{k=-\infty}^{+\infty}(k \neq 0)$ in $X \times X$ is not complete.

Lemma 3.2.5 *If X is a separable Hilbert space, the infinite-dimensional Hamiltonian operator H is invertible and H^{-1} is a compact operator, then the eigenfunction system of the operator JH is complete in the space $X \times X$.*

Proof When H^{-1} is a compact operator, $H^{-1}J$ is a compact self-adjoint operator, where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. According to the Hilbert-Schmidt theorem, the eigenfunction system of $H^{-1}J$ is complete. The eigenfunction system of $H^{-1}J$ is the same as that of JH, and thus the eigenfunction system of JH is complete. □

3.3 Completeness of Eigenfunction System of 2 × 2 Infinite-Dimensional Hamiltonian Operators

In the process of studying the spectrum of Hamiltonian operators, we found a very interesting phenomenon that the eigenvalues of many infinite-dimensional Hamiltonian operators are distributed on the real or imaginary axis, appearing in positive

and negative pairs. Therefore, although it is a bit difficult to achieve the completeness of the eigenfunction system in the general sense, it is relatively easy to achieve completeness in the sense of the Cauchy principal value. After several partial differential equations, such as the harmonic equation, the free vibration equation of a bounded string, and the forced vibration equation of a bounded string, are directed to the Hamiltonian systems, we conduct the separation of variables method in the sense of Cauchy principal value (i.e., the separation of variables method adopting the superposition principle in the sense of Cauchy principal value), and the same results as the traditional separation variable method are obtained. Just as known by calculation, the eigenfunction system of these infinite-dimensional Hamiltonian operators is complete in the sense of the Cauchy principal value. Therefore, it is very important to study the completeness of the eigenfunction system of the infinite-dimensional Hamiltonian operators in the sense of the Cauchy principal value.

3.3.1 Completeness of Eigenfunction System of 2×2 Infinite-Dimensional Hamiltonian Operators in the Sense of Cauchy Principal Value

Definition 3.3.1 The vector system $\{u_n\}_{n=-\infty}^{+\infty}(n \neq 0)$ in a separable Hilbert space X, is said to be complete in the sense of Cauchy principal value, if there exists sequences $\{C_k\}_{k=1}^{+\infty}$, $\{C_{-k}\}_{k=1}^{+\infty}$ of constants such that

$$x = \sum_{k=1}^{+\infty}(C_k u_k + C_{-k}u_{-k})$$

for any $x \in X$.

Lemma 3.3.2 *Let T be a densely defined closed linear operator in a Hilbert space X, if there exists $v \in \mathscr{N}(T^* - \overline{\lambda})^k$ such that $(u, v) \neq 0$ for any $u \in \mathscr{N}(T - \lambda)$, then the zero chain length of T at λ does not exceed k. In particular, for $k = 1$, the zero chain length of T at λ is 1.*

Proof Suppose $P_\lambda(T) = k + 1$, then there exists $u_0 \in X$ such that

$$(T - \lambda)^{k+1}u_0 = 0, \quad (T - \lambda)^k u_0 \neq 0,$$

that is to say, $(T - \lambda)^k u_0 \in N(T - \lambda)$. According to the given conditions, there exists $v \in N(T^* - \overline{\lambda})^k$ such that

$$((T - \lambda)^k u_0, v) \neq 0,$$

that is, $(u_0, (T^* - \overline{\lambda})^k v) \neq 0$ contradicting $(T^* - \overline{\lambda})^k v = 0$, so $P_\lambda(T) \leq k$. □

Theorem 3.3.3 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If* B *is a uniformly positive definite operator,* $\mathscr{D}(B) \subset \mathscr{D}(A^*)$ *and* $B^{-1}A$ *are symmetric operators, then we have:*
(i) $\sigma_p(H) \subset \mathbb{R} \cup i\mathbb{R}$*;*
(ii) $\lambda \in \sigma_p(H)$ *if and only if* $-\lambda \in \sigma_p(H)$*;*
(iii) For any $\lambda \in \sigma_p(H)\backslash\{0\}$, $P_\lambda(H) = 1$ *(that is, the generalized eigenfunction system coincides with the eigenfunction system).*

Proof (i) For any $u = \begin{bmatrix} f & g \end{bmatrix}^T \in \mathscr{N}(H - \lambda)$, note that

$$Af + Bg = \lambda f,$$

$$Cf - A^*g = \lambda g,$$

and B are positive definite operators. It is easy to get that $f \neq 0$ and

$$\lambda^2(B^{-1}f, f) + \lambda(A^*B^{-1}f, f) - \lambda(B^{-1}Af, f) - (Cf, f) - (A^*B^{-1}Af, f) = 0.$$

Since $(B^{-1}Af, f) \in \mathbb{R}$,

$$\lambda(A^*B^{-1}f, f) - \lambda(B^{-1}Af, f) = 0$$

and

$$\lambda^2 = \frac{(A^*B^{-1}Af, f) + (Cf, f)}{(B^{-1}f, f)},$$

that is, $\sigma_p(H) \subset \mathbb{R} \cup \mathtt{i}\mathbb{R}$.

(ii) If $\lambda \in \sigma_p(H)$ and let $\begin{bmatrix} f & g \end{bmatrix}^T$ be the corresponding eigenfunction, then

$$g = \lambda B^{-1}f - B^{-1}Af$$

and

$$(H + \lambda)\begin{bmatrix} -f \\ \lambda B^{-1}f + B^{-1}Af \end{bmatrix} = 0,$$

that is, $-\lambda \in \sigma_p(H)$. Conversely, the proof is similar.

(iii) If $\sigma_p(H) \subset \mathbb{R}\backslash\{0\}$, then taking $v = \begin{bmatrix} -\lambda B^{-1}f - B^{-1}Af \\ -f \end{bmatrix}$ we get

$$(H^* - \overline{\lambda})v = (H^* - \lambda)v = J(H + \lambda)Jv = 0,$$

and

$$(u, v) = \left(\begin{bmatrix} f \\ \lambda B^{-1}f - B^{-1}Af \end{bmatrix}, \begin{bmatrix} -\lambda B^{-1}f - B^{-1}Af \\ -f \end{bmatrix}\right) = -2\lambda(B^{-1}f, f) \neq 0.$$

From Lemma 3.3.2, we have $P_\lambda(H) = 1$.

If $\sigma_p(H) \subset \mathrm{i}\mathbb{R}\backslash\{0\}$, then taking $v = \begin{bmatrix} -\lambda B^{-1} f + B^{-1} A f \\ f \end{bmatrix}$ we have

$$(H^* - \overline{\lambda})v = (H^* + \lambda)v = J(H - \lambda)Jv = 0,$$

and

$$(u, v) = \left(\begin{bmatrix} f \\ \lambda B^{-1} f - B^{-1} A f \end{bmatrix}, \begin{bmatrix} -\lambda B^{-1} f + B^{-1} A f \\ f \end{bmatrix} \right) = 2\lambda (B^{-1} f, f) \neq 0.$$

Again, by Lemma 3.3.2, $P_\lambda(H) = 1$. The proof is completed. □

Remark 3.3.4 In Theorem 3.3.3, if C is uniformly positive definite, $\mathscr{D}(C) \subset \mathscr{D}(A)$ and $C^{-1}A^*$ is a symmetric operator, we can similarly get $P_\lambda(H) = 1$.

Recall that an operator T in a Hilbert space is said to be simple if $dim\mathscr{N}(T - \lambda) = 1$ for any $\lambda \in \sigma_p(T)$.

Theorem 3.3.5 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a simple invertible infinite-dimensional Hamiltonian operator with countable eigenvalues. If B is a uniformly positive definite operator with $\mathscr{D}(B) \subset \mathscr{D}(A^*)$, $C + A^* B^{-1} A$ is a semi-definite operator, $B^{-1}A$ is a symmetric operator, and $(B^{-1}Af_n, f_m) = 0$ for $n \neq m$ where $u_n = \begin{bmatrix} f_n & g_n \end{bmatrix}^T$ $(n = \pm 1, \pm 2, \ldots)$ is the eigenfunction corresponding to eigenvalue λ_n $(n = \pm 1, \pm 2, \ldots)$, then*
(i) $Span\{f_n\}_{n=1}^{+\infty} = Span\{f_{-n}\}_{n=1}^{+\infty}$;
(ii) If $\{f_n\}_{n=1}^{+\infty}$ is a complete orthogonal vector system in X, then the eigenfunction system $\{u_n\}_{n=\pm 1}^{\pm\infty}$ is complete in the sense of Cauchy principal value in $X \times X$.

Proof (i) From the proof of Lemma 3.3.3 and the quasi-definiteness of $C + A^* B^{-1} A$, it follows that H has only real eigenvalues or only pure imaginary eigenvalues, which appear in positive and negative pairs. We may let $\lambda_{-n} = -\lambda_n \in \mathbb{R}$, then it is easy to get

$$u_n = \begin{bmatrix} f_n \\ \lambda_n B^{-1} f_n - B^{-1} A f_n \end{bmatrix},$$

and the eigenfunction corresponding to λ_{-n} is

$$u_{-n} = \begin{bmatrix} -f_n \\ \lambda_n B^{-1} f_n + B^{-1} A f_n \end{bmatrix}.$$

Therefore, $Span\{f_n\}_{n=1}^{+\infty} = Span\{f_{-n}\}_{n=1}^{+\infty}$.

(ii) Let $\{f_n\}_{n=1}^{+\infty}$ be an orthogonal basis of X. For any $x \in X$, we have the Parseval equation

$$x = \sum_{n=1}^{+\infty} \frac{(x, f_n)}{(f_n, f_n)} f_n.$$

Let $\sigma_p(H) \subset \mathbb{R}$. In order to prove $u_n = \begin{bmatrix} f_n & g_n \end{bmatrix}^T, n = \pm 1, \pm 2, \ldots$ is complete in the sense of Cauchy principal value in $X \times X$, for any $\begin{bmatrix} x & y \end{bmatrix}^T \in X \times X$ taking

$$C_k = \frac{\left(\begin{bmatrix} x \\ y \end{bmatrix}, J u_{-k} \right)}{(u_k, J u_{-k})}, \quad C_{-k} = \frac{\left(\begin{bmatrix} x \\ y \end{bmatrix}, J u_k \right)}{(u_{-k}, J u_k)}, k = 1, 2, \ldots,$$

we then have

$$\sum_{k=1}^{+\infty} C_k u_k + C_{-k} u_{-k} = \sum_{k=1}^{+\infty} \begin{bmatrix} \frac{(B^{-1} f_k, x)}{(B^{-1} f_k, f_k)} f_k \\ \frac{(x, B^{-1} A f_k) + (y, f_k)}{(B^{-1} f_k, f_k)} B^{-1} f_k - \frac{(B^{-1} f_k, x)}{(B^{-1} f_k, f_k)} B^{-1} A f_k \end{bmatrix}. \quad (3.10)$$

If $n \neq m$, then $\lambda_n \neq \lambda_m$ and hence

$$(u_n, J u_m) = 0,$$

which implies $(B^{-1} f_n, f_m) = 0 \, (n \neq m)$, so

$$(B^{-1} f_k - \frac{(B^{-1} f_k, f_k)}{(f_k, f_k)} f_k, f_j) = 0, \quad j = 1, 2, \ldots.$$

As a result, $B^{-1} f_k = \frac{(B^{-1} f_k, f_k)}{(f_k, f_k)} f_k, k = 1, 2, \ldots$ and

$$\sum_{k=1}^{+\infty} \frac{(B^{-1} f_k, x)}{(B^{-1} f_k, f_k)} f_k = \sum_{k=1}^{+\infty} \frac{(x, f_k)}{(f_k, f_k)} f_k = x.$$

Similarly, if $n \neq m$, then $x = \sum_{k=1}^{+\infty} \frac{(f_k, x)}{(f_k, f_k)} f_k$ and $(B^{-1} A f_n, f_m) = 0$, and hence

$$\begin{aligned} &\sum_{k=1}^{+\infty} \frac{(x, B^{-1} A f_k) + (y, f_k)}{(B^{-1} f_k, f_k)} B^{-1} f_k - \frac{(B^{-1} f_k, x)}{(B^{-1} f_k, f_k)} B^{-1} A f_k \\ =&\sum_{k=1}^{+\infty} \frac{(x, B^{-1} A f_k) B^{-1} f_k - (B^{-1} f_k, x) B^{-1} A f_k}{(B^{-1} f_k, f_k)} + \frac{(y, f_k)}{(B^{-1} f_k, f_k)} B^{-1} f_k = y, \end{aligned}$$

the conclusion is established.

If $\sigma_p(H) \subset \mathtt{i}\mathbb{R}$, take

$$C_k = \frac{\left(\begin{bmatrix} x \\ y \end{bmatrix}, Ju_k\right)}{(u_k, Ju_k)}, \quad C_{-k} = \frac{\left(\begin{bmatrix} x \\ y \end{bmatrix}, Ju_{-k}\right)}{(u_{-k}, Ju_{-k})}, k = 1, 2, \ldots.$$

Then similar to the proof of the case $\sigma_p(H) \subset \mathbb{R}$, we can obtain

$$\sum_{k=1}^{+\infty} C_k u_k + C_{-k} u_{-k} = \begin{bmatrix} x \\ y \end{bmatrix}$$

for any $\begin{bmatrix} x & y \end{bmatrix}^T \in X \times X$, that is, $\{u_n = [f_n\ g_n]^T\}$ is complete in the sense of Cauchy principal value in $X \times X$. The proof is completed. □

Similar to the proof of the above theorem, the following theorem can also be proved.

Theorem 3.3.6 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a simple invertible infinite-dimensional Hamiltonian operator with countable eigenvalues, C is a positive definite invertible operator with* $\mathscr{D}(C) \subset \mathscr{D}(A)$, $C^{-1}A^*$ *is a symmetric operator,* $AC^{-1}A^* + B$ *is a quasi-definite operator, and* $(C^{-1}A^*g_n, g_m) = 0$ *for* $n \neq m$ *where* $u_n = \begin{bmatrix} f_n & g_n \end{bmatrix}^T$ $(n = \pm 1, \pm 2, \ldots)$ *is the eigenfunction corresponding to eigenvalue* λ_n $(n = \pm 1, \pm 2, \ldots)$. *If the vector system* $\{g_n\}_{n=\pm\infty}^{\pm\infty}$ *is a conjugate orthogonal complete vector system in X, then* $u_n = \begin{bmatrix} f_n & g_n \end{bmatrix}^T$, $n = \pm 1, \pm 2, \ldots$ *is complete in the sense of Cauchy principal value in* $X \times X$.

The following will illustrate the effectiveness of the criterion with some examples.

Example 3.3.7 For the boundary value problem of the harmonic equation

$$\begin{cases} \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0, \\ u|_{x=0} = 0, u|_{x=1} = 0, \\ u|_{y=0} = \varphi(x), u|_{y=1} = \phi(x), \end{cases}$$

the corresponding Hamiltonian canonical equation is given by

$$\frac{\partial}{\partial y}\begin{bmatrix} u \\ \sigma \end{bmatrix} = \begin{bmatrix} 0 & I \\ -\dfrac{\partial^2}{\partial x^2} & 0 \end{bmatrix}\begin{bmatrix} u \\ \sigma \end{bmatrix}$$

with $\sigma = \frac{\partial u}{\partial y}$, and

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} = \begin{bmatrix} 0 & I \\ -\dfrac{\mathrm{d}^2}{\mathrm{d}x^2} & 0 \end{bmatrix}$$

is an infinite-dimensional Hamiltonian operator defined in $X \times X$, where $\mathscr{D}(C) = \{x \in X : x, x' \text{ are absolutely continuous}, x'' \in X, x(0) = x(1) = 0\}$ and $X = L^2[0, 1]$ (see Example 3.2.4).

It is easy to know that H^{-1} exists, $(H^{-1})^2$ is a compact operator, and H has countable eigenvalues and satisfies other conditions of Theorem 3.3.6. Thus the eigenfunction system of the operator H is complete in the sense of Cauchy principal value in the Hilbert space $X \times X$.

On the other hand, through calculation we get $\lambda_k = k\pi$ with the corresponding eigenfunction $u_k = \begin{bmatrix} \sin k\pi x \\ k\pi \sin k\pi x \end{bmatrix}$, $k = \pm 1, \pm 2, \ldots$, and

$$(u_k, Ju_{-k}) = k\pi, \quad k = \pm 1, \pm 2, \ldots.$$

Then, for any $\begin{bmatrix} f(x) & g(x) \end{bmatrix}^T \in X \times X$, taking

$$C_{-k} = \frac{(\begin{bmatrix} f \\ g \end{bmatrix}, Ju_k)}{-k\pi}, \quad C_k = \frac{(\begin{bmatrix} f \\ g \end{bmatrix}, Ju_{-k})}{k\pi}, k = 1, 2, \ldots$$

and substituting them into (3.10), we get

$$\begin{bmatrix} f(x) \\ g(x) \end{bmatrix} = \sum_{k=1}^{+\infty} \begin{bmatrix} 2(f(x), \sin k\pi x) \sin k\pi x \\ 2(g(x), \sin k\pi x) \sin k\pi x \end{bmatrix},$$

whose components are exactly the Fourier series expansion of $f(x)$, $g(x)$ in X and thus are convergent. Therefore, the eigenfunction system of the operator H is complete in the sense of Cauchy principal value in the Hilbert space $X \times X$, which coincides with the conclusion shown by Theorem 3.3.6.

Example 3.3.8 Consider the initial-boundary value problem of the wave equation

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} = \dfrac{\partial^2 u}{\partial x^2}, \\ u(0, t) = u(1, t) = 0, \\ u(x, 0) = \varphi(x), u_t(x, 0) = \phi(x) \end{cases}$$

and introduce the Hamiltonian system. Then we can get the Hamiltonian canonical equation

$$\frac{\partial}{\partial t} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & I \\ \dfrac{\partial^2}{\partial x^2} & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix},$$

where $\frac{\partial v}{\partial t} = \frac{\partial^2 u}{\partial x^2}$. Let $X = L^2[0, 1]$, then the Hamiltonian operator derived from $X \times X$ is

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} = \begin{bmatrix} 0 & I \\ \frac{\mathrm{d}^2}{\mathrm{d}x^2} & 0 \end{bmatrix},$$

where $\mathscr{D}(C) = \{x \in X : x, x' \text{ are absolutely continuous}, x'' \in X, x(0) = x(1) = 0\}$.

It is easy to know that H^{-1} exists, $(H^{-1})^2$ is compact operator, and H has countable eigenvalues and satisfies other conditions of Theorem 3.3.6, and thus the eigenfunction system of the operator H is complete in the sense of Cauchy principal value in the Hilbert space $X \times X$.

On the other hand, through calculation we get $\lambda_k = \mathrm{i}k\pi$ with the corresponding eigenfunction $u_k = \begin{bmatrix} \sin k\pi x \\ \mathrm{i}k\pi \sin k\pi x \end{bmatrix}, k = \pm 1, \pm 2, \ldots,$ and

$$(u_k, Ju_k) = ik\pi, \quad k = \pm 1, \pm 2, \ldots.$$

So, for any $\begin{bmatrix} f(x) & g(x) \end{bmatrix}^T \in X \times X$, taking

$$C_k = \frac{(\begin{bmatrix} f \\ g \end{bmatrix}, Ju_k)}{ik\pi}, \quad C_{-k} = \frac{(\begin{bmatrix} f \\ g \end{bmatrix}, Ju_{-k})}{ik\pi}, k = 1, 2, \ldots$$

and substituting them into (3.10), we get

$$\begin{bmatrix} f(x) \\ g(x) \end{bmatrix} = \sum_{k=1}^{+\infty} \begin{bmatrix} 2(f(x), \sin k\pi x) \sin k\pi x \\ 2(g(x), \sin k\pi x) \sin k\pi x \end{bmatrix},$$

whose components are exactly the Fourier series expansions of $f(x)$, $g(x)$ in X and thus are convergent. Then the eigenfunction system of the operator H is complete in the sense of Cauchy principal value of the Hilbert space $X \times X$, which coincides with the conclusion drawn by Theorem 3.3.6.

Remark 3.3.9 It is not difficult to find that the eigenfunction systems of the above two infinite-dimensional Hamiltonian operators are not complete in the general sense in $L^2[0, 1] \times L^2[0, 1]$. In fact, take $f = 1, g = 0$ and substitute them into the equation

$$\begin{bmatrix} f \\ g \end{bmatrix} = \sum_{k=-\infty, k\neq 0}^{+\infty} C_k u_k \tag{3.11}$$

After that, taking the second component into account, we get

$$\sum_{k=-\infty, k\neq 0}^{+\infty} \frac{\lambda_k (f, B^{-1} f_k) + (f, B^{-1} A f_k)}{2(B^{-1} f_k, f_k)} = \sum_{k=-\infty, k\neq 0}^{+\infty} (1 - \cos k\pi),$$

in which the general term $1-\cos k\pi$ doesn't tend to 0 as $k\to\infty$, and the series do not converge. This implies that the eigenfunction system of the operator H is not complete in the Hilbert space $X\times X$. Similarly, the eigenfunction system of the infinite-dimensional Hamiltonian operator H corresponding to the second example is also incomplete in the Hilbert space $X\times X$.

3.3.2 Divergence Problem of Eigenfunction Expansions for 2 × 2 Infinite-Dimensional Hamiltonian Operators

It is natural to ask of which infinite-dimensional Hamiltonian operators the eigenfunction systems must be incomplete in the general sense? In order to answer this question, we must give the definition of k-compact operator first.

Definition 3.3.10 A bounded linear operator T in a Hilbert space X is said to be k-compact, if there exists a positive integer k such that T^k is a compact operator.

Theorem 3.3.11 *Let* $H=\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}:\mathscr{D}(H)\subset X\times X\to X\times X$ *be an invertible infinite-dimensional Hamiltonian operator, and its inverse is a k-compact operator. Then the eigenfunction system of the operator H is incomplete in the general sense in Hilbert space* $X\times X$*, if* B, C *are non-negative operators and satisfy one of the following conditions:*
(i) B *is a invertible operator with* $\mathscr{D}(B)\subset\mathscr{D}(A^*)$*,* $B^{-1}A$ *is a symmetric operator and* $\mathscr{N}(A)\neq\emptyset$*;*
(ii) C *is a invertible operator with* $\mathscr{D}(C)\subset\mathscr{D}(A)$*,* $C^{-1}A^*$ *is a symmetric operator and* $\mathscr{N}(A^*)\neq\emptyset$*.*

Proof The following is the proof of the case (i), and that of case (ii) is completely similar. It is easy to prove that the operator H has only real eigenvalues appearing in positive and negative pairs. Since H is invertible and its inverse is a k-compact operator, H has at most countable eigenvalues. Then all eigenvalues may be set as $\{\lambda_{\pm n}\}_{n=1}^{+\infty}$, $\lambda_{-n}=-\lambda_n$, and let $u_n=\begin{bmatrix} f_n & g_n \end{bmatrix}^T$ be the corresponding eigenfunction. Thus

$$u_n=\begin{bmatrix} f_n \\ \lambda_n B^{-1}f_n-B^{-1}Af_n \end{bmatrix}$$

is the corresponding eigenfunction corresponding to eigenvalue λ_n, and

$$u_{-n}=\begin{bmatrix} -f_n \\ \lambda_n B^{-1}f_n+B^{-1}Af_n \end{bmatrix}$$

is the corresponding eigenfunction corresponding to eigenvalue λ_{-n}.

Without loss of generality, the first component $\{f_n\}$ of the eigenfunction system can be assumed to be complete in X, otherwise the eigenfunction system of the

infinite-dimensional Hamiltonian operator H is incomplete in the Hilbert space $X \times X$. The conclusion holds obviously. Now suppose that the eigenfunction system of the infinite-dimensional Hamiltonian operator H is complete in the space $X \times X$ in the general sense. Then, for any $\begin{bmatrix} f & g \end{bmatrix}^T \in X \times X$, there exists a constant sequence $\{C_k\}_{k=-\infty}^{+\infty}$ such that

$$\begin{bmatrix} f \\ g \end{bmatrix} = \sum_{k=-\infty k\neq 0}^{+\infty} C_k u_k. \tag{3.12}$$

By calculation we can know

$$(u_n, Ju_m) = \begin{cases} 0, & m \neq -n, \\ 2\lambda_n(B^{-1}f_n, f_n), & m = -n, \end{cases}$$

which combines with $B^{-1}A = A^*B^{-1}|_{\mathrm{d}(A)}$ implies that

$$(B^{-1}f_n, f_m) = 0, \quad m \neq -n, \quad n, m = \pm 1, \pm 2, \ldots.$$

Therefore,

$$C_k = \frac{\left(\begin{bmatrix} f \\ g \end{bmatrix}, Ju_{-k}\right)}{2\lambda_k(B^{-1}f_k, f_k)} = \frac{\lambda_k(f, B^{-1}f_k) + (f, B^{-1}Af_k) + (g, f_k)}{2\lambda_k(B^{-1}f_k, f_k)}$$

can be obtained by taking the inner product by Ju_{-k} in (3.12). Since $\mathscr{N}(A) \neq \emptyset$, we get

$$\begin{aligned} \begin{bmatrix} f \\ 0 \end{bmatrix} &= \begin{bmatrix} \sum\limits_{k=-\infty k\neq 0}^{+\infty} \frac{\lambda_k(f, B^{-1}f_k) + (f, B^{-1}Af_k)}{2\lambda_k(B^{-1}f_k, f_k)} f_k \\ \sum\limits_{k=-\infty k\neq 0}^{+\infty} \frac{\lambda_k(f, B^{-1}f_k) + (f, B^{-1}Af_k)}{2(B^{-1}f_k, f_k)} B^{-1}f_k - B^{-1}Af \end{bmatrix} \\ &= \begin{bmatrix} \sum\limits_{k=-\infty k\neq 0}^{+\infty} \frac{(f, B^{-1}f_k)}{2(B^{-1}f_k, f_k)} f_k \\ \sum\limits_{k=-\infty k\neq 0}^{+\infty} \frac{\lambda_k(f, B^{-1}f_k)}{2(B^{-1}f_k, f_k)} B^{-1}f_k \end{bmatrix} \end{aligned}$$

by taking $0 \neq f \in \mathscr{N}(A)$, $g = 0$.

Taking the inner product for both sides of the equation

$$0 = \sum_{k=-\infty k\neq 0}^{+\infty} \frac{\lambda_k(f, B^{-1}f_k)}{2(B^{-1}f_k, f_k)} B^{-1}f_k \tag{3.13}$$

by f_j, $j = \pm 1, \pm 2, \ldots$, we can know:

$$(f, B^{-1}f_j) = (B^{-1}f, f_j) = 0, \quad j = \pm 1, \pm 2, \ldots.$$

According to the assumption that $\{f_k\}$ is complete in X, so $B^{-1}f = 0$, that is, $f = 0$, which is contradictory to $f \neq 0$. Therefore, the eigenfunction system of the infinite-dimensional Hamiltonian operator H in Hilbert space $X \times X$ is incomplete. The proof is completed. □

In order to discuss the case that the infinite-dimensional Hamiltonian operator has only pure imaginary eigenvalues and the eigenfunction system is incomplete, the following lemma will be given firstly.

Lemma 3.3.12 (Mixed Schwarz Inequality) *Let T be a linear operator in Hilbert space X, and denote $|T| = (T^*T)^{\frac{1}{2}}$, $|T^*| = (TT^*)^{\frac{1}{2}}$, then*

$$|(Tx, y)|^2 \leq (|T|x, x)(|T^*|y, y)$$

for any $x \in \mathscr{D}(T)$, $y \in \mathscr{D}(T^*)$.

Proof According to the properties of Polar decomposition, it can be seen that $T = U|T|$, where U is the isometric operator from $\overline{\mathscr{R}(|T|)}$ to $\overline{\mathscr{R}(|T|)}$. Since $|T^*| = U|T|U^*$,

$$\begin{aligned}
|(Tx, y)|^2 =& |(U|T|x, y)|^2 \\
=& |(|T|^{\frac{1}{2}}x, |T|^{\frac{1}{2}}U^*y)|^2 \\
\leq& (|T|^{\frac{1}{2}}x, |T|^{\frac{1}{2}}x)(|T|^{\frac{1}{2}}U^*y, |T|^{\frac{1}{2}}U^*y) \\
=& (|T|x, x)(U|T|U^*y, y) \\
=& (|T|x, x)(|T^*|y, y).
\end{aligned}$$

The proof is completed. □

Theorem 3.3.13 *Let $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ be an invertible infinite-dimensional Hamiltonian operator and its inverse is a k-compact operator, then the eigenfunction system of the infinite-dimensional Hamiltonian operator H in Hilbert space $X \times X$ is incomplete in the general sense, if one of the conditions is satisfied:*
(i) B is an invertible non-negative operator with $\mathscr{D}(B) \subset \mathscr{D}(A^*)$, $-C$ is a non-negative operator, $B^{-1}A$ is a symmetric operator, $N(A) \neq \emptyset$, $B > (A^*A)^{\frac{1}{2}}$ and $-C > (AA^*)^{\frac{1}{2}}$;
(ii) C is an invertible non-negative operator with $\mathscr{D}(C) \subset \mathscr{D}(A)$, $-B$ is a non-negative operator, $C^{-1}A^*$ is symmetric operator, $\mathscr{N}(A^*) \neq \emptyset$, $-B > (A^*A)^{\frac{1}{2}}$ and $C > (AA^*)^{\frac{1}{2}}$.

Proof The following only proves the case (i), and the proof of (ii) is completely similar. Let $\lambda \in \sigma_p(H)$ and let $u = \begin{bmatrix} f & g \end{bmatrix}^T$ be the corresponding eigenfunction, then

$$Af + Bg = \lambda f,$$

$$Cf - A^*g = \lambda g.$$

Taking the inner product of two sides of the first equation and of two sides of the second equation by g and f respectively, and then making the difference for the obtained results, we have

$$(Bg, g) - (Cf, f) + 2Re(Af, g) = 2\mathrm{i}\lambda Im(f, g).$$

By combining $B > (A^*A)^{\frac{1}{2}}$, $-C > (AA^*)^{\frac{1}{2}}$ and the mixed Schwarz inequality, it can be derived that

$$2\mathrm{i}\lambda Im(f, g) > 0.$$

Furthermore, we can get that $Im(f, g) \neq 0$ and λ is a pure imaginary number. The rest of the proof is completely similar to the above theorem. The proof is completed. □

Example 3.3.14 Consider the following initial-boundary value problem

$$\begin{cases} s(x)\dfrac{\partial^2 u}{\partial t^2} = \dfrac{\partial u}{\partial x}[p(x)\dfrac{\partial u}{\partial x}], \\ u(0, t) = u(1, t) = 0, \\ u(x, 0) = f(x), \quad u_t(x, 0) = g(x), \end{cases}$$

where $p(x), s(x)$ are positive real-valued functions on $[0, 1]$ satisfying $p(x) \in C^1[0, 1], s(x) \in C[0, 1]$ and introduce the Hamiltonian system. Then we can get the Hamiltonian canonical equation

$$\frac{\partial}{\partial t}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & I \\ s(x)^{-1}\dfrac{\partial}{\partial x}(p(x)\dfrac{\partial}{\partial x}) & 0 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix}$$

where $\frac{\partial v}{\partial t} = s(x)^{-1}\frac{\partial u}{\partial x}(p(x)\frac{\partial u}{\partial x})$. Let $X = L_s^2(0, 1)$, then the Hamiltonian operator can be derived as

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} = \begin{bmatrix} 0 & I \\ s(x)^{-1}\dfrac{\mathrm{d}}{\mathrm{d}x}(p(x)\dfrac{d}{dx}) & 0 \end{bmatrix}$$

in $X \times X$, where $\mathscr{D}(C) = \{u \in X : u, u'$are absolutely continuous, $u'' \in X, u(0) = u(1) = 0\}$. Because the inverse of the Sturm-Liouville operator is a Hilbert-Schmidt operator, H^{-1} exists and is a 2-compact operator and satisfies the other conditions of the above theorem. Therefore, the eigenfunction system of the operator H in Hilbert space $X \times X$ is incomplete.

On the other hand, the eigenvalue of operator H can be calculated to be $\lambda_k = i\tau_k$, where τ_k^2 is the eigenvalue of the Sturm-Liouville operator $C = s(x)^{-1}\frac{\mathrm{d}}{\mathrm{d}x}(p(x)\frac{\mathrm{d}}{\mathrm{d}x})$, and $\tau_k = O(k)$. Because the general term of the series of the right side of (3.13) does not tend toward 0 as $k \to \infty$, the series does not converge, which shows that the eigenfunction system of the operator H is incomplete in the Hilbert space $X \times X$.

3.4 Completeness of the Eigenfunction System of 4 × 4 Infinite-Dimensional Hamiltonian Operators

In addition to 2×2 infinite-dimensional Hamiltonian operators, there are many 4×4 infinite-dimensional Hamiltonian operators in practical applications. So it is necessary to study the completeness of the eigenfunction system and the root vector system for 4×4 infinite-dimensional Hamiltonian operators. The completeness problem of the eigenfunction system or the root vector system for the general 4×4 infinite-dimensional Hamiltonian operators is relatively complicated. Note that the form of 4×4 infinite-dimensional Hamiltonian operators appearing in the plate bending equation and plane elasticity problem can be described as

$$H = \begin{bmatrix} 0 & A_1 & B_1 & 0 \\ A_2 & 0 & 0 & B_2 \\ C_1 & 0 & 0 & -A_2^* \\ 0 & C_2 & -A_1^* & 0 \end{bmatrix},$$

where $C_i, B_i, i = 1, 2$ are self-adjoint operators. For example, an infinite-dimensional Hamiltonian operator of the plate bending equation (see [5])

$$\frac{\partial^2 M_x}{\partial x^2} - 2\frac{\partial^2 M_{xy}}{\partial x \partial y} + \frac{\partial^2 M_y}{\partial y^2} = 0 \tag{3.14}$$

can be given by

$$H = \begin{bmatrix} 0 & v\frac{\mathrm{d}}{\mathrm{d}y} & D(1-v^2) & 0 \\ -\frac{\mathrm{d}}{\mathrm{d}y} & 0 & 0 & 2D(1-v) \\ 0 & 0 & 0 & -\frac{\mathrm{d}}{\mathrm{d}y} \\ 0 & -\frac{\mathrm{d}^2}{D\mathrm{d}y^2} & v\frac{\mathrm{d}}{\mathrm{d}y} & 0 \end{bmatrix},$$

where $D > 0$ is a constant and

$$\mathrm{d}(H) = \left\{ \begin{bmatrix} \psi_1(y) \\ \psi_2(y) \\ \psi_3(y) \\ \psi_4(y) \end{bmatrix} \in X : \begin{array}{c} \psi_1(0) = \psi_1(b) = 0,\ \frac{\mathrm{d}\psi_2(0)}{D\mathrm{d}x} - v\psi_3(0) = 0, \\ \frac{\mathrm{d}\psi_2(b)}{D\mathrm{d}x} - v\psi_3(b) = 0 \\ \psi_i(y) \text{ is absolutely continuous and } \psi_i'(x) \in L^2(0,b),\ i = 1,3,4 \\ \psi_2(y),\ \psi_2'(y) \text{ are absolutely continuous and } \psi_2''(x) \in L^2(0,b) \end{array} \right\}.$$

Therefore, we will discuss the completeness of the eigenfunction system or the root vector system for 4×4 infinite-dimensional Hamiltonian operators mentioned above in the sense of Cauchy principal value.

3.4.1 Completeness of Eigenfunction System of 4×4 Infinite-Dimensional Hamiltonian Operators in the Sense of Cauchy Principal Value

Firstly, we will study the completeness problem of the eigenfunction system of 4×4 infinite-dimensional Hamiltonian operators in the sense of Cauchy principal value.

Theorem 3.4.1 *Let* $H = \begin{bmatrix} 0 & A_1 & B_1 & 0 \\ A_2 & 0 & 0 & B_2 \\ C_1 & 0 & 0 & -A_2^* \\ 0 & C_2 & -A_1^* & 0 \end{bmatrix}$ *be an infinite-dimensional Hamiltonian operator. Then*
(i) H is similar to $-H$;
(ii) H is similar to $\widetilde{H}$, where $\widetilde{H} = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$, $B = \begin{bmatrix} -C_1 & A_2^* \\ A_2 & B_2 \end{bmatrix}$, $C = \begin{bmatrix} -B_1 & A_1 \\ A_1^* & C_2 \end{bmatrix}$.

Proof (i) Let $\mathfrak{J}_1 = \begin{bmatrix} -I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \\ 0 & 0 & 0 & -I \end{bmatrix}$, then it is easy to prove $\mathfrak{J}_1 = \mathfrak{J}_1^{-1}$ and

$$H = \mathfrak{J}_1(-H)\mathfrak{J}_1^{-1},$$

that is, H is similar to $-H$. Furthermore, we have $\lambda \in \sigma_p(H)$ if and only if $-\lambda \in \sigma_p(H)$. Let $\begin{bmatrix} f_1 & f_2 & g_1 & g_2 \end{bmatrix}^T \in \mathscr{N}(H - \lambda I)$, then $\begin{bmatrix} -f_1 & f_2 & g_1 & -g_2 \end{bmatrix}^T \in \mathscr{N}(H + \lambda I)$.

(ii) Let $\mathfrak{J}_2 = \begin{bmatrix} 0 & 0 & -I & 0 \\ 0 & I & 0 & 0 \\ I & 0 & 0 & 0 \\ 0 & 0 & 0 & I \end{bmatrix}$, then

$$\mathfrak{J}_2 H \mathfrak{J}_2^{-1} = \widetilde{H},$$

that is, H is similar to $\widetilde{H}$. □

Theorem 3.4.2 *Let* $H = \begin{bmatrix} 0 & A_1 & B_1 & 0 \\ A_2 & 0 & 0 & B_2 \\ C_1 & 0 & 0 & -A_2^* \\ 0 & C_2 & -A_1^* & 0 \end{bmatrix}$ *be an infinite-dimensional Hamiltonian operator and*

$$B = \begin{bmatrix} -C_1 & A_2^* \\ A_2 & B_2 \end{bmatrix},\ C = \begin{bmatrix} -B_1 & A_1 \\ A_1^* & C_2 \end{bmatrix}.$$

If B, C are invertible and B^{-1} or C^{-1} is a compact positive definite operator, then we have the following conclusions:
(i) $\sigma_p(H)$ is an at most countable set and $\sigma_p(H) \subset \mathbb{R} \cup i\mathbb{R}$; if B, C are both positive definite (or $-B, -C$ are positive definite), then $\sigma_p(H) \subset \mathbb{R}$; if $B, -C$ are positive definite (or $-B, C$ are positive definite), then $\sigma_p(H) \subset i\mathbb{R}$.
(ii) For any $\lambda_n \in \sigma_p(\widetilde{H})$, let $u_n = \begin{bmatrix} f_n & g_n \end{bmatrix}^T$ be the corresponding eigenfunction. If B, C are positive definite(or $-B, -C$ are positive definite), then the fact $\{f_{-1}, f_{-2}, \ldots\}$ and $\{f_1, f_2, \ldots\}$ are conjugate orthogonal bases of X implies that $\{u_{-1}, u_{-2}, \ldots\} \cup \{u_1, u_2, \ldots\}$ is a symplectic conjugate orthogonal basis of $X \times X$ in the sense of Cauchy principal value, that is,

$$(u_n, Ju_m) = \begin{cases} 0, & m \neq -n, \\ \tau \neq 0, & m = -n, \end{cases}$$

and there exist $\{C_k\}_{k=1}^{+\infty}$ and $\{C_{-k}\}_{k=1}^{+\infty}$ such that

$$u = \sum_{k=1}^{+\infty} (C_k u_k + C_{-k} u_{-k})$$

for any $u = \begin{bmatrix} f & g \end{bmatrix}^T \in X \times X$; If $B, -C$ are positive definite (or $-B, C$ are positive definite), then the fact $\{f_{-1}, f_{-2}, \ldots\}$ and $\{f_1, f_2, \ldots\}$ are conjugate orthogonal bases of X implies that $\{u_{-1}, u_{-2}, \ldots\} \cup \{u_1, u_2, \ldots\}$ is a symplectic orthogonal basis of $X \times X$ in the sense of Cauchy principal value, that is,

$$(u_n, Ju_m) = \begin{cases} 0, & m \neq n, \\ \tau \neq 0, & m = n, \end{cases}$$

and there exist $\{C_k\}_{k=1}^{+\infty}$ and $\{C_{-k}\}_{k=1}^{+\infty}$ such that

$$u = \sum_{k=1}^{+\infty} (C_k u_k + C_{-k} u_{-k})$$

for any $u = \begin{bmatrix} f & g \end{bmatrix}^T \in X \times X$.

Proof (i) Let $\lambda \in \sigma_p(H)$ and $u = \begin{bmatrix} f & g \end{bmatrix}^T$ be the corresponding eigenfunction. We may suppose B^{-1} be positive definite, then $(B^{-1} f, f) \neq 0$ and

$$\lambda^2 = \frac{(Cf, f)}{(B^{-1} f, f)}.$$

So $\sigma_p(H) \subset \mathbb{R} \cup \mathtt{i}\mathbb{R}$. Because $(H^{-1})^2 = \begin{bmatrix} C^{-1}B^{-1} & 0 \\ 0 & B^{-1}C^{-1} \end{bmatrix}$ is a compact operator, H has at most countable eigenvalues.

(ii) If B, C are positive definite (or $-B, -C$ are positive definite), then $\sigma_p(\widetilde{H}) \subset \mathbb{R}$, and since $\lambda_n \in \sigma_p(\widetilde{H})$ and $u_n = \begin{bmatrix} f_n & g_n \end{bmatrix}^T$ is the corresponding eigenfunction, $\lambda_{-n} = -\lambda_n \in \sigma_p(\widetilde{H})$ and

$$u_{-n} = \begin{bmatrix} f_{-n} & \lambda_{-n}B^{-1}f_{-n} \end{bmatrix}^T = \begin{bmatrix} f_n & -\lambda_n B^{-1}f_n \end{bmatrix}^T$$

is the corresponding eigenfunction, that is, $Span\{f_n\}_{n=1}^{+\infty} = Span\{f_{-n}\}_{n=1}^{+\infty}$. Therefore, if $\{f_{-1}, f_{-2}, \ldots\}$ and $\{f_1, f_2, \ldots\}$ are conjugate orthogonal bases in X, $\{f_1, f_2, \ldots\}$ is an orthogonal bases in X and

$$(u_n, Ju_m) = \begin{cases} 0, & m \neq -n, \\ -2\lambda_n(B^{-1}f_n, f_n) \neq 0, & m = -n. \end{cases}$$

For any $u = \begin{bmatrix} f & g \end{bmatrix}^T \in X \times X$, take

$$C_k = \frac{(\begin{bmatrix} f \\ g \end{bmatrix}, Ju_{-k})}{(u_k, Ju_{-k})}, C_{-k} = \frac{(\begin{bmatrix} f \\ g \end{bmatrix}, Ju_k)}{(u_{-k}, Ju_k)}, k = 1, 2, \ldots.$$

Similar to the proof of Theorem 3.3.5, we can show

$$u = \sum_{k=1}^{+\infty}(C_k u_k + C_{-k}u_{-k}),$$

that is, $\{u_{-1}, u_{-2}, \ldots\} \cup \{u_1, u_2, \ldots\}$ is a symplectic conjugate orthogonal basis of $X \times X$ in the sense of Cauchy principal value. In the same way, if $B, -C$ are positive definite (or $-B, C$ are positive definite), then

$$(u_n, Ju_m) = \begin{cases} 0, & m \neq n, \\ -2\lambda_n(B^{-1}f_n, f_n) \neq 0, & m = n. \end{cases}$$

For any $u = \begin{bmatrix} f & g \end{bmatrix}^T \in X \times X$, let

$$C_k = \frac{\left(\begin{bmatrix} f \\ g \end{bmatrix}, Ju_k\right)}{(u_k, Ju_k)}, C_{-k} = \frac{\left(\begin{bmatrix} f \\ g \end{bmatrix}, Ju_{-k}\right)}{(u_{-k}, Ju_{-k})}, k = 1, 2, \ldots.$$

Then we have

$$u = \sum_{k=1}^{+\infty}(C_k u_k + C_{-k}u_{-k}),$$

that is, $\{u_{-1}, u_{-2}, \ldots\} \cup \{u_1, u_2, \ldots\}$ is a symplectic orthogonal bases of $X \times X$ in the sense of Cauchy principal value. □

3.4.2 Completeness of Root Vector System of 4 × 4 Infinite-Dimensional Hamiltonian Operators in the Sense of Cauchy Principal Value

In the following, we will study the completeness of the root vector system of 4×4 infinite-dimensional Hamiltonian operators in the sense of Cauchy principal value. If $P_\lambda(H) = 2$, then the operator H has a generalized eigenfunction, and the completeness of its root vector system (i.e., all eigenfunctions and generalized eigenfunctions) can be considered in this case. The definition of the completeness of the root vector system in the sense of the Cauchy principal value when $P_\lambda(H) = 2$ will be given next, and the definition is similar when $P_\lambda(H) > 2$.

Definition 3.4.3 Let X be a separable Hilbert space and $P_\lambda(T) = 2$ for every $\lambda \in \sigma_p(T)$. If there exist constant sequences $\{C_k\}_{k=1}^{+\infty}$, $\{C_{-k}\}_{k=1}^{+\infty}$ and $\{D_k\}_{k=1}^{+\infty}$, $\{D_{-k}\}_{k=1}^{+\infty}$ such that

$$x = \sum_{k=1}^{+\infty} (C_k U_k + C_{-k} U_{-k} + D_k V_k + D_{-k} V_{-k}) \tag{3.15}$$

for any $x \in X$, then the root vector system of T is said to be complete in X in the sense of Cauchy principal value, where $\{U_n\}$, $\{V_n\}$ denote the eigenfunction system and the first-order generalized eigenfunction system, respectively.

Theorem 3.4.4 *Let* $H = \begin{bmatrix} 0 & A_1 & B_1 & 0 \\ A_2 & 0 & 0 & B_2 \\ C_1 & 0 & 0 & -A_2^* \\ 0 & C_2 & -A_1^* & 0 \end{bmatrix}$ *be an infinite-dimensional Hamiltonian operator, satisfying* $B_i \geq M > 0, C_i \geq 0, i = 1, 2$ *or* $B_i \leq -M < 0, C_i \leq 0, i = 1, 2$. *If for any* $\lambda \in \sigma_p(H)$ *and the corresponding eigenfunction* $u = \begin{bmatrix} f_1 & f_2 & g_1 & g_2 \end{bmatrix}^T$ *there exists* $\alpha^2 > \frac{1}{4}$ *such that*

$$(A_2^* B_2^{-1} - B_1^{-1} A_1) f_2 = \frac{2\lambda}{2\alpha - 1} B_1^{-1} f_1, \tag{3.16}$$

$$(A_1^* B_1^{-1} - B_2^{-1} A_2) f_1 = \frac{2\lambda}{-2\alpha - 1} B_2^{-1} f_2, \tag{3.17}$$

then we have the following conclusions:
(i) The length of Jordan chain of λ *is 2;*
(ii) If H *is a simple operator* $(0 \notin \sigma_p(H))$, $\sigma_p(H)$ *is at most countable, and* $(f_2^{(n)}, B_2^{-1} A_2 f_1^{(m)}) = (f_1^{(n)}, B_1^{-1} A_1 f_2^{(m)}) = 0$ *for* $n \neq m$, *where*

$\{f_1^{(n)}\}_{n=-\infty}^{+\infty}, \{f_2^{(n)}\}_{n=-\infty}^{+\infty}$ are orthogonal bases of X, then the root vector system of H is complete in the sense of Cauchy principal value in $X \times X \times X \times X$, where $\{f_1^{(n)}\}$ and $\{f_2^{(n)}\}$ are the first and second components of the eigenfunction of λ_k.

Proof (i) If $0 \in \sigma_p(H)$, let $U = \begin{bmatrix} f_1 & f_2 & g_1 & g_2 \end{bmatrix}^T$ be the corresponding eigenfunction, then take

$$V = \begin{bmatrix} f_1 \\ f_2 \\ B_1^{-1} f_1 - B_1^{-1} A_1 f_2 \\ B_2^{-1} f_2 - B_2^{-1} A_2 f_1 \end{bmatrix}.$$

Considering (3.16) and (3.17), we get

$$HV = U,$$

that is to say, $P_0(H) \geq 2$.

On the other hand, taking

$$\widetilde{U} = \begin{bmatrix} -g_1 \\ g_2 \\ -f_1 \\ f_2 \end{bmatrix}, \quad \widetilde{V} = \begin{bmatrix} -B_1^{-1} f_1 + B_1^{-1} A_1 f_2 \\ B_2^{-1} f_2 - B_2^{-1} A_2 f_1 \\ -f_1 \\ f_2 \end{bmatrix},$$

then we have

$$H^* \widetilde{U} = JHJ\widetilde{U} = 0.$$

Again, from (3.16) and (3.17), we get

$$(A_2^* B_2^{-1} - B_1^{-1} A_1) f_2 = 0, \ (A_1^* B_1^{-1} - B_2^{-1} A_2) f_1 = 0.$$

Therefore,

$$H^* \widetilde{V} = JHJ\widetilde{V} = \widetilde{U},$$

namely,

$$(H^*)^2 \widetilde{V} = 0.$$

Because

$$(U, \widetilde{V}) = -(B_1^{-1} f_1, f_1) - (B_2^{-1} f_2, f_2) \neq 0,$$

we get $P_0(H) \leq 2$ by Lemma 3.3.2, so $P_0(H) = 2$.

If $0 \neq \lambda \in \sigma_p(H)$ and $U = \begin{bmatrix} f_1 & f_2 & g_1 & g_2 \end{bmatrix}^T$ are corresponding eigenfunctions, it follows from $(H - \lambda)U = 0$ and the invertibility of $B_i, i = 1, 2$ that

$$g_1 = \lambda B_1^{-1} f_1 - B_1^{-1} A_1 f_2, \quad g_2 = \lambda B_2^{-1} f_2 - B_2^{-1} A_2 f_1,$$

so we have

$$(C_1 - \lambda^2 B_1^{-1} + A_2^* B_2^{-1} A_2) f_1 = \lambda (A_2^* B_2^{-1} - B_1^{-1} A_1) f_2, \tag{3.18}$$

$$(C_2 - \lambda^2 B_2^{-1} + A_1^* B_1^{-1} A_1) f_2 = \lambda (A_1^* B_1^{-1} - B_2^{-1} A_2) f_1. \tag{3.19}$$

Selecting $V = \begin{bmatrix} \frac{\alpha}{\lambda} f_1 \\ -\frac{\alpha}{\lambda} f_2 \\ B_1^{-1} f_1 + \alpha B_1^{-1} f_1 + \frac{\alpha}{\lambda} B_1^{-1} A_1 f_2 \\ B_2^{-1} f_2 - \alpha B_2^{-1} f_2 - \frac{\alpha}{\lambda} B_2^{-1} A_2 f_1 \end{bmatrix}$, we get

$$(H - \lambda) V = U$$

by Eqs. (3.16)–(3.19), that is to say, the zero chain length of λ satisfies $P_\lambda(H) \geq 2$.

On the other hand, from (3.16), (3.17), it follows that $\sigma_p(H) \subset \mathbb{R}$. Taking

$$\widetilde{U} = \begin{bmatrix} -g_1 \\ g_2 \\ -f_1 \\ f_2 \end{bmatrix} \text{ and } \widetilde{V} = \begin{bmatrix} -B_1^{-1} f_1 - \alpha B_1^{-1} f_1 - \frac{\alpha}{\lambda} B_1^{-1} A_1 f_2 \\ B_2^{-1} f_2 - \alpha B_2^{-1} f_2 - \frac{\alpha}{\lambda} B_2^{-1} A_2 f_1 \\ -\frac{\alpha}{\lambda} f_1 \\ -\frac{\alpha}{\lambda} f_2 \end{bmatrix},$$

we have

$$(H^* - \overline{\lambda}) \widetilde{U} = J(H + \lambda) J \widetilde{U} = 0,$$

$$(H^* - \overline{\lambda}) \widetilde{V} = J(H + \lambda) J \widetilde{V} = \widetilde{U},$$

so $\widetilde{V} \in \mathscr{N}((H^* - \overline{\lambda})^2)$. Again by Eqs. (3.16)–(3.19) we get

$$(U, \widetilde{V}) = (-2\alpha - 1)(B_1^{-1} f_1, f_1) + (1 - 2\alpha)(B_2^{-1} f_2, f_2).$$

Because $\alpha^2 > \frac{1}{4}$, $(-2\alpha - 1)(1 - 2\alpha) > 0$, and $(U, \widetilde{V}) \neq 0$. By Lemma 3.3.2 we can know $P_\lambda(H) \leq 2$, which proves $P_\lambda(H) = 2$.

(ii) As $\sigma_p(H) \subset \mathbb{R}$ is symmetric about the real axis, let

$$U_k = \begin{bmatrix} f_1^{(k)} \\ f_2^{(k)} \\ \lambda_k B_1^{-1} f_1^{(k)} - B_1^{-1} A_1 f_2^{(k)} \\ \lambda_k B_2^{-1} f_2^{(k)} - B_2^{-1} A_2 f_1^{(k)} \end{bmatrix}, \quad U_{-k} = \begin{bmatrix} f_1^{(k)} \\ -f_2^{(k)} \\ -\lambda_k B_1^{-1} f_1^{(k)} + B_1^{-1} A_1 f_2^{(k)} \\ \lambda_k B_2^{-1} f_2^{(k)} - B_2^{-1} A_2 f_1^{(k)} \end{bmatrix}$$

be the eigenfunctions corresponding to λ_k and $\lambda_{-k} (\lambda_{-k} = -\lambda_k)$, so

$$V_k = \begin{bmatrix} \frac{\alpha}{\lambda_k} f_1^{(k)} \\ -\frac{\alpha}{\lambda_k} f_2^{(k)} \\ (\alpha+1)B_1^{-1} f_1^{(k)} + \frac{\alpha}{\lambda_k} B_1^{-1} A_1 f_2^{(k)} \\ (1-\alpha)B_2^{-1} f_2^{(k)} - \frac{\alpha}{\lambda_k} B_2^{-1} A_2 f_1^{(k)} \end{bmatrix}, \quad V_{-k} = \begin{bmatrix} -\frac{\alpha}{\lambda_k} f_1^{(k)} \\ -\frac{\alpha}{\lambda_k} f_2^{(k)} \\ (\alpha+1)B_1^{-1} f_1^{(k)} + \frac{\alpha}{\lambda_k} B_1^{-1} A_1 f_2^{(k)} \\ (\alpha-1)B_2^{-1} f_2^{(k)} + \frac{\alpha}{\lambda_k} B_2^{-1} A_2 f_1^{(k)} \end{bmatrix}$$

are the first-order generalized eigenfunctions corresponding to λ_k and $\lambda_{-k}(\lambda_{-k} = -\lambda_k)$.

For any $U = \begin{bmatrix} x_1 & x_2 & y_1 & y_2 \end{bmatrix}^T \in X \times X \times X \times X$, taking

$$C_k = \frac{(U, JV_{-k})}{(U_k, JV_{-k})}, C_{-k} = \frac{(U, JV_k)}{(U_{-k}, JV_k)}, k = 1, 2, \ldots,$$

$$D_k = \frac{(U, JU_{-k})}{(V_k, JU_{-k})}, D_{-k} = \frac{(U, JU_k)}{(V_{-k}, JU_k)}, k = 1, 2, \ldots,$$

we have

$$\sum_{k=1}^{+\infty} C_k U_k + C_{-k} U_{-k} + D_k V_k + D_{-k} V_{-k} = \sum_{k=1}^{+\infty} \begin{bmatrix} \frac{(x_1, B_1^{-1} f_1^{(k)})}{(f_1^{(k)}, B_1^{-1} f_1^{(k)})} f_1^{(k)} \\ \frac{(x_2, B_2^{-1} f_2^{(k)})}{(f_2^{(k)}, B_2^{-1} f_2^{(k)})} f_2^{(k)} \\ \frac{(y_1, f_1^{(k)})}{(f_1^{(k)}, B_1^{-1} f_1^{(k)})} B_1^{-1} f_1^{(k)} + \triangle_1 \\ \frac{(y_2, f_2^{(k)})}{(f_2^{(k)}, B_2^{-1} f_2^{(k)})} B_2^{-1} f_2^{(k)} + \triangle_2 \end{bmatrix},$$

where

$$\triangle_1 = \frac{(x_2, B_2^{-1} A_2 f_1^{(k)})}{(f_1^{(k)}, B_1^{-1} f_1^{(k)})} B_1^{-1} f_1^{(k)} - \frac{(x_2, B_2^{-1} f_2^{(k)})}{(f_2^{(k)}, B_2^{-1} f_2^{(k)})} B_1^{-1} A_1 f_2^{(k)} - \frac{4\lambda (x_2, B_2^{-1} f_2^{(k)})}{(4\alpha - 2)(f_2^{(k)}, B_2^{-1} f_2^{(k)})} B_1^{-1} f_1^{(k)},$$

$$\triangle_2 = \frac{(x_1, B_1^{-1} A_1 f_2^{(k)})}{(f_2^{(k)}, B_2^{-1} f_2^{(k)})} B_2^{-1} f_2^{(k)} - \frac{(x_1, B_1^{-1} f_1^{(k)})}{(f_1^{(k)}, B_1^{-1} f_1^{(k)})} B_2^{-1} A_2 f_1^{(k)} + \frac{4\lambda (x_1, B_1^{-1} f_1^{(k)})}{(4\alpha + 2)(f_2^{(k)}, B_2^{-1} f_2^{(k)})} B_2^{-1} f_2^{(k)}.$$

Because $B_1^{-1} f_1^{(k)} = \frac{(B_1^{-1} f_1^{(k)}, f_1^{(k)})}{(f_1^{(k)}, f_1^{(k)})} f_1^{(k)}$ and $B_2^{-1} f_2^{(k)} = \frac{(B_2^{-1} f_2^{(k)}, f_2^{(k)})}{(f_2^{(k)}, f_2^{(k)})} f_2^{(k)} (k = 1, 2, \ldots)$,

$$\sum_{k=1}^{+\infty} \frac{(x_1, B_1^{-1} f_1^{(k)})}{(f_1^{(k)}, B_1^{-1} f_1^{(k)})} f_1^{(k)} = x_1, \quad \sum_{k=1}^{+\infty} \frac{(x_2, B_2^{-1} f_2^{(k)})}{(f_2^{(k)}, B_2^{-1} f_2^{(k)})} f_2^{(k)} = x_2.$$

By the Parseval equation, we obtain

$$\sum_{k=1}^{+\infty} \frac{(y_1, f_1^{(k)})}{(f_1^{(k)}, B_1^{-1} f_1^{(k)})} B_1^{-1} f_1^{(k)} = y_1, \quad \sum_{k=1}^{+\infty} \frac{(y_2, f_2^{(k)})}{(f_2^{(k)}, B_2^{-1} f_2^{(k)})} B_2^{-1} f_2^{(k)} = y_2.$$

Due to $n \neq m$, $(f_2^{(n)}, B_2^{-1} A_2 f_1^{(m)}) = 0$ and $(f_1^{(n)}, B_1^{-1} A_1 f_2^{(m)}) = 0$. From Eqs. (3.16) and (3.17), it follows that

$$\begin{aligned}(\triangle_1, f_2^{(j)}) &= (x_2, B_2^{-1} A_2 f_1^{(k)}) - \frac{(x_2, B_2^{-1} f_2^{(k)})}{(f_2^{(k)}, B_2^{-1} f_2^{(k)})} (f_2^{(k)}, B_2^{-1} A_2 f_1^{(k)}) \\ &= (x_2, B_2^{-1} A_2 f_1^{(k)}) - (x_2, B_2^{-1} A_2 f_1^{(k)}) = 0\end{aligned}$$

for any $j = 1, 2, \ldots$, which implies $\triangle_1 = 0$. Similarly, $\triangle_2 = 0$, so

$$\sum_{k=1}^{+\infty} C_k U_k + C_{-k} U_{-k} + D_k V_k + D_{-k} V_{-k} = U.$$

The proof is completed. □

The following is an example to illustrate the effectiveness of Theorem 3.4.4.

Example 3.4.5 Consider the infinite-dimensional Hamiltonian operator corresponding to a class of plate bending equations

$$H = \begin{bmatrix} 0 & v\frac{\mathrm{d}}{\mathrm{d}y} & D(1-v^2) & 0 \\ -\frac{\mathrm{d}}{\mathrm{d}y} & 0 & 0 & 2D(1-v) \\ 0 & 0 & 0 & -\frac{\mathrm{d}}{\mathrm{d}y} \\ 0 & -\frac{\mathrm{d}^2}{D\mathrm{d}y^2} & v\frac{\mathrm{d}}{\mathrm{d}y} & 0 \end{bmatrix}$$

where $X = L^2(0, b) \times L^2(0, b) \times L^2(0, b) \times L^2(0, b)$, $D > 0$,

$$\mathscr{D}(H) = \left\{ \begin{bmatrix} \psi_1(y) \\ \psi_2(y) \\ \psi_3(y) \\ \psi_4(y) \end{bmatrix} \in X : \begin{array}{c} \psi_1(0) = \psi_1(b) = 0,\ \frac{\mathrm{d}\psi_2(0)}{D\mathrm{d}x} - v\psi_3(0) = 0, \\ \frac{\mathrm{d}\psi_2(b)}{D\mathrm{d}x} - v\psi_3(b) = 0 \\ \psi_i(y) \text{ is absolutely continuous and } \psi_i'(x) \in L^2(0, b),\ i = 1, 3, 4 \\ \psi_2(y), \psi_2'(y) \text{ are absolutely continuous and } \psi_2''(x) \in L^2(0, b) \end{array} \right\},$$

the eigenfunction and the eigenvalue of the operator can be given by

$$U_k = \begin{bmatrix} \frac{D(1-v)}{\lambda_k} \sin \lambda_k y \\ \frac{D(1-v)}{\lambda_k} \cos \lambda_k y \\ \sin \lambda_k y \\ \cos \lambda_k y \end{bmatrix}, \quad \lambda_k = \frac{k\pi}{b} \ (k = \pm 1, \pm 2, \ldots).$$

Take $\alpha = \frac{3+v}{2(v-1)}$, then $\alpha^2 > \frac{1}{4}$ (because Poisson's ratio v satisfies $0 \le v \le 0.5$) and

$$(A_2^* B_2^{-1} - B_1^{-1} A_1) f_2 = \frac{1-v}{2(1+v)} \sin \lambda_k y = \frac{2\lambda}{2\alpha - 1} B_1^{-1} f_1,$$

$$(A_1^* B_1^{-1} - B_2^{-1} A_2) f_1 = \frac{1-v}{2(v+1)} \cos \lambda_k y = \frac{2\lambda}{-2\alpha - 1} B_2^{-1} f_2.$$

According to the theorem above, the length of the Jordan chain of all eigenvalues equal to 2 and the root vector system of the operator H is complete in X in the sense of Cauchy principal value (see [60]).

Chapter 4
Symplectic Self-adjointness of Infinite-Dimensional Hamiltonian Operators



4.1 Definition of Symplectic Self-adjoint Operators

As far as we know, the eigenvalues of the complex Hamiltonian matrix are symmetric about the imaginary axis, and the eigenvalues of the real Hamiltonian matrix are symmetric about the imaginary axis and symmetric about the real axis. Then, does the point spectrum of the infinite-dimensional Hamiltonian operator have the property of imaginary axis symmetry? The answer is no.

Example 4.1.1 Let $X = L^2[0, +\infty)$ and $A : \mathscr{D}(A) \subset X \to X$ be

$$Ax = x',$$

where $\mathscr{D}(A) = \{x \in X : x \text{ is absolutely continuous}, x(0) = 0, x' \in X\}$. Considering the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix},$$

by calculation,

$$\begin{aligned} \sigma_p(H) &= \{\lambda \in \mathbb{C} : Re\lambda < 0\}, \\ \sigma_r(H) &= \{\lambda \in \mathbb{C} : Re\lambda > 0\}, \\ \sigma_c(H) &= \{\lambda \in \mathbb{C} : Re\lambda = 0\}, \end{aligned}$$

that is, the point spectrum is not symmetry about the imaginary axis.

In order to answer the question of when the point spectrum of the infinite-dimensional Hamiltonian operator is symmetric about the imaginary axis, we need to study when the infinite-dimensional Hamiltonian operator is symplectic self-adjoint.

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Definition 4.1.2 If the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$$

satisfies

$$(JH)^* = JH,$$

then the infinite-dimensional Hamiltonian operator H is symplectic self-adjoint, where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$.

The symplectic self-adjoint infinite-dimensional Hamiltonian operator has good properties.

Lemma 4.1.3 *If the infinite-dimensional Hamiltonian operator* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *is symplectic self-adjoint, then*

(i) $\lambda \in \sigma(H)$ *if and only if* $-\overline{\lambda} \in \sigma(H)$;
(ii) $\lambda \in \sigma_c(H)$ *if and only if* $-\overline{\lambda} \in \sigma_c(H)$;
(iii) $\lambda \in \sigma_p(H)$ *if and only if* $-\overline{\lambda} \in \sigma_r(H) \cup \sigma_p(H)$, *namely, the point spectrum of the infinite-dimensional Hamiltonian operator is symmetric about the imaginary axis if and only if the residual spectrum is an empty set;*
(iv) $\sigma_r(H)$ *has no intersection with the imaginary axis.*

Proof (i) Since $\lambda \in \sigma(H)$ if and only if $\overline{\lambda} \in \sigma(H^*)$. Taking $(JH)^* = JH$ into account we get

$$H^* = JHJ.$$

Also because $J^2 = -I$, then $\lambda \in \sigma(H)$ if and only if $-\overline{\lambda} \in \sigma(H)$.

(ii) Due to $\lambda \in \sigma_c(H)$ if and only if $\overline{\lambda} \in \sigma_c(H^*)$, then by $(JH)^* = JH$, we get $\lambda \in \sigma_c(H)$ if and only if $-\overline{\lambda} \in \sigma_c(H)$.

(iii) If $\lambda \in \sigma_p(H)$, then $\overline{\lambda} \in \sigma_r(H^*) \cup \sigma_p(H^*)$. Again by $H^* = JHJ$, we can get $-\overline{\lambda} \in \sigma_r(H) \cup \sigma_p(H)$.

Conversely, when $-\overline{\lambda} \in \sigma_r(H)$, according to Lemma 2.1.17 we have $-\lambda \in \sigma_p(H^*)$. Through $H^* = JHJ$, we can get $\lambda \in \sigma_p(H)$.

(iv) Assume that there exists $\lambda \in \mathtt{i}\mathbb{R}$ such that $\lambda \in \sigma_r(H)$, then $\overline{\lambda} \in \sigma_p(H^*)$. By $H^* = JHJ$, $-\overline{\lambda} \in \sigma_p(H)$ is obtained, namely, $\lambda \in \sigma_p(H)$, This contradicts with $\lambda \in \sigma_r(H)$. The proof is completed. □

Proposition 4.1.4 *If the infinite-dimensional Hamiltonian operator* H *satisfies* $(JH)^* = JH$, *then the operator* H *is similar to the operator* $-H^*$, *and thus they have the same spectral structures.*

Proof When $(JH)^* = JH$, and $\mathfrak{J} = \begin{bmatrix} 0 & iI \\ -iI & 0 \end{bmatrix}$, then, $\mathfrak{J} = \mathfrak{J}^* = \mathfrak{J}^{-1}$ and

$$-H^* = \mathfrak{J}H\mathfrak{J}^{-1},$$

that is, H is similar to the operator $-H^*$. Because the similar operators have the same spectral structures, H and $-H^*$ have the same spectral structures. □

When the infinite-dimensional Hamiltonian operator is symplectic self-adjoint, the bounded inverse is equivalent to the invertibility.

Lemma 4.1.5 *If the infinite-dimensional Hamiltonian operator H satisfies $(JH)^* = JH$, then H has a bounded inverse if and only if H is invertible.*

Proof When H is invertible, the existence of a bounded inverse is trivial. Conversely, if H has bounded inverse, then $0 \in \rho(H) \cup \sigma_{r,1}(H)$. Considering that $\sigma_r(H)$ has no intersection with the imaginary axis, we have $0 \in \rho(H)$. □

From the above conclusions, the symplectic self-adjointness of the infinite-dimensional Hamiltonian operator plays an important role in characterizing its spectral symmetry. However, for the general densely defined infinite-dimensional Hamiltonian operator,

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$$

it can only satisfy

$$\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}^* \supset \begin{bmatrix} A^* & C \\ B & -A \end{bmatrix},$$

that is,

$$(JH)^* \supset JH.$$

This shows that the general densely defined infinite-dimensional Hamiltonian operator is a symplectic symmetric operator, not necessarily a symplectic self-adjoint operator.

Example 4.1.6 Let the infinite-dimensional Hamiltonian operator H be

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} = \begin{bmatrix} A & -A \\ A & -A \end{bmatrix},$$

where A is an unbounded self-adjoint operator in a Hilbert space X. Then

$$\begin{bmatrix} A & -A \\ A & -A \end{bmatrix}^* \neq \begin{bmatrix} A & A \\ -A & -A \end{bmatrix}.$$

The operator $\begin{bmatrix} A & -A \\ A & -A \end{bmatrix}^*$ is a closed operator, but $\begin{bmatrix} A & A \\ -A & -A \end{bmatrix}$ is not a closed operator.

4.2 The Adjoint of the Sum of Two Operators

We can decompose some special infinite-dimensional Hamiltonian operators into the sum of two operators and then characterize their adjoint operators. For example, given an infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X,$$

then the infinite-dimensional Hamiltonian operator H can be written as

$$H = T + S,$$

where $T = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix}$, $S = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ and

$$T^* = \begin{bmatrix} A^* & 0 \\ 0 & -A \end{bmatrix}, S^* = \begin{bmatrix} 0 & C \\ B & 0 \end{bmatrix}.$$

It is worth noting that, in general,

$$(T + S)^* \neq T^* + S^*.$$

For example, let A be an unbounded self-adjoint operator in a Hilbert space X, $T = \begin{bmatrix} A & 0 \\ 0 & -A \end{bmatrix}$, $S = \begin{bmatrix} 0 & -A \\ A & 0 \end{bmatrix}$. Then

$$T^* + S^* = \begin{bmatrix} A & A \\ -A & -A \end{bmatrix} \neq \begin{bmatrix} A & -A \\ A & -A \end{bmatrix}^* = (T + S)^*.$$

Therefore, after the infinite-dimensional Hamiltonian operator is decomposed into the sum of two operators, it is necessary to discuss the adjoint operator properties of the sum of two operators to characterize adjoint operators.

Definition 4.2.1 Let H_1, H_2, H_3 be Hilbert spaces and $T : \mathscr{D}(T) \subset H_1 \to H_2$, $S : \mathscr{D}(S) \subset H_1 \to H_3$ be linear operators. If $\mathscr{D}(T) \subset \mathscr{D}(S)$ and there exist non-negative constants a, b satisfying

$$\|Sx\|_3 \leq a\|x\|_1 + b\|Tx\|_2$$

for any $x \in \mathscr{D}(T)$, we say that operator S is relatively bounded to T, shortened as T-bounded; the infimum of all b such that the above equation holds is called the relative bound of the operator S concerning T, shortened as T-bounded.

Remark 4.2.2 If S is a bounded linear operator satisfying $\mathscr{D}(T) \subset \mathscr{D}(S)$, then there exists $M > 0$ such that for any $x \in \mathscr{D}(S)$, we have

$$\|Sx\| \leq M\|x\|,$$

and thus $b = 0$, that is, S is bounded relative to T and the T-bound is 0. Furthermore, if T is a closed operator, S is a closable operator, then S is bounded relative to T if and only if $\mathscr{D}(T) \subset \mathscr{D}(S)$. When $\rho(T) \neq \emptyset$, S is bounded relative to T if and only if, there exists $\lambda \in \rho(T)$ such that $S(T - \lambda I)^{-1}$ is bounded.

Given decomposition

$$S = T + A,$$

if $\mathscr{D}(T) \subset \mathscr{D}(A)$, then T is a dominant operator and A is a perturbation operator. When the relative bound of the perturbation operator is small to a certain extent, the closedness (or closability) of the operator S is determined by the closedness (or closability) of the operator T.

Lemma 4.2.3 *Let T, A be linear operators from Hilbert space X to Y. If A is relatively bounded to T and the bound of $T-$ is less than 1, then*

(i) $S = T + A$ is closable (or closed) if and only if T is closable (or closed).
(ii) If T, A are closable, then $\overline{A} + \overline{T} = \overline{A + T}$.

Proof (i) According to the given conditions, let

$$\|Au\| \leq a\|u\| + b\|Tu\|, b < 1, u \in \mathscr{D}(T),$$

then

$$-\frac{a}{1+b}\|u\| + \frac{1}{1+b}\|Su\| \leq \|Tu\| \leq \frac{a}{1-b}\|u\| + \frac{1}{1-b}\|Su\|, u \in \mathscr{D}(T) \tag{4.1}$$

and

$$-a\|u\| + (1-b)\|Tu\| \leq \|Su\| \leq a\|u\| + (1+b)\|Tu\|, u \in \mathscr{D}(T). \tag{4.2}$$

When T is closable, taking arbitrary $\{u_n\} \subset \mathscr{D}(S) = \mathscr{D}(T)$, such that

$$u_n \to 0, Su_n \to v,$$

then, in Eq. (4.1), replace u with $u_n - u_m$ to obtain Tu_n convergence, and let $Tu_n \to \widetilde{u}$. Since T is closable, $\widetilde{u} = 0$, that is, $Tu_n \to 0$. $Su_n \to 0$ is obtained from (4.2), that is, S is closed.

When S is closable, taking arbitrary $\{u_n\} \subset \mathscr{D}(S) = \mathscr{D}(T)$ such that $u_n \to 0$, $Tu_n \to v$, then in (4.2), we replace u with $u_n - u_m$, then obtain $\{Su_n\}$ convergence, and we may let $Su_n \to \widetilde{u}$. Since S is closed, $\widetilde{u} = 0$, that is, $Su_n \to 0$. From Eq. (4.1), we obtain $Tu_n \to 0$, that is, T is closed. $S = T + A$ is closed if and only if T is closed. The proof is completely similar.

(ii) If T, A are closable, then by (i) we know that $T + A$ is closable and $\overline{A}$ is bounded with respect to $\overline{T}$ and the bound of $\overline{T}$-bound is less than 1. Therefore, $\overline{A} + \overline{T}$ is closed operator and

$$A + T \subset \overline{A} + \overline{T}.$$

Then, $\overline{A + T} \subset \overline{A} + \overline{T}$. On the other hand, let $f \in \mathscr{D}(\overline{A} + \overline{T})$, then there exists $\{f_n\} \subset \mathscr{D}(T) = \mathscr{D}(T + A)$ such that

$$f_n \to f \text{ and } Tf_n \to \overline{T} f.$$

Because

$$\|Af_n\| \leq a\|f_n\| + b\|Tf_n\|,$$

$\{Af_n\}$ converges and $Af_n \to \overline{A} f$, so $f \in \mathscr{D}(\overline{A + T})$, that is, $\overline{A} + \overline{T} = \overline{A + T}$. The proof is completed. □

Definition 4.2.4 Let H_1, H_2, H_3 be Hilbert spaces and $T : \mathscr{D}(T) \subset H_1 \to H_2$, $S : \mathscr{D}(S) \subset H_1 \to H_3$ be linear operators. If $\mathscr{D}(T) \subset \mathscr{D}(S)$ and for any $u_n \in \mathscr{D}(T)$, $\{Su_n\}$ contains convergent subsequences when $\{u_n\}$, $\{Tu_n\}$ is bounded. Then, the operator S is said to be relatively compact concerning T , shortened as $T-$ compact.

Remark 4.2.5 Linear operator S is a relatively compact with respect to T, which implies that S is bounded relative to T.

Theorem 4.2.6 *Let T, A be linear operators from a Hilbert space X to Y. If A is a relatively compact with respect to T and T is a closable operator, then $S = T + A$ is a closable operator. In particular, if T is a closed operator, then $S = T + A$ is a closed operator.*

Proof When T is a closable operator, we first prove that A is a compact operator with respect to S. Let $\{u_n\}, \{Su_n\}$ be a bounded sequence, then we only need to prove that $\{Au_n\}$ contains convergent subsequences. Since A is T-compact, it only needs to prove that $\{Tu_n\}$ contains bounded sequences. If not, let $\|Tu_n\| \to \infty$, let $u_n' = \frac{u_n}{\|Tu_n\|}$, then $u_n' \to 0$, $Su_n' \to 0$ and $\{Tu_n'\}$ are bounded. Therefore, $\{Au_n'\}$ contains convergent subsequences, and we may let $Au_n' \to w$, then

$$Tu_n' = Su_n' - Au_n' \to -w.$$

Because $u_n' \to 0$ and T is closed, $w = 0$. This is in contradiction with the fact that $-w$ is the limit of $\{Tu_n'\}$, where $\|Tu_n'\| = 1$.

Next, we prove that S is closable. Let $u_n \to 0$, $Su_n \to v$, then we have that A is S-compact, $\{Au_n\}$ contains a convergent subsequence. We may assume $Au_n \to w$, then

$$Tu_n = Su_n - Au_n \to v - w.$$

Because T is closable, $v - w = 0$, and also because A is T-bounded, $Au_n \to 0$, so $v = w = 0$. The proof is completed. □

Remark 4.2.7 If S, T are densely defined closed operators satisfying $\mathscr{D}(T) \subset \mathscr{D}(S)$, and if $\rho(T) \neq \emptyset$, then S is T-compact if and only if there exists $\lambda \in \rho(T)$ such that $S(T - \lambda)^{-1}$ is a compact operator.

Lemma 4.2.8 *Let T, A be densely defined closed operators in a Hilbert space X. If T is a self-adjoint operator and A is T-compact operator, then*

$$\lim_{\lambda \to \infty} \|A(T - i\lambda I)^{-1}\| = 0,$$

where $\lambda \in \mathbb{R}$.

Proof Because

$$A(T - \mathrm{i}\lambda I)^{-1} = A(T - \mathrm{i}I)^{-1}(T - \mathrm{i}I)(T - \mathrm{i}\lambda I)^{-1},$$

let $T_{\lambda I} = (T - \mathrm{i}I)(T - \mathrm{i}\lambda I)^{-1}$, then T_λ is a normal operator and for any $a \in \mathbb{R}$

$$\lim_{\lambda \to \infty} \frac{a \pm \mathrm{i}}{a \pm \mathrm{i}\lambda} = 0.$$

Thus, when $\lambda \to \infty$, T_λ strongly converges to 0, while $A(T - \mathrm{i}I)^{-1}$ is a compact operator, so $A(T - \mathrm{i}\lambda I)^{-1}$ converges to 0 in norm. □

The following lemma shows that the semi-Fredholm (or Fredholm) property and the index of the closed operator remain unchanged under compact perturbations (see [54]). Next, we will give a different proof method from [54].

Lemma 4.2.9 *Let T be a closed operator in a Hilbert space X, and A be a $T-$compact operator. If T is a semi-Fredholm operator, then $S = T + A$ is also a semi-Fredholm operator and $\mathtt{ind}(T) = \mathtt{ind}(S)$.*

Proof According to Remark IV1.12 of [54], let T be a Fredholm operator and A be a compact operator. From the properties of compact perturbations, it is easy to know that $T + A$ is a closed operator. In order to prove that $\mathscr{R}(T + A)$ is closed, we only need to prove that for any $\|x_n\| = 1 (n = 1, 2, \ldots)$ if

$$(T + A)x_n \to 0,$$

then $\{x_n\}$ contains a convergence subsequence. Since $\{Ax_n\}$ contains convergent subsequences, you might want to let

$$Ax_n \to w,$$

then

$$Tx_n \to -w.$$

Because $\mathscr{R}(T)$ is closed, there exists $u \in \mathscr{D}(T)$ such that $Tu = -w$. Thus,

$$T(x_n - u) \to 0,$$

that is, $\{x_n - u\}$ contains convergent subsequence, so $\{x_n\}$ contains convergent subsequence.

We will prove that $\mathscr{N}(T + A)$ is finite-dimensional. Under the space decomposition $X = \mathscr{N}(T) \oplus \mathscr{N}(T)^\perp = \mathscr{R}(T)^\perp \oplus \mathscr{R}(T)$, the operators T, A have the following block forms:

$$T = \begin{bmatrix} 0 & 0 \\ 0 & T_1 \end{bmatrix} : \mathscr{N}(T) \oplus \mathscr{N}(T)^\perp \cap \mathscr{D}(T) \to \mathscr{R}(T)^\perp \oplus \mathscr{R}(T),$$

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} : \mathscr{N}(T) \oplus \mathscr{N}(T)^\perp \to \mathscr{R}(T)^\perp \oplus \mathscr{R}(T).$$

Since

$$\mathscr{N}(T + A) = \{\begin{bmatrix} x \\ y \end{bmatrix} : A_{11}x + A_{12}y = 0, A_{21}x + (T_1 + A_{22})y = 0\},$$

it is only necessary to prove that the second component is finite-dimensional. Assuming that the second component is infinite-dimensional, and since $\dim\mathscr{N}(T) < \infty$, we have a unit orthogonal sequence $\{y_n\}_{n=1}^{+\infty} \subset \mathscr{N}(T)^\perp \cap \mathscr{D}(T)$ and a vector sequence $\{x_n\} \subset \mathscr{N}(T)$ such that

$$A_{11}x_n + A_{12}y_n = 0 \tag{4.3}$$

and

$$A_{21}x_n + (T_1 + A_{22})y_n = 0. \tag{4.4}$$

Because A_{22} is a compact operator, $A_{22}y_n \to 0$. By $\dim\mathscr{N}(T) < \infty$, we know that $\{A_{21}x_n\}$ is convergent, so $\{T_1y_n\}$ is convergent by Eq. (4.4), and $\{y_n\}$ convergence is obtained by the invertibility of T_1. This contradicts $\{y_n\}$ being an orthogonal sequence. The proof of $\dim\mathscr{R}(T + A)^\perp < \infty$ is similar. □

Lemma 4.2.10 *Let T be a densely defined closed operator in a Hilbert space X. Then T is a Fredholm operator if and only if T^* is a Fredholm operator and $ind(T) = -ind(T^*)$.*

Proof It is easy to know

$$\mathscr{N}(T) = \mathscr{R}(T^*)^{\perp}, \mathscr{N}(T^*) = \mathscr{R}(T)^{\perp},$$

then $\mathtt{nul}(T) = \mathtt{def}(T^*), \mathtt{nul}(T^*) = \mathtt{def}(T)$, that is, T is a Fredholm operator if and only if T^* is a Fredholm operator, and

$$\mathtt{ind}(T) = \mathtt{nul}(T) - \mathtt{def}(T) = \mathtt{def}(T^*) - \mathtt{nul}(T^*) = -\mathtt{ind}(T^*).$$

The proof is completed. □

Lemma 4.2.11 *If T, S are Fredholm operators and satisfy $S \subset T$ such that $\mathtt{ind}(T) \leq \mathtt{ind}(S)$ and $T = S$ if and only if $\mathtt{ind}(T) = \mathtt{ind}(S)$.*

Proof Since $S \subset T$, so $\mathscr{R}(S) \subset \mathscr{R}(T), \mathscr{N}(S) \subset \mathscr{N}(T)$, thus $\mathtt{ind}(T) \leq \mathtt{ind}(S)$ is obvious. When $\mathtt{ind}(T) = \mathtt{ind}(S)$ is satisfied, $R(S) = \mathscr{R}(T), \mathscr{N}(S) = \mathscr{N}(T)$, so $T = S$. □

4.2.1 Representation of the Adjoint of the Sum of Two Operators

The adjoint operator of the sum of two operators has critical applications in quantum mechanics. For example, when A, B are self-adjoint operators, whether $A + B$ is a self-adjoint operator is one of the fundamental problems in quantum mechanics (see [62]).

Lemma 4.2.12 *Let A be a Fredholm operator in a Hilbert space X, $B : \mathscr{D}(B) \subset X \to X$ and B be $A-$ compact operator, B^* be A^*- compact operator. Then $(A + B)^* = A^* + B^*$.*

Proof By Lemmas 4.2.9 and 4.2.10, we know that $(A + B)^*$ is a Fredholm operator and $A^* + B^*$ is also a Fredholm operator. According to Lemma 4.2.11, in order to prove $(A + B)^* = A^* + B^*$ we only need to prove

$$\mathtt{ind}((A + B)^*) = \mathtt{ind}(A^* + B^*).$$

In fact,

$$\mathtt{ind}(A^* + B^*) = \mathtt{ind}(A^*) = -\mathtt{ind}(A) = -\mathtt{ind}(A + B) = \mathtt{ind}((A + B)^*),$$

then the conclusion holds. □

Lemma 4.2.13 *Let $A : \mathscr{D}(A) \subset X \to Y$ be a densely defined closed linear operator. If B is A-bounded, B^* is A^*-bounded, and the relative bounds are less than 1, then $A + B$ is a closed operator and satisfies $(A + B)^* = A^* + B^*$.*

Proof It can be seen from the given conditions that there are non-negative constants $a_1, a_2, b_1, b_2 (b_1 < 1, b_2 < 1)$ such that

$$\|Bx\| \leq a_1\|x\| + b_1\|Ax\|, x \in \mathscr{D}(A),$$

$$\|B^*y\| \leq a_2\|y\| + b_2\|A^*y\|, y \in \mathscr{D}(A^*).$$

So for any $z \in [0, 1]$, we have

$$\frac{1}{1+b_1z}\|(A+zB)x\| - \frac{za_1}{1+b_1z}\|x\| \leq \|Ax\| \leq \frac{1}{1-b_1z}\|(A+zB)x\| + \frac{za_1}{1-b_1z}\|x\| \tag{4.5}$$

and

$$\frac{1}{1+b_2z}\|(A^*+zB^*)y\| - \frac{za_2}{1+b_2z}\|y\| \leq \|A^*y\| \leq \frac{1}{1-b_2z}\|(A^*+zB^*)y\| + \frac{za_2}{1-b_2z}\|y\|. \tag{4.6}$$

Thus $\{(A+zB)x_n\}, \{(A^*+zB^*)y_n\}$ are Cauchy sequences implying $\{Ax_n\}, \{A^*y_n\}$ is Cauchy sequence, so A is a closed operator implying $A+zB$ is a closed operator and A^*+zB^* is a closed operator. It is known from Theorem 5.27 of [43]

$$(A+zB)^* = A^* + zB^*.$$

Let $z = 1$, we get $(A+B)^* = A^* + B^*$. □

Proposition 4.2.14 *Let $A : \mathscr{D}(A) \subset X \to X$ be self-adjoint operators. If B is a symmetric operator with A-bound less than 1, then $A+B$ is a self-adjoint operator.*

Proof Since the relative bound of B with respect to A is less than 1, there exist non-negative numbers $a, b, b < 1$ such that

$$\|Bx\| \leq a\|x\| + b\|Ax\|, \forall x \in \mathscr{D}(A).$$

Because B is a symmetric operator, $\mathscr{D}(A) \subset \mathscr{D}(B^*)$ and for any $x \in \mathscr{D}(A)$, we have

$$\|B^*x\| = \|Bx\| \leq a\|x\| + b\|Ax\|, \forall x \in \mathscr{D}(A).$$

That is, B^* is A^* bounded, and the relative bound is less than 1. By Lemma 4.2.13, we know that

$$(A+B)^* = (A^*+B^*)|_{\mathscr{D}(A)} = A+B.$$

The proof is completed. □

According to Lemma 4.2.13, the following corollary is obvious.

Proposition 4.2.15 *Let $A : \mathscr{D}(A) \subset X \to X$ be a closed operator. If B is a bounded linear operator defined on the whole space, then $(A+B)^* = A^* + B^*$.*

4.2.2 Characterization of Symplectic Self-adjointness of Infinite-Dimensional Hamiltonian Operators Based on Operator Perturbation Theory

In order to study the symplectic self-adjointness of the infinite-dimensional Hamiltonian operator, the following lemmas are given.

Lemma 4.2.16 *Let A, B, C, D be densely defined closed operators in a Hilbert space X and the relative bound of C with respect to A is less than 1 and the relative bound of B with respect to D is less than 1. Let*

$$S = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}, T = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix},$$

then the relative bound of S with respect to T is less than 1.

Proof Under given conditions, there exist constants a_1, a_2 and $b_1 < 1, b_2 < 1$ such that

$$\|Cu\| \leq a_1\|u\| + b_1\|Au\|, \forall u \in \mathscr{D}(A),$$

$$\|Bv\| \leq a_2\|v\| + b_2\|Dv\|, \forall v \in \mathscr{D}(D).$$

According to Cauchy inequality, for any $r > 0$, we have

$$\|Cu\|^2 \leq (1 + \frac{1}{r})a_1^2\|u\|^2 + (1 + r)b_1^2\|Au\|^2, \forall u \in \mathscr{D}(A),$$

$$\|Bv\|^2 \leq (1 + \frac{1}{r})a_2^2\|v\|^2 + (1 + r)b_2^2\|Dv\|^2, \forall v \in \mathscr{D}(D).$$

Let $\widetilde{a} = \max\{a_1, a_2\}, \widetilde{b} = \max\{b_1, b_2\}$, then for any $U = \begin{bmatrix} u & v \end{bmatrix}^T \in \mathscr{D}(T)$, we have

$$\|SU\|^2 = \|Cu\|^2 + \|Bv\|^2 \leq (1 + \frac{1}{r})\widetilde{a}^2\|U\|^2 + (1 + r)\widetilde{b}^2\|TU\|^2,$$

that is,

$$\|SU\| \leq \widetilde{a}\sqrt{1 + \frac{1}{r}}\|U\| + \widetilde{b}\sqrt{1 + r}\|TU\|.$$

Because of $\widetilde{b} < 1$, there exists $0 < \varepsilon < 1$ such that $\widetilde{b} < 1 - \varepsilon$. And because $r > 0$ is arbitrary, take $r = \frac{1}{2}(\frac{1}{(1-\varepsilon)^2} - 1)$, then we have

$$\widetilde{b}\sqrt{1 + r} < \sqrt{\frac{1}{2}(\frac{1}{(1 - \varepsilon)^2} + 1))(1 - \varepsilon)^2} < 1,$$

so the operator S is a relative operator and the relative bound of T is less than 1. The proof is completed. □

Theorem 4.2.17 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator, if one of the following conditions is satisfied*

(i) $\mathscr{D}(H) = \mathscr{D}(A) \times \mathscr{D}(A^*)$, *the relative bounds of* C *with respect to* A *and the relative bounds of* B *with respect to* A^* *are less than 1.*

(ii) $\mathscr{D}(H) = \mathscr{D}(C) \times \mathscr{D}(B)$, *the relative bounds of* A *with respect to* C *and the relative bounds of* A^* *with respect to* B *are less than 1,*

then $(JH)^* = JH$.

Proof (i) Let

$$T = \begin{bmatrix} 0 & -A^* \\ -A & 0 \end{bmatrix}, S = \begin{bmatrix} C & 0 \\ 0 & -B \end{bmatrix},$$

then

$$JH = T + S.$$

When the relative bounds of C with respect to A and B with respect to A^* are both less than 1, by Lemma 4.2.16, we know that S is bounded relative to T and the $T-$ bound is less than 1. Also because, S and T are self-adjoint operators, S^* is bounded relative to T^* and the relative bound is also less than 1, it can be known from Lemma 4.2.13 that

$$(T + S)^* = T^* + S^* = T + S,$$

that is, $(JH)^* = JH$.

The proof of (ii) is completely similar. □

Proposition 4.2.18 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If there exists* $\lambda \in \rho(A)$ *such that* $\|C(A - \lambda I)^{-1}\| < 1$, $\|B(A^* - \overline{\lambda} I)^{-1}\| < 1$, *then* $(JH)^* = JH$.

Proof If there exists $\lambda \in \rho(A)$ such that $\|C(A - \lambda I)^{-1}\| < 1$, $\|B(A^* - \overline{\lambda} I)^{-1}\| < 1$, then the relative bounds of C with respect to A and B with respect to A^* are less than 1, so $(JH)^* = JH$. □

Theorem 4.2.19 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If* A *is a compact operator relative to* C *and* A^* *is a compact operator relative to* B, *then* $(JH)^* = JH$.

Proof When A is a compact operator relative to C and A^* is a compact operator relative to B, according to Lemma 4.2.8

$$\lim_{\lambda\to\infty} \|A(C - \mathrm{i}\lambda I)^{-1}\| = 0 \text{ and } \lim_{\lambda\to\infty} \|A^*(B - \mathrm{i}\lambda I)^{-1}\| = 0.$$

That is, for any $\varepsilon > 0$, there exists $N > 0$, when $|\lambda| > N$

$$\|A(C - \mathrm{i}\lambda I)^{-1}\| < \varepsilon \text{ and } \|A^*(B - \mathrm{i}\lambda I)^{-1}\| < \varepsilon.$$

For any $x \in \mathscr{D}(C)$, we have

$$\|Ax\| = \|A(C - \mathrm{i}\lambda I)^{-1}(C - \mathrm{i}\lambda I)x\| < \varepsilon\|(C - \mathrm{i}\lambda I)x\| \le \varepsilon|\lambda|\|x\| + \varepsilon\|Cx\|.$$

Since $\varepsilon > 0$ is arbitrary, A is bounded relative to C and the relative bound is 0. Similarly, we can prove that A^* is bounded relative to B, and the relative bound is 0. Therefore, let $S = \begin{bmatrix} 0 & -A^* \\ -A & 0 \end{bmatrix}$, $T = \begin{bmatrix} C & 0 \\ 0 & -B \end{bmatrix}$, then S and T are self-adjoint operators and S is bounded relative to T, and $T-$ bound is less than 1. From Lemma 4.2.13, we can see that

$$(T + S)^* = T^* + S^* = T + S,$$

$(JH)^* = JH$. The proof is completed. □

Proposition 4.2.20 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a non-negative Hamiltonian operator. If* A *is a compact operator relative to* C *and* A^* *is a compact operator relative to* B, *and* $0 \in \rho(B) \cap \rho(C)$, *then* $0 \in \rho(H)$.

Proof When A is compact relative to C and A^* is compact relative to B, $(JH)^* = JH$ is known by Theorem 4.2.17, thus $0 \notin \sigma_r(H)$. On the other hand, if $0 \in \rho(B) \cap \rho(C)$, then H has bounded inverse. So, $0 \in \rho(H)$. □

Proposition 4.2.21 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant infinite-dimensional Hamiltonian operator. If* $B \in \mathscr{B}(X)$ *or* $C \in \mathscr{B}(X)$, *then* $(JH)^* = JH$ *holds.*

Proof It can be assumed that $C \in \mathscr{B}(X)$, $T = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix}$, $S = \begin{bmatrix} 0 & 0 \\ C & 0 \end{bmatrix}$, then

$$H = T + S.$$

Since S is a bounded operator on the whole space, we get

$$H^* = T^* + S^*.$$

From Lemma 4.2.16, we can know that

$$T^* = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix}^* = \begin{bmatrix} A^* & 0 \\ B & -A \end{bmatrix}.$$

Then, $(JH)^* = JH$. □

4.3 The Adjoint of Product of Operators

Some special Hamiltonian matrices can be decomposed into the product of several simple matrices. For example, given a Hamiltonian matrix

$$H_{2n} = \begin{bmatrix} A_n & B_n \\ C_n & -A_n^* \end{bmatrix} \in \mathbb{C}^{2n\times 2n},$$

if the set of eigenvalues of H_{2n} has no intersection with the imaginary axis, then there exists a unitary matrix $U_{2n} = \begin{bmatrix} W_n & V_n \\ -V_n & W_n \end{bmatrix}$, $W_n, V_n \in \mathbb{C}^{n\times n}$ such that

$$H_{2n} = U_{2n}^* \begin{bmatrix} T_n & R_n \\ 0 & -T_n^* \end{bmatrix} U_{2n}, T_n, R_n \in \mathbb{C}^{n\times n},$$

that is, a Hamiltonian matrix H_{2n} and an upper triangular Hamiltonian matrix $\begin{bmatrix} T_n & R_n \\ 0 & -T_n^* \end{bmatrix}$ are unitary similar (see [63]). At this time, the spectrum of H_{2n} can be determined by $\begin{bmatrix} T_n & R_n \\ 0 & -T_n^* \end{bmatrix}$. For unbounded block operator matrices, a certain decomposition can also be introduced to express the block operator matrices as the product of several simple operator matrices. Then, the properties of the spectrum can be studied.

Lemma 4.3.1 *Let T, S be densely defined linear operators in a Hilbert space X, when $\rho(T) \neq \emptyset$, $(T-\lambda I)^{-1}S$ is bounded, if and only if $\mathscr{D}(T^*) \subset \mathscr{D}(S^*)$, where $\lambda \in \rho(T)$.*

Proof If $\mathscr{D}(T^*) \subset \mathscr{D}(S^*)$, then $S^*(T^* - \overline{\lambda} I)^{-1}$ is bounded on the whole space and

$$(S^*(T^* - \overline{\lambda} I)^{-1})^* \supset (T-\lambda I)^{-1}S,$$

so $(T-\lambda I)^{-1}S$ is bounded.

Conversely, if $(T-\lambda I)^{-1}S$ is bounded on $\mathscr{D}(S)$, then $\overline{(T-\lambda I)^{-1}S}$ is a bounded operator defined on the whole space and

$$(S^*(T^*-\overline{\lambda}I)^{-1})^* \supset \overline{(T-\lambda I)^{-1}S},$$

therefore, $\mathscr{D}(T^*) \subset \mathscr{D}(S^*)$. The proof is completed. □

Lemma 4.3.2 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} : \mathscr{D}(\mathscr{A}) \subset X \times X \to X \times X$ *be a densely defined block operator matrix. If* $\rho(A) \neq \emptyset$, C *is closable and* $\mathscr{D}(A) \subset \mathscr{D}(C)$, $\mathscr{D}(A^*) \subset \mathscr{D}(B^*)$, *then for any* $\lambda \in \rho(A)$, *we have*

$$\mathscr{A}-\lambda I = \begin{bmatrix} I & 0 \\ C(A-\lambda I)^{-1} & I \end{bmatrix} \begin{bmatrix} A-\lambda I & 0 \\ 0 & S_1(\lambda) \end{bmatrix} \begin{bmatrix} I & \overline{(A-\lambda I)^{-1}B} \\ 0 & I \end{bmatrix}, \tag{4.7}$$

where $S_1(\lambda) = D-\lambda I - C(A-\lambda I)^{-1}B$.

Proof For any $\lambda \in \rho(A)$, let

$$\mathscr{A}_\lambda = \begin{bmatrix} I & 0 \\ C(A-\lambda I)^{-1} & I \end{bmatrix} \begin{bmatrix} A-\lambda I & 0 \\ 0 & S_1(\lambda) \end{bmatrix} \begin{bmatrix} I & \overline{(A-\lambda I)^{-1}B} \\ 0 & I \end{bmatrix},$$

then for any $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(\mathscr{A})$, we have

$$\mathscr{A}_\lambda \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (A-\lambda)x + By \\ Cx + (D-\lambda)y \end{bmatrix} = (\mathscr{A}-\lambda I)\begin{bmatrix} x \\ y \end{bmatrix},$$

that is, $\mathscr{A}-\lambda I \subset \mathscr{A}_\lambda$.

On the other hand, let $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(\mathscr{A}_\lambda)$, then

$$x + \overline{(A-\lambda I)^{-1}B}y \in \mathscr{D}(A),\ y \in \mathscr{D}(S_1(\lambda)).$$

By $y \in \mathscr{D}(S_1(\lambda))$, we know $y \in \mathscr{D}(B)$, $\overline{(A-\lambda I)^{-1}B}y = (A-\lambda I)^{-1}By \in \mathscr{D}(A)$, so $x \in \mathscr{D}(A)$, $\mathscr{D}(\mathscr{A}_\lambda) \subset \mathscr{D}(\mathscr{A})$, that is, $\mathscr{A}_\lambda = \mathscr{A}-\lambda$. The proof is completed. □

Similarly, when B is closable and $\rho(D) \neq \emptyset$, $\mathscr{D}(D) \subset \mathscr{D}(B)$, $\mathscr{D}(D^*) \subset \mathscr{D}(C^*)$, for any $\lambda \in \rho(D)$, we have

$$\mathscr{A}-\lambda I = \begin{bmatrix} I & B(D-\lambda I)^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} S_2(\lambda) & 0 \\ 0 & D-\lambda I \end{bmatrix} \begin{bmatrix} I & 0 \\ \overline{(D-\lambda I)^{-1}C} & I \end{bmatrix}, \tag{4.8}$$

where $S_2(\lambda) = A-\lambda I - B(D-\lambda I)^{-1}C$.

Similar to Schur complement and Schur decomposition of bounded block operator matrices, Eqs. (4.7) and (4.8) are called Frobenius-Schur decomposition of $\mathscr{A}-\lambda I$, while $S_1(\lambda)$ and $S_2(\lambda)$ are called Schur complement.

Theorem 4.3.3 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} : \mathscr{D}(\mathscr{A}) \subset X \times X \to X \times X$ *be a densely defined block operator matrix. If* $\rho(A) \neq \emptyset$, C *is closable, and* $\mathscr{D}(A) \subset \mathscr{D}(C)$, $\mathscr{D}(A^*) \subset \mathscr{D}(B^*)$, *then*

(i) $\mathscr{A}$ *is closed, if and only if there exists* $\lambda \in \rho(A)$ *such that* $S_1(\lambda)$ *is closed;*
(ii) $\mathscr{A}$ *is closable, if and only if there exists* $\lambda \in \rho(A)$ *such that* $S_1(\lambda)$ *is closable and we have*

$$\overline{\mathscr{A}} = \lambda I + \begin{bmatrix} I & 0 \\ C(A-\lambda I)^{-1} & I \end{bmatrix} \begin{bmatrix} A-\lambda I & 0 \\ 0 & \overline{S_1(\lambda)} \end{bmatrix} \begin{bmatrix} I & \overline{(A-\lambda I)^{-1}B} \\ 0 & I \end{bmatrix}.$$

Proof (i) For any $\lambda \in \rho(A)$ there exists

$$\mathscr{A} - \lambda I = \begin{bmatrix} I & 0 \\ C(A-\lambda I)^{-1} & I \end{bmatrix} \begin{bmatrix} A-\lambda I & 0 \\ 0 & S_1(\lambda) \end{bmatrix} \begin{bmatrix} I & \overline{(A-\lambda I)^{-1}B} \\ 0 & I \end{bmatrix}.$$

$\mathscr{A}$ is closed and if and only if $\mathscr{A} - \lambda$ is closed, $\mathscr{A} - \lambda$ is closed and if and only if $\begin{bmatrix} A-\lambda I & 0 \\ 0 & S_1(\lambda) \end{bmatrix}$ is closed by Lemma 2.6.2. Because $A - \lambda I$ is closed, $\begin{bmatrix} A-\lambda I & 0 \\ 0 & S_1(\lambda) \end{bmatrix}$ is closed and if and only if $S_1(\lambda)$ is closed.

(ii) Because

$$\overline{\begin{bmatrix} A-\lambda I & 0 \\ 0 & S_1(\lambda) \end{bmatrix}} = \begin{bmatrix} A-\lambda I & 0 \\ 0 & \overline{S_1(\lambda)} \end{bmatrix},$$

it can be known from Lemma 2.6.2,

$$\overline{\mathscr{A}} = \lambda + \begin{bmatrix} I & 0 \\ C(A-\lambda I)^{-1} & I \end{bmatrix} \begin{bmatrix} A-\lambda I & 0 \\ 0 & \overline{S_1(\lambda)} \end{bmatrix} \begin{bmatrix} I & \overline{(A-\lambda I)^{-1}B} \\ 0 & I \end{bmatrix}.$$

The proof is completed. □

Similarly, the following theorem can be proved.

Theorem 4.3.4 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} : \mathscr{D}(\mathscr{A}) \subset X \times X \to X \times X$ *be a densely defined block operator matrix. If* $\rho(D) \neq \emptyset$, B *is closable, and* $\mathscr{D}(D) \subset \mathscr{D}(B)$, $\mathscr{D}(D^*) \subset \mathscr{D}(C^*)$, *then*

(i) $\mathscr{A}$ *is closed if and only if there exists* $\lambda \in \rho(D)$ *such that* $S_2(\lambda)$ *is closed operator;*
(ii) $\mathscr{A}$ *is closable if and only if there exists* $\lambda \in \rho(D)$ *such that* $S_2(\lambda)$ *is closable and we have*

$$\overline{\mathscr{A}} = \lambda I + \begin{bmatrix} I & B(D-\lambda I)^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} \overline{S_2(\lambda)} & 0 \\ 0 & D-\lambda I \end{bmatrix} \begin{bmatrix} I & 0 \\ \overline{(D-\lambda I)^{-1}C} & I \end{bmatrix}.$$

Let $J = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$, then $J\mathscr{A}$ is closed (closable) if and only if $\mathscr{A}$ is closed (closable). Therefore, we use the Frobenius-Schur decomposition of $J\mathscr{A}$, the following conclusions can be proved.

Theorem 4.3.5 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} : \mathscr{D}(\mathscr{A}) \subset X \times X \to X \times X$ *be a densely defined block operator matrix. If* $\rho(C) \neq \emptyset$, *A is closable, and* $\mathscr{D}(C) \subset \mathscr{D}(A)$, $\mathscr{D}(C^*) \subset \mathscr{D}(D^*)$, *then*

(i) $\mathscr{A}$ *is closed if and only if there exists* $\lambda \in \rho(C)$ *such that* $S_3(\lambda)$ *is closed;*
(ii) $\mathscr{A}$ *is closable if and only if there exists* $\lambda \in \rho(C)$ *such that* $S_3(\lambda)$ *is closable and we have*

$$\overline{\mathscr{A}} = J(\lambda I + \begin{bmatrix} I & 0 \\ A(C-\lambda I)^{-1} & I \end{bmatrix} \begin{bmatrix} C-\lambda I & 0 \\ 0 & \overline{S_3(\lambda)} \end{bmatrix} \begin{bmatrix} I & \overline{(C-\lambda I)^{-1}D} \\ 0 & I \end{bmatrix}),$$

where $S_3(\lambda) = B - \lambda - A(C-\lambda I)^{-1}D$.

Theorem 4.3.6 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} : \mathscr{D}(\mathscr{A}) \subset X \times X \to X \times X$ *be a densely defined block operator matrix. If* $\rho(B) \neq \emptyset$, *D is closable, and* $\mathscr{D}(B) \subset \mathscr{D}(D)$, $\mathscr{D}(B^*) \subset \mathscr{D}(A^*)$, *then*

(i) $\mathscr{A}$ *is closed if and only if there exists* $\lambda \in \rho(B)$ *such that* $S_4(\lambda)$ *is closed;*
(ii) $\mathscr{A}$ *is closable, if and only if there exists* $\lambda \in \rho(B)$ *such that* $S_4(\lambda)$ *is closable and we have*

$$\overline{\mathscr{A}} = J(\lambda I + \begin{bmatrix} I & D(B-\lambda I)^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} \overline{S_4(\lambda)} & 0 \\ 0 & B-\lambda I \end{bmatrix} \begin{bmatrix} I & 0 \\ \overline{(B-\lambda I)^{-1}A} & I \end{bmatrix}),$$

where $S_4(\lambda) = C - \lambda - D(B-\lambda)^{-1}A$.

Frobenius-Schur decomposition can also be used to solve the invertibility problem of the block operator matrix.

Theorem 4.3.7 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ *be a* 2×2 *densely defined closed block operator matrix.*

(i) If A is invertible, C is closable, and $\mathscr{D}(A) \subset \mathscr{D}(C)$, $\mathscr{D}(A^*) \subset \mathscr{D}(B^*)$, *then* $\mathscr{A}$ *is invertible, if and only if* $D - CA^{-1}B$ *is invertible.*
(ii) If D is invertible, B is closable, and $\mathscr{D}(D) \subset \mathscr{D}(B)$, $\mathscr{D}(D^*) \subset \mathscr{D}(C^*)$, *then* $\mathscr{A}$ *is invertible, if and only if* $A - BD^{-1}C$ *is invertible.*
(iii) If C is invertible, A is closable, and $\mathscr{D}(C) \subset \mathscr{D}(A)$, $\mathscr{D}(C^*) \subset \mathscr{D}(D^*)$, *then* $\mathscr{A}$ *is invertible, if and only if* $B - AC^{-1}D$*is invertible.*
(iv) If B is invertible, D is closable, and $\mathscr{D}(B) \subset \mathscr{D}(D)$, $\mathscr{D}(B^*) \subset \mathscr{D}(A^*)$, *then* $\mathscr{A}$ *is invertible, if and only if* $C - DB^{-1}A$ *is invertible.*

Proof The conclusion (i) is only proved, and the proof of other conclusions is completely similar. If A is invertible, C is closable, and $\mathscr{D}(A) \subset \mathscr{D}(C)$, $\mathscr{D}(A^*) \subset \mathscr{D}(B^*)$, then

$$\mathscr{A} = \begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I & \overline{A^{-1}B} \\ 0 & I \end{bmatrix} \triangleq RST.$$

Since R, T are invertible operators defined on the whole space, and when A is invertible, $\mathscr{A}$ is invertible, if and only if $D - CA^{-1}B$ is invertible. □

Using the Frobenius-Schur decomposition, the spectral point distribution of the block operator matrix can also be characterized.

Theorem 4.3.8 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ *be a densely defined closed block operator matrix.*

(i) If $\rho(A) \neq \emptyset$, *C is closable, and* $\mathscr{D}(A) \subset \mathscr{D}(C)$, $\mathscr{D}(A^*) \subset \mathscr{D}(B^*)$, *then there exists* $\Sigma(\mathscr{A} - \lambda I) = \Sigma(S_1(\lambda))$ *for any* $\lambda \in \rho(A)$.

(ii) If $\rho(D) \neq \emptyset$, *B is closable, and* $\mathscr{D}(D) \subset \mathscr{D}(B)$, $\mathscr{D}(D^*) \subset \mathscr{D}(C^*)$, *then there exists* $\Sigma(\mathscr{A} - \lambda I) = \Sigma(S_2(\lambda))$ *for any* $\lambda \in \rho(D)$.

(iii) If $\rho(C) \neq \emptyset$, *A is closable, and* $\mathscr{D}(C) \subset \mathscr{D}(A)$, $\mathscr{D}(C^*) \subset \mathscr{D}(D^*)$, *then there exists* $\Sigma(\mathscr{A} - \lambda I) = \Sigma(S_3(\lambda))$ *for any* $\lambda \in \rho(C)$.

(iv) If $\rho(B) \neq \emptyset$, *D is closable, and* $\mathscr{D}(B) \subset \mathscr{D}(D)$, $\mathscr{D}(B^*) \subset \mathscr{D}(A^*)$, *then there exists* $\Sigma(\mathscr{A} - \lambda I) = \Sigma(S_4(\lambda))$ *for any* $\lambda \in \rho(B)$.

Here, $\Sigma \in \{\sigma_p, \sigma_r, \sigma_c, \sigma\}$, $S_i(\lambda)$, $i = 1, 2, 3, 4$ *are shown in the previous theorem.*

Proof The conclusion (i) is only proved, and the proof of other conclusions is completely similar. If $\rho(A) \neq \emptyset$, C is closable, and $\mathscr{D}(A) \subset \mathscr{D}(C)$, $\mathscr{D}(A^*) \subset \mathscr{D}(B^*)$, then for any $\lambda \in \rho(A)$, we have

$$\mathscr{A} - \lambda I = \begin{bmatrix} I & 0 \\ C(A - \lambda I)^{-1} & I \end{bmatrix} \begin{bmatrix} A - \lambda I & 0 \\ 0 & S_1(\lambda) \end{bmatrix} \begin{bmatrix} I & \overline{(A - \lambda I)^{-1}B} \\ 0 & I \end{bmatrix} \triangleq RST.$$

Since R, T are invertible operators defined on the whole space, and when $A - \lambda$ is invertible, $\mathscr{A} - \lambda$ is injective, if and only if $S_1(\lambda)$ is injective; $\mathscr{R}(\mathscr{A} - \lambda)$ is dense, if and only if $\mathscr{R}(S_1(\lambda))$ is dense. So,

$$\Sigma(\mathscr{A} - \lambda I) = \Sigma(S_1(\lambda)).$$

The proof is completed. □

Proposition 4.3.9 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ *be a diagonally dominant dense closed block operator matrix. If* $\rho(A) \cap \rho(D) \neq \emptyset$, $\mathscr{D}(A^*) \subset \mathscr{D}(B^*)$, *and* $\mathscr{D}(D^*) \subset \mathscr{D}(C^*)$, *the following conclusions are equivalent for* $\lambda \in \rho(A) \cap \rho(D)$*:*

(i) $\lambda \in \rho(\mathscr{A})$;

(ii) $1 \in \rho(B(D-\lambda)^{-1}C(A-\lambda)^{-1})$;
(iii) $1 \in \rho(C(A-\lambda)^{-1}B(D-\lambda)^{-1})$.

Proof For $\lambda \in \rho(A) \cap \rho(D)$, we have

$$\begin{aligned}\mathscr{A}-\lambda I &= \begin{bmatrix} I & 0 \\ C(A-\lambda I)^{-1} & I \end{bmatrix}\begin{bmatrix} A-\lambda I & 0 \\ 0 & S(\lambda) \end{bmatrix}\begin{bmatrix} I & \overline{(A-\lambda I)^{-1}B} \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} I & B(D-\lambda I)^{-1} \\ 0 & I \end{bmatrix}\begin{bmatrix} R(\lambda) & 0 \\ 0 & D-\lambda I \end{bmatrix}\begin{bmatrix} \overline{I} & 0 \\ \overline{(D-\lambda I)^{-1}C} & I \end{bmatrix},\end{aligned}$$

where

$$S(\lambda) = (I - C(A-\lambda I)^{-1}B(D-\lambda I)^{-1})(D-\lambda I),$$

$$R(\lambda) = (I - B(D-\lambda I)^{-1}C(A-\lambda I)^{-1})(C-\lambda I),$$

so the result is satisfied. □

Proposition 4.3.10 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ *be a skew-diagonally dominant dense closed block operator matrix. If* $0 \in \rho(B) \cap \rho(C)$ *and when* $\mathscr{D}(C^*) \subset \mathscr{D}(D^*)$ *and* $\mathscr{D}(B^*) \subset \mathscr{D}(A^*)$*, the following conclusions are equivalent:*

(i) $\lambda \in \rho(\mathscr{A})$;
(ii) $1 \in \rho((A-\lambda)C^{-1}(D-\lambda)B^{-1})$;
(iii) $1 \in \rho((D-\lambda)B^{-1}(A-\lambda)C^{-1})$.

4.3.1 Representation of the Adjoint of the Product of Two Operators

For two unbounded densely defined operators T, S, if TS is densely defined, then

$$(TS)^* \supset S^*T^*.$$

However, the equal sign does not necessarily hold in general. For example, let A be an unbounded self-adjoint operator in a Hilbert space X, and

$$T = \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix},\ S = \begin{bmatrix} I & I \\ I & I \end{bmatrix},$$

then $T^* = T$, $S^* = S$ and

$$(TS)^* \neq ST.$$

In fact, $ST = \begin{bmatrix} A & A \\ A & A \end{bmatrix}$ is a non-closed operator, and $(TS)^*$ is a closed operator. Therefore, to characterize the adjoint of the product operator, we first discuss the properties of the product of two operators.

Lemma 4.3.11 *Let $B : \mathscr{D}(B) \subset X_1 \to X_2$ and $A : \mathscr{D}(A) \subset X_2 \to X_3$ be densely defined closed linear operators and AB also be densely defined. Then we have the following conclusions:*

(i) If $A \in \mathscr{B}(X_1, X_2)$, then $(AB)^* = B^*A^*$.
(ii) If $B \in \mathscr{B}(X_1, X_2)$, then $(AB)^* = B^*A^*$, if and only if B^*A^* is a closed operator.

Proof (i) $(AB)^* \supset B^*A^*$ is obvious. The following is the proof of $(AB)^* \subset B^*A^*$. If $x^* \in \mathscr{D}((AB)^*)$, then there exists $f \in X_1$ such that for any $x \in \mathscr{D}(AB)$, we have

$$(ABx, x^*) = (x, f).$$

Since A is a bounded operator defined on the whole space, $\mathscr{D}(AB) = \mathscr{D}(B)$ and A^*x^* are meaningful such that

$$(Bx, A^*x^*) = (x, f)$$

for any $x \in \mathscr{D}(B)$, thus $A^*x^* \in \mathscr{D}(B^*)$, that is, $x^* \in \mathscr{D}(B^*A^*)$. So, $(AB)^* \subset B^*A^*$.

(ii) Considering that

$$\begin{aligned}(AB)^* &= (A^{**}B^{**})^* \\ &= ((B^*A^*)^*)^* = \overline{B^*A^*}.\end{aligned}$$

The conclusion holds. □

Lemma 4.3.12 *Let $B : \mathscr{D}(B) \subset X_1 \to X_2$ and $A : \mathscr{D}(A) \subset X_2 \to X_3$ be densely defined closed linear operators. If AB is also densely defined and $B^{-1} \in \mathscr{B}(X_2, X_1)$. Then $(AB)^* = B^*A^*$.*

Proof We only need to prove $\mathscr{D}((AB)^*) \subset \mathscr{D}(B^*A^*)$. If $x^* \in \mathscr{D}((AB)^*)$, then there exists $f \in X$ such that for any $x \in \mathscr{D}(AB)$, we have

$$(ABx, x^*) = (x, f).$$

Taking into account $\mathscr{D}(AB) = B^{-1}\mathscr{D}(A)$, that is, for any $g \in \mathscr{D}(A)$, we have

$$(Ag, x^*) = (g, (B^{-1})^* f).$$

This contains $x^* \in \mathscr{D}(A^*)$ and

$$A^*x^* = (B^{-1})^* f = (B^*)^{-1} f,$$

so $A^*x^* \in \mathscr{D}(B^*)$, that is, $\mathscr{D}((AB)^*) \subset \mathscr{D}(B^*A^*)$. □

The following conclusions show that for some special operators, $(AB)^* = B^*A^*$ can also hold even if B^{-1} is not a bounded operator on the whole space.

Lemma 4.3.13 *Let A, B be self-adjoint operators in a Hilbert space X. If the product operator AB is densely defined, $\mathscr{D}(A) \subset \mathscr{R}(B)$ and $\mathscr{R}(B^*) \supset \mathscr{R}((AB)^*)$. Then $(AB)^* = B^*A^* = BA$.*

Proof We only need to prove $(AB)^* \subset B^*A^*$. Since $\mathscr{D}(A) \subset \mathscr{R}(B)$, $0 \in \rho(B) \cup \sigma_c(B)$ holds. When $0 \in \rho(B)$, by Lemma 4.3.12, the conclusion obviously holds. When $0 \in \sigma_c(B)$, B^{-1} exists and

$$A = (AB)B_A^{-1},$$

where B_A^{-1} represents the restriction of B^{-1} on $\mathscr{D}(A)$. Therefore,

$$A^* = ((AB)B_A^{-1})^* \supset (B_A^{-1})^*(AB)^*.$$

And because of $B_A^{-1} \subset B^{-1}$, $(B_A^{-1})^* \supset (B^{-1})^*$ and

$$(B_A^{-1})^*(AB)^* \supset (B^{-1})^*(AB)^*,$$

that is,

$$A^* \supset (B^{-1})^*(AB)^*.$$

After the left side interacts with B^*, we get

$$B^*A^* \supset B^*(B^{-1})^*(AB)^*.$$

Taking into account $\mathscr{R}(B) \supset \mathscr{R}((AB)^*)$, we get

$$B^*A^* \supset B^*(B^{-1})^*(AB)^* \supset (AB)^*.$$

The proof is completed. □

Remark 4.3.14 If the condition $\mathscr{R}(B^*) \supset \mathscr{R}((AB)^*)$ in Lemma 4.3.13 is not satisfied, the conclusion does not necessarily hold. For example, if A is an unbounded self-adjoint operator in a Hilbert space X and $0 \in \sigma_c(A)$, $B = A^{-1}$, then AB is densely defined and $\mathscr{D}(A) \subset \mathscr{R}(B)$. However,

$$(AB)^* = I.$$

But,

$$B^*A^* = I_A,$$

where I_A denotes the restriction of identity operator I on $\mathscr{D}(A)$. Obviously, $(AB)^* \neq B^*A^*$.

Lemma 4.3.15 *Let A be a densely defined closed operator in a Hilbert space X, $B \in \mathscr{B}(X)$. If the product operator AB is densely defined, $\mathscr{R}(B)$ is a closed set and $\mathtt{dim}\mathscr{N}(B^*) < \infty$, then $(AB)^* = B^*A^*$.*

Proof If $\mathscr{R}(B)$ is a closed set and $\mathtt{dim}\mathscr{N}(B^*) < \infty$, then since A is a closed operator, B^*A^* is a closed operator (see Proposition XVII3.2 of [55]). Furthermore, the conclusion can be proved by Lemma 4.3.11. □

Lemma 4.3.16 *If A, B are Fredholm operators in a Hilbert space X and AB is dense, then $(AB)^* = B^*A^*$.*

Proof Since A, B are Fredholm operators, AB, B^*A^* are also Fredholm operators. By Lemma 4.2.10, we only need to prove $\mathtt{ind}((AB)^*) = \mathtt{ind}(B^*A^*)$. In fact,

$$\begin{aligned}\mathtt{ind}((AB)^*) &= -\mathtt{ind}(AB)\\ &= -\mathtt{ind}(A) - \mathtt{ind}(B)\\ &= \mathtt{ind}(A^*) + \mathtt{ind}(B^*) = \mathtt{ind}(B^*A^*).\end{aligned}$$

The proof is completed. □

4.3.2 Characterization of the Symplectic Self-adjointness of Infinite-Dimensional Hamiltonian Operators Based on Schur Complement

Theorem 4.3.17 *Let $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ be a diagonally dominant infinite-dimensional Hamiltonian operator. If $\rho(A) \neq \emptyset$, then the following propositions are equivalent:*

(i) $(JH)^* = JH$, *where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$;

(ii) $(A^* + C(A - \lambda I)^{-1}B)^* = A + B(A^* - \overline{\lambda} I)^{-1}C$, *where* $\lambda \in \rho(A)$;

(iii) $(A + B(A^* - \overline{\lambda} I)^{-1}C)^* = A^* + C(A - \lambda I)^{-1}B$, *where* $\lambda \in \rho(A)$.

Proof We only need to prove the equivalence between (i) and (ii), and the proof of (i) and (iii) is completely similar. When $\lambda \in \rho(A)$,

$$H-\lambda I=\begin{bmatrix} I & 0 \\ C(A-\lambda I)^{-1} & I \end{bmatrix}\begin{bmatrix} A-\lambda I & 0 \\ 0 & S_1(\lambda) \end{bmatrix}\begin{bmatrix} I & \overline{(A-\lambda I)^{-1}B} \\ 0 & I \end{bmatrix}=LTR,$$

where $S_1(\lambda)=-A^*-\lambda I-C(A-\lambda I)^{-1}B$. From Lemmas 4.3.11 and 4.3.12, we can see that

$$H^*-\overline{\lambda}I=R^*T^*L^*.$$

And because

$$T^*=\begin{bmatrix} A^*-\overline{\lambda}I & 0 \\ 0 & (S_1(\lambda))^* \end{bmatrix},$$

then $(JH)^*=JH$, if and only if

$$(S_1(\lambda))^*=-A-\overline{\lambda}I-B(A^*-\overline{\lambda}I)^{-1}C,$$

that is,

$$(A^*+C(A-\lambda I)^{-1}B)^*=A+B(A^*-\overline{\lambda}I)^{-1}C.$$

The proof is completed. □

Proposition 4.3.18 *Let* $H=\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}:\mathscr{D}(A)\times\mathscr{D}(A^*)\to X\times X$ *be a diagonally dominant infinite-dimensional Hamiltonian operator. If* $\rho(A)\neq\emptyset$, *and one of the following conditions is met:*

(i) C *is bounded relative to* A *and the relative bound is 0;*
(ii) B *is bounded relative to* A^* *and the relative bound is 0,*

then $(JH)^*=JH$, *where* $J=\begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$.

Proof We only need to prove the case satisfying (i), and the proof of satisfying (ii) is completely similar. By Theorem 4.3.18, we only need to prove $(A^*+C(A-\lambda I)^{-1}B)^*=A+B(A^*-\overline{\lambda}I)^{-1}C$. Because C is bounded concerning A and the relative bound is 0, $a(\varepsilon)\geq 0$ exists for any $\varepsilon>0$. Such that for any $x\in\mathscr{D}(A)$, there exists

$$\|Cx\|\leq a(\varepsilon)\|x\|+\varepsilon\|(A-\lambda I)x\|,$$

then for any $y\in\mathscr{D}(A^*)$, we have

$$\|C(A-\lambda I)^{-1}By\|\leq a(\varepsilon)\|(A-\lambda I)^{-1}By\|+\varepsilon\|By\|.$$

From $\mathscr{D}(B)\subset\mathscr{D}(A^*)$, it can be seen that there exist $b,c\geq 0$, such that for any $y\in\mathscr{D}(A^*)$, we have

$$\|By\|\leq b\|y\|+c\|A^*y\|,$$

that is,

$$\|C(A-\lambda I)^{-1}By\| \le (a(\varepsilon)\|(A-\lambda I)^{-1}B\| + \varepsilon b)\|y\| + \varepsilon c\|A^*y\|.$$

Considering the arbitrariness of $\varepsilon > 0$, we get $C(A-\lambda I)^{-1}B$ is bounded relative to A^* and A^*-bound is 0. On the other hand, for any $x \in \mathscr{D}(A)$, we have

$$\begin{aligned}\|(C(A-\lambda I)^{-1}B)^*x\| =& \|\overline{B(A^*-\bar{\lambda}I)^{-1}Cx}\| \\ =& \|B(A^*-\bar{\lambda}I)^{-1}Cx\| \\ \le& \|B(A^*-\bar{\lambda}I)^{-1}\|\|Cx\| \\ \le& \|B(A^*-\bar{\lambda}I)^{-1}\|(a(\varepsilon)\|x\| + \varepsilon\|(A-\lambda I)x\|).\end{aligned}$$

Then, $(C(A-\lambda I)^{-1}B)^*$ is relatively bounded with respect to A, and the relative bound is 0, so it can be seen from Lemma 4.2.13 that

$$(A^* + C(A-\lambda I)^{-1}B)^* = A + B(A^*-\bar{\lambda}I)^{-1}C.$$

The proof is completed. □

Similarly, the following conclusions can be proved.

Theorem 4.3.19 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a skew-diagonally dominant infinite-dimensional Hamiltonian operator. Then the following propositions are equivalent:*

(i) $(JH)^* = JH$, *where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$;
(ii) $(B + A(C-\mathrm{i}I)^{-1}A^*)^* = B + A^*(C+\mathrm{i}I)^{-1}A$;
(iii) $(C + A^*(B-\mathrm{i}I)^{-1}A)^* = C + A(B+\mathrm{i}I)^{-1}A^*$.

Proposition 4.3.20 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a skew-diagonally dominant infinite-dimensional Hamiltonian operator. If one of the following conditions is satisfied:*

(i) A is bounded relative to C and the relative bound is 0;
(ii) A^* is bounded relative to B and the relative bound is 0,

then $(JH)^* = JH$, *where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$.

In summary, for the diagonally dominant case, if both relative bounds are less than one or one of them is 0, the infinite-dimensional Hamiltonian operator is symplectic self-adjoint. At this time, it is natural to ask whether the product of two relative bounds can characterize the symplectic self-adjointness of the infinite-dimensional Hamiltonian operator.

Theorem 4.3.21 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant infinite-dimensional Hamiltonian operator. If* $\rho(A) \neq \emptyset$ *and the product of the relative bounds of* C *with respect to* A *and* B *with respect to* A^* *is less than 1, then* $(JH)^* = JH$, *where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$.

Proof By Theorem 4.3.18, we only need to prove $(A^* + C(A - \lambda I)^{-1}B)^* = A + B(A^* - \overline{\lambda} I)^{-1}C$. The relative bounds of C with respect to A and B with respect to A^* are set as b_1, b_2. Since C is relatively bounded with respect to A, we have a constant a_1 such that

$$\|Cx\| \leq a_1\|x\| + b_1\|(A - \lambda I)x\|, x \in \mathscr{D}(A),$$

that is,

$$\|C(A - \lambda I)^{-1}Bx\| \leq a_1\|(A - \lambda I)^{-1}Bx\| + b_1\|Bx\|, x \in \mathscr{D}(B).$$

Since B is relatively bounded with respect to A^*, there exists a constant a_2 such that

$$\|Bx\| \leq a_2\|x\| + b_2\|A^*x\|, x \in \mathscr{D}(A^*),$$

so

$$\|C(A - \lambda I)^{-1}Bx\| \leq a_1\|(A - \lambda I)^{-1}B\| + b_1a_2)\|x\| + b_1b_2\|A^*x\|, x \in \mathscr{D}(A^*).$$

Because $b_1b_2 < 1$, $C(A - \lambda I)^{-1}B$ is bounded with A^* and the relative bound is less than 1. On the other hand, for any $x \in \mathscr{D}(A)$, we have

$$\begin{aligned} \|(C(A - \lambda I)^{-1}B)^*x\| =& \|\overline{B(A^* - \overline{\lambda} I)^{-1}C}x\| \\ =& \|B(A^* - \overline{\lambda} I)^{-1}Cx\| \\ \leq& \|B(A^* - \overline{\lambda} I)^{-1}\|\|Cx\| \\ \leq& \|B(A^* - \overline{\lambda} I)^{-1}\|(a_1\|x\| + b_1\|(A - \lambda I)x\|). \end{aligned}$$

Since B is relatively bounded with respect to $A^* - \overline{\lambda} I$, there exists a constant a_3 such that

$$\|Bx\| \leq a_3\|x\| + b_2\|(A^* - \overline{\lambda} I)x\|, x \in \mathscr{D}(A^*),$$

that is,

$$\|B(A^* - \overline{\lambda} I)^{-1}y\| \leq a_3\|(A^* - \overline{\lambda} I)^{-1}y\| + b_2\|y\|, y \in X.$$

Since $0 \in \sigma_{ap}((A^* - \overline{\lambda} I)^{-1})$, $(A^* - \overline{\lambda} I)^{-1})x_n \to 0$, thus

$$\|B(A^* - \overline{\lambda} I)^{-1}\| \leq b_2.$$

Therefore, $(C(A-\lambda I)^{-1}B)^*$ is relatively bounded with respect to A, and the relative bound is less than 1, so it can be seen from Lemma 4.2.13 that

$$(A^* + C(A-\lambda I)^{-1}B)^* = A + B(A^* - \overline{\lambda} I)^{-1}C.$$

The proof is completed. □

Similarly, the following conclusions can be proved.

Theorem 4.3.22 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a skew-diagonally dominant infinite-dimensional Hamiltonian operator. If the product of the relative bounds of* A *with respect to* C *and* A^* *with respect to* B *is less than 1, then* $(JH)^* = JH$, *where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$.

Theorem 4.3.23 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonally dominant infinite-dimensional Hamiltonian operator. If there exists* $\lambda \in \rho(A)$ *such that bounded* 2×2 *block operator matrix*

$$T_\lambda = \begin{bmatrix} \lambda(A-\lambda I)^{-1} + I & \overline{(A-\lambda I)^{-1}B} \\ -\overline{(A^* - \overline{\lambda} I)^{-1}C} & \overline{\lambda}(A^* - \overline{\lambda} I)^{-1} + I \end{bmatrix}$$

satisfies the condition that $\mathscr{R}(T_\lambda)$ *is a closed set,* $\dim \mathscr{N}(T_\lambda^*) < \infty$, *then* $(JH)^* = JH$, *where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$.

Proof Because

$$H = \begin{bmatrix} A - \lambda I & 0 \\ 0 & -A^* + \overline{\lambda} \end{bmatrix} T_\lambda,$$

the conclusion can be proved by Lemma 4.3.15. □

Theorem 4.3.24 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a skew-diagonally dominant infinite-dimensional Hamiltonian operator. If* 2×2 *bounded block operator matrix*

$$T_\lambda = \begin{bmatrix} i(C - iI)^{-1} + I & -\overline{(C - iI)^{-1}A^*} \\ \overline{(B - iI)^{-1}A} & i(B - iI)^{-1} + I \end{bmatrix}$$

satisfies the condition that $\mathscr{R}(T_\lambda)$ *is a closed set,* $\dim \mathscr{N}(T_\lambda^*) < \infty$, *then* $(JH)^* = JH$, *where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$.

4.4 Characterization of Symplectic Self-adjointness by Using the Spectral Set of Infinite-Dimensional Hamiltonian Operators

The perturbation theory and Schur decomposition method only apply to some special infinite-dimensional Hamiltonian operators, such as diagonally dominant and skew-diagonally dominant. However, infinite-dimensional Hamiltonian operators do not have dominant properties in practical problems. For example, if T, S are unbounded self-adjoint operators in a Hilbert space X and $\mathscr{D}(T) \subsetneqq \mathscr{D}(S)$ and $A = \begin{bmatrix} 0 & S \\ T & 0 \end{bmatrix}$, $B = \begin{bmatrix} S & 0 \\ 0 & T \end{bmatrix}$, $C = \begin{bmatrix} T & 0 \\ 0 & S \end{bmatrix}$, then the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$$

does not have the dominant property. Therefore, we will use some properties of the spectrum of the infinite-dimensional Hamiltonian operators to characterize the problem of when the infinite-dimensional Hamiltonian operator is a symplectic self-adjoint operator.

4.4.1 Point Spectra and Symplectic Self-adjointness of Infinite-Dimensional Hamiltonian Operators

Theorem 4.4.1 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If one of the following conditions is satisfied:*

(i) $\sigma_{p,1}(H) \cap \mathrm{i}\mathbb{R} \neq \emptyset$;
(ii) $\sigma_{p,2}(H) \cap \mathrm{i}\mathbb{R} \neq \emptyset$;
(iii) $\sigma_{p,1}(H)$ *is symmetric about the imaginary axis;*
(iv) $\sigma_{p,2}(H)$ *is symmetric about the imaginary axis;*
(v) $\sigma_{p,1}(H) \cup \sigma_{p,2}(H)$ *is symmetric about the imaginary axis,*

then $JH \neq (JH)^*$.

Proof (i) Assume $JH = (JH)^*$ and $\lambda \in \sigma_{p,1}(H) \cap \mathrm{i}\mathbb{R}$. Then $\overline{\lambda} \in \sigma_r(H^*)$, thus $-\overline{\lambda} \in \sigma_r(H)$, that is, $\lambda \in \sigma_r(H)$, the contradiction is derived. The proof of (ii) is similar.

(iii) Let $\lambda \in \sigma_{p,1}(H)$, then, as $\sigma_{p,1}(H)$ is symmetric about the imaginary axis, we can know $-\overline{\lambda} \in \sigma_{p,1}(H)$. On the other hand, assuming $JH = (JH)^*$, then $-\overline{\lambda} \in \sigma_{r,1}(H)$ is known from $\lambda \in \sigma_{p,1}(H)$, and the contradiction is derived.

(iv) Since $\lambda \in \sigma_{p,2}(H)$ if and only if $\bar{\lambda} \in \sigma_{r,2}(H^*)$, similar to (ii), $JH \neq (JH)^*$ can be proved.

(v) Since $\lambda \in (\sigma_{p,1}(H) \cup \sigma_{p,2}(H))$ if and only if $\bar{\lambda} \in \sigma_r(H^*)$, similar to (iii), $JH \neq (JH)^*$ can be proved. □

4.4.2 Residual Spectra and Symplectic Self-adjointness of Infinite-Dimensional Hamiltonian Operators

In order to reach the main conclusion, we first provide the following lemma.

Lemma 4.4.2 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator, where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. *Then* $\sigma_r(H) = \emptyset$ *if and only if* $\sigma_r(JHJ) = \emptyset$.

Proof If $\mathfrak{J} = \mathtt{i}J$, then $\mathfrak{J} = \mathfrak{J}^{-1} = \mathfrak{J}^*$ and

$$JHJ = \mathfrak{J}(-H)\mathfrak{J}.$$

Therefore, JHJ is similar to $-H$. So, they have the same spectral properties, that is, $\sigma_r(JHJ) = \sigma_r(-H)$. So, $\sigma_r(H) = \emptyset$ if and only if $\sigma_r(JHJ) = \emptyset$. □

Theorem 4.4.3 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator, satisfying* $\rho(H) \neq \emptyset$ *and* $\sigma_r(H) = \emptyset$. *We have* $JH = (JH)^*$.

Proof From the symplectic symmetry of the infinite-dimensional Hamiltonian operators, we can know that, in order to prove $JH = (JH)^*$, it only needs to prove $\mathscr{D}(H^*) \subset \mathscr{D}(JHJ)$. Since $\rho(H) \neq \emptyset$, we take $\lambda \in \rho(H)$ arbitrary, then $\bar{\lambda} \in \rho(H^*)$. From the definition of the resolvent set, we know that $H^* - \bar{\lambda}I$ is injective and boundedly invertible, and $JHJ \subset H^*$, $H^*|_{\mathscr{D}(JHJ)} = JHJ$, that is, $JHJ - \bar{\lambda}I$ is also injective and $(JHJ - \bar{\lambda})^{-1}$ is bounded. Thus $\bar{\lambda} \in \sigma_{r,1}(JHJ) \cup \rho(JHJ)$.

By Lemma 4.4.2, we know that when $\sigma_r(H) = \emptyset$, $\sigma_r(JHJ) = \emptyset$, thus $\sigma_{r,1}(JHJ) = \emptyset$, this means that $\bar{\lambda} \in \rho(JHJ)$.

Therefore, for any $x^* \in \mathscr{D}(H^*)$, there exists $x \in \mathscr{D}(JHJ)$ such that

$$(JHJ - \bar{\lambda}I)x = (H^* - \bar{\lambda}I)x^*.$$

Since $(H^* - \bar{\lambda}I)|_{\mathscr{D}(JHJ)} = JHJ - \bar{\lambda}I$, we get $(H^* - \bar{\lambda}I)(x - x^*) = 0$. Then by $H^* - \bar{\lambda}I$ is injective, so $x - x^* = 0$, that is, $x^* = x \in \mathscr{D}(JHJ)$. Therefore, $\mathscr{D}(H^*) \subseteq \mathscr{D}(JHJ)$. In summary, $JH = (JH)^*$. □

Remark 4.4.4 From the proof of Theorem 4.4.3, we know that the condition $\sigma_r(H) = \emptyset$ can be replaced by $\sigma_{r,1}(H) = \emptyset$ or $\sigma_{r,1}(JHJ) = \emptyset$.

Theorem 4.4.5 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be a closed infinite-dimensional Hamiltonian operator. If* $\rho(JH) \neq \emptyset$ *and* $\sigma_r(JH) = \emptyset$, *then* $(JH)^* = JH$.

Proof Let $\lambda \in \rho((JH)^*)$, then we can assert that $\lambda \in \rho(JH)$. In fact, $\lambda \notin \sigma_p(JH)$ is obvious. Otherwise, taking into account $JH \subset (JH)^*$, it contradicts $\lambda \in \rho((JH)^*)$.

Assume that $\lambda \in \sigma_c(JH)$, then the operator JH is closed, $(JH - \lambda)^{-1}$ is unbounded. On the other hand, because $\lambda \in \rho((JH)^*)$, there exists $M > 0$ such that for any $x \in \mathscr{D}((JH)^*)$, we have

$$\| ((JH)^* - \lambda)x \| \geq M \| x \| .$$

Due to $(JH)^*|_{\mathscr{D}(JH)} = JH$,

$$\| (JH - \lambda)x \| \geq M \| x \|$$

holds, for all $x \in \mathscr{D}(JH)$, which implies $(JH - \lambda I)^{-1}$ is bounded, being contradictory to unboundedness. Therefore, $\lambda \notin \sigma_c(JH)$. Since $\sigma_r(JH) = \emptyset$, then $\lambda \in \rho(JH)$.

Take $x^* \in \mathscr{D}((JH)^*)$, since $JH - \lambda$ is surjective, there exists $x \in \mathscr{D}(JH)$ such that

$$(JH - \lambda)x = ((JH)^* - \lambda)x^*,$$

because $(JH)^*|_{\mathscr{D}(JH)} = JH$, we have

$$((JH)^* - \lambda I)(x - x^*) = 0,$$

because of $\lambda \in \rho((JH)^*)$, then $x - x^* = 0$, which implies $x^* \in \mathscr{D}(JH)$, that is, $(JH)^* \subset JH$. The proof is completed. □

4.4.3 Characterization of Symplectic Self-adjointness by Using the Invertibility of Infinite-Dimensional Hamiltonian Operators

Lemma 4.4.6 *Suppose that* T *is a symmetric operator and there exists* $\lambda \in \mathbb{R}$ *such that* $\lambda \in \rho(T)$. *Then* T *is a self-adjoint operator.*

Proof We only need to prove $\mathscr{D}(T^*) \subset \mathscr{D}(T)$. If we have $\lambda \in \mathbb{R}$ such that $\lambda \in \rho(T)$, then $\lambda \in \rho(T^*)$. Let $x^* \in \mathscr{D}(T^*)$, because $\mathscr{R}(T - \lambda) = X$, there exists $x \in \mathscr{D}(T)$ such that

$$(T^* - \lambda I)x^* = (T - \lambda I)x.$$

Because $T^*|_{\mathscr{D}(T)} = T$, so $(T^* - \lambda I)x^* = (T^* - \lambda I)x$, that is, $(T^* - \lambda I)(x^* - x) = 0$. Since $\lambda \in \rho(T^*)$, we get $x^* = x$ such that $\mathscr{D}(T^*) \subset \mathscr{D}(T)$. □

Theorem 4.4.7 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If* $0 \in \rho(H)$, *then* $(JH)^* = JH$.

Proof When $0 \in \rho(H)$, it is easy to get $0 \in \rho(JH)$. But, JH is a symmetric operator, by Lemma 4.4.6, we can get $(JH)^* = JH$. □

Proposition 4.4.8 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If there exists* $M > 0$ *such that for any* $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$, *we have*

$$|(By, y) + (Cx, x)| \geq M \| \begin{bmatrix} x \\ y \end{bmatrix} \|^2,$$

and $0 \in \rho(A)$, *operator* $I + (CA^{-1}B(A^*)^{-1})^*$ *is injective, then* $(JH)^* = JH$.

Proof From Theorem 2.5.28, we can know $0 \in \rho(H)$, and then by Theorem 4.4.7, we can get $(JH)^* = JH$. □

Proposition 4.4.9 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If there exists* $M > 0$ *such that for any* $\begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$,

$$|(By, y) + (Cx, x)| \geq M \| \begin{bmatrix} x \\ y \end{bmatrix} \|^2,$$

and the operator $I + (AC^{-1}A^*B^{-1})^*$ *is injective, then* $(JH)^* = JH$.

Proof From Theorem 2.5.29, we can know $0 \in \rho(H)$, and then by Theorem 4.4.7 we can get $(JH)^* = JH$. □

Chapter 5
Numerical Range Theory of Infinite-Dimensional Hamiltonian Operators



5.1 Numerical Range and Its Definition

The systematic study of quadratic forms originated from the discussion of the classification of quadratic curves and quadratic surfaces. Mathematicians such as Hilbert, Hellinger, Toeplitz, and Hausdorff have systematically studied quadratic forms. The classical real quadratic form is a function of n variables as

$$f(x) = x^T Ax \tag{5.1}$$

where x^T is the transpose of vector x and A is a real symmetric matrix. Quadratic forms are widely used in mathematics and other disciplines. For example, Fermat's theorem in the field of number theory, Witt's theorem in the field of group theory, Riemann metric in differential geometry, and Cartan type in Lie algebra are all related to quadratic forms. In addition to quadratic forms, another important mathematical concept that needs to be mentioned is the Rayleigh quotient. The form of Rayleigh quotient is the multivariate function like

$$R(A, x) = \frac{x^* Ax}{x^* x}, x \neq 0 \tag{5.2}$$

which has important applications in the min-max theorem of functional analysis, the Poincaré separation theorem, and the Sturm-Liouville problem. However, Eq. (5.1) and Eq. (5.2) can extend to infinite-dimensional complex Hilbert space,

$$W(A) = \left\{ \frac{(Ax, x)}{(x, x)} : x \neq 0 \right\}. \tag{5.3}$$

The equation $W(A)$ defined by Eq. (5.3) is called the numerical range of bounded linear operator A in Hilbert space. Therefore, the numerical range of bounded linear operators in Hilbert space is a generalization of quadratic form and Rayleigh quotient

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from finite dimension to infinite dimension, which has a profound theoretical basis. When A is an unbounded linear operator, the numerical range is defined as

$$W(A) = \left\{ \frac{(Ax, x)}{(x, x)} : x \neq 0, x \in \mathscr{D}(A) \right\}. \tag{5.4}$$

Like the spectral set of linear operators, the numerical range of linear operators in Hilbert space is also a subset of the complex plane, which also contains information about linear operators, and can even provide some information that the spectral set cannot provide. For example, the numerical range of a linear operator is a real number set, which implies that the linear operator is symmetric, and the spectral sets are real numbers, which are not necessarily symmetric operators. From the perspective of academic research, the study of the numerical range of linear operators involves many branches of pure theory and applied science such as operator theory, functional analysis, C^*-algebra, Banach algebra, numerical analysis, perturbation theory, control theory, and quantum physics. In addition, the research methods on the numerical range of linear operators are also very rich. Algebra, analysis, geometry, combination theory, and computer programming are very useful research tools. Therefore, the research on the numerical range of linear operators and related problems has received extensive attention from many scholars. The following are some examples of numerical ranges.

Example 5.1.1 Let X be a Hilbert space, I be a unit operator in X, and define block operator matrix in Hilbert space $X \times X$ (without confusion, its inner product is still denoted by $(\cdot, \cdot)$) as the following:

$$T = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix},$$

then $W(T) = \{\lambda \in \mathbb{C} : |\lambda| \leq \frac{1}{2}\}$. In fact, setting $x = \begin{bmatrix} f & g \end{bmatrix}^T \in X \times X$, $\|f\|^2 + \|g\|^2 = 1$, we have

$$(Tx, x) = (g, f).$$

Notice,

$$|(g, f)| \leq \|g\|\|f\| \leq \frac{1}{2}(\|g\|^2 + \|f\|^2) = \frac{1}{2}.$$

So, $W(T) \subset \{\lambda \in \mathbb{C} : |\lambda| \leq \frac{1}{2}\}$.

On the other hand, for any $\lambda = re^{\mathrm{i}\theta}$, $0 \leq r \leq \frac{1}{2}$, take $x = \begin{bmatrix} f\cos\alpha & fe^{\mathrm{i}\theta}\sin\alpha \end{bmatrix}^T \in X \times X$, where $\sin(2\alpha) = 2r \leq 1$, $0 \leq \alpha \leq \frac{\pi}{4}$, $f \in X$ and $\|f\| = 1$, then $\|x\| = 1$ and

$$(Tx, x) = (fe^{\mathrm{i}\theta}\sin\alpha, f\cos\alpha) = re^{\mathrm{i}\theta}.$$

So, $W(T) = \{\lambda \in \mathbb{C} : |\lambda| \leq \frac{1}{2}\}$.

Example 5.1.2 Let $X = \ell^2$ be the set of complex-valued sequences $x = (x_1, x_2, \ldots)$ of $\sum_{i=1}^{\infty} |x_i|^2 < \infty$. The inner product is defined as

$$(x, y) = \sum_{i=1}^{\infty} x_i \overline{y_i},\ x = (x_1, x_2, \ldots),\ y = (y_1, y_2, \ldots) \in X.$$

The shift operator S_l in X space is defined as

$$S_l x =: (x_2, x_3, \ldots),$$

then $W(T) = \{\lambda \in \mathbb{C} : |\lambda| < 1\}$. In fact, for any $x = (x_1, x_2, \ldots) \in X$, $\|x\| = 1$,

$$(S_l x, x) = x_2\overline{x_1} + x_3\overline{x_2} + \cdots + x_n\overline{x_{n-1}} + \cdots .$$

Notice that $|x_1|^2 + |x_2|^2 + \cdots = 1$, and assume that $|x_1| \neq 0$, then

$$\begin{aligned} |(S_l x, x)| \leq & |x_2||x_1| + |x_3||x_2| + \cdots + |x_n||x_{n-1}| + \cdots \\ \leq & \frac{1}{2}(|x_1|^2 + 2|x_2|^2 + \cdots + 2|x_n|^2 + \cdots) \\ < & \frac{1}{2}(2 - |x_1|^2). \end{aligned}$$

Therefore, $W(S_l) \subset \{\lambda \in \mathbb{C} : |\lambda| < 1\}$.

On the other hand, for any $\lambda = re^{\mathrm{i}\theta}$, $0 \leq r < 1$, take

$$x = (\sqrt{1 - r^2}, r\sqrt{1 - r^2}e^{\mathrm{i}\theta}, r^2\sqrt{1 - r^2}e^{2\mathrm{i}\theta}, \ldots),$$

then

$$\|x\|^2 = 1 - r^2 + r^2(1 - r^2) + r^4(1 - r^2) + \cdots = 1,$$

and

$$\begin{aligned} (S_l x, x) &= e^{\mathrm{i}\theta} r(1 - r^2) + e^{\mathrm{i}\theta} r^3(1 - r^2) + \cdots \\ &= re^{\mathrm{i}\theta}. \end{aligned}$$

So, $W(S_l) = \{\lambda \in \mathbb{C} : |\lambda| < 1\}$.

According to the definition of numerical range, it is easy to obtain the following properties.

Corollary 5.1.3 *Let T be a bounded linear operator in Hilbert space X. Then the numerical range $W(T)$ is a bounded set.*

Proof When T is a bounded linear operator, considering that

$$|(Tx, x)| \leq \|T\| \cdot \|x\|^2,$$

the numerical range $W(T)$ is a bounded set. □

Remark 5.1.4 When the linear operator T is an unbounded linear operator in Hilbert space, its domain is not necessarily the whole space. The numerical range $W(T)$ is defined as

$$W(T) = \{(Tx, x) : x \in \mathscr{D}(T), (x, x) = 1\}.$$

For the unbounded linear operator, its numerical range is not necessarily bounded sets, and may even be the whole plane. For example, let

$$AC[0, 1] = \{x(t) \in L^2[0, 1] : x(t) \text{is absolutely continuous in [0,1] and} x'(t) \in L^2[0, 1]\}.$$

Define linear operator

$$Tx(t) = \mathrm{i}x'(t),$$

where

$$\mathscr{D}(T) = \{x(t) : x(t) \in AC[0, 1]\}.$$

For any $\lambda \in \mathbb{C}$, take $x(t) = e^{-\mathrm{i}\lambda t}$, then $x(t) \in \mathscr{D}(T)$, $\|x(t)\| = 1$ and

$$\begin{aligned}(Tx, x) &= \int_0^1 \mathrm{i}x'(t)\overline{x(t)}dt \\ &= \int_0^1 \mathrm{i}(-\mathrm{i}\lambda)e^{-\mathrm{i}\lambda t}e^{\mathrm{i}\lambda t}dt \\ &= \lambda,\end{aligned}$$

that is, $W(T) = \mathbb{C}$.

The following properties show that the numerical range of linear operators has a certain translatability and remains unchanged under the unitary transformation.

Corollary 5.1.5 *Let T, S be linear operators (not necessarily bounded) in a Hilbert space X. Then*

(i) $W(\alpha T + \beta I) = \alpha W(T) + \beta$;
(ii) $W(T + S) \subset W(T) + W(S)$;
(iii) *if* $\mathscr{R}(U) \subset \mathscr{D}(T)$, *then* $W(U^*TU) = W(T)$, *where U is unitary operator;*
(iv) *if* $\mathscr{D}(T) = \mathscr{D}(T^*)$, *then* $\lambda \in W(T)$ *if and only if* $\overline{\lambda} \in W(T^*)$;
(v) *if* $\mathscr{D}(T) = \mathscr{D}(T^*)$, *then* $W(Re(T)) = Re(W(T))$ *and* $W(Im(T)) = Im(W(T))$, *where* $Re(T) = \frac{T+T^*}{2}$, $Im(T) = \frac{T-T^*}{2i}$.

Remark 5.1.6 When $\mathscr{D}(T) \neq \mathscr{D}(T^*)$, the (iv) of property Corollary 5.1.5 may not hold. For example, defining linear operator

$$Tx(t) = \mathrm{i}x'(t),$$

where

$$\mathscr{D}(T) = \{x(t) \in L^2[0,1] : x(t) \in AC[0,1], x'(t) \in L^2[0,1]\},$$

$W(T) = \mathbb{C}$ is known from Remark 5.1.4. On the other hand, it is easy to obtain by calculation,

$$T^*x(t) = ix'(t),$$

where

$$\mathscr{D}(T^*) = \{x(t) \in L^2[0,1] : x(t) \in AC[0,1], x'(t) \in L^2[0,1], x(0) = x(1) = 0\}.$$

Therefore, for any $x(t) \in \mathscr{D}(T^*)$,

$$\begin{aligned}(T^*x, x) &= \int_0^1 \mathrm{i}x'(t)\overline{x(t)}\mathrm{d}t \\ &= \mathrm{i}x(t)\overline{x(t)}|_0^1 - \mathrm{i}\int_0^1 x(t)\overline{x'(t)}\mathrm{d}t \\ &= \int_0^1 x(t)\overline{\mathrm{i}x'(t)}\mathrm{d}t \\ &= (x, T^*x).\end{aligned}$$

This shows that $W(T^*) \subset \mathbb{R}$. Therefore, there exists $\lambda \in W(T)$ such that $\overline{\lambda} \notin W(T^*)$.

Theorem 5.1.7 *The numerical range $W(T)$ of the linear operators T (not necessarily bounded) in Hilbert space X is a convex set.*

Proof By the property Corollary 5.1.5, assuming that $(Tx_1, x_1) = 0$, $(Tx_2, x_2) = 1$, $\|x_i\| = 1$, $i = 1, 2$. Set $x = \alpha x_1 + \beta x_2$, $\alpha, \beta \in \mathbb{R}$, in order to prove that $W(T)$ is convex set, it suffices to find $\alpha, \beta \in \mathbb{R}$ such that

$$\|x\|^2 = \alpha^2 + \beta^2 + 2\alpha\beta Re(x_1, x_2) = 1, \tag{5.5}$$

and

$$(Tx, x) = \beta^2 + \alpha\beta[(Tx_1, x_2) + (Tx_2, x_1)] = t \tag{5.6}$$

for any $t \in (0, 1)$.

Let $B(x_1, x_2) = (Tx_1, x_2) + (Tx_2, x_1)$. If $B(x_1, x_2) \in \mathbb{R}$, then considering $|Re(x_1, x_2)| \leq 1$, the system (5.5) and the system (5.6) formulate an ellipse (the points of intersection with the coordinate axis are $(1, 0)$, $(0, 1)$, $(-1, 0)$, $(0, -1)$) and a hyperbola (the points of intersection with the coordinate axis are $(0, \sqrt{t})$, $(0, -\sqrt{t})$).

When $B(x_1, x_2) \notin \mathbb{R}$, x_1 can be selected appropriately such that $B(x_1, x_2) \in \mathbb{R}$. In fact, set $x_1' = \mu x_1$, where $\mu = a + \mathrm{i}b$,

$$|a|^2 + |b|^2 = 1, \tag{5.7}$$

and

$$Im(B(x_1', x_2)) = aIm(B(x_1, x_2)) + bRe((Tx_1, x_2) - (Tx_2, x_1)) = 0. \tag{5.8}$$

Obviously, the system (5.7) and the system (5.8) formulate the circle with the center of the circle at the origin and the straight line passing through the origin, so there is a common point of intersection. In summary, $W(T)$ is a convex set. □

For linear operators in Hilbert spaces, an important property of the numerical range is that the numerical range contains the point spectrum and the residual spectrum. If it is a bounded linear operator, then the numerical range closure contains the spectral set.

Lemma 5.1.8 *Let T be a linear operator in Hilbert space X. The following statements hold:*

(i) $Conv(\sigma_p(T)) \subset W(T)$ *and* $Conv(\sigma_{ap}(T)) \subset \overline{W(T)}$.
(ii) If $\mathscr{D}(T^*) \subset \mathscr{D}(T)$, *then* $Conv(\sigma_p(T) \cup \sigma_r(T)) \subset W(T)$ *and* $Covn(\sigma(T)) \subset \overline{W(T)}$.
(iii) If $\sigma(T)) \subset \overline{W(T)}$, *then for any* $\lambda \notin \overline{W(T)}$ *we have* $\|(T - \lambda I)^{-1}\| \leq \frac{1}{dist(\lambda, W(T))}$.

Proof (i) If $\lambda \in \sigma_p(T)$, then by the definition of the point spectrum, there exists $x \neq 0$ such that $Tx = \lambda x$. Taking the inner product with x on both sides, we have

$$\lambda = \frac{(Tx, x)}{(x, x)},$$

that is, $\sigma_p(T) \subset W(T)$. Considering the convexity of the numerical range, we attain $Conv(\sigma_p(T)) \subset W(T)$. When $\lambda \in \sigma_{ap}(T)$, there exists $\|x_n\| = 1, n = 1, 2, \ldots$ such that

$$(T - \lambda I)x_n \to 0, n \to \infty,$$

that is, $(Tx_n, x_n) \to \lambda$, thus $\sigma_{ap}(T) \subset \overline{W(T)}$.

(ii) When $\lambda \in \sigma_r(T)$, $\overline{\lambda} \in \sigma_p(T^*)$ can be known from Lemma 2.1.12, thus $\overline{\lambda} \in W(T^*)$. Considering $\mathscr{D}(T^*) \subset \mathscr{D}(T)$, we obtain $\lambda \in W(T)$. So $Conv(\sigma_p(T) \cup \sigma_r(T)) \subset W(T)$. According to

$$\sigma(T) = \sigma_r(T) \cup \sigma_{ap}(T),$$

and from Conclusion (i), $Covn(\sigma(T)) \subset \overline{W(T)}$ can be attained.

(iii) For any $x \in \mathscr{D}(T)$, $\|x\| = 1$, we have

$$\begin{aligned}\|(T - \lambda I)x\| \geq & |(Tx, x) - \lambda| \\ \geq & \inf\{|\boldsymbol{w} - \lambda| : \boldsymbol{w} \in W(T)\} \\ = & dist(\lambda, W(T)).\end{aligned}$$

Thus, the conclusion holds. The proof is completed. □

If S, T be bounded linear operators in a Hilbert space X, then

$$[S, T] = ST - TS$$

is the commutator of S, T. In particular, $[T^*, T] = T^*T - TT^*$ is called the self-commutator of T. The commutator plays an important role in the field of Lie algebra and group theory. For example, the Lie bracket is a commutator. In addition, in the field of operator theory, normal (or regular) operators, hyponormal (or subregular) operators, or seminormal (or semi-regular) operators can be defined by self-commutators of linear operators. Moreover, the self-commutators of the well-known self-adjoint operators ($T = T^*$) and the unitary operators ($TT^* = T^*T = I$) are equal to zero. So the commutators of bounded linear operators not only have profound theoretical significance but also have a wide range of practical applications.

Definition 5.1.9 Let T be a bounded linear operator in Hilbert space X. If $T^*T \geq TT^*$ (that is, $[T^*, T] \geq 0$ or $(T^*Tx, x) \geq (TT^*, x)$), then T is called a hyponormal operator. If T is hyponormal or T^* is hyponormal (that is, $T^*T \geq TT^*$ or $T^*T \leq TT^*$), then T is called a seminormal operator. In particular, if $T^*T = TT^*$ (that is, $[T^*, T] = 0$), then T is called a normal (or regular) operator.

Remark 5.1.10 Obviously, non-negative operators $\subset$ self-adjoint operators $\subset$ normal operators $\subset$ hyponormal operators $\subset$ seminormal operators. In addition, we can also define p-hyponormal operators ($(T^*T)^p \geq (TT^*)^p$, where $0 < p \leq 1$) and log-hyponormal operators, that is, T is invertible and $\log(T^*T) \geq \log(TT^*)$, and so on. In addition, for any $\alpha, \beta \in \mathbb{C}$,

$$[(\alpha T + \beta I)^*, \alpha T + \beta I] = |\alpha|^2[T^*, T].$$

Thus $\alpha T + \beta I$ is the normal operator, hyponormal operator, or seminormal operator if and only if T is the normal operator, hyponormal operator, or seminormal operator, respectively.

Definition 5.1.11 z is called the extreme point of the point set S on the complex plane if $z \in S$ and there exists a closed half-plane such that there are no other points of S except z. All extreme points of S are denoted by $Ex(S)$.

Remark 5.1.12 Obviously, the extreme point is a boundary point, and there exists a support line through which the set is completely located on the side of the support line. For example, all the boundary points of the unit disk are extreme points, and each tangent is a support line.

Definition 5.1.13 z is called a corner of a point set S in the complex plane, if $z \in \overline{S}$ and there exists a half cone inclusion the set S with z vertices and half vertex angle less than $\frac{\pi}{2}$.

Remark 5.1.14 Obviously, the corner point is a boundary point. If the corner point belongs to the set and is also a extreme point, and there are two intersecting support lines through the point so that the set is completely located in the area surrounded by two support lines.

Theorem 5.1.15 *If T is a seminormal operator in Hilbert space X, then*

(i) $\overline{W(T)} = Conv(\sigma(T))$.
(ii) The extreme points of $W(T)$ are the eigenvalues of T.
(iii) The necessary and sufficient condition for $W(T)$ to be closed is that the extreme points of $Conv(\sigma(T))$ belong to the point spectra.

Proof (i) We just need to prove $\overline{W(T)} \subset Conv(\sigma(T))$ according to the spectral inclusion relation. Let L be any support line of $Conv(\sigma(T))$ when T are hyponormal operators. Since the hyponormal property is closed to translation, it can be assumed that L is the imaginary axis and $Conv(\sigma(T))$ is on the left side of L. To prove $\overline{W(T)} \subset Conv(\sigma(T))$, we only need to prove that there is $a \leq 0$ for any $a + ib \in W(T)$. If not, then there exists $\|x\| = 1$, $Tx = (a + ib)x + y$, $(x, y) = 0$, $a > 0$. According to Theorem 5.1.8, for any $c > 0$ we have

$$c^2 \leq \|(T - cI)x\|^2 = (a - c)^2 + b^2 + \|y\|^2,$$

that is,

$$2ac \leq a^2 + b^2 + \|y\|^2,$$

which is impossible for sufficiently large c.

When T^* are hyponormal operators, as shown above, we can prove $\overline{W(T^*)} \subset Conv(\sigma(T^*))$, and we can also attain

$$\overline{W(T)} \subset Conv(\sigma(T)).$$

Thus, the conclusion is proved.

(ii) Assume that $0 \in Ex(W(T))$ and $Re(Tx, x) \leq 0$, then $-T - T^* \geq 0$. From $(Tx_0, x_0) = 0$, it can be known that

$$(T^*x_0, x_0) = 0,$$

thus

$$((-T - T^*)x_0, x_0) = 0.$$

Again from $-T - T^* \geq 0$, we attain

$$Tx_0 = -T^*x_0.$$

Set $N = \{x : Tx = -T^*x\}$, then $T|_N$ is an anti-self-adjoint operator, thus $W(T|_N) \subset \mathrm{i}\mathbb{R}$. And because, $W(T|_N) \subset W(T)$ and $W(T) \cap \mathrm{i}\mathbb{R} = \{0\}$, hence $W(T|_N) = \{0\}$, thus $T|_N = 0$ and $Tx = T|_N x = 0$, that is, $0 \in \sigma_p(T)$.

(iii) When $W(T)$ is a closed set, $W(T) = \overline{W(T)}$, and hence $W(T) = Conv(\sigma(T))$ and the extreme points of $Conv(\sigma(T))$ are the extreme points of $W(T)$. From (ii), we know that the extreme points of $Conv(\sigma(T))$ belong to the point spectra.

When the extreme points of $Conv(\sigma(T))$ belong to the point spectra, considering that the point spectra are contained in the numerical range, and hence $Conv(\sigma(T)) \subset W(T)$. From (i) it can be known that $W(T) = \overline{W(T)}$, that is, $W(T)$ is a closed set. □

Proposition 5.1.16 *Let T be hyponormal operators, or normaloid operators, or unitary operators, or self-adjoint operators in a Hilbert space X. Then*

(i) $\overline{W(T)} = Conv(\sigma(T))$.
(ii) The extreme points of $W(T)$ are the eigenvalues of T.
(iii) The necessary and sufficient condition for $W(T)$ to be closed is that the extreme points of $Conv(\sigma(T))$ belong to the point spectra.

Proof According to the definition, hyponormal operators, normal operators, unitary operators, or self-adjoint operators must be seminormal operators, so it is proved by Theorem 5.1.15. □

Proposition 5.1.17 *If T is a unitary operator, then $W(T)$ is closed if and only if $\sigma_c(T) = \emptyset$.*

Proof Since the spectral points of unitary operators are distributed on the unit circle, the extreme point set of $Conv(\sigma(T))$ is $\sigma(T)$. Because the residual spectra of unitary operators are empty, and hence $\sigma(T) = \sigma_p(T)$ if and only if $\sigma_c(T) = \emptyset$. Therefore, it is proved completely by Theorem 5.1.15. □

Proposition 5.1.18 *If T is a bounded self-adjoint operator and $\sigma(T) = [m, M]$, then $m, M \in \sigma_p(T)$.*

Proof Since m, M are the extreme points of $Conv(\sigma(T))$, it is proved completely by Theorem 5.1.15. □

Proposition 5.1.19 *Let T be a bounded seminormal operator and X be a separable Hilbert space. Then $W(T)$ have at most countable extreme points.*

Proof When T is a bounded hyponormal operator, the eigenvectors corresponding to different eigenvalues are orthogonal, and because X is separable, there are at most countable orthogonal bases, so T has at most countable eigenvalues, so from Theorem 5.1.15, it can be seen that $W(T)$ has at most countable extreme points. When T^* is hyponormal operators, $W(T^*)$ has at most countable extreme points. However, $\lambda \in W(T)$ if and only if $\overline{\lambda} \in W(T^*)$, so $W(T)$ also has at most countable extreme points. □

Proposition 5.1.20 *If T is a bounded seminormal operator, then T is a self-adjoint operator if and only if $\sigma(T) \subset \mathbb{R}$.*

Proof When T is a self-adjoint operator, the proof of $\sigma(T) \subset \mathbb{R}$ is trivial. Conversely, by Theorem 5.1.15, we know $\overline{W(T)} = Conv(\sigma(T)) \subset \mathbb{R}$, $W(T) \subset \mathbb{R}$, so T is a self-adjoint operator. □

Theorem 5.1.21 *If T is a bounded seminormal operator and $AT = T^*A$, where $0 \notin \overline{W(A)}$, then T is a self-adjoint operator.*

Proof Assume that T is the hyponormal operator. According to Corollary 5.1.22, it is only necessary to prove $\sigma(T) \subset \mathbb{R}$. Because $0 \notin \overline{W(A)}$, A are invertible $ATA^{-1} = T^*$, and hence

$$\sigma(T) = \sigma(T^*).$$

Considering $\sigma_r(T^*) = \emptyset$, so

$$\sigma_{ap}(T^*) = \sigma(T) = \sigma(T^*).$$

Assume that there exists $\lambda \neq \overline{\lambda}$ such that $\lambda \in \sigma(T)$, then there exists $\{x_n\}_{n=1}^{\infty}$, $\|x_n\| = 1, n = 1, 2, \ldots,$ such that

$$\|(T^* - \overline{\lambda}I)x_n\| \leq \|(T - \lambda I)x_n\| \to 0.$$

Because $\|(T^* - \overline{\lambda}I)x_n\| = \|A(T - \overline{\lambda}I)A^{-1}x_n\|$, so $\|(T - \overline{\lambda}I)A^{-1}x_n\| \to 0$. Setting $y_n = \frac{A^{-1}x_n}{\|A^{-1}x_n\|}$, then $\|y_n\| = 1, n = 1, 2, \ldots$, from Theorem 5.1.8 we can see that when $\lambda \neq \overline{\lambda}$

$$(Ay_n, y_n) = \frac{(x_n, A^{-1}x_n)}{\|A^{-1}x_n\|^2} \to 0,$$

which contradicts with $0 \notin \overline{W(A)}$, so $\sigma(T) \subset \mathbb{R}$. □

Lemma 5.1.22 *Let T be a densely defined linear operator. If $W(T) \neq \mathbb{C}$, then T is a closable operator.*

Proof Considering that $W(T) \neq \mathbb{C}$ and $W(T)$ is a convex set, it can be assumed that

$$W(T) \subset \{\lambda \in \mathbb{C} : Re(\lambda) \geq 0\}.$$

Set $\{u_n\} \subset \mathscr{D}(T)$, $u_n \to 0$, $Tu_n \to u$, then for any $v \in \mathscr{D}(T)$ and $\lambda > 0$ we have

$$\begin{aligned} 0 \leq & Re(T(u_n + \lambda v), u_n + \lambda v) \\ = & Re((Tu_n, u_n) + \lambda(Tu_n, v) + \lambda(Tv, u_n) + \lambda^2(Tv, v)) \\ \to & Re(\lambda(u, v) + \lambda^2(Tv, v)) \\ = & \lambda Re((u, v) + \lambda(Tv, v)). \end{aligned}$$

Then, for any $\lambda > 0$, we have $Re((u, v) + \lambda(Tv, v)) \geq 0$, and letting $\lambda \to 0$, we get

$$Re(u, v) \geq 0.$$

Because v is arbitrary,

$$(u, v) = 0.$$

Then $u = 0$ is attained from the density of T. The proof is completed. □

5.2 Numerical Range of Infinite-Dimensional Hamiltonian Operators

5.2.1 *Description of Numerical Range of Infinite-Dimensional Hamiltonian Operators*

In this section, we will discuss the numerical range of infinite-dimensional Hamiltonian operators.

Theorem 5.2.1 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. Then* $W(H) = Conv(W(A) \cup W(-A^*))$.

Proof If $\lambda \in W(A)$, then there exists $x \in \mathscr{D}(A)$, $\|x\| = 1$ such that

$$(Ax, x) = \lambda.$$

Set $u = \begin{bmatrix} x & 0 \end{bmatrix}^T$, then $u \in \mathscr{D}(H)$, $\|u\| = 1$ and

$$(Hu, u) = \lambda,$$

that is, $W(A) \subset W(H)$. Similarly, $W(-A^*) \subset W(A)$, so $(W(A) \cup W(-A^*)) \subset W(H)$, $Conv(W(A) \cup W(-A^*)) \subset W(H)$.

When $\lambda \in W(H)$, there exists $\|x\|^2 + \|y\|^2 = 1$ such that

$$\lambda = (Ax, x) + (-A^* y, y).$$

If $x = 0$ or $y = 0$, $\lambda \in W(A) \cup W(-A^*)$, thus $W(H) \subset Conv(W(A) \cup W(-A^*))$.

If $x \neq 0$ and $y \neq 0$,

$$\lambda = \|x\|^2 \frac{(Ax, x)}{(x, x)} + (1 - \|x\|^2) \frac{(-A^* y, y)}{(y, y)},$$

thus $\lambda \in Conv(W(A) \cup W(-A^*))$, so $W(H) \subset Conv(W(A) \cup W(-A^*))$. The proof is completed. □

Similarly, the following conclusion can be proved.

Theorem 5.2.2 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal dominant infinite-dimensional Hamiltonian operator. Then* $Conv(W(A) \cup W(-A^*)) \subset W(H)$.

Remark 5.2.3 From the above conclusions, it is easy to know that if $\mathscr{D}(A^*) \subset \mathscr{D}(A)$ or $\mathscr{D}(A) \subset \mathscr{D}(A^*)$ is satisfied, the intersection of the numerical range and the imaginary axis of diagonal dominant infinite-dimensional Hamiltonian operators H is non-empty.

The following will give some specific examples to justify Theorem 5.2.1.

Example 5.2.4 Let infinite-dimensional Hamiltonian operator H be

$$H = \begin{bmatrix} S_l & 0 \\ 0 & -S_l^* \end{bmatrix},$$

where the definition of S_l operator is shown in Example 5.1.1. Because

$$S_l^* x = (0, x_1, x_2, \ldots),$$

we have

$$W(A) = W(-A^*) = \{\lambda \in \mathbb{C} : |\lambda| < 1\}.$$

Thus, it can be seen from Theorem 5.2.1 that

$$W(H) = Conv(W(A) \cup W(-A^*)) = \{\lambda \in \mathbb{C} : |\lambda| < 1\}.$$

On the other hand, for any $x = (x_1, x_2, \ldots), y = (y_1, y_2, \ldots) \in X$, $\|x\|^2 + \|y\|^2 = 1$, we have

$$\left(H \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right) = \sum_{i=1}^{+\infty} x_{i+1} x_i - \sum_{i=1}^{+\infty} y_i y_{i+1}.$$

Notice that $|x_1|^2 + |x_2|^2 + \cdots = 1$, and assume that $|x_1| \neq 0$ or $|y_1| \neq 0$, then

$$\begin{aligned} \left| \left(H \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right) \right| &\leq |x_2||x_1| + |x_3||x_2| + \cdots + |y_1||y_2| + |y_2||y_3| + \cdots \\ &\leq \frac{1}{2}(|x_1|^2 + 2|x_2|^2 + \cdots + |y_1|^2 + 2|y_2|^2 + \cdots) \\ &\leq \frac{1}{2}(2 - |x_1|^2 - |y_1|^2). \end{aligned}$$

Thus, $W(H) \subset \{\lambda \in \mathbb{C} : |\lambda| < 1\}$. Conversely, for any $\lambda = re^{i\theta}, 0 \le r < 1$, take

$$u = \begin{bmatrix} x \\ 0 \end{bmatrix},$$

where

$$x = (\sqrt{1-r^2}, r\sqrt{1-r^2}e^{i\theta}, r^2\sqrt{1-r^2}e^{2i\theta}, \ldots),$$

then

$$\|u\|^2 = 1 - r^2 + r^2(1-r^2) + r^4(1-r^2) + \cdots = 1,$$

and

$$\begin{aligned}(Hu, u) &= e^{i\theta}r(1-r^2) + e^{i\theta}r^3(1-r^2) + \cdots \\ &= re^{i\theta}.\end{aligned}$$

Thus, $W(H) = \{\lambda \in \mathbb{C} : |\lambda| < 1\}$. This is completely consistent with the conclusion obtained by Theorem 5.2.1.

Example 5.2.5 Let $X = L^2[0, \infty)$, $A = \frac{\mathrm{d}}{\mathrm{d}t}$, $Ax = \frac{\mathrm{d}x}{\mathrm{d}t}$,

$$\mathscr{D}(A) = \{x \in X : x \text{ is absolutely continuous}, x' \in X, x(0) = 0\}.$$

Then $A^* = -\frac{\mathrm{d}}{\mathrm{d}t}$,

$$\mathscr{D}(A^*) = \{x \in X : x \text{ is absolutely continuous}, x' \in X\}.$$

Set the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} \frac{\mathrm{d}}{\mathrm{d}t} & 0 \\ 0 & \frac{\mathrm{d}}{\mathrm{d}t} \end{bmatrix},$$

then for any $u = \begin{bmatrix} x & y \end{bmatrix}^T \in \mathscr{D}(H)$, $\|u\| = 1$, we have

$$2Re(W(Hu, u)) = -|y(0)|^2 \le 0.$$

On the other hand, it can be attained by calculation

$$\sigma_p(H) = \{\lambda \in \mathbb{C} : Re(\lambda) < 0\}.$$

Reconsidering that $\sigma_p(H) \subset W(H)$, we have $W(H) = \{\lambda \in \mathbb{C} : Re(\lambda) \le 0\}$.

The following discusses the properties of the numerical range of skew-diagonal infinite-dimensional Hamiltonian operators.

Theorem 5.2.6 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then* $\lambda \in W(H)$ *if and only if* $-\lambda \in W(H)$. *Thus, the numerical range of skew-diagonal infinite-dimensional Hamiltonian operator contains the origin.*

Proof In fact, if $\lambda \in W(H)$, then there exists $u = \begin{bmatrix} f & g \end{bmatrix}^T \in \mathscr{D}(H)$, $\|u\| = 1$ such that

$$(Hu, u) = (Bg, f) + (Cf, g) = \lambda.$$

Set $v = \begin{bmatrix} f & -g \end{bmatrix}^T$, then $v \in \mathscr{D}(H)$, $\|v\| = 1$ and

$$(Hv, v) = -\lambda.$$

When $-\lambda \in W(H)$, it is similar to prove that $\lambda \in W(H)$. □

The following is the connection between the numerical range of skew-diagonal infinite-dimensional Hamiltonian operators and the numerical range of its adjoint operators.

Theorem 5.2.7 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then* $W(H) = W(-H) = W(H^*) = W(-H^*)$.

Proof By Theorem 2.3.2, we can know that H is unitary similar to $-H$, H^* and $-H^*$, so the conclusion is obvious. □

Theorem 5.2.8 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then*

$$\{\lambda \in \mathbb{C} : \lambda^2 \in W(BC) \cup W(CB)\} \subset W(H).$$

Proof If $\lambda^2 \in W(BC)$, then there exists $f \in \mathscr{D}(BC)$, $\|f\| = 1$ such that

$$(BCf, f) = \lambda^2.$$

Without loss of generality, set $\lambda \neq 0$, then $Cf \neq 0$, take $u = \begin{bmatrix} \alpha_1 f & \frac{\alpha_2 Cf}{\|Cf\|} \end{bmatrix}^T$, where

$$\alpha_1 = \frac{\lambda}{\sqrt{\|Cf\|^2 + |\lambda|^2}}, \ \alpha_2 = \frac{\|Cf\|}{\sqrt{\|Cf\|^2 + |\lambda|^2}},$$

then $u \in \mathscr{D}(H)$, $\|u\| = 1$ and

$$(Hu, u) = \lambda.$$

So $\lambda \in W(H)$. The proof about $\{\lambda \in \mathbb{C} : \lambda^2 \in W(CB)\} \subset W(H)$ is exactly similar. □

Theorem 5.2.9 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then* $Conv(W(B+C) \cup W(-B-C)) \subset W(H)$.

Proof Taking $U = \frac{1}{\sqrt{2}} \begin{bmatrix} I & I \\ -I & I \end{bmatrix}$, then U is a unitary operator and satisfies

$$U^* \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} U = \frac{1}{2} \begin{bmatrix} -C-B & -C+B \\ C-B & C+B \end{bmatrix}.$$

So

$$Conv\left(W\left(\frac{B+C}{2}\right) \cup W\left(\frac{-B-C}{2}\right)\right) \subset W(H).$$

Considering that $C+B$ are symmetric operators, when $\mathscr{D}(C) \cap \mathscr{D}(B) \neq \{0\}$, $W(C+B) \neq \emptyset$, and $\lambda \in W(C+B)$ if and only if $-\lambda \in W(-C-B)$, thus we have $0 \in W(H)$. □

Theorem 5.2.10 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : (\mathscr{D}(A) \cap \mathscr{D}(C)) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*)) \to X \times X$ *be an infinite-dimensional Hamiltonian operator.*

(i) If $B = C$, $A = -A^*$, *then* $W(H) = Conv(W(A-C) \cup W(A+C))$.
(ii) If $B = -C$, $A = -A^*$, *then* $W(H) = Conv(W(A - \mathrm{i}C) \cup W(A + \mathrm{i}C))$.

Proof (i) Set $U = \frac{1}{\sqrt{2}} \begin{bmatrix} I & I \\ -I & I \end{bmatrix}$, then U is a unitary operator and satisfies

$$U^* \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} U = \begin{bmatrix} A-C & 0 \\ 0 & A+C \end{bmatrix}.$$

So, $W(H) = Conv(W(A-C) \cup W(A+C))$.

(ii) Taking $U = \frac{1}{\sqrt{2}} \begin{bmatrix} I & \mathrm{i}I \\ \mathrm{i}I & I \end{bmatrix}$, then U is a unitary operator and satisfies

$$U^* \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} U = \begin{bmatrix} A - \mathrm{i}C & -0 \\ 0 & A + \mathrm{i}C \end{bmatrix}.$$

So, $W(H) = Conv(W(A - \mathrm{i}C) \cup W(A + \mathrm{i}C))$. □

For the general infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : (\mathscr{D}(A) \cap \mathscr{D}(C)) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*)) \to X \times X,$$

considering that

$$\frac{1}{2}\begin{bmatrix} I & I \\ -I & I \end{bmatrix}^* \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} \begin{bmatrix} I & I \\ -I & I \end{bmatrix} = \frac{1}{2}\begin{bmatrix} A-B-C-A^* & A+B-C+A^* \\ A-B+C+A^* & A+B+C-A^* \end{bmatrix}$$

and

$$W\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) = \frac{1}{2} W\left(\begin{bmatrix} I & I \\ -I & I \end{bmatrix}^* \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} \begin{bmatrix} I & I \\ -I & I \end{bmatrix}\right),$$

the following conclusions are attained.

Theorem 5.2.11 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : (\mathscr{D}(A) \cap \mathscr{D}(C)) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*)) \to X \times X$ *be an infinite-dimensional Hamiltonian operator. Then* $Conv(W(\frac{A-B-C-A^*}{2}) \cup W(\frac{A+B+C-A^*}{2})) \subset W(H)$.

Theorem 5.2.12 *If* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : (\mathscr{D}(A) \cap \mathscr{D}(C)) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*)) \to X \times X$ *is a non-negative Hamiltonian operator, then* $ReW(\mathfrak{J}H) \geq 0$, *where* $\mathfrak{J} = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$.

5.2.2 Symmetry of Numerical Range of Infinite-Dimensional Hamiltonian Operators

As far as we know, when the infinite-dimensional Hamiltonian operator H is a symplectic self-adjoint operator, that is,

$$(JH)^* = JH.$$

Then its spectral set is symmetric about the imaginary axis and the union of the point spectra and the residual spectra is also symmetric about the imaginary axis, where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. However, for the numerical range of infinite-dimensional Hamiltonian operators, even if it is symplectic self-adjoint, it is not necessarily symmetric about the imaginary axis (see Example 5.2.5), but $W(H)$ and $W(H^*)$ are symmetric about the origin.

Theorem 5.2.13 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : (\mathscr{D}(A) \cap \mathscr{D}(C)) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*)) \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If it satisfies*

$$(JH)^* = JH,$$

where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$, *then* $\lambda \in W(H)$ *if and only if* $-\lambda \in W(H^*)$.

Proof Considering $H^* = JHJ$ and $J^* = J^{-1} = -J$, the proof of the conclusion is obvious. □

At the same time, it is natural to ask a question when the numerical range of symplectic self-adjoint infinite-dimensional Hamiltonian operators is about the imaginary axis symmetry?

Theorem 5.2.14 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : (\mathscr{D}(A) \cap \mathscr{D}(C)) \times (\mathscr{D}(B) \cap \mathscr{D}(A^*)) \to X \times X$ *be a symplectic self-adjoint infinite-dimensional Hamiltonian operator. If* $J\mathscr{D}(H) = \mathscr{D}(H)$, *where* $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$, *then* $W(H)$ *is symmetric about the imaginary axis.*

Proof If $\lambda \in W(H)$, then there exists $u \in \mathscr{D}(H)$, $\|u\| = 1$ such that

$$(Hu, u) = \lambda.$$

Considering that $H^* = JHJ$ and $J^* = J^{-1} = -J$, we get

$$(H^*Ju, Ju) = -\lambda.$$

And because $J\mathscr{D}(H) = \mathscr{D}(H)$, so $Ju \in \mathscr{D}(H)$, $\|Ju\| = 1$ and

$$(HJu, Ju) = -\overline{\lambda},$$

that is, $-\overline{\lambda} \in W(H)$. The proof is completed. □

According to Theorem 5.3.34, the following corollary is obvious.

Proposition 5.2.15 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator. If* $\mathscr{D}(A) = \mathscr{D}(A^*)$, *then* $W(H)$ *is symmetric about the imaginary axis.*

5.2.3 Spectral Inclusion of Numerical Range of Infinite-Dimensional Hamiltonian Operators

As far as we know, the numerical range of bounded linear operators has spectral inclusion properties, so the numerical range can formulate the distribution range of spectral points. For unbounded linear operators, the numerical range can only formulate the scope of approximate point spectra, and the characterization range of spectra can only be formulated under certain conditions. For the infinite-dimensional Hamiltonian operators, even the numerical range of symplectic self-adjoint infinite-dimensional Hamiltonian operators may not be able to formulate the distribution of spectral points.

Example 5.2.16 Let $X = L^2[0, \infty)$, $A = \frac{\mathrm{d}}{\mathrm{d}t}$, $Ax = \frac{\mathrm{d}x}{\mathrm{d}t}$,

$$\mathscr{D}(A) = \{x \in X : x \text{ is absolutely continuous}, x' \in X, x(0) = 0\}.$$

Then $A^* = -\frac{\mathrm{d}}{\mathrm{d}t}$,

$$\mathscr{D}(A^*) = \{x \in X : x \text{ is absolutely continuous}, x' \in X\}.$$

Set the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} \frac{\mathrm{d}}{\mathrm{d}t} & 0 \\ 0 & \frac{\mathrm{d}}{\mathrm{d}t} \end{bmatrix},$$

then for any $u = [x \ \ y]^T \in \mathscr{D}(H)$, $\|u\| = 1$, we have

$$2Re(W(Hu, u)) = -|y(0)|^2 \leq 0.$$

On the other hand, it can be calculated that

$$\sigma_p(H) = \{\lambda \in \mathbb{C} : Re(\lambda) < 0\}.$$

Reconsidering that $(JH)^* = JH$, that is,

$$\sigma_r(H) = \{\lambda \in \mathbb{C} : Re(\lambda) > 0\}.$$

In addition, there is

$$\sigma_c(H) = \{\lambda \in \mathbb{C} : Re(\lambda) = 0\}.$$

So $\sigma(H) = \mathbb{C}$. In summary, the numerical range closure of infinite-dimensional Hamiltonian operator H does not contain a spectral set, that is, the spectral inclusion relationship does not hold.

Firstly, the spectral inclusion problem of the numerical range of diagonal infinite-dimensional Hamiltonian operators is discussed.

Theorem 5.2.17 *Let* $H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal infinite-dimensional Hamiltonian operator. If*

$$(\sigma_{r,1}(A) \cup \sigma_{r,1}(-A^*)) \subset Conv(\overline{W(A)} \cup \overline{W(-A^*)}),$$

then $\sigma(H) \subset \overline{W(H)}$ *holds.*

Proof Considering that

$$\sigma(H) = (\sigma_{ap}(A) \cup \sigma_{ap}(-A^*)) \cup (\sigma_{r,1}(A) \cup \sigma_{r,1}(-A^*)),$$
$$\overline{W(H)} = Conv(\overline{W(A)}, \overline{W(-A^*)}),$$

and

$$\sigma_{ap}(A) \subset \overline{W(A)}, \sigma_{ap}(-A^*) \subset \overline{W(-A^*)},$$

the conclusion is obvious. □

Secondly, the numerical range spectral inclusion problem of skew-diagonal infinite-dimensional Hamiltonian operators is discussed.

Theorem 5.2.18 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. If* $\nabla_1 \subset \nabla_2$, *then*

$$\sigma(H) \subset \overline{W(H)},$$

where

$$\nabla_1 = \{\lambda \in \mathbb{C} : \overline{\lambda^2} \in \sigma_{p,1}(CB) \cap \sigma_{p,1}(BC)\},$$
$$\nabla_2 = \{\lambda \in \mathbb{C} : \lambda^2 \in W(BC) \cup W(CB)\}.$$

Proof Combining Corollary 2.3.20 and Theorem 5.2.8, the conclusion is immediately proved. □

Proposition 5.2.19 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. If* $\sigma_{p,1}(CB) \cap \sigma_{p,1}(BC) = \emptyset$, *then* $\sigma(H) \subset \overline{W(H)}$.

Theorem 5.2.20 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal dominant upper triangular infinite-dimensional Hamiltonian operator. If* $\lambda \in \sigma_{p,1}(A) \cup \sigma_{p,1}(-A^*)$ *contains* $-\overline{\lambda} \in Conv(\overline{W(A)} \cup \overline{W(-A^*)})$, *then* $\sigma(H) \subset \overline{W(H)}$ *holds.*

Proof Considering that

$$\sigma(H) \subset (\sigma_{ap}(A) \cup \sigma_{ap}(-A^*)) \cup (\sigma_{r,1}(A) \cup \sigma_{r,1}(-A^*))$$

and

$$(\sigma_{ap}(A) \cup \sigma_{ap}(-A^*)) \subset (\overline{W(A)} \cup \overline{W(-A^*)}),$$

from $Conv(\overline{W(A)} \cup \overline{W(-A^*)}) \subset \overline{W(H)}$, it can be known that the conclusion is obvious. □

Then, when do the general infinite-dimensional Hamiltonian operators have a spectral inclusion relationship?

Theorem 5.2.21 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator and satisfies* $(JH)^* = JH$. *If* $\sigma_p(H)$ *is empty set or symmetric about the imaginary axis, then* $\sigma(H) \subset \overline{W(H)}$ *holds.*

Proof Since $\sigma_p(H)$ is empty, $\sigma_r(H)$ is empty set in view of $(JH)^* = JH$, so $\sigma(H) \subset \overline{W(H)}$ holds.

When $\sigma_p(H)$ is non-empty set, considering that $\sigma_p(H)$ is symmetric about the imaginary axis, that is, $\sigma_r(H)$ is still empty set, so $\sigma(H) \subset \overline{W(H)}$ holds. □

Theorem 5.2.22 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator and satisfies* $(JH)^* = JH$. *If* $W(H)$ *is symmetric with respect to the imaginary axis, then* $\sigma(H) \subset \overline{W(H)}$ *holds.*

Proof When $W(T)$ is symmetric about the imaginary axis, take any $\lambda \in \sigma(H)$, if $\lambda \in \sigma_{r,1}(H)$ $-\overline{\lambda} \in \sigma_{p,1}(H)$. So, $-\overline{\lambda} \in W(H)$. Considering that $W(H)$ is symmetric about the imaginary axis, we can get $\lambda \in W(H)$. And because $\sigma(H) = \sigma_{ap}(H) \cup \sigma_{r,1}(H)$ and $\sigma_{ap}(H) \subset \overline{W(H)}$ is trivial, so the spectral inclusion relation holds. □

The following will give specific examples to illustrate the effectiveness of the criterion.

Example 5.2.23 Let $X = L^2[0, \infty)$, $A = \mathrm{i}\frac{\mathrm{d}}{\mathrm{d}t}$, $Ax = \mathrm{i}\frac{\mathrm{d}x}{\mathrm{d}t}$,

$$\mathscr{D}(A) = \{x \in X : x \text{ is absolutely continuous}, x' \in X, x(0) = 0\}.$$

Then $A^* = \mathrm{i}\frac{\mathrm{d}}{\mathrm{d}t}$,

$$\mathscr{D}(A^*) = \{x \in X : x \text{ is absolutely continuous}, x' \in X\}.$$

Set the infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} \mathrm{i}\frac{\mathrm{d}}{\mathrm{d}t} & 0 \\ 0 & -\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}t} \end{bmatrix},$$

then it is calculated that $W(H) = \{\lambda \in \mathbb{C} : Im(\lambda) \geq 0\}$, that is, $W(T)$ is symmetric with respect to the imaginary axis, so the spectral inclusion relation is held by Theorem 5.2.22.

On the other hand, it can be calculated that

$$\sigma_p(H) = \{\lambda \in \mathbb{C} : Im(\lambda) > 0\}.$$

Reconsidering that $(JH)^* = JH$, that is, $\sigma_r(H) = \emptyset$. In addition,

$$\sigma_c(H) = \{\lambda \in \mathbb{C} : Im(\lambda) = 0\}.$$

Hence, $\sigma(H) = \{\lambda \in \mathbb{C} : Im(\lambda) \geq 0\}$, that is, the spectral inclusion relation is held, and the actual operation is consistent with the theoretical results.

In the following, we will give some conditions for the numerical range $W(T)$ of infinite-dimensional Hamiltonian operators to be symmetric to the imaginary axis.

Theorem 5.2.24 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator and satisfies* $(JH)^* = JH$. *If* $\mathscr{D}(H) = J\mathscr{D}(H)$, *then* $W(H)$ *is symmetric about imaginary axis.*

Proof For any $\lambda \in W(H)$, then there exists $u \in \mathscr{D}(H)$, $\|u\| = 1$ such that

$$\lambda = (Hu, u).$$

Considering that $(JH)^* = JH$, that is, $\overline{\lambda} = (u, JH^*Ju) = -(Ju, H^*Ju)$. And because $\mathscr{D}(H) = J\mathscr{D}(H)$, hence $Ju \in \mathscr{D}(H)$, $\|Ju\| = 1$ and

$$-\overline{\lambda} = (HJu, Ju),$$

that is, $W(H)$ is symmetric about the imaginary axis. □

Proposition 5.2.25 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If* $\mathscr{D}(A) = \mathscr{D}(A^*)$ *and* $B, C \in \mathscr{B}(X)$, *then* $W(H)$ *is symmetric about the imaginary axis.*

Proof If $\mathscr{D}(A) = \mathscr{D}(A^*)$ and $B, C \in \mathscr{B}(X)$, then $(JH)^* = JH$ and $\mathscr{D}(H) = J\mathscr{D}(H)$, hence $W(H)$ is symmetric about the imaginary axis. □

Proposition 5.2.26 *Let* $H_0 = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. If* B *or* C *is a semi-definite operator, then* $\sigma(H_0) \subset \overline{W(H_0)}$.

Proof If B or C is a semi-definite operator, then $\sigma_r(H_0) = \emptyset$. From Theorem 5.2.22, it is known immediately that $\sigma(H_0) \subset \overline{W(H_0)}$. □

5.3 Numerical Radius of Infinite-Dimensional Hamiltonian Operators

5.3.1 Numerical Radius of Bounded Linear Operators

For bounded linear operators T, for any $\|x\| = 1$, we have

$$|(Tx, x)| \leq \|T\|,$$

that is, the numerical range is a bounded set. Therefore, the concept of numerical radius can be introduced.

Definition 5.3.1 For a given bounded linear operator T in the Hilbert space X, its numerical radius $\boldsymbol{w}(T)$ is defined as

$$\boldsymbol{w}(T) = \sup\{|\lambda| : \lambda \in W(T)\}.$$

The numerical radius describes the radius of the smallest circle whose center is at the origin and contains the closure of the numerical range. Moreover, the numerical radius also has important applications in the field of stability theory of finite difference approximate solutions of hyperbolic initial value problems. For example, considering hyperbolic initial value problems

$$\begin{cases} u_t = Au_x + Bu_y, \ -\infty < x < +\infty, -\infty < y < +\infty, t \geq 0, \\ u(x, y, 0) = f(x, y) \in L^2(-\infty, +\infty), \ -\infty < x < +\infty, -\infty < y < +\infty, \end{cases} \tag{5.9}$$

where the coefficient matrices A, B are the $n \times n$ Hermitian matrices. To use the finite difference method to solve the problem of the system (5.9), the fixed ratios $\lambda = \frac{\Delta t}{\Delta x}, \mu = \frac{\Delta t}{\Delta y}$ of $\Delta x, \Delta y, \Delta t$ are introduced, if

$$\lambda^2[\boldsymbol{w}(A)]^2 + \mu^2[\boldsymbol{w}(B)]^2 \leq \frac{1}{4},$$

then the Lax-Wendroff difference scheme of the system (5.9) is stable, as described in the cited work [66].

The following is an example of a numerical radius.

Example 5.3.2 Let X be a Hilbert space, and I be a unit operator in Hilbert space X and define the block operator matrix in Hilbert space $X \times X$ (without confusion, its inner product is still denoted by $(\cdot, \cdot)$) as the following

$$T = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix},$$

then $W(T) = \{\lambda \in \mathbb{C} : |\lambda| \leq \frac{1}{2}\}$. Hence, $\boldsymbol{w}(T) = \frac{1}{2}$.

Example 5.3.3 Define the matrix T_3 in $\mathbb{C}^3$ as follows:

$$T_3 = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Then for any $x = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^T \in \mathbb{C}^3$, we have

$$|(T_3x, x)| = |x_1\overline{x_2} + x_2\overline{x_3}| \leq |x_1||x_2| + |x_2||x_3|.$$

In order to calculate $\boldsymbol{w}(T)$, it is only needed to calculate $\sup\{|x_1||x_2| + |x_2||x_3|\}$ under the condition of $|x_1|^2 + |x_2|^2 + |x_3|^2 = 1$. Let $r_i = |x_i|$, $i = 1, 2, 3$, and construct the Lagrangian function

$$f(r_1, r_2, r_3, \lambda) = r_1 r_2 + r_2 r_3 + \lambda\left(\sum_{i=1}^{3} r_i^2 - 1\right).$$

From the necessary condition of extreme value, it can be attained that

$$\begin{aligned} r_2 + 2\lambda r_1 &= 0, \\ r_1 + r_3 + 2\lambda r_2 &= 0, \\ r_2 + 2\lambda r_3 &= 0. \end{aligned}$$

It is calculated that $r_1 = r_3 = \frac{1}{2}$, $r_2 = \frac{\sqrt{2}}{2}$. Hence, $\boldsymbol{w}(T_3) = \frac{\sqrt{2}}{2}$.

Remark 5.3.4 Similar to the above example, it can be calculated that $\boldsymbol{w}(T_n) = \cos\frac{\pi}{n+1}$, where

$$T_n = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ I & 0 & \cdots & 0 & 0 \\ 0 & I & \cdots & 0 & 0 \\ & \cdots & & \cdots & \\ 0 & 0 & \cdots & I & 0 \end{bmatrix}_{n\times n}.$$

Thus, the numerical radius of the right (or left) shift operator S_r(or S_l) in the infinite-dimensional space is $\boldsymbol{w}(S_r) = \lim_{n\to\infty} \boldsymbol{w}(T_n) = 1$.

According to the definition of numerical radius, it is easy to prove the following properties.

Corollary 5.3.5 *Let T, S be bounded linear operators in a complex Hilbert space X. Then*

(i) $\boldsymbol{w}(T) = \boldsymbol{w}(U^*TU)$, *where U is a unitary operator;*
(ii) $\boldsymbol{w}(T) = \boldsymbol{w}(T^*) = \boldsymbol{w}(\alpha T)$, *where $|\alpha| = 1$;*
(iii) $\boldsymbol{w}(T) = \sup\{|(Tx, x)| : x \in X, \|x\| \le 1\}$;
(iv) $\boldsymbol{w}\left(\begin{bmatrix} T & 0 \\ 0 & S \end{bmatrix}\right) = \max\{\boldsymbol{w}(T), \boldsymbol{w}(S)\}$;
(v) $\boldsymbol{w}\left(\begin{bmatrix} 0 & T \\ e^{i\theta}S & 0 \end{bmatrix}\right) = \boldsymbol{w}\left(\begin{bmatrix} 0 & T \\ S & 0 \end{bmatrix}\right)$, *where $\theta \in \mathbb{R}$;*
(vi) $\boldsymbol{w}\left(\begin{bmatrix} 0 & T \\ S & 0 \end{bmatrix}\right) = \boldsymbol{w}\left(\begin{bmatrix} 0 & S \\ T & 0 \end{bmatrix}\right)$;
(vii) $\boldsymbol{w}\left(\begin{bmatrix} T & S \\ S & T \end{bmatrix}\right) = \max\{\boldsymbol{w}(T + S), \boldsymbol{w}(T - S)\}$;
(viii) $\boldsymbol{w}\left(\begin{bmatrix} T & S \\ -S & T \end{bmatrix}\right) = \max\{\boldsymbol{w}(T + iS), \boldsymbol{w}(T - iS)\}$.

Proof The proofs of conclusions (i), (ii), and (iii) are trivial. The following is the proof of the conclusion (iv). If $\lambda \in W\left(\begin{bmatrix} T & 0 \\ 0 & S \end{bmatrix}\right)$, then there exists $\|x\|^2 + \|y\|^2 = 1$ such that

$$(Tx, x) + (Sy, y) = \lambda,$$

that is,

$$\frac{\|x\|^2}{\|x\|^2 + \|y\|^2}\left(\frac{Tx}{\|x\|}, \frac{x}{\|x\|}\right) + \frac{\|y\|^2}{\|x\|^2 + \|y\|^2}\left(\frac{Sy}{\|y\|}, \frac{y}{\|y\|}\right) = \lambda.$$

Thus $W(T) = Conv(W(T) \cup W(S))$, where $Conv(W(T) \cup W(S))$ denotes the convex hull of $W(T) \cup W(S)$. Then we have $\boldsymbol{w}\left(\begin{bmatrix} T & 0 \\ 0 & S \end{bmatrix}\right) = \max\{\boldsymbol{w}(A), \boldsymbol{w}(B)\}$.

(v) Select the unitary operator $U = \begin{bmatrix} I & 0 \\ 0 & e^{\frac{\mathrm{i}}{2}\theta} I \end{bmatrix}$, then

$$\begin{aligned} \boldsymbol{w}\left(U^* \begin{bmatrix} 0 & T \\ e^{\mathrm{i}\theta} S & 0 \end{bmatrix} U\right) &= \boldsymbol{w}\left(\begin{bmatrix} I & 0 \\ 0 & e^{-\frac{\mathrm{i}}{2}\theta} \end{bmatrix} \begin{bmatrix} 0 & T \\ e^{\mathrm{i}\theta} S & 0 \end{bmatrix} \begin{bmatrix} I & 0 \\ 0 & e^{\frac{\mathrm{i}}{2}\theta} \end{bmatrix}\right) \\ &= \boldsymbol{w}\left(\begin{bmatrix} 0 & e^{\frac{\mathrm{i}}{2}\theta} T \\ e^{\frac{\mathrm{i}}{2}\theta} S & 0 \end{bmatrix}\right) \\ &= |e^{\frac{\mathrm{i}}{2}\theta}| \boldsymbol{w}\left(\begin{bmatrix} 0 & T \\ S & 0 \end{bmatrix}\right) = \boldsymbol{w}\left(\begin{bmatrix} 0 & T \\ S & 0 \end{bmatrix}\right). \end{aligned}$$

(vi) Select the unitary operator $U = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$, then

$$\begin{aligned} \boldsymbol{w}\left(U^* \begin{bmatrix} 0 & T \\ S & 0 \end{bmatrix} U\right) &= \boldsymbol{w}\left(\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} 0 & T \\ S & 0 \end{bmatrix} \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}\right) \\ &= \boldsymbol{w}\left(\begin{bmatrix} 0 & S \\ T & 0 \end{bmatrix}\right). \end{aligned}$$

(vii) Select the unitary operator $U = \frac{1}{\sqrt{2}} \begin{bmatrix} I & I \\ -I & I \end{bmatrix}$, then

$$\begin{aligned} \boldsymbol{w}\left(U^* \begin{bmatrix} T & S \\ S & T \end{bmatrix} U\right) &= \boldsymbol{w}\left(\begin{bmatrix} T - S & 0 \\ 0 & T + S \end{bmatrix}\right) \\ &= \max\{\boldsymbol{w}(T + S), \boldsymbol{w}(T - S)\}. \end{aligned}$$

(viii) Select the unitary operator $U = \frac{1}{\sqrt{2}} \begin{bmatrix} I & \mathrm{i}I \\ \mathrm{i}I & I \end{bmatrix}$, then

$$\boldsymbol{w}\left(U^*\begin{bmatrix} T & S \\ -S & T \end{bmatrix}U\right) = \boldsymbol{w}\left(\begin{bmatrix} T+\mathrm{i}S & 0 \\ 0 & T-\mathrm{i}S \end{bmatrix}\right)$$
$$= \max\{\boldsymbol{w}(T+\mathrm{i}S), \boldsymbol{w}(T-\mathrm{i}S)\}.$$

The proof is completed.

(viii) Select the unitary operator $U = \frac{1}{\sqrt{2}}\begin{bmatrix} I & \mathrm{i}I \\ \mathrm{i}I & I \end{bmatrix}$, then

$$\boldsymbol{w}\left(U^*\begin{bmatrix} T & S \\ -S & T \end{bmatrix}U\right) = \boldsymbol{w}\left(\begin{bmatrix} T+\mathrm{i}S & 0 \\ 0 & T-\mathrm{i}S \end{bmatrix}\right)$$
$$= \max\{\boldsymbol{w}(T+\mathrm{i}S), \boldsymbol{w}(T-\mathrm{i}S)\}.$$

The proof is completed. □

Another important property of the numerical radius of bounded linear operators is that the numerical radius is between the spectral radius and the operator norm and is equivalent to the operator norm.

Corollary 5.3.6 *Let T be a bounded linear operator in Hilbert space X. Then*

(i) $r(T) \leq \boldsymbol{w}(T) \leq \|T\|$;
(ii) $\boldsymbol{w}(T) \leq \|T\| \leq 2\boldsymbol{w}(T)$.

Proof (i) $r(T) \leq \boldsymbol{w}(T)$ is obtained by the spectral inclusion property $\sigma(T) \subset \overline{W(T)}$, and $|(Tx, x)| \leq \|T\|\|x\|^2$ is obtained by the Schwarz inequality $|(Tx, x)| \leq \|Tx\|\|x\|$, that is, $\boldsymbol{w}(T) \leq \|T\|$.

(ii) It is only need to prove $\|T\| \leq 2\boldsymbol{w}(T)$. Considering the polarization identity

$$4(Tx, y) = (T(x+y), x+y) - (T(x-y), x-y) + \mathrm{i}(T(x+\mathrm{i}y), x+iy)$$
$$- \mathrm{i}(T(x-\mathrm{i}y), x-\mathrm{i}y),$$

that is,

$$4|(Tx, y)| \leq \boldsymbol{w}(T)[\|x+y\|^2 + \|x-y\|^2 + \|x+\mathrm{i}y\|^2 + \|x-\mathrm{i}y\|^2]$$
$$= 4\boldsymbol{w}(T)[\|x\|^2 + \|y\|^2].$$

Set $\|x\|^2 = \|y\|^2 = 1$, then $|(Tx, y)| \leq 2\boldsymbol{w}(T)$. Considering that

$$\|T\| = \sup_{\|x\|=1, \|y\|=1} |(Tx, y)|,$$

that is, $\|T\| \leq 2\boldsymbol{w}(T)$. The proof is completed. □

According to the spectral mapping theorem of linear operators (see [67]): $\lambda \in \sigma(T)$ if and only if $\lambda^n \in \sigma(T^n)$, the spectral radius of the bounded linear operators

satisfies $r(T^n) = [r(T)]^n$. In addition, the operator norm satisfies $\|T^n\| \leq \|T\|^n$. However, numerical radius and operator norm are equivalent norms and is there an inequality similar to the operator norm? The answer is yes, and the inequality is called the power inequality of numerical radius . The following proof of power inequality is given by Pearcy (see [68]).

Theorem 5.3.7 *Let T be a bounded linear operator in the Hilbert space X. Then for any positive integers n,*

$$\boldsymbol{w}(T^n) \leq [\boldsymbol{w}(T)]^n$$

holds.

To prove Theorem 5.3.7, the following lemma should be proved first.

Lemma 5.3.8 *Let T be a bounded linear operator in Hilbert space X. Then for any positive integers m, $\boldsymbol{w}(T^m) \leq [\boldsymbol{w}(T)]^m$ holds, if and only if $\boldsymbol{w}(T) \leq 1$ implies $\boldsymbol{w}(T^m) \leq 1$ for any positive integers m.*

Proof When $\boldsymbol{w}(T^m) \leq [\boldsymbol{w}(T)]^m$, the proof that $\boldsymbol{w}(T) \leq 1$ implies $\boldsymbol{w}(T^m) \leq 1$ is trivial. Conversely, assume that $T \neq 0$ and let $S = \frac{T}{\boldsymbol{w}(T)}$, then $\boldsymbol{w}(S) = 1$, so $\boldsymbol{w}(S^m) \leq 1$, that is, $\boldsymbol{w}(T^m) \leq [\boldsymbol{w}(T)]^m$. The proof is completed. □

The proof of Theorem 5.3.7: According to Lemma 5.3.8, we only need to prove that $\boldsymbol{w}(T) \leq 1$ implies $\boldsymbol{w}(T^m) \leq 1$ for any positive integers m. Let $\boldsymbol{w}_j = e^{\frac{2\pi \mathrm{i}}{m} j}$, $j = 1, 2, \ldots, m$, be the m roots of the polynomial $z^m - 1 = 0$ of degree m. Then $\overline{\boldsymbol{w}_j} = \boldsymbol{w}_j^{-1}$ and $(-1)^m \prod_{j=1}^m \overline{\boldsymbol{w}_j} = -1$. Thus, we have

$$\begin{aligned} 1 - z^m &= -\prod_{j=1}^{m}(z - \boldsymbol{w}_j) \\ &= (-1)^m \prod_{j=1}^{m} \overline{\boldsymbol{w}_j} \prod_{j=1}^{m}(z - \boldsymbol{w}_j) \\ &= \prod_{j=1}^{m}(1 - \boldsymbol{w}_j z), \end{aligned}$$

$$1 = \frac{1}{m} \sum_{j=1}^{m} \prod_{k=1 k \neq j}^{m} (1 - \boldsymbol{w}_k z).$$

Obviously, the above two equations also hold even if z is substituted by any bounded linear operator S, that is,

$$I - S^m = \prod_{j=1}^{m}(I - \boldsymbol{w}_j S), \tag{5.10}$$

$$I = \frac{1}{m}\sum_{j=1}^{m}\prod_{k=1k\neq j}^{m}(I - \boldsymbol{w}_k S). \tag{5.11}$$

For any unit vectors $x \in X$, setting $x_j = [\prod_{k=1k\neq j}^{m}(I - \boldsymbol{w}_k S)]x$, then

$$\sum_{j=1}^{m} x_j = mx \tag{5.12}$$

and

$$\begin{aligned}\frac{1}{m}\sum_{j=1}^{m}\|x_j\|^2\left[1 - \boldsymbol{w}_j\left(\frac{Sx_j}{\|x_j\|}, \frac{x_j}{\|x_j\|}\right)\right] &= \frac{1}{m}\sum_{j=1}^{m}((I - \boldsymbol{w}_j S)x_j, x_j)\\ &= \frac{1}{m}\sum_{j=1}^{m}\left(\left[\prod_{k=1}^{m}(I - \boldsymbol{w}_k S)\right]x, x_j\right)\\ &= \frac{1}{m}\sum_{j=1}^{m}((I - S^m)x, x_j)\\ &= \left((I - S^m)x, \frac{1}{m}\sum_{j=1}^{m} x_j\right)\\ &= 1 - (S^m x, x).\end{aligned}$$

Take $S = e^{\mathrm{i}\theta}T$, where θ is an arbitrary real number, then it is attained from the above formula

$$\frac{1}{m}\sum_{j=1}^{m}\|x_j\|^2\left[1 - \boldsymbol{w}_j e^{\mathrm{i}\theta}\left(\frac{Tx_j}{\|x_j\|}, \frac{x_j}{\|x_j\|}\right)\right] = 1 - e^{\mathrm{i}m\theta}(T^m x, x).$$

According to the hypothesis, $\boldsymbol{w}(T) \leq 1$, thus

$$Re\left[1 - \boldsymbol{w}_j e^{\mathrm{i}\theta}\left(\frac{Tx_j}{\|x_j\|}, \frac{x_j}{\|x_j\|}\right)\right] \geq 0,$$

so

$$Re[1 - e^{\mathrm{i}m\theta}(T^m x, x)] \geq 0.$$

Because θ is an arbitrary real number,

$$1 - |(T^m x, x)| \geq 0,$$

that is, $\boldsymbol{w}(T^m) \leq 1$. The proof is completed.

According to the power inequality, the following corollary is obvious.

Proposition 5.3.9 *Let T be a bounded linear operator in Hilbert space X. If $\boldsymbol{w}(T) \leq 1$, then for any $m = 1, 2, \ldots$, we have $\|T^m\| \leq 2$.*

Definition 5.3.10 For a bounded linear operator, T is called a spectraloid operator if $\boldsymbol{w}(T) = r(T)$ is satisfied.

Theorem 5.3.11 *Let T be a bounded linear operator in Hilbert space X. Then T is a spectraloid operator if and only if for any positive integers n there is $\boldsymbol{w}(T^n) = [\boldsymbol{w}(T)]^n$.*

Proof When T is a spectraloid operator, for any positive integers n there is

$$[\boldsymbol{w}(T)]^n = [r(T)]^n = r(T^n) \leq \boldsymbol{w}(T^n).$$

Then according to Theorem 5.3.7, it can be attained that $\boldsymbol{w}(T^n) = [\boldsymbol{w}(T)]^n$.

Conversely, when there is $\boldsymbol{w}(T^n) = [\boldsymbol{w}(T)]^n$ for any positive integers n,

$$[\boldsymbol{w}(T)]^n = \boldsymbol{w}(T^n) \leq \|T^n\|.$$

Thus, $\boldsymbol{w}(T) \leq \|T^n\|^{\frac{1}{n}}$. Considering that $r(T) = \lim_{n\to\infty} \|T^n\|^{\frac{1}{n}}$, we can attain $\boldsymbol{w}(T) \leq r(T)$. Following from property Corollary 5.3.6 that $\boldsymbol{w}(T) = r(T)$. The proof is completed. □

Proposition 5.3.12 *If T is a spectraloid operator, then T^n is also a spectraloid operator for any positive integers n.*

Proof If T is a spectraloid operator, then $\boldsymbol{w}(T) = r(T)$, that is, $[\boldsymbol{w}(T)]^n = [r(T)]^n = r(T^n)$. Furthermore, $\boldsymbol{w}(T^n) = r(T^n)$ is attained by Theorem 5.3.7, that is, T^n is also a spectraloid operator. □

Remark 5.3.13 In general, there is no necessary connection between the numerical range $W(T)$ and $W(T^n)$ of the bounded linear operators T and T^n. For example,

$$T = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix},$$

$W(T)$ is the line segment [−i, i] on the imaginary axis. However, $T^2 = \begin{bmatrix} -I & 0 \\ 0 & -I \end{bmatrix}$, so $W(T^2) = \{-1\}$.

Definition 5.3.14 For a bounded linear operator, if it satisfies $\overline{W(T)} = Conv(\sigma(T))$, then T is called a convexoid operator .

Theorem 5.3.15 *Let T be a bounded linear operator in Hilbert space X. Then T is a convexoid operator if and only if for any $\lambda \in \mathbb{C}$, there is $T - \lambda I$ is a spectraloid operator.*

Proof When T is a convexoid operator, $r(T) = \boldsymbol{w}(T)$, T is a spectraloid operator, so for any $\lambda \in \mathbb{C}$, $T - \lambda I$ is a spectraloid operator.

Conversely, because the convex compact subset M of the complex plane $\mathbb{C}$ can be written as the intersection of all closed disks containing M, then

$$\begin{aligned}\overline{W(T)} &= \bigcap_{\alpha}\{\lambda : |\lambda - \alpha| \leq \sup_{\mu \in \overline{W(T)}} |\mu - \alpha|\} \\ &= \bigcap_{\alpha}\{\lambda : |\lambda - \alpha| \leq \boldsymbol{w}(T - \alpha I)\}.\end{aligned}$$

Similarly,

$$Conv(\sigma(T)) = \bigcap_{\alpha}\{\lambda : |\lambda - \alpha| \leq r(T - \alpha I)\}.$$

Since there is $\boldsymbol{w}(T - \alpha I) = r(T - \alpha I)$ for any $\alpha \in \mathbb{C}$, $\overline{W(T)} = Conv(\sigma(T))$, T is a convexoid operator. □

Definition 5.3.16 For a bounded linear operator, if it satisfies $\boldsymbol{w}(T) = \|T\|$ (or equivalently $r(T) = \|T\|$), then T is called a normaloid operator .

Remark 5.3.17 By comparing the definitions of the normaloid operator, spectraloid operator, and convexoid operator, it is easy to find that the normaloid operator must be a spectraloid operator and the convexoid operator is also a spectraloid operator. Otherwise, it is not true in general.

Example 5.3.18 Normaloid operator is not necessarily a convexoid operator. Take $X = \mathbb{C}^3$,

$$T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

then for any $x = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^T$, $|x_1|^2 + |x_2|^2 + |x_3|^2 = 1$, there is $Tx = \begin{bmatrix} x_2 & 0 & x_3 \end{bmatrix}^T$, so

$$\|T\|^2 = \sup_{\|x\|=1} (|x_2|^2 + |x_3|^2) = 1.$$

On the other hand, $(Tx, x) = x_2\overline{x_1} + |x_3|^2$, so

$$\boldsymbol{w}(T) = \sup_{\|x\|=1} |x_2\overline{x_1} + |x_3|^2| = 1.$$

Thus, T is a normaloid operator. However, T is not a convexoid operator. In fact,

$$T = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \oplus [1],$$

so

$$\overline{W(T)} = Conv(S_{\frac{1}{2}} \cup \{1\}),$$

where $S_{\frac{1}{2}} = \{\lambda : |\lambda| \leq \frac{1}{2}\}$. On the other hand, considering that $\sigma(T) = \{0, 1\}$, that is,

$$Conv(\sigma(T)) = [0, 1].$$

Hence, T is not a convexoid operator.

Example 5.3.19 Convexoid operator is not necessarily a normaloid operator. Let $\{x_1, x_2, \ldots\}$ be the orthogonal basis of the space $X = \ell^2$. Define

$$z_n = x_{2n+1},\ n = 0, 1, 2, \ldots,$$
$$z_{-n} = x_{2n},\ n = 1, 2, \ldots,$$

then each $x \in X$ can be written as

$$x = \sum_{-\infty}^{+\infty} \alpha_k z_k.$$

Define the linear operator S as

$$Sx = \frac{1}{2} \sum_{-\infty}^{+\infty} \alpha_k z_{k+1},$$

then it is easy to attain by calculation that $W(S) = \{\lambda \in \mathbb{C} : |\lambda| \leq \frac{1}{2}\}$. Again, define linear operators T as

$$T = S \oplus L,$$

where $L = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, then

$$\overline{W(T)} = Conv(\sigma(T)),$$

that is, T is a convexoid operator. However, T is not a normaloid operator. In fact, $\|T\| = 1$, $w(T) = \frac{1}{2}$.

Example 5.3.20 Spectraloid operator is not necessarily a convexoid operator. Set $X = \mathbb{C}^3$,

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

then $\sigma(T) = \{0, 1\}$ and

$$(Tx, x) = |x_1|^2 + x_2\overline{x_3},$$

thus, $r(T) = \boldsymbol{w}(T) = 1$, T is a spectraloid operator. However,

$$\begin{aligned} Conv(\sigma(T)) &= [0, 1], \\ \overline{W(T)} &= Conv(S_{\frac{1}{2}} \cup \{1\}), \end{aligned}$$

where $S_{\frac{1}{2}} = \{\lambda : |\lambda| \leq \frac{1}{2}\}$. Hence, T is not a convexoid operator.

Example 5.3.21 Spectraloid operator is not necessarily a normaloid operator. Set $X = \mathbb{C}^3$,

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix},$$

then $\sigma(T) = \{0, 1\}$ and

$$(Tx, x) = |x_1|^2 + 2x_2\overline{x_3},$$

thus, $r(T) = \boldsymbol{w}(T) = 1$, T is a spectraloid operator. However, $Tx = \begin{bmatrix} x_2 & 0 & 2x_2 \end{bmatrix}^T$, so

$$\|T\|^2 = \sup_{\|x\|=1} (|x_2|^2 + 4|x_2|^2) = 4.$$

Thus, $\|T\| = 2$, T is not a normaloid operator.

The following conclusions are about the discriminant methods of convexoid operators.

Theorem 5.3.22 *Let T be a bounded linear operator in Hilbert space X. If for any $\lambda \notin Conv(\sigma(T))$, we have*

$$\|(T - \lambda I)^{-1}\| \leq \frac{1}{dist(\lambda, Conv(\sigma(T)))},$$

then T is a convexoid operator.

Proof Let $Conv(\sigma(T))$ be located in the left half-plane and supported by the imaginary axis. To prove $\overline{W(T)} \subset Conv(\sigma(T))$, we only need to prove that $W(T)$ is also located in the left half-plane. According to the given conditions, for any $\lambda > 0$, there is

$$\|(T - \lambda I)^{-1}\| \leq \frac{1}{\lambda}.$$

Take $x = (T - \lambda I)y$, $\|x\| = 1$, then

$$\lambda\|y\| \leq \|(T - \lambda I)y\|.$$

And then it can be attained that

$$2\lambda Re(Ty, y) \leq \|y\|^2.$$

Considering the arbitrariness of $\lambda > 0$, that is, $Re(Ty, y) \leq 0$, $W(T)$ is also located in the left half-plane, so T is a convexoid operator. □

The equivalent form of property Corollary 5.3.6 is $\frac{\|T\|}{2} \leq \boldsymbol{w}(T) \leq \|T\|$. Recently, Kittaneh [71] attained the generalization of this inequality.

Theorem 5.3.23 *Let T be a bounded linear operator in Hilbert space X. Then the inequality*

$$\frac{1}{4}\|T^*T + TT^*\| \leq [\boldsymbol{w}(T)]^2 \leq \frac{1}{2}\|T^*T + TT^*\|$$

holds.

Proof Set $T = \frac{T+T^*}{2} + \mathtt{i}\frac{T-T^*}{2\mathtt{i}} = A + \mathtt{i}B$, then

$$\begin{aligned} |(Tx, x)|^2 =& (Ax, x)^2 + (Bx, x)^2 \\ \geq& \frac{1}{2}(|(Ax, x)| + |(Bx, x)|)^2 \\ \geq& \frac{1}{2}((A \pm B)x, x)^2. \end{aligned}$$

Therefore,

$$\begin{aligned} [\boldsymbol{w}(T)]^2 =& \sup\{|(Tx, x)|^2 : \|x\| = 1\} \\ \geq& \frac{1}{2}\sup\{((A \pm B)x, x)^2 : \|x\| = 1\} \\ =& \frac{1}{2}\|A \pm B\|^2 \\ \geq& \frac{1}{2}\|(A \pm B)^2\|. \end{aligned}$$

Hence,

$$\begin{aligned} [\boldsymbol{w}(T)]^2 \geq& \frac{1}{4}(\|(A + B)^2\| + \|(A - B)^2\|) \\ \geq& \frac{1}{4}\|(A + B)^2 + \|(A - B)^2\|\| \\ =& \frac{1}{4}\|T^*T + TT^*\|. \end{aligned}$$

On the other hand, for any $\|x\| = 1$, there is

$$\begin{aligned}|(Tx, x)|^2 =&(Ax, x)^2 + (Bx, x)^2\\ \le&\|Ax\|^2 + \|Bx\|^2\\ =&(A^2x, x) + (B^2x, x)\\ =&((A^2 + B^2)x, x).\end{aligned}$$

Hence,

$$\begin{aligned}[\boldsymbol{w}(T)]^2 =& \sup\{|(Tx, x)|^2 : \|x\| = 1\}\\ \le& \sup\{((A^2 + B^2)x, x) : \|x\| = 1\}\\ =&\|A^2 + B^2\|\\ =&\frac{1}{2}\|T^*T + TT^*\|.\end{aligned}$$

The proof is completed. □

Remark 5.3.24 Theorem 5.3.23 is a further generalization of property Corollary 5.3.6, because

$$\begin{aligned}\|T\| \le&\|T^*T + TT^*\|^{\frac{1}{2}},\\ \|T\| \ge&\frac{1}{\sqrt{2}}\|T^*T + TT^*\|^{\frac{1}{2}},\end{aligned}$$

that is,

$$\frac{\|T\|}{2} \le \frac{1}{2}\|T^*T + TT^*\|^{\frac{1}{2}} \le \boldsymbol{w}(T) \le \frac{1}{\sqrt{2}}\|T^*T + TT^*\|^{\frac{1}{2}} \le \|T\|.$$

To attain a further generalization of the relation $\boldsymbol{w}(T) \le \|T\|$, the following lemma should be first given.

Lemma 5.3.25 (Mixed Schwarz Inequality) *Let T be a bounded linear operator in Hilbert space X, $|T| = (T^*T)^{\frac{1}{2}}$ and $|T^*| = (TT^*)^{\frac{1}{2}}$. Then the inequality*

$$|(Tx, y)|^2 \le (|T|x, x)(|T^*|y, y)$$

holds.

Proof According to the properties of polar decomposition, $T = U|T|$, where U is a isometric operator from $\overline{\mathscr{R}(|T|)}$ to $\overline{\mathscr{R}(|T|)}$, and there is

$$\begin{aligned}|(Tx, y)|^2 =&|(U|T|x, y)|^2\\ =&|(|T|^{\frac{1}{2}}x, |T|^{\frac{1}{2}}U^*y)|^2\\ \leq&(|T|^{\frac{1}{2}}x, |T|^{\frac{1}{2}}x)(|T|^{\frac{1}{2}}U^*y, |T|^{\frac{1}{2}}U^*y)\\ =&(|T|x, x)(U|T|U^*y, y)\\ =&(|T|x, x)(|T^*|y, y).\end{aligned}$$

The relation $|T^*| = U|T|U^*$ is used here, and its proof is obvious. In fact, it is known from $T = U|T|$, $T^* = |T|U^*$, and hence

$$[U|T|U^*]^2 = U|T|U^*U|T|U^* = U|T||T|U^* = TT^* = |T^*|^2.$$

Then, $|T^*| = U|T|U^*$ can be known from the uniqueness of the square root operator. □

Lemma 5.3.26 *Let T, S be bounded linear operators in Hilbert space X. If TS are self-adjoint operators, then the inequality*

$$\|Re(ST)\| \geq \|TS\|$$

holds, where $Re(ST) = \frac{1}{2}(ST + (ST)^*)$.

Proof When A is a bounded linear operator in Hilbert space X, letting $Re(\sigma(A)) = \sup\{|Re(\lambda)| : \lambda \in \sigma(A)\}$, then it is easy to prove

$$\|Re(A)\| \geq Re(\sigma(A)).$$

In fact, let $\gamma = Re(\sigma(A))$, then there exists $\lambda \in \sigma(A)$ such that $|Re(\lambda)| = \gamma$. From the spectral inclusion relation of the numerical range, it is known that there exists $\{\lambda_n\} \subset \overline{W(A)}$ and $\{x_n\}_{n=1}^{\infty}$, $\|x_n\| = 1$, $n = 1, 2, \ldots$, such that $(Ax_n, x_n) = \lambda_n$ and when $n \to \infty$, $|\lambda_n| \to |\lambda|$. Then considering that $Re(A)$ are self-adjoint operators, the operator norm is equal to the numerical radius, so

$$\begin{aligned}\|Re(A)\| =& \sup_{\|x\|=1} \frac{1}{2}|(Ax, x) + (x, Ax)|\\ \geq&\frac{1}{2}|(Ax_n, x_n) + (x_n, Ax_n)|\\ =&|Re(\lambda_n)| \to |Re(\lambda)|.\end{aligned}$$

Therefore, $\|Re(A)\| \geq \gamma$ is attained from the order preservation of the limit, that is, $\|Re(A)\| \geq Re(\sigma(A))$. This conclusion is applied to bounded operators ST, and reconsider $\sigma(ST)\backslash\{0\} = \sigma(TS)\backslash\{0\}$ and the self-adjointness of TS it is attained that

$$\begin{aligned}\|Re(ST)\| \geq & Re(\sigma(ST)) \\ = & Re(\sigma(TS)) \\ = & r(TS) = w(TS) \\ = & \|TS\|.\end{aligned}$$

The proof is completed. □

Lemma 5.3.27 *Let T, S be non-negative bounded linear operators in Hilbert space X. Then for any $t \geq 0$ and $s \geq 0$, the inequality*

$$\|T^{\frac{t+s}{2}} S^{\frac{t+s}{2}}\| \leq \|T^{t+s} S^{t+s}\|^{\frac{1}{2}}$$

holds. In particular, if $t = s = \frac{1}{2}$, then $\|T^{\frac{1}{2}} S^{\frac{1}{2}}\| \leq \|TS\|^{\frac{1}{2}}$.

Proof In fact,

$$\begin{aligned}\|T^{\frac{t+s}{2}} S^{\frac{t+s}{2}}\|^2 = & \sup_{\|x\|=1} \|T^{\frac{t+s}{2}} S^{\frac{t+s}{2}} x\|^2 \\ = & \sup_{\|x\|=1} (S^{\frac{t+s}{2}} T^{t+s} S^{\frac{t+s}{2}} x, x) \\ = & \|S^{\frac{t+s}{2}} T^{t+s} S^{\frac{t+s}{2}}\| \\ = & r(S^{\frac{t+s}{2}} T^{t+s} S^{\frac{t+s}{2}}) \\ = & r(T^{t+s} S^{\frac{t+s}{2}} S^{\frac{t+s}{2}}) \leq \|T^{t+s} S^{t+s}\|.\end{aligned}$$

The proof is completed. □

Lemma 5.3.28 *Let T, S be non-negative bounded linear operators in Hilbert space X, and M, N be bounded self-adjoint operators. Then the inequality*

$$\begin{aligned}&\|T^{\frac{1}{2}} M T^{\frac{1}{2}} + S^{\frac{1}{2}} N S^{\frac{1}{2}}\| \\ \leq & \frac{1}{4}(\|TM + MT\| + \|SN + NS\|) \\ + & \frac{1}{4}\left(\sqrt{(\|TM + MT\| - \|SN + NS\|)^2 + 4\|T^{\frac{1}{2}} S^{\frac{1}{2}} N + M T^{\frac{1}{2}} S^{\frac{1}{2}}\|^2}\right)\end{aligned}$$

holds. In particular, when $M = N = I$,

$$\|T + S\| \leq \frac{1}{2}((\|T\| + \|S\|) + \sqrt{(\|T\| - \|S\|)^2} + 4\|T^{\frac{1}{2}} S^{\frac{1}{2}}\|^2).$$

Proof

$$L = \begin{bmatrix} T^{\frac{1}{2}} & S^{\frac{1}{2}} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & T^{\frac{1}{2}} & S^{\frac{1}{2}} \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & 0 & M & 0 \\ 0 & 0 & 0 & N \\ M & 0 & 0 & 0 \\ 0 & N & 0 & 0 \end{bmatrix},$$

then

$$LRL^* = \begin{bmatrix} 0 & 0 & T^{\frac{1}{2}}MT^{\frac{1}{2}} + S^{\frac{1}{2}}NS^{\frac{1}{2}} & 0 \\ 0 & 0 & 0 & 0 \\ T^{\frac{1}{2}}MT^{\frac{1}{2}} + S^{\frac{1}{2}}NS^{\frac{1}{2}} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$L^*LR = \begin{bmatrix} 0 & 0 & TM & T^{\frac{1}{2}}S^{\frac{1}{2}}N \\ 0 & 0 & S^{\frac{1}{2}}T^{\frac{1}{2}}M & SN \\ TM & T^{\frac{1}{2}}S^{\frac{1}{2}}N & 0 & 0 \\ S^{\frac{1}{2}}T^{\frac{1}{2}}M & SN & 0 & 0 \end{bmatrix},$$

and

$$Re(L^*LR) = \frac{1}{2}\begin{bmatrix} 0 & K \\ K & 0 \end{bmatrix},$$

where $K = \begin{bmatrix} TM + MT & T^{\frac{1}{2}}S^{\frac{1}{2}}N + MT^{\frac{1}{2}}S^{\frac{1}{2}} \\ S^{\frac{1}{2}}T^{\frac{1}{2}}M + NS^{\frac{1}{2}}T^{\frac{1}{2}} & SN + NS \end{bmatrix}$. Because LRL^* are self-adjoint operators, it is known by Lemma 5.3.26 that

$$\|Re(L^*LR)\| \geq \|L^*RL\|,$$

that is,

$$\left\| \begin{bmatrix} TM + MT & T^{\frac{1}{2}}S^{\frac{1}{2}}N + MT^{\frac{1}{2}}S^{\frac{1}{2}} \\ S^{\frac{1}{2}}T^{\frac{1}{2}}M + NS^{\frac{1}{2}}T^{\frac{1}{2}} & SN + NS \end{bmatrix} \right\| \geq 2\|T^{\frac{1}{2}}MT^{\frac{1}{2}} + S^{\frac{1}{2}}NS^{\frac{1}{2}}\|. \quad (5.13)$$

Note that

$$\left\| \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \right\| \leq \left\| \begin{bmatrix} \|A_{11}\| & \|A_{21}\| \\ \|A_{21}\| & \|A_{22}\| \end{bmatrix} \right\|.$$

In fact,

$$\begin{aligned} \left\| \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \right\|^2 &= \sup_{\|x\|=1} \left(\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}^* \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} x, x \right) \\ &= \sup_{\|x_1\|^2+\|x_2\|^2=1} \Big| \sum_{i,j,k=1}^{2} (A_{k,j}x_j, A_{k,i}x_i) \Big| \\ &\leq \sup_{\|x_1\|^2+\|x_2\|^2=1} \Big| \sum_{i,j,k=1}^{2} \|A_{k,j}\|\|x_j\|\|A_{k,i}\|\|x_i\| \Big| \\ &= \sup_{\|x\|=1} \left(\begin{bmatrix} \|A_{11}\| & \|A_{21}\| \\ \|A_{21}\| & \|A_{22}\| \end{bmatrix}^* \begin{bmatrix} \|A_{11}\| & \|A_{21}\| \\ \|A_{21}\| & \|A_{22}\| \end{bmatrix} \begin{bmatrix} \|x_1\| \\ \|x_2\| \end{bmatrix}, \begin{bmatrix} \|x_1\| \\ \|x_2\| \end{bmatrix} \right) \\ &= \left\| \begin{bmatrix} \|A_{11}\| & \|A_{21}\| \\ \|A_{21}\| & \|A_{22}\| \end{bmatrix} \right\|^2. \end{aligned}$$

Moreover, it is known from (5.13) that

$$\left\| \begin{bmatrix} \|TM+MT\| & \|T^{\frac{1}{2}}S^{\frac{1}{2}}N+MT^{\frac{1}{2}}S^{\frac{1}{2}}\| \\ \|S^{\frac{1}{2}}T^{\frac{1}{2}}M+NS^{\frac{1}{2}}T^{\frac{1}{2}}\| & \|SN+NS\| \end{bmatrix} \right\| \geq 2 \left\| T^{\frac{1}{2}}MT^{\frac{1}{2}}+S^{\frac{1}{2}}NS^{\frac{1}{2}} \right\|.$$

Considering that M, N are self-adjoint operators, we can obtain

$$(T^{\frac{1}{2}}S^{\frac{1}{2}}N+MT^{\frac{1}{2}}S^{\frac{1}{2}})^* = S^{\frac{1}{2}}T^{\frac{1}{2}}M+NS^{\frac{1}{2}}T^{\frac{1}{2}},$$

so the matrix $\mathscr{A} = \begin{bmatrix} \|TM+MT\| & \|T^{\frac{1}{2}}S^{\frac{1}{2}}N+MT^{\frac{1}{2}}S^{\frac{1}{2}}\| \\ \|S^{\frac{1}{2}}T^{\frac{1}{2}}M+NS^{\frac{1}{2}}T^{\frac{1}{2}}\| & \|SN+NS\| \end{bmatrix}$ is a symmetric matrix, so

$$\|\mathscr{A}\| = r(\mathscr{A}),$$

where $r(\mathscr{A})$ represents the spectral radius (the maximum eigenvalue) of $\mathscr{A}$ and

$$\begin{aligned} r(\mathscr{A}) = \frac{1}{2}[&\|TM+MT\| + \|SN+NS\| \\ &+ \sqrt{(\|TM+MT\| - \|SN+NS\|)^2 + 4\|T^{\frac{1}{2}}S^{\frac{1}{2}}N+MT^{\frac{1}{2}}S^{\frac{1}{2}}\|^2}]. \end{aligned}$$

Thus, the conclusion is proved. □

Theorem 5.3.29 *Let T be a bounded linear operator in Hilbert space X. Then the inequality*

$$w(T) \leq \frac{1}{2}(\|T\| + \|T^2\|^{\frac{1}{2}})$$

holds.

Proof It is known by the Mixed Schwarz Inequality (Lemma 5.3.25) that

$$\begin{aligned} w(T) &= \sup_{\|x\|=1} |(Tx,x)| \\ &\leq \sup_{\|x\|=1} \sqrt{(|T|x,x)(|T^*|x,x)} \\ &\leq \sup_{\|x\|=1} \frac{1}{2}((|T|+|T^*|)x,x) \\ &\leq \frac{1}{2}\||T|+|T^*|\|. \end{aligned}$$

It is known by Lemmas 5.3.27, 5.3.28, and $\|T\| = \||T|\| = \||T^*|\|$ that

$$\begin{aligned} \||T|+|T^*|\| &\leq \frac{1}{2}(\||T|\| + \||T^*|\|) + \sqrt{(\||T|\| - \||T^*|\|)^2 + 4\||T|^{\frac{1}{2}}|T^*|^{\frac{1}{2}}\|^2)} \\ &\leq \|T\| + \|T^2\|^{\frac{1}{2}}. \end{aligned}$$

Thus, the conclusion is proved. □

Remark 5.3.30 Theorem 5.3.29 was first attained by Kittaneh [72], which generalizes $\boldsymbol{w}(T) \leq \|T\|$. In fact, it is easy to derive $\boldsymbol{w}(T) \leq \|T\|$ from the inequality $\boldsymbol{w}(T) \leq \frac{1}{2}(\|T\| + \|T^2\|^{\frac{1}{2}})$.

Proposition 5.3.31 *Let T be a bounded linear operator in Hilbert space X and satisfy $T^2 = 0$. Then we have $\boldsymbol{w}(T) = \frac{1}{2}\|T\|$.*

Proof When $T^2 = 0$, it is known by Theorem 5.3.29 that

$$\boldsymbol{w}(T) \leq \frac{1}{2}\|T\|.$$

Combining that $\frac{1}{2}\|T\| \leq \boldsymbol{w}(T)$, the conclusion can be proved. □

Proposition 5.3.32 *Let T be a bounded linear operator in Hilbert space X and satisfy $\boldsymbol{w}(T) = \|T\|$. Then we have $\|T^2\| = \|T\|^2$.*

Proof When $\boldsymbol{w}(T) = \|T\|$, it is known by Theorem 5.3.29 that

$$\|T\| \leq \|T^2\|^{\frac{1}{2}}.$$

Considering that $\|T^2\| \leq \|T\|^2$, the conclusion can be proved. □

5.3.2 *Inequality of Numerical Radius of Infinite-Dimensional Hamiltonian Operators*

In this section, we will discuss the properties of the numerical radius of bounded infinite-dimensional Hamiltonian operators.

Theorem 5.3.33 *Let $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ be a bounded infinite-dimensional Hamiltonian operator. Then*

$$\boldsymbol{w}(H) \geq \boldsymbol{w}\left(\begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix}\right)$$

and

$$\boldsymbol{w}(H) \geq \boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)$$

hold.

Proof Considering $W(A) \subset W(H)$, the conclusion $\boldsymbol{w}(H) \geq \boldsymbol{w}\left(\begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix}\right)$ is trivial. Because

$$\begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} = \begin{bmatrix} A & -B \\ -C & -A^* \end{bmatrix},$$

thus

$$\boldsymbol{w}\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) = \boldsymbol{w}\left(\begin{bmatrix} -A & B \\ C & A^* \end{bmatrix}\right).$$

From the triangle inequality of the numerical radius, it can be known that

$$\begin{aligned} \boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) &= \frac{1}{2}\boldsymbol{w}\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} + \begin{bmatrix} -A & B \\ C & A^* \end{bmatrix}\right) \\ &\leq \frac{1}{2}\boldsymbol{w}\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) + \frac{1}{2}\boldsymbol{w}\left(\begin{bmatrix} -A & B \\ C & A^* \end{bmatrix}\right) \\ &= \boldsymbol{w}\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right). \end{aligned}$$

The proof is completed. □

The following is an estimate of the numerical radius of skew-diagonal bounded infinite-dimensional Hamiltonian operators.

Theorem 5.3.34 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal bounded infinite-dimensional Hamiltonian operator. Then for any* $\theta \in R$, *we have*

$$\boldsymbol{w}(H) \geq \frac{\max\{\boldsymbol{w}(B + e^{i\theta}C), \boldsymbol{w}(B - e^{i\theta}C\}}{2} \tag{5.14}$$

and

$$\boldsymbol{w}(H) \leq \frac{\boldsymbol{w}(B + e^{i\theta}C) + \boldsymbol{w}(B - e^{i\theta}C)}{2}. \tag{5.15}$$

Proof Firstly, we prove Inequality (5.14). Considering that

$$\begin{aligned} \boldsymbol{w}(B + e^{\mathrm{i}\theta}C) =& \boldsymbol{w}\left(\begin{bmatrix} 0 & B + e^{\mathrm{i}\theta}C \\ B + e^{\mathrm{i}\theta}C & 0 \end{bmatrix}\right) \\ =& \boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ e^{\mathrm{i}\theta}C & 0 \end{bmatrix} + \begin{bmatrix} 0 & e^{\mathrm{i}\theta}C \\ B & 0 \end{bmatrix}\right) \\ \leq& \boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ e^{\mathrm{i}\theta}C & 0 \end{bmatrix}\right) + \boldsymbol{w}\left(\begin{bmatrix} 0 & e^{\mathrm{i}\theta}C \\ B & 0 \end{bmatrix}\right) \\ =& 2\boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ e^{\mathrm{i}\theta}C & 0 \end{bmatrix}\right) = 2\boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right). \end{aligned}$$

Similarly, $e^{\mathrm{i}\theta}C$ is replaced by $-e^{\mathrm{i}\theta}C$ to attain

$$w(B - e^{\mathrm{i}\theta}C) \leq 2w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right).$$

Thus, $w(H) \geq \frac{\max\{w(B+e^{\mathrm{i}\theta}C), w(B-e^{\mathrm{i}\theta}C\}}{2}$ holds.

Secondly, we prove Inequality (5.15). Select unitary operator $U = \frac{1}{\sqrt{2}}\begin{bmatrix} I & -I \\ I & I \end{bmatrix}$, then

$$\begin{aligned} w\left(\begin{bmatrix} 0 & B \\ e^{\mathrm{i}\theta}C & 0 \end{bmatrix}\right) &= w\left(U^*\begin{bmatrix} 0 & B \\ e^{\mathrm{i}\theta}C & 0 \end{bmatrix}U\right) \\ &= \frac{1}{2}w\left(\begin{bmatrix} B + e^{\mathrm{i}\theta}C & B - e^{\mathrm{i}\theta}C \\ -(B - e^{\mathrm{i}\theta}C) & -(B + e^{\mathrm{i}\theta}C) \end{bmatrix}\right) \\ &= \frac{1}{2}w\left(\begin{bmatrix} B + e^{\mathrm{i}\theta}C & 0 \\ 0 & -(B + e^{\mathrm{i}\theta}C) \end{bmatrix} + \begin{bmatrix} 0 & B - e^{\mathrm{i}\theta}C \\ -(B - e^{\mathrm{i}\theta}C) & 0 \end{bmatrix}\right) \\ &\leq \frac{1}{2}w\left(\begin{bmatrix} B + e^{\mathrm{i}\theta}C & 0 \\ 0 & -(B + e^{\mathrm{i}\theta}C) \end{bmatrix}\right) + \frac{1}{2}w\left(\begin{bmatrix} 0 & B - e^{\mathrm{i}\theta}C \\ -(B - e^{\mathrm{i}\theta}C) & 0 \end{bmatrix}\right) \\ &= \frac{1}{2}w(B + e^{\mathrm{i}\theta}C + \frac{1}{2}w(B - e^{\mathrm{i}\theta}C). \end{aligned}$$

The proof is completed. □

Proposition 5.3.35 *Let T be a bounded linear operator in Hilbert space X and let $T = Re(T) + \mathrm{i}Im(T)$. Then the relation*

$$\frac{w(T)}{2} \leq w\left(\begin{bmatrix} 0 & Re(T) \\ Im(T) & 0 \end{bmatrix}\right) \leq w(T)$$

holds.

Proof It can be known from Theorem 5.3.34 that

$$\max\{\frac{w(Re(T) + \mathrm{i}Im(T))}{2}, \frac{w(Re(T) - \mathrm{i}Im(T))}{2}\} \leq w\left(\begin{bmatrix} 0 & Re(T) \\ Im(T) & 0 \end{bmatrix}\right)$$

and

$$w\left(\begin{bmatrix} 0 & Re(T) \\ Im(T) & 0 \end{bmatrix}\right) \leq \frac{w(Re(T) + \mathrm{i}Im(T)) + w(Re(T) - \mathrm{i}Im(T))}{2}.$$

Considering that $Re(T) - \mathrm{i}Im(T) = T^*$, we have $w(Re(T) + \mathrm{i}Im(T)) = w(Re(T) - \mathrm{i}Im(T))$. Thus, the conclusion holds. □

In the following, we will give another estimation of the numerical radius of skew-diagonal bounded infinite-dimensional Hamiltonian operators.

Theorem 5.3.36 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a skew-diagonal bounded infinite-dimensional Hamiltonian operator. Then*

$$w(H) \geq |\frac{1}{2}\max\{w(B+C), w(B-C)\} - \min\{w(B), w(C)\}| \tag{5.16}$$

and

$$w(H) \leq \min\{w(B), w(C)\} + \frac{1}{2}\min\{w(B+C), w(B-C)\}. \tag{5.17}$$

Proof Considering that B, C are bounded self-adjoint operators, $B+C$ are also self-adjoint operators and $\begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix}^2 = 0$. It is known by Corollary 5.3.32 that

$$\frac{1}{2}\left\| \begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix} \right\| = w\left(\begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix} \right) = \|B+C\| = w(B+C).$$

Select unitary operator $U = \frac{1}{\sqrt{2}} \begin{bmatrix} I & -I \\ I & I \end{bmatrix}$, then

$$\frac{1}{2} \begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix} = U^* \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} U - \begin{bmatrix} 0 & -C \\ C & 0 \end{bmatrix},$$

thus

$$\begin{aligned} \frac{1}{2}w(B+C) =& \frac{1}{2}w\left(\begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix} \right) \\ =& w\left(U^* \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} U - \begin{bmatrix} 0 & -C \\ C & 0 \end{bmatrix} \right) \\ \leq& w\left(U^* \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} U \right) + w\left(\begin{bmatrix} 0 & -C \\ C & 0 \end{bmatrix} \right) \\ =& w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \right) + w(C), \end{aligned}$$

that is,

$$\frac{1}{2}w(B+C) \leq w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \right) + w(C). \tag{5.18}$$

In the formula (5.18), replacing C with $-C$, and considering $w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \right) = w\left(\begin{bmatrix} 0 & B \\ -C & 0 \end{bmatrix} \right)$, it can be attained that

$$\frac{1}{2}w(B-C) \leq w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \right) + w(C). \tag{5.19}$$

It can be attained from Eqs. (5.18) and (5.19) that

$$\frac{1}{2}\max\{w(B-C), w(B+C)\} \le w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) + w(C). \tag{5.20}$$

Similarly, we can have

$$\frac{1}{2}\max\{w(B-C), w(B+C)\} \le w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) + w(B). \tag{5.21}$$

Thus, from Eqs. (5.20) and (5.21), we obtain

$$\frac{1}{2}\max\{w(B-C), w(B+C)\} - \min\{w(B), w(C)\} \le w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right). \tag{5.22}$$

Secondly, reconsider that

$$\begin{bmatrix} 0 & -C \\ C & 0 \end{bmatrix} = U^* \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} U - \frac{1}{2}\begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix},$$

it can be also attained similarly that

$$\min\{w(B), w(C)\} - \frac{1}{2}\max\{w(B-C), w(B+C)\} \le w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right). \tag{5.23}$$

From (5.22) and (5.23), the inequality (5.16) can be proved.

Let's prove the second inequality. Select unitary operator $U = \frac{1}{\sqrt{2}}\begin{bmatrix} I & -I \\ I & I \end{bmatrix}$, then

$$\begin{aligned}
w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) &= w\left(U^* \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} U\right) \\
&= \frac{1}{2} w\left(\begin{bmatrix} B+C & B+C \\ -(B-C) & -(B+C) \end{bmatrix}\right) \\
&= \frac{1}{2} w\left(\begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix} + \begin{bmatrix} 0 & -2C \\ 2C & 0 \end{bmatrix}\right) \\
&\le \frac{1}{2} w\left(\begin{bmatrix} B+C & B+C \\ -(B-C) & -(B+C) \end{bmatrix}\right) \\
&= \frac{1}{2} w\left(\begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix}\right) + \frac{1}{2} w\left(\begin{bmatrix} 0 & -2C \\ 2C & 0 \end{bmatrix}\right) \\
&= \frac{1}{2} w(B+C) + w(C).
\end{aligned}$$

Similarly, replacing C with $-C$, we obtain

$$w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) \leq w(B-C) + w(C).$$

Thus,

$$w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) \leq \frac{1}{2}\min\{w(B-C), w(B+C)\} + w(C).$$

Similarly, we can also have

$$w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) \leq \frac{1}{2}\min\{w(B-C), w(B+C)\} + w(B).$$

To sum up, it can be known that

$$w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) \leq \frac{1}{2}\min\{w(B-C), w(B+C)\} + \min\{w(C), w(B)\}.$$

The proof is completed. □

The following is an estimation of the numerical radius of the general bounded infinite-dimensional Hamiltonian operators. For bounded block operator matrices, it is easy to prove the Pinching inequality

$$w\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right) \geq \max\left\{w\left(\begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix}\right), w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)\right\}. \tag{5.24}$$

In fact, $w\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right) \geq w\left(\begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix}\right)$ is obvious.

The following is about the proof of the inequality

$$w\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right) \geq w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right).$$

Because

$$\begin{aligned} w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) &= w\left(\frac{1}{2}\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} + \frac{1}{2}\begin{bmatrix} -A & B \\ C & A^* \end{bmatrix}\right) \\ &\leq \frac{1}{2}w\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) + \frac{1}{2}w\left(\begin{bmatrix} -A & B \\ C & A^* \end{bmatrix}\right) \\ &= w\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right). \end{aligned}$$

Here, we use the relation $\boldsymbol{w}\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) = \boldsymbol{w}\left(\begin{bmatrix} -A & B \\ C & A^* \end{bmatrix}\right)$.

Considering Theorem 5.3.34 and the relation

$$\begin{aligned}\boldsymbol{w}\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) =& \boldsymbol{w}\left(\begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} + \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) \\ \leq& \boldsymbol{w}\left(\begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix}\right) + \boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right),\end{aligned}$$

the following conclusions are obvious.

Proposition 5.3.37 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be a bounded infinite-dimensional Hamiltonian operator. Then*

$$\boldsymbol{w}(H) \leq \max\{\boldsymbol{w}(A), \frac{1}{2}\boldsymbol{w}(B+C), \frac{1}{2}\boldsymbol{w}(B-C)\} \tag{5.25}$$

and

$$\boldsymbol{w}(H) \geq \boldsymbol{w}(A) + \frac{1}{2}(\boldsymbol{w}(B+C) + \boldsymbol{w}(B-C)). \tag{5.26}$$

Similarly, considering Theorem 5.3.36, it is easy to prove the following conclusions.

Theorem 5.3.38 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be a bounded infinite-dimensional Hamiltonian operator. Then*

$$\boldsymbol{w}(H) \leq \max\{\boldsymbol{w}(A), |\frac{1}{2}\max\{\boldsymbol{w}(B+C), \boldsymbol{w}(B-C)\} - \min\{\boldsymbol{w}(B), \boldsymbol{w}(C)\}|\} \tag{5.27}$$

and

$$\boldsymbol{w}(H) \geq \boldsymbol{w}(A) + \min\{\boldsymbol{w}(B), \boldsymbol{w}(C)\} + \frac{1}{2}\min\{\boldsymbol{w}(B+C), \boldsymbol{w}(B-C)\}. \tag{5.28}$$

Observing the above conclusions, we can find that $\boldsymbol{w}(B \pm C)$ plays a crucial role in estimating the numerical radius of the bounded infinite-dimensional Hamiltonian operators. Therefore, we will discuss the estimation of $\boldsymbol{w}(B \pm C)$ for the general bounded linear operators B, C. Firstly, several lemmas are given.

Lemma 5.3.39 *Let* A, B, C *be bounded linear operators in Hilbert space* X. *If* A, B *are non-negative operators, then*

$$T = \begin{bmatrix} A & C^* \\ C & B \end{bmatrix}$$

are non-negative and if and only if $|(Cx, y)|^2 \leq (Ax, x)(By, y)$ *holds for any* $x, y \in X$.

Proof When $T = \begin{bmatrix} A & C^* \\ C & B \end{bmatrix}$ is non-negative, according to the mixed Schwarz inequality (see Lemma 5.3.25), for any $x, y \in X$, there is

$$|\left(\begin{bmatrix} A & C^* \\ C & B \end{bmatrix}\begin{bmatrix} x \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ y \end{bmatrix}\right)|^2 \leq \left(\begin{bmatrix} A & C^* \\ C & B \end{bmatrix}\begin{bmatrix} x \\ 0 \end{bmatrix}, \begin{bmatrix} x \\ 0 \end{bmatrix}\right)\left(\begin{bmatrix} A & C^* \\ C & B \end{bmatrix}\begin{bmatrix} 0 \\ y \end{bmatrix}, \begin{bmatrix} 0 \\ y \end{bmatrix}\right),$$

after simplification, $|(Cx, y)|^2 \leq (Ax, x)(By, y)$ can be attained.

Conversely, for any $x, y \in X$, there is

$$\begin{aligned}\left(\begin{bmatrix} A & C^* \\ C & B \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix}\right) =& (Ax, x) + (By, y) + 2Re(Cx, y) \\ \geq& 2[(Ax, x)(By, y)]^{\frac{1}{2}} + 2Re(Cx, y) \\ \geq& 2|(Cx, y)| + 2Re(Cx, y) \geq 0.\end{aligned}$$

The proof is completed. □

The following lemma is a generalization of the mixed Schwarz inequality.

Lemma 5.3.40 *Let* $A \in \mathscr{B}(X)$, *and* $0 \leq \alpha \leq 1$. *Then for any* $x, y \in X$,

$$|(Ax, y)|^2 \leq (|A|^{2\alpha}x, x)(|A^*|^{2(1-\alpha)}y, y)$$

holds.

Proof According to Lemma 5.3.39, we only need to prove that $\begin{bmatrix} |A|^{2\alpha} & A^* \\ A & |A^*|^{2(1-\alpha)} \end{bmatrix} \geq 0$. It can be known by $A|A|^2 = |A^*|^2A$ that $A|A| = |A^*|A$, thus $Af(|A|) = f(|A^*|)A$, where $f(t), t \geq 0$ is a continuous function. Then considering that

$$\begin{bmatrix} |A|^{2\alpha} & A^* \\ A & |A^*|^{2(1-\alpha)} \end{bmatrix} = \begin{bmatrix} |A|^{\alpha-\frac{1}{2}} & 0 \\ 0 & |A^*|^{\frac{1}{2}-\alpha} \end{bmatrix}\begin{bmatrix} |A| & A^* \\ A & |A^*| \end{bmatrix}\begin{bmatrix} |A|^{\alpha-\frac{1}{2}} & 0 \\ 0 & |A^*|^{\frac{1}{2}-\alpha} \end{bmatrix},$$

again, from Lemma 5.3.39 and the mixed Schwarz inequality, we can know that $\begin{bmatrix} |A| & A^* \\ A & |A^*| \end{bmatrix} \geq 0$, thus the conclusion is proved. □

The following lemma is about the Jensen inequality of non-negative operators.

Lemma 5.3.41 *Let* $A \in \mathscr{B}(X)(A \geq 0)$, *and* $x \in X$, $\|x\| = 1$. *Then the following statements hold.*

(i) *When* $r \geq 1$, $(Ax, x)^r \leq (A^r x, x)$.
(ii) *When* $0 < r \leq 1$, $(Ax, x)^r \geq (A^r x, x)$.

Proof Let $E(t)$ be the spectral family of non-negative operator A. Then we have

$$A = \int_0^{+\infty} t \, \mathrm{d}E(t).$$

Combined with the concave-convex function and Jensen inequality, the conclusion can be proved. □

The following lemma is based on the results of Jensen's inequality and Young's inequality. The specific proof can be seen in [77, 78].

Lemma 5.3.42 *Let* $a, b \geq 0$, $0 \leq \alpha \leq 1$, *and* $p, q > 1$ *such that* $\frac{1}{p} + \frac{1}{q} = 1$. *Then*

(i) $a^\alpha b^{1-\alpha} \leq \alpha a + (1-\alpha)b \leq [\alpha a^r + (1-\alpha)b]^{\frac{1}{r}}$, *for* $r \geq 1$;
(ii) $ab \leq \frac{a^p}{p} + \frac{b^q}{q} \leq (\frac{a^{pr}}{p} + \frac{b^{qr}}{q})^{\frac{1}{r}}$, *for* $r \geq 1$.

The next lemma is the *Grüss* inequality .

Lemma 5.3.43 *Let* X *be an infinite-dimensional Hilbert space,* $\alpha \in \mathbb{C}$, *and* $|\alpha - 1| = 1$. *Then for any* $e \in X$, $\|e\| = 1$, $x, y \in X$, *there is*

$$|(x, y) - \alpha(x, e)(e, y)| \leq \|x\|\|y\|.$$

In particular, if $\alpha = 2$, *then*

$$|(x, e)(e, y)| \leq \frac{1}{2}[|(x, y)| + ||x||||y||].$$

Proof Because

$$|(u, v)| \leq \|u\|\|v\|,$$

for any $\|e\| = 1$ and $|\alpha - 1| = 1$, set $u = \alpha(x, e)e - x$, $v = y$, after putting the above formula, it can be attained that

$$|(x, y) - \alpha(x, e)(e, y)| \leq \|\alpha(x, e)e - x\|\|y\|.$$

Considering that $\|\alpha(x, e)e - x\| = \|x\|$ it can be attained that

$$|(x, y) - \alpha(x, e)(e, y)| \leq \|x\|\|y\|.$$

Also, because

$$|\alpha(x,e)(e,y)| - |(x,y)| \le \|x\|\|y\|,$$

so when $\alpha = 2$,

$$|(x,e)(e,y)| \le \frac{1}{2}[|(x,y)| + ||x||||y||].$$

The proof is completed. □

Theorem 5.3.44 *Let B, C be bounded linear operators in Hilbert space X, and α be any positive real number such that $0 \le \alpha \le 1$. Then for $r \ge 1$, we have*

$$\boldsymbol{w}(B \pm C) \ge \frac{|\boldsymbol{w}(C) - \boldsymbol{w}(B)|}{2} \tag{5.29}$$

and

$$\boldsymbol{w}(B \pm C) \le \min\{a, b, c\}, \tag{5.30}$$

where

$$a = 2^{1-\frac{1}{r}} \||B|^{2\alpha r} + |C|^{2\alpha r}\|^{\frac{1}{2r}} \||B^*|^{2(1-\alpha)r} + |C^*|^{2(1-\alpha)r}\|^{\frac{1}{2r}},$$

$$b = 2^{1-\frac{1}{r}} \||B|^{2\alpha r} + |C^*|^{2(1-\alpha)r}\|^{\frac{1}{2r}} \||B^*|^{2(1-\alpha)r} + |C|^{2\alpha r}\|^{\frac{1}{2r}},$$

$$c = 2^{1-\frac{1}{r}} \||B^*|^{2(1-\alpha)r} + |C|^{2\alpha r}\|^{\frac{1}{2r}} \||B|^{2\alpha r} + |C^*|^{2(1-\alpha)r}\|^{\frac{1}{2r}}.$$

Proof Select the unitary operator $U = \frac{1}{\sqrt{2}}\begin{bmatrix} I & -I \\ I & I \end{bmatrix}$, on the one hand, we consider that

$$\begin{aligned}
\boldsymbol{w}\left(\begin{bmatrix} 0 & -C \\ C & 0 \end{bmatrix}\right) &= \boldsymbol{w}\left(U^*\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}U - \frac{1}{2}\begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix}\right) \\
&\le \boldsymbol{w}\left(U^*\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}U\right) + \frac{1}{2}\boldsymbol{w}\left(\begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix}\right) \\
&= \boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) + \frac{1}{2}\boldsymbol{w}\left(\begin{bmatrix} B+C & 0 \\ 0 & -(B+C) \end{bmatrix} + \begin{bmatrix} 0 & B+C \\ -(B+C) & 0 \end{bmatrix}\right) \\
&\le \boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) + \frac{1}{2}[\boldsymbol{w}(B+C) + \boldsymbol{w}(B+C)] \\
&= \boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) + \boldsymbol{w}(B+C),
\end{aligned}$$

so

$$\boldsymbol{w}(C) - \boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) \le \boldsymbol{w}(B+C). \tag{5.31}$$

On the other hand,

$$\begin{aligned}
w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) =& w\left(U^* \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} U\right) \\
=& \frac{1}{2} w\left(\begin{bmatrix} B+C & B-C \\ -(B-C) & -(B+C) \end{bmatrix}\right) \\
=& \frac{1}{2} w\left(\begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix} + \begin{bmatrix} 0 & -2C \\ 2C & 0 \end{bmatrix}\right) \\
\leq& \frac{1}{2} w\left(\begin{bmatrix} B+C & B+C \\ -(B+C) & -(B+C) \end{bmatrix}\right) + w\left(\begin{bmatrix} 0 & -C \\ C & 0 \end{bmatrix}\right) \\
=& \frac{1}{2} w\left(\begin{bmatrix} B+C & 0 \\ 0 & -(B+C) \end{bmatrix} + \begin{bmatrix} 0 & B+C \\ -(B+C) & 0 \end{bmatrix}\right) + w(C) \\
\leq& \frac{1}{2} w\left(\begin{bmatrix} B+C & 0 \\ 0 & -(B+C) \end{bmatrix}\right) + \frac{1}{2} w\left(\begin{bmatrix} 0 & B+C \\ -(B+C) & 0 \end{bmatrix}\right) + w(C) \\
=& w(B+C) + w(C),
\end{aligned}$$

so

$$w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) - w(C) \leq w(B+C). \tag{5.32}$$

Therefore, following from Eqs. (5.31) and (5.32) yields that

$$\left| w(C) - w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)\right| \leq w(B+C). \tag{5.33}$$

Similarly, exchanging B and C in Eq. (5.33), we can obtain that

$$\left| w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) - w(B)\right| \leq w(B+C). \tag{5.34}$$

From Eqs. (5.33) and (5.34), we can obtain that

$$\frac{|w(C) - w(B)|}{2} \leq \frac{\left| w(C) - w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)\right| + \left| w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) - w(B)\right|}{2}, \tag{5.35}$$

and

$$\frac{\left| w(C) - w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)\right| + \left| w\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) - w(B)\right|}{2} \leq w(B+C). \tag{5.36}$$

Therefore,

$$\frac{|w(C) - w(B)|}{2} \leq w(B+C). \tag{5.37}$$

In the inequality (5.37), it can be known with C replaced by $-C$ that

$$\frac{|\boldsymbol{w}(C)-\boldsymbol{w}(B)|}{2}\leq \boldsymbol{w}(B-C). \tag{5.38}$$

According to Inequalities (5.37) and (5.38), the proof of (5.29) is completed. In addition, we use the following basic inequality:

$$(ab+cd)^2\leq (a^2+c^2)(b^2+d^2),\ (a+b)^r\leq 2^{r-1}(a^r+b^r).$$

Then it can be known by Lemmas 5.3.40 and 5.3.41,

$$\begin{aligned}
|((B+C)x,x)|^r &= |(Bx,x)+(Cx,x)|^r\\
&\leq (|(Bx,x)|+|(Cx,x)|)^r\\
&\leq 2^{r-1}(|(Bx,x)|^r+|(Cx,x)|^r)\\
&\leq 2^{r-1}[(|B|^{2\alpha}x,x)^{\frac{r}{2}}(|B^*|^{2(1-\alpha)}x,x)^{\frac{r}{2}}+(|C|^{2\alpha}x,x)^{\frac{r}{2}}(|C^*|^{2(1-\alpha)}x,x)^{\frac{r}{2}}]\\
&\leq 2^{r-1}[(|B|^{2\alpha}x,x)^r+(|C|^{2\alpha}x,x)^r]^{\frac{1}{2}}[(|B^*|^{2(1-\alpha)}x,x)^r+(|C^*|^{2(1-\alpha)}x,x)^r]^{\frac{1}{2}}\\
&\leq 2^{r-1}[(|B|^{2\alpha r}x,x)+(|C|^{2\alpha r}x,x)]^{\frac{1}{2}}[(|B^*|^{2(1-\alpha)r}x,x)+(|C^*|^{2(1-\alpha)r}x,x)]^{\frac{1}{2}}\\
&= 2^{r-1}[((|B|^{2\alpha r}+|C|^{2\alpha r})x,x)]^{\frac{1}{2}}[((|B^*|^{2(1-\alpha)r}+|C^*|^{2(1-\alpha)})x,x)]^{\frac{1}{2}}\\
&= 2^{r-1}||(|B|^{2\alpha r}+|C|^{2\alpha r}||^{\frac{1}{2}}||(|B^*|^{2(1-\alpha)r}+|C^*|^{2(1-\alpha)}||^{\frac{1}{2}}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
(\boldsymbol{w}(B+C))^r &= \sup\{|((B+C)x,x)|^r : x\in H, ||x||=1\}\\
&\leq 2^{r-1}||(|B|^{2\alpha r}+|C|^{2\alpha r}||^{\frac{1}{2}}||(|B^*|^{2(1-\alpha)r}+|C^*|^{2(1-\alpha)}||^{\frac{1}{2}}.
\end{aligned}$$

Similarly, by exchanging a, b, c, d, we can obtain different forms

$$(\boldsymbol{w}(B+C))^r\leq 2^{r-1}||(|B|^{2\alpha r}+|C^*|^{2(1-\alpha)r}||^{\frac{1}{2}}||(|B^*|^{2(1-\alpha)r}+|C|^{2\alpha r}||^{\frac{1}{2}}.$$

So

$$\boldsymbol{w}(B+C)\leq \min\{a,b\}, \tag{5.39}$$

where

$$a=2^{1-\frac{1}{r}}||(|B|^{2\alpha r}+|C|^{2\alpha r}||^{\frac{1}{2r}}||(|B^*|^{2(1-\alpha)r}+|C^*|^{2(1-\alpha)r}||^{\frac{1}{2r}},$$

$$b=2^{1-\frac{1}{r}}||(|B|^{2\alpha r}+|C^*|^{2(1-\alpha)r}||^{\frac{1}{2r}}||(|B^*|^{2(1-\alpha)r}+|C|^{2\alpha r}||^{\frac{1}{2r}}.$$

After replacing C with $-C$ in (5.39), we can obtain that

$$\boldsymbol{w}(B-C)\leq \min\{a,b,c\}. \tag{5.40}$$

According to Inequalities (5.39) and (5.40), the proof of Inequality (5.30) is completed. The proof is completed. □

Proposition 5.3.45 *If taking* $r = 1$, $\alpha = \frac{1}{2}$ *in Theorem 5.3.44, we can obtain*

$$\boldsymbol{w}(B + C) \leq \min\{a', b'\},$$

where

$$a' = |||B| + |C|||^{\frac{1}{2}}|||B^*| + |C^*|||^{\frac{1}{2}},$$

$$b' = |||B^*| + |C|||^{\frac{1}{2}}|||B| + |C^*|||^{\frac{1}{2}}.$$

Theorem 5.3.46 *Let* B, C *be bounded linear operators in Hilbert space* X, *and* α *be any positive real numbers such that* $0 \leq \alpha \leq 1$. *Then for* $r \geq 1$ *we have*

$$\boldsymbol{w}(B \pm C) \leq 2^{1-\frac{1}{r}} \left[\left|\left|\alpha(|B|^r + |C|^r) + (1-\alpha)(|B^*|^r + |C^*|^r)\right|\right|\right]^{\frac{1}{r}}.$$

Proof According to Theorem 5.3.46 and Lemma 5.3.41, we can obtain

$$\begin{aligned}
&|((B + C)x, x)|^r \\
&= |(Bx, x) + (Cx, x)|^r \\
&\leq (|(Bx, x)| + |(Cx, x)|)^r \\
&\leq 2^{r-1}(|(Bx, x)|^r + |(Cx, x)|^r) \\
&\leq 2^{r-1}[(|B|^{2\alpha}x, x)^{\frac{r}{2}}(|B^*|^{2(1-\alpha)}x, x)^{\frac{r}{2}} + (|C|^{2\alpha}x, x)^{\frac{r}{2}}(|C^*|^{2(1-\alpha)}x, x)^{\frac{r}{2}}] \\
&\leq 2^{r-1}\left[(|B|^r x, x)^{\alpha}(|B^*|^r x, x)^{(1-\alpha)} + (|C|^r x, x)^{\alpha}(|C^*|^r x, x)^{(1-\alpha)}\right] \\
&\leq 2^{r-1}\left[\alpha(|B|^r x, x) + (1-\alpha)(|B^*|^r x, x) + \alpha(|C|^r x, x) + (1-\alpha)(|C^*|^r x, x)\right] \\
&= 2^{r-1}\left[\alpha(|B|^r + |C|^r)x, x) + (1-\alpha)((|B^*|^r + |C^*|^r)x, x)\right] \\
&\leq 2^{r-1}\left|\left|\alpha(|B|^r + |C|^r) + (1-\alpha)(|B^*|^r + |C^*|^r)\right|\right|.
\end{aligned}$$

Therefore,

$$\begin{aligned}
(\boldsymbol{w}(B + C))^r &= \sup\{|((B + C)x, x)|^r : x \in H, ||x|| = 1\} \\
&\leq 2^{r-1}\left|\left|\alpha(|B|^r + |C|^r) + (1-\alpha)(|B^*|^r + |C^*|^r)\right|\right|.
\end{aligned}$$

Hence,

$$\boldsymbol{w}(B + C) \leq 2^{1-\frac{1}{r}}\left|\left|\alpha(|B|^r + |C|^r) + (1-\alpha)(|B^*|^r + |C^*|^r)\right|\right|^{\frac{1}{r}}.$$

In the above equation, replacing C with $-C$, we can obtain

$$\boldsymbol{w}(B - C) \leq 2^{1-\frac{1}{r}}\left|\left|\alpha(|B|^r + |C|^r) + (1-\alpha)(|B^*|^r + |C^*|^r)\right|\right|^{\frac{1}{r}}.$$

The proof is completed. □

Theorem 5.3.47 *Let B, C be bounded linear operators in Hilbert space X, and α be an arbitrary positive real number such that $0 \le \alpha \le 1$. Then for $r \ge 1$, $p \ge q > 1$, $m \ge n > 1$, $\frac{1}{p} + \frac{1}{q} = 1$, $\frac{1}{m} + \frac{1}{n} = 1$ and $pr \ge 1$, $mr \ge 1$, $qr \ge 1$, $nr \ge 1$, we have*

$$w(B \pm C) \le 2^{1-\frac{1}{r}} \left[|| \frac{1}{p}|B|^{\alpha pr} + \frac{1}{q}|B^*|^{(1-\alpha)rq} + \frac{1}{m}|C|^{\alpha rm} + \frac{1}{n}|C^*|^{(1-\alpha)nr} || \right]^{\frac{1}{r}}.$$

Proof According to Theorem 5.3.44 and Lemma 5.3.41, we can attain

$$\begin{aligned}
&|((B+C)x,x)|^r \\
&= |(Bx,x) + (Cx,x)|^r \\
&\le (|(Bx,x)| + |(Cx,x)|)^r \\
&\le 2^{r-1}(|(Bx,x)|^r + |(Cx,x)|^r) \\
&\le 2^{r-1}[(|B|^{2\alpha}x,x)^{\frac{r}{2}}(|B^*|^{2(1-\alpha)}x,x)^{\frac{r}{2}} + (|C|^{2\alpha}x,x)^{\frac{r}{2}}(|C^*|^{2(1-\alpha)}x,x)^{\frac{r}{2}}] \\
&\le 2^{r-1}\left[\frac{1}{p}(|B|^{2\alpha}x,x)^{\frac{pr}{2}} + \frac{1}{q}(|B^*|^{2(1-\alpha)}x,x)^{\frac{qr}{2}} + \frac{1}{m}(|C|^{2\alpha}x,x)^{\frac{mr}{2}} + \frac{1}{n}(|C^*|^{2(1-\alpha)}x,x)^{\frac{nr}{2}}\right] \\
&\le 2^{r-1}\left[\frac{1}{p}(|B|^{\alpha r}x,x)^{p} + \frac{1}{q}(|B^*|^{(1-\alpha)r}x,x)^{q} + \frac{1}{m}(|C|^{\alpha r}x,x)^{m} + \frac{1}{n}(|C^*|^{(1-\alpha)r}x,x)^{n}\right] \\
&\le 2^{r-1}\left[\frac{1}{p}(|B|^{\alpha rp}x,x) + \frac{1}{q}(|B^*|^{(1-\alpha)qr}x,x) + \frac{1}{m}(|C|^{\alpha rm}x,x) + \frac{1}{n}(|C^*|^{(1-\alpha)rn}x,x)\right] \\
&= 2^{r-1}\| \frac{1}{p}|B|^{\alpha rp} + \frac{1}{q}|B^*|^{(1-\alpha)qr} + \frac{1}{m}|C|^{\alpha rm} + \frac{1}{n}|C^*|^{(1-\alpha)rn}\|.
\end{aligned}$$

Therefore,

$$\begin{aligned}
(w(B+C))^r &= \sup\{|((B+C)x,x)|^r : x \in X, ||x|| = 1\} \\
&\le 2^{r-1}\left[\| \frac{1}{p}|B|^{\alpha rp} + \frac{1}{q}|B^*|^{(1-\alpha)qr} + \frac{1}{m}|C|^{\alpha rm} + \frac{1}{n}|C^*|^{(1-\alpha)rn}\|\right].
\end{aligned}$$

Hence,

$$w(B+C) \le 2^{1-\frac{1}{r}}\left[\| \frac{1}{p}|B|^{\alpha rp} + \frac{1}{q}|B^*|^{(1-\alpha)qr} + \frac{1}{m}|C|^{\alpha rm} + \frac{1}{n}|C^*|^{(1-\alpha)rn}\|\right]^{\frac{1}{r}}.$$

In the above equations, replacing C with $-C$, we can obtain

$$w(B-C) \le 2^{1-\frac{1}{r}}\left[\| \frac{1}{p}|B|^{\alpha rp} + \frac{1}{q}|B^*|^{(1-\alpha)qr} + \frac{1}{m}|C|^{\alpha rm} + \frac{1}{n}|C^*|^{(1-\alpha)rn}\|\right]^{\frac{1}{r}}.$$

The proof is completed. □

Theorem 5.3.48 *Let B, C be bounded linear operators in Hilbert space X, and α be any positive real number such that $0 \le \alpha \le 1$. Then for $r \ge 1$ we have*

$$w(B \pm C) \le 2^{1-\frac{1}{r}}[(|||B|^{\alpha}|||B^*|^{(1-\alpha)}||)^{2r} + (|||C|^{\alpha}|||C^*|^{(1-\alpha)}||)^{2r}]^{\frac{1}{r}}.$$

Proof According to Theorem 5.3.44 and Lemma 5.3.41, we can attain

$$
\begin{aligned}
&|((B+C)x,x)|^r \\
&= |(Bx,x)+(Cx,x)|^r \\
&\le (|(Bx,x)|+|(Cx,x)|)^r \\
&\le 2^{r-1}(|(Bx,x)|^r+|(Cx,x)|^r) \\
&\le 2^{r-1}[(|B|^{2\alpha}x,x)^{\frac{r}{2}}(|B^*|^{2(1-\alpha)}x,x)^{\frac{r}{2}}+(|C|^{2\alpha}x,x)^{\frac{r}{2}}(|C^*|^{2(1-\alpha)}x,x)^{\frac{r}{2}}] \\
&= 2^{r-1}[((|B|^{2\alpha}x,x)(x,|B^*|^{2(1-\alpha)}x))^{\frac{r}{2}}+((|C|^{2\alpha}x,x)(x,|C^*|^{2(1-\alpha)}x))^{\frac{r}{2}}] \\
&\le 2^{r-1}\cdot(\frac{1}{2})^{\frac{r}{2}}[(|(|B|^{2\alpha}x,|B^*|^{2(1-\alpha)}x)|+|||B|^{2\alpha}|||||B^*|^{2(1-\alpha)}||)^{\frac{r}{2}} \\
&+(|(|C|^{2\alpha}x,|C^*|^{2(1-\alpha)}x)|+|||C|^{2\alpha}|||||C^*|^{2(1-\alpha)}||)^{\frac{r}{2}}] \\
&\le 2^{\frac{r}{2}-1}[(|||B|^{2\alpha}|||||B^*|^{2(1-\alpha)}||+|||B|^{2\alpha}|||||B^*|^{2(1-\alpha)}||)^{\frac{r}{2}} \\
&+(|||C|^{2\alpha}|||||C^*|^{2(1-\alpha)}||+|||C|^{2\alpha}|||||C^*|^{2(1-\alpha)}||)^{\frac{r}{2}}] \\
&= 2^{\frac{r}{2}-1}\cdot 2^{\frac{r}{2}}[(|||B|^{2\alpha}|||||B^*|^{2(1-\alpha)}||)^{\frac{r}{2}}+(|||C|^{2\alpha}|||||C^*|^{2(1-\alpha)}||)^{\frac{r}{2}}] \\
&= 2^{r-1}[(|||B|^{2\alpha}|||||B^*|^{2(1-\alpha)}||)^{\frac{r}{2}}+(|||C|^{2\alpha}|||||C^*|^{2(1-\alpha)}||)^{\frac{r}{2}}] \\
&= 2^{r-1}[(|||B|^{\alpha}|||||B^*|^{(1-\alpha)}||)^{2r}+(|||C|^{\alpha}|||||C^*|^{(1-\alpha)}||)^{2r}].
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
(\boldsymbol{w}(B+C))^r &= \sup\{|((B+C)x,x)|^r : x\in X, ||x||=1\} \\
&\le 2^{r-1}[(|||B|^{\alpha}|||||B^*|^{(1-\alpha)}||)^{2r}+(|||C|^{\alpha}|||||C^*|^{(1-\alpha)}||)^{2r}].
\end{aligned}
$$

Then it can be obtained that

$$
\boldsymbol{w}(B+C)\le 2^{1-\frac{1}{r}}[(|||B|^{\alpha}|||||B^*|^{(1-\alpha)}||)^{2r}+(|||C|^{\alpha}|||||C^*|^{(1-\alpha)}||)^{2r}]^{\frac{1}{r}}.
$$

In the above equations, replacing C with $-C$, we can obtain

$$
\boldsymbol{w}(B-C)\le 2^{1-\frac{1}{r}}[(|||B|^{\alpha}|||||B^*|^{(1-\alpha)}||)^{2r}+(|||C|^{\alpha}|||||C^*|^{(1-\alpha)}||)^{2r}]^{\frac{1}{r}}.
$$

The proof is completed. □

5.4 Quadratic Numerical Range of Infinite-Dimensional Hamiltonian Operators

Convexity is an extremely important property for the numerical range of linear operators in Hilbert space, that is, if a support line of the numerical range of bounded linear operators is found, it means that the half-plane of the distribution of the spectral set of

linear operators is found. However, due to the connectivity of convex sets, it is sometimes impossible to characterize the distribution of spectral sets more accurately, for example, when the spectral sets are a union of several disjoint subsets, the numerical range cannot characterize this case exactly. In view of this, Tretter [69] and Langer [70] and other scholars are studying a 2×2 block operator matrix, the concept of a quadratic numerical range is introduced. By using the quadratic numerical range, the variational principle of a self-adjoint 2×2 block operator matrix can be established. Furthermore, we estimate the eigenvalues of the operators. It is worth noting that for bounded linear operators, the quadratic numerical range is a subset of the numerical range, but it is not necessarily connected, and the quadratic numerical range also has the property of spectral inclusion. Hence in terms of the spectral characterization of bounded linear operators, the quadratic numerical range can provide more accurate information than the numerical range.

5.4.1 *Definition of Quadratic Numerical Range of Linear Operators*

Firstly, the definition of a quadratic numerical range is given.

Definition 5.4.1 Let X be a Hilbert space, for $f \in \mathscr{D}(A) \cap \mathscr{D}(C)$, $g \in \mathscr{D}(B) \cap \mathscr{D}(D)$, $f, g \neq 0$, introducing a 2×2 matrix

$$\mathscr{A}_{f,g} = \begin{bmatrix} \frac{(Af,f)}{\|f\|^2} & \frac{(Bg,f)}{\|f\|\|g\|} \\ \frac{(Cf,g)}{\|f\|\|g\|} & \frac{(Dg,g)}{\|g\|^2} \end{bmatrix} \in M_2(\mathbb{C}),$$

then the set

$$\mathscr{W}^2(\mathscr{A}) = \{\lambda \in \mathbb{C} : \exists f \in \mathscr{D}(A) \cap \mathscr{D}(C), g \in \mathscr{D}(B) \cap \mathscr{D}(D), f, g \neq 0, \det(\mathscr{A}_{f,g} - \lambda I) = 0\}$$

is called the quadratic numerical range of the following 2×2 block operator matrix in the Hilbert space $X \times X$

$$\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} : \mathscr{D}(\mathscr{A}) \subset X \times X \to X \times X.$$

Equivalently we can see

$$\begin{aligned} \mathscr{W}^2(\mathscr{A}) &= \bigcup_{f \neq 0, g \neq 0} \sigma_p(\mathscr{A}_{f,g}) \\ &= \bigcup_{\|f\| = \|g\| = 1} \sigma_p(\mathscr{A}_{f,g}). \end{aligned}$$

From the definition, it is not difficult to find that the quadratic numerical range is also a subset of the complex field $\mathbb{C}$, when

$$\mathscr{D}(\mathscr{A}) = \mathscr{D}(A) \times \mathscr{D}(D)$$

and if $B = 0$ or $C = 0$, then

$$\mathscr{W}^2(\mathscr{A}) = W(A) \cup W(D),$$

where $W(A)$, $W(D)$ represent the numerical ranges of A, D, respectively. Thus, the quadratic numerical range does not necessarily have the convexity of the numerical range. The following are some examples of the quadratic numerical range.

Example 5.4.2 Let X be a Hilbert space, I be a unit operator in X. Define a block operator matrix in Hilbert space $X \times X$

$$\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix},$$

then $\mathscr{W}^2(\mathscr{A}) = \{0\}$. In fact, if $x = \begin{bmatrix} f & g \end{bmatrix}^T \in X \times X$, $f, g \neq 0$, then

$$\mathscr{A}_{f,g} = \begin{bmatrix} \frac{(Af,f)}{\|f\|^2} & \frac{(Bg,f)}{\|f\|\|g\|} \\ \frac{(Cf,g)}{\|f\|\|g\|} & \frac{(Dg,g)}{\|g\|^2} \end{bmatrix} = \begin{bmatrix} 0 & \frac{(g,f)}{\|f\|\|g\|} \\ 0 & 0 \end{bmatrix},$$

take

$$\det(\mathscr{A}_{f,g} - \lambda) = 0,$$

then $\lambda = 0$. So $\mathscr{W}^2(\mathscr{A}) = \{0\} = \sigma(\mathscr{A})$.

Example 5.4.3 Let X be a Hilbert space, I be a unit operator in X, B be a linear operator in X (not necessarily bounded). Define a block operator matrix in the Hilbert space $X \times X$

$$\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} I & B \\ 0 & -I \end{bmatrix},$$

then $\mathscr{W}^2(\mathscr{A}) = \{1\} \cup \{-1\}$. In fact, set $x = \begin{bmatrix} f & g \end{bmatrix}^T \in \mathscr{D}(\mathscr{A})$, $f, g \neq 0$, then

$$\mathscr{A}_{f,g} = \begin{bmatrix} \frac{(Af,f)}{\|f\|^2} & \frac{(Bg,f)}{\|f\|\|g\|} \\ \frac{(Cf,g)}{\|f\|\|g\|} & \frac{(Dg,g)}{\|g\|^2} \end{bmatrix} = \begin{bmatrix} 1 & \frac{(Bg,f)}{\|f\|\|g\|} \\ 0 & -1 \end{bmatrix}.$$

Take

$$\det(\mathscr{A}_{f,g} - \lambda) = 0,$$

then it can be gotten that $\lambda = \pm 1$. So $\mathscr{W}^2(\mathscr{A}) = \{1\} \cup \{-1\}$, which shows that $\mathscr{W}^2(\mathscr{A})$ is not necessary a convex set.

The quadratic numerical range also has the following basic properties.

Lemma 5.4.4 *Given a block operator matrix*

$$\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} : \mathscr{D}(\mathscr{A}) \subset X \times X \to X \times X,$$

then the following statements are satisfied:

(i) *For any* $\alpha, \beta \in \mathbb{C}$, *we can have* $\mathscr{W}^2(\alpha\mathscr{A} + \beta I) = \alpha\mathscr{W}^2(\mathscr{A}) + \beta$.

(ii) $\mathscr{W}^2(U^{-1}\mathscr{A}U) = \mathscr{W}^2(\mathscr{A})$, *where* $U = \begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix}$, U_1, U_2 *are unitary operators in Hilbert space* X.

(iii) $\mathscr{W}^2(U^{-1}\mathscr{A}U) = \mathscr{W}^2(\mathscr{A})$, *where* $U = \begin{bmatrix} 0 & U_1 \\ U_2 & 0 \end{bmatrix}$, U_1, U_2 *are unitary operators in Hilbert space* X.

(iv) *If* A, B, C, D *are bounded linear operators on the whole space, then* $\mathscr{W}^2(\mathscr{A}^*) = (\mathscr{W}^2(\mathscr{A}))^* = \{\overline{\lambda} \in \mathbb{C} : \lambda \in \mathscr{W}^2(\mathscr{A})\}$.

(v) *If* A, D *are self-adjoint operators,* $B = C^*$, *that is,* $\mathscr{A}$ *is a symmetric operator, then* $\mathscr{W}^2(\mathscr{A}) \subset \mathbb{R}$, *but the reverse is not true.*

Proof (i) If $\lambda \in \mathscr{W}^2(\alpha\mathscr{A} + \beta I)$, then there exist $f \in \mathscr{D}(A) \cap \mathscr{D}(C)$, $g \in \mathscr{D}(B) \cap \mathscr{D}(D)$, $\|f\| = 1$, $\|g\| = 1$, such that

$$\det \begin{bmatrix} \alpha(Af, f) - (\lambda - \beta) & \alpha(Bg, f) \\ \alpha(Cf, g) & \alpha(Dg, g) - (\lambda - \beta) \end{bmatrix} = 0.$$

There is no harm in supposing that $\alpha \neq 0$. From the above equation, $\frac{\lambda-\beta}{\alpha} \in \mathscr{W}^2(\mathscr{A})$, reconsidering that $\lambda = \alpha\frac{\lambda-\beta}{\alpha} + \beta$, it can be obtained that $\lambda \in \alpha\mathscr{W}^2(\mathscr{A}) + \beta$, that is, $\mathscr{W}^2(\alpha\mathscr{A} + \beta I) \subset \alpha\mathscr{W}^2(\mathscr{A}) + \beta$. Similarly, the anti-inclusion relationship can be proved.

(ii) Considering that

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix} = \begin{bmatrix} AU_1 & BU_2 \\ CU_1 & DU_2 \end{bmatrix}, \tag{5.41}$$

moreover,

$$U^{-1}\mathscr{A}U = \begin{bmatrix} U_1^{-1}AU_1 & U_1^{-1}BU_2 \\ U_2^{-1}CU_1 & U_2^{-1}DU_2 \end{bmatrix}.$$

So, set $\lambda \in \mathscr{W}^2(U^{-1}\mathscr{A}U)$, then there exist $f, g \neq 0$ such that

$$\det \begin{bmatrix} \frac{(U_1^{-1}AU_1 f, f)}{\|f\|^2} - \lambda & \frac{(U_1^{-1}BU_2 g, f)}{\|f\|\|g\|} \\ \frac{(U_2^{-1}CU_1 f, g)}{\|f\|\|g\|} & \frac{(U_2^{-1}DU_2 g, g)}{\|g\|^2} - \lambda \end{bmatrix} = 0,$$

combining that U_1, U_2 are unitary operators, $\|U_1 f\| = \|f\|$, $\|U_2 g\| = \|g\|$, that is, $\lambda \in \mathscr{W}^2(\mathscr{A})$, so $\mathscr{W}^2(U^{-1}\mathscr{A}U) \subset \mathscr{W}^2(\mathscr{A})$. Similarly, the anti-inclusion relationship can be proved. The proof of (iii) is completely similar to (ii).

(iv) When A, B, C, D are linear operators on the whole space,

$$\mathscr{A}^* = \begin{bmatrix} A^* & C^* \\ B^* & D^* \end{bmatrix}.$$

If $\lambda \in \mathscr{W}^2(\mathscr{A})$, then there exist $\|f\| = 1$, $\|g\| = 1$ such that

$$\det \begin{bmatrix} (Af, f) - \lambda & (Bg, f) \\ (Cf, g) & (Dg, g) - \lambda \end{bmatrix} = 0,$$

after taking the conjugation, it can be obtained that

$$\det \begin{bmatrix} (f, Af) - \overline{\lambda} & (f, Bg) \\ (g, Cf) & (g, Dg) - \overline{\lambda} \end{bmatrix} = \det \begin{bmatrix} (A^* f, f) - \overline{\lambda} & (C^* g, f) \\ (B^* f, g) & (D^* g, g) - \overline{\lambda} \end{bmatrix} = 0,$$

that is, $\overline{\lambda} \in \mathscr{W}^2(\mathscr{A}^*)$, and hence $(\mathscr{W}^2(\mathscr{A}))^* \subset \mathscr{W}^2(\mathscr{A}^*)$. Similarly, $\mathscr{W}^2(\mathscr{A}^*) \subset (\mathscr{W}^2(\mathscr{A}))^*$.

(v) If A, D are self-adjoint operators, $B = C^*$, then for any $f \in \mathscr{D}(A) \cap \mathscr{D}(C)$, $g \in \mathscr{D}(C^*) \cap \mathscr{D}(D)$, $\|f\| = 1$, $\|g\| = 1$, $\mathscr{A}_{f,g}$ satisfies

$$\mathscr{A}_{f,g} = \mathscr{A}_{f,g}^*,$$

namely, $\mathscr{A}_{f,g}$ is a symmetric matrix and its eigenvalues are real numbers. So $\mathscr{W}^2(\mathscr{A}) \subset \mathbb{R}$. On the contrary, the conclusion does not necessarily hold. For example, take

$$\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} I & I \\ 0 & -I \end{bmatrix},$$

then $\mathscr{W}^2(\mathscr{A}) = \{1\} \cup \{-1\} \subset \mathbb{R}$, but $\mathscr{A}$ is not a symmetric operator. □

Remark 5.4.5 It is worth noting that the equality does not necessarily hold when the bounded operator is non-diagonal or skew-diagonal in (5.41) right side. For example, let

$$\mathscr{A}_1 = \begin{bmatrix} A & A \\ A & A \end{bmatrix} \begin{bmatrix} \frac{1}{2}I & \frac{1}{2}I \\ \frac{1}{2}I & \frac{1}{2}I \end{bmatrix}$$

and

$$\mathscr{A}_2 = \begin{bmatrix} A & A \\ A & A \end{bmatrix},$$

then

$$\mathscr{D}(\mathscr{A}_1) = \{ \begin{bmatrix} x \\ y \end{bmatrix} : x + y \in \mathscr{D}(A) \},$$

$$\mathscr{D}(\mathscr{A}_2) = \{\begin{bmatrix} x \\ y \end{bmatrix} : x, y \in \mathscr{D}(A)\}.$$

Obviously, $\mathscr{D}(\mathscr{A}_1) \neq \mathscr{D}(\mathscr{A}_2)$, so

$$\begin{bmatrix} A & A \\ A & A \end{bmatrix} \begin{bmatrix} \frac{1}{2}I & \frac{1}{2}I \\ \frac{1}{2}I & \frac{1}{2}I \end{bmatrix} \neq \begin{bmatrix} A & A \\ A & A \end{bmatrix}.$$

Lemma 5.4.6 *Given a block operator matrix*

$$\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(D) \to X \times X,$$

if $\dim X \geq 2$, *then* $W(A) \cup W(D) \subset \mathscr{W}^2(\mathscr{A})$.

Proof If $\lambda \in W(A)$, then there exists $f \in \mathscr{D}(A)$, $\|f\| = 1$ such that

$$(Af, f) = \lambda.$$

Considering that $\mathscr{D}(A) \subset \mathscr{D}(C)$ and the density of $\mathscr{D}(D)$ and $\dim X \geq 2$, we have $f \in \mathscr{D}(C)$ and there exists $g \in \mathscr{D}(D)$, $\|g\| = 1$ such that

$$(Cf, g) = 0.$$

So

$$\det \begin{bmatrix} (Af, f) - \lambda & (Bg, f) \\ (Cf, g) & (Dg, g) - \lambda \end{bmatrix} = 0,$$

thus $\lambda \in \mathscr{W}^2(\mathscr{A})$, that is, $W(A) \subset \mathscr{W}^2(\mathscr{A})$. Similarly, we can prove that $W(D) \subset \mathscr{W}^2(\mathscr{A})$. The proof is completed. □

For a 2×2 block operator matrix (not necessarily bounded), the quadratic numerical range is a subset of the numerical range.

Lemma 5.4.7 *Given a block operator matrix*

$$\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} : \mathscr{D}(\mathscr{A}) \subset X \times X \to X \times X,$$

then $\mathscr{W}^2(\mathscr{A}) \subset W(\mathscr{A})$.

Proof If $\lambda \in \mathscr{W}^2(\mathscr{A})$, then there exist $\|f\| = 1$, $\|g\| = 1$, $f \in \mathscr{D}(A) \cap \mathscr{D}(C)$, $g \in \mathscr{D}(B) \cap \mathscr{D}(D)$ such that

$$\det(\mathscr{A}_{f,g} - \lambda) = 0,$$

that is, λ is the eigenvalue of a 2×2 matrix $\mathscr{A}_{f,g}$, so there exists $\begin{bmatrix} \alpha_1 & \alpha_2 \end{bmatrix}^T \in \mathbb{C}^2$, $|\alpha_1|^2 + |\alpha_2|^2 = 1$ such that

$$\mathscr{A}_{f,g}\begin{bmatrix}\alpha_1\\ \alpha_2\end{bmatrix}=\lambda\begin{bmatrix}\alpha_1\\ \alpha_2\end{bmatrix}.$$

Taking the inner product of both sides of the above equation with $\begin{bmatrix}\alpha_1 & \alpha_2\end{bmatrix}^T$ yields that

$$(A\alpha_1 f,\alpha_1 f)+(B\alpha_2 g,\alpha_2 f)+(C\alpha_1 f,\alpha_1 g)+(D\alpha_2 g,\alpha_2 g)=\lambda,$$

that is,

$$\left(\mathscr{A}\begin{bmatrix}\alpha_1 f\\ \alpha_2 g\end{bmatrix},\begin{bmatrix}\alpha_1 f\\ \alpha_2 g\end{bmatrix}\right)=\lambda.$$

Since $\|\alpha_1 f\|^2+\|\alpha_2 g\|^2=|\alpha_1|^2+|\alpha_2|^2=1$, so $\lambda\in W(\mathscr{A})$. The proof is completed. □

5.4.2 *Properties of the Quadratic Numerical Range of Infinite-Dimensional Hamiltonian Operators*

Firstly, the properties of the quadratic numerical range of a skew-diagonal infinite-dimensional Hamiltonian operator are discussed. For a skew-diagonal infinite-dimensional Hamiltonian operator

$$H=\begin{bmatrix}0 & B\\ C & 0\end{bmatrix}:\mathscr{D}(C)\times\mathscr{D}(B)\to X\times X,$$

it is easy to prove that $\lambda\in\mathscr{W}^2(H)$ if and only if $-\lambda\in\mathscr{W}^2(H)$, that is, the quadratic numerical range is symmetric about the origin. When $\dim X\geq 2, 0\in\mathscr{W}^2(H)$. Moreover, when $B, C\in\mathscr{B}(X)$, a skew-diagonal infinite-dimensional Hamiltonian operator is not only symmetric about the origin, but also is symmetric about the imaginary axis and the real axis, and it is a connected set.

The following is the relationship between the quadratic numerical range of a skew-diagonal infinite-dimensional Hamiltonian operator and the quadratic numerical range of its adjoint operator.

Theorem 5.4.8 *Let* $H=\begin{bmatrix}0 & B\\ C & 0\end{bmatrix}:\mathscr{D}(C)\times\mathscr{D}(B)\to X\times X$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then* $\mathscr{W}^2(H)=\mathscr{W}^2(-H)=\mathscr{W}^2(H^*)=\mathscr{W}^2(-H^*)$.

Proof It can be seen from Theorem 2.3.2 that H_0 and $-H_0$, H_0^* and $-H_0^*$ are unitary similar, so the conclusion is obvious. □

Theorem 5.4.9 *Let* $H=\begin{bmatrix}0 & B\\ C & 0\end{bmatrix}:\mathscr{D}(C)\times\mathscr{D}(B)\to X\times X$ *be a skew-diagonal infinite-dimensional Hamiltonian operator. Then*

$$\{\lambda \in \mathbb{C} : \lambda^2 \in W(BC)\} \cup \{\lambda \in \mathbb{C} : \lambda^2 \in W(CB)\} \subset \mathscr{W}^2(H).$$

Proof If $\lambda^2 \in W(BC)$, then there exists $f \in \mathscr{D}(BC)$, $\|f\| = 1$ such that

$$(BCf, f) = \lambda^2.$$

Without losing the generality, assume that $\lambda = 0$. When $Cf = 0$, we can take $g \in \mathscr{D}(B)$, $\|g\| = 1$, then

$$\det(H_{f,g}) = \det\left(\begin{bmatrix} 0 & (Bg, f) \\ (Cf, g) & 0 \end{bmatrix}\right) = 0,$$

that is, $0 \in \mathscr{W}^2(H)$. When $Cf \neq 0$, we can take $g = \frac{Cf}{\|Cf\|}$, then $Cf \in \mathscr{D}(B)$, $\|g\| = 1$ and

$$\det(H_{f,g}) = \det\left(\begin{bmatrix} 0 & \frac{1}{\|Cf\|}(BCf, f) \\ \frac{1}{\|Cf\|}(Cf, Cf) & 0 \end{bmatrix}\right) = (BCf, f) = 0.$$

So $0 \in \mathscr{W}^2(H)$. About the proof of $\{\lambda \in \mathbb{C} : \lambda^2 \in W(CB)\} \subset \mathscr{W}^2(H)$ is completely similar. □

According to the definition of quadratic numerical range, the quadratic numerical range of a diagonal dominant upper triangular infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$$

is

$$\mathscr{W}^2(H) = W(A) \cup W(-A^*).$$

However, the quadratic numerical range of a diagonal dominant upper triangular infinite-dimensional Hamiltonian operator is not symmetric about the imaginary axis in general.

Example 5.4.10 Let $X = L^2[0, \infty)$, $A = \frac{\mathrm{d}}{\mathrm{d}t}$, $Ax = \frac{\mathrm{d}x}{\mathrm{d}t}$,

$$\mathscr{D}(A) = \{x \in X : x \text{ is absolutely continuous}, x' \in X, x(0) = 0\}.$$

Then $A^* = -\frac{\mathrm{d}}{\mathrm{d}t}$,

$$\mathscr{D}(A^*) = \{x \in X : x \text{ is absolutely continuous}, x' \in X\}.$$

Set an infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix} = \begin{bmatrix} \frac{\mathrm{d}}{\mathrm{d}t} & 0 \\ 0 & \frac{\mathrm{d}}{\mathrm{d}t} \end{bmatrix},$$

then

$$\mathscr{W}^2(H) \subset \{\lambda \in \mathbb{C} : Re(\lambda) \leq 0\}.$$

In fact, for any $x(t) \in \mathscr{D}(A)$, $\|x(t)\| = 1$, we can see

$$\begin{aligned}(Ax, x) &= \int_0^{+\infty} x'(t)\overline{x}(t) \\ &= -\int_0^{+\infty} x(t)\overline{x}'(t) = -(x, Ax).\end{aligned}$$

So

$$W(A) \subset \mathtt{i}\mathbb{R}.$$

Similarly, for any $x(t) \in \mathscr{D}(A^*)$, $\|x(t)\| = 1$, we have

$$\begin{aligned}(A^*x, x) &= \int_0^{+\infty} x'(t)\overline{x}(t) \\ &= -|x(0)|^2 - \int_0^{+\infty} x(t)\overline{x}'(t) = -|x(0)|^2 - (x, A^*x).\end{aligned}$$

So

$$W(A^*) \subset \{\lambda \in \mathbb{C} : Re(\lambda) \leq 0\}.$$

To sum up, $\mathscr{W}^2(H) \subset \{\lambda \in \mathbb{C} : Re(\lambda) \leq 0\}$ is not symmetric about the imaginary axis.

Then, when is the quadratic numerical range of an infinite-dimensional Hamiltonian operator symmetric about the imaginary axis?

Theorem 5.4.11 *Let* $H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be an upper triangular infinite-dimensional Hamiltonian operator and satisfy* $\mathscr{D}(A) = \mathscr{D}(A^*)$. *Then* $\mathscr{W}^2(H)$ *is symmetric about the imaginary axis.*

Proof When $\mathscr{D}(A) = \mathscr{D}(A^*)$, $\lambda \in W(A)$ if and only if $\overline{\lambda} \in W(A^*)$, so considering that

$$\mathscr{W}^2(H) = W(A) \cup W(-A^*),$$

the conclusion is obvious. □

In the following, we discuss when the quadratic numerical range of a general infinite-dimensional Hamiltonian operator is symmetric about the imaginary axis.

Theorem 5.4.12 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X$ *be a diagonal dominant infinite-dimensional Hamiltonian operator. If* $\mathscr{D}(A) = \mathscr{D}(A^*)$ *is satisfied, then* $\mathscr{W}^2(H)$ *is symmetric about the imaginary axis.*

Proof Considering that $\lambda \in \mathscr{W}^2(H)$ if and only if $f, g \in \mathscr{D}(A)$, $\|f\| = \|g\| = 1$ such that

$$\begin{aligned}\det\begin{bmatrix}(Af,f)-\lambda & (Bg,f)\\ (Cf,g) & -(A^*g,g)-\lambda\end{bmatrix} &= \det\begin{bmatrix}(f,Af)-\overline{\lambda} & (f,Bg)\\ (g,Cf) & -(g,A^*g)-\overline{\lambda}\end{bmatrix}\\ &= \det\begin{bmatrix}(A^*f,f)-\overline{\lambda} & -(Bf,g)\\ -(Cg,f) & -(Ag,g)-\overline{\lambda}\end{bmatrix}\\ &= \det\begin{bmatrix}(Ag,g)+\overline{\lambda} & (Bf,g)\\ (Cg,f) & -(A^*f,f)+\overline{\lambda}\end{bmatrix} = 0.\end{aligned}$$

So $\mathscr{W}^2(H)$ is symmetric about the imaginary axis. □

Proposition 5.4.13 *Let* $H = \begin{bmatrix}A & B\\ C & -A^*\end{bmatrix}$ *be a bounded infinite-dimensional Hamiltonian operator. Then* $\mathscr{W}^2(H)$ *is symmetric about the imaginary axis.*

5.4.3 Spectral Inclusion of Quadratic Numerical Range of Infinite-Dimensional Hamiltonian Operators

Firstly, we discuss the spectral inclusion properties of the quadratic numerical range of a diagonal infinite-dimensional Hamiltonian operator.

Theorem 5.4.14 *Given a diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix}A & 0\\ 0 & -A^*\end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X,$$

the following conclusions hold.

(i) $\sigma_{ap}(H) \subset \overline{\mathscr{W}^2(H)}$.

(ii) If $(\sigma_{r,1}(A) \cup \sigma_{r,1}(-A^*)) \subset (W(A) \cup W(-A^*))$, *then* $\sigma(H) \subset \overline{\mathscr{W}^2(H)}$.

Proof (i) Considering that

$$\sigma_{ap}(H) = \sigma_{ap}(A) \cup \sigma_{ap}(-A^*),$$

$$\mathscr{W}^2(H) = W(A) \cup W(-A^*),$$

and

$$\sigma_{ap}(A) \subset \overline{W(A)}, \sigma_{ap}(-A^*) \subset \overline{W(-A^*)},$$

the conclusion is obvious.

(ii) Because

$$\sigma(H) = \sigma_{ap}(H) \cup (\sigma_{r,1}(A) \cup \sigma_{r,1}(-A^*)).$$

Therefore, combined with (i), the conclusion is immediately proved. □

Secondly, the spectral inclusion properties of the quadratic numerical range of a skew-diagonal infinite-dimensional Hamiltonian operator are discussed.

Theorem 5.4.15 *given a skew-diagonal infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X,$$

then the following conclusions are satisfied:

(i) If B, C are invertible, then $\sigma_{ap}(H) \subset \overline{\mathscr{W}^2(H)}$.
(ii) If B, C are invertible and $\nabla_1 \subset \nabla_2$, then $\sigma(H) \subset \overline{\mathscr{W}^2(H)}$,

where

$$\nabla_1 = \{\lambda \in \mathbb{C} : \overline{\lambda^2} \in \sigma_{p,1}(CB) \cap \sigma_{p,1}(BC)\},$$
$$\nabla_2 = \{\lambda \in \mathbb{C} : \lambda^2 \in W(BC) \cup W(CB)\}.$$

Proof (i) If $\lambda \in \sigma_{ap}(H)$, then there exist $\begin{bmatrix} f_n & g_n \end{bmatrix}^T \in \mathscr{D}(H)$, $\|f_n\|^2 + \|g_n\|^2 = 1, n = 1, 2, 3, \ldots$, such that

$$-\lambda f_n + B g_n \to 0, n \to \infty, \tag{5.42}$$

and

$$C f_n - \lambda g_n \to 0, n \to \infty. \tag{5.43}$$

If B, C are invertible, then it can be asserted that $\liminf_{n\to\infty} \|f_n\| > 0$ and $\liminf_{n\to\infty} \|g_n\| > 0$. In fact, it is assumed that there exists a subsequence $\{f_{n_k}\}$ such that $\lim_{n\to\infty} \|f_{n_k}\| = 0$, then it is known by Eq. 5.42 that

$$\lim_{n\to\infty} \|B g_{n_k}\| = 0,$$

considering that B is invertible, $\lim_{n\to\infty} \|g_{n_k}\| = 0$, which is in contradiction with $\|f_{n_k}\|^2 + \|g_{n_k}\|^2 = 1$. Proving by the same methods $\liminf_{n\to\infty} \|g_n\| > 0$, we can assume that $f_n \neq 0, g_n \neq 0, n = 1, 2, 3, \ldots$, and set $\widehat{f_n} = \frac{f_n}{\|f_n\|}, \widehat{g_n} = \frac{g_n}{\|g_n\|}$, then

$$\begin{aligned} \det(H_{\widehat{f_n}, \widehat{g_n}} - \lambda I) &= \det\left(\begin{bmatrix} -\lambda & (B\widehat{g_n}, \widehat{f_n}) \\ (C\widehat{f_n}, \widehat{g_n}) & -\lambda \end{bmatrix} \right) \\ &= \lambda^2 - \frac{(Bg_n, f_n)(Cf_n, g_n)}{\|f_n\|^2 \|g_n\|^2} \to 0, \end{aligned}$$

so $\lambda \in \overline{\mathscr{W}^2(H)}$.

(ii) If $\nabla_1 \subset \nabla_2$, then $\sigma_{r,1}(H) \subset \mathscr{W}^2(H)$. It is immediately proved by Conclusion (i). □

Theorem 5.4.16 *Given a diagonal dominant upper triangular infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & B \\ 0 & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X,$$

then the following conclusions are satisfied.

(i) $\sigma_{ap}(H) \subset \overline{\mathscr{W}^2(H)}$.
(ii) If $\lambda \in \sigma_p(A) \cup \sigma_p(-A^*)$ *including* $-\overline{\lambda} \in W(A) \cup W(-A^*)$, *then* $\sigma(H) \subset \overline{\mathscr{W}^2(H)}$.

Proof (i) Considering that

$$\begin{aligned} \sigma_{ap}(H) \subset & \sigma_{ap}(A) \cup \sigma_{ap}(-A^*) \\ \subset & \overline{W(A)} \cup \overline{W(-A^*)} = \overline{\mathscr{W}^2(H)}, \end{aligned}$$

the conclusion is obvious.

(ii) If $\lambda \in \sigma_r(H)$, then $-\overline{\lambda} \in \sigma_p(-H^*) \subset (\sigma_p(A) \cup \sigma_p(-A^*))$. Thus, it can be known from the given conditions

$$\sigma_r(H) \subset \mathscr{W}^2(H).$$

Considering $\sigma(H) = \sigma_r(H) \cup \sigma_{ap}(H)$ and (i), the conclusions can be proved. □

Theorem 5.4.17 *Given a diagonal dominant infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(A) \times \mathscr{D}(A^*) \to X \times X,$$

if the relative bounds of B with respect to A^* *and C with respect to A are both 0, then* $\sigma_{ap}(H) \subset \overline{\mathscr{W}^2(H)}$.

Proof If $\lambda \in \sigma_{ap}(H)$, then there exist $\begin{bmatrix} f_n & g_n \end{bmatrix}^T \in \mathscr{D}(H)$, $\|f_n\|^2 + \|g_n\|^2 = 1, n = 1, 2, 3, \ldots$, such that when $n \to \infty$,

$$(A - \lambda) f_n + B g_n \to 0 \tag{5.44}$$

and

$$C f_n - (A^* + \lambda) g_n \to 0, \tag{5.45}$$

that is,

$$(A f_n, f_n) - \lambda (f_n, f_n) + (B g_n, f_n) \to 0 \tag{5.46}$$

and

$$(Cf_n, g_n) - (A^* g_n, g_n) - \lambda(g_n, g_n) \to 0. \tag{5.47}$$

Considering that the relative bounds of B with respect to A^* and the relative bounds of C with respect to A are both 0, the operator $S = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ is relatively bounded about H, so $\{Bg_n\}$ and $\{Cf_n\}$ are bounded, $\{(A^* + \lambda)g_n\}$ and $\{(A - \lambda)f_n\}$ are also bounded.

When $\liminf_{n\to\infty} \|f_n\| > 0$ and $\liminf_{n\to\infty} \|g_n\| > 0$, set $\widehat{f_n} = \frac{f_n}{\|f_n\|}, \widehat{g_n} = \frac{g_n}{\|g_n\|}$, then

$$\begin{aligned} \det(H_{\widehat{f_n},\widehat{g_n}} - \lambda I) &= \det\left(\begin{bmatrix} (A\widehat{f_n}, \widehat{f_n}) - \lambda & (B\widehat{g_n}, \widehat{f_n}) \\ (C\widehat{f_n}, \widehat{g_n}) & -(A^*\widehat{g_n}, \widehat{g_n}) - \lambda \end{bmatrix}\right) \\ &= \frac{1}{\|f_n\|^2\|g_n\|^2} \det\left(\begin{bmatrix} (Af_n, f_n) - \lambda(f_n, f_n) & (Bg_n, f_n) \\ (Cf_n, g_n) & -(A^*g_n, g_n) - \lambda(g_n, g_n) \end{bmatrix}\right) \\ &\to 0, \end{aligned}$$

so $\lambda \in \overline{\mathscr{W}^2(H)}$.

When $\liminf_{n\to\infty} \|f_n\| > 0$ and $\liminf_{n\to\infty} \|g_n\| = 0$, considering that the relative bound of B with respect to A^* is 0.

$$\liminf_{n\to\infty} \|Bg_n\| = 0,$$

derived from Eq. (5.44)

$$\liminf_{n\to\infty} \|(A - \lambda)f_n\| = 0,$$

that is, $\lambda \in \sigma_{ap}(A) \subset \overline{W(A)} \subset \overline{\mathscr{W}^2(H)}$.

When $\liminf_{n\to\infty} \|f_n\| = 0$ and $\liminf_{n\to\infty} \|g_n\| > 0$, the proof is completely similar. □

Theorem 5.4.18 *Given a skew-diagonal dominant infinite-dimensional Hamiltonian operator*

$$H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(C) \times \mathscr{D}(B) \to X \times X,$$

if the relative bounds of A^* with respect to B and A with respect to C are both 0 and B, C are invertible, then $\sigma_{ap}(H) \subset \mathscr{W}^2(H)$.

Proof If $\lambda \in \sigma_{ap}(H)$, then there exist $\begin{bmatrix} f_n & g_n \end{bmatrix}^T \in \mathscr{D}(H)$, $\|f_n\|^2 + \|g_n\|^2 = 1, n = 1, 2, 3, \ldots$, such that when $n \to \infty$,

$$(A - \lambda)f_n + Bg_n \to 0 \tag{5.48}$$

and

$$Cf_n - (A^* + \lambda)g_n \to 0, \tag{5.49}$$

that is,

$$(Af_n, f_n) - \lambda(f_n, f_n) + (Bg_n, f_n) \to 0 \tag{5.50}$$

and

$$(Cf_n, g_n) - (A^*g_n, g_n) - \lambda(g_n, g_n) \to 0. \tag{5.51}$$

Considering that the relative bounds of A^* with respect to B and A with respect to C are both 0, so the operator $S = \begin{bmatrix} A - \lambda I & 0 \\ 0 & -A^* - \lambda I \end{bmatrix}$ is bounded about H, so $\{(A^* + \lambda)g_n\}$ and $\{(A - \lambda)f_n\}$ are bounded, thus $\{Bg_n\}$ and $\{Cf_n\}$ are also bounded. It can be asserted that $\liminf_{n\to\infty} \|f_n\| > 0$ and $\liminf_{n\to\infty} \|g_n\| > 0$. In fact, assuming that $\liminf_{n\to\infty} \|f_n\| = 0$, then there exists a subsequence $\{f_{n_k}\}$ such that

$$\lim_{k\to\infty} \|f_{n_k}\| = 0.$$

Let $h_n = (A - \lambda)f_n + Bg_n$, then when $k \to \infty$,

$$g_{n_k} = B^{-1}h_{n_k} - B^{-1}(A - \lambda I)f_{n_k} \to 0.$$

This is in contradiction with $\liminf_{n\to\infty} \|g_n\| > 0$. Thus similar to Theorem 5.4.17, the conclusion can be proved to hold. □

Remark 5.4.19 In the above theorem, if B, C are not invertible operators, then $\sigma_{ap}(H) \subset \overline{\mathscr{W}^2(H)}$ does not necessarily hold. For example, let

$$H = \begin{bmatrix} I & B \\ 0 & -I \end{bmatrix},$$

where B is an unbounded operator in space X. Then it is obtained by calculation that

$$\sigma(H) = \sigma_{ap}(H) = \mathbb{C}.$$

However,

$$\overline{\mathscr{W}^2(H)} = \{-1, 1\},$$

that is, $\sigma_{ap}(H) \subset \overline{\mathscr{W}^2(H)}$ does not hold.

5.4.4 Dichotomy and Connectedness of the Quadratic Numerical Range of Infinite-Dimensional Hamiltonian Operators

If the spectra of a linear operator have no intersection with the imaginary axis, then it is said that the spectra of the linear operator are dichotomous. When the spectra of a linear operator are dichotomous, its spectral set can be decomposed into two subsets σ_1, σ_2 such that

$$\sigma_1 \subset C_+, \sigma_2 \subset C_-,$$

where C_-, C_+ denote the left and right open half-planes, respectively. The dichotomy of the operator's spectra can characterize the exponential dichotomy of differential equations. For example, we consider the evolution equation in Hilbert (or Banach space)

$$u'(t) = Au(t), t \in [0, +\infty). \tag{5.52}$$

If there is an invariant subspace L_1, L_2 of A such that $X = L_1 \oplus L_2$ and for $u(t) \in L_2$ (or $u(t) \notin L_2$), when $t \to \infty$, the exponential decay (or exponential increase) of $\|u(t)\|$, the evolution equation is said to have exponential dichotomy. If A is a bounded dichotomous operator, then L_1, L_2 can be selected as the spectral subspace corresponding to the spectral sets σ_1, σ_2, that is,

$$L_1 = \mathscr{R}(P_{\sigma_1}), L_2 = \mathscr{R}(P_{\sigma_2}),$$

where

$$P_{\sigma_i} = -\frac{1}{2\pi \mathrm{i}} \int_{\Gamma_i} (A - zI)^{-1} \mathrm{d}z, i = 1, 2$$

represent the Riesz mapping, $\sigma(A) = \sigma_1 \dot{\cup} \sigma_2$.

When the quadratic numerical range of an infinite-dimensional Hamiltonian operator has the property of spectral inclusion, the dichotomy of the quadratic numerical range implies the dichotomy of the spectra of an infinite-dimensional Hamiltonian operator. Therefore, we will discuss the dichotomy of the quadratic numerical range of an infinite-dimensional Hamiltonian operator.

Theorem 5.4.20 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be a bounded infinite-dimensional Hamiltonian operator. If* $\dim X \geq 2$ *and*

$$dist(W(A), W(-A^*)) > \sqrt{2}\sqrt{w(BC) + w(B)w(C)},$$

then $\mathscr{W}^2(H)$ *consists of two disjoint subsets.*

Proof Assume that $W(A)$, $W(-A^*)$ are located on both sides of the line L and let $dist(W(A), L) = dist(W(-A^*), L) = a$. Any $\lambda \in L$ is taken, then it can be seen from Lemma 5.3.43 that for any $\|f\| = \|g\| = 1$,

$$\begin{aligned}|\det(H_{f,g} - \lambda I)| &= |((Af, f) - \lambda)((-A^*g, g) - \lambda) - (Bg, f)(Cf, g)| \\ &\geq a^2 - |(Bg, f)(f, Cg)| \\ &\geq [\frac{dist(W(A), W(-A^*))}{2}]^2 - \frac{1}{2}(\|Bg\|\|Cg\| + |(Bg, Cg)|) \\ &\geq [\frac{dist(W(A), W(-A^*))}{2}]^2 - \frac{1}{2}(\|B\|\|C\| + w(BC)) \\ &> 0.\end{aligned}$$

So $L \cap \mathscr{W}^2(H) = \emptyset$ and consider that $W(A) \cup W(-A^*) \subset \mathscr{W}^2(H)$, the conclusion holds immediately. □

The following conclusion is obvious.

Proposition 5.4.21 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be a bounded infinite-dimensional Hamiltonian operator. If* $\dim X \geq 2$ *and*

$$dist(W(A), i\mathbb{R}) > \frac{\sqrt{2}}{2}\sqrt{w(BC) + w(B)w(C)},$$

then $\mathscr{W}^2(H) \cap i\mathbb{R} = \emptyset$.

Considering Lemma 5.3.40, a more general condition for $\mathscr{W}^2(H)$ to be dichotomous is obtained.

Theorem 5.4.22 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be a bounded infinite-dimensional Hamiltonian operator. If* $\dim X \geq 2$ *and there is* $r \geq 1$ *and* $0 \leq \alpha \leq 1$ *such that*

$$dist(W(A), W(-A^*)) > 2^{\frac{2r-1}{2r}}[w(|B|^{4r\alpha} + |C|^{4r(1-\alpha)})w(|C|^{4r\alpha} + |B|^{4r(1-\alpha)})]^{\frac{1}{4r}},$$

then $\mathscr{W}^2(H)$ *consists of two disjoint subsets.*

Proof Considering Lemma 5.3.40, it can be obtained that

$$\begin{aligned}|(Bg, f)(Cf, g)| &\leq \frac{1}{2}[|(Bg, f)|^2 + |(Cf, g)|^2] \\ &\leq \frac{1}{2}[(|B|^{2\alpha}g, g)(|B|^{2-2\alpha}f, f) + (|C|^{2\alpha}f, f)(|C|^{2-2\alpha}g, g)] \\ &\leq \frac{1}{2}[(|B|^{2\alpha}g, g)^2 + (|C|^{2-2\alpha}g, g)^2]^{\frac{1}{2}}[(|C|^{2\alpha}f, f)^2 + (|B|^{2-2\alpha}f, f)^2]^{\frac{1}{2}} \\ &\leq [\frac{(|B|^{4\alpha}g, g) + (|C|^{4-4\alpha}g, g)}{2}]^{\frac{1}{2}}[\frac{(|C|^{4\alpha}f, f) + (|B|^{4-4\alpha}f, f)}{2}]^{\frac{1}{2}}\end{aligned}$$

$$
\begin{aligned}
&\le [\frac{(|B|^{4\alpha}g,g)^r+(|C|^{4-4\alpha}g,g)^r}{2}]^{\frac{1}{2r}}[\frac{(|C|^{4\alpha}f,f)^r+(|B|^{4-4\alpha}f,f)^r}{2}]^{\frac{1}{2r}}\\
&\le [\frac{(|B|^{4r\alpha}g,g)+(|C|^{4r-4r\alpha}g,g)}{2}]^{\frac{1}{2r}}[\frac{(|C|^{4r\alpha}f,f)+(|B|^{4r-4r\alpha}f,f)}{2}]^{\frac{1}{2r}}\\
&\le (\frac{1}{2})^{\frac{1}{r}}((|B|^{4r\alpha}+|C|^{4r(1-\alpha)})g,g)^{\frac{1}{2r}}((|C|^{4r\alpha}+|B|^{4r(1-\alpha)})g,g)^{\frac{1}{2r}}\\
&\le (\frac{1}{2})^{\frac{1}{r}}[\boldsymbol{w}(|B|^{4r\alpha}+|C|^{4r(1-\alpha)})\boldsymbol{w}(|C|^{4r\alpha}+|B|^{4r(1-\alpha)})]^{\frac{1}{2r}}.
\end{aligned}
$$

Without loss of generality, assume that $W(A)$, $W(-A^*)$ be located on both sides of the line L and $dist(W(A),L)=dist(W(-A^*),L)=a$. Take $\lambda\in L$, then for any $\|f\|=\|g\|=1$, we can have

$$
\begin{aligned}
|\det(H_{f,g}-\lambda I)| &= |((Af,f)-\lambda)((-A^*g,g)-\lambda)-(Bg,f)(Cf,g)|\\
&\ge a^2-|(Bg,f)(f,Cg)|\\
&\ge a^2-(\frac{1}{2})^{\frac{1}{r}}[\boldsymbol{w}(|B|^{4r\alpha}+|C|^{4r(1-\alpha)})\boldsymbol{w}(|C|^{4r\alpha}+|B|^{4r(1-\alpha)})]^{\frac{1}{2r}}\\
&> 0.
\end{aligned}
$$

So $L\cap\mathscr{W}^2(H)=\emptyset$ and considering that $W(A)\cup W(-A^*)\subset\mathscr{W}^2(H)$, the conclusion holds immediately. □

In general, the quadratic numerical range $\mathscr{W}^2(H)$ of an infinite-dimensional Hamiltonian operator is composed of two subsets $\wedge_-(f,g)$ and $\wedge_+(f,g)$, that is,

$$\mathscr{W}^2(H)=\wedge_+(f,g)\cup\wedge_-(f,g),$$

where

$$\wedge_\pm(f,g)=\{\lambda\in\mathbb{C}:\lambda=\frac{1}{2}[(Af,f)-(A^*g,g)\pm\sqrt{\triangle_{f,g}}e^{\frac{\mathrm{i}}{2}\theta_{f,g}}]\},$$

$\theta_{f,g}$ represents the argument principal value of the complex number $\triangle_{f,g}$,

$$\triangle_{f,g}=[(Af,f)+(A^*g,g)]^2+4(Bg,f)(f,Cg).$$

If $A,B,C\in\mathscr{B}(X)$, then $\bigwedge_-(f,g)$ and $\bigwedge_+(f,g)$ are connected, respectively. It is worth noting that even if $H_{f,g}$ have two different eigenvalues, which does not mean that $\mathscr{W}^2(H)$ consists of two disjoint subsets. For example, let

$$A=\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix},\ B=\begin{bmatrix}1&0&0\\0&1&1\\0&1&1\end{bmatrix},$$

and take an infinite-dimensional Hamiltonian operator

$$H = \begin{bmatrix} A & B \\ B & -A \end{bmatrix}.$$

Then it is easy to obtain that 0 is the eigenvalue of A and for any $f_0 \in \mathcal{N}(A)$, we can have $(Bf_0, f_0) \neq 0$. Therefore, for any unit vectors f, g,

$$\wedge_+(f, g) - \wedge_-(f, g) = 2\sqrt{(\frac{(Af, f) + (Ag, g)}{2})^2 - |(Bg, f)|^2} > 0.$$

On the other hand, take

$$f_1 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T,$$

$$f_2 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T,$$

$$f_3 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T,$$

then

$$\wedge_+(f_2, f_1) = \wedge_-(f_1, f_3) = 0.$$

So $\mathcal{W}^2(H)$ is connected.

The following will discuss when the quadratic numerical range $\mathcal{W}^2(H)$ of an infinite-dimensional Hamiltonian operator is connected.

Theorem 5.4.23 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathcal{D}(H) \subset X \times X \to X \times X$ *be an infinite-dimensional Hamiltonian operator. If one of the following conditions is satisfied:*

(i) There exist unit vectors $f, g \in X$ *such that* $[(Af, f) + (A^*g, g)]^2 + 4(Bg, f)(Cf, g) = 0$.
(ii) $\mathcal{N}(C) \cap \Sigma_1 \neq \emptyset$, *where* $\Sigma_1 = \{f : (Af, f) \in W(-A^*), \|f\| = 1\}$.
(iii) $\mathcal{N}(B) \cap \Sigma_2 \neq \emptyset$, *where* $\Sigma_2 = \{g : -(A^*f, f) \in W(A), \|g\| = 1\}$,

then $\mathcal{W}^2(H)$ *is a connected set.*

Proof If it satisfies condition (i), then there exist unit vectors $f, g \in X$ such that

$$\wedge_+(f, g) = \wedge_-(f, g).$$

So $\mathcal{W}^2(H)$ is a connected set.

(ii) If $\mathcal{N}(C) \cap \Sigma_1 \neq \emptyset$, then there exists a unit vector f such that $Cf = 0$ and $(Af, f) \in W(-A^*)$. Then there exists a unit vector g such that

$$(Af, f) + (A^*g, g) = 0.$$

Thus it satisfies the condition (i).

The proof of (iii) is completely similar to (ii). □

Theorem 5.4.24 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be a bounded infinite-dimensional Hamiltonian operator. If one of the following conditions is satisfied:*

(i) $\wedge_+(f, g) \cap \mathbb{R} \neq \emptyset$;
(ii) $\wedge_-(f, g) \cap \mathbb{R} \neq \emptyset$,

then $\mathscr{W}^2(H)$ *is a connected set.*

Proof We only need to prove (i), and the proof of (ii) is exactly similar. When $A, B, C \in \mathscr{B}(X)$, it is easy to attain

$$\overline{\wedge_\pm(f, g)} = \wedge_\pm(g, f).$$

Thus, when $\wedge_+(f, g) \cap \mathbb{R} \neq \emptyset$, we assume $\wedge_+(f_0, g_0) \in \mathbb{R}$, $\|f_0\| = \|g_0\| = 1$, then

$$\wedge_+ (f_0, g_0) = \wedge_-(g_0, f_0).$$

Then $\mathscr{W}^2(H)$ is a connected set. □

5.5 Quadratic Numerical Radius of Infinite-Dimensional Hamiltonian Operator

5.5.1 Quadratic Numerical Radius of Bounded 2 × 2 Block Operator Matrices

Similar to the numerical radius of the bounded linear operators, the definition of the quadratic numerical radius of a bounded 2×2 block operator matrix can be introduced.

Definition 5.5.1 Given a bounded 2×2 block operator matrix $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, where $A, B, C, D \in \mathscr{B}(X)$, then

$$\boldsymbol{w}_2(\mathscr{A}) = \sup\{|\lambda| : \lambda \in \mathscr{W}^2(\mathscr{A})\}$$

is called the quadratic numerical radius of a block operator matrix $\mathscr{A}$.

Remark 5.5.2 For a bounded 2×2 block operator matrix $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, it is obvious that

$$r(\mathscr{A}) \leq \boldsymbol{w}_2(\mathscr{A}) \leq \boldsymbol{w}(\mathscr{A})$$

holds.

The following is an example of a quadratic numerical radius.

Example 5.5.3 Let X be the Hilbert space, I be a unit operator in X, and define a block operator matrix in Hilbert space $X \times X$

$$\mathscr{A} = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix}.$$

Then $\mathscr{W}^2(\mathscr{A}) = \{0\}$ and then $\boldsymbol{w}_2(\mathscr{A}) = 0$.

Example 5.5.4 Let X be the Hilbert space, I be a unit operator in X, B be a bounded linear operator in X, and define a block operator matrix in Hilbert space $X \times X$,

$$\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} I & B \\ 0 & -I \end{bmatrix}.$$

Then $\boldsymbol{w}_2(\mathscr{A}) = 1$.

The quadratic numerical radius remains unchanged under adjoint operation and special unitary similarity transformation.

Corollary 5.5.5 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ *be a bounded* 2×2 *block operator matrix in Hilbert space* $X \times X$. *Then*

(i) $\boldsymbol{w}_2(\mathscr{A}) = \boldsymbol{w}_2(U^*\mathscr{A}U)$, *where* $U = \begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix}$, U_1, U_2 *are the unitary operators;*

(ii) $\boldsymbol{w}_2(\mathscr{A}) = \boldsymbol{w}_2(U^*\mathscr{A}U)$, *where* $U = \begin{bmatrix} 0 & U_1 \\ U_2 & 0 \end{bmatrix}$, U_1, U_2 *are the unitary operators;*

(iii) $\boldsymbol{w}_2(\mathscr{A}) = \boldsymbol{w}_2(\mathscr{A}^*) = \boldsymbol{w}_2(\alpha\mathscr{A})$, *where* $|\alpha| = 1$.

The following is about the properties of the quadratic numerical radius of some 2×2 block operator matrices.

Lemma 5.5.6 *If* $A, B, C, D \in \mathscr{B}(X)$, *then*

(i) $\boldsymbol{w}_2\left(\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}\right) = \boldsymbol{w}_2\left(\begin{bmatrix} A & 0 \\ C & D \end{bmatrix}\right) = \boldsymbol{w}_2\left(\begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix}\right) = \max\{\boldsymbol{w}(A), \boldsymbol{w}(D)\}$;

(ii) $\boldsymbol{w}_2\left(\begin{bmatrix} 0 & B \\ e^{i\theta}C & 0 \end{bmatrix}\right) = \boldsymbol{w}_2\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)$;

(iii) $\boldsymbol{w}_2\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) = \boldsymbol{w}_2\left(\begin{bmatrix} 0 & C \\ B & 0 \end{bmatrix}\right)$;

(iv) $\boldsymbol{w}_2\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}\right) = \max\{\boldsymbol{w}(A+B), \boldsymbol{w}(A-B)\}$;

(v) $\boldsymbol{w}_2\left(\begin{bmatrix} A & B \\ -B & A \end{bmatrix}\right) = \max\{\boldsymbol{w}(A+iB), \boldsymbol{w}(A-iB)\}$;

(vi) $\boldsymbol{w}_2\left(\begin{bmatrix} 0 & B \\ B & 0 \end{bmatrix}\right) = \boldsymbol{w}\left(\begin{bmatrix} 0 & B \\ B & 0 \end{bmatrix}\right) = \boldsymbol{w}(B).$

Proof (i) When $\mathscr{A} = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix}$ or $\mathscr{A} = \begin{bmatrix} A & 0 \\ C & D \end{bmatrix}$ or $\mathscr{A} = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix}$, it is easy to prove that

$$\mathscr{W}^2(\mathscr{A}) = W(A) \cup W(D),$$

so we have $\boldsymbol{w}_2(\mathscr{A}) = \max\{\boldsymbol{w}(A), \boldsymbol{w}(D)\}$.

(ii) Selecting a unitary matrix $U = \begin{bmatrix} I & 0 \\ 0 & e^{\frac{\mathrm{i}\theta}{2}} I \end{bmatrix}$, then

$$U^* \begin{bmatrix} 0 & B \\ e^{\mathrm{i}\theta} C & 0 \end{bmatrix} U = e^{\frac{\mathrm{i}\theta}{2}} \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix},$$

by the property Corollary 5.5.5, the conclusion is proved.

(iii) Selecting a unitary matrix $U = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$, then

$$U^* \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} U = \begin{bmatrix} 0 & C \\ B & 0 \end{bmatrix},$$

by the property Corollary 5.5.5, the conclusion is proved.

(iv) Firstly, it is known by the property Corollary 5.3.5,

$$\boldsymbol{w}\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}\right) = \max\{\boldsymbol{w}(A + B), \boldsymbol{w}(A - B)\},$$

so $\boldsymbol{w}_2\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}\right) \leq \max\{\boldsymbol{w}(A + B), \boldsymbol{w}(A - B)\}$. On the other hand, according to the definition of quadratic numerical radius,

$$\begin{aligned} \boldsymbol{w}_2\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}\right) &= \sup_{\|f\|=\|g\|=1} \frac{|(Af, f) + (Ag, g) \pm \sqrt{[(Af, f) - (Ag, g)]^2 + 4(Bg, f)(Bf, g)}|}{2} \\ &\geq \sup_{\|f\|=1} |((A \pm B)f, f)| = \boldsymbol{w}(A \pm B). \end{aligned}$$

So the conclusion holds.

(v) It is known by the property Corollary 5.3.5 that

$$\boldsymbol{w}\left(\begin{bmatrix} A & B \\ -B & A \end{bmatrix}\right) = \max\{\boldsymbol{w}(A + \mathrm{i}B), \boldsymbol{w}(A - \mathrm{i}B)\},$$

so $\boldsymbol{w}_2\left(\begin{bmatrix} A & B \\ -B & A \end{bmatrix}\right) \leq \max\{\boldsymbol{w}(A + \mathrm{i}B), \boldsymbol{w}(A - \mathrm{i}B)\}$. On the other hand, according to the definition of quadratic numerical radius

$$w_2\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}\right) = \sup_{\|f\|=\|g\|=1} \frac{|(Af,f)+(Ag,g) \pm \sqrt{[(Af,f)-(Ag,g)]^2 - 4(Bg,f)(Bf,g)|}}{2}$$
$$\geq \sup_{\|f\|=1} |((A \pm \mathrm{i}B)f,f)| = w(A \pm B).$$

So the conclusion holds.

(vi) If $A = 0$, then it can be obtained $w_2\left(\begin{bmatrix} 0 & B \\ B & 0 \end{bmatrix}\right) = w\left(\begin{bmatrix} 0 & B \\ B & 0 \end{bmatrix}\right) = w(B)$. The proof is completed. □

There are significant differences between the properties of the quadratic numerical radius and the properties of the numerical radius. For example, the quadratic numerical radius does not have a triangular inequality, so it is not a norm in general.

Example 5.5.7 Taking

$$\mathscr{A}_1 = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix}, \mathscr{A}_2 = \begin{bmatrix} 0 & 0 \\ I & 0 \end{bmatrix},$$

then $w_2(\mathscr{A}_1) = w_2(\mathscr{A}_2) = 0$ and $w_2(\mathscr{A}_1 + \mathscr{A}_2) = 1$ does not have a triangular inequality.

A very important property of the numerical radius is the power inequality. The quadratic numerical radius of some special 2×2 block operator matrices also has the power inequality.

Theorem 5.5.8 *If $A, B, C, D \in \mathscr{B}(X)$, then*

(i) $w_2\left(\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}^n\right) \leq [w_2\left(\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}\right)]^n, n = 1, 2, \ldots;$

(ii) $w_2\left(\begin{bmatrix} A & 0 \\ C & D \end{bmatrix}^n\right) \leq [w_2\left(\begin{bmatrix} A & 0 \\ C & D \end{bmatrix}\right)]^n, n = 1, 2, \ldots;$

(iii) $w_2\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}^n\right) \leq [w_2\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)]^n, n = 2k, k = 1, 2, \ldots;$

(iv) $w_2\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}^n\right) \leq [w_2\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}\right)]^n, n = 1, 2, \ldots;$

(v) $w_2\left(\begin{bmatrix} A & B \\ -B & A \end{bmatrix}^n\right) \leq [w_2\left(\begin{bmatrix} A & B \\ -B & A \end{bmatrix}\right)]^n, n = 1, 2, \ldots$

Proof (i) From Lemma 5.5.6 and the numerical radius power inequality, it can be known that

$$w_2\left(\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}^n\right) = w_2\left(\begin{bmatrix} A^n & * \\ 0 & D^n \end{bmatrix}\right)$$
$$= \max\{w(A^n), w(D^n)\}$$
$$\leq \max\{[w(A)]^n, [w(D)]^n\} = [w_2\left(\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}\right)]^n.$$

The proof of (ii) is completely similar to (i).

(iii) Firstly,

$$\begin{aligned}[\boldsymbol{w}_2\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)]^{2k} &= \sup_{\|f\|=\|g\|=1} |(Bg, f)(Cf, g)|^k \\ &\geq \max\{[\boldsymbol{w}(BC)]^k, [\boldsymbol{w}(CB)]^k\} \\ &\geq \max\{\boldsymbol{w}((BC)^k), \boldsymbol{w}((CB)^k)\}.\end{aligned}$$

On the other hand,

$$\begin{aligned}\boldsymbol{w}_2\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}^{2k}\right) &= \boldsymbol{w}_2\left(\begin{bmatrix} (BC)^k & 0 \\ 0 & (CB)^k \end{bmatrix}\right) \\ &= \max\{[\boldsymbol{w}(BC)]^k, [\boldsymbol{w}(CB)]^k\}.\end{aligned}$$

So the conclusion holds.

(iv) Selecting a unitary matrix $U = \frac{1}{\sqrt{2}}\begin{bmatrix} I & -I \\ I & I \end{bmatrix}$, then

$$\begin{aligned}\boldsymbol{w}_2\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}^n\right) &\leq \boldsymbol{w}\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}^n\right) \\ &= \boldsymbol{w}\left(U^*\begin{bmatrix} A & B \\ B & A \end{bmatrix}^n U\right) \\ &= \boldsymbol{w}((U^*\begin{bmatrix} A & B \\ B & A \end{bmatrix} U)^n) \\ &\leq \max\{[\boldsymbol{w}(A+B)]^n, [\boldsymbol{w}(A-B)]^n\} = [\boldsymbol{w}_2\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}\right)]^n.\end{aligned}$$

The proof of (v) is completely similar to that of (iv). □

Remark 5.5.9 When B, C are bounded, the spectral radii of BC, CB are equal, but $\boldsymbol{w}(BC), \boldsymbol{w}(CB)$ are not necessarily equal. For example, letting

$$B = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix},\ C = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix},$$

then $\boldsymbol{w}(BC) = 0$ and $\boldsymbol{w}(CB) = \frac{1}{2}$.

5.5.2 Estimate of Quadratic Numerical Radius of Bounded Infinite-Dimensional Hamiltonian Operators

Firstly, the quadratic numerical radius estimate of a skew-diagonal bounded infinite-dimensional Hamiltonian operator is given.

Theorem 5.5.10 *Let* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}$ *be a bounded skew-diagonal infinite-dimensional Hamiltonian operator. Then*

$$w_2(H) \geq \sqrt{w(BC)} = \sqrt{w(CB)}$$

and

$$w_2(H) \leq \sqrt{\frac{w(B)w(C) + w(BC)}{2}} = \sqrt{\frac{w(B)w(C) + w(CB)}{2}}.$$

Proof Considering that B, C are bounded self-adjoint operators, $w(BC) = w(CB)$ are obvious. Firstly, the first inequality is proved.

$$\begin{aligned} w_2(H) &= \sup_{\|f\|=\|g\|=1} \sqrt{|(Bg, f)(Cf, g)|} \\ &\geq \sup_{\|f\|=1, g=\frac{Cf}{\|Cf\|}} \sqrt{|(BCf, f)|} \\ &= \sqrt{w(BC)}. \end{aligned}$$

Secondly, the second inequality is proved. Considering Lemma 5.3.43,

$$\begin{aligned} w_2(H) &= \sup_{\|f\|=\|g\|=1} \sqrt{|(Bg, f)(Cf, g)|} \\ &= \sup_{\|f\|=\|g\|=1} \sqrt{|(Bg, f)(f, C^*g)|} \\ &\leq \sup_{\|f\|=\|g\|=1} \sqrt{\frac{\|Bg\|\|C^*f\| + |(Bg, C^*f)|}{2}} \\ &\leq \sqrt{\frac{\|B\|\|C\| + w(CB)}{2}} = \sqrt{\frac{w(B)w(C) + w(CB)}{2}}. \end{aligned}$$

The proof is completed. □

Theorem 5.5.11 *Let* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *be a bounded infinite-dimensional Hamiltonian operator. Then*

$$w_2(H) \geq \max\{w(A), \sqrt{|[w(A)]^2 - w(B)w(C)|}\}$$

and

$$w_2(H) \le \min\{2w(A) + \sqrt{\frac{w(B)w(C) + w(BC)}{2}}, w(A) + w(B) + w(C)\}.$$

Proof The relation $w_2(H) \ge w(A)$ is obvious. Next, it is to prove that $w_2(H) \ge \sqrt{|[w(A)]^2 - w(B)w(C)|}$. Taking

$$a = \sup_{\|f\|=\|g\|=1} \frac{|(Af, f) - (A^*g, g) + \sqrt{[(Af, f) + (A^*g, g)]^2 + 4(Bg, f)(Cf, g)|}}{2},$$

$$b = \sup_{\|f\|=\|g\|=1} \frac{|(Af, f) - (A^*g, g) - \sqrt{[(Af, f) + (A^*g, g)]^2 + 4(Bg, f)(Cf, g)|}}{2},$$

then

$$\begin{aligned} w_2(H) &= \max\{a, b\} \\ &\ge \sqrt{ab} \\ &\ge \sqrt{\sup_{\|f\|=\|g\|=1} |(Af, f)(A^*g, g) - (Bg, f)(Cf, g)|} \\ &\ge \sqrt{|\sup_{\|f\|=\|g\|=1} |(Af, f)(A^*g, g)| - \sup_{\|f\|=\|g\|=1} |(Bg, f)(Cf, g)||} \\ &= \sqrt{|\sup_{\|f\|=1} |(Af, f)| \sup_{\|g\|=1} |(A^*g, g)| - \sup_{\|f\|=\|g\|=1} |(Bg, f)(Cf, g)||} \\ &\ge \sqrt{|[w(A)]^2 - w(B)w(C)|}. \end{aligned}$$

Secondly, we prove the second inequality. Considering the definition of quadratic numerical radius and Lemma 5.3.43 and Theorem 5.3.36, we obtain

$$\begin{aligned} w_2(H) &= \sup_{\|f\|=\|g\|=1} \frac{|(Af, f) - (A^*g, g) \pm \sqrt{[(Af, f) + (A^*g, g)]^2 + 4(Bg, f)(Cf, g)|}}{2} \\ &\le \frac{\sup_{\|f\|=1} |(Af, f)| + \sup_{\|g\|=1} |(A^*g, g)|}{2} \\ &+ \frac{\sup_{\|f\|=\|g\|=1} \sqrt{|(Af, f) + (A^*g, g)|^2 + 4|(Bg, f)(Cf, g)|}}{2} \\ &\le 2w(A) + \sqrt{\frac{w(B)w(C) + w(BC)}{2}}. \end{aligned}$$

On the other hand,

$$\begin{aligned}\boldsymbol{w}_2(H) \leq & \boldsymbol{w}(H)\\ \leq & \boldsymbol{w}\left(\begin{bmatrix} A & 0\\ 0 & -A^* \end{bmatrix}\right) + \boldsymbol{w}\left(\begin{bmatrix} 0 & B\\ C & 0 \end{bmatrix}\right)\\ \leq & \boldsymbol{w}(A) + \frac{1}{2}\min\{\boldsymbol{w}(B+C), \boldsymbol{w}(B-C)\} + \min\{\boldsymbol{w}(B), \boldsymbol{w}(C)\}\\ \leq & \boldsymbol{w}(A) + \boldsymbol{w}(B) + \boldsymbol{w}(C).\end{aligned}$$

So the conclusion is proved. □

As far as we know, there is the Pinching inequality for the numerical radius (see Eq. (5.24)), but the Pinching inequality for the quadratic numerical radius does not necessarily hold. In general, when $\dim X \geq 2$, considering that Lemma 5.4.6, we only have

$$\boldsymbol{w}_2\left(\begin{bmatrix} A & B\\ C & D \end{bmatrix}\right) \geq \boldsymbol{w}_2\left(\begin{bmatrix} A & 0\\ 0 & D \end{bmatrix}\right).$$

However, the inequality

$$\boldsymbol{w}_2\left(\begin{bmatrix} A & B\\ C & D \end{bmatrix}\right) \geq \boldsymbol{w}_2\left(\begin{bmatrix} 0 & B\\ C & 0 \end{bmatrix}\right)$$

does not necessarily hold (see the following example).

Example 5.5.12 If $\mathscr{A} = \begin{bmatrix} A & B\\ C & D \end{bmatrix} = \begin{bmatrix} \frac{\mathrm{i}}{2}I & I\\ I & -\frac{\mathrm{i}}{2}I \end{bmatrix}$, then

$$\mathscr{A}_{f,g} = \begin{bmatrix} \frac{\mathrm{i}}{2} & (g,f)\\ (f,g) & -\frac{\mathrm{i}}{2} \end{bmatrix},$$

so

$$\begin{aligned}\boldsymbol{w}_2(\mathscr{A}) = & \sup_{\|f\|=\|g\|=1} \sqrt{|(f,g)|^2 - \frac{1}{4}|}\\ = & \frac{\sqrt{3}}{2}.\end{aligned}$$

On the other hand,

$$\begin{aligned}\boldsymbol{w}_2\left(\begin{bmatrix} 0 & B\\ C & 0 \end{bmatrix}\right) = & \sup_{\|f\|=\|g\|=1} |(f,g)|\\ = & 1.\end{aligned}$$

So the Pinching inequality does not hold.

5.6 Essential Numerical Range of Infinite-Dimensional Hamiltonian Operators

As far as we know, the bounded linear operator T in an infinite-dimensional separable Hilbert space X is compact if and only if any orthogonal basis $\{e_n\}$ of X satisfies $\lim_{n\to\infty}(Te_n, e_n) = 0$. In this case, it is natural to ask the question: what conditions for the bounded linear operator T does satisfy then there exists an orthogonal basis $\{e_n\}$ of X such that

$$\lim_{n\to\infty}(Te_n, e_n) = 0? \tag{5.53}$$

Obviously, compact operators are included in this class of operators. For the non-compact operator T, there may be an orthogonal basis making (5.53) hold. For example, the right shift operator S_r in $l^2[1,\infty)$ space for the orthogonal basis $\{e_n = (0,\ldots,0,1,0,\ldots)\}_{n=1}^{+\infty}$, there is

$$\lim_{n\to\infty}(S_r e_n, e_n) = 0,$$

but S_r is not a compact operator. Therefore, to completely solve the above problems, Williams [79] and other people first introduced the concept of essential numerical range, and then Gustafson [80], Wolf [81–83], and other people carried out the systematic research.

5.6.1 Definition and Properties of Essential Numerical Range

Definition 5.6.1 Given a bounded linear operator T in Hilbert space X, and its essential numerical range $W_e(T)$ is defined as

$$W_e(T) = \bigcap_{K\in\mathscr{K}(X)} \overline{W(T+K)},$$

where $\mathscr{K}(X)$ denotes the ideal of all compact operators.

From the definition, it is not difficult to find that the essential numerical range is a non-empty convex and closed subset of the complex field $\mathbb{C}$ and satisfies $W_e(T) \subset \overline{W(T)}$. The following is an example of the essential numerical range.

Example 5.6.2 Let T be a bounded linear operator from l^2 space to l^2 space and be defined

$$Tx = \left(\frac{x_1}{1}, \frac{x_2}{2}, \frac{x_3}{3}, \ldots\right),$$

where $x = (x_1, x_2, x_3, \ldots)$, then $W_e(T) = \{0\}$. In fact, set $\{e_n\}_{n=1}^{+\infty}$ be a set of orthogonal basis of l^2, and note that T is a compact operator, so $T + K$ is also a compact operator for any $K \in \mathscr{K}(X)$, then when $n \to \infty$

$$\|(T+K)e_n\| \to 0.$$

Thus $((T+K)\frac{e_n}{\|e_n\|}, \frac{e_n}{\|e_n\|}) \to 0$, that is, $0 \in \bigcap_{K\in\mathscr{K}(X)} \overline{W(T+K)} = W_e(T)$.

On the other hand, take $K = -T$, then $\bigcap_{K\in\mathscr{K}(X)} \overline{W(T+K)} \subset \overline{W(T+(-T))} = \{0\}$. So, $W_e(T) = \{0\}$.

According to the definition, it is easy to obtain the following properties.

Corollary 5.6.3 *If T, S are bounded linear operators in a Hilbert space X, then*

(i) $W_e(\alpha T + \beta I) = \alpha W_e(T) + \beta$;
(ii) $\lambda \in W_e(T)$ *if and only if* $\overline{\lambda} \in W_e(T^*)$;
(iii) $W_e(U^*TU) = W_e(T)$, *where U is a unitary operator;*
(iv) $W_e(T+S) \subset W_e(T) + W_e(S)$;
(v) $W_e(T) = \{0\}$ *if and only if T is a compact operator;*
(vi) $W_e(T+K) = W_e(T)$, *where K is a compact operator.*

Proof (i) When $\alpha = 0$, we just need to prove $W_e(\beta I) = \{\beta\}$. Because $W_e(T) \subset W(T)$, the conclusion is obvious.

When $\alpha \neq 0$, set $\lambda \in W_e(\alpha T + \beta I)$, then for any $K \in \mathscr{K}(X)$, there exists $\{x_n\}_{n=1}^{\infty} \subset X$ and $\|x_n\| = 1$ such that

$$\begin{aligned}\Big((\alpha T + \beta I + K)x_n, x_n\Big) =& \alpha\Big((T + \frac{K}{\alpha})x_n, x_n\Big) + \beta \\ \to& \lambda,\end{aligned}$$

that is, $\lambda \in \alpha W_e(T) + \beta$, so $W_e(\alpha T + \beta I) \subset \alpha W_e(T) + \beta$.

Conversely, when $\lambda \in \alpha W_e(T) + \beta$, for any $K \in \mathscr{K}(X)$ there exists $\{x_n\}_{n=1}^{\infty} \subset X$ and $\|x_n\| = 1$ such that

$$\begin{aligned}\alpha\Big((T+K)x_n, x_n\Big) + \beta =& \Big((\alpha T + \beta I + \alpha K)x_n, x_n\Big) \\ \to& \lambda,\end{aligned}$$

that is, $\lambda \in W_e(\alpha T + \beta I)$, so $\alpha W_e(T) + \beta \subset W_e(\alpha T + \beta I)$.

(ii) Since T is a bounded linear operator in the whole space X, T^* is also a bounded linear operator in the whole space X. Therefore, for any $K \in \mathscr{K}(X)$, there exists $\{x_n\}_{n=1}^{\infty} \subset X$ and $\|x_n\| = 1$, we can see

$$\Big((T+K)x_n, x_n\Big) = \Big((T^* + K^*)x_n, x_n\Big)^*,$$

that is, $\lambda \in W_e(T)$ if and only if $\overline{\lambda} \in W_e(T^*)$.

(iii) U and U^* are invertible operators due to $UU^* = U^*U = I$. Therefore, when $\lambda \in W_e(U^*TU)$, for any $K \in \mathscr{K}(X)$, there exists $\{x_n\}_{n=1}^{\infty} \subset X$ and $\|x_n\| = 1$ such that

$$\begin{aligned}\Big((U^*TU+K)x_n,x_n\Big)&\\ =&\Big((T+(U^*)^{-1}KU^{-1})Ux_n,Ux_n\Big)\\ \to&\lambda.\end{aligned}$$

It is easy to know that $(U^*)^{-1}KU^{-1}$ is also a compact operator, and there is $\|Ux_n\|^2 = (x_n,x_n)=1$, that is, $\lambda\in W_e(T)$, thus $W_e(U^*TU)\subset W_e(T)$.

Conversely, when $\lambda\in W_e(T)$, due to $R(U)=X$, there exists $\{y_n\}$ such that $x_n = Uy_n$, and there is

$$\begin{aligned}\Big((T+K)x_n,x_n\Big)&=\Big((T+K)Uy_n,Uy_n\Big)\\ &=\Big((U^*TU+U^*KU)y_n,y_n\Big)\to\lambda.\end{aligned}$$

It is easy to know that U^*KU is a compact operator, because $y_n = U^{-1}x_n$, $\|y_n\|^2 = (U^{-1}x_n,U^{-1}x_n)=(x_n,x_n)=1$. This shows that $\lambda\in W_e(U^*TU)$, that is, $W_e(T)\subset W_e(U^*TU)$.

(iv) When $\lambda\in W_e(T+S)$, for any $K\in\mathscr{K}(X)$, there exists $\{x_n\}_{n=1}^{\infty}\subset X$ and $\|x_n\|=1$ such that

$$\Big((T+S+K)x_n,x_n\Big)\to\lambda,$$

that is,

$$\Big((T+\frac{K}{2})x_n,x_n\Big)+\Big((S+\frac{K}{2})x_n,x_n\Big)\to\lambda.$$

Because $\{A_n=\Big((T+\frac{K}{2})x_n,x_n\Big)\}_{n=1}^{\infty}$ is a bounded infinite sequence on the complex plane, there is a convergent subsequence. Assume that $k\to\infty$, $\Big((T+\frac{K}{2})x_{n_k},x_{n_k}\Big)\to\lambda_1$. Then $\lambda_1\in W_e(T)$, $\lambda-\lambda_1\in W_e(S)$, and thus there is $W_e(T+S)\subset W_e(T)+W_e(S)$.

(v) If T is the compact operator, then $\bigcap\limits_{K\in\mathscr{K}(X)}\overline{W(T+K)}\subset\overline{W(T+(-T))}=\{0\}$. So, $W_e(T)=\{0\}$.

On the contrary, let $W_e(T)=\bigcap\limits_{K\in\mathscr{K}(X)}\overline{W(T+K)}=\{0\}$. Since K is a compact operator if and only if for any orthogonal bases $\{e_n\}_{n=1}^{\infty}$ there are $\lim\limits_{n\to\infty}(Ke_n,e_n)=0$, according to the definition of essential numerical range, we can attain

$$\Big((T+K)e_n,e_n\Big)\in\bigcap_{K\in\mathscr{K}(X)}\overline{W(T+K)}=\{0\}.$$

Therefore, T is a compact operator.

(vi) This conclusion is easily obtained by applying the definition of an essential numerical range. □

In the following, we will give some equivalent descriptions of the essential numerical range [84, 85].

Theorem 5.6.4 *Let T be a bounded linear operator in a separable Hilbert space X. Then the following statements are equivalent to each other:*

(i) $0 \in W_e(T)$.
(ii) There exists a set of orthogonal vector group $\{e_n\}$ *in* X *such that* $\lim_{n\to\infty}(Te_n, e_n) = 0$.
(iii) There exists a set of orthogonal basis $\{e_n\}$ *of* X *such that* $\lim_{n\to\infty}(Te_n, e_n) = 0$.
(iv) There exists an infinite-dimensional projection operator P *such that* PTP *is a compact operator.*

Proof (i) → (ii): Taking $\varepsilon_n \to 0 (n \to \infty)$, assuming that there exist k orthogonal vectors $e_1, e_2, \ldots, e_k$ such that

$$(Te_k, e_k) < \varepsilon_k.$$

Let $M = Span\{e_1, e_2, \ldots, e_k\}$, $P : X \to M$ be an orthogonal projection operator. In order to prove that there is a vector $e_{k+1} \in M^{\perp}$ such that

$$(Te_{k+1}, e_{k+1}) < \varepsilon_{k+1},$$

we only need to prove that $0 \in \overline{W((I-P)T|_{M^{\perp}})}$. Setting $\mu \in W((I-P)T|_{M^{\perp}})$, defining the operator K as

$$K = \mu P - TP - PT + PTP,$$

then K is a finite rank operator, so it is also a compact operator and

$$T + K = \mu P + (I-P)T(I-P) = \mu I_M \oplus (I-P)T|_{M^{\perp}}.$$

Considering that $\overline{W(T+K)} = Conv(W(\mu I_M), W((I-P)T|_{M^{\perp}}))$, $W(\mu I_M) = \{\mu\} \subset W((I-P)T|_{M^{\perp}})$, it can be obtained that

$$\overline{W(T+K)} = \overline{W((I-P)T|_{M^{\perp}})}.$$

Therefore, $0 \in \overline{W((I-P)T|_{M^{\perp}})}$ is known by $0 \in W_e(T)$, that is, there exists a set of orthogonal vector group $\{e_n\}$ in X such that $\lim_{n\to\infty}(Te_n, e_n) = 0$.

(ii) → (iii): The orthogonal vector group in separable Hilbert space can be extended to an orthogonal basis, so the conclusion is obvious.

(iii)→ (iv): If $\{e_n\}$ be a set of orthogonal basis of X, then it can be known from $(Te_n, e_n) \to 0$ that there is a subsequence such that $\sum_{k=1}^{+\infty} |(Te_{n_k}, e_{n_k})|^2$ converges. So it is assumed that

$$\sum_{k=1}^{+\infty} |(Te_n, e_n)|^2 < \infty.$$

According to the Bessel inequality $\sum_{n=1}^{+\infty} |(x, e_n)|^2 \leq \|x\|^2$, and taking $n_1 = 1$, then we can have

$$\sum_{n=1}^{+\infty} |(Te_{n_1}, e_n)|^2 \leq \|Te_{n_1}\|^2, \ \sum_{n=1}^{+\infty} |(Te_n, e_{n_1})|^2 \leq \|T^* e_{n_1}\|^2,$$

then there exists $n_2 \in N$ such that

$$\sum_{n=n_2}^{+\infty} |(Te_{n_1}, e_n)|^2 \leq \frac{1}{2}, \ \sum_{n=n_2}^{+\infty} |(Te_n, e_{n_1})|^2 \leq \frac{1}{2}.$$

By analogy, the orthogonal sequence $\{e_{n_k}\}$ is obtained such that

$$\sum_{n=n_k}^{+\infty} |(Te_{n_1}, e_n)|^2 \leq \frac{1}{2^k}, \ \sum_{n=n_k}^{+\infty} |(Te_n, e_{n_1})|^2 \leq \frac{1}{2^k}.$$

Let $M = Span\{e_k\}_{k=1}^{+\infty}$ and $P : X \to M$ be an orthogonal projection operator. Then P is an infinite-dimensional projection operator and combines with $\sum_{k=1}^{+\infty} |(Te_n, e_n)|^2 < \infty$, we can obtain

$$\sum_{i=1}^{\infty} \|PTPe_{n_i}\|^2 = \sum_{i=1, j=1}^{\infty} |(PTPe_{n_i}, e_{n_j})|^2 = \sum_{i=1, j=1}^{\infty} |(Te_{n_i}, e_{n_j})|^2 < \infty.$$

That is, PTP is a Hilbert-Schmidt operator, so it is a compact operator.

(iv) → (i): Taking $M = \mathscr{R}(P)$, then $\dim M = \infty$ and $\{e_n\}_{n=1}^{\infty}$ is taken as a orthogonal set of M. So $\{e_n\}_{n=1}^{\infty}$ converges weakly to 0 and

$$(PTPe_n, e_n) = (Te_n, e_n) \to 0.$$

Therefore, for any $K \in \mathscr{K}(X)$, there is

$$((T + K)e_n, e_n) = (Te_n, e_n) + (Ke_n, e_n) \to 0.$$

That is, $0 \in W_e(T)$. The proof is completed. □

5.6.2 Relationship Between Essential Numerical Range and Numerical Range

According to the definition of the essential numerical range, we can easily attain $W_e(T) \subset \overline{W(T)}$. In fact, letting $\lambda \in W_e(T)$, then for any compact operators K, there is

$$\lambda \in W_e(T + K).$$

By Theorem 5.6.4, we know that there exists a group of orthogonal vector $\{e_n\}$ in X such that

$$\lim_{n\to\infty} ((T + K)e_n, e_n) = \lambda.$$

The orthogonal vector group converges weakly to zero, K is a compact operator, $Ke_n \to 0$. So $(Te_n, e_n) \to \lambda$, that is, $\lambda \in \overline{W(T)}$.

In addition, the essential numerical range also has important applications in characterizing the closure of the numerical range of a bounded linear operator and the closure of the numerical range.

Theorem 5.6.5 *If T is a bounded linear operator in a Hilbert space X, then*

$$Ex(\overline{W(T)}) \subset W_e(T) \cup W(T).$$

In particular, $\overline{W(T)} = Conv(W(T) \cup W_e(T))$.

Proof If $\lambda \in Ex(\overline{W(T)})$, then consider $W(\alpha T + \beta I) = \alpha W(T) + \beta$, assume that $Re(W(T)) \geq 0$ and $\lambda = 0$. From $0 \in \overline{W(T)}$, there exists a sequence $\{x_n\}_{n=1}^{\infty}$, $\|x_n\| = 1 (n = 1, 2, 3 \ldots)$ such that

$$(Tx_n, x_n) \to 0, n \to \infty.$$

Since the closed unit ball in a Hilbert space is weakly compact, there is a subsequence of $\{x_n\}$ such that it converges weakly to $x(\|x\| \leq 1)$. Assume that $\{x_n\}$ converges weakly to x, then there are three cases:

(i) When $\|x\| = 0$, for any $K \in \mathscr{K}(X)$, we can see $\{Kx_n\}$ converges strongly to 0 and

$$((T + K)x_n, x_n) = (Tx_n, x_n) + (Kx_n, x_n) \to 0,$$

that is, $0 \in W_e(T)$.

(ii) When $\|x\| = 1$, it is easy to know

$$\begin{aligned}\|x_n - x\|^2 =& (x_n - x, x_n - x)\\ =& (x_n, x_n) - 2Re(x_n, x) + (x, x)\\ =& 2 - 2Re(x_n, x)\\ \to& 2 - 2Re(x, x) = 0.\end{aligned}$$

So

$$\begin{aligned}|(Tx,x)| \leq & |(T(x-x_n),x)| + |(Tx_n,x-x_n)| + |(Tx_n,x_n)| \\ \leq & \|x-x_n\|\|T^*x\| + \|T\|\|x-x_n\| \\ \to & 0,\end{aligned}$$

thus $(Tx,x)=0$, it is said that $0 \in W(T)$.

(iii) When $0 < \|x\| < 1$, it is easy to know

$$\begin{aligned}\|x_n - x\|^2 = & (x_n - x, x_n - x) \\ = & (x_n,x_n) - 2Re(x_n,x) + (x,x) \\ = & 1 + \|x\|^2 - 2Re(x_n,x) \\ \to & 1 - \|x\|^2 > 0.\end{aligned}$$

That is to say, when n is sufficiently large, $x_n - x \neq 0$, setting $y_n = \frac{x_n - x}{\|x_n - x\|}$, then $n \to \infty$

$$(Ty_n, y_n) \to -\frac{(Tx,x)}{1-\|x\|^2},$$

that is, $-\frac{(Tx,x)}{1-\|x\|^2} \in \overline{W(T)}$. While

$$Re(-\frac{(Tx,x)}{1-\|x\|^2}) = -\frac{Re(Tx,x)}{1-\|x\|^2} \leq 0.$$

$$Re(Tx,x) = 0$$

can be obtained from $Re(W(T)) \geq 0$. Assume that $Im(Tx,x) \neq 0$, then the intersection of $\overline{W(T)}$ and the imaginary axis is not unique, which contradicts that 0 is the extreme point of $\overline{W(T)}$, so $Im(Tx,x)=0$, that is, $0 \in W(T)$. In summary, $Ex(\overline{W(T)}) \subset (W(T) \cup W_e(T))$. Then by $\overline{W(T)} = Conv(Ex(\overline{W(T)}))$ (Krein-Milman theorem), it can be proved that the equality $\overline{W(T)} = Conv(W(T) \cup W_e(T))$ holds. □

According to the above theorem, it is easy to obtain the following corollaries.

Proposition 5.6.6 *If T is a bounded linear operator in a Hilbert space X, then $W(T)$ is closed if and only if $W_e(T) \subset W(T)$.*

Proof When $W(T)$ is closed, $W(T) = \overline{W(T)}$. Considering $W_e(T) = \overline{W(T)}$, we can get $W_e(T) \subset W(T)$.

Conversely, when $W_e(T) \subset W(T)$,

$$\overline{W(T)} = Conv(W(T) \cup W_e(T)) = Conv(W(T)) = W(T),$$

that is, $W(T)$ is a closed set. □

Proposition 5.6.7 *Let T be a bounded linear operator in a Hilbert space X. Then $W_e(T) = \overline{W(T)}$ if and only if $Ex(W(T)) \subset W_e(T)$; in particular, when $Ex(W(T)) = \emptyset$, $W_e(T) = \overline{W(T)}$.*

Proof When $W_e(T) = \overline{W(T)}$, the proof of $Ex(W(T)) \subset W_e(T)$ is obvious.

On the contrary, when $Ex(W(T)) \subset W_e(T)$,

$$Ex(\overline{W(T)}) \subset \overline{W_e(T)} = W_e(T).$$

After taking convex hulls on both sides, it can be obtained that $W_e(T) = \overline{W(T)}$. □

Proposition 5.6.8 *Let T be a bounded linear operator in a Hilbert space X and λ be the corner of $\overline{W(T)}$. Then $\lambda \in W_e(T)$ or the finitely isolated eigenvalue of T.*

Proof When $\lambda \notin W_e(T)$, considering that $\overline{W(T)} = Conv(W(T) \cup W_e(T))$ there must exist that λ is the corner point of $W(T)$. From Theorem 5.1.15, we can see that $\lambda \in \sigma_p(T)$. Because $W_e(T)$ contains the essential spectrum, and $T - \lambda I$ is a Fredholm operator. Therefore, $\lambda \in \sigma_p(T)$ is finite multiple. It is proved that λ is isolated. Because $\mathbb{N}(T - \lambda I) = \mathbb{N}(T^* - \bar{\lambda} I) = R(T - \lambda I)^{\perp}$, with the decomposition of $X = \mathbb{N}(T - \lambda I) \oplus R(T - \lambda I)$, $T - \lambda I$ can be expressed as

$$T - \lambda = \begin{bmatrix} 0 & 0 \\ 0 & T_1 \end{bmatrix},$$

where $T_1(\lambda) : R(T - \lambda I) = \mathbb{N}(T - \lambda I)^{\perp} \to R(T - \lambda I)$ is bijective, and therefore it is invertible. Since the regular set is an open set, when ε is sufficiently small, there is $z \in \rho(T - \lambda I)$ for any $z \in \{z : 0 < |z| < \varepsilon\}$, which means that 0 is an isolated eigenvalue of $T - \lambda I$, that is, λ is an isolated eigenvalue of T. The proof is completed. □

Definition 5.6.9 If λ is a corner point of a closed convex set G, and there exists $\varepsilon > 0$ such that $B_\varepsilon(\lambda) \cap \partial(G)$ is composed of the rays emitted by λ, then λ is said to be lineal , where $B_\varepsilon(\lambda) = \{z \in \mathbb{C} : |z - \lambda| < \varepsilon\}$.

Proposition 5.6.10 *Let T be a bounded linear operator in a Hilbert space X. If λ is the corner point of $\overline{W(T)}$ and $\lambda \notin W_e(T)$, then λ is a lineal.*

Proof When λ is the corner point of $\overline{W(T)}$ and $\lambda \notin W_e(T)$, λ is a finitely multiple isolated eigenvalue of T by Corollary 5.6.8. Letting $X = \mathbb{N}(T - \lambda I) \oplus \mathbb{N}(T - \lambda I)^{\perp}$, then T can be expressed as

$$T = \begin{bmatrix} \lambda I & 0 \\ 0 & T_1 \end{bmatrix}.$$

So

$$W(T) = Conv(\{\lambda\} \cup W(T_1)),$$

and $\lambda \notin \overline{W(T_1)}$. In fact, assuming that $\lambda \in \overline{W(T_1)}$, then considering $W(T_1) \subset W(T)$, it can be known that λ is the corner point of $\overline{W(T_1)}$, so λ is the regular finitely multiple

isolated eigenvalue of T_1, which contradicts with $\mathbb{N}(T-\lambda I)\cap\mathbb{N}(T-\lambda I)^{\perp}=\{0\}$. Therefore, $B_{\varepsilon}(\lambda)\cap\overline{W(T)}$ is composed of straight line segments, that is, λ is lineal. □

Remark 5.6.11 From the above corollary, it can be seen that the closed semicircle is not the numerical range of compact operators. It is assumed that the radius of the closed semicircle is 1 and the center of the circle is at the origin, then $\lambda=1$ and $\lambda=-1$ are the corner points of the closed semicircle. On the other hand, if the closed semicircle is the numerical range of a compact operator T, then $\pm1\notin W_e(T)$, from Corollary 5.6.10, it can be seen that $\lambda=\pm1$ is lineal, which is contrary to the fact.

To use the essential numerical range to characterize the closure of the numerical range, we first give the following lemma.

Lemma 5.6.12 *If* $Im(W(T))\geq 0$, *then* $W(T)\cap\mathbb{R}=\{(Tx,x):\|x\|=1, x\in\mathbb{N}(Im(T))\}$.

Proof The proof of the relation $W(T)\cap\mathbb{R}\supset\{(Tx,x):\|x\|=1, x\in\mathbb{N}(Im(T))\}$ is trivial. Conversely, if $\lambda\in W(T)\cap\mathbb{R}$, then there exists $\|x\|=1$ such that

$$\lambda=(Tx,x)=(Re(T)x,x)+\mathrm{i}(Im(T)x,x).$$

Considering that $\lambda\in\mathbb{R}$, there is $(Im(T)x,x)=0$. $Im(W(T))\geq 0$, so $Im(T)x=0$, that is, $x\in\mathbb{N}(Im(T))$. The proof is completed. □

Similarly, the following conclusions can be proved.

Lemma 5.6.13 *If* $Re(W(T))\geq 0$, $i\mathbb{R}$ *denotes the imaginary axis, then* $W(T)\cap i\mathbb{R}=\{(Tx,x):\|x\|=1, x\in\mathbb{N}(Re(T))\}$.

Theorem 5.6.14 *If* T *be a bounded linear operator in a Hilbert space* X, *then*

$$\overline{W(T)}=(W(T)\cup W_e(T))\dot{\cup}L(T),$$

where $L(T)$ *denotes the union of countable disjoint open line segments. One endpoint of these open line segments is located in* $Ex(W(T))\setminus W_e(T)$, *and the other endpoint is located in* $Ex(W_e(T))\setminus W(T)$, *and* $\dot{\cup}$ *denotes the disjoint union.*

Proof We only need to prove $\overline{W(T)}\subset(W(T)\cup W_e(T))\dot{\cup}L(T)$. Set $\lambda\in\overline{W(T)}$ and $\lambda\notin W(T)\cup W_e(T)$, then $\lambda\in\partial\overline{W(T)}$ and $\lambda\notin Ex(\overline{W(T)})$ can be known by Theorem 5.6.5. Therefore, there is a line L such that

$$L\cap\overline{W(T)}=[\alpha,\beta].$$

Obviously, $\alpha,\beta\in Ex(\overline{W(T)})\subset W_e(T)\cup W(T)$. Consider that $\lambda\notin W(T)\cup W_e(T)$, α,β cannot be included in $W(T)$ or $W_e(T)$. Assume that $\alpha\in Ex(W(T))\setminus W_e(T)$, $\beta\in Ex(W_e(T))\setminus W(T)$, then considering the closure of

$W_e(T)$, the intersection of the closed line segments $[\alpha, \beta]$ and $W_e(T)$ is a closed set, that is, there is $\mu \in (Ex(W_e(T)) \setminus W(T)) \cap (\lambda, \beta]$ such that

$$[\alpha, \beta] \cap W_e(T) = [\mu, \beta].$$

If there is also $\eta \in (Ex(W(T)) \setminus W_e(T)) \cap [\alpha, \lambda)$ such that

$$[\alpha, \beta] \cap W(T) = [\alpha, \eta] \tag{5.54}$$

holds, then $\lambda \in (\eta, \mu)$, that is, the open line segment (η, μ) is the set needed to prove the conclusion.

To prove (5.54) holds, setting $Im(T) \geq 0$, $\lambda = 0$, the line segment $[\mu, \beta]$, $\beta > \mu$ is located on the positive semi-real axis, and $W(T) \cap \mathbb{R}$ is located on the negative semi-real axis. Then it can be known from Lemma 5.6.12 that

$$[\alpha, \beta] \cap W(T) = \mathbb{R} \cap W(T) = \{(Tx, x) : \|x\| = 1, x \in \mathbb{N}(Im(T))\}.$$

Taking $P : X \to \mathbb{N}(Im(T))$ be an orthogonal projection operator, then

$$\{(Tx, x) : \|x\| = 1, x \in \mathbb{N}(Im(T))\} = W(PT|_{\mathscr{R}(P)}).$$

It is easy to prove that $\mathscr{R}(P)$ is finite-dimensional. In fact, assuming that $\mathscr{R}(P)$ is infinite-dimensional, there exists an orthogonal sequence $\{x_n\}_n^\infty \subset \mathbb{N}(Im(T))$, $\|x_n\| = 1, n = 1, 2, \ldots$, such that

$$(Tx_n, x_n) \in [\alpha, \beta] \cap W(T), n = 1, 2, \ldots.$$

Because $[\alpha, \beta]$ is a bounded interval on the real axis, $\{(Tx_n, x_n)\}$ has a convergent subsequence and the limit point is located on the negative semi-real axis. On the other hand, by Theorem 5.6.4, the limit point of the convergent subsequence of $\{(Tx_n, x_n)\}$ belongs to $W_e(T) \cap \mathbb{R}$, while $W_e(T) \cap \mathbb{R}$ is located on the positive semi-real axis, which leads to a contradiction. Therefore, $\mathscr{R}(P)$ is finite-dimensional and $W(PT|_{\mathscr{R}(P)})$ is closed, so there is $\eta \in (Ex(W(T)) \setminus W_e(T)) \cap [\alpha, \lambda)$ such that Eq. (5.54) holds. □

The essential numerical range can also be used to characterize the structural properties of the boundary of the compact operator's numerical range.

Theorem 5.6.15 *If T is a compact operator in an infinite-dimensional Hilbert space X, then $\partial W(T) \setminus W(T)$ consisting of a line segment of the form $[0, \lambda)$ with at most two segments, where $[0, \lambda)$ can be reduced to a point or an empty set.*

Proof When $0 \in W(T)$, $W(T)$ is a closed set, so $\partial W(T) \setminus W(T) = \emptyset$, the conclusion holds. When $0 \notin W(T)$, due to $W_e(T) = \{0\}$, $Ex(W_e(T)) \setminus W(T) = \{0\}$, it can be known from Theorem 5.6.14 that $L(T)$ is composed of linear segments like $(0, \lambda)$, and since $0 \in \partial W(T)$ and $W(T)$ are convex sets, there is no more than two straight line segments like $(0, \lambda)$ on the boundary of $W(T)$. The proof is completed. □

Remark 5.6.16 It can solve the problem of whether a given convex set is a numerical range of a compact operator by using the above conclusion. For example, letting $\triangle$ be the closed triangular area with vertex $A(1, 0)$, $B(0, 1)$, $C(-1, 0)$, then the convex sets $\triangle_1 = \triangle\backslash[-1, 0]$, $\triangle_2 = \triangle\backslash[-1, 1]$, and $\triangle_3 = \triangle\backslash[-1, 1)$ are not numerical ranges of a compact operator. In fact, it is easy to know

$$\begin{aligned}\partial\triangle_1\backslash\triangle_1 &= [-1, 0],\\ \partial\triangle_2\backslash\triangle_2 &= [-1, 1],\\ \partial\triangle_3\backslash\triangle_3 &= [-1, 1).\end{aligned}$$

These sets are not linear segments of the form $[0, \lambda)$ (semi-open and semi-closed), so they are not numerical range of a compact operator from the above theorem.

5.6.3 Essential Numerical Radius

Similar to the numerical radius, the concept of essential numerical radius can also be introduced.

Definition 5.6.17 Let T be a bounded linear operator in Hilbert space X. The essential numerical radius $\boldsymbol{w}_e(T)$ is defined as

$$\boldsymbol{w}_e(T) = \sup\{|\ \lambda\ |: \lambda \in W_e(T)\}.$$

According to the properties of the essential numerical range, the proof of the following conclusion is trivial.

Corollary 5.6.18 *If T and S are bounded linear operators in Hilbert space X, then*

(i) $0 \leq \boldsymbol{w}_e(T) \leq \boldsymbol{w}(T)$;
(ii) $\boldsymbol{w}_e(\lambda T) =|\ \lambda\ |\ \boldsymbol{w}_e(T)$;
(iii) $\boldsymbol{w}_e(U^*TU) = \boldsymbol{w}_e(T)$, *where U is a unitary operator;*
(iv) $\boldsymbol{w}_e(T + S) \leq \boldsymbol{w}_e(T) + \boldsymbol{w}_e(S)$;
(v) $\boldsymbol{w}_e(T) = 0$ *if and only if T is a compact operator;*
(vi) $\boldsymbol{w}_e(T + K) = \boldsymbol{w}_e(T)$, *where K is a compact operator.*

Theorem 5.6.19 *If T is a bounded linear operator in separable Hilbert space X, then there exist a compact operator K_0 such that $\boldsymbol{w}_e(T) = \boldsymbol{w}(T + K_0)$.*

Proof According to Corollary 4.1 of [86], there exist a compact operator K_0 such that

$$W_e(T) = \overline{W(T + K_0)}.$$

Hence the conclusion holds. □

Theorem 5.6.20 *If $T \in \mathscr{B}(X)$, then*

$$\frac{1}{2}\|T\|_K \leq \boldsymbol{w}_e(T) \leq \|T\|_K,$$

where $\|T\|_K = \inf\limits_{K\in\mathscr{K}(X)} \|T+K\|$.

Proof According to Corollary 4.1 of [86], there exist a compact operator K such that $\overline{W(T+K)} = W_e(T)$, so

$$\inf_{K\in\mathscr{K}(X)} \boldsymbol{w}(T+K) = \boldsymbol{w}_e(T).$$

Setting $\|T\|_K = \inf\limits_{K\in\mathscr{K}(X)} \|T+K\|$, then from

$$\frac{1}{2}\|T\| \leq \boldsymbol{w}(T) \leq \|T\|,$$

it can be known that

$$\frac{1}{2}\|T\|_K \leq \inf_{K\in\mathscr{K}(X)} \boldsymbol{w}(T+K) \leq \|T\|_K,$$

that is, $\frac{1}{2}\|T\|_K \leq \boldsymbol{w}_e(T) \leq \|T\|_K$. The proof is completed. □

5.6.4 Essential Numerical Range and Essential Numerical Radius of Bounded Infinite-Dimensional Hamiltonian Operators

In the following, we will discuss the properties of the essential numerical range and the essential numerical radius of a bounded infinite-dimensional Hamiltonian operator.

Theorem 5.6.21 *If* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$ *is a bounded infinite-dimensional Hamiltonian operator, then* $Conv(W_e(A) \cup W_e(-A^*)) \subset W_e(H)$.

Proof Take $\lambda \in W_e(A)$, then for any $K \in \mathscr{K}(X)$ there exists $\|f_n\| = 1, n = 1, 2, \ldots,$ such that

$$\Big((A+K)f_n, f_n)\Big) \to \lambda.$$

Because $\left\| \begin{bmatrix} f_n \\ 0 \end{bmatrix} \right\| = 1$, then for any $S = \begin{bmatrix} K & K_2 \\ K_3 & K_4 \end{bmatrix} \in \mathscr{K}(X \times X)$, there is

$$\left(\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} + \begin{bmatrix} K & K_2 \\ K_3 & K_4 \end{bmatrix}\right)\begin{bmatrix} f_n \\ 0 \end{bmatrix}, \begin{bmatrix} f_n \\ 0 \end{bmatrix}\right) = \Big((A+K)f_n, f_n\Big) \to \lambda.$$

That is, $W_e(A) \subset W_e(H)$. The same reasoning leads to $W_e(-A^*) \subset W_e(H)$. Therefore, from the convexity of the essential numerical range, it can be known that $Conv(W_e(A) \cup W_e(-A^*)) \subset W_e(H)$. □

Theorem 5.6.22 *For the bounded infinite-dimensional Hamiltonian operator* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}$, *the relation*

$$Conv(W_e(A) \cup W_e(-A^*)) = W_e(H)$$

holds if B, C *are compact operators.*

Proof We only need to prove $W_e(H) \subset Conv(W_e(A) \cup W_e(-A^*))$. If $\lambda \in W_e(H)$, then for any $S = \begin{bmatrix} K_1 & K_2 \\ K_3 & K_4 \end{bmatrix} \in \mathscr{K}(X \times X)$, there exists a sequence $\{\begin{bmatrix} x_n \\ y_n \end{bmatrix}\}_{n=1}^{\infty}$ and $\|x_n\|^2 + \|y_n\|^2 = 1 (n = 1, 2, \ldots)$ such that

$$\left(\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} + \begin{bmatrix} K_1 & K_2 \\ K_3 & K_4 \end{bmatrix}\right)\begin{bmatrix} x_n \\ y_n \end{bmatrix}, \begin{bmatrix} x_n \\ y_n \end{bmatrix}\right) \to \lambda.$$

Selecting $K' = \begin{bmatrix} K_1 & -B \\ -C & K_4 \end{bmatrix}$, then

$$W_e(H) \subset \bigcap_{K_1, K_4 \in \mathscr{K}(X)} \overline{W(H + K')}.$$

Therefore,

$$\begin{aligned} &\left(\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} + \begin{bmatrix} K_1 & -B \\ -C & K_4 \end{bmatrix}\right)\begin{bmatrix} x_n \\ y_n \end{bmatrix}, \begin{bmatrix} x_n \\ y_n \end{bmatrix}\right) \\ =&\Big((A + K_1)x_n, x_n\Big) + \Big((-A^* + K_4)y_n, y_n\Big) \to \lambda. \end{aligned}$$

When $\liminf_{n\to\infty} \|x_n\| = 0$ or $\liminf_{n\to\infty} \|y_n\| = 0$, there exists $\lambda \in W_e(A) \cup W_e(-A^*)$, thus it can be known that

$$W_e(H) \subset \bigcap_{K_1, K_4 \in \mathscr{K}(X)} \overline{W(\mathscr{A} + K')} \subset Conv(W_e(A) \cup W_e(-A^*)).$$

When $\liminf_{n\to\infty} \|x_n\| \neq 0$ and $\liminf_{n\to\infty} \|y_n\| \neq 0$, without losing generality, we assume that $x_n \neq 0$ and $y_n \neq 0 (n = 1, 2, \ldots)$, then we can see

$$\|x_n\|^2 \frac{\Big((A + K_1)x_n, x_n\Big)}{\|x_n\|^2} + (1 - \|x_n\|^2)\frac{\Big((-A^* + K_4)y_n, y_n\Big)}{\|y_n\|^2} \to \lambda.$$

Obviously, $\lambda \in Conv(W_e(A) \cup W_e(-A^*))$. Therefore,

$$W_e(H) \subset \bigcap_{K_1, K_4 \in \mathscr{K}(X)} \overline{W(H + K')} \subset Conv(W_e(A) \cup W_e(-A^*)).$$

. In summary, $W_e(H) = Conv(W_e(A) \cup W_e(-A^*))$. The proof is completed. □

Proposition 5.6.23 *$H = \begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix}$ is a bounded infinite-dimensional Hamiltonian operator, then $Conv(W_e(A) \cup W_e(-A^*)) = W_e(H)$.*

Theorem 5.6.24 *If $A, B, C \in \mathscr{B}(X)$, then the following conclusions can be drawn.*

(i) If B and C are compact operators, then $\boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) = \boldsymbol{w}(A)$, especially

$$\boldsymbol{w}_e\left(\begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix}\right) = \boldsymbol{w}(A);$$

(ii) $\boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ e^{i\theta}C & 0 \end{bmatrix}\right) = \boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right), \forall \theta \in \mathbb{R};$

(iii) $\boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) = \boldsymbol{w}_e\left(\begin{bmatrix} 0 & C \\ B & 0 \end{bmatrix}\right);$

(iv) $\boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ B & A \end{bmatrix}\right) = \max\{\boldsymbol{w}_e(A + B), \boldsymbol{w}_e(A - B)\}$; *especially,* $\boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ B & 0 \end{bmatrix}\right) = \boldsymbol{w}_e(B)$;

(v) $\boldsymbol{w}_e\left(\begin{bmatrix} A & -B \\ B & A \end{bmatrix}\right) = \max\{\boldsymbol{w}_e(A + iB), \boldsymbol{w}_e(A - iB)\}.$

Proof (i) When B and C are compact operators, it can be obtained from Theorem 5.6.22 that

$$W_e\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) = Conv(W_e(A) \cup W_e(-A^*)).$$

And hence

$$\boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) = \max\{\boldsymbol{w}_e(A), \boldsymbol{w}_e(-A^*)\} = \boldsymbol{w}(A).$$

(ii) Selecting a unitary operator $U = \begin{bmatrix} I & 0 \\ 0 & e^{\frac{-i\theta}{2}}I \end{bmatrix}$, then we have

$$\begin{bmatrix} 0 & e^{\frac{-i\theta}{2}}B \\ e^{\frac{i\theta}{2}}C & 0 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & e^{\frac{i\theta}{2}}I \end{bmatrix}\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\begin{bmatrix} I & 0 \\ 0 & e^{\frac{-i\theta}{2}}I \end{bmatrix}.$$

This conclusion is easily obtained by the definition of the essential numerical radius and the unitary similarity invariance of the essential numerical radius.

(iii) Selecting a unitary operator $U = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$, then

$$\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} 0 & C \\ B & 0 \end{bmatrix} \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}.$$

Therefore, the conclusion can be obtained according to the unitary similarity invariance of the essential numerical radius.

(iv) Selecting a unitary operator $U_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} I & -I \\ I & I \end{bmatrix}$, then

$$\begin{bmatrix} A+B & 0 \\ 0 & A-B \end{bmatrix} = \frac{1}{2} \begin{bmatrix} I & I \\ -I & I \end{bmatrix} \begin{bmatrix} A & B \\ B & A \end{bmatrix} \begin{bmatrix} I & -I \\ I & I \end{bmatrix}.$$

The conclusion is proved by the unitary similarity invariance of the essential numerical range.

(v) Selecting a unitary operator $U_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} I & I \\ \mathrm{i}I & -\mathrm{i}I \end{bmatrix}$, then

$$\begin{bmatrix} A-\mathrm{i}B & 0 \\ 0 & A+\mathrm{i}B \end{bmatrix} = \frac{1}{2} \begin{bmatrix} I & -\mathrm{i}I \\ I & \mathrm{i}I \end{bmatrix} \begin{bmatrix} A & -B \\ B & A \end{bmatrix} \begin{bmatrix} I & I \\ \mathrm{i}I & -\mathrm{i}I \end{bmatrix}.$$

The conclusion is proved by a unitary similarity invariance of the essential numerical range. □

Theorem 5.6.25 *If* $H = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \in \mathscr{B}(X \times X)$, *then*

$$\frac{\max\{w_e(B+C), w_e(B-C)\}}{2} \le w_e(\mathscr{A}) \le \frac{w_e(B+C) + w_e(B-C)}{2}.$$

Proof Firstly, we prove the first inequality. From Theorem 5.6.24, it can be obtained

$$\begin{aligned} w_e(B+C) =& w_e\left(\begin{bmatrix} 0 & B+C \\ B+C & 0 \end{bmatrix}\right) \\ =& w_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} + \begin{bmatrix} 0 & C \\ B & 0 \end{bmatrix}\right) \\ \le& w_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) + w_e\left(\begin{bmatrix} 0 & C \\ B & 0 \end{bmatrix}\right) \\ =& w_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) + w_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) \\ =& 2w_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right), \end{aligned}$$

that is, we have

$$\frac{\boldsymbol{w}_e(B+C)}{2} \leq \boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right).$$

The same reasoning leads to

$$\frac{\boldsymbol{w}_e(B-C)}{2} \leq \boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ -C & 0 \end{bmatrix}\right).$$

Since $\boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ -C & 0 \end{bmatrix}\right) = \boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)$, so

$$\frac{\max\{\boldsymbol{w}_e(B+C), \boldsymbol{w}_e(B-C)\}}{2} \leq \boldsymbol{w}_e(\mathscr{A}).$$

In order to prove the second inequality, we select the operator $U = \frac{1}{\sqrt{2}}\begin{bmatrix} I & -I \\ I & I \end{bmatrix}$, then it is easy to prove that U is a unitary operator. From Theorem 5.6.24, it is easy to obtain that

$$\begin{aligned}
\boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) &= \boldsymbol{w}_e\left(U^*\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}U\right) \\
&= \frac{1}{2}\boldsymbol{w}_e\left(\begin{bmatrix} B+C & B-C \\ -(B-C) & -(B+C) \end{bmatrix}\right) \\
&= \frac{1}{2}\boldsymbol{w}_e\left(\begin{bmatrix} B+C & 0 \\ 0 & -(B+C) \end{bmatrix} + \begin{bmatrix} 0 & B-C \\ -(B-C) & 0 \end{bmatrix}\right) \\
&\leq \frac{1}{2}\left(\boldsymbol{w}_e\left(\begin{bmatrix} B+C & 0 \\ 0 & -(B+C) \end{bmatrix}\right) + \boldsymbol{w}_e\left(\begin{bmatrix} 0 & B-C \\ -(B-C) & 0 \end{bmatrix}\right)\right) \\
&= \frac{\boldsymbol{w}_e(B+C) + \boldsymbol{w}_e(B-C)}{2}.
\end{aligned}$$

To sum up, the conclusion is proved. □

The following conclusions show that the essential numerical radius satisfies the Pinching inequality.

Theorem 5.6.26 *Let* $\mathscr{A} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \in \mathscr{B}(X \times X)$. *Then*

(i) $\boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right) \geq \boldsymbol{w}_e\left(\begin{bmatrix} A & B' \\ C' & D \end{bmatrix}\right)$, *where* B', C' *are compact operators; in particular,*

$$\boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right) \geq \boldsymbol{w}_e\left(\begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix}\right);$$

(ii) $\boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right) \geq \boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)$.

Proof (i) By the property of the essential numerical range of a 2×2 block operator, when B' and C' are compact operators,

$$W_e\left(\begin{bmatrix} A & B' \\ C' & D \end{bmatrix}\right) = Conv(W_e(A) \cup W_e(D)) \subset W_e\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right).$$

By the definition of the essential numerical range, it can be obtained that

$$\boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right) \geq \boldsymbol{w}_e\left(\begin{bmatrix} A & B' \\ C' & D \end{bmatrix}\right).$$

(ii) Selecting a unitary operator $U = \begin{bmatrix} -I & 0 \\ 0 & I \end{bmatrix}$, then

$$\begin{bmatrix} A & -B \\ -C & D \end{bmatrix} = \begin{bmatrix} -I & 0 \\ 0 & I \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix}\begin{bmatrix} -I & 0 \\ 0 & I \end{bmatrix}.$$

Therefore,

$$\begin{aligned} \boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right) &= \frac{1}{2}\boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix} - \begin{bmatrix} A & -B \\ -C & D \end{bmatrix}\right) \\ &\leq \frac{1}{2}\left(\boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right) + \boldsymbol{w}_e\left(\begin{bmatrix} A & -B \\ -C & D \end{bmatrix}\right)\right) \\ &= \boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right). \end{aligned}$$

The conclusion can be proved. □

Proposition 5.6.27 *If* $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} \in \mathscr{B}(X \times X)$ *is a bounded infinite-dimensional Hamiltonian operator, then*

$$\boldsymbol{w}_e(H) \geq \max\{\boldsymbol{w}_e(A), \frac{\boldsymbol{w}_e(B+C)}{2}, \frac{\boldsymbol{w}_e(B-C)}{2}\},$$

and

$$\boldsymbol{w}_e(H) \leq \boldsymbol{w}_e(A) + \frac{\boldsymbol{w}_e(B+C) + \boldsymbol{w}_e(B-C)}{2}.$$

Proof It is easy to know that

$$\begin{aligned} \boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) &\geq \max\{\boldsymbol{w}_e\left(\begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix}\right), \boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)\} \\ &= \max\{\boldsymbol{w}_e(A), \boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right)\} \\ &\geq \max\{\boldsymbol{w}_e(A), \frac{\boldsymbol{w}_e(B+C)}{2}, \frac{\boldsymbol{w}_e(B+C)}{2}\}. \end{aligned}$$

And hence

$$\boldsymbol{w}_e(H) \geq \max\{\boldsymbol{w}_e(A), \frac{\boldsymbol{w}_e(B+C)}{2}, \frac{\boldsymbol{w}_e(B-C)}{2}\}.$$

In addition, it can be known from the triangle inequality that

$$\boldsymbol{w}_e\left(\begin{bmatrix} A & B \\ C & -A^* \end{bmatrix}\right) \leq \boldsymbol{w}_e\left(\begin{bmatrix} A & 0 \\ 0 & -A^* \end{bmatrix}\right) + \boldsymbol{w}_e\left(\begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix}\right).$$

It can be obtained by Theorem 5.6.24 that

$$\boldsymbol{w}_e(H) \leq \boldsymbol{w}_e(A) + \frac{\boldsymbol{w}_e(B+C) + \boldsymbol{w}_e(B-C)}{2}.$$

The conclusion is proved. □

Chapter 6
Spectral Theory of Infinite-Dimensional Hamiltonian Operators in Complete Indefinite Inner Product Spaces



The operator theory on the indefinite inner product spaces is not a simple logical extension of the operator theory in the Hilbert space but has a profound foundation. Its application involves physics, mathematics, and mechanics. The space of "time-space" in relativity is an indefinite inner product space. The indefinite inner product space first appeared in the article of P. A. M. Dirac (see [88]) about quantum field theory and was then widely used in the field of quantum field theory. The Li-Wick theory proposed by Zhengdao Li and G. C. Wick is a theory that uses the indefinite inner product to eliminate the divergence difficulty in quantum field theory. Later, L. S. Pontryagin (see [89]) explored the operator theory in indefinite inner product spaces from a mathematical perspective due to the need for the study of mechanical problems. Since then, M. G. Krein, I. S. Iokhvidov, H. Langer, T. Y. Azizov, R. S. Phillips, J. Bognár, S. L. Sobolev, Daoxing Xia, Shaozong Yan, and other scholars have also studied operator theory in the indefinite inner product space. Due to the special structure of the infinite-dimensional Hamiltonian operator, if we study its spectral theory in the indefinite inner product spaces, we may get the twice results with the half effort.

6.1 Krein Spaces

6.1.1 Definition and Properties of Krein Spaces

To study the spectral theory of infinite-dimensional Hamiltonian operator in complete indefinite inner product spaces, the definition of complete indefinite inner product space (that is, Krein space) is given first.

Definition 6.1.1 Let K_0 be a linear space in complex number field $\mathbb{C}$, and $[\cdot, \cdot]$ be a bilinear Hermite functional on K_0 having the following conditions:

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(i) For any $x, y \in K_0$ satisfies $[x, y] = \overline{[y, x]}$.
(ii) For any $\alpha, \beta \in \mathbb{C}, x, y, z \in K_0$ satisfies $[\alpha x + \beta y, z] = \alpha[x, z] + \beta[y, z]$,
then $[\cdot, \cdot]$ is called a quasi-inner product on K_0 and $\{K_0, [\cdot, \cdot]\}$ is called the quasi-indefinite inner product space. If quasi-inner product $[\cdot, \cdot]$ is non-degenerate, then the inner product is called a quasi-indefinite inner product, correspondingly, $\{K_0, [\cdot, \cdot]\}$ is called an indefinite inner product space.

Definition 6.1.2 Let $\{K, [\cdot, \cdot]\}$ be an indefinite inner product space and $x \in K$. If $[x, x] \geq 0$, then x is called a non-negative vector, and the subspace composed of non-negative vectors is called non-negative subspace. If $[x, x] > 0$, then x is called a positive vector, and the subspace consisting of positive vectors is called positive subspace. If $[x, x] = 0$, then x is said to be a null vector, also known as the isotropic vector. The subspace constituted by isotropic vectors is called isotropic subspace. In the same way, we can define non-positive subspaces, negative subspaces, etc.

We will introduce a particularly important class of indefinite inner product spaces, which is complete indefinite inner product spaces. The operator theory in this chapter is based on this type of space.

Definition 6.1.3 Let $\{K, [\cdot, \cdot]\}$ be an indefinite inner product space. If there exist positive subspace K_+ and negative subspace K_- of K such that

$$K = K_+ \oplus K_-, \tag{6.1}$$

where $[\cdot, \cdot]$ is restricted to K_+, $\{K_+, [\cdot, \cdot]\}$ becomes a Hilbert space; $-[\cdot, \cdot]$ is restricted to K_-, $\{K_-, -[\cdot, \cdot]\}$ also becomes a Hilbert space, then $\{K, [\cdot, \cdot]\}$ is a complete indefinite inner product space, also called Krein space and hence (6.1) is called the canonical factorization of K.

Let $\{K, [\cdot, \cdot]\}$ be a complete indefinite inner product space, and $K = K_+ \oplus K_-$ be a canonical factorization. Then there is a unique factorization $x = x_+ + x_-$, $y = y_+ + y_-$, $x_\pm, y_\pm \in K_\pm$ for $x, y \in K$, and we introduce new inner product onto K

$$(x, y) = [x_+, y_+] - [x_-, y_-], \tag{6.2}$$

then $(\cdot, \cdot)$ is called the inner product induced by canonical factorization $K = K_+ \oplus K_-$. Denote the norm on K derived from $(\cdot, \cdot)$ as

$$\|x\| = (x, x)^{\frac{1}{2}}, x \in K. \tag{6.3}$$

Obviously, the space $\{K, (\cdot, \cdot)\}$ induced by canonical factorization is not only an inner product space but also a Hilbert space. $K_\pm$ is a closed subspace of K if we use $P_\pm$ to represent the projection operator of Hilbert space K on $K_\pm$, respectively. Setting $J = P_+ - P_-$, then (6.2) can also be written as

$$(x, y) = [Jx, y] = [x, Jy], \tag{6.4}$$

where J is called the metric operator (or canonical symmetry operator) having the following properties:

$$J = J^* = J^{-1}. \tag{6.5}$$

For example, all the following operators

$$J_1 = \begin{bmatrix} 0 & \mathrm{i}I \\ -\mathrm{i}I & 0 \end{bmatrix}, \ J_2 = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}, \ J_3 = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix},$$

satisfy (6.5).

6.1.2 Linear Operators in Krein Spaces

According to the definition, the complete indefinite inner product space exists at least one canonical factorization which can induce an inner product $(\cdot, \cdot)$ to ensure $\{K, (\cdot, \cdot)\}$ forms a Hilbert space, therefore there is a topology on K. From the norm equivalence theorem, it is known that the topology on K does not depend on the selection of canonical factorization, but is determined by the complete indefinite inner product space itself. In this way, the concepts of "boundedness," "continuity," and "closedness" of sets can be introduced on the complete indefinite inner product space, which is the same as the usual Hilbert space, so we will give no more discussion on this. In addition, the definition of the adjoint operator of the linear operator is similar to that of the Hilbert space. The adjoint operator in the sense of indefinite inner product is denoted as $T^\dagger$ in order not to be confused with linear operator T. The adjoint operator in the sense of the inner product is still denoted as T^*. There are also some connections between adjoint operators in complete indefinite inner product spaces and adjoint operators in Hilbert space.

For example, let $\mathfrak{H} = (K, [\cdot, \cdot])$ be a complete indefinite inner product space, the standard symmetric operator (metric operator) induced by $[\cdot, \cdot]$ be set to $\mathfrak{J}$, and $T : \mathscr{D}(T) \subset \mathfrak{H} \to \mathfrak{H}$ be a densely defined linear operator, then there are the following connections between the adjoint operator $T^\dagger$ in the sense of indefinite inner product and the adjoint operator T^* in the sense of Hilbert space

$$T^\dagger = \mathfrak{J} T^* \mathfrak{J}. \tag{6.6}$$

Therefore, (6.6) is a bridge between the concept of the adjoint operator in Hilbert space and the concept of the adjoint operator in complete indefinite space, and it is easy to prove the following lemma.

Lemma 6.1.4 *Let T, S be a densely defined closed linear operator in Krein space $\{K, [\cdot, \cdot]\}$, and its corresponding metric operator be $\mathfrak{J}$. Then the following statements are satisfied:*

(i) $T^\dagger$ *is a closed operator.*
(ii) If $\mathscr{D}(T+S)$ *is dense, then* $(T+S)^\dagger \supset T^\dagger + S^\dagger$.
(iii) If $\mathscr{D}(TS)$ *is dense, then* $(TS)^\dagger \supset S^\dagger T^\dagger$.
(iv) If $T \subset S$, *then* $S^\dagger \subset T^\dagger$.

Definition 6.1.5 Let the corresponding metric operator to Krein space $\{K, [\cdot,\cdot]\}$ be $\mathfrak{J}$. Then linear operator T in K is called $\mathfrak{J}-$ symmetric, if $T \subset T^\dagger$; it is called a $\mathfrak{J}-$ self-adjoint operator if $T = T^\dagger$. If $[Tx, x] \geq 0$ for any $x \in \mathscr{D}(T)$, then T is called a $\mathfrak{J}-$ non-negative operator.

Because the metric is indefinite, its unique difficulties are formed. Many obvious facts in Hilbert spaces are not necessarily true in indefinite inner product spaces. For example, let X be a Hilbert space, $T : \mathscr{D}(T) \subset X \to X$ is a densely closed linear operator. Then T^*T is a self-adjoint operator in Hilbert space X, which is generated by von Neumann. However, the above properties are not necessarily true in the indefinite metric space, that is, $T^\dagger T$ is not necessarily a self-adjoint operator in the sense of indefinite inner product space.

Example 6.1.6 Let the canonical factorization of indefinite inner product space K be

$$K = K_- \oplus K_+,$$

and the corresponding metric operator to $K_\pm = L^2[0,1]$ be $J_1 = \begin{bmatrix} 0 & \mathrm{i}I \\ -\mathrm{i}I & 0 \end{bmatrix}$. Define linear operator $A_0 : L^2[0,1] \to L^2[0,1]$ as follows:

$$A_0 f(t) = \frac{1}{t} f(t),$$

then

$$\mathscr{D}(A_0) = \{f(t) | f(t) \in L^2[0,1], \frac{1}{t} f(t) \in L^2[0,1]\}.$$

Take the linear operator T of $K \to K$ as follows:

$$T = \begin{bmatrix} A & B \\ A & B \end{bmatrix},$$

where $A = V_1 A_0$, $B = V_2 A_0$, V_1, V_2 are the isometric operators in $L^2[0,1]$, and satisfy $\mathscr{R}(V_1) \perp \mathscr{R}(V_2)$. If the adjoint operator in the indefinite inner product sense of the linear operator T is denoted by $T^\dagger$, then it can be proved that $T^\dagger T$ is only the J_1- symmetric operator and not closed, and thus it is not a J_1- self-adjoint operator. In fact, by considering $\mathscr{R}(V_1) \perp \mathscr{R}(V_2)$ we can get $\mathscr{R}(A) \perp \mathscr{R}(B)$, and because A, B are densely defined closed operators, T is a closed operator. However,

$$T^\dagger T = J_3 T^* J_3 T \supset \begin{bmatrix} B^* & -B^* \\ -A^* & A^* \end{bmatrix} \begin{bmatrix} A & B \\ A & B \end{bmatrix}.$$

Let $\mathscr{D}_0 = \{f(t)|f(t) \in L^2[0,1], \frac{1}{t^2}f(t) \in L^2[0,1]\}$. Then $\overline{\mathscr{D}_0} = L^2[0,1]$ and

$$\begin{bmatrix} B^* & -B^* \\ -A^* & A^* \end{bmatrix} \begin{bmatrix} A & B \\ A & B \end{bmatrix} |_{\mathscr{D}_0} = 0|_{\mathscr{D}_0},$$

thus

$$(T^\dagger T)^\dagger \subset (0|_{\mathscr{D}_0})^\dagger = 0.$$

Hence $T^\dagger T$ is not closed and not a J_1- self-adjoint operator.

Lemma 6.1.7 *Let T be a densely defined $\mathfrak{J}-$ non-negative operator in Krein space $\{K, [\cdot,\cdot]_\mathfrak{J}\}$. If there exists $x_0 \in \mathscr{D}(A)$ such that $[Tx_0, x_0]_\mathfrak{J} = 0$, then $Tx_0 = 0$.*

Proof

$$|[Tx_0, y]_\mathfrak{J}| \le |[Tx_0, x_0]_\mathfrak{J}^{\frac{1}{2}}||[Ty, y]_\mathfrak{J}^{\frac{1}{2}}| = 0$$

for any $y \in \mathscr{D}(T)$. Therefore, by considering $\overline{\mathscr{D}(T)} = K$ we can get $Tx_0 = 0$. □

The following will discuss the properties of the spectra of densely defined closed linear operators in Krein spaces.

Lemma 6.1.8 *Let T be a densely defined closed linear operator in Krein space $\{K, [\cdot,\cdot]_\mathfrak{J}\}$. Then*

(i) $\lambda \in \rho(T)$ *if and only if* $\overline{\lambda} \in \rho(T^\dagger)$;
(ii) $\lambda \in \sigma_r(T)$ *contains* $\overline{\lambda} \in \sigma_p(T^\dagger)$;
(iii) $\lambda \in \sigma_p(T)$ *contains* $\overline{\lambda} \in \sigma_p(T^\dagger) \cup \sigma_r(T^\dagger)$;
(iv) $\lambda \in \sigma_c(T)$ *if and only if* $\overline{\lambda} \in \sigma_c(T^\dagger)$.

Proof We only prove conclusion (i), and the other proofs are similar. Let $\lambda \in \rho(T)$, then $\overline{\lambda} \in \rho(T^*)$, and again by

$$T^\dagger = \mathfrak{J}T^*\mathfrak{J}$$

and $\mathfrak{J}^2 = I$ we know that $\overline{\lambda} \in \rho(T^\dagger)$. Conversely, it is the same. □

Lemma 6.1.9 *Let T be a $\mathfrak{J}-$ self-adjoint operator in Krein space $\{K, [\cdot,\cdot]_\mathfrak{J}\}$. Then $\sigma_r(T) \cap \mathbb{R} = \emptyset$.*

Proof Assume that there exists $\lambda \in \mathbb{R}$ such that $\lambda \in \sigma_r(T)$, then by Lemma 6.1.8 we get $\lambda \in \sigma_p(T^\dagger) = \sigma_p(T)$, contradiction is derived. □

Lemma 6.1.10 *Let T be a $\mathfrak{J}-$ non-negative self-adjoint operator in Krein space $\{K, [\cdot,\cdot]_\mathfrak{J}\}$. Then*

(i) $\sigma_p(T) \cap (\mathbb{C}^+ \cup \mathbb{C}^-) = \emptyset$.
(ii) $\sigma_r(T) = \emptyset$.
(iii) If $\rho(T) \neq \emptyset$, *then* $\mathbb{C}^+ \cup \mathbb{C}^- \subset \rho(T)$,

where $\mathbb{C}^+, \mathbb{C}^-$ represent the upper and lower open half-planes, respectively.

Proof (i) Let $\lambda \in \sigma_p(T) \cap (\mathbb{C}^+ \cup \mathbb{C}^-)$, and the corresponding eigenvector be x_0. Then

$$[Tx_0, x_0]_{\mathfrak{J}} = \lambda [x_0, x_0]_{\mathfrak{J}}.$$

Because $[Tx_0, x_0]_{\mathfrak{J}} \geq 0$, $\lambda \in \mathbb{C}^+ \cup \mathbb{C}^-$, $[x_0, x_0]_{\mathfrak{J}} = 0$, that is, $[Tx_0, x_0]_{\mathfrak{J}} = 0$. By Lemma 6.1.7 we know that $Tx_0 = 0$, so $\lambda = 0$, then contradiction is derived.

(ii) If $\lambda \in \sigma_r(T)$, then $\overline{\lambda} \in \sigma_p(T^\dagger) = \sigma_p(T)$. By (i) we know $\lambda \in \mathbb{R}$, which is contradictory to Lemma 6.1.9.

(iii) Firstly we prove that if T is continuous operator, then $\mathbb{C}^+ \cup \mathbb{C}^- \subset \rho(T)$. Suppose there exists $\overline{\lambda}_0 \neq \lambda_0 \in \sigma(T)$, then by (i) and (ii) we know that

$$\lambda_0 \in \sigma_c(T).$$

Therefore, there exists an orthogonalized sequence $\{x_n\}(\|x_n\| = 1, n = 1, 2, \ldots)$ such that

$$(T - \lambda_0 I)x_n \to 0.$$

Because

$$[(T - \lambda_0 I)x_n, x_n]_{\mathfrak{J}} = [(T - Re(\lambda_0))x_n, x_n]_{\mathfrak{J}} - i\,Im(\lambda_0)[x_n, x_n]_{\mathfrak{J}},$$

by $Im(\lambda_0) \neq 0$ we can get $[x_n, x_n]_{\mathfrak{J}} \to 0$, that is, $[Tx_n, x_n]_{\mathfrak{J}} \to 0$. Also because

$$[Tx_n, x_n]_{\mathfrak{J}} = (\mathfrak{J}Tx_n, x_n) = ((\mathfrak{J}T)^{\frac{1}{2}}x_n, (\mathfrak{J}T)^{\frac{1}{2}}x_n),$$

therefore $(\mathfrak{J}T)^{\frac{1}{2}}x_n \to 0$, and by the continuity of $(\mathfrak{J}T)^{\frac{1}{2}}$ we know that

$$Tx_n \to 0,$$

which is contradictory to $\|x_n\| = 1$. So $\mathbb{C}^+ \cup \mathbb{C}^- \subset \rho(T)$.

Lastly we prove that the general T has $\mathbb{C}^+ \cup \mathbb{C}^- \subset \rho(T)$. By $\rho(T) \neq \emptyset$ we know that $(\mathbb{C}^+ \cup \mathbb{C}^-) \cap \rho(T) \neq \emptyset$, and taking $\overline{\mu}_0 \neq \mu_0 \in \rho(T)$, then we know that by Lemma 6.1.8

$$\overline{\mu}_0 \in \rho(T).$$

Setting $B_0 = (T - \overline{\mu}_0 I)^{-1} T (T - \mu_0 I)^{-1}$, and then it is easy to prove B_0 is continuous and $\mathfrak{J}-$ non-negative self-adjoint. So

$$\sigma(B_0) \cap (\mathbb{C}^+ \cup \mathbb{C}^-) = \emptyset.$$

From the spectral mapping theorem,

$$\sigma(B_0) = \{\mu \in \mathbb{C} : \mu = \frac{\lambda}{(\lambda - \overline{\mu}_0)(\lambda - \mu_0)}, \lambda \in \sigma(T)\},$$

and if $|\overline{\lambda_0}| \neq |\mu_0|$ then $\dfrac{\lambda_0}{(\lambda_0-\overline{\mu}_0)(\lambda_0-\mu_0)}$ is not real number for $\overline{\lambda}_0 \neq \lambda_0$, therefore

$$\frac{\lambda_0}{(\lambda_0-\overline{\mu}_0)(\lambda_0-\mu_0)} \in \rho(B_0) \Rightarrow \lambda_0 \in \rho(T),$$

that is,

$$(\mathbb{C}^+ \cup \mathbb{C}^-) \setminus \{\lambda \in \mathbb{C} : |\lambda| = |\mu_0|\} \subset \rho(T). \tag{6.7}$$

On the other hand, taking $\overline{\mu}_1 \neq \mu_1 \in \rho(T), |\mu_1| \neq |\mu_0|$, we get

$$(\mathbb{C}^+ \cup \mathbb{C}^-) \setminus \{\lambda \in \mathbb{C} : |\lambda| = |\mu_1|\} \subset \rho(T). \tag{6.8}$$

Combining (6.7) and (6.8), the conclusion is proved. □

Remark 6.1.11 Krein space and complete symplectic space have an intrinsic connection. If $\{K, [\cdot,\cdot]_{\mathfrak{J}}\}$ is a Krein space, we define

$$< x, y >= \mathrm{i}[x, y]_{\mathfrak{J}},$$

then $\{K, <\cdot,\cdot>\}$ is a complete symplectic space.

6.2 Spectra of Infinite-Dimensional Hamiltonian Operators in Krein Spaces

6.2.1 Self-adjointness of a Class of Infinite-Dimensional Hamiltonian Operators in Krein Spaces

According to the definition, the infinite-dimensional Hamiltonian operator in Hilbert space is generally a non-self-adjoint operator. Therefore, the study of the spectra of the infinite-dimensional Hamiltonian operators in Hilbert space is not as proficient as the study of self-adjoint operators. However, due to the special structure of the infinite-dimensional Hamiltonian operator, the infinite-dimensional Hamiltonian operator will become in the sense of an indefinite inner product an anti-self-adjoint operator which is a constant multiple different from the self-adjoint operator after the introduction of the appropriate indefinite inner product space, which makes it relatively easy to study the spectral properties. Therefore, it is necessary to study the spectral theory of the infinite-dimensional Hamiltonian operator in the indefinite inner product spaces.

The following lemma will tell us how to extend the concept of infinite-dimensional Hamiltonian operator in Hilbert space to complete indefinite inner product space.

Lemma 6.2.1 *Let $\{X, (\cdot,\cdot)\}$ be a Hilbert space, densely defined closed linear operator $H : \mathscr{D}(H) \subset X \oplus X \to X \oplus X$ be symplectic self-adjoint infinite-dimensional Hamiltonian operator if and only if $\mathtt{i}H$ is a J_1- self-adjoint operator in Krein space K_{J_1}, where $K_{J_1} = \{X \oplus X, [\cdot,\cdot]_{J_1}\}$, $[\cdot,\cdot]_{J_1} = (J_1\cdot,\cdot)$ and $J_1 = \begin{bmatrix} 0 & iI \\ -iI & 0 \end{bmatrix}$.*

Proof Adjoint operator of H in Krein space K_{J_1} is denoted as $H^\dagger$, then there is

$$H^\dagger = J_1 H^* J_1.$$

According to the definition of symplectic self-adjoint infinite-dimensional Hamiltonian operator,

$$(\mathtt{i}H)^* = J_1(\mathtt{i}H)J_1.$$

After substituting the equation above we get

$$(\mathtt{i}H)^\dagger = (\mathtt{i}H),$$

therefore $\mathtt{i}H$ is a J_1- self-adjoint operator in Krein space K_{J_1}.

On the contrary, when $\mathtt{i}H$ is a J_1- self-adjoint operator in Krein space K_{J_1}, then using relation

$$H^\dagger = J_1 H^* J_1$$

we can prove in the same way that H is a symplectic self-adjoint infinite-dimensional Hamiltonian operator. The proof is completed. □

Theorem 6.2.2 *Let $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \oplus X \to X \oplus X$ be an infinite-dimensional Hamiltonian operator, then in Krein space $K_{\mathfrak{J}}$*

(i) $\lambda \in \sigma_{p,1}(H)$ *if and only if* $\overline{\lambda} \in \sigma_{r,1}(H^\dagger)$;
(ii) $\lambda \in \sigma_{p,2}(H)$ *if and only if* $\overline{\lambda} \in \sigma_{r,2}(H^\dagger)$;
(iii) $\lambda \in \sigma_{p,3}(H)$ *if and only if* $\overline{\lambda} \in \sigma_{p,3}(H^\dagger)$;
(vi) $\lambda \in \sigma_{p,4}(H)$ *if and only if* $\overline{\lambda} \in \sigma_{p,4}(H^\dagger)$;
(v) $\lambda \in \sigma_c(H)$ *if and only if* $\overline{\lambda} \in \sigma_c(H^\dagger)$;
(v) $\lambda \in \rho(H)$ *if and only if* $\overline{\lambda} \in \sigma_c(H^\dagger)$.

Proof Since $H^\dagger = \mathfrak{J}H^*\mathfrak{J}$, the proof is trivial. □

Proposition 6.2.3 *Let $H = \begin{bmatrix} A & B \\ C & -A^* \end{bmatrix} : \mathscr{D}(H) \subset X \oplus X \to X \oplus X$ be a symplectic self-adjoint infinite-dimensional Hamiltonian operator, then in Krein space K_{J_1}, $J_1 = \begin{bmatrix} 0 & iI \\ -iI & 0 \end{bmatrix}$*

(i) $\lambda \in \sigma_{p,1}(H)$ *if and only if* $-\overline{\lambda} \in \sigma_{r,1}(H)$;
(ii) $\lambda \in \sigma_{p,2}(H)$ *if and only if* $-\overline{\lambda} \in \sigma_{r,2}(H)$;

(iii) $\lambda \in \sigma_{p,3}(H)$ *if and only if* $-\overline{\lambda} \in \sigma_{p,3}(H)$;
(vi) $\lambda \in \sigma_{p,4}(H)$ *if and only if* $-\overline{\lambda} \in \sigma_{p,4}(H)$;
(v) $\lambda \in \sigma_c(H)$ *if and only if* $-\overline{\lambda} \in \sigma_c(H)$;
(v) $\lambda \in \rho(H)$ *if and only if* $-\overline{\lambda} \in \rho(H)$.

6.2.2 *Existence of Maximal Definite Invariant Subspaces in Krein Spaces*

In the theory of linear operators in indefinite inner product spaces, the existence of maximal quasi-definite invariant subspaces is its central content. In this regard, I. S. Iokhvidov, H. Langer, T. Y. Azizov, J. Bognár et al. have achieved significant results (see [91, 92]). For example, in 1944 Pontryagin found that there exists an invariant subspace of self-adjoint operator in Pontryagin space. M. G. Krein found that there is an invariant subspace for unitary operators in Pontryagin space. H. Langer gave a sufficient condition for the existence of invariant subspaces of self-adjoint operators in Krein spaces. T. Y. Azizov gave sufficient conditions for maximal dissipative operators in invariant subspaces in Krein spaces, and so on. Known from [25], the problem of solving a class of Riccati equations can be transformed into the problem of solving the infinite-dimensional Hamiltonian operator's angle operator, and the existence of the angle operator is equivalent to the existence of an invariant subspace of an infinite-dimensional Hamiltonian operator. Based on the above considerations, this section is devoted to the study of the existence of maximal quasi-invariant subspaces of infinite-dimensional Hamiltonian operator, and a series of conclusions are given.

Definition 6.2.4 The linear operator T in an indefinite inner product space K is called plus operator, if the domain of T is the whole space and the non-negative vector is mapped into a non-negative vector.

Definition 6.2.5 A subspace L in an indefinite inner product space K is called a definite subspace if L is either a positive definite subspace or a negative definite space. The definite subspace that overlaps with itself after extension is called maximal definite subspace.

On the invariant subspace problem of the plus operator, Bognár [91] gave the following conclusions.

Lemma 6.2.6 *Let* $K = K_+ \oplus K_-$ *be a Krein space,* P_+, P_- *be the projection operators corresponding to the canonical factorization.* $V \in \mathscr{B}(K)$ *is a plus operator, if operator* $P_+ V P_-$ *is a compact operator, then satisfies the condition that the closed positive subspace* $L \subset K$ *of* $\overline{VL} = L$ *can be expanded into a maximal positive subspace* $L_1 \subset K$ *such that* $VL_1 \subset L_1$, *that is,* $L \subset K$ *can be expanded into a maximal invariant positive subspace of* V.

Lemma 6.2.7 *Let $K = K_+ \oplus K_-$ be a Krein space, P_+, P_- be the projection operators corresponding to the canonical factorization. $V \in \mathscr{B}(K)$ is a plus operator, if operator $P_+ V P_-$ is a compact operator, then operator V has a maximal invariant positive subspace.*

Proof By Lemma 6.2.6, take $L = \{0\}$, then $\overline{VL} = L$ and L is a non-negative subspace, thus it can be extended to the maximal invariant positive subspace of V. □

Lemma 6.2.8 *Let K be a Krein space, and L be a maximal positive subspace (maximal negative subspace) of K, then $L^\perp$ is a maximal negative subspace (maximal positive subspace).*

Proof If L is a maximal positive subspace, then it is easy to know that $L^\perp$ is closed and negative. Suppose $L^\perp$ is not maximal, then consider the canonical factorization

$$K = K_+ \oplus K_-,$$

and let the corresponding projection operators be P_+ and P_-, then $P_- L^\perp \neq K^-$. So there exists $x \in K_-$ which satisfies $x \perp P_- L^\perp$, $x \neq 0$. However, $[x, x] < 0$ and $x \in L^{\perp\perp}$. But this is contradictory to the fact that $L^{\perp\perp}$ is a positive subspace. □

Theorem 6.2.9 *Let $H\,(\rho(H) \neq \emptyset)$ be a symplectic self-adjoint infinite-dimensional Hamiltonian operator, and K_{J_1} be Krein space. If $K_+ \subset \mathscr{D}(H)$, where $K_{J_1} = K_+ \oplus K_-$ is the canonical factorization corresponding to inner product operator $J_1 = \begin{bmatrix} 0 & iI \\ -iI & 0 \end{bmatrix}$ and*

(i) $Im[Hx, x]_{J_1} \leq 0$ is satisfied for any $x \in \mathscr{D}(H)$;
(ii) operator $P_+ H P_-$ is a $P_- H P_-$ − compact operator,

then the infinite-dimensional Hamiltonian operator H has a maximal positive subspace and a maximal negative subspace.

Proof If $\rho(H) \neq \emptyset$ and satisfies $Im[Hx, x]_{J_1} \leq 0$ for any $x \in \mathscr{D}(H)$, $\mathtt{i}H$ is a J_1− non-negative self-adjoint, and by Lemma 6.1.10 it is easy to get

$$\mathbb{C}_+ \cup \mathbb{C}_- \subset \rho(H),$$

where $\mathbb{C}_+$, $\mathbb{C}_-$ represent the open right half-plane and the open left half-plane, respectively.

Let

$$H = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix}$$

be 2×2 block matrix form of infinite-dimensional Hamiltonian operator H in Krein space K_{J_1}. Then due to the boundedness of H_{21}, there exists $Im(\xi) < 0$, $Re(\xi) \neq 0$ and $|Im(\xi)| > \|H_{21}\|$ such that $\xi \in \rho(H)$. Let $V = (H - \overline{\xi} I)(H - \xi I)^{-1}$, then

$$[Vx, Vx]_{J_1} = [x, x]_{J_1} + 4Im(\xi)Im[Hy, y]_{J_1},$$

where $y = (H - \xi I)^{-1}x$.

Therefore, we get

$$[Vx, Vx]_{J_1} \geq [x, x]_{J_1},$$

that is, V is a plus operator.

On the other hand, since $\mathtt{i}H$ is a self-adjoint in Krein space K_{J_1} and H_{21} is a bounded operator on the whole space, it is easy to prove that $\xi \in \rho(H_{22})$ and H_{22} is a bounded linear operator on the whole space, therefore

$$H - \xi I = \begin{bmatrix} I & T_1 \\ 0 & I \end{bmatrix} \begin{bmatrix} T_2 & 0 \\ 0 & H_{22} - \xi \end{bmatrix} \begin{bmatrix} I & 0 \\ T_3 & I \end{bmatrix},$$

where

$$T_1 = H_{12}(H_{22} - \xi)^{-1}, T_2 = H_{11} - \xi - H_{12}(H_{22} - \xi)^{-1}H_{21},$$

$$T_3 = (H_{22} - \xi)^{-1}H_{21}.$$

Let

$$V = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix}$$

be a 2×2 block matrix form of the operator V in Krein space K_{J_1}. Considering the Cayley transformation

$$V = (H - \overline{\xi})(H - \xi)^{-1} = I + (\xi - \overline{\xi})(H - \xi)^{-1}, \tag{6.9}$$

we can get

$$v_{11} = I + (\xi - \overline{\xi})T_2^{-1}, v_{12} = -(\xi - \overline{\xi})T_2^{-1}T_1.$$

Because $H_{12} = P^+HP^-$ is a compact operator compared to operator $H_{22} = P^-HP^-$, v_{12} is a compact operator. Therefore, by Lemma 6.2.7 there exists a maximal positive subspace $L \subset K$ such that $VL \subset K$.

It is easy to prove that $(V - I)L \subset L$. The following will prove $(V - I)L = L$. As $L \subset K$ is a maximal positive subspace if and only if $P_+L = K_+$, $(V - I)L = L$ if and only if

$$P_+(V - I)L = K_+.$$

If A be a corresponding angular operator to subspace L in K_+ in K_+, then L can be expressed as

$$L = \left\{ \begin{bmatrix} x_+ \\ Ax_+ \end{bmatrix}, x_+ \in K_+ \right\},$$

and

$$P_+(V-I)L = (v_{11} - I + v_{12}A)K_+.$$

Furthermore, we get

$$P_+(V-I)L = (\xi - \overline{\xi})T_2^{-1}(I - T_1A)K_+.$$

As $\mathscr{D}(T_2) = \mathscr{D}(H_{11}) = \mathscr{D}(H_{21}) = K_+$ and satisfies $Im[Hx, x]_{J_1} \leq 0$ for any $x \in \mathscr{D}(H)$,

$$Im\left(\begin{bmatrix} P^+ & 0 \\ 0 & -P^- \end{bmatrix}\begin{bmatrix} P^+HP^+ & P^+HP^- \\ P^-HP^+ & P^-HP^- \end{bmatrix}\begin{bmatrix} 0 \\ x^- \end{bmatrix}, \begin{bmatrix} 0 \\ x^- \end{bmatrix}\right) \leq 0.$$

As $x_- \in K_-$, we have

$$Im(H_{22}x_-, x_-) \geq 0$$

in Hilbert space K_-. We consider the following equation

$$\|(H_{22} - \xi)x_-\|^2 = \|(H_{22} - Re(\xi))x_-\|^2 + (Im(\xi))^2\|x_-\|^2,$$

with regard to $x_- \in K_-$ and get

$$\|(H_{22} - \xi)^{-1}\| \leq \frac{1}{|Im(\xi)|}.$$

And by $|Im(\xi)| > \|H_{21}\|$ we know $\|T_1\| < 1$. Because A is an angular operator of $L \subset K^+$ corresponding to K^+, we have $\|A\| \leq 1$. Furthermore we get

$$\|T_1A\| < 1,$$

and operator $I - T_1A$ is invertible. So $\mathscr{R}(T_2^{-1}(I - T_1A)) = K^+$, that is,

$$(V-I)L = L.$$

By solving (6.9) we can get

$$H = (\xi - \overline{\xi})(V-I)^{-1} + \xi = (\xi V - \overline{\xi})(V-I)^{-1}.$$

Again by $(V-I)L = L$ we can get

$$L \subset \mathscr{D}(H), HL \subset L,$$

so there exists a maximal positive subspace.

Let $\widetilde{L} = L^{\perp}$, then by Lemma 6.2.8 we know that $\widetilde{L}$ is a maximal negative subspace. In addition,

$$[Hx, y]_{J_1} = -[x, Hy]_{J_1} = 0,$$

for any $x \in \widetilde{L} \cap \mathscr{D}(H)$ and $y \in L$, that is, subspace $\widetilde{L}$ concerning H is invariable, as a result, $\widetilde{L}$ is a maximal negative subspace. The proof is completed. □

Theorem 6.2.10 *Let $H(\rho(H) \neq \phi)$ be a symplectic self-adjoint infinite-dimensional Hamiltonian operator, K_{J_1} be a Krein space. If $K_+ \subset \mathscr{D}(H)$, then $K_{J_1} = K_+ \oplus K_-$ is the canonical factorization corresponding to the inner product operator $\mathfrak{J}$, and satisfies*

(i) $Im[Hx, x]_{J_1} \geq 0$ is satisfied for any $x \in \mathscr{D}(H)$;
(ii) $P_+ H P_-$ is a $P_- H P_-$ − compact operator,

then the infinite-dimensional Hamiltonian operator H has a maximal positive subspace and a maximal negative subspace.

Proof According to the given conditions, it is easy to know that $-H$ satisfies Theorem 6.2.9. Thus $-H$ has a maximal positive subspace and a maximal negative subspace. Furthermore, H has a maximal positive subspace and a maximal negative subspace. The proof is completed. □

6.3 $\Im-$Numerical Range in Krein Spaces

6.3.1 Definition of $\Im-$Numerical Range in Krein Spaces

Similar to the classical numerical range in Hilbert space, the numerical range in the sense of indefinite inner product can be introduced in the indefinite inner product space.

Definition 6.3.1 The numerical range of bounded operator A in indefinite inner product space $\{K, [\cdot, \cdot]_{\Im}\}$ can be defined as

$$W_{\Im}(A) = \{\frac{[Ax, x]_{\Im}}{[x, x]_{\Im}} : [x, x]_{\Im} \neq 0\},$$

and is called $\Im$-numerical range, where $[\cdot, \cdot]_{\Im} = (\Im\cdot, \cdot)$. Equivalently

$$W_{\Im}(A) = \{[Ax, x]_{\Im} : [x, x]_{\Im} = 1\} \cup \{-[Ax, x]_{\Im} : [x, x]_{\Im} = -1\}.$$

The $\Im$-numerical range of unbounded operators is defined as

$$W_{\Im}(A) = \{\frac{[Ax, x]_{\Im}}{[x, x]_{\Im}} : x \in \mathscr{D}(A), [x, x]_{\Im} \neq 0\},$$

and it is easy to prove

$$W_{\mathfrak{J}}(A) = \{[Ax, x]_{\mathfrak{J}} : x \in \mathscr{D}(A), [x, x]_{\mathfrak{J}} = 1\} \cup \{-[Ax, x]_{\mathfrak{J}} : x \in \mathscr{D}(A), [x, x]_{\mathfrak{J}} = -1\}.$$

The following examples illustrate that the eigenvalue inclusion relation of linear operators in Hilbert space is not true for the $\mathfrak{J}$-numerical range.

Example 6.3.2 Let $X = L^2[0, 1]$, $K = X \times X$, $K_{\mathfrak{J}} = \{K, [\cdot, \cdot]_{J_1}\}$, where $\mathfrak{J} = J_1 = \begin{bmatrix} 0 & \mathrm{i}I \\ -\mathrm{i}I & 0 \end{bmatrix}$. Define the following block operator matrix

$$T = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 0 & I \\ -\dfrac{\mathrm{d}^2}{\mathrm{d}x^2} & 0 \end{bmatrix},$$

where $\mathscr{D}(C) = \{x \in X : x' \text{ absolutely continuous }, x', x'' \in X, x(0) = x(1) = 0\}$, then it is easy to prove that $\mathrm{i}T$ is a $\mathfrak{J}-$ self-adjoint operator in space $K_{\mathfrak{J}}$, therefore $W_{\mathfrak{J}}(T) \subset \mathrm{i}\mathbb{R}$.

On the other hand, we get

$$\sigma(T) = \sigma_p(T) = \{k\pi : k = \pm 1, \pm 2, \ldots\}$$

by calculation which shows that the eigenvalue inclusion relation $\sigma_p(T) \subset W_{\mathfrak{J}}(T)$ doesn't hold.

The above examples demonstrate that there is a big difference between the numerical range in Hilbert space and that in the indefinite inner product space. Next, we will discuss the problem of when the $\mathfrak{J}-$numerical range in the indefinite inner product space has a spectral inclusion relation.

Theorem 6.3.3 *Let A be a bounded $\mathfrak{J}-$non-negative self-adjoint operator (that is, $A = A^{\dagger} \geq 0$) in indefinite inner product space $\{K, [\cdot, \cdot]_{\mathfrak{J}}\}$. Then $\sigma_p(A)\backslash\{0\} \subset W_{\mathfrak{J}}(A)$ and $\sigma(A)\backslash\{0\} \subset \overline{W_{\mathfrak{J}}(A)}$; if $A\mathfrak{J} = \mathfrak{J}A$ is satisfied, then $\sigma(A)\backslash\{0\} \subset \overline{W_{\mathfrak{J}}(A)}$.*

Proof Setting $\lambda \in \sigma_p(A)\backslash\{0\}$, then there exists $x_0 \neq 0$ such that

$$Ax_0 = \lambda x_0,$$

therefore $[Ax_0, x_0]_{\mathfrak{J}} = \lambda[x_0, x_0]_{\mathfrak{J}}$. It can be asserted that $[x_0, x_0]_{\mathfrak{J}} \neq 0$. In fact, if $[x_0, x_0]_{\mathfrak{J}} = 0$, then $[Ax_0, x_0]_{\mathfrak{J}} = 0$. By Lemma 6.1.7 we know that $Ax_0 = 0$, and $\lambda = 0$, then contradiction is derived. Therefore,

$$\lambda = \frac{[Ax_0, x_0]_{\mathfrak{J}}}{[x_0, x_0]_{\mathfrak{J}}} \in W_{\mathfrak{J}}(A),$$

that is, $\sigma_p(A)\backslash\{0\} \subset W_{\mathfrak{J}}(A)$.

The following proves $\sigma(A)\backslash\{0\} \subset \overline{W_{\Im}(A)}$. Because $\sigma(A) \subset \mathbb{R}$ and $\sigma_r(A) = \emptyset$, $\sigma(A) = \sigma_{ap}(A)$, where $\sigma_{ap}(A)$ is an approximate point spectrum, that is,

$$\sigma_{ap}(A) = \{\lambda \in C : (A-\lambda)x_n \to 0, \{x_n\}_{n=1}^{+\infty} \subset \mathscr{D}(A), \|x_n\| = 1, n = 1, 2, \ldots\}.$$

Let $\lambda \in \sigma_{ap}(A)\backslash\{0\}$, then there exists $\|x_n\| = 1$ such that

$$(A-\lambda)x_n \to 0,$$

then we have

$$[Ax_n, x_n]_{\Im} - \lambda[x_n, x_n]_{\Im} \to 0.$$

If the lower limit of $\{[x_n, x_n]_{\Im}\}$ is 0, we might as well let $[x_n, x_n]_{\Im} \to 0$, then

$$[Ax_n, x_n]_{\Im} = (\Im Ax_n, x_n) = ((\Im A)^{\frac{1}{2}}x_n, (\Im A)^{\frac{1}{2}}x_n) \to 0,$$

and $(\Im A)^{\frac{1}{2}}x_n \to 0$ is obtained. As $(\Im A)^{\frac{1}{2}}$ is bounded, $\Im Ax_n = (\Im A)^{\frac{1}{2}}(\Im A)^{\frac{1}{2}}x_n \to 0$, thus

$$Ax_n \to 0, \quad \lambda x_n \to 0,$$

which is contradictory to $\|x_n\| = 1$. Therefore $\sigma_{ap}(A)\backslash\{0\} \subset \overline{W_{\Im}(A)}$.

If $A\Im = \Im A$, let $0 \in \sigma_{ap}(A)$, then there exists sequence $\{x_n\}(\|x_n\| = 1, n = 1, 2, \ldots)$ such that

$$Ax_n \to 0.$$

If the lower limit $[x_n, x_n]_{\Im}$ is not 0, then obviously $0 \in \overline{W_{\Im}(A)}$. If the lower limit of $[x_n, x_n]_{\Im}$ is 0, we might as well let $[x_n, x_n]_{\Im} \to 0$ and $y_n = x_n + \Im x_n$, then

$$\begin{aligned}[y_n, y_n]_{\Im} =& [x_n, x_n]_{\Im} + [\Im x_n, x_n]_{\Im} + [x_n, \Im x_n]_{\Im} + [\Im x_n, \Im x_n]_{\Im} \\ \to& 2,\end{aligned}$$

and

$$\begin{aligned}[Ay_n, y_n]_{\Im} =& [Ax_n, x_n]_{\Im} + [A\Im x_n, x_n]_{\Im} + [Ax_n, \Im x_n]_{\Im} + [A\Im x_n, \Im x_n]_{\Im} \\ =& [Ax_n, x_n]_{\Im} + [\Im x_n, Ax_n]_{\Im} + [Ax_n, \Im x_n]_{\Im} + [\Im Ax_n, \Im x_n]_{\Im} \\ \to& 0.\end{aligned}$$

So $0 \in \overline{W_{\Im}(A)}$. The proof is completed. □

Remark 6.3.4 Conclusion $\sigma_p(A) \subset W_{\Im}(A)$ doesn't hold in general conditions. In addition, if A is unbounded, $\sigma(A) \subset \overline{W_{\Im}(A)}$ doesn't necessarily hold.

Example 6.3.5 Let X be a Hilbert space and its inner produce can be denoted as $(\cdot,\cdot)$. Set $K = X \times X$ and

$$[\cdot,\cdot]_{\Im} = (\Im\cdot,\cdot),$$

where $\Im = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$. Firstly, take $A = \begin{bmatrix} 0 & 0 \\ I & 0 \end{bmatrix}$, then

$$W_{\Im}(A) = \{\frac{(x,x)}{2Re(x,y)} : \begin{bmatrix} x \\ y \end{bmatrix} \in X \times X, Re(x,y) \neq 0\},$$

and $0 \in \sigma_p(A)$, $0 \notin W_{\Im}(A)$. Secondly, let B be an unbounded non-negative self-adjoint operator in Hilbert space X. Take

$$A = \begin{bmatrix} 0 & 0 \\ B & 0 \end{bmatrix},$$

then operator A is a $\Im-$ self-adjoint operator in Krein space $\{K, [\cdot,\cdot]_{\Im}\}$, $W_{\Im}(A) \subset \mathbb{R}$.

On the other hand, select $x_0 \notin \mathscr{D}(B)$, then vector $\begin{bmatrix} x_0 & 0 \end{bmatrix}^T \in X \times X$ is not included in $\mathscr{R}(A - \lambda I)$ for any $\lambda \in \mathbb{C}$, so $\sigma(A) = \mathbb{C}$, and hence the conclusion $\sigma(A) \subset \overline{W_{\Im}(A)}$ doesn't hold.

Theorem 6.3.6 *Let A be a $\Im-$ non-negative self-adjoint operator possibly unbounded in Krein space $\{K, [\cdot,\cdot]_{\Im}\}$. If $0 \in \rho(A)$, then $\sigma_p(A) \subset W_{\Im}(A)$, $\sigma(A) \subset \overline{W_{\Im}(A)}$.*

Proof Similar to the proof of Theorem 6.3.3, it is easy to prove $\sigma_p(A) \subset W_{\Im}(A)$ and $\sigma_r(A) = \emptyset$. If $\lambda \in \sigma_c(A)$, then there exists $\|x_n\| = 1$ such that $(A - \lambda)x_n \to 0$, which contains

$$[Ax_n, x_n]_{\Im} \to \lambda[x_n, x_n]_{\Im}.$$

By $0 \in \rho(A)$, we know that $[Ax_n, x_n]_{\Im} \geq M[x_n, x_n]_{\Im}$, and there exists $N > 0$ such that

$$\lambda[x_n, x_n]_{\Im} \geq M \text{ for any } n > N,$$

which means in detail that

$$\lambda_n = \frac{[Ax_n, x_n]_{\Im}}{[x_n, x_n]_{\Im}} \to \lambda,$$

so $\sigma_c(A) \subset \overline{W_{\Im}(A)}$, that is, $\sigma(A) \subset \overline{W_{\Im}(A)}$. The proof is completed. □

6.3.2 Boundedness and Convexity of $\Im-$Numerical Range

The numerical range of a bounded operator in a Hilbert space is bounded, but the $\Im-$numerical range of a bounded operator in an indefinite inner product space is not

necessarily bounded. The following will discuss when the $\mathfrak{J}-$numerical range of a bounded operator in an indefinite inner product space is bounded.

Lemma 6.3.7 *Let T, S be bounded linear operators in Hilbert space X. If there exists a bounded operator P such that $T^* = SP$, then there exists $\lambda \geq 0$ so that $T^*T \leq \lambda SS^*$.*

Proof If there exists bounded operator P such that $T^* = SP$,

$$\begin{aligned} T^*T =& SPP^*S^* \\ =& \|P\|^2 SS^* - S(\|P\|^2 I - PP^*)S^* \\ \leq& \|P\|^2 SS^*, \end{aligned}$$

so, take $\lambda = \|P\|^2$, then $T^*T \leq \lambda SS^*$. □

Theorem 6.3.8 *Let T be a bounded operator in Krein space $\{K, [\cdot,\cdot]_{\mathfrak{J}}\}$. If one of the following conditions is satisfied*

(i) $T = \lambda I$;
(ii) canonical symmetric operator $\mathfrak{J}$ is definite (that is, $\mathfrak{J} > 0$ or $\mathfrak{J} < 0$),

then $W_{\mathfrak{J}}(T)$ is bounded.

Proof If $T = \lambda I$, it is obvious that $W_{\mathfrak{J}}(T)$ is bounded. If $\mathfrak{J}$ is a definite operator, we might as well let $\mathfrak{J} > 0$, then due to the fact that $\mathfrak{J} = \mathfrak{J}^* = \mathfrak{J}^{-1}$ we have

$$T^*\mathfrak{J}^{\frac{1}{2}} = \mathfrak{J}^{\frac{1}{2}}(\mathfrak{J}^{\frac{3}{2}}T^*\mathfrak{J}^{\frac{1}{2}}).$$

By Lemma 6.3.7 we know there exists $\lambda \geq 0$ such that $T^*\mathfrak{J}T \leq \lambda\mathfrak{J}$. Also because

$$\begin{aligned} \frac{[Tx,x]_{\mathfrak{J}}}{[x,x]_{\mathfrak{J}}} =& \frac{(\mathfrak{J}^{\frac{1}{2}}Tx, \mathfrak{J}^{\frac{1}{2}}x)}{(\mathfrak{J}x,x)} \\ \leq& \frac{(\mathfrak{J}^{\frac{1}{2}}Tx, \mathfrak{J}^{\frac{1}{2}}Tx)^{\frac{1}{2}}(\mathfrak{J}^{\frac{1}{2}}x, \mathfrak{J}^{\frac{1}{2}}x)^{\frac{1}{2}}}{(\mathfrak{J}x,x)} \\ =& \frac{(T^*\mathfrak{J}Tx,x)^{\frac{1}{2}}}{(\mathfrak{J}x,x)^{\frac{1}{2}}} \leq \sqrt{\lambda}[\frac{(\mathfrak{J}x,x)}{(\mathfrak{J}x,x)}]^{\frac{1}{2}}. \end{aligned}$$

So $W_{\mathfrak{J}}(T)$ is bounded. □

An important property of classical numerical range is the convexity. The $\mathfrak{J}$numerical range of a bounded operator in an indefinite inner product space is not necessarily a convex set. The following will solve the problem of when the $\mathfrak{J}$numerical range is convex.

Theorem 6.3.9 *Let T be a bounded operator in Krein space $\{K, [\cdot,\cdot]_{\mathfrak{J}}\}$. If the standard symmetric operator $\mathfrak{J}$ is determined, then $W_{\mathfrak{J}}(T)$ is a convex set.*

Proof If $\Im$ is a determined operator, we might as well let $\Im > 0$, then

$$\frac{[Tx, x]_{\Im}}{[x, x]_{\Im}} = \frac{(\Im^{\frac{1}{2}} T \Im^{\frac{3}{2}} \Im^{\frac{1}{2}} x, \Im^{\frac{1}{2}} x)}{(\Im^{\frac{1}{2}} x, \Im^{\frac{1}{2}} x)},$$

that is, $W_{\Im}(T) = W(\Im^{\frac{1}{2}} T \Im^{\frac{3}{2}})$ is convex. □

References

1. Hamilton, W. R. 1940. *The mathematical papers of Sir William Rowan Hamilton*, vol. 11. Cambridge: Cambridge Uniuersity Press.
2. Johnson, R., R. Obaya, S. Novo, C. Núñez, and R. Fabbri. 2016. *Nonautonomous linear hamiltonian systems: oscillation, spectral theory and control*. Switzerland: Springer International Publishing.
3. Chernoff, P. R., and J. E. Marsden. 1974. In *Properties of infinite dimensional hamiltonian systems*, vol. 425. Lecture notes in mathemathics. New York: Springer.
4. Lancaster, P., and L. Rodman. 1995. *Algebraic Riccati equations*. Oxford: Clarendon Press.
5. Zhong, W. X. 1995. *On new system of elasticity solution*. Dalian Dalian University of Technology Press.
6. Feng, K., and M. Z. Qing. 2003. *Hamiltonian symplectic geometric algorithm systems*, 2nd ed. Hangzhou: Zhejiang Science and Technology Press.
7. Yao, W. A., and W. X. Zhong. 2002. *Symplectic elasticity*. Beijing: Higher Education Press.
8. Sun, J., and Z. Wang. 2005. *Spectra analysis of linear operators*. Beijing: Science Press.
9. Wu, D. Y., and A. Chen. 2013. *Spectral theory of on block operator matrices and application*. Beijing: Science Press.
10. Wu, D. Y., A. Chen, J. J. Huang, and G. J. Hai. 2018. *Numerical range of linear operators in hilbert spaces and its application*. Beijing: Science Press.
11. Magri, F. 1978. A simple model of the integrable Hamiltonian equation. *Journal of Mathematical Physics* 19: 1156–1162.
12. Magri, F. 1980. In *A geometrical approach to the nonlinear solvable equation*, vol. 120. Lecture notes in physics. New York: Springer.
13. Vinogradov, A. M. 1978. Hamiltonian structures in field theory. *Soviet Mathematics. Doklady* 19: 790–794.
14. Kupershmidt, B. A. 1980. In *Geometry of jet bundles and the structure of lagrangian and hamiltonian formalisms*, vol. 775. Lecture notes in mathematics, 162–218. New York: Springer.
15. Manin, Y. I. 1979. Algebraic aspects of nonlinear differential equations. *Journal of Soviet Mathematics* 11: 1–122.
16. Gel'fand, I. M., and I. Y. Dorfman. 1979. Hamiltonian operators and algebraic structures related to them. *Functional Analysis and Its Applications* 13: 248–262.
17. Olver, P. J. 1980. On the Hamiltonian structure of evolution equations. *Mathematical Proceedings of the Cambridge Philosophical Society* 88: 71–88.
18. Olver, P. J. 1982. A nonlinear Hamiltonian structures for the Euler equations. *Journal of Mathematical Analysis and Applications* 89: 233–250.

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19. Olver, P. J. 1984. Conservation law in elasticity. I: General results. *The Archive for Rational Mechanics and Analysis* 85: 111–129.
20. Olver, P. J. 1986. *Applications of lie groups to differential equations*. New York: Springer.
21. Olver, P. J. 1988. Darboux' theorem for Hamiltonian differential operators. *The Journal of Differential Equations* 71 (1): 10–33.
22. Kosmann-Schwarzbach, Y. 1981. Hamiltonian systems on fibered manifolds. *Letters in Mathematical Physics* 5: 229–237.
23. Vainberg, M. M. 1964. *Variational methods for the study of nonlinear operators, translated from the 1956 Russian Monograph*. San Francisco: Holden-Day.
24. Curtain, R. F., and H. J. Zwart. 1995. *An introduction to infinite-dimensional linear systems theory*. New York: Springer.
25. Langer, H., A. C. M. Ran, and B. A. van de Rotten. 2001. Invariant subspaces of infinite dimensional Hamiltonians and solutions of the corresponding Riccati equations. *Operator Theory: Advances and Applications* 130: 235–254.
26. Zhong, W. X. 1991. Method of variable separation and hamiltonian systems. *Computational Structural Mechanics and Applications* 8 (3): 229–240.
27. Zhong, W. X. 1991. On elastic plane problem of strip domain and Hamiltonian systems. *Journal of Dalian University of Technology* 31 (4): 373–384.
28. Zhong, W. X., and W. A. Yao. 1999. A new system for solving plank bending and its application. *Journal of Mechanics* 31 (2): 173–184.
29. Zhong, W. X. 2003. Developmental hamiltonian kernel integral equations. *Journal of Dalian University of Technology* 43 (1): 1–11.
30. Zhong, W. X. 2004. Completeness of hamiltonian eigensolutions for elliptic equations. *Journal of Dalian University of Technology* 44 (1): 1–6.
31. Shi, H. 1998. *Introduction to mechanized mathematics*. Changsha: Hunan Education Press.
32. Wu, W. T. 1986. On zeros of algebraic equations An application of Ritt principle. *Chinese Science Bulletin* 31 (1): 1–5.
33. David, C., L. John, and O. S. Donal. 1997. *Ideals, variables, and algorithms*, 3rd ed. New York: Springer.
34. Lu, B., and H. Q. Zhang. 2002. Mechanized algorithm for constructing general solutions of partial differential equations of first kind and automatic reasoning of mechanical equations. *Journal of Shandong University of Science and Technology (Natural Science Edition)* 21 (1): 18–24.
35. Cao, L. N., and H. Q. Zhang. 2002. Mechanized algorithm for general solutions of a class of linear partial differential equations. *Journal of Jinzhou Normal University (Natural Science Edition)* 23 (1): 57–59.
36. Chen, A., H. Q. Zhang, and W. X. Zhong. 2000. Pseudo-division algorithm of multivariate polynomial matrices and its application. *Applied Mathmatics and Mechanics* 21 (7): 661–668.
37. Qiu, Z. Y. 1994. *Mathematica application of mathematical software systems and its program design*. Beijing: Beijing University Press.
38. Li, S. Z., F. L. Cheng, and Y. H. Zhang. 1999. *Mathematics experiment*. Beijing: Beijing University Press.
39. Chen, A., H. Q. Zhang, and W. X. Zhong. 1999. Infinite dimensional hamiltonian canonical representation of a class partial differential equations. *Journal of Mechanics* 31 (3): 347–357.
40. Zheng, Y., and H. Q. Zhang. 1996. Hamiltonian canonical representation in solid mechanics. *Journal of Mechanics* 28 (1): 119–125.
41. Song, H. S. 2004. *Quantum mechanics*. Dalian: Dalian University of Technology Press.
42. Li, D. Q., and T. H. Qing. 2005. *On physics and partial differential equations*, 2nd ed. Beijing: Higher Education Press.
43. Weidmann, J. 1980. *Linear operator theory in hilbert space*. New York: Springer.
44. Wang, H., A. Chen, and J. J. Huang. 2009. Symmetry of the point spectrum of upper triangular infinite dimensional hamiltonian operators. *Journal of Mathematical Research and Exposition* 29 (5): 907–912.

45. Weyl, H. 1909. Uber beschrankte quadratische Formen, deren differenz vollstelig ist. *Rendiconti del Circolo Matematico di Palermo* 27 (1): 373–392.
46. Jerbi, A., and M. Mnif. 2005. Fredholm operators, essential spectra and application to transport equations. *Acta Applicandae Mathematicae* 89 (1–3): 155–176.
47. Jeribi, A., and N. Moalla. 2009. A characterization of some subsets of Schechter's essential spectrum and application to singular transport equation. *Journal of Mathematical Analysis and Applications* 358 (2): 434–444.
48. Stampfly, J. G., and J. P. Williams. 1968. Growth conditions and the numerical range in a Banach algebra. *Tohoku Mathematical Journal* 20 (4): 417–424.
49. Schemoeger, C. 1997. The spectral mapping theorem for the essential approximate point spectrum. *Colloquium Mathematicum* 74 (2): 167–176.
50. Hundertmark, D., and Y. Lee. 2007. Exponential decay of eigenfunctions and generalized eigenfunctions of a non-self-adjoint matrix Schrödinger operator related to NLS. *Bulletin of the London Mathematical Society* 39 (5): 709–720.
51. Gustafson, K. 1972. Necessary and sufficient conditions for Weyl's theorem. *Michigan Mathematical Journal* 19 (1): 71–81.
52. Jeribi, A., and I. Walha. 2011. Gustafson, weidmann, kato, wolf, schechter and browder essential spectra of some matrix operator and application to two-group transport equation. *Mathematische Nachrichten* 284 (1): 67–86.
53. Hai, G. J., and A. Chen. 2018. *Generalized inverse operators and fredholm operator in hilbert spaces*. Beijing: Higher Education Press.
54. Kato, T. 1984. *Perturbation Theory for Linear Operators (Second Corrected Printing of the Second*. Nd. New York, Tokyo: Springer.
55. Gohberg, I., S. Goldberg, and M. A. Kaashoek. 1990. *Classes of linear operators*, vol. I. Boston/Berlin: Birkhäuser Verlag.
56. Schechter, M. 2002. *Principles of functional analysis*, 2nd ed. Rhode Island: American Mathematical Society Providence.
57. Gohberg, I., S. Goldberg, and M. A. Kaashoek. 1990. *Classes of linear operators*, vol. I. Boston/Berlin: Birkhäuser Verlag.
58. Denisov, M. S. 2005. Invertibility of linear operators in a Krein space (Vestnik Voronezhskogo gosudarstvennogo universiteta). *Mathematical Physics* 1 (2): 130–134.
59. Anosov, D. V. 1960. On limit cycles of systems of differential equations with small parameter at the highest derivatives. *Matematicheskii Sbornik* 92 (3): 299–334.
60. Bai, E., and A. Chen. 2010. On Feasibility of variable separation method based on hamiltonian systems for a class of plate bending equations. *Communications in Theoretical Physics* 53 (3): 569–574.
61. Hess, P., and T. Kato. 1970. Perturbation of closed operators and their adjoints. *Commentarii Mathematici Helvetici* 45 (1): 524–529.
62. Reed, M., and B. Simon. 1978. *Methods of modern mathematical physics. II. Fourier analysis, self-adjointness*. New York/London: Academic.
63. Paige, C., and C. V. Loan. 1981. A Schur decomposition for Hamiltonian Mareices. *Linear Algebra and Its Applications* 41: 11–32.
64. Gustafson, K. 2011. On operator sum and product adjoints and closures. *Canadian Mathematical Bulletin* 54 (3): 456–463.
65. Gustafson, K. 1969. On projections of selfadjoint operators and operator product adjoints. *Bulletin of the American Mathematical Society* 75 (4): 739–741.
66. Goldberg, M., and E. Tadmor. 1982. On the numerical radius and its applications. *Linear Algebra and Its Applications* 42: 263–284.
67. Radjavi, H., and P. Rosenthal. 1973. *Invariant subspaces*. Berlin, New York: Springer.
68. Pearcy, C. 1966. An elementary proof of the power inequality for the numerical radius. *The Michigan Mathematical Journal* 13 (3): 289–291.
69. Tretter, C. 2008. *Spectral theory of block operator matrices and applications*. London: Imperial College Press.

70. Langer, H., A. Markus, V. Matsaev, and C. Tretter. 2001. A new concept for block operator matrices:the quadratic numerical range. *Linear Algebra and Its Applications* 330 (1–3): 89–112.
71. Kittaneh, F. 2005. Numerical radius inequalities for Hilbert space operators. *Studia Mathematica* 168 (1): 73–80.
72. Kittaneh, F. 2003. A numerical radius inequality and an estimate for the numerical radius of the Frobenius companion matrix. *Studia Mathematica* 158 (1): 11–17.
73. Dragomir, S. S. 2008. Reverse inequalities for the numerical radius of linear operators in hilbert spaces. *Tamkang Journal of Mathematics* 73 (1): 255–262.
74. Holbrook, J. A. R. 1969. Multiplicative properties of the numerical radius in operator theory. *Journal für die Reine und Angewandte Mathematik* 237: 166–174.
75. Deeds, J. B. 1969. A proof of Fuglede's theorem. *Journal of Mathematical Analysis and Applications* 27 (1): 101–102.
76. Suen, C. Y. 1995. The minimum norm of certain completely positive maps. *Proceedings of the American Mathematical Society* 123 (8): 2407–2416.
77. Hardy, G. H., J. E. Littlewood, and G. Polya. 1988. *Inequalities*, 2nd ed. Cambridge: Cambridge University Press.
78. Hajmohamadi, M., R. Lashkaripour, and M. Bakherad. 2018. Some generalizations of numerical radius on off-diagonal part of 2×2 operator matrices. *Journal of Mathematics Inequalites* 12 (2): 447–457.
79. Fillmore, P. A., J. G. Stampfli, and J. P. Williams. 1972. On the essential numerical range, the essential spectrum, and a problem of Halmos. *Acta Scientiarum Mathematicarum (Szeged)* 33 (197): 179–192.
80. Gustafson, K., and J. Weidmann. 1969. On the essential spectrum. *Journal of Mathematical Analysis and Applications* 25 (1): 121–127.
81. Wolf, F. 1959. On the invariance of the essential spectrum under a change of the doundary conditions of partial differential operators. *Indagations Mathematicae* 21: 142–147.
82. Schechter, M. 1965. Invariance of essential spectrum. *Bulletin of the American Mathematical Society* 71 (2): 365–367.
83. Barraa, M., and V. Mller. 2005. On the essential numerical range. *Acta Scientiarum Mathematicarum* 71 (1): 285–298.
84. Bakić, D. 1998. Compact operators, the essential spectrum and the essential numerical range. *Mathematical Communications* 3 (1): 103–108.
85. Bakić, D., and B. Guljaš. 1998. Which operators approximately annihilate orthonormal bases? *Acta Scientiarum Mathematicarum* 64 (3): 601–607.
86. Chui, C., P. Smith, and R. R. Smith. 1977. L-ideals and numerical range preservation. *Illinois Journal of Mathematics* 21 (2): 365–373.
87. Legg, D. A., and D. W. Townsend. 1981. Essential Numerical Range in $B(l_1)$. *Proceedings of the American Mathematical Society* 81 (4): 541–545.
88. Dirac, P. A. M. 1942. The physical interpretation of quantum mechanics. *Proceedings of the Royal Society* 180: 1–40.
89. Pontrjagin, L. S. 1944. Hermitian operators in space with indefinite metric. *Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya* 1: 243–280.
90. Xia, D. X., and S. Z. Yan. 1987. *Spectral theory of linear operators II operator theory on indefinite metric space*. Beijing: Science Press.
91. Bognár, J. 1974. *Indefinite inner product spaces*. Berlin: Springer.
92. Azizov, T. Y., and I. S. Iokhvidov. 1989. *Linear operators in spaces with an indefinite metric*. New York: Wiley.

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